STRUCTURAL ENGINEERING

Mechanics of Continua

A. CEMAL ERINGEN
Professor, Aerospace and Mechanical Sciences
Princeton University

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to Jean, Meva, Peri, Lisa, Leyla

Preface

Students of engineering, applied mathematics, and physics are usually exposed in differing doses to instructions in various branches of classical continuum mechanics. A sound training in this field is often planned by subjecting the learner to several courses in fluid and solid mechanics both at the undergraduate and graduate levels. Hopefully, at the end, the student recognizes the common ground upon which are built the diverse fields of continuum mechanics.

This book was planned some eight years ago to bring efficiency to this process by exposing students in the latter part of their undergraduate training to a unified approach, one that emphasizes the basic principles of continuum mechanics and the central role these principles play in the formulation of the fundamental equations of elasticity, fluid dynamics, heat conduction, thermoelasticity, and viscoelastic materials. The students can then take some courses of specialization to learn the mathematical methods for solving more sophisticated boundary-value problems. In this sequence, the foundation is not repeated in various fragmentary forms with each course but is used to tie the vast fields of mechanics of continua together. The plan has been in practice at Purdue University since 1954, and has proved to be successful in spite of its evolving nature.

During the last decade or so, the pace of significant research on the foundations of continuum physics has been so fast that even the experts are finding it difficult to keep abreast of recent advances. The mechanics of continua is now on firm ground and the need for teaching this integrated point of view is unquestioned in nearly all major universities. It is my hope that this book will provide a text for such an exposition. It can be covered in a sequence of two courses at senior undergraduate or junior graduate level, or as one course covering Chapters 1 through 5 on basic principles,

along with some selected problems from the remaining chapters. It should be suitable for use in various schools of engineering, engineering sciences, applied mechanics, applied mathematics, and applied physics, where a definite program to cultivate the mathematical and physical foundations of the mechanics of continua exists.

The book consists of nine chapters and three appendices. Chapter 1 deals with the study of local deformation, strain, and rotation. Chapter 2 is devoted to a discussion of motion, a subject that centers around the kinematics of deformable bodies The concept of mass and global balance laws are postulated Chapter 3 is concerned with the concept of stress and the balance of momenta. Chapter 4 presents a discussion of the thermodynamics of continuous media The law of conservation of energy and the most recent formulation of the principle of entropy are given. The basic laws of motion of thermomechanics are now complete. Chapter 5 is an exposition of the recent unified approach to constitutive theory. Based on physical foundations, the fundamental axioms of constitution are laid down and used to obtain the constitutive equations of a large class of thermomechanical materials Here, the power and the aesthetics of the integrated approach comes to the surface. Constitutive equations of nonlinear elastic solids, viscous fluids, thermoelasticity, and thermoviscous fluids are developed to illustrate the method. With Chapter 5 the formulation is complete.

The later chapters, 6 to 9, are written for the purpose of applying various theories to specific problems. In each chapter, first the linear theory is developed, then the basic field equations, jump conditions, and boundary and initial conditions are summarized; and then the application is made to the solution of certain problems for illustrative purposes. Chapters 6, 7, and 9 also contain accounts of the experimental determination of material moduli. The problems are carefully selected so as to illustrate a variety of physical phenomena and yet permit exact solutions. The discussion of approximate methods of solutions is considered to be beyond the scope of this book, belonging instead, to specialized treatises. Similar types of problems are treated for each material medium in order to *demonstrate the differences arising from the material constitution.* Thus, for example, half space, or a cylindrical medium subject to pressure loading and spherical blast, appears in Chapter 6 on elasticity, in Chapter 7 on fluid dynamics, in Chapter 8 on thermoelasticity, and in Chapter 9 on viscoelasticity. Most of the problems treated are of a dynamical nature. Various problems selected from the fields of wave propagations, fluid dynamics, shock waves, heat conduction, thermoelasticity, and viscoelasticity should provide sufficient illustrations of the common origin of a variety of physical phenomena and of the strength of the unified theory.

Chapter 9 returns to formulation, this time for memory-dependent materials. Mechanical and thermodynamic foundations of viscoelastic materials are covered before the illustrative examples. In this way, the student is not given too large a dose while studying Chapter 5 on constitutive theory.

Cartesian tensors are used throughout the book with transitions indicated to curvilinear coordinates in later chapters. Appendices on vectors, matrices, and tensors are provided for help in this direction as well as for self-study.

The approach and composition of the book are believed to be novel. In addition, a number of new results that have not appeared elsewhere are included. This book should provide a sound prerequisite for later studies in the nonlinear theory of continuous media.

Over the years, many different versions of the text have been written and revised as changes have occurred in the field. Professor B. Boley was kind enough to read and criticize several chapters of an early manuscript. I am indebted to him for the help I received from these criticisms when I came to make later compositions. The first three chapters of an early manuscript were read by Professor C. Truesdell, and a chapter on irreversible thermodynamics (subsequently abandoned) was read by Professor C. F. Curtiss. I am indebted to both gentlemen for certain valuable points of view. Professors S. L. Koh and L. V. Kline have used portions of the manuscript at its various stages and commented on several chapters. Some chapters were distributed privately to students and interested colleagues in other universities. Their encouragements for publication are most appreciated. Over the years, many of my past and present students have read parts of the manuscript. I gratefully remember Drs. J. W. Dunkin, N. F. Jordan, J. D. Ingram, E. Suhubi, R. Beil, and V. Parfitt for reading parts of various early versions and R. Grot and C. Thiel for reading and criticizing parts of the present. Without the careful proofreading and checking of this entire manuscript by Messrs. W. D. Claus Jr. and C. B. Kafadar, the work would not have gone to press. To them goes my deepest appreciation. Considerable assistance in drafting was rendered by Dr. T. Ariman.

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Mechanics of Continua

Introduction

Continuum mechanics is a branch of the physical sciences concerned with the deformations and motions of continuous material media (called bodies) under the influence of external effects. External effects that influence bodies appear in the form of forces, displacements, and velocities which arise from contact with other bodies, gravitational forces, thermal changes, chemical interactions, electromagnetic effects, and other environmental changes. In this book we are concerned only with bodies subject to forces of mechanical origin and/or thermal changes. The nature of these forces and their physical measurements are considered to be integral parts of any program in this discipline.

Physical bodies appear in the form of a collection of material points interconnected by some internal forces. In certain branches of physics, for example, statistical mechanics and atomic physics, the physical phenomena are considered to result from the interaction of a large number of small particles, atoms, and molecules. The various types of statistical averages of certain notions give rise to global phenomena. For example, the mean intrinsic kinetic energy of a large number of particles appears as the temperature. This point of view has certainly contributed a great deal to our knowledge of the physical phenomena. In continuum mechanics, however, this is not the point of view taken. Because, in general, the subject matter is concerned with the macroscopic phenomena in which the smallest characteristic length is much larger than the size of an atom, the fundamental assumption of continuity of mass represents an excellent approximation in support of mathematical idealization. The strength of the statistical laws in the molecular chaos of particles is basic in this outcome.

Moreover, most instruments measure the statistical averages that characterize the global phenomena The mechanics of continua thus makes sense as an independent discipline on these grounds, even if we were to disregard its long list of successes

Response of a body to external loads depends upon its material constitution. The characterization of the physical nature of the body is essential in the investigation of its reponse to external effects. All solids (steel, aluminum, rubber, wood, stone, bone, etc.) deform when subjected to forces or heat; all fluids (water, olive oil, castor oil, blood, etc.) flow and take the shape of their contaioers as do all gases (air, water vapor, argon, oitrogen, oxygen, etc.). Such terms as solids, fluids, and gases or plasma are only crude descriptions of bodies. Generally, in more careful characterization of tbe media, no definite line of demarcation exists. Nevertheless, we benefit from these intuitive notions concerning solids, fluids, and gases in obtaining simple theories about a limited class of bodies. In other branches of physics the molecular nature of materials is used to acbieve the same objective In this latter case, because of our ignorance of the nature of molecular forces and of the necessity for gross mathematical approximations, the region of applicability of the theory is often limited and/or not clearly marked A precise continuum theory, here, provides the necessary guidance and security. The statistical, mechanical, and transport theories, on the other hand, can, at this meeting ground, supply the relations between the material properties (defined by continuum theory) and the molecular parameters. Consequently, the benefit in this joint venture is the improvement of material properties by readjustment of the material composition

In continuum mechanics, the material characterization is made by the establishment of some functional relations (called constitutive equations) among the constitutive variables. The selection of tbese variables and the restrictions on these equations are based on certaio axioms of constitution as well as oo the basic laws of mechanics and thermodynamics. It is remarkable that under some very general axioms we can obtain definite and concrete material theories.

The foregoing considerations indicate that tbe theory of continuous media is built upon two strong foundations: (1) basic laws of motion aod (2) a constitutive theory.

The Basic Laws of Motion. These are the fundamental axioms of motion that are valid for all bodies irrespective of their constitution. They are the result of our experience with the physical world. In this book we are concerned ooly with the thermomechanical phenomena governed by the followiog five basic laws: (1) Conservation of mass, (2) Balance of momen-

tum, (3) Balance of moment of momentum, (4) Conservation of energy, (5) Entropy. Within the realm of classical mechanics, excluding the relativistic and quantum phenomena, these laws are unchallenged.

Constitutive Theory. The responses of different material bodies of the same geometry to identical loads are, in general, different. To take account of the nature of different materials, a set of constitutive equations is constructed. These equations depend upon the nature and the range of physical effects desired. Certain basic axioms (cf. Chapter 5) are employed in the construction and restrictions of the constitutive equations. The resulting equations nevertheless contain some unknown material parameters that must be determined through experiments and/or statistical mechanical considerations. Once this is done, the theory is fully constructed and ready for use.

The first four chapters of this book are devoted to a systematic study of the above five laws of motion. The fifth chapter discusses the constitutive theory and its applications to various types of media. Chapters 6 to 9 contain widely varied applications to and solutions of some illustrative problems for different types of bodies.

CHAPTER 1

Strain

1.1. SCOPE OF THE CHAPTER

All substances under the influence of external loads undergo deformation and motion. The study of the local geometrical changes and motion of the points of continua is the purpose of this chapter. The relationship between the initial position of the material points of a body and their subsequent places is essential in the description of the local length and angle changes and translations and rotations of elements of the body. In this chapter we are concerned with such changes and their measures, irrespective of the type of substance and the external effects.

To describe the positions of material points we introduce in Section 1.2 two sets of rectangular coordinate systems, one for the undeformed body and one for the deformed body. The deformation of a point is then described by the relation of the coordinates of the same material point in the undeformed and the deformed states. Axioms of continuity and indestructibility of matter are discussed. Section 1.3 introduces base vectors and shifters that are essential to the component representation of vectors and their parallel transports. In Section 1.4 we discuss the deformation gradients and deformation tensors that are basic in the determination of local length and angle changes and their measures. Essential to these calculations are the Cauchy and Green tensors introduced in this section and the lagrangian and eulerian strain tensors given in Section 1.5. The relations of these tensors to deformation vectors are presented. The infinitesimal strains and rotations fundamental to the linear theory are presented in Section 1.6. The geometrical meaning of strains and rotations

is understood by the study of length and angle changes; this constitutes the main discussion of Section 1.7. Transformation laws that are obeyed by strains and rotations are given in the succeeding section. Through a geometrical study in Section 1.9 we learn how a small sphere is deformed into an ellipsoid. Thus the study of the strain ellipsoid of Cauchy clarifies the directional dependence of deformation at a point of the body. The strain invariants are discussed in Section 1.10 Section 1.11 presents a discussion of finite rotation and gives the fundamental theorem of deformation of an infiottesimal neighborhood of a material point.

In Section 1.12 we discuss area and volume changes that occur with deformation To achieve further iosight into various classes of deformations, wc present in Section 1.13 some simple examplcs of strain, for example, rigid deformations, potential dcformation, isochoric deformation, uniform dilatation, and simple shear. For the plane strain, we prcsent the geometrical construction of Mohr. The conditions under which a symmetrie oonnegative second-order tensor may be a deformation tensor are known as the compatibility conditions. Thcse are discussed in Section 1.14

1.2. COORDINATES

The material points of a contiouous medium at time $t = 0$ occupy a region B which consists of the material volume $\mathscr{V}$ and its surface $\mathscr{S}$. The position of a material poiot P io this region may be denotcd by a rectangular coordinate system X_1, X_2, and X_3 (or simply by X_K, $K = 1, 2, 3$), or by a vector $\mathbf{P}$ that exteods from an origin O of the coordinates to the point P (Figure 1.2.1). After deformation takes place, at time t the material points of $\mathscr{V} + \mathscr{S}$ occupy a region b consisting of a spatial volume v and its surface s In this deformed state, a material point may occupy a spatial point p. We may locate p by a vector $\mathbf{p}$ extending from the origin o of a new rectangular coordinate frame or by a set of rectangular coordinates $x_k (k = 1, 2, 3)$. Sometimes it may be advantageous to select these two reference frames to be nonidentical. The choice of two distinct coordinate systems, one for the undeformed body and one for the deformed body, is particularly suited in the case where the curvilinear coordinate systems are used. A typical example is the deformation of a rectangular block into a circular cylinder. The use of rectangular coordinates for the undeformed block and the use of cylindrical coordinates for the dcformed body may prove to be particularly advantageous.

In the description of the deformation and motion of a continuous body, the use of two sets of coordinates (even when both are chosen to be rectangular) makes many subtle points clearly understood. When it becomes advantageous to express iodividual components, we write $X_1 = X, X_2 = Y, X_3 = Z$ and $x_1 = x, x_2 = y, x_3 = z$. Followiog the current

terminology, we shall call X_K the *material* or *lagrangian* coordinates and x_k the *spatial* or *eulerian* coordinates.

The deformation and motion of the body carries various material points through various spatial positions.

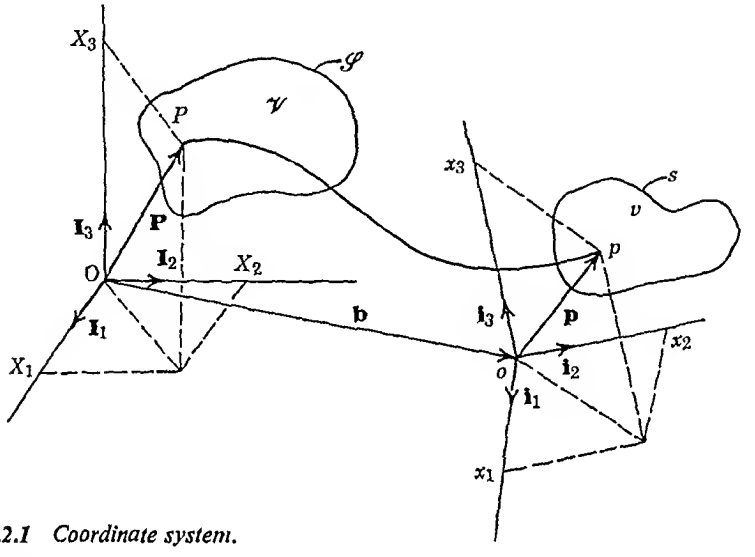


Fig. 1.2.1 Coordinate system.

This is expressed by

$$x_k = x_k(X_1, X_2, X_3, t) \quad \text{or} \quad x_k = x_k(X_K, t) \qquad (k = 1, 2, 3) \tag{1.2.1}$$

or conversely

$$X_K = X_K(x_1, x_2, x_3, t) \quad \text{or} \quad X_K = X_K(x_k, t) \qquad (K = 1, 2, 3). \tag{1.2.2}$$

For brevity we write these as

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t), \qquad \mathbf{X} = \mathbf{X}(\mathbf{x}, t) \tag{1.2.3}$$

Thus, according to (1.2.1), the motion takes a material point P in B at $t = 0$ to a spatial position p in b at time t. The inverse motion (1.2.2) states the inverse phenomena, namely, that we can trace the material point occupying the spatial position $\mathbf{p}$ at time t to its original position $\mathbf{P}$.

We assume that the mappings (1.2.3) are single-valued and possess continuous partial derivatives with respect to their arguments for whatever order is desired, except possibly at some singular points, curves, and surfaces. Moreover, each member of (1.2.3) is the unique inverse of the other in a neighborhood of the material point P. This assumption is known as the *axiom of continuity*. It expresses the fact that the matter is *indestructible*, that is, no region of positive, finite volume of matter is deformed into a zero or infinite volume. Another implication of this axiom is that the

matter is *impenetrable*, that is, the motion carries every region into a region, every surface into a surface, and every curve into a curve. One portion of matter never penetrates into another. In practice, there are cases in which this axiom is violated. For example, materials may break or may transmit shock and other types of discontinuities. Special attention must be given to these. The axiom of continuity is secured through a well-known theorem of calculus known as the implicit function theorem.[1]

THEOREM (IMPLICIT FUNCTION). *If for a fixed time t the functions $x_k(X_K, t)$ are continuous and possess continuous first-order partial derivatives with respect to X_K in a neighborhood $|X'_K - X_K| < \Delta$ of the point P, and if the jacobian*

$$j \equiv \left| \frac{\partial x_k}{\partial X_K} \right| = \begin{vmatrix} \partial x_1/\partial X_1 & \partial x_1/\partial X_2 & \partial x_1/\partial X_3 \\ \partial x_2/\partial X_1 & \partial x_2/\partial X_2 & \partial x_2/\partial X_3 \\ \partial x_3/\partial X_1 & \partial x_3/\partial X_2 & \partial x_3/\partial X_3 \end{vmatrix} \tag{1.2.4}$$

does not vanish there, then a unique inverse of the form (1.2.2) *exists in a neighborhood $|x'_k - x_k| < \delta$ of a point p at time t.*

If we determine p as a function of time for each material point P (that is, given P), we can construct the new shape and position of the body at each time t relative to that at $t = 0$. This enables us to calculate the change of length between any two points and the change of angle between any two directions. The ultimate goal is to relate these deformations to the external effects (for example, external forces, thermal changes). With the possession of such a knowledge, one may hope to design machines and buildings or analyse the existing natural or man-made materials and structures so that not only can failure be avoided, but also maximum performance can be achieved. Thus, the subject matter of continuous media deals, in essence, with the determination of the explicit form of (1.2.1) when the external effects and the initial and boundary conditions of the body are known.

The quantities associated with the undeformed body B will be denoted by majuscules, and those associated with the deformed body b by minuscules. When these quantities are referred to coordinates X_K, their indices will be majuscules; and when they are referred to x_k, their indices will be minuscules. For example, a vector V in B referred to X_K will have components V_K, and referred to x_k will have the components V_k. Conversely, a vector v in b referred to X_K and x_k will have components denoted by v_K and v_k,[2] respectively.

[1] Cf. Widder [1950, p. 47].

[2] The notation used here closely follows that of Eringen [1962].

1.3. BASE VECTORS, SHIFTERS

The position vectors $\mathbf{P}$ of a point P in B and $\mathbf{p}$ of p in b, respectively, referred to rectangular coordinates X_K and x_k are given by

$$\mathbf{P} = X_K \mathbf{I}_K, \qquad \mathbf{p} = x_k \mathbf{i}_k \tag{1.3.1}$$

where $\mathbf{I}_K$ and $\mathbf{i}_k$ are, respectively, the *unit base vectors* in Figure (1.2.1). Henceforth we employ the usual summation convention over the repeated indices, that is,

$$\mathbf{P} = X_K \mathbf{I}_K = X_1 \mathbf{I}_1 + X_2 \mathbf{I}_2 + X_3 \mathbf{I}_3$$

Infinitesimal vectors $d\mathbf{P}$ in B and $d\mathbf{p}$ in b may be expressed as

$$d\mathbf{P} = dX_K \mathbf{I}_K, \qquad d\mathbf{p} = dx_k \mathbf{i}_k \tag{1.3.2}$$

The square of the elements of length in B and b are, respectively,

$$\begin{aligned} dS^2 &= d\mathbf{P} \cdot d\mathbf{P} = \delta_{KL}\, dX_K\, dX_L = dX_K\, dX_K \\ ds^2 &= d\mathbf{p} \cdot d\mathbf{p} = \delta_{kl}\, dx_k\, dx_l = dx_k\, dx_k \end{aligned} \tag{1.3.3}$$

where

$$\delta_{KL} = \mathbf{I}_K \cdot \mathbf{I}_L, \qquad \delta_{kl} = \mathbf{i}_k \cdot \mathbf{i}_l \tag{1.3.4}$$

are the *Kronecker symbols*. Whenever the two indices are equal, they have the value 1; otherwise they are equal to zero.

When the two rectangular coordinates are not identical, we shall also be using symbols with mixed indices such as

$$\delta_{Kk} = \delta_{kK} = \mathbf{I}_K \cdot \mathbf{i}_k \tag{1.3.5}$$

These are called *shifters*. They play an important role in expressing a vector in one frame (say x_k) in terms of its projections in another frame (say X_K). Thus, suppose $\mathbf{v}$ is a vector in the frame x_k; then

$$\mathbf{v} = v_k(\mathbf{x})\mathbf{i}_k$$

The components V_K of this vector in X_K are related to v_k by

$$V_K = \mathbf{v} \cdot \mathbf{I}_K = v_k(\mathbf{x})\mathbf{i}_k \cdot \mathbf{I}_K = \delta_{Kk} v_k \tag{1.3.6}$$

When the two frames are identical, δ_{Kk} becomes a Kronecker delta so that the vector $\mathbf{v}$ is simply shifted from a point $\mathbf{p}$ to another point $\mathbf{P}$ by parallel transport. Thus, the name shifter. Note that the dual of (1.3.6) is

$$v_k = \delta_{Kk} V_K \tag{1.3.7}$$

so that we have

$$\mathbf{v} = v_k \mathbf{i}_k = V_K \mathbf{I}_K \tag{1.3.8}$$

By carrying one of (1.3.6), (1.3.7) into the other we find

$$\delta_{Kk}\delta_{Lk} = \delta_{KL}, \qquad \delta_{Kk}\delta_{Kl} = \delta_{kl} \tag{1.3.9}$$

We emphasize again the fact that δ_{Kk} is not, in general, a Kronecker symbol (that is, in general, $\delta_{Kk} \neq 1$ if K and k are the same numbers, and nonzero if they are different). They are the cosine directors of $\mathbf{i}_k$ with respect to $\mathbf{I}_K$. The importance of these symbols will soon appear in the course of our study.

1.4. DEFORMATION GRADIENTS AND DEFORMATION TENSORS

Through the equations of motion (1.2.1) and (1.2.2) we have

$$dx_k = x_{k,K}\, dX_K, \qquad dX_K = X_{K,k}\, dx_k \tag{1.4.1}$$

where indices after a comma indicate partial differentiation with respect to X_K when they are majuscules, and with respect to x_k when they are minuscules, that is,

$$x_{k,K} \equiv \frac{\partial x_k}{\partial X_K}, \qquad X_{K,k} \equiv \frac{\partial X_K}{\partial x_k} \tag{1.4.2}$$

The two sets of quantities defined by (1.4.2) are called *deformation gradients.* Through the chain rule of partial differentiation it is clear that

$$x_{k,K}X_{K,l} = \delta_{kl}, \qquad X_{K,k}x_{k,L} = \delta_{KL} \tag{1.4.3}$$

Each one of these two sets is a set of nine linear equations for the nine unknowns $x_{k,K}$ or $X_{K,k}$. A unique solution exists, since the jacobian of the transformation is assumed not to vanish. Using Cramer's rule of determinants, the solution for $X_{K,k}$ may be obtained in terms of $x_{k,K}$. Thus

$$X_{K,k} = \frac{\text{cofactor } x_{k,K}}{j} = \frac{1}{2j}\, e_{KLM}e_{klm}x_{l,L}x_{m,M} \tag{1.4.5}$$

where e_{KLM} and e_{klm} are the permutation symbols (cf. App. A5) and

$$j \equiv |x_{k,K}| = \tfrac{1}{6}e_{KLM}e_{klm}x_{k,K}x_{l,L}x_{m,M} \tag{1.4.6}$$

is the jacobian. By differentiating (1.4.6), we obtain an important identity of Jacobi

$$\frac{\partial j}{\partial x_{k,K}} = \text{cofactor } x_{k,K} = jX_{K,k} \tag{1.4.7}$$

If we substitute (1.4.1) into (1.3.2), we obtain

$$d\mathbf{P} = \mathbf{c}_k\, dx_k, \qquad d\mathbf{p} = \mathbf{C}_K\, dX_K \tag{1.4.8}$$

where

$$\mathbf{c}_k(\mathrm{x}, t) \equiv X_{K,k}\mathbf{I}_K, \qquad \mathbf{C}_K(\mathrm{X}, t) \equiv x_{k,K}\mathbf{i}_k \tag{1.4.9}$$

From these, we may solve $\mathbf{I}_K$ and $\mathbf{i}_k$. For example, multiply $(1.4.9)_1$ by $x_{k,L}$ and use $(1.4.3)_2$ to solve for $\mathbf{I}_K$, and similarly multiply $(1.4.9)_2$ by $X_{K,l}$ and use $(1.4.3)_1$ to solve $\mathbf{i}_k$. Thus

$$\mathbf{I}_K = x_{k,K}\mathbf{c}_k, \qquad \mathbf{i}_k = X_{K,k}\mathbf{C}_K \tag{1.4.10}$$

In the same way as in (1.3.3), by use of (1.4.8), we see that

$$dS^2 = c_{kl}\, dx_k\, dx_l, \qquad ds^2 = C_{KL}\, dX_K\, dX_L \tag{1.4.11}$$

where

$$\begin{aligned} c_{kl}(\mathrm{x}, t) &\equiv \mathbf{c}_k \cdot \mathbf{c}_l = \delta_{KL}X_{K,k}X_{L,l} = X_{K,k}X_{K,l} \\ C_{KL}(\mathrm{X}, t) &\equiv \mathbf{C}_K \cdot \mathbf{C}_L = \delta_{kl}x_{k,K}x_{l,L} = x_{k,K}x_{k,L} \end{aligned} \tag{1.4.12}$$

are, respectively, *Cauchy's deformation tensor and Green's deformation tensor. Both of these quantities are symmetric, that is,* $c_{kl} = c_{lk}$, $C_{KL} = C_{LK}$ *and are positive definite.*

Two other vectors, $\overset{-1}{\mathbf{c}}_k$ and $\overset{-1}{\mathbf{C}}_K$, may be defined by[3]

$$\overset{-1}{\mathbf{c}}_k(\mathrm{x}, t) \equiv x_{k,K}\mathbf{I}_K, \qquad \overset{-1}{\mathbf{C}}_K(\mathrm{X}, t) \equiv X_{K,k}\mathbf{i}_k \tag{1.4.13}$$

of which the first satisfies relations

$$\begin{aligned} \mathbf{c}_k \cdot \overset{-1}{\mathbf{c}}_l &= (X_{K,k}\mathbf{I}_K) \cdot (x_{l,L}\mathbf{I}_L) = X_{K,k}x_{l,L}\delta_{KL} \\ &= X_{K,k}x_{l,K} = \delta_{kl} \end{aligned} \tag{1.4.14}$$

and the second a similar one. Thus $\overset{-1}{\mathbf{c}}_k$ and $\overset{-1}{\mathbf{C}}_K$ are *reciprocal vectors*, respectively, to $\mathbf{c}_k$ and $\mathbf{C}_K$.

From reciprocal vectors we may form

$$\begin{aligned} \overset{-1}{c}_{kl} &\equiv \overset{-1}{\mathbf{c}}_k \cdot \overset{-1}{\mathbf{c}}_l = \delta_{KL}x_{k,K}x_{l,L} \\ \overset{-1}{C}_{KL} &\equiv \overset{-1}{\mathbf{C}}_K \cdot \overset{-1}{\mathbf{C}}_L = \delta_{kl}X_{K,k}X_{L,l} \end{aligned} \tag{1.4.15}$$

[3] Superscript -1 is part of the symbol; it does not denote $1/\mathbf{c}_k$.

We now see that

$$\begin{aligned}\overset{-1}{c}_{kl}c_{lm} &= (\delta_{KL}x_{k,K}x_{l,L})(\delta_{MN}X_{M,l}X_{N,m})\\ &= x_{k,K}(x_{l,K}X_{M,l})X_{M,m} = x_{k,K}\delta_{KM}X_{M,m}\\ &= x_{k,K}X_{K,m} = \delta_{km}\end{aligned}$$

Since c_{kl} is a nonsingular matrix, we see that the tensor $\overset{-1}{c}_{kl}$ is the *reciprocal tensor* to c_{kl}. Similarly, it can be shown that $\overset{-1}{C}_{KL}$ is the reciprocal tensor to C_{KL}. In fact, by carrying (1.4.10)$_1$ into (1.4.13)$_1$ and the result into (1.4.15) it is not difficult to show that

$$\begin{aligned}\overset{-1}{c}_{k} &= X_{k,K}X_{l,K}c_{l}\\ \overset{-1}{c}_{kl} &= X_{k,K}X_{m,K}X_{l,M}X_{n,M}c_{mn} = \overset{-1}{c}_{km}\overset{-1}{c}_{ln}c_{mn}\end{aligned} \tag{1.4.16}$$

We have been using the word *tensor* for quantities such as C_{KL} and c_{kl}. This term refers to a set of quantities that transform according to a certain definite law upon coordinate transformation. Suppose that coordinates X_K are transformed into X'_K according to

$$X_K = X_K(X'_1, X'_2, X'_3) \tag{1.4.17}$$

The left-hand side of (1.4.11)$_2$ is independent of the coordinate transformations. If on the right-hand side we put

$$dX_K = \frac{\partial X_K}{\partial X'_M}\,dX'_M$$

we get

$$ds^2 = C_{KL}\frac{\partial X_K}{\partial X'_M}\frac{\partial X_L}{\partial X'_N}\,dX'_M\,dX'_N = C'_{MN}\,dX'_M\,dX'_N$$

Hence

$$C'_{MN}(X', t) = C_{KL}(X, t)\frac{\partial X_K}{\partial X'_M}\frac{\partial X_L}{\partial X'_N} \tag{1.4.18}$$

since dX'_M is arbitrary and $C_{KL} = C_{LK}$. Thus, knowing C_{KL} in one set of coordinates X_K, we can find the corresponding quantities C'_{KL} in another set X'_K once the relations (1.4.17) between X_K and X'_K are given. Quantities that transform according to the law of transformation (1.4.18) are known as *absolute tensors*. It is not difficult to show that c_{kl}, $\overset{-1}{c}_{kl}$, and $\overset{-1}{C}_{KL}$ are all absolute tensors.

Similarly, an absolute vector $V_K(X)$ transforms according to

$$V'_M(X') = V_K(X)\frac{\partial X_K}{\partial X'_M} \tag{1.4.19}$$

For further detail on this subject, see Appendix C.

1.5. STRAIN TENSORS AND DISPLACEMENT VECTOR

From (1.3.3) and (1.4.11) we have two different expressions of the squares of element of length, dS^2 in the undeformed body and ds^2 in the deformed body

$$\begin{aligned} dS^2 &= \delta_{KL}\, dX_K\, dX_L = c_{kl}\, dx_k\, dx_l \\ ds^2 &= C_{KL}\, dX_K\, dX_L = \delta_{kl}\, dx_k\, dx_l \end{aligned} \tag{1.5.1}$$

The difference $ds^2 - dS^2$ for the same material points in B and b is a measure of the change of length. When this difference vanishes for any two neighboring points, the deformation has not changed the distance between the pair. When it is zero for all points in the body, the body has undergone only a *rigid displacement*.

Thus, from (1.5.1) for this difference, in the same coordinate system, we obtain

$$ds^2 - dS^2 = 2E_{KL}(\mathbf{X}, t)\, dX_K\, dX_L = 2e_{kl}(\mathbf{x}, t)\, dx_k\, dx_l \tag{1.5.2}$$

where

$$\begin{aligned} 2E_{KL} &\equiv C_{KL}(\mathbf{X}, t) - \delta_{KL} \\ 2e_{kl} &\equiv \delta_{kl} - c_{kl}(\mathbf{x}, t) \end{aligned} \tag{1.5.3}$$

are called *lagrangian* and *eulerian strain tensors*, respectively. Clearly, when either vanishes, $ds^2 = dS^2$.

From (1.5.2) we can see that

$$E_{KL} = e_{kl} x_{k,K} x_{l,L}, \qquad e_{kl} = E_{KL} X_{K,k} X_{L,l} \tag{1.5.4}$$

We may express the strain components in terms of the *displacement vector* $\mathbf{u}$ that extends from a material point P in the undeformed body to its spatial position in the deformed body (Figure 1.5.1):

$$\mathbf{u} \equiv \mathbf{p} - \mathbf{P} + \mathbf{b} = x_l \mathbf{i}_l - X_L \mathbf{I}_L + \mathbf{b} \tag{1.5.5}$$

Components of $\mathbf{u}$ in X_K or in x_k are denoted by U_K and u_k, respectively, so that

$$\mathbf{u} = U_L \mathbf{I}_L = u_l \mathbf{i}_l$$

By taking the inner product of both sides of (1.5.5) by $\mathbf{i}_k$ or $\mathbf{I}_K$, we obtain

$$\begin{aligned} u_k &= x_k - \delta_{Lk} X_L + b_k \\ U_K &= \delta_{Kl} x_l - X_K + B_K \end{aligned} \tag{1.5.6}$$

where $b_k = \mathbf{b} \cdot \mathbf{i}_k$ and $B_K = \mathbf{b} \cdot \mathbf{I}_K$. Here again we see the appearance of shifters.

We now calculate the strain tensors. First from (1.4.8) and (1.5.5) we have

$$\mathbf{C}_K = \frac{\partial \mathbf{p}}{\partial X_K} = \frac{\partial \mathbf{P}}{\partial X_K} + \frac{\partial \mathbf{u}}{\partial X_K} = \mathbf{I}_K + U_{M,K}\mathbf{I}_M$$

(1.5.7)

$$\mathbf{c}_k = \frac{\partial \mathbf{P}}{\partial x_k} = \frac{\partial \mathbf{p}}{\partial x_k} - \frac{\partial \mathbf{u}}{\partial x_k} = \mathbf{i}_k - u_{m,k}\mathbf{i}_m$$

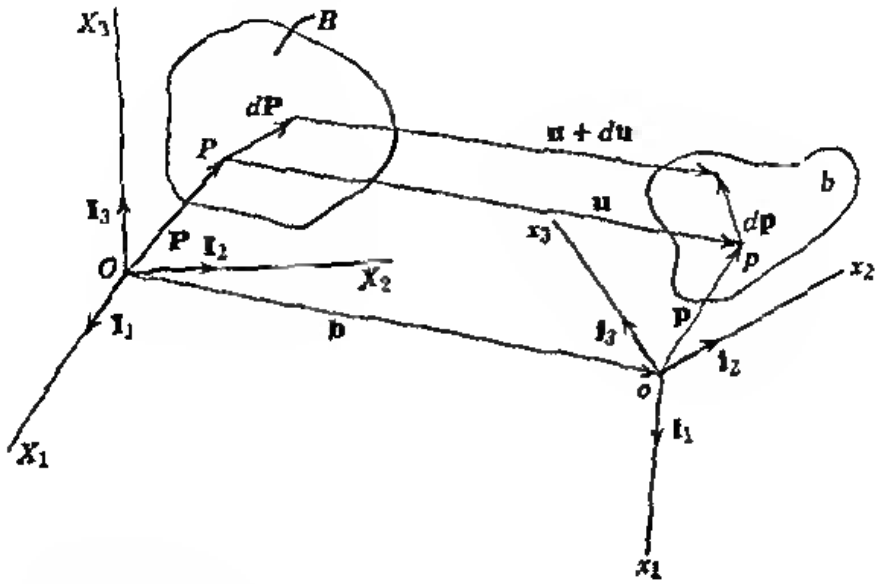


Fig. 1.5.1 Displacement vector

Now calculate

$$C_{KL} = \mathbf{C}_K \cdot \mathbf{C}_L = (\mathbf{I}_K + U_{M,K}\mathbf{I}_M) \cdot (\mathbf{I}_L + U_{N,L}\mathbf{I}_N)$$
$$= \delta_{KL} + U_{K,L} + U_{L,K} + \delta_{NM}U_{M,K}U_{N,L}$$

Substituting this into $(1.5.3)_1$, we obtain

(1.5.8) $$2E_{KL} = U_{K,L} + U_{L,K} + \delta_{MN}U_{M,K}U_{N,L}$$

A similar procedure starting with $\mathbf{c}_k$ of (1.5.7) gives

(1.5.9) $$2e_{kl} = u_{k,l} + u_{l,k} - \delta_{mn}u_{m,k}u_{n,l}$$

From (1.5.7) it is clear that

(1.5.10)
$$d\mathbf{p} = \mathbf{C}_K\, dX_K = (\delta_{MK} + U_{M,K})\mathbf{I}_M\, dX_K$$
$$d\mathbf{P} = \mathbf{c}_k\, dx_k = (\delta_{mk} - u_{m,k})\mathbf{i}_m\, dx_k$$

Since the deformation carries $d\mathbf{P}$ into $d\mathbf{p}$, referred to coordinates X_K, the rectangular parallelepiped with side vectors $\mathbf{I}_1\, dX_1$, $\mathbf{I}_2\, dX_2$, and $\mathbf{I}_3\, dX_3$ after

deformation is carried into a curvilinear parallelepiped with side vectors $\mathbf{C}_1\,dX_1$, $\mathbf{C}_2\,dX_2$, and $\mathbf{C}_3\,dX_3$ (Figure 1.5.2). Similarly, in the eulerian picture of the deformation the inverse motion carries $\mathbf{i}_1\,dx_1$, $\mathbf{i}_2\,dx_2$, and

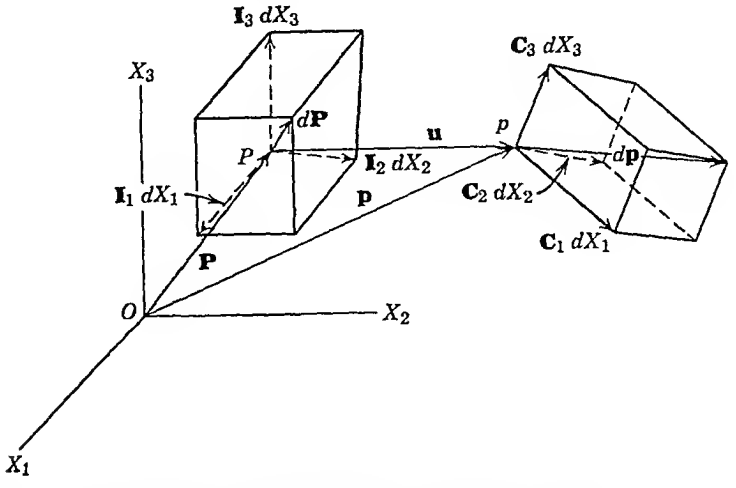


Fig. 1.5.2 Deformation of a parallelepiped (lagrangian representation).

$\mathbf{i}_3\,dx_3$ into $\mathbf{c}_1\,dx_1$, $\mathbf{c}_2\,dx_2$, and $\mathbf{c}_3\,dx_3$, respectively (Figure 1.5.3). By writing $d\mathbf{p} = dx_k\mathbf{i}_k$ and $d\mathbf{P} = dX_K\mathbf{I}_K$ in (1.5.10) we also find two useful

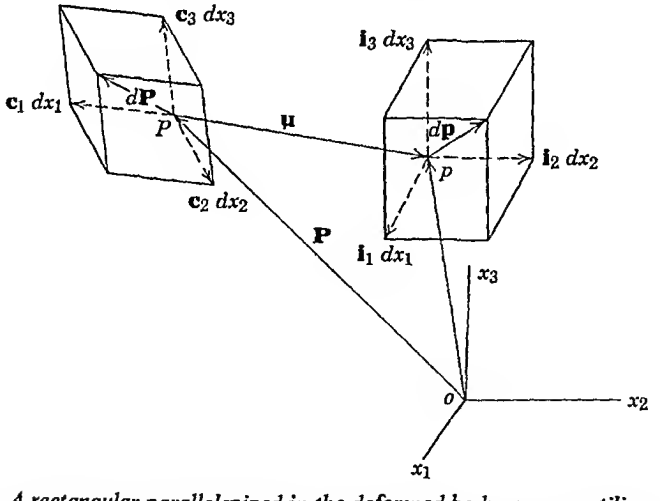


Fig. 1.5.3 A rectangular parallelepiped in the deformed body was a rectilinear one in the undeformed body (eulerian representation).

expressions

$$dx_k = (\delta_{MK} + U_{M,K})\delta_{Mk}\,dX_K$$
$$dX_K = (\delta_{mk} - u_{m,k})\delta_{mK}\,dx_k \tag{1.5.11}$$

Returning to strain tensors, we see that both E_{KL} and e_{kl} are symmetric tensors, that is,

$$E_{KL} = E_{LK}, \qquad e_{kl} = e_{lk} \tag{1.5.12}$$

Therefore, in three dimensions there are only six independent components for each of these tensors, for example, $E_{11}, E_{22}, E_{33}, E_{12} = E_{21}, E_{23} = E_{32}$, and $E_{31} = E_{13}$. The first three components E_{11}, E_{22}, and E_{33} are called *normal strains* and the last three E_{12}, E_{23}, and E_{31} are called *shear strains*. The reason for this will be discussed in Section 1.7.

Sometimes it is convenient to employ matrix notation. In this notation we write

$$E = \|E_{KL}\| = \begin{bmatrix} E_{11} & E_{12} & E_{13} \\ E_{21} & E_{22} & E_{23} \\ E_{31} & E_{32} & E_{33} \end{bmatrix} \tag{1.5.13}$$

This matrix is symmetric with respect to the main diagonal. A similar representation is valid for the eulerian strain tensor e.[4]

To indicate the advantage of the present compressed notation over the usual ones used in some textbooks, we now write (1.5.8) and (1.5.9) in their usual expanded forms. Thus, writing X, Y, Z for X_1, X_2, X_3; x, y, z for x_1, x_2, x_3; U, V, W for U_1, U_2, U_3, u, v, w for u_1, u_2, u_3; and $E_{XX}, E_{XY} \ldots$ for E_{11}, E_{12}, etc., we see that the six lagrangian strains E_{KL} and six eulerian strains e_{kl} read.

$$\begin{aligned}
2E_{XX} &= C_{XX} - 1 = 2\frac{\partial U}{\partial X} + \left(\frac{\partial U}{\partial X}\right)^2 + \left(\frac{\partial V}{\partial X}\right)^2 + \left(\frac{\partial W}{\partial X}\right)^2 \\
2E_{YY} &= C_{YY} - 1 = 2\frac{\partial V}{\partial Y} + \left(\frac{\partial U}{\partial Y}\right)^2 + \left(\frac{\partial V}{\partial Y}\right)^2 + \left(\frac{\partial W}{\partial Y}\right)^2 \\
2E_{ZZ} &= C_{ZZ} - 1 = 2\frac{\partial W}{\partial Z} + \left(\frac{\partial U}{\partial Z}\right)^2 + \left(\frac{\partial V}{\partial Z}\right)^2 + \left(\frac{\partial W}{\partial Z}\right)^2 \\
2E_{XY} &= C_{XY} = \frac{\partial U}{\partial Y} + \frac{\partial V}{\partial X} + \frac{\partial U}{\partial X}\frac{\partial U}{\partial Y} + \frac{\partial V}{\partial X}\frac{\partial V}{\partial Y} + \frac{\partial W}{\partial X}\frac{\partial W}{\partial Y} \\
2E_{YZ} &= C_{YZ} = \frac{\partial V}{\partial Z} + \frac{\partial W}{\partial Y} + \frac{\partial U}{\partial Y}\frac{\partial U}{\partial Z} + \frac{\partial V}{\partial Y}\frac{\partial V}{\partial Z} + \frac{\partial W}{\partial Y}\frac{\partial W}{\partial Z} \\
2E_{ZX} &= C_{ZX} = \frac{\partial W}{\partial X} + \frac{\partial U}{\partial Z} + \frac{\partial U}{\partial X}\frac{\partial U}{\partial Z} + \frac{\partial V}{\partial X}\frac{\partial V}{\partial Z} + \frac{\partial W}{\partial X}\frac{\partial W}{\partial Z}
\end{aligned} \tag{1.5.14}$$

[4] For convenience, we shall use boldface letters to denote both vectors and tensors. This should not cause any confusion since from the context it is deducible what they are.

$$
\begin{aligned}
2e_{xx} &= 1 - c_{xx} = 2\frac{\partial u}{\partial x} - \left(\frac{\partial u}{\partial x}\right)^2 - \left(\frac{\partial v}{\partial x}\right)^2 - \left(\frac{\partial w}{\partial x}\right)^2 \\
2e_{yy} &= 1 - c_{yy} = 2\frac{\partial v}{\partial y} - \left(\frac{\partial u}{\partial y}\right)^2 - \left(\frac{\partial v}{\partial y}\right)^2 - \left(\frac{\partial w}{\partial y}\right)^2 \\
2e_{zz} &= 1 - c_{zz} = 2\frac{\partial w}{\partial z} - \left(\frac{\partial u}{\partial z}\right)^2 - \left(\frac{\partial v}{\partial z}\right)^2 - \left(\frac{\partial w}{\partial z}\right)^2 \\
2e_{xy} &= -c_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial w}{\partial x}\frac{\partial w}{\partial y} \\
2e_{yz} &= -c_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial u}{\partial z} - \frac{\partial v}{\partial y}\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y}\frac{\partial w}{\partial z} \\
2e_{zx} &= -c_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} - \frac{\partial u}{\partial x}\frac{\partial u}{\partial z} - \frac{\partial v}{\partial x}\frac{\partial v}{\partial z} - \frac{\partial w}{\partial x}\frac{\partial w}{\partial z}
\end{aligned}
\tag{1.5.15}
$$

1.6. INFINITESIMAL STRAINS AND ROTATIONS

In the study of small deformation theory, various nonlinear terms of the strain tensor are either dropped or approximated. The linear theory, for example, employs only the *infinitesimal strain tensors* $\tilde{E}_{KL}$, $\tilde{e}_{kl}$ and *infinitesimal rotation tensors* $\tilde{R}_{KL}$, $\tilde{r}_{kl}$ defined by

$$
\begin{aligned}
\tilde{E}_{KL} &\equiv \tfrac{1}{2}(U_{K,L} + U_{L,K}) \equiv U_{(K,L)}, \qquad \tilde{e}_{kl} \equiv \tfrac{1}{2}(u_{k,l} + u_{l,k}) \equiv u_{(k,l)} \\
\tilde{R}_{KL} &\equiv \tfrac{1}{2}(U_{K,L} - U_{L,K}) \equiv U_{[K,L]}, \qquad \tilde{r}_{kl} \equiv \tfrac{1}{2}(u_{k,l} - u_{l,k}) \equiv u_{[k,l]}
\end{aligned}
\tag{1.6.1}
$$

where we have employed parentheses and brackets enclosing indices to indicate the symmetric and antisymmetric parts of the quantities enclosed.

From (1.6.1) by addition we obtain

$$
U_{K,L} = \tilde{E}_{KL} + \tilde{R}_{KL}, \qquad u_{k,l} = \tilde{e}_{kl} + \tilde{r}_{kl} \tag{1.6.2}
$$

It is clear that the infinitesimal strains are symmetric, and the infinitesimal rotations are skew-symmetric, that is,

$$
\tilde{e}_{kl} = \tilde{e}_{lk}, \qquad \tilde{r}_{kl} = -\tilde{r}_{lk} \tag{1.6.3}
$$

In matrix form they are respectively represented by

$$
\|\tilde{e}_{kl}\| = \begin{bmatrix} \tilde{e}_{11} & \tilde{e}_{12} & \tilde{e}_{13} \\ \tilde{e}_{21} & \tilde{e}_{22} & \tilde{e}_{23} \\ \tilde{e}_{31} & \tilde{e}_{32} & \tilde{e}_{33} \end{bmatrix}, \qquad \|\tilde{r}_{kl}\| = \begin{bmatrix} 0 & \tilde{r}_{12} & \tilde{r}_{13} \\ -\tilde{r}_{12} & 0 & \tilde{r}_{23} \\ -\tilde{r}_{13} & -\tilde{r}_{23} & 0 \end{bmatrix} \tag{1.6.4}
$$

The infinitesimal strain tensor has six independent components, whereas the infinitesimal rotation tensor has only three nonzero components.

Since in a three-dimensional space a vector will, in general, have three independent components, we may form a vector $\tilde{\mathbf{r}}$, called the *infinitesimal rotation vector*, by setting

$$\tilde{r}_1 = \tilde{r}_{32}, \qquad \tilde{r}_2 = \tilde{r}_{13}, \qquad \tilde{r}_3 = \tilde{r}_{21} \tag{1.6.5}$$

These expressions and the corresponding ones for $\tilde{R}_{KL}$ may be expressed in compressed notation as

$$2\tilde{r}_k = e_{klm}\tilde{r}_{ml}, \qquad 2\tilde{R}_K = e_{KLM}\tilde{R}_{ML} \tag{1.6.6}$$

where e_{klm} and e_{KLM} are the permutation symbols.

Substituting (1.6.2) into (1.5.8) and (1.5.9) we find

$$\begin{aligned} E_{KL} &= \tilde{E}_{KL} + \tfrac{1}{2}(\tilde{E}_{MK} + \tilde{R}_{MK})(\tilde{E}_{ML} + \tilde{R}_{ML}) \\ e_{kl} &= \tilde{e}_{kl} - \tfrac{1}{2}(\tilde{e}_{mk} + \tilde{r}_{mk})(\tilde{e}_{ml} + \tilde{r}_{ml}) \end{aligned} \tag{1.6.7}$$

From (1.6.7), it is clear that in order that $E_{KL} \cong \tilde{E}_{KL}$, not only must $\tilde{E}_{KL}$ be small, but $\tilde{R}_{KL}$ must also be small so that products such as $\tilde{E}_{MK}\tilde{E}_{ML}$, $\tilde{E}_{MK}\tilde{R}_{ML}$, and $\tilde{R}_{MK}\tilde{R}_{ML}$ will be negligible as compared to $\tilde{E}_{KL}$. Various other theories are also suggested here. One may, for example, have small $\tilde{E}_{KL}$ but large $\tilde{R}_{KL}$ so that, while $\tilde{E}_{MK}\tilde{E}_{ML}$ is dropped, not all of the remaining terms can be disregarded. Such situations occur in large deflections of bars, plates, and shells. The *infinitesimal deformation theory*, however, assumes $E_{KL} = \tilde{E}_{KL}$ and $e_{kl} = \tilde{e}_{kl}$. In this case, (1.5.4) by linearization gives

$$\tilde{E}_{KL} = \tilde{e}_{kl}\delta_{kK}\delta_{lL}, \qquad \tilde{e}_{kl} = \tilde{E}_{KL}\delta_{Kk}\delta_{Ll} \tag{1.6.8}$$

where δ_{Kk} is the Kronecker delta. *Thus in the infinitesimal deformation theory the distinction between the lagrangian and eulerian strains disappears.*

An important conclusion which follows from $(1.6.7)_2$ is that in general $\tilde{e}_{kl} = 0$ does not imply $e_{kl} = 0$, that is, vanishing infinitesimal strains are not sufficient for rigid deformation. We emphasize the fact that $\tilde{e}_{kl}$ is *not* a strain measure. It is only approximately so in the infinitesimal deformation theory.

1.7. LENGTH AND ANGLE CHANGES, GEOMETRICAL MEANING OF STRAINS AND ROTATIONS

A geometrical meaning for the strains and rotations is provided by considering the length and angle changes as a result of the deformation. Referred to the same rectangular coordinates X_K, an infinitesimal rectangular parallelepiped having edge vectors $\mathbf{I}_1\, dX_1$, $\mathbf{I}_2\, dX_2$, and $\mathbf{I}_3\, dX_3$ after deformation becomes a rectilinear parallelepiped with corresponding

edge vectors $\mathbf{C}_1\, dX_1$, $\mathbf{C}_2\, dX_2$, and $\mathbf{C}_3\, dX_3$, that is, a vector $d\mathbf{X}$ at $\mathbf{X}$ after deformation becomes $d\mathbf{x}$ at $\mathbf{x}$ (Figure 1.7.1).

$$d\mathbf{X} = \mathbf{I}_K\, dX_K, \qquad d\mathbf{x} = \mathbf{C}_K\, dX_K \tag{1.7.1}$$

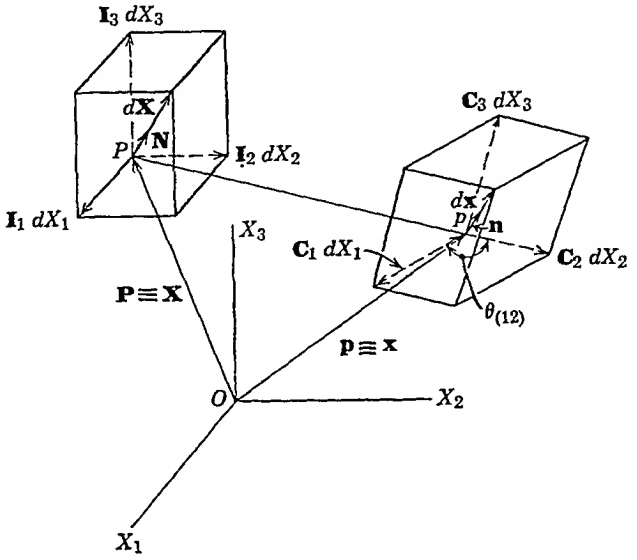


Fig. 1.7.1 Deformation of an infinitesimal rectangular parallelepiped.

Let $\mathbf{N}$ and $\mathbf{n}$ be the unit vectors along $d\mathbf{X}$ and $d\mathbf{x}$, respectively, that is,

$$N_K \equiv \frac{dX_K}{|d\mathbf{X}|} = \frac{dX_K}{dS}, \qquad n_k \equiv \frac{dx_k}{|d\mathbf{x}|} = \frac{dx_k}{ds} \tag{1.7.2}$$

We denote the length of $d\mathbf{X}$ and that of $d\mathbf{x}$ by dS and ds, respectively.

The ratio ds/dS of the lengths of $d\mathbf{x}$ and $d\mathbf{X}$ is called the *stretch*. This ratio may be expressed in terms of either $\mathbf{N}$ or $\mathbf{n}$. To indicate these dependences, we denote the stretch either by $\Lambda_{(\mathbf{N})}$ or by $\lambda_{(\mathbf{n})}$. They are, of course, the same physical quantity expressed differently, that is, $\Lambda_{(\mathbf{N})} = \lambda_{(\mathbf{n})}$.

$$\Lambda_{(\mathbf{N})} = \frac{ds}{dS} = \sqrt{C_{KL} N_K N_L}$$

$$\lambda_{(\mathbf{n})} = \frac{ds}{dS} = \frac{1}{\sqrt{c_{kl} n_k n_l}} \tag{1.7.3}$$

From (1.7.3) it is clear that *the normal components of* $\mathbf{C}$ *and* $\mathbf{c}$ *in the directions of* $\mathbf{N}$ *and* $\mathbf{n}$ *are, respectively, the squares and inverse squares of stretches in these directions.* This point is clarified further if we select $\mathbf{N}$ and $\mathbf{n}$ along

one of the coordinate axes (say X_1 and x_1). Then, $N_1 = 1$, $N_2 = N_3 = 0$, and $n_1 = 1$, $n_2 = n_3 = 0$ and we get

$$\Lambda_{(1)} = \sqrt{C_{11}}, \qquad \lambda_{(1)} = \frac{1}{\sqrt{c_{11}}} \tag{1.7.4}$$

The *extension* $E_{(N)} = e_{(n)}$ *is defined by*

$$E_{(N)} = e_{(n)} \equiv \Lambda_{(N)} - 1 = \frac{ds - dS}{dS} \tag{1.7.5}$$

For the case N taken along the X_1 axis, we get

$$E_{(1)} = \Lambda_{(1)} - 1 = \sqrt{C_{11}} - 1 = \sqrt{1 + 2E_{11}} - 1 \tag{1.7.6}$$

From this it follows that

$$2E_{11} = (1 + E_{(1)})^2 - 1 = \Lambda_{(1)}^2 - 1 \tag{1.7.7}$$

Thus, the normal component of the lagrangian strain is one half of the square of the stretch, minus one. When the extension is small, $E_{(1)} \ll 1$, by expanding (1.7.7), neglecting the square of $E_{(1)}$, we get

$$E_{11} \cong E_{(1)} \cong \tilde{E}_{11} \tag{1.7.8}$$

Similar results are of course valid for E_{22} and E_{33}, which indicates that *the infinitesimal normal strains are approximately the extensions of the fibers along the coordinate axes when the deformation is small.* The corresponding results for the eulerian strains are

$$\begin{aligned} e_{(1)} &= \lambda_{(1)} - 1 = \frac{1}{\sqrt{1 - 2e_{11}}} - 1 \\ 2e_{11} &= 1 - (1 + e_{(1)})^{-2} \end{aligned} \tag{1.7.9}$$

so that for small extensions as compared to unity, we obtain

$$e_{11} \cong e_{(1)} \cong \tilde{e}_{11} \tag{1.7.10}$$

The geometrical meaning of the shear strains E_{12}, E_{23}, E_{31} is found by considering the angle changes between two directions N_1 and N_2. To simplify the presentation, we consider two vectors $I_1\, dX_1$ and $I_2\, dX_2$ which after deformation become $C_1\, dX_1$ and $C_2\, dX_2$, having an angle $\theta_{(12)}$ given by

$$\cos \theta_{(12)} = \frac{(C_1\, dX_1) \cdot (C_2\, dX_2)}{|C_1\, dX_1|\, |C_2\, dX_2|}$$

or

$$\cos \theta_{(12)} = \frac{C_{12}}{\sqrt{C_{11} C_{22}}} = \frac{2E_{12}}{\sqrt{1 + 2E_{11}}\sqrt{1 + 2E_{22}}} \tag{1.7.11}$$

Using (1.7.6), we may solve from this for E_{12}.

$$2E_{12} = (1 + E_{(1)})(1 + E_{(2)}) \cos \theta_{(12)} \tag{1.7.12}$$

When the extensions are small compared to unity, retaining the lowest-order terms, this gives

$$2E_{12} \cong 2\tilde{E}_{12} \cong \cos \theta_{(12)} \tag{1.7.13}$$

For an infinitesimal deformation, the angle change

$$\frac{\pi}{2} - \theta_{(12)} \equiv \Gamma_{(12)}$$

is small. Hence, writing $\cos \theta_{(12)} = \sin \Gamma_{(12)} \cong \Gamma_{(12)}$, we have

$$2E_{12} \cong 2\tilde{E}_{12} \cong \Gamma_{(12)} \tag{1.7.14}$$

Therefore, *the infinitesimal shearing strains are approximately one half of the angle change between the coordinate axes for small deformations.*

From the rigorous equation (1.7.11), it is also clear that *the necessary and sufficient condition for the angle change to be zero is the vanishing of the shearing strains.* Further, from (1.7.6) we deduce that *for the extension to vanish it is necessary and sufficient that* $E_{11} = E_{22} = E_{33} = 0$, or $C_{11} = C_{22} = C_{33} = 1$.

Thus, *the necessary and sufficient condition for rigid deformation at each point* is

$$C_{KL} = \delta_{KL} \quad \text{or} \quad c_{kl} = \delta_{kl} \quad \text{or} \quad E_{KL} = e_{kl} = 0 \tag{1.7.15}$$

Geometrically, the infinitesimal normal and shear strains are clearly observed by considering planar deformations of an infinitesimal plane, rectangular lamina $OABC$, having sides dX_1 and dX_2 before deformation (Figure 1.7.2). If this lamina is stretched infinitesimally along the X_1-axis

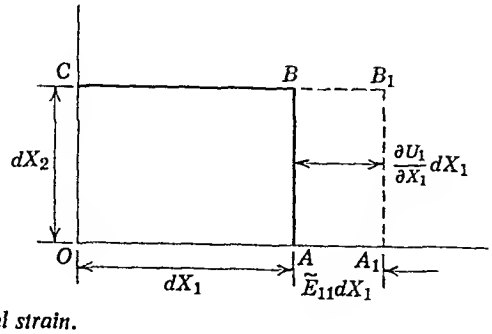


Fig. 1.7.2 Normal strain.

so that, in the deformed position, it takes the shape OA_1B_1C, then

$$\tilde{E}_{11} = E_{(1)} = \frac{OA_1 - OA}{OA} = \frac{\partial U_1}{\partial X_1}$$

If, on the other hand, the lamina is sheared slightly so that in the final position it becomes $OA_2B_2C_2$ (Figure 1.7.3), then the angle Γ_1 that OA_2 makes with OA is $\partial U_2/\partial X_1$ and the angle Γ_1 that OC_2 makes with OC is

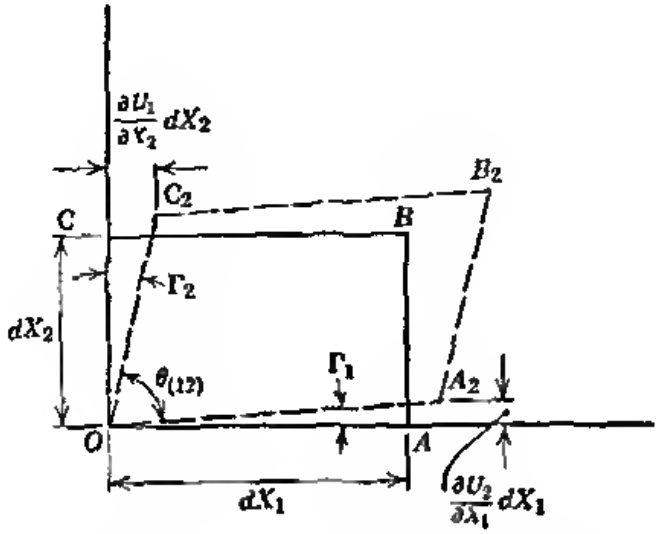


Fig. 1.7.3 Shear strain.

$\partial U_1/\partial X_2$ for the infinitesimal shears.[5] Hence, the angle change is

$$\frac{\pi}{2} - \theta_{(12)} \cong \Gamma_1 + \Gamma_2 \simeq \frac{\partial U_1}{\partial X_2} + \frac{\partial U_2}{\partial X_1} = 2\tilde{E}_{12}$$

A rigorous interpretation of the infinitesimal strain tensor $\tilde{\mathbf{E}}$ exists. To this end, consider an infinitesimal fiber $d\mathbf{X}$ at $\mathbf{X}$ that deforms into $d\mathbf{x}$ at $\mathbf{x}$. Now bring $d\mathbf{x}$ to $\mathbf{X}$ remaining parallel to itself (Figure 1.7.4).

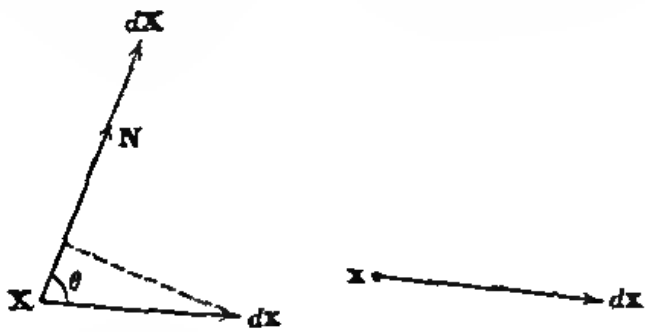


Fig. 1.7.4

The *elongation* $\varepsilon_{(\mathbf{N})}$ in the direction of $d\mathbf{X}$ is defined by

$$\varepsilon_{(\mathbf{N})} = \frac{|d\mathbf{x}| \cos\theta - |d\mathbf{X}|}{|d\mathbf{X}|} \tag{1.7.16}$$

$$= \frac{d\mathbf{x} \cdot d\mathbf{X}}{|d\mathbf{X}|^2} - 1 = \frac{d\mathbf{x} \cdot \mathbf{N}}{|d\mathbf{X}|} - 1 = \frac{\delta_{k\kappa}\, dx_k N_\kappa}{|d\mathbf{X}|} - 1$$

[5] For finite extensions and shears, see Eringen [1962, Art. 15].

where θ is the angle between $d\mathbf{x}$ and $d\mathbf{X}$ and $\mathbf{N} \equiv d\mathbf{X}/|d\mathbf{X}|$ is the unit vector along $d\mathbf{X}$. According to $(1.5.11)_1$ and (1.6.2) we have

$$dx_k = (\delta_{ML} + \tilde{E}_{ML} + \tilde{R}_{ML})\delta_{Mk}\, dX_L \tag{1.7.17}$$

If we substitute this into (1.7.16) it simplifies to

$$\varepsilon_{(\mathbf{N})} = \tilde{E}_{KL} N_K N_L \tag{1.7.18}$$

Hence, *the normal component of* $\tilde{\mathbf{E}}$ *in the direction of* $\mathbf{N}$ *is the elongation in that direction.* This result is clarified further by choosing $\mathbf{N}$ in the direction of any one of the coordinate axes, say X_1. In this case $N_1 = 1, N_2 = N_3 = 0$, and (1.7.18) gives $\varepsilon_{(1)} = \tilde{E}_{11}$. *Thus the infinitesimal normal strains* $\tilde{E}_{11}, \tilde{E}_{22}, \tilde{E}_{33}$ *are none other than elongations along the* X_1, X_2 *and* X_3 *axes.*

The geometrical meaning of the infinitesimal rotation is understood by calculating the mean rotation at a point.[6]

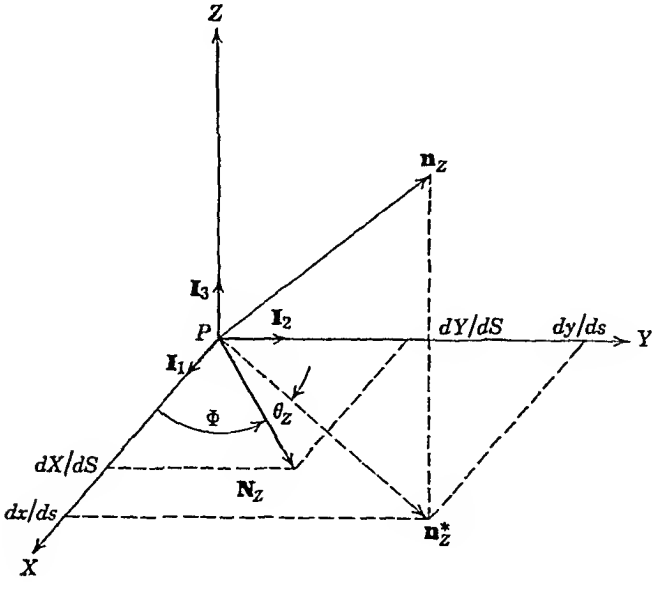


Fig. 1.7.5 Mean rotation.

Let $\mathbf{N}_Z$ be a unit vector in the XY plane of a rectangular frame of reference X, Y, Z at P (Figure 1.7.5). After deformation, $\mathbf{N}_Z$ becomes $\mathbf{n}_Z$ at p. We bring $\mathbf{n}_Z$ to P by parallel transport. Let the angle between $\mathbf{N}_Z$ and the

[6] This description is from Novozhilov [1948, Art. 7].

projection n_Z^* of n_Z on the XY plane be denoted by θ_Z. From

$$\tan(\Phi + \theta_Z) = \frac{\tan\Phi + \tan\theta_Z}{1 - \tan\Phi \tan\theta_Z}$$

$$= \frac{dy}{dx} = \frac{y_{,X}\,dX + y_{,Y}\,dY}{x_{,X}\,dX + x_{,Y}\,dY}$$

by use of $dX = dS\cos\Phi$, $dY = dS\sin\Phi$, we solve for

$$\tan\theta_Z = \frac{y_{,X}\cos^2\Phi + (y_{,Y} - x_{,X})\sin\Phi\cos\Phi - x_{,Y}\sin^2\Phi}{x_{,X}\cos^2\Phi + y_{,Y}\sin^2\Phi + (y_{,X} + x_{,Y})\sin\Phi\cos\Phi} \tag{1.7.19}$$

$$= \frac{-\bar{R}_{XY} + \bar{E}_{XY}\cos 2\Phi + \tfrac{1}{2}(\bar{E}_{YY} - \bar{E}_{XX})\sin 2\Phi}{1 + \tfrac{1}{2}(\bar{E}_{XX} + \bar{E}_{YY}) - \tfrac{1}{2}(\bar{E}_{YY} - \bar{E}_{XX})\cos 2\Phi + \bar{E}_{XY}\sin 2\Phi}$$

where the second line is obtained from the first by putting $y_{,X} = V_{,X} = \bar{E}_{XY} - \bar{R}_{XY}$, $y_{,Y} = 1 + V_{,Y} = 1 + \bar{E}_{YY}$, ... Expression (1.7.19) is periodic in Φ with period π. Therefore, θ_Z subject to $0 < \theta_Z < \pi$, except for indistinguishable angles $\theta_Z = 0$ and $\theta_Z = \pi$, is well defined. Novozhilov defines the mean value of (1.7.19) over all Φ in $(0, 2\pi)$ as his measure of rotation, that is,

$$\langle\tan\theta_Z\rangle = \frac{1}{2\pi}\int_0^{2\pi} \tan\theta_Z(\Phi)\,d\Phi \tag{1.7.20}$$

We now carry $(1.7.19)_2$ into this expression and perform the integration. In this process we note that, excluding the term $-\bar{R}_{XY}$, the numerator of $(1.7.19)_2$ is the derivative of the denominator. Therefore, this part of the fraction integrates to a logarithm that upon substitution of the limits vanishes. Thus,

$$\langle\tan\theta_Z\rangle \tag{1.7.21}$$

$$= -\frac{\bar{R}_{XY}}{2\pi}\int_0^{2\pi} \frac{d\Phi}{1 + \tfrac{1}{2}(\bar{E}_{YX} + \bar{E}_{YY}) + \tfrac{1}{2}(\bar{E}_{XX} - \bar{E}_{YY})\cos 2\Phi + \bar{E}_{XY}\sin 2\Phi}$$

$$= \frac{-\bar{R}_{XY}}{\sqrt{(1 + \bar{E}_{XX})(1 + \bar{E}_{YY}) - \bar{E}_{XY}^2}}$$

We therefore see that the mean rotation, so defined, is proportional to $\bar{R}_{XY}$. For small strains, $\bar{E}_{XX}$, $\bar{E}_{YY}$, and $\bar{E}_{XY}$ are negligible as compared to unity, and we find

$$\langle\tan\theta_Z\rangle \cong \langle\theta_Z\rangle \cong -\bar{R}_{XY} \tag{1.7.22}$$

Similar expressions are, of course, obtained for $\langle\tan\theta_X\rangle$ and $\langle\tan\theta_Y\rangle$ by mere *cyclic change* of X to Y, Y to Z, and Z to X in the above expressions

Since $\tilde{\mathbf{R}}$ is a tensor, if it vanishes in one coordinate system, then it vanishes in all coordinates. Thus, if at a point the mean rotation in three perpendicular planes is zero or π radians, then in any plane through that point it is zero or π radians. The mean rotation is zero if and only if there exists a displacement potential F such that

$$U_K = F_{,K} \tag{1.7.23}$$

In this case $\tilde{\mathbf{R}} = \mathbf{0}$, and such deformations are called *potential deformations.*

1.8. TRANSFORMATION OF STRAINS AND ROTATIONS UNDER RIGID MOTION OF RECTANGULAR FRAME

It is important to know what new values the strains and rotations take when the rectangular frame of reference X_K is translated and rigidly rotated. Let the new rectangular coordinates be denoted by X'_K (Figure 1.8.1).

Referred to each system of coordinates, we have

$$\begin{aligned} \mathbf{P} &= X_K \mathbf{I}_K, \qquad \mathbf{B} = B_K \mathbf{I}_K \\ \mathbf{P}' &= X'_M \mathbf{I}'_M, \qquad \mathbf{B} = B'_M \mathbf{I}'_M \end{aligned} \tag{1.8.1}$$

where $\mathbf{P}$, $\mathbf{P}'$ are the position vectors of the same material point P in the

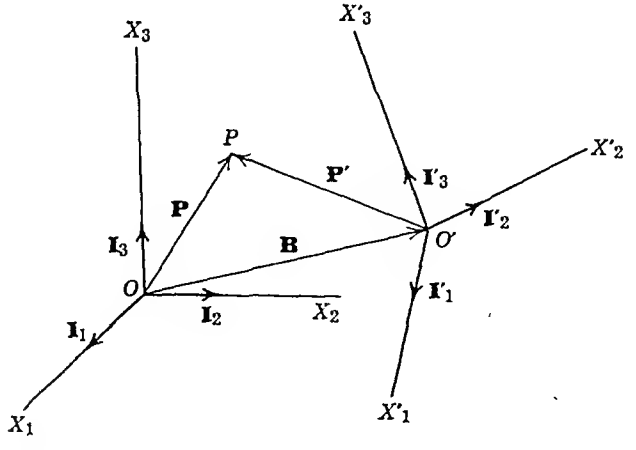


Fig. 1.8.1 Rigid translation and rotation of rectangular coordinates.

body, $\mathbf{B}$ is the position vector of the origin O' of X'_K, and $\mathbf{I}_K$ and $\mathbf{I}'_K$ are the unit base vectors in X_K and X'_K, respectively. We have

$$\mathbf{P} = \mathbf{P}' + \mathbf{B} \tag{1.8.2}$$

The scalar multiplication of (1 8.2) by $\mathbf{I}_M$ and $\mathbf{I}'_M$ yields

$$X_K = Q_{MK}X'_M + Q_{MK}B'_M$$
$$X'_M = Q_{MK}X_K - Q_{MK}B_K \tag{1.8 3}$$

where Q_{MK} are the *cosine directors* of the new axes X'_M with respect to X_K (Table 1.8.1).

Table 1.8.1. Cosine directors

	X_1	X_2	X_3
X'_1	Q_{11}	Q_{12}	Q_{13}
X'_2	Q_{21}	Q_{22}	Q_{23}
X'_3	Q_{31}	Q_{32}	Q_{33}

From (1.8.3) we have

$$\frac{\partial X_K}{\partial X'_M} = Q_{MK} = \mathbf{I}'_M \cdot \mathbf{I}_K, \qquad \frac{\partial X'_M}{\partial X_K} = Q_{MK} \tag{1.8.4}$$

Through the chain rule of partial differentiation we have

$$\frac{\partial X_K}{\partial X'_M}\frac{\partial X'_M}{\partial X_L} = \frac{\partial X_K}{\partial X_L} = \delta_{KL}$$

and similarly for $\partial X'_K/\partial X'_L = \delta_{KL}$ which upon use of (1.8.4) gives

$$Q_{MK}Q_{ML} = Q_{KM}Q_{LM} = \delta_{KL} \tag{1.8.5}$$

Since the determinant of the products of matrices is equal to the product of the determinants of matrices, from (1.8.5) it also follows that

$$|Q_{MK}|\,|Q_{ML}| = |Q_{MK}|^2 = 1 \quad \text{or} \quad |Q_{MK}| = \pm 1$$

For a rigid motion, a right-hand system of coordinates remains right-handed so that we must have

$$|Q_{MK}| = 1 \tag{1.8 6}$$

When a reflection of axes is allowed, then both signs $\pm$ may have to be included in the determinant of Q_{MK}.

We now recall expression (1.5.2). Upon substituting $dX_K = Q_{MK}\,dX'_M$ into this expression we obtain

$$ds^2 - dS^2 = 2E_{KL}Q_{MK}Q_{NL}\,dX'_M\,dX'_N = 2E'_{MN}\,dX'_M\,dX'_N$$

where

$$E'_{MN} = E_{KL}Q_{MK}Q_{NL} \tag{1.8.7}$$

Conversely, we find

$$E_{MN} = E'_{KL}Q_{KM}Q_{LN} \tag{1.8.8}$$

Equations (1.8.7) and (1.8.8) relate strain components in two rectangular coordinates X_K and X'_K to each other. They are a special form of (1.4.18), which is valid for the tensor $\mathbf{E}$ in general curvilinear coordinates. The transformation of the infinitesimal strains is similar to (1.8.7) and (1.8.8). It is instructive perhaps to obtain these expressions by using the defining equations (1.6.1). The displacement vector in rectangular coordinates is expressed as

$$\mathbf{u} = U_L \mathbf{I}_L = U'_N \mathbf{I}'_N \tag{1.8.9}$$

By scalar multiplication of (1.8.9) by $\mathbf{I}_K$ and $\mathbf{I}'_M$ we obtain

$$U_K = Q_{MK} U'_M, \qquad U'_M = Q_{MK} U_K \tag{1.8.10}$$

From these, by partial differentiation, we get

$$\frac{\partial U'_M}{\partial X'_N} = \frac{\partial}{\partial X_L}(Q_{MK} U_K) \frac{\partial X_L}{\partial X'_N} = Q_{MK} Q_{NL} \frac{\partial U_K}{\partial X_L}$$

Hence

$$2\tilde{E}'_{MN} = \frac{\partial U'_M}{\partial X'_N} + \frac{\partial U'_N}{\partial X'_M} = Q_{MK} Q_{NL} \left(\frac{\partial U_K}{\partial X_L} + \frac{\partial U_L}{\partial X_K} \right)$$

and

$$\tilde{E}'_{MN} = \tilde{E}_{KL} Q_{MK} Q_{NL} \tag{1.8.11}$$

which is identical in form to (1.8.7). Similarly, an expression for $\tilde{\mathbf{e}}$, identical in form to (1.8.7), can be obtained for the rigid translations and rotations of the spatial frame of reference.

The law of transformation for $\tilde{R}_{KL}$ and $\tilde{r}_{kl}$ is similar to (1.8.11). By the same method, in fact, we find

$$\tilde{R}'_{MN} = \tilde{R}_{KL} Q_{MK} Q_{NL}, \qquad \tilde{R}_{KL} = \tilde{R}'_{MN} Q_{MK} Q_{NL} \tag{1.8.12}$$

It is not difficult to deduce that the rotation vectors $\tilde{R}_K$ should obey the transformation law of axial vectors, namely

$$\tilde{R}'_K = \pm Q_{KM} \tilde{R}_M, \qquad \tilde{R}_K = \pm Q_{MK} \tilde{R}'_M \tag{1.8.13}$$

Here the + sign is valid for $\det Q_{KL} = 1$, and the − sign for $\det Q_{KL} = -1$. Consider

$$2\tilde{R}'_K = e_{KMN} \tilde{R}'_{NM} = e_{KMN} Q_{MP} Q_{NR} \tilde{R}_{RP} \tag{1.8.14}$$

To simplify this, we need the lemma:

$$e_{KMN} Q_{MP} Q_{NR} = \pm e_{PRS} Q_{KS} \tag{1.8.15}$$

To show this, we consider the cross products of unit base vectors:

$$\mathbf{I}_P \times \mathbf{I}_R = e_{PRS} \mathbf{I}_S \tag{1.8.16}$$

But since

$$\mathbf{I}_P = Q_{LP}\mathbf{I}'_L \tag{1.8.17}$$

substituting this expression into (1.8.16), we get

$$e_{PRS}Q_{KS}\mathbf{I}'_K = Q_{MP}Q_{NR}\mathbf{I}'_M \times \mathbf{I}'_N$$

Using (1.8.16) now for the vector $\mathbf{I}'_K$, we get

$$e_{PRS}Q_{KS}\mathbf{I}'_K = \pm Q_{MP}Q_{NR}e_{MNK}\mathbf{I}'_K$$

The coefficients of $\mathbf{I}'_K$ on both sides must be equal. This gives (1.8.15), since the replacement of e_{MNK} by e_{KMN} is permissible. This completes the proof of our lemma.

We now substitute (1.8.15) into (1.8.14), yielding

$$2\tilde{R}'_K = \pm e_{PRS}Q_{KS}\tilde{R}_{RP} = \pm Q_{KS}(e_{SPR}\tilde{R}_{RP})$$

Using $(1.6.6)_2$ on the extreme right, we obtain $(1.8.13)_1$.

1.9. STRAIN ELLIPSOID

To throw more light onto the picture of local deformations in the neighborhood of a point P of the undeformed body, we now study the deformation of an infinitesimal sphere $dS = \text{const.}$ at X. It is of interest to know what happens to this sphere after deformation. The sphere of radius K at X swept by a vector $d\mathbf{X}$,

$$\delta_{KL}\,dX_K\,dX_L = dS^2 = K^2 \tag{1.9.1}$$

after deformation becomes a quadric surface swept by $d\mathbf{x}$ at x, that is,

$$c_{kl}\,dx_k\,dx_l = dS^2 = K^2 \tag{1.9.2}$$

Since (1.9.1) is positive definite, (1.9.2) is also positive definite. Consequently, the quadric surface that this sphere deforms into, (1.9.2), is an ellipsoid. Similarly, a sphere swept by dx_k at x, before deformation, was an ellipsoid, that is, the sphere at p of the deformed body

$$\delta_{kl}\,dx_k\,dx_l = ds^2 = k^2 \tag{1.9.3}$$

was an ellipsoid in the undeformed body at P

$$C_{KL}\,dX_K\,dX_L = ds^2 = k^2 \tag{1.9.4}$$

These ellipsoids are called the *strain ellipsoids* of Cauchy. The ellipsoid (1.9.2) is known as the *material ellipsoid* of Cauchy, and (1.9.4) as the *spatial ellipsoid* of Cauchy. The study of the basic characteristics of one of

these ellipsoids will clarify the character of the deformation in the neighborhood of a point.

It is interesting to know how an orthogonal triad of radii of the sphere at P deforms. To this end, we have

THEOREM 1 (*Cauchy's First Theorem*). *Perpendicular diameters of an infinitesimal sphere at* $\mathbf{X}$ *are deformed into conjugate diameters of the material ellipsoid at* $\mathbf{x}$.

To prove this theorem, consider the perpendicular vectors $d\mathbf{X}^1$ and $d\mathbf{X}^2$ at the point $\mathbf{X}$. We have

$$d\mathbf{X}^1 \cdot d\mathbf{X}^2 = 0$$

Now according to (1.4.8), we have $d\mathbf{X}^1 = \mathbf{c}_k \, dx^1{}_k$. Thus, the above orthogonality condition reads

$$(\mathbf{c}_k \, dx^1{}_k) \cdot (\mathbf{c}_l \, dx^2{}_l) = c_{kl} \, dx^1{}_k \, dx^2{}_l = 0 \tag{1.9.5}$$

This shows that the gradient vector $2c_{kl} \, dx^1{}_k$ of the ellipsoid at the end point of the vector $d\mathbf{x}^1$ is perpendicular to the vector $d\mathbf{x}^2$ (Figure 1.9.1).

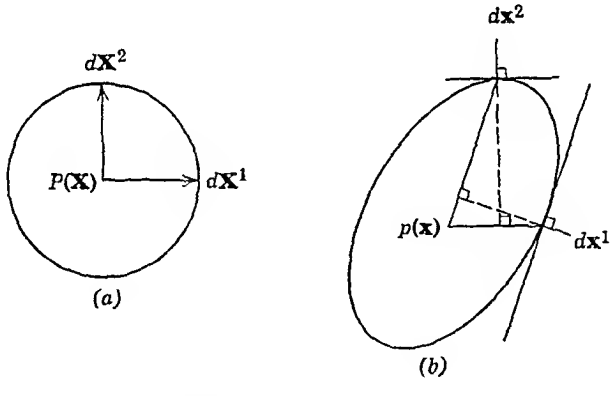


Fig. 1.9.1 (*a*) *Sphere.* (*b*) *Material ellipsoid.*

Hence, the proof of the theorem.

An ellipsoid has at least three diameters that are perpendicular to their conjugate planes. These mutually orthogonal diameters are the axes of the ellipsoid. Thus, according to Cauchy's first theorem, the axes of the strain ellipsoid are the material lines that were originally three perpendicular diameters of the sphere. Hence,

COROLLARY 1. *At a point* $P(\mathbf{X})$ *of* B *there exist at least three mutually perpendicular directions which, after deformation, remain orthogonal to each other, constituting the principal axes of the strain ellipsoid at* $p(\mathbf{x})$.

A rectangular parallelepiped built upon an orthogonal triad at P goes into the parallelepiped built upon the principal axes of the strain ellipsoid at p. The angles of the faces of this particular parallelepiped are not altered from 90°. Therefore,

COROLLARY 2 *Referred to the principal axes of strain at* P, c_{23}, c_{31}, *and* c_{12}, *that is, the shearing strains* e_{23}, e_{31}, *and* e_{12} *vanish.*

All of these, of course, are valid for the spatial strain ellipsoid at P. According to Cauchy's first theorem, there exists at least one orthogooal triad at X, which deforms into another orthogonal triad at x, and vice versa. The triad at X, considered as the orthogonal diameters of a sphere, deforms into the triad at x which constitutes the principal axes of the material strain ellipsoid at x. Conversely, this triad of the sphere at x was the triad of principal axes of the strain ellipsoid at X. Consequently,

COROLLARY 3 *The deformation rotates the principal axes of the strain ellipsoid at* X *into the principal axes of the strain ellipsoid at* x (Figure 1.9.2). This result assumes that the leogths of the principal axes of the strain ellipsoid at x are unequal

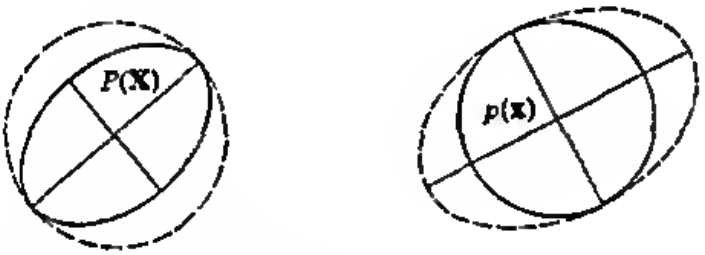


Fig. 1.9.2 Spatial and material ellipsoids

Under special conditions, the strain ellipsoid may degenerate into an ellipsoid of rotation or a sphere. In the former case, there is a plane through p perpendicular to the axis of rotation. In this plane, every pair of perpendicular directioos constitutes two principal axes, the third being the axis of rotatioo of the ellipsoid. When the ellipsoid degenerates into a sphere then, again an infinite number of orthogonal triads constructed from the diameters of the sphere are principal axes.

We have seen that the diameters of a sphere at P are deformed into diameters of an ellipsoid at p. Hence, *the ratios of the diameters of the ellipsoid to the corresponding diameters of the sphere are the stretches.* Thus, the magnitudes of the stretches in different undeformed original directions at P vary as does the distance from P to the surface of the ellipsoid.

In an ellipsoid the lengths of the priocipal axes are such that at least one is the greatest, one is the smallest, and a third is in between. Hence,

THEOREM 2 (*Cauchy's Second Theorem*). *At any point P there exist at least three mutually perpendicular directions for one of which the stretch is not less than any other, for another one it is not greater than any other, and for the third one it is a minimax. When they are ordered as* $\Lambda_{(1)} \geq \Lambda_{(2)} \geq \Lambda_{(3)}$, *then they stand in the ratio* $\Lambda_{(1)} : \Lambda_{(2)} : \Lambda_{(3)}$ *to the lengths of the axes of the strain ellipsoid.*

The maximum extension $E_{(1)} = \Lambda_{(1)} - 1$ occurs along the major axis of the ellipsoid, the minimum $E_{(3)}$ along the minor axis. The extensions that occur along the principal axes of the strain ellipsoid are called *principal extensions.*

The picture of local deformation is now clear. The material points located on a sphere at P and on its diameters acquire their final positions on an ellipsoid at p and on its diameters in three steps.

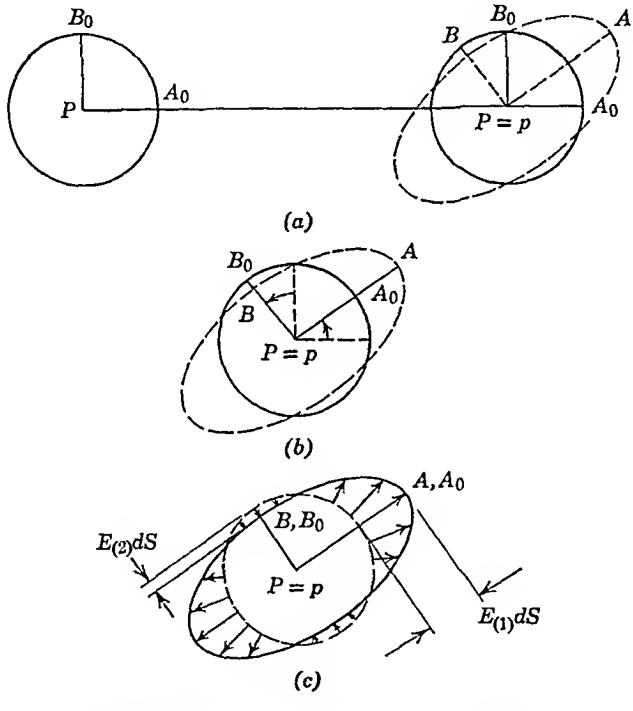


Fig. 1.9.3 Deformation of the neighborhood of point P. (a) Rigid translation. (b) Rigid rotation. (c) Pure strain.

(a) The infinitesimal sphere having center P is *rigidly translated* to the point p (Figure 1.9.3*a*).

(b) The sphere is *rigidly rotated* about an axis through p so that the principal axes of the strain ellipsoid at P coincide with their deformed directions at p (Figure 1.9.3*b*).

(c) Finally, these diameters are *stretched* as much as the principal stretches along the principal axes of the strain ellipsoid at p (Figure 1.9.3*c*).

The first two steps, (a) and (b), preserve the original distances between the material points; therefore they are called a *rigid body displacement.* In step (c) the actual distance changes. Hence, the body in this step is said to undergo a *pure strain.* The foregoing process may be taken in any order. For example, the sphere at P may first be rigidly rotated, then stretched, and afterwards may be translated to reach its final destination. We therefore have

THEOREM 3. *The deformation of any line element at a point may be regarded as resulting from a rigid translation, a rigid rotation of the principal axes of strain, and stretches along these axes.*

The analytical proof of this theorem in the case of infinitesimal deformation is simple and is given below For finite deformations, see Section 1.11.

We select the same rectangular frame of reference for X_K and x_k. Two neighboring points X and $\mathrm{X} + d\mathrm{X}$ in the undeformed body after deformation go to $\mathrm{X} + \mathrm{U}^0(\mathrm{X})$ and $\mathrm{X} + d\mathrm{X} + \mathrm{U}^0(\mathrm{X} + d\mathrm{X})$, respectively We have

$$U_K = U_k(\mathrm{X} + d\mathrm{X}) = U^0{}_K + U^0{}_{K,L}\, dX_L$$

where $U^0{}_K$ and $U^0{}_{K,L}$ are evaluated at X. Now, by adding and subtracting $\frac{1}{2}U^0{}_{L,K}\, dX_L$, this expression can be written as

$$U_K = U^0{}_K + \tfrac{1}{2}(U^0{}_{K,L} - U^0{}_{L,K})\, dX_L + \tfrac{1}{2}(U^0{}_{K,L} + U^0{}_{L,K})\, dX_L$$

or

$$U_K = U^0{}_K + \bar{R}^0{}_{KL}\, dX_L + \bar{E}^0{}_{KL}\, dX_L \tag{1.9.6}$$

This shows that an oriented vector $d\mathrm{X}$ at X undergoes a translation $U^0{}_K$, an infinitesimal rotation $\bar{R}^0{}_{KL}\, dX_L$ about X, and a straining $\bar{E}^0{}_{KL}\, dX_L$ of the material at X. Expression (1.9.6) is rigorous. Only the physical interpretations of the second and the third terms are approximate and valid for infinitesimal deformations.

For a rectangular parallelepiped at P the deformation picture was studied in Section 1.5. We learned that an infinitesimal rectangular parallelepiped having the main diagonal vector $d\mathrm{X} = dX_K \mathbf{I}_K$ after deformation becomes a rectilinear one with the diagonal vector $d\mathbf{x} = dx_k \mathbf{i}_k = dX_K \mathbf{C}_K$. The length of the sides have become $(1 + E_{(1)})\, dX_1$, $(1 + E_{(2)})\, dX_2$, and $(1 + E_{(3)})\, dX_3$

(Figure 1.9.4). The angles $\theta_{(12)}$, $\theta_{(23)}$, $\theta_{(13)}$ between the edges are no longer 90°. These angles and the elongations $E_{(K)}$ will be known as soon as the strain components E_{KL} are given at the point P. [See equation (1.7.6)

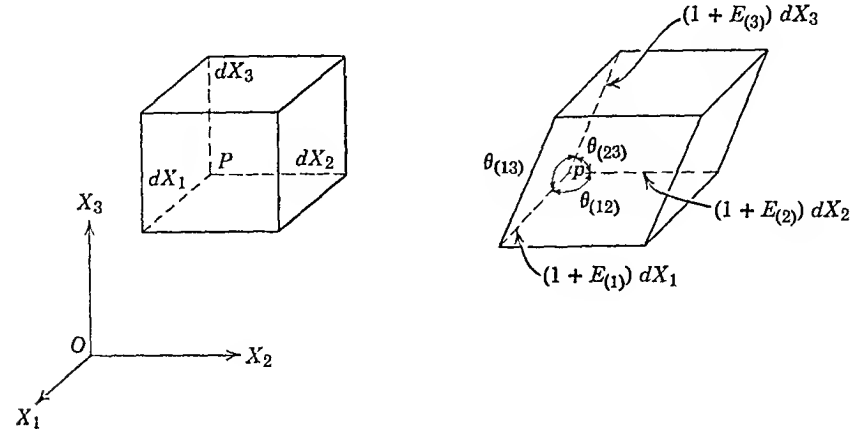


Fig. 1.9.4 Deformation of a parallelepiped.

and (1.7.11).] When the deformation is infinitesimal, from (1.7.8) and (1.7.14) we have

$$E_{(1)} \cong \tilde{E}_{11}, \qquad E_{(2)} \cong \tilde{E}_{22}, \qquad E_{(3)} \cong \tilde{E}_{33}$$

$$\frac{\pi}{2} - \theta_{(12)} \cong 2\tilde{E}_{12}, \qquad \frac{\pi}{2} - \theta_{(23)} \cong 2\tilde{E}_{23}, \qquad \frac{\pi}{2} - \theta_{(13)} \cong 2\tilde{E}_{13}$$

The parallelepiped whose edges before deformation coincide with the principal axes at the point in question is still rectangular after deformation.

1.10. STRAIN INVARIANTS AND PRINCIPAL DIRECTIONS

In the previous section we found that the deformation rotates the principal directions of the strain ellipsoid at P to those at p. The strain components referred to the principal axes of the strain ellipsoids may be expressed in terms of those referred to any other axes once the cosine directors of the principal directions are known. Here we would like to find the principal directions and primitive functions of strains that remain unaltered during this or any other coordinate transformation. To this end, we begin with the analytical determinations of the principal directions. This requires that we find the extremum values of the stretches $\Lambda_{(N)}$ since, according to Cauchy's second theorem, along these axes $\Lambda_{(N)}$ is stationary. Thus,

$$(1.10.1) \qquad \Lambda^2_{(N)} = C_{KL}N_KN_L = 2E_{KL}N_KN_L + 1, \qquad N_K = \frac{dX_K}{dS}$$

must be minimized with respect to N_K subject to the condition that N is a unit vector, that is,

$$\delta_{KL}N_K N_L = 1 \tag{1.10.2}$$

Lagrange's method of multipliers is best suited for this purpose. We set

$$\frac{\partial}{\partial N_M}(\Lambda^2_{(N)}) = \frac{\partial}{\partial N_M}[C_{KL}N_K N_L - C(\delta_{KL}N_K N_L - 1)] = 0$$

where C is the unknown Lagrange multiplier. This gives three linear homogeneous equations for N_L

$$(C_{KL} - C\delta_{KL})N_L = 0 \tag{1.10.3}$$

Upon using $C_{KL} = 2E_{KL} + \delta_{KL}$, we may also write this as

$$(E_{KL} - E\delta_{KL})N_L = 0 \tag{1.10.4}$$

where

$$2E \equiv C - 1 \tag{1.10.5}$$

We may solve either set (1.10.3) or (1.10.4). Using the relations of C_{KL} to E_{KL} and C to E, one can always convert the E-language to the C-language.

A nontrivial solution of (1.10.4) exists when the determinant of the coefficients of (1.10.4) is equal to zero, that is,

$$|E_{KL} - E\delta_{KL}| = \begin{vmatrix} E_{11} - E & E_{12} & E_{13} \\ E_{21} & E_{22} - E & E_{23} \\ E_{31} & E_{32} & E_{33} - E \end{vmatrix} = 0 \tag{1.10.6}$$

This determinant is known as the *characteristic determinant* of the strain tensor. When it is expanded, a cubic in E results

$$-E^3 + \mathrm{I}_E E^2 - \mathrm{II}_E E + \mathrm{III}_E = 0 \tag{1.10.7}$$

where

(1.10.8)

$$\mathrm{I}_E \equiv E_{KK} = E_{11} + E_{22} + E_{33}$$

$$\mathrm{II}_E \equiv E_{22}E_{33} + E_{33}E_{11} + E_{11}E_{22} - {E_{23}}^2 - {E_{31}}^2 - {E_{12}}^2$$

$$= \begin{vmatrix} E_{22}E_{23} \\ E_{32}E_{33} \end{vmatrix} + \begin{vmatrix} E_{11}E_{31} \\ E_{13}E_{33} \end{vmatrix} + \begin{vmatrix} E_{11}E_{12} \\ E_{21}E_{22} \end{vmatrix}$$

$$\mathrm{III}_E \equiv E_{11}E_{22}E_{33} + 2E_{12}E_{23}E_{31} - E_{11}E_{23}^2 - E_{22}E_{31}^2 - E_{33}E_{12}^2$$

$$= \begin{vmatrix} E_{11} & E_{12} & E_{13} \\ E_{21} & E_{22} & E_{23} \\ E_{31} & E_{32} & E_{33} \end{vmatrix}$$

The *characteristic equation* (1.10.7) possesses three roots, E_α ($\alpha = 1, 2, 3$), called *principal strains.* The coefficients I_E, II_E, and III_E of the characteristic equation are the sums of products of those roots taken one, two, and three at a time, that is,

$$\begin{aligned} \mathrm{I}_E &= E_1 + E_2 + E_3 \\ \mathrm{II}_E &= E_2E_3 + E_3E_1 + E_1E_2 \\ \mathrm{III}_E &= E_1E_2E_3 \end{aligned} \tag{1.10.9}$$

The set (1.10.4) for each principal strain E_α determines a *principal direction* (*or proper direction*) $\mathbf{N}_\alpha$ ($\alpha = 1, 2, 3$).

If the principal strains E_1, E_2, E_3 are *real* and *distinct*, then directions N_{1K}, N_{2K}, and N_{3K} are real and uniquely determined. We now prove the following lemmas:

LEMMA 1. *All principal strains E_α are real.*

Proof: We start with the hypothesis that the contrary is true, that is, one of the roots of the cubic equation (1.10.7), say E_1, is complex. Then, of course, the complex conjugate $\overset{*}{E}_1$ of E_1 is also a root. Corresponding to these, we obtain a complex direction $\mathbf{N}_1$ and its conjugate $\overset{*}{\mathbf{N}}_1$ satisfying (1.10.4), that is,

$$E_{KL}N_{1L} = E_1N_{1K}, \qquad E_{KL}\overset{*}{N}_{1L} = \overset{*}{E}_1\overset{*}{N}_{1K}$$

Multiplying the first of these by $\overset{*}{N}_{1K}$ and the second by N_{1K}, subtracting the second from the first, and remembering that $E_{KL} = E_{LK}$, we get

$$0 = (E_1 - \overset{*}{E}_1)N_{1K}\overset{*}{N}_{1K}$$

Now since $N_{1K}\overset{*}{N}_{1K} > 0$ we must have

$$E_1 - \overset{*}{E}_1 = 2\,Im\,E_1 = 0$$

Consequently, $Im\,E_1 = 0$, provided that E_1, contrary to our hypothesis, is real. We note that the crucial point in proving this lemma is the symmetry of the strain tensor.

LEMMA 2. *The principal directions* $\mathbf{N}_1$ *and* $\mathbf{N}_2$ *corresponding to two distinct principal strains* E_1 *and* E_2 *are orthogonal to each other.*

To prove this lemma, we must show that

$$\mathbf{N}_1 \cdot \mathbf{N}_2 = N_{1K}N_{2K} = 0 \tag{1.10.10}$$

The two directions corresponding to *distinct* principal strains E_1 and E_2 ($E_1 \neq E_2$) satisfy (1.10.4), that is,

$$E_{KL}N_{1L} = E_1N_{1K}, \qquad E_{KL}N_{2L} = E_2N_{2K}$$

Multiply the first by N_{2K} and the second by N_{1K} and subtract one from the other Hence

$$0 = (E_1 - E_2)N_{1K}N_{2K}$$

In view of our hypothesis we have $E_1 \neq E_2$. Thus

$$N_{1K}N_{2K} = \mathbf{N}_1 \cdot \mathbf{N}_2 = 0$$

which proves the lemma.

To arrive at this result, we assumed that $E_1 \neq E_2$, that is, all roots of (1.10.7) are *distinct*. The question arises as to what happens when (1.10 7) has multiple roots. In this case, the associated directions become indeterminate. For a double root, there will be an infinite number of directions in a plane satisfying (1.10.4). Any two orthogonal directions in this plane may be chosen as the proper direction, the third being perpendicular to this plane. The strain ellipsoid (1.9.4) in this case will have rotational symmetry; the axis of revolution, being a proper direction, and any two other orthogonal directions in the plane through P and perpendicular to this axis will make up the orthogonal triad. When all roots of (1.10.7) are identical, the quadric surface is a sphere. Hence, any three mutually orthogonal radii may be selected as the principal directions.

The state of strain takes a particularly simple form when the reference frame X_K is selected to coincide with the principal directions. In this case $N_{LK} = 0$ whenever $L \neq K$ We may write

$$N_{LK} = \delta_{LK} \tag{1.10.11}$$

From (1 10.4) we have

$$E_{KL}N_{ML} = E_{\underline{M}}N_{\underline{M}K} \qquad \text{(not summed over } M\text{)}$$

Here an underscore is used to indicate that the summation is suspended. Upon using (1.10 11), we find that this equation becomes

$$E_{KM} = E_{\underline{M}}\delta_{\underline{M}K} \tag{1.10.12}$$

For $M = 1$, 2, and 3 we get

$$E_{11} = E_1, \qquad E_{22} = E_2, \qquad E_{33} = E_3, \qquad E_{KL} = 0 \quad (K \neq L)$$

which shows that *the E_K are normal strains associated with the principal axes of strain at* X. Thus, referred to these axes, we may write

$$\begin{bmatrix} E_{11} & E_{12} & E_{13} \\ E_{21} & E_{22} & E_{23} \\ E_{31} & E_{32} & E_{33} \end{bmatrix} = \begin{bmatrix} E_1 & 0 & 0 \\ 0 & E_2 & 0 \\ 0 & 0 & E_3 \end{bmatrix} \tag{1.10.13}$$

Hence the *determination of principal directions* $\mathbf{N}_\alpha$ *and principal strains of a tensor* E_{KL} *is simply equivalent to finding a rectangular coordinate frame in which the matrix* $\|E_{KL}\|$ *will reduce to a diagonal form.* Thus

LEMMA 3. *Principal strains are the normal components of the strain tensor in the principal triad, and the shear components in this frame of reference vanish.*

Clearly, the principal directions are the principal axes of the strain ellipsoid. If coordinates X_K are chosen to be a *principal triad,* then the strain ellipsoid referred to these axes takes the simple form

$$ds^2 = k^2 = C_{KL}\, dX_K\, dX_L = \sum_\alpha C_\alpha (dX_\alpha)^2 \tag{1.10.14}$$

where according to (1.10.5)

$$C_\alpha = 1 + 2E_\alpha \tag{1.10.15}$$

We shall call C_α *proper numbers.*

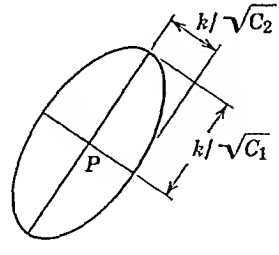


Fig. 1.10.1

Now C_{11}, C_{22}, and C_{33} are positive, hence $C_\alpha > 0$. Therefore (1.10.14) is an ellipsoid having semi-axis lengths $a_\alpha = k/\sqrt{C_\alpha}$ (Figure 1.10.1). The stretches $\Lambda_{(\alpha)} = \lambda_{(\alpha)} = ds/dS$ along the principal axes are given by

$$\Lambda_{(\alpha)} = \frac{k}{dX_\alpha} = \sqrt{C_\alpha} = \frac{k}{a_\alpha} \tag{1.10.16}$$

Had we considered the material ellipsoid, we would have found

$$\lambda_{(\alpha)} = \frac{dx_\alpha}{K} = \frac{1}{\sqrt{c_\alpha}} = \frac{a_\alpha}{K} \tag{1.10.17}$$

where c_α are the proper numbers for c_{kl}. Equating (1.10.16) to (1.10.17) we find that

$$C_\alpha = \frac{1}{c_\alpha} = \lambda^2_{(\alpha)} \tag{1.10.18}$$

Hence the proof of the following theorem:

THEOREM 1. *The lengths of the semi-axes of the strain ellipsoid at P are reciprocal to those at p and the proper numbers are reciprocals of each other. The proper numbers C_α are equal to the square of the stretches along the principal directions of the strain ellipsoid at P. Along the principal directions the stretches have extremal values.*

The foregoing constitutes also an algebraic proof of Cauchy's second theorem.

LEMMA 4 *The principal directions of the material and spatial ellipsoids are related to each other by*

$$N_{\alpha K} = \frac{X_{K,k} n_{\underline{\alpha} k}}{\sqrt{c_{\underline{\alpha}}}} \tag{1.10.19}$$

$$n_{\alpha k} = \sqrt{c_{\underline{\alpha}}}\, x_{k,K} N_{\underline{\alpha} K} \tag{1.10.20}$$

To prove these relations we must show (a) that $\mathbf{N}_\alpha$ are mutually orthogonal whenever $\mathbf{n}_\alpha$ are, and (b) that $N_{\alpha K}$ satisfy (1.10.3) whenever $n_{\alpha k}$ satisfy the dual of (1.10.3) for c_{kl}, namely,

$$(c_{kl} - c\delta_{kl}) n_l = 0 \tag{1.10.21}$$

To prove (a), we form

$$\mathbf{N}_\alpha \cdot \mathbf{N}_\beta = N_{\alpha K} N_{\beta K} = \frac{1}{\sqrt{c_{\underline{\alpha}} c_{\underline{\beta}}}} X_{K,k} X_{K,l} n_{\underline{\alpha} k} n_{\underline{\beta} l}$$

$$= \frac{c_{kl} n_{\underline{\alpha} k} n_{\underline{\beta} l}}{\sqrt{c_{\underline{\alpha}} c_{\underline{\beta}}}}$$

Using (1.10.21), we obtain

$$\mathbf{N}_\alpha \cdot \mathbf{N}_\beta = \frac{\sqrt{c_{\underline{\alpha}}}}{\sqrt{c_{\underline{\beta}}}}\, n_{\underline{\alpha} l} n_{\underline{\beta} l} = \delta_{\alpha\beta}$$

whenever the unit triad $\mathbf{n}_\alpha$ are mutually orthogonal. To prove (b) we form

$$C_{KL}N_{\underline{\alpha}L} = \frac{1}{\sqrt{c_{\underline{\alpha}}}}\, x_{k,K}x_{k,L}X_{L,m}n_{\underline{\alpha}m} = \frac{1}{\sqrt{c_{\underline{\alpha}}}}\, x_{k,K}n_{\underline{\alpha}k}$$

Using (1.10.21)

$$\begin{aligned} C_{KL}N_{\underline{\alpha}L} &= \frac{c_{kl}n_{\underline{\alpha}l}x_{k,K}}{c_{\underline{\alpha}}\sqrt{c_{\underline{\alpha}}}} \\ &= \frac{X_{M,k}X_{M,l}x_{k,K}n_{\underline{\alpha}l}}{c_{\underline{\alpha}}\sqrt{c_{\underline{\alpha}}}} = \frac{X_{K,l}n_{\underline{\alpha}l}}{c_{\underline{\alpha}}\sqrt{c_{\underline{\alpha}}}} \\ &= \frac{1}{c_{\underline{\alpha}}}N_{\underline{\alpha}K} \end{aligned}$$

where in the second line we used $(1.4.12)_1$ to replace c_{kl} and in the third line (1.10.19). Since $c_\alpha = 1/C_\alpha$ we get

$$C_{KL}N_{\underline{\alpha}L} = C_{\underline{\alpha}}N_{\underline{\alpha}K}$$

and this shows that $N_{\alpha K}$ given by (1.10.19) satisfy (1.10.3) whenever $n_{\alpha k}$ satisfy (1.10.21). The proof of (1.10.21) is similar to this.

This lemma is in fact the analytical proof of Corollary 3 of Cauchy's first theorem.

We may interpret these results, geometrically, as follows: *The Green and Cauchy deformation tensors* **C** *and* **c** *may be characterized as the unique symmetric tensors whose principal axes are the principal axes of the strain ellipsoids at* **X** *and* **x** *and whose proper numbers are the squares and the reciprocals of the squares of the principal stretches, respectively.*

The physical lengths remain unchanged in a coordinate transformation. Consequently C_α and by (1.10.5) E_α remain unchanged. This through (1.10.9) implies that I_E, II_E, and III_E remain unchanged. Hence,

THEOREM 2. *Quantities* I_E, II_E, *and* III_E *are invariant with respect to any coordinate transformation at* X_K. Moreover, the number of principal axes that are needed to characterize uniquely a quadric surface is not more than three. Hence,

THEOREM 3. *In three dimensions the number of independent invariants of a second-order tensor is not more than three.*[7]

[7] For analytical proof see App. B7.

Invariants may be formed from strain tensors $\mathring{E}_{KL}$, e_{kl}, $\tilde{e}_{kl}$, C_{KL}, c_{kl}, c^{-1}_{kl}, etc. These are obtained from (I.10.8) and (1.10.9) by replacement of E by $\mathring{\mathbf{E}}$, e, $\tilde{\mathbf{e}}$, C, c, $\mathbf{c}^{-1}$, etc. Upon using

$$2E_{KL} = C_{KL} - \delta_{KL}, \qquad 2e_{kl} = \delta_{kl} - c_{kl} \tag{1.10.22}$$

in (1.10.8) we can show that

$$\begin{aligned}
&1_C = 3 + 2\mathrm{I}_E, && 2\mathrm{I}_E = -3 + 1_C \\
&1\mathrm{I}_C = 3 + 4\mathrm{I}_E + 4\mathrm{II}_E, && 4\mathrm{II}_E = 3 - 2\mathrm{I}_C + \mathrm{II}_C \\
&\mathrm{III}_C = 1 + 2\mathrm{I}_E + 4\mathrm{II}_E + 8\mathrm{III}_E, && 8\mathrm{III}_E = -\mathrm{I} + \mathrm{I}_C - \mathrm{II}_C + \mathrm{III}_C
\end{aligned} \tag{1.10.23}$$

$$\begin{aligned}
&\mathrm{I}_c = 3 - 2\mathrm{I}_e \\
&1\mathrm{I}_c = 3 - 4\mathrm{I}_e + 4\mathrm{II}_e \\
&\mathrm{III}_c = \mathrm{I} - 2\mathrm{I}_e + 4\mathrm{II}_e - 8\mathrm{III}_e
\end{aligned} \tag{1.10.24}$$

In terms of principal stretches $\lambda_{(a)}$ we also have

$$\begin{aligned}
&\mathrm{I}_C = \mathrm{I}_{\overset{-1}{c}} = \lambda^2_{(1)} + \lambda^2_{(2)} + \lambda^2_{(3)} \\
&\mathrm{II}_C = \mathrm{II}_{\overset{-1}{c}} = \lambda^2_{(1)}\lambda^2_{(2)} + \lambda^2_{(2)}\lambda^2_{(3)} + \lambda^2_{(3)}\lambda^2_{(1)} \\
&\mathrm{III}_C = 1\mathrm{I}_{\overset{-1}{c}} = \lambda^2_{(1)}\lambda^2_{(2)}\lambda^2_{(3)}
\end{aligned} \tag{1.10.25}$$

$$\begin{aligned}
&\mathrm{I}_c = \mathrm{I}_{\overset{-1}{C}} = \frac{\mathrm{I}}{\lambda^2_{(1)}} + \frac{\mathrm{I}}{\lambda^2_{(2)}} + \frac{1}{\lambda^2_{(3)}} \\
&\mathrm{II}_c = \mathrm{II}_{\overset{-1}{C}} = \frac{\mathrm{I}}{\lambda^2_{(1)}\lambda^2_{(2)}} + \frac{\mathrm{I}}{\lambda^2_{(2)}\lambda^2_{(3)}} + \frac{\mathrm{I}}{\lambda^2_{(3)}\lambda^2_{(1)}} \\
&\mathrm{III}_c = 1\mathrm{I}_{\overset{-1}{C}} = \frac{\mathrm{I}}{\lambda^2_{(1)}\lambda^2_{(2)}\lambda^2_{(3)}}
\end{aligned} \tag{1.10.26}$$

From these follow the fundamental identities:

$$1_c = \frac{1\mathrm{I}_C}{\mathrm{III}_C}, \qquad \mathrm{II}_c = \frac{\mathrm{I}_C}{\mathrm{III}_C}, \qquad \mathrm{III}_c = \frac{\mathrm{I}}{\mathrm{III}_C} \tag{1.10.27}$$

Since $0 < \lambda_{(a)} < \infty$, we see that

$$0 < \mathrm{I}_C, 1\mathrm{I}_C, \mathrm{III}_C < \infty, \qquad 0 < \mathrm{I}_c, \mathrm{II}_c, \mathrm{III}_c < \infty \tag{I.10.28}$$

For a rigid deformation $\lambda_{(1)} = \lambda_{(2)} = \lambda_{(3)} = \mathrm{I}$. Hence

$$\mathrm{I}_C = \mathrm{II}_C = 3, \qquad \mathrm{III}_C = 1 \tag{1.10.29}$$

which is the necessary and sufficient condition for a local rigid deformation.

In much of the literature the quadric surface of Cauchy is studied in terms of the strain tensor **E**, that is,

$$E_{KL}\,dX_K\,dX_L = \tfrac{1}{2}(ds^2 - dS^2) = h = \text{const.} \tag{1.10.30}$$

Clearly h may be positive, negative, or zero. Thus this quadric surface is not necessarily an ellipsoid such as (1.9.4) is. Consequently the study based on (1.10.30) is more complicated. In the finite deformation theory C_{KL} plays the central role instead of E_{KL}.

1.11. ROTATION

The finite rotations of the fibers of a body resulting from a deformation cannot be characterized by the infinitesimal rotation tensors $\tilde{\mathbf{R}}$ and $\tilde{\mathbf{r}}$. For the description of the local rotation of a given fiber, we introduce a rotation tensor **R**, as follows.

Let $\mathbf{N}_\alpha$ be an orthogonal unit triad along the principal axes of strain at **X**. According to Corollary 3 of Cauchy's first theorem (Section 1.9) the deformation rotates $\mathbf{N}_\alpha$ into an orthogonal triad $\mathbf{n}_\alpha$ along the principal axes of strain at **x**. If we shift $\mathbf{N}_\alpha$ to **x** by parallel transport, we can define a unique orthogonal tensor **R** which rotates $\mathbf{N}_\alpha$ into $\mathbf{n}_\alpha$ (Figure 1.11.1).

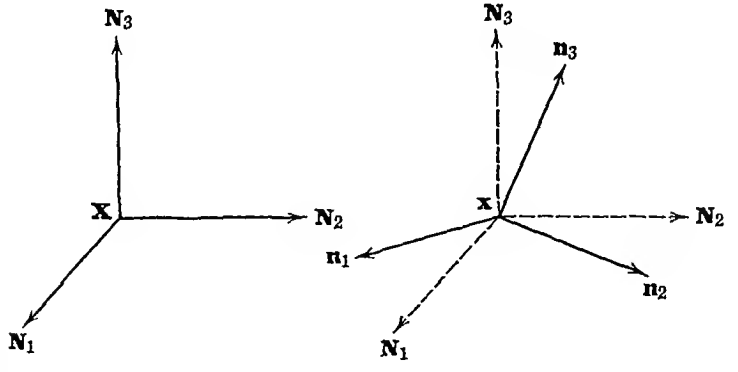


Fig. 1.11.1 Rotation of principal axes.

$$n_{\alpha k} = R_{kK} N_{\alpha K} \tag{1.11.1}$$

$$N_{\alpha K} = \overset{-1}{R}_{Kk} n_{\alpha k} \tag{1.11.2}$$

By substituting one of these equations into another, we can see that

$$\overset{-1}{R}_{kK} R_{Kl} = \delta_{kl}, \qquad \overset{-1}{R}_{Kk} R_{kL} = \delta_{KL} \tag{1.11.3}$$

Furthermore, we see that

$$R_{kK} = \overset{-1}{R}_{Kk} = N_{\alpha K} n_{\alpha k} \tag{1.11.4}$$

Hence $\overset{-1}{\mathbf{R}}$ is the reciprocal of **R** and it rotates $\mathbf{n}_\alpha$ back into $\mathbf{N}_\alpha$.

The case in which

$$\mathbf{R} = \mathbf{I}$$

where **I** is the unit tensor, is called *pure strain.*

LEMMA 1. *The* n*th powers of the Cauchy and Green deformation tensors are related to each other by*

$$\overset{-n}{C}_{KL} = \overset{-1}{R}_{Kk} \overset{n}{c}_{kl} R_{lL}, \qquad \overset{-n}{c}_{kl} = R_{kK} \overset{n}{C}_{KL} \overset{-1}{R}_{Ll} \tag{1.11.5}$$

To prove $(1.11.5)_1$, we recall that C_{KL} is a nonsingular symmetric matrix. Moreover, any positive, negative, and fractional power of this matrix has the proper numbers of C_{KL} raised to the same power, that is,

$$\overset{-n}{C}_{KL} N_{\alpha L} = (C_\alpha)^{-n} N_{\alpha K} \tag{1.11.6}$$

From this we solve for

$$\overset{-n}{C}_{KL} = \sum_\alpha (C_\alpha)^{-n} N_{\alpha K} N_{\alpha L} \tag{1.11.7}$$

Similarly, for the powers of $\overset{-n}{c}_{kl}$ we obtain

$$\overset{-n}{c}_{kl} = \sum_\alpha (c_\alpha)^{-n} n_{\alpha k} n_{\alpha l} \tag{1.11.8}$$

Now substitute N_α from (1.11.2) into (1.11.7) and write $C_\alpha = 1/c_\alpha$ in accordance with (1.10.18). Hence

$$\overset{-n}{C}_{KL} = \sum_\alpha \overset{-1}{R}_{Kk} (c_\alpha)^n n_{\alpha k} n_{\alpha l} R_{lL}$$

Upon using (1.11.8) we obtain (1.11.5). The proof of $(1.11.5)_2$ follows from (1.11.8) by use of (1.11.1).

LEMMA 2. *The displacement gradients, the Cauchy and Green deformation tensors, are related to each other by*

$$x_{k,K} = R_{kL} \overset{\frac{1}{2}}{C}_{LK} = \overset{-\frac{1}{2}}{R}_{lK} c_{kl} \tag{1.11.9}$$

$$X_{K,k} = \overset{-1}{R}_{Kl} \overset{\frac{1}{2}}{c}_{lk} = \overset{-1}{R}_{Lk} \overset{-\frac{1}{2}}{C}_{KL} \tag{1.11.10}$$

To prove these results we solve for $x_{k,K}$ from (1.10.20), that is

$$x_{k,K} = \sum_{\alpha} \sqrt{C_\alpha}\, N_{\alpha K} n_{\alpha k}$$

Substituting $n_{\alpha k}$ from (1.11.1) and using (1.11.7) with $n = -\frac{1}{2}$ we get

$$x_{k,K} = \sum_{\alpha} \sqrt{C_\alpha}\, R_{kL} N_{\alpha L} N_{\alpha K} = \overset{\frac{1}{2}}{C}_{LK} R_{kL}$$

which is the proof of $(1.11.9)_1$. We get $(1.11.9)_2$ if we use $(1.11.5)_1$ with $n = -\frac{1}{2}$ to replace C_{LK} in this equation. The proof of (1.11.10) is similar to this.

From (1.11.9) and (1.11.10) we also obtain

$$R_{kK} = x_{k,L} \overset{-\frac{1}{2}}{C}_{LK}, \qquad R_{Kk} = X_{K,l} \overset{-\frac{1}{2}}{c}_{lk} \tag{1.11.11}$$

As a corollary to this lemma we have

$$U_{K,M} = R_{KL} \overset{\frac{1}{2}}{C}_{LM} - \delta_{KM} \tag{1.11.12}$$

where

$$R_{KL} \equiv R_{kL} \delta_{kK} \tag{1.11.13}$$

To prove (1.11.12) we recall (1.5.11) in the form

$$x_{k,K} = (\delta_{MK} + U_{M,K}) \delta_{Mk} \tag{1.11.14}$$

which may be solved for $U_{M,K}$ to give

$$U_{M,K} = \delta_{kK} x_{k,M} - \delta_{MK} \tag{1.11.15}$$

Substituting $x_{k,M}$ from (1.11.9) we obtain (1.11.12).

Using (1.11.12), $(1.6.1)_1$, and $(1.6.1)_2$ we also find

$$\tilde{E}_{KM} = R_{(KL} \overset{\frac{1}{2}}{C}_{M)L} - \delta_{MK}, \qquad \tilde{R}_{KM} = R_{[KL} \overset{\frac{1}{2}}{C}_{M]L} \tag{1.11.16}$$

Another result of interest is

$$R_{KM} = (\delta_{KL} + \tilde{E}_{KL} + \tilde{R}_{KL}) \overset{-\frac{1}{2}}{C}_{LM} \tag{1.11.17}$$

which follows from using (1.6.3) in (1.11.12) and then solving for R_{KM}.

Dual results of the above formulas in the eulerian representation are not difficult to find.

If the deformation gradients are small, retaining only the lowest-order terms in (1.11.17), we find that

$$R_{KM} - \delta_{KM} \simeq \tilde{R}_{KM}, \qquad \tilde{R}_{KM} \simeq \delta_{kK} \delta_{mM} \tilde{r}_{km} \tag{1.11.18}$$

since through $(1\ 5.3)_1$ we have the approximation

$$\overset{\pm\frac{1}{2}}{C}_{KM} \simeq \delta_{KM} \pm \bar{E}_{KM}$$

These results then justify the terminology "infinitesimal rotation" for $\bar{R}_{KM}$ and $\bar{r}_{km}$.

We are now ready to prove the fundamental theorem of rotation (Theorem 3, Section 1.9).

THEOREM *The deformation of any line element at a point may be considered as resulting from a rigid translation, a rigid rotation of the principal axes of strain and stretches along these axes.* The translation, rotation, and stretch may be applied in any order.

To prove this theorem, consider the vector dX_K at X. After deformation this vector is carried to

$$dx_k = x_{k,K}\, dX_K$$

Upon using (1.11.9) we have

$$(1\ 11.19) \qquad dx_k = \delta_{kL} R_{LM} \overset{\frac{1}{2}}{C}_{MK}\, dX_K = \delta_{mK} R_{lm} \overset{-\frac{1}{2}}{c}_{kl}\, dX_K$$

where

$$(1\ 11.20) \qquad R_{km} \equiv R_{kK}\delta_{Km}$$

Equation (1.11.19) may be decomposed as follows (Figure 1.11.2):

a *The vector dX_K is rigidly translated in parallel displacement to $dx_{(T)k}$*

$$(1\ 11\ 21) \qquad dx_{(T)k} = \delta_{kK}\, dX_K$$

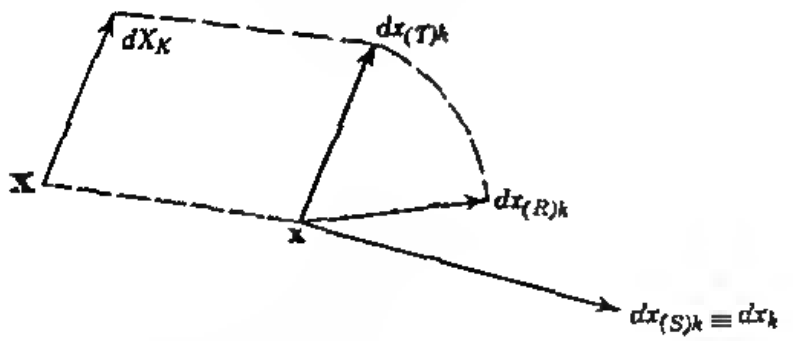


Fig. 1.11.2 Decomposition of deformation.

b. *The vector $dx_{(T)k}$ is rigidly rotated into $dx_{(R)k}$*

$$(1.11.22) \qquad dx_{(R)k} = R_{kl}\, dx_{(T)l}$$

c. *The vector $dx_{(R)k}$ is stretched into $dx_{(S)k} \equiv dx_k$*

$$(1.11.23) \qquad dx_k = \overset{-\frac{1}{2}}{c}_{kl}\, dx_{(R)l}$$

Successive substitution of (1.11.21) and (1.11.22) into (1.11.23) gives $(1.11.19)_2$.

Note that if and only if dX_K is a proper vector of C_{KL} will the stretching not involve a further rotation of the vector $dx_{(R)k}$. It can be seen also that the order of the operation of translation, rotation, and stretching is unimportant.

The foregoing description of the deformation of an infinitesimal neighborhood of a point X may be summarized as follows: An infinitesimal sphere with center at X is rigidly translated to x carrying the principal triad of the strain ellipsoid at X to x. The sphere is then rotated until the translated principal triad coincides with that of the strain ellipsoid at x. Finally, all diameters of the sphere are stretched, but only those lying along the principal directions at x are stretched without further rotation (see also Figure 1.9.3).

1.12. AREA AND VOLUME CHANGES

In this article we investigate the change of area and volume with the deformation. We have already found that (cf. Section 1.5) an infinitesimal rectangular parallelepiped with edge vectors $\mathbf{I}_1\, dX_1$, $\mathbf{I}_2\, dX_2$, and $\mathbf{I}_3\, dX_3$ after deformation becomes a rectilinear parallelepiped with edge vectors $\mathbf{C}_1\, dX_1$, $\mathbf{C}_2\, dX_2$, and $\mathbf{C}_3\, dX_3$ (Figure 1.12.1) where

$$\mathbf{C}_K = x_{k,K}\mathbf{i}_k$$

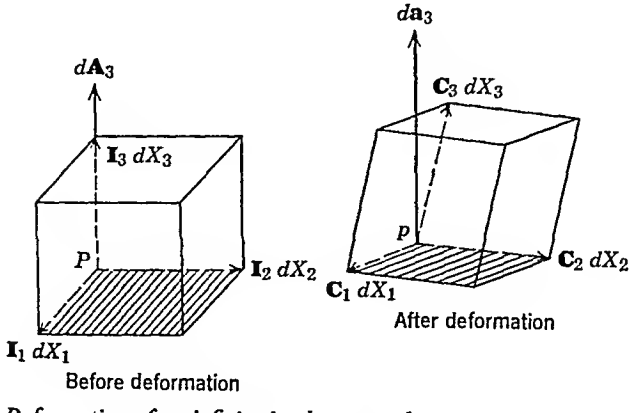


Fig. 1.12.1 Deformation of an infinitesimal rectangular parallelepiped.

We calculate the area vector $d\mathbf{a}_3$ according to

$$d\mathbf{a}_3 = (\mathbf{C}_1\, dX_1) \times (\mathbf{C}_2\, dX_2)$$
$$= x_{k,1}x_{l,2}\mathbf{i}_k \times \mathbf{i}_l\, dX_1\, dX_2$$

or since $dA_3 = dX_1\, dX_2$ and $\mathbf{i}_k \times \mathbf{i}_l = e_{klm}\mathbf{i}_m$, this reads

$$d\mathbf{a}_3 = e_{klm}x_{k,1}x_{l,2}\mathbf{i}_m\, dA_3 \tag{1.12 1}$$

This may be expressed in terms of the jacobian

$$j = |x_{k,K}| = e_{klm}x_{k,1}x_{l,2}x_{m,3} \tag{1.12.2}$$

For if we multiply both sides of this expression by $X_{3,n}$, we get

$$jX_{3,m} = e_{klm}x_{k,1}x_{l,2}$$

which upon substitution into (1.12.1) gives

$$d\mathbf{a}_3 = jX_{3,m}\mathbf{i}_m\, dA_3$$

Similar expressions are obtained for $d\mathbf{a}_1$ and $d\mathbf{a}_2$ by replacing the index 3 by 1 and 2, respectively. The area vector $d\mathbf{a}$ is therefore given by

$$\begin{aligned} d\mathbf{a} &= d\mathbf{a}_1 + d\mathbf{a}_2 + d\mathbf{a}_3 \\ &= j(X_{1,k}\, dA_1 + X_{2,k}\, dA_2 + X_{3,k}\, dA_3)\mathbf{i}_k \end{aligned}$$

or

$$d\mathbf{a} = jX_{K,k}\, dA_K\mathbf{i}_k, \quad \text{or} \quad da_k = jX_{K,k}\, dA_K \tag{1.12 3}$$

The magnitude of the area, da, is calculated by

$$da^2 = d\mathbf{a}\cdot d\mathbf{a} = j^2X_{K,k}X_{L,k}\, dA_K\, dA_L \tag{1.12 4}$$

A more concise expression results if we recall that

$$\overset{-1}{C}_{KL} = X_{K,k}X_{L,k} \tag{1.12.5}$$

and

$$|C_{KL}| = |x_{k,K}x_{k,L}| = |x_{k,K}|^2 = j^2$$

or in short

$$\mathrm{III}_C = j^2 \tag{1.12.6}$$

Substitution of (1.12 5) and (1.12.6) into (1.12.4) gives

$$da^2 = \mathrm{III}_C\overset{-1}{C}_{KL}\, dA_K\, dA_L \tag{1.12.7}$$

an expression which is analogous to the expression of the element of length

$$ds^2 = C_{KL}\, dX_K\, dX_L$$

Thus we conclude that, *with respect to the measure of area,* $\mathrm{III}_C\overset{-1}{C}_{KL}$ *plays*

a role analogous to that which the tensor C_{KL} plays with respect to the measure of length.

A dual result corresponding to (1.12.3) for the lagrangian area is

$$d\mathbf{A} = j^{-1} x_{k,K}\, da_k \mathbf{I}_K \tag{1.12.8}$$

Another useful expression is obtained through

$$|d\mathbf{a}_3|/dA_3 = |\mathbf{C}_1||\mathbf{C}_2| \sin(\mathbf{C}_1, \mathbf{C}_2)$$

Now since

$$|\mathbf{C}_1| = \sqrt{C_{11}}, \qquad |\mathbf{C}_2| = \sqrt{C_{22}}$$

$$\sin(\mathbf{C}_1, \mathbf{C}_2) = \sqrt{1 - \cos^2(\mathbf{C}_1, \mathbf{C}_2)} = \left[1 - \frac{(\mathbf{C}_1 \cdot \mathbf{C}_2)^2}{C_{11}C_{22}}\right]^{1/2}$$

by substitution we get

(1.12.9)

$$|d\mathbf{a}_3|/dA_3 = \sqrt{C_{11}C_{22} - C_{12}^2} = [(1 + 2E_{11})(1 + 2E_{22}) - 4E_{12}^2]^{1/2}$$

When the strains are small as compared to unity, a binomial expansion of the radical gives, for the lowest-order terms,

$$\frac{|d\mathbf{a}_3| - dA_3}{dA_3} \cong \tilde{E}_{11} + \tilde{E}_{22} \tag{1.12.10}$$

Similar results are valid, of course, for the other components of the area vector.

To calculate the volume dv of the deformed element we form

$$dv = d\mathbf{a}_3 \cdot \mathbf{C}_3\, dX_3$$

Using (1.12.1) and the expression of $\mathbf{C}_3$, we obtain

$$\begin{aligned} dv &= (jX_{3,k}\mathbf{i}_k) \cdot (x_{m,3}\mathbf{i}_m)\, dA_3\, dX_3 \\ &= jX_{3,k}x_{m,3}\delta_{km}\, dV \end{aligned}$$

Hence

$$dv = j\, dV = \sqrt{\mathrm{III}_C}\, dV \tag{1.12.11}$$

the extreme right being the result of use of (1.12.6). Alternative expressions are

$$\frac{dv}{dV} = j = \sqrt{\mathrm{III}_C} = \frac{1}{\sqrt{\mathrm{III}_c}} = (1 + 2\mathrm{I}_E + 4\mathrm{II}_E + 8\mathrm{III}_E)^{1/2} \tag{1.12.12}$$

$$= (1 - 2\mathrm{I}_e + 4\mathrm{II}_e - 8\mathrm{III}_e)^{-1/2}$$

which follows from using $(1.10.23)_3$, $(1.10.24)_3$, and $(1.10.27)_3$.

For infinitesimal strains a binomial expansion gives

$$\frac{dv - dV}{dV} \cong \mathrm{I}_E \cong \mathrm{I}_e \tag{1.12.13}$$

In this case the difference between lagrangian and eulerian strains vanishes. In the linear theory, sometimes I_e is called *dilatation*. *Only* in infinitesimal strain theory is the volume change per unit of the initial volume given by the dilatation.

1.13. SOME SIMPLE EXAMPLES OF STRAIN

In this section we consider some special deformations in detail. Because of simplicity in the geometry of deformations, these special examples provide a clear understanding of the nature of local deformations. We consider here only two types of deformations.

1. Deformations resulting from restrictions on the displacements, irrespective of the shape of the body and the reference frame. To this class belong *rigid deformations*, *potential deformations*, *isochoric deformations*, etc.

2. Deformations resulting from specification of the displacements in a chosen reference frame include *uniform dilatation*, *simple extensions*, *simple shear*, and *plane strain*.

Rigid Deformation. The deformation of a body is said to be *rigid* if and only if the distance between every pair of material points remains unchanged. The necessary and sufficient condition for this is

$$\mathbf{C} = \mathbf{c} = \mathbf{I} \quad \text{or} \quad \mathbf{E} = \mathbf{e} = 0 \tag{1.13.1}$$

at each point. In this case we also have

$$C_{\alpha} = \lambda_{\alpha}^{2} = 1, \qquad \mathrm{I}_C = \mathrm{II}_C = 3, \qquad \mathrm{III}_C = 1 \tag{1.13.2}$$

Potential Deformation. In Section I.7 we saw that when the displacement field is derivable from a potential, the mean rotation vanishes. The deformations for which this is valid are called *potential deformations*. Thus, for potential deformations the necessary and sufficient condition is

$$\mathbf{U} = \operatorname{grad} F \tag{1.13.3}$$

Isochoric Deformation. Deformations in which the volume element remains unaltered are called *isochoric deformations*. According to (1 12.12) the necessary and sufficient condition for this case is given by any one of

(1.13.4)
$$j = 1, \qquad \mathrm{III}_C = 1, \qquad \mathrm{III}_c = 1, \qquad 1 + 2\mathrm{I}_E + 4\mathrm{II}_E + 8\mathrm{III}_E = 1, \ldots$$

We recall the fact that any scalar invariant function of C is reducible to a function of I_C, II_C, III_C. In the case of isochoric deformations such a function would only be a function of I_C and II_C alone. From (1.10.25) we have $\lambda_{(1)}\lambda_{(2)}\lambda_{(3)} = 1$; thus

(1.13.5)
$$\mathrm{I}_C = \lambda_{(1)}^2 + \lambda_{(2)}^2 + \frac{1}{\lambda_{(1)}^2\lambda_{(2)}^2}$$
$$\mathrm{II}_C = \frac{1}{\lambda_{(1)}^2} + \frac{1}{\lambda_{(2)}^2} + \lambda_{(1)}^2\lambda_{(2)}^2$$

For infinitesimal deformations the volume is unchanged whenever

(1.13.6)
$$\mathrm{I}_E = 0 \quad \text{or} \quad \mathrm{I}_e = 0$$

This follows from (1.13.4) by neglecting all second-order terms in $(1.13.4)_4$ (see also 1.12.13).

Deformation Resulting from the Specifications of Displacements in a Chosen Frame of Reference. A large subclass of important deformations of this class is included in a homogeneous *affine* transformation carrying a material point X before deformation to x after deformation in accordance with

(1.13.7)
$$x_k = D_{kK}X_K$$

where D_{kK} is a constant matrix admitting an inverse matrix $\overset{-1}{D}_{Kk}$ so that

(1.13.8)
$$X_K = \overset{-1}{D}_{Kk}x_k$$

where D_{kK} and its inverse $\overset{-1}{D}_{Kk}$ satisfy

(1.13.9)
$$D_{kK}\overset{-1}{D}_{Kl} = \delta_{kl}, \qquad \overset{-1}{D}_{Kk}D_{kL} = \delta_{KL}$$

Deformations described by (1.13.7) carry straight lines into straight lines, ellipses into ellipses, and ellipsoids into ellipsoids. The class of such deformations is called *homogeneous strain*. Special selection of the matrix D_{kK} leads to special deformations of which a few are described below.

Uniform Dilatation. Suppose that we select $D_{11} = D_{22} = D_{33} = \lambda$ and all other $D_{kK} = 0$, that is,

(1.13.10)
$$\mathbf{D} = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}, \qquad 0 < \lambda < \infty$$

The state of deformation so obtained is called *uniform dilatation*. In this case we have

$$\mathbf{C} = \overset{-1}{\mathbf{c}} = \lambda^2\mathbf{I}, \qquad \mathbf{c} = \overset{-1}{\mathbf{C}} = \frac{\mathbf{I}}{\lambda^2}, \qquad 2\mathbf{E} = (\lambda^2 - 1)\mathbf{I} \tag{1.13.11}$$

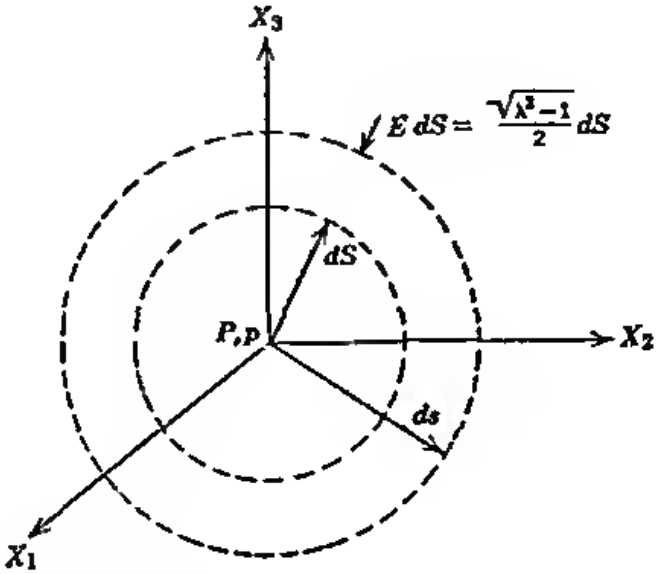


Fig. 1.13.1 Uniform dilatation

where I is the unit tensor. Thus the principal stretches (or extensions) are identical and the invariants of **C** and **E** are given by

$$\begin{aligned} &\mathrm{I}_C = 3\lambda^2, \qquad \mathrm{II}_C = 3\lambda^4, \qquad \mathrm{III}_C = \lambda^6 \\ &\mathrm{I}_E = \tfrac{3}{2}(\lambda^2 - 1), \qquad \mathrm{II}_E = \tfrac{3}{4}(\lambda^2 - 1)^2, \qquad \mathrm{III}_E = \tfrac{1}{8}(\lambda^2 - 1)^3 \end{aligned} \tag{1.13.12}$$

Clearly $\lambda > 1$ is uniform extension and $0 < \lambda < 1$ is uniform compression. The spatial ellipsoid (1.9.4) is a sphere (Figure 1.13.1)

$$\lambda^2(dX_1^2 + dX_2^2 + dX_3^2) = k^2 \tag{1.13.13}$$

Thus a uniform dilatation deforms a sphere into a sphere.

Uniaxial Strain and Simple Extension. The homogeneous strain characterized by

$$\mathbf{D} = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad 0 < \lambda < \infty \tag{1.13.14}$$

gives for the deformation and strain tensors through (1.4.12), (1.4.15), and (1.5.3)

$$\mathbf{C} = \overset{-1}{\mathbf{c}} = \begin{bmatrix} \lambda^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{c} = \overset{-1}{\mathbf{C}} = \begin{bmatrix} 1/\lambda^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{E} = \frac{1}{2}\begin{bmatrix} \lambda^2 - 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \mathbf{e} = \frac{1}{2}\begin{bmatrix} 1 - \lambda^{-2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{1.13.15}$$

For various invariants we have

$$\mathrm{I}_C = 2 + \lambda^2, \qquad \mathrm{II}_C = 1 + 2\lambda^2, \qquad \mathrm{III}_C = \lambda^2$$

$$\mathrm{I}_E = \frac{\lambda^2 - 1}{2}, \qquad \mathrm{II}_E = \mathrm{III}_E = 0 \tag{1.13.16}$$

The spatial ellipsoid is a spheroid

$$\lambda^2\, dX_1^{\,2} + dX_2^{\,2} + dX_3^{\,2} = k^2 \tag{1.13.17}$$

Geometrically, the situation is clarified by considering a rectangular parallelepiped that is stretched uniformly along one of its axes (X_1-axis) with the cross section perpendicular to it remaining unchanged (Figure 1.13.2). Note that for $\lambda > 1$ we have extension along the X_1-axis, and for $0 < \lambda < 1$ compression. The principal extensions are 27607

$$E_1 = \lambda^2 - 1 = \sqrt{1 + 2E_{11}} - 1$$

$$E_2 = E_3 = 0 \tag{1.13.18}$$

For small E_{11} as compared to unity this gives $E_1 \cong E_{11}$.

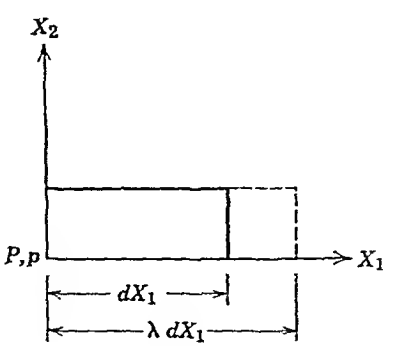


Fig. 1.13.2 Uniaxial strain.

In general, when a bar is stretched along its axis, experiment indicates that its transverse dimensions also change unless the bar is held fixed in the transverse directions. The state of strain in this case is represented by

$$\mathbf{C} = \overset{-1}{\mathbf{c}} = \begin{bmatrix} \lambda^2 & 0 & 0 \\ 0 & K^2\lambda^2 & 0 \\ 0 & 0 & K^2\lambda^2 \end{bmatrix} \tag{1 13 19}$$

where K is a constant. In this case we have

$$\lambda_{(1)} = \lambda, \qquad \lambda_{(2)} = \lambda_{(3)} = K\lambda \tag{1.13.20}$$

Therefore the *transverse contraction* ratio ν is given by

$$\nu = \frac{1 - K\lambda}{\lambda - 1} \tag{1.13.21}$$

For $K = 1$ or $\nu = -1$, we get uniform dilatation. For $K = 1/\lambda$ or $\nu = 0$, we obtain uniaxial strain. For $\nu > 0$, the cross section perpendicular to the axis of stretch contracts, and for $\nu < 0$ it swells. For actual materials, when a bar is stretched, the case $\nu < 0$ is not observed.

Simple Shear. This results when $\mathbf{D}$ has the form

$$\mathbf{D} = \begin{bmatrix} 1 & S & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad -\infty < S < \infty \tag{1.13.22}$$

where S is constant. In this case (1.13.7) has the simple form

$$x = X + SY, \qquad y = Y, \qquad z = Z \tag{1.13.23}$$

Thus, simple shear rotates $X = \text{const.}$ planes rigidly about their lines of intersection with the $Y = 0$ plane, in an amount equal to the shear angle γ given by

$$\gamma = \arctan S \tag{1 13.24}$$

The $Y = \text{const}$ and $Z = \text{const.}$ planes remain unchanged (Figure 1.13.3). Simple shear may be approximated by sliding a deck of cards from a rectangular parallelepiped form to a rectilinear parallelepiped form. In this case the Green and Cauchy deformation and strain tensors are

$$\begin{aligned} \mathbf{C} &= \begin{bmatrix} 1 & S & 0 \\ S & 1+S^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, & \mathbf{E} &= \begin{bmatrix} 0 & S/2 & 0 \\ S/2 & S^2/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \mathbf{c} &= \begin{bmatrix} 1 & -S & 0 \\ -S & 1+S^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, & \mathbf{e} &= \begin{bmatrix} 0 & S/2 & 0 \\ S/2 & -S^2/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned} \tag{1.13 25}$$

The invariants are

$$\begin{aligned} &\mathrm{I}_C = \mathrm{II}_C = 3 + S^2, \qquad \mathrm{III}_C = 1 \\ &-\mathrm{I}_E = \mathrm{I}_e = 2\mathrm{II}_E = 2\mathrm{II}_e = -\frac{S^2}{2}, \qquad \mathrm{III}_E = \mathrm{III}_e = 0 \end{aligned} \tag{1.13.26}$$

Since $\mathrm{III}_C = 1$, simple shear is isochoric. The proper numbers C_α and principal strains E_α are obtained by solving

$$-C^3 + \mathrm{I}_C C^2 - \mathrm{II}_C C + \mathrm{III}_C = 0$$

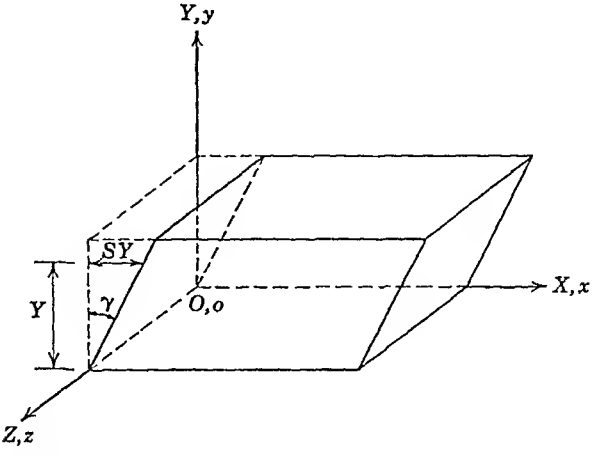


Fig. 1.13.3 Simple shear.

Using (1.13.26) we get

$$C_1 = \lambda^2_{(1)} = 1 + \tfrac{1}{2}S^2 + S\sqrt{1 + \tfrac{1}{4}S^2} = 1 + 2E_1$$

(1.13.27) $$C_2 = \lambda^2_{(2)} = 1 + \tfrac{1}{2}S^2 - S\sqrt{1 + \tfrac{1}{4}S^2} = 1 + 2E_2$$

$$C_3 = \lambda^2_{(3)} = 1 = 1 + 2E_3$$

Proper directions N_α corresponding to $\mathbf{C}$, according to Section 10, are obtained by solving

$$(C_{KL} - C\delta_{KL})N_L = 0$$

where for C we use C_1, C_2, and C_3 given above

(1.13.28) $$\left.\begin{matrix}\mathrm{N}_1\\ \mathrm{N}_2\end{matrix}\right\} = \pm \frac{\mathrm{I}_1 + (\tfrac{1}{2}S \pm \sqrt{1 + \tfrac{1}{4}S^2})\mathrm{I}_2}{\sqrt{2 + \tfrac{1}{2}S^2 \pm S\sqrt{1 + \tfrac{1}{4}S^2}}}, \qquad \mathrm{N}_3 = \mathrm{I}_3$$

In the $Z = 0$ plane, according to (1.13.23), a point $A(-S/2, 1, 0)$ after deformation occupies the position $a(S/2, 1, 0)$ (Figure 1.13.4). We have $OA = oa$; the fibers parallel to OA making an angle $\phi = \arctan(S/2)$ with the Y-axis do not alter their length. From (1.13.28), we have

(1.13.29) $$\tan\theta = \tfrac{1}{2}S + \sqrt{1 + \tfrac{1}{4}S^2}$$

The angle θ is the angle between $\mathbf{N}_1$ and X. By calculating $\tan(N_1OA) = \tan(\pi/2 - \theta + \phi)$ with the use of (1.13.29) we find that the angle (N_1OA) is identical to θ. Hence, *the principal axis* $\mathbf{N}_1$ *bisects the angle* XOA; $\mathbf{N}_2$ *is*

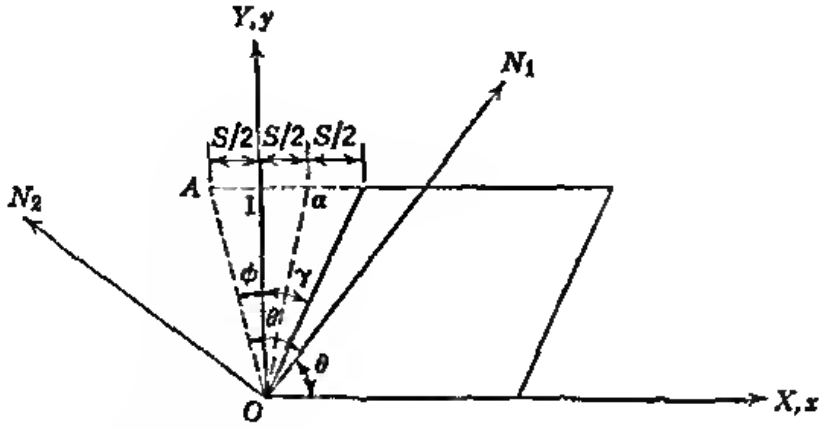


Fig. 1.13 4 Plane of shear (finite).

perpendicular to it in the $Z = const.$ *plane and* $\mathbf{N}_3$ *is along the* Z *axis. It is not difficult also to show that the greatest shear occurs at the elements bisecting the principal axes.* One of these elements makes an angle $\frac{1}{2}$ arc tan $(S/2)$ with the X-axis. Thus the maximum shear strain is given by

$$\tfrac{1}{2}S\sqrt{1 + S^2/4} \tag{1.13.30}$$

For small shear S as compared to unity, from (1.13.27)–(1.13 28), we find that

$$E_1 = -E_2 \cong \frac{S}{2}, \qquad E_3 = 0$$

$$\left.\begin{matrix}\mathbf{N}_1\\ \mathbf{N}_2\end{matrix}\right\} \cong \frac{1}{\sqrt{2}}(\pm\mathbf{I}_1 + \mathbf{I}_2), \qquad \mathbf{N}_3 = \mathbf{I}_3 \tag{1.13.31}$$

Therefore, *for the infinitesimal strain, the principal directions bisect the* X- *and* Y*-axes* (Figure 1.13.5) *and the principal strains are equal in magnitude and opposite in sign.* Consequently, a rectangular block compressed on one of its faces, and stretched equally on a perpendicular face, is equivalent to a simple shear when the extensions are infinitesimal.

Plane Strain. The plane strain is characterized by identical deformations in a family of parallel planes and no deformation in the direction of their normals. Thus, if we select X_3 as the normal of these planes, then the deformation may be expressed as

$$x_k = x_k(X_1, X_2, t) \quad (k = 1, 2), \qquad x_3 = X_3 \tag{1.13.32}$$

Since the X_3-axis is a principal axis of strain, the stretch in this direction is

unity, that is, $\lambda_{(3)} = 1$. The invariants of strain are

$$\text{(1.13.33)} \qquad \mathrm{I}_E = E_{11} + E_{22}, \qquad \mathrm{II}_E = E_{11}E_{22} - {E_{12}}^2, \qquad \mathrm{III}_E = 0$$

The principal strains follow from the solution of the characteristic equations (1.10.7).

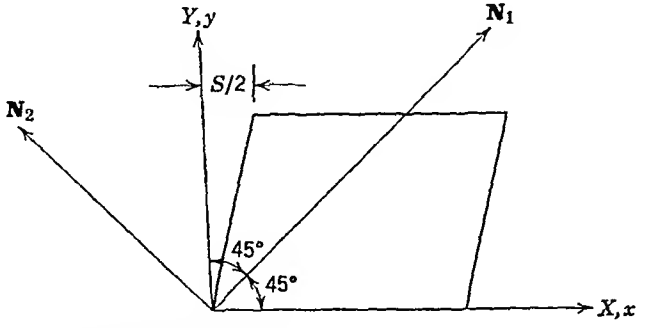


Fig. 1.13.5 Plane of shear (infinitesimal).

$$\left.\begin{matrix} E_1 \\ E_2 \end{matrix}\right\} = \tfrac{1}{2}\mathrm{I}_E \pm \sqrt{\tfrac{1}{4}{\mathrm{I}_E}^2 - \mathrm{II}_E}$$

$$\text{(1.13.34)} \qquad = \tfrac{1}{2}(E_{11} + E_{22}) \pm [\tfrac{1}{4}(E_{11} - E_{22})^2 + {E_{12}}^2]^{1/2}$$

$$E_3 = 0$$

The principal directions are now obtained by solving the three simultaneous equations (1.10.4) for each of $E = E_\alpha$ ($\alpha = 1, 2, 3$). This gives

$$\text{(1.13.35)} \qquad \frac{N_{\alpha 1}}{N_{\alpha 2}} = \frac{E_{12}}{E_\alpha - E_{11}}, \qquad N_{\alpha 3} = 0$$

Had we eliminated E_α, we would have obtained

$$\text{(1.13.36)} \qquad \frac{2N_{\alpha 1}N_{\alpha 2}}{N_{\alpha 1}^2 - N_{\alpha 2}^2} = \frac{2E_{12}}{E_{11} - E_{22}}$$

In the textbooks on strength of materials the foregoing results appear in a somewhat different form. From Figure 1.13.6 $N_{\alpha L}$ are the direction cosines of the N_α axis with rectangular coordinates. Thus, if the angle between the N_1 axis and the X axis is θ, then (1.13.36) reduces to the familiar form

$$\text{(1.13.37)} \qquad \tan 2\theta = \frac{2E_{12}}{E_{11} - E_{22}}$$

Finally, we briefly give the technique of Mohr's circle for any state of plane strain. Let X'_K be a new rectangular frame of reference having the same origin as X_K with X'_3 taken as identical to X_3. We denote the angle between X' and X by ϕ (Figure 1.13.6). Thus, we have for the cosine

directors Q_{KL}

$$Q_{11} = Q_{22} = \cos\phi, \qquad Q_{12} = -Q_{21} = \sin\phi$$
$$Q_{13} = Q_{31} = Q_{23} = Q_{32} = 0, \qquad Q_{33} = 1$$

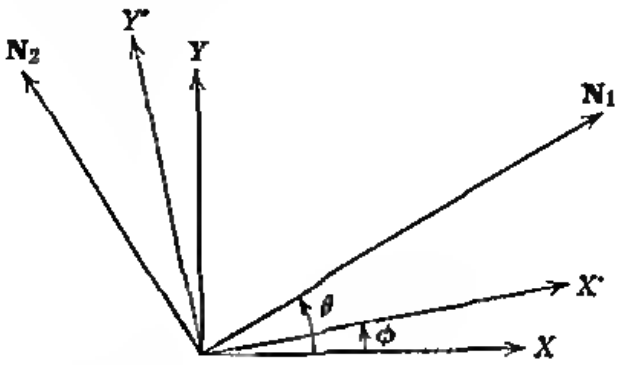


Fig 1.13.6 Rotation of axes

Substituting these into the transformation formula (1.8 7) for strains, we have

$$E'_{XX} = \tfrac{1}{2}(E_{XX} + E_{YY}) + \tfrac{1}{2}(E_{XX} - E_{YY})\cos 2\phi + E_{XY}\sin 2\phi$$
$$E'_{YY} = \tfrac{1}{2}(E_{XX} + E_{YY}) - \tfrac{1}{2}(E_{XX} - E_{YY})\cos 2\phi - E_{XY}\sin 2\phi \tag{1.13.38}$$
$$E'_{XY} = -\tfrac{1}{2}(E_{XX} - E_{YY})\sin 2\phi + E_{XY}\cos 2\phi$$

On the abscissa (marked as the E_{XX} or E_{YY} axis) of a rectangular frame, we mark $OA = E_{XX}$ and $OB = E_{YY}$ from the origin O (Figure 1.13.7), and at A and B we take $AH = E_{XY}$, $BH' = -E_{XY}$ parallel to the ordinate axis, marked as the E_{XY}-axis. The line HH' crosses the E_{XX}-axis at a point P. Taking P as the center, and PH as radius, we draw a circle We will now show that any point M and its antipode M' on this circle mark the plane strain E' with their abscissas, ordinates, and the angle made with HH', that is,

$$OA' = E'_{XX}, \qquad OB' = E'_{YY}, \qquad A'M = E'_{XY} \tag{1.13 39}$$

and $\phi = \frac{1}{2}$ angle (MPH) is the angle that X' makes with the X-axis.

These results can be seen as follows:

$$OA' = OP + PM\cos(2\theta - 2\phi) \tag{1.13.40}$$

where 2θ is the angle that PH makes with the E_{XX}-axis. But now

$$OP = \frac{1}{2}(E_{XX} + E_{YY}), \qquad PM = PH = \frac{PA}{\cos 2\theta} = \frac{E_{XX} - E_{YY}}{2\cos 2\theta}$$

$$\tan 2\theta = \frac{AH}{PA} = \frac{2E_{XY}}{E_{XX} - E_{YY}}$$

Expanding $\cos(2\theta - 2\phi)$ in (1.13.40) and using the foregoing relations we obtain

$$OA' = \tfrac{1}{2}(E_{XX} + E_{YY}) + \tfrac{1}{2}(E_{XX} - E_{YY})\cos 2\phi + E_{XY}\sin 2\phi$$

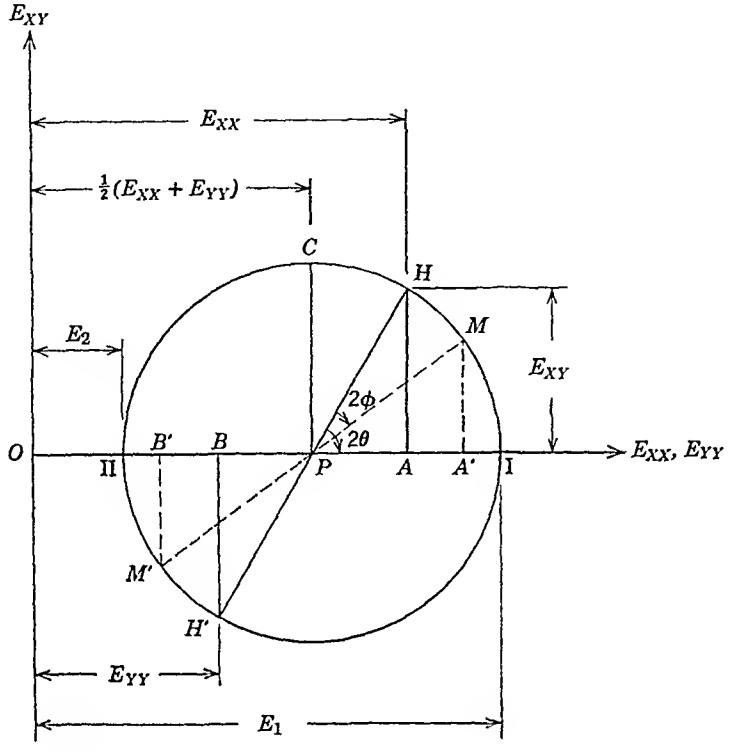


Fig. 1.13.7 Mohr's circle.

The right side of this is identical to the right side of $(1.13.38)_1$. Thus the proof of $(1.13.39)_1$. We can likewise show that the remaining expressions of (1.13.39) are valid. Also from Figure 1.13.7 it is clear that

$$\tan 2\theta = \frac{2E_{XY}}{E_{XX} - E_{YY}}$$

This agrees with (1.13.37), indicating the fact that $\theta = \frac{1}{2}$ angle (APH) is the angle that one of the principal directions makes with the X-axis. To determine the principal strain E_1, we measure

$$OI = OP + PI = OP + PH = OP + (\overline{AP}^2 + \overline{AH}^2)^{1/2}$$

$$= \tfrac{1}{2}(E_{XX} + E_{YY}) + [\tfrac{1}{4}(E_{XX} - E_{YY})^2 + E_{XY}{}^2]^{1/2}$$

This expression agrees with E_1 given by (1.13.34). Likewise, we find that $OII = E_2$. Note also that $PC = (E'_{XY})_{\max}$ and that it occurs in the coordinates

that make a 45° angle with the principal axes, since $\frac{1}{2}$ angle $(IPC) = 45°$. For other details of Mohr's circle, the reader is referred to textbooks on strength of materials.

1.14. COMPATIBILITY CONDITIONS

In a three-dimensional space the deformation tensor C_{KL} and the strain tensor E_{KL} each have six components expressible in terms of the three components U_K of the displacement vector. According to $(1.5.3)_1$ and (1.5.8) we have

$$2E_{KL} = C_{KL} - \delta_{KL} = U_{K,L} + U_{L,K} + U_{M,K}U_{M,L} \tag{1.14.1}$$

If the displacement vector possesses continuous first-order partial derivatives, by mere substitution into (1.14.1) we obtain the strain components. If, on the other hand, the six strains E_{KL} are given, then a question arises as to the existence of a *single-valued continuous displacement field* corresponding to strains. It is clear that the six partial differential equations (1.14.1) are an overdetermined system, and may not possess such a solution for the *three* unknowns U_K, unless certain *integrability* conditions are satisfied. These conditions constitute a set of partial differential equations involving E_{KL} alone, and are known as the *compatibility conditions*. When the compatibility conditions are violated, the corresponding displacement field is not unique. The body may then possess *dislocations*.

An obvious way of finding the compatibility conditions is the elimination of the displacements U_K from the six equations (1.14.1) by partial differentiation. This method (for finite strain) is, however, very tedious if not extremely awkward. An alternative method makes use of a theorem of Riemann. To prepare for this theorem, let us remember that we have referred the material points of the undeformed body to a rectangular frame of reference. This means that the three-dimensional space in which we operate is *euclidean*. Elements of arc length dS and ds of the undeformed and deformed bodies are given by (1.5.1), that is,

$$\begin{aligned} dS^2 &= \delta_{KL}\, dX_K\, dX_L = c_{kl}\, dx_k\, dx_l \\ ds^2 &= C_{KL}\, dX_K\, dX_L = \delta_{kl}\, dx_k\, dx_l \end{aligned} \tag{1.14.2}$$

Both the undeformed and deformed bodies are imbedded in an euclidean space. In the coordinates x_k the original length is calculated when all six components of c_{kl} are known, and for the calculation of the final length we need all six components of C_{KL}. Thus, if we look at the motions

$$x_k = x_k(X_1, X_2, X_3, t) \qquad (k = 1, 2, 3)$$

as the coordinate transformation from rectangular coordinates X_K to curvilinear coordinates x_k at a fixed time t, then c_{kl} plays the role of a metric tensor in the curvilinear coordinates x_k, and the same is valid for C_{KL} for the inverse motion. Now in an euclidean space any six quantities cannot be a metric tensor unless they satisfy the theorem of Riemann, namely: *For a symmetric tensor a_{kl} to be a metric tensor for an euclidean space, it is necessary and sufficient that a_{kl} be a nonsingular positive definite tensor and the Riemann-Christoffel tensor $R^{(a)}_{klmn}$ formed from it vanish identically.* By definition (App. C6) we have

$$R^{(a)}_{klmn} \equiv \tfrac{1}{2}(a_{kn,lm} + a_{lm,kn} - a_{km,ln} - a_{ln,km}) + \overset{-1}{a}_{rs}([lm, s][kn, r] - [ln, s][km, r]) \tag{1.14.3}$$

where

$$[kl, m] \equiv \tfrac{1}{2}(a_{km,l} + a_{lm,k} - a_{kl,m})$$

$$\overset{-1}{a}_{ns}a_{sl} = a_{ns}\overset{-1}{a}_{sl} = \delta_{nl} \tag{1.14.4}$$

Both C_{KL} and c_{kl} are nonsingular symmetric, positive definite tensors of the euclidean three-dimensional space. Therefore, we must have

$$R^{(C)}_{KLMN} = 0, \qquad R^{(c)}_{klmn} = 0 \tag{1.14.5}$$

In the former, the partial differentiation is understood to be with respect to X_K and in the latter with respect to x_k. In a three-dimensional space only six of the eighty-one components of R_{klmn} are algebraically independent and nonvanishing. The first of (1.14.5) gives six partial differential equations, which constitute the compatibility conditions for C_{KL}, and the second gives six for c_{kl}. Since we have

$$C_{KL} = \delta_{KL} + 2E_{KL}, \qquad c_{kl} = \delta_{kl} - 2e_{kl} \tag{1.14.6}$$

by substituting (1.14.6) for C and c in (1.14.5), we get compatibility conditions for E_{KL} and e_{kl}. Below we give one of these:

$$e_{kn,lm} + e_{lm,kn} - e_{km,ln} - e_{ln,km} - \overset{-1}{c}_{rs}[(e_{kr,n} + e_{nr,k} - e_{kn,r})(e_{ls,m} + e_{ms,l} - e_{lm,s}) - (e_{kr,m} + e_{mr,k} - e_{km,r})(e_{ls,n} + e_{ns,l} - e_{ln,s})] = 0$$

For the case of infinitesimal strains we drop all products of e_{kl}, thus obtaining the compatibility conditions for infinitesimal strain

$$\tilde{e}_{kn,lm} + \tilde{e}_{lm,kn} - \tilde{e}_{km,ln} - \tilde{e}_{ln,km} = 0 \tag{1.14.7}$$

When these conditions are satisfied, then the single-valued integral of

$$\bar{e}_{kl} = \tfrac{1}{2}(u_{k,l} + u_{l,k}) \tag{1.14 8}$$

exists and is given by

$$u_k = u_k^{\circ} + \bar{R}_{lk}x_l + b_k \tag{1.14.9}$$

where u_k° is any solution of (1 14.8), $\bar{R}_{lk}$ is a skew-symmetric tensor independent of the rectangular coordinate x_k, and b_k is a vector independent of x_k. Physically, the solution (1.14.9) of (1.148) implies that the displacement field corresponding to $\bar{e}_{kl}$ is single valued and uniquely determined to within a rigid motion.[8]

Below we give the explicit expressions of compatibility conditions (1.14,7) for $\bar{\mathbf{e}}$ in rectangular coordinates.

$$\begin{aligned}
&\frac{\partial^2\bar{e}_{xx}}{\partial y\partial z} + \frac{\partial}{\partial x}\left(\frac{\partial\bar{e}_{yz}}{\partial x} - \frac{\partial\bar{e}_{zx}}{\partial y} - \frac{\partial\bar{e}_{xy}}{\partial z}\right) = 0\\
&\frac{\partial^2\bar{e}_{yy}}{\partial z\partial x} + \frac{\partial}{\partial y}\left(\frac{\partial\bar{e}_{zx}}{\partial y} - \frac{\partial\bar{e}_{xy}}{\partial z} - \frac{\partial\bar{e}_{yz}}{\partial x}\right) = 0\\
&\frac{\partial^2\bar{e}_{zz}}{\partial x\partial y} + \frac{\partial}{\partial z}\left(\frac{\partial\bar{e}_{xy}}{\partial z} - \frac{\partial\bar{e}_{yz}}{\partial x} - \frac{\partial\bar{e}_{zx}}{\partial y}\right) = 0\\
&2\frac{\partial^2\bar{e}_{xy}}{\partial x\partial y} - \frac{\partial^2\bar{e}_{xx}}{\partial y^2} - \frac{\partial^2\bar{e}_{yy}}{\partial x^2} = 0\\
&2\frac{\partial^2\bar{e}_{yz}}{\partial y\partial z} - \frac{\partial^2\bar{e}_{yy}}{\partial z^2} - \frac{\partial^2\bar{e}_{zz}}{\partial y^2} = 0\\
&2\frac{\partial^2\bar{e}_{zx}}{\partial z\partial x} - \frac{\partial^2\bar{e}_{zz}}{\partial x^2} - \frac{\partial^2\bar{e}_{xx}}{\partial z^2} = 0
\end{aligned} \tag{1.14 10}$$

PROBLEMS

1.1. The displacement field in a body is given by

$$x_1 = (2AX_1 + B)^{1/2}, \qquad x_2 = CX_2, \qquad x_3 = DX_3$$

where (X_1, X_2, X_3) and (x_1, x_2, x_3) are rectangular coordinates, and A, B, C, and D are constants.

[8] For infinitesimal strains, an alternative derivation may be found in Sokolnikoff [1956] For a large collection of papers on the subject, see Truesdell and Toupin [1960, footnotes, Art. 34].

(*a*) Find the expression of vectors $\mathbf{c}_k$ and $\mathbf{C}_K$.

(*b*) Determine the deformation tensors c_{kl} and C_{KL}.

(*c*) Find the expressions $\overset{-1}{\mathbf{c}}_k$ and $\overset{-1}{\mathbf{C}}_K$.

(*d*) Determine the deformation tensors $\overset{-1}{c}_{kl}$ and $\overset{-1}{C}_{KL}$.

1.2. The coordinates x_1, x_2, x_3, in problem 1.1 are taken to be cylindrical coordinates having the same origin as X_K. Find a geometrical meaning for the displacement field.

1.3. The cylindrical coordinates X'_K are related to rectangular coordinates by

$$X_1 = X'_1 \cos X'_2, \qquad X_2 = X'_1 \sin X'_2, \qquad X_3 = X'_3$$

(*a*) Find the expression of the Green deformation tensor C'_{KL} in terms of the components of the Green deformation tensor C_{KL} in rectangular coordinates.

(*b*) Express the components of a vector in cylindrical coordinates in terms of the components of the same vector in rectangular coordinates.

1.4. For the displacement field given in problem 1.1, determine:

(*a*) The Lagrangian strain tensor, E_{KL}, and find the change in the square of arc length.

(*b*) The infinitesimal strains and rotations $\tilde{E}_{KL}$ and $\tilde{R}_{KL}$.

1.5. The displacement field of a body is described by

$$u_1 = A\frac{x_1 x_3}{r^3}, \qquad u_2 = A\frac{x_2 x_3}{r^3}, \qquad u_3 = A\left(\frac{x_3{}^2}{r^3} + \frac{\lambda+3\mu}{\lambda+\mu}\frac{1}{r}\right)$$

where $r \equiv (x_1{}^2 + x_2{}^2 + x_3{}^3)^{1/2}$ and A, λ, and μ are constants.

(*a*) Determine the infinitesimal strains $\tilde{e}_{kl}$ and infinitesimal rotations $\tilde{r}_{kl}$.

(*b*) Sketch the deformed shape of a spherical cavity $r = r_0$.

(*c*) Find the differences between finite strain components e_{kl} and the corresponding infinitesimal strains $\tilde{e}_{kl}$.

1.6. The rectangular frame of reference x_k is rotated about the x_3-axis by an angle measured from the x_1-axis. Find the relationships between the strains and the rotations in the two reference frames.

1.7. The parabolic cylinder coordinates x'_k are related to the rectangular coordinates x_k by

$$x_1 = c(x'_2 - x'_1), \qquad x_2 = 2c\sqrt{x'_1 x'_2}, \qquad x_3 = x'_3$$

(*a*) Sketch the coordinate surfaces and lines.

(*b*) Express the deformation tensor $\overset{-1}{\mathbf{c}}$ referred to parabolic coordinates in terms of $\overset{-1}{\tilde{C}}$, referred to rectangular coordinates.

(*c*) Find the relations between the rotation vectors $\tilde{\mathbf{r}}'$ and $\tilde{\mathbf{r}}$.

1.8. At a point in a body, principal strains are

$$E_1 = 3E, \qquad E_2 = -3E, \qquad E_3 = 7E$$

where E is constant.

(*a*) Determine the principal axes of strain **E**.

(*b*) Find the maximum shear strain and the direction in which the shear strain is maximum.

1.9. A body is in a state of plane strain in the X_1-, X_2-plane. The components of the strain tensor referred to the X_1-, X_2-axes are

$$E_{11} = -4E, \qquad E_{22} = -E, \qquad E_{33} = -3E$$

where E is a constant. Sketch Mohr's circle and determine the maximum normal and shear strains and their directions.

1.10. A body is deformed isochorically with $\mathrm{I}_C = \mathrm{II}_C$.

(*a*) Determine the principal strains.

(*b*) Determine the principal axes of strain.

(*c*) Find the stretches and sketch Cauchy's strain ellipsoid

(*d*) Draw Mohr's circle of strain

1.11. For the displacement field given in problem 1.5, determine the principal strains and principal axes

1.12. A body undergoes a displacement field given by

$$r = f(X_1), \qquad \theta = g(X_2), \qquad z = h(X_3),$$

where (r, θ, z) are the cylindrical coordinates of a point which is located at (X_1, X_2, X_3) in a rectangular frame of reference before deformation If the deformation is isochoric, determine

(*a*) The principal stretches.

(*b*) The principal directions

(*c*) The rotation tensor, R_{iI}

1.13. The uniform twist of a circular cylinder along its length may be represented by the deformation

$$X_1 = r\cos(\theta - Kz), \qquad X_2 = r\sin(\theta - Kz), \qquad X_3 = z,$$

where K is a constant, (X_1, X_2, X_3) are rectangular coordinates of a point before deformation, and (r, θ, z) are the cylindrical coordinates of the same point after deformation

(*a*) Find the deformation tensor C_{KL}.

(*b*) Determine the invariants I_C, II_C, III_C

(*c*) Find the principal axes of C_{KL} and principal strains.

(*d*) Draw the Cauchy strain ellipsoid.

1.14. At a point of a body, find the directions for which the shear strains attain the stationary values.

1.15. At a point of a body, find the directions for which the area change is stationary.

1.16. Show that the twist of a circular cylinder described in problem 1.13 is isochoric. Find the area changes and the magnitude of the element of surface

1.17. Express the ratio of the deformed area vector to the undeformed area vector in terms of eulerian strain measures.

1.18. Prove that at each point of a deformable body there exists at least one direction which is preserved upon the deformation of the body.

1.19. For the case of two-dimensional state of an infinitesimal strain field, obtain the explicit forms of the compatibility conditions by elimination of displacement components.

1.20. For the infinitesimal strains and rotations we can calculate the displacement field by integrating

$$U_{K,L} = \tilde{E}_{KL} + \tilde{R}_{KL}$$

Show that, in a simply connected domain, U_K is single-valued if, and only if, the compatibility conditions

$$\tilde{E}_{KN,LM} + \tilde{E}_{LM,KN} - \tilde{E}_{KM,LN} - \tilde{E}_{LN,KM} = 0$$

are satisfied. Determine the general solution U_K when these conditions are fulfilled.

1.21. Find the conditions of compatibility for multiply connected domains.

1.22. If, in a simply connected body, the infinitesimal strain field is given by

$$\tilde{e}_{kl} = a_{klmn} x_m x_n$$

where a_{klmn} are constants, what are the conditions that must be imposed upon a_{klmn} so that the compatibility conditions are satisfied? Find the displacement field of u_k for this case.

1.23. In a hollow circular cylinder whose axis is the z-axis, a possible two-dimensional infinitesimal strain field is given by

$$\tilde{e}_{11} = \frac{A}{2\mu}\frac{x}{r^2}\left(1 - \frac{\lambda+\mu}{\lambda+2\mu}\frac{2y^2}{r^2}\right)$$

$$\tilde{e}_{22} = \frac{A}{2(\lambda+2\mu)}\frac{x}{r^2}\left(1 - \frac{\lambda+\mu}{\mu}\frac{x^2-y^2}{r^2}\right)$$

$$\tilde{e}_{12} = \frac{A(\lambda+\mu)}{\mu(\lambda+2\mu)}\frac{yx^2}{r^4}$$

where A, λ, and μ are constants and (x, y, z) are rectangular coordinates.

(*a*) Does this strain field satisfy the compatibility conditions?

(*b*) Determine the general displacement field corresponding to these strain fields, subject to compatibility conditions.

(*c*) Is the displacement field so obtained single-valued and continuous? If not, under what condition can it be made single-valued?

1.24. (*Short Term Paper.*) Write a short paper on the relative importance of various strain measures and their uses in different fields of continuum mechanics. Present a systematic discussion on the approximations of strain and deformation tensors in terms of displacement gradients.

1.25. (*Short Term Paper.*) Make a literature search on the invariants of:

(*a*) Two symmetric tensors.

(*b*) One symmetric tensor and two vectors.

(*c*) Two symmetric tensors and one vector.

CHAPTER 2

Motion

2.1. SCOPE OF THE CHAPTER

This chapter deals with the kinematics and global kinetics of continuous media. In Chapter 1 we considered the picture of deformation at two distinct instants. Here we analyze the relationships of the varying configurations that the elements of the body acquire with the passage of time, that is, we look at a continuous film of the deformations. In Section 2.2 the material derivative of vectors, velocity and acceleration vectors are introduced, and in Section 2.3, path, stream, and streak lines.

The time rates of an element of arc, surface, and volume are essential in determining the speed with which the body deforms, and in carrying out time differentiation of integrals extended over the deforming body. These are discussed in Sections 2.4 and 2.5. In all these dynamical processes the deformation rate and vorticity tensors play a central role. These are given in Section 2.6. The strain rates are found in Section 2.7. The concepts of mass, momentum, moment of momentum, and energy, and the four basic axioms of mechanics—the conservation of mass, balance of momentum, balance of moment of momentum, and conservation of energy—are introduced in Sections 2.8 and 2.9. The discussion of global kinetics is thus complete. Finally, in Section 2.10, we discuss objective tensors, which are of central importance in the constitutive theory.

2.2. MOTION, TIME RATES OF VECTORS, VELOCITY, AND ACCELERATION

A material point $\mathbf{X}$ with rectangular coordinates X_K at time $t = 0$ in body B is carried to a spatial place $\mathbf{x}$ with rectangular coordinates x_k at time t. The motion is defined by three equations

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t) \quad \text{or} \quad x_k = x_k(X_K, t) \tag{2.2.1}$$

where t is real. The single-valued inverse of (2.2.1) is assumed to exist according to the axiom of continuity, so that the inverse motion is expressed by

$$\mathbf{X} = \mathbf{X}(\mathbf{x}, t), \qquad X_K = X_K(x_k, t) \tag{2.2.2}$$

It is assumed that both (2.2.1) *and* (2.2.2) *possess continuous partial derivatives with respect to their arguments to whatever order needed, except possibly at some singular points, lines, or surfaces.* Generally, we shall not need more than second- or third-order derivatives. Single-valuedness of (2.2.1) and (2.2.2) is essential for the axiom of *impenetrability of matter.* Otherwise the body may split, or two disjoint bodies may intermingle.

In the kinematics of continuous media, time rates of vectors and tensors are encountered. The most important among various time rates is the concept of the *material derivative.*

DEFINITION 1. *The material time rate of change of a vector (or tensor)* $\mathbf{f}$ *is defined as*

$$\frac{d\mathbf{f}}{dt} = \frac{\partial \mathbf{f}}{\partial t}\bigg|_{\mathbf{X}} \tag{2.2.3}$$

where the subscript $\mathbf{X}$ accompanying the vertical bar indicates that $\mathbf{X}$ is held constant in the differentiation of $\mathbf{f}$. If $\mathbf{f}$ is a material function, for example,

$$\mathbf{f} = \mathbf{f}(\mathbf{X}, t) = F_K(\mathbf{X}, t)\mathbf{I}_K$$

then it is clear that

$$\frac{d\mathbf{f}}{dt} = \frac{\partial F_K}{\partial t}\mathbf{I}_K$$

since the $\mathbf{I}_K$ are constant unit vectors. If, on the other hand, $\mathbf{f}$ is a spatial function, for example,

$$\mathbf{f} = \mathbf{f}(\mathbf{x}, t) = f_k(\mathbf{x}, t)\mathbf{i}_k$$

then

$$\frac{d\mathbf{f}}{dt} = \left(\frac{\partial f_k}{\partial t}\bigg|_{\mathbf{x}} + \frac{\partial f_k}{\partial x_l}\frac{\partial x_l}{\partial t}\right)\mathbf{i}_k$$

since the variable x in f is a function of X through (2.2.1) The above expression can also be written as

$$\frac{d\mathbf{f}}{dt} \equiv \dot{\mathbf{f}} = \frac{Df_k}{Dt}\mathbf{i}_k = \dot{f}_k\mathbf{i}_k \tag{2.2.4}$$

where

$$\frac{Df_k}{Dt} \equiv \dot{f}_k \equiv \frac{\partial f_k}{\partial t} + f_{k,l}\frac{\partial x_l}{\partial t} \tag{2.2.5}$$

is called the *material derivative* of f_k. The first term on the extreme right of (2.2.5) is sometimes called the *local* or *nonstationary rate* and the second term is known as the *convective time rate.*

If we wish, we may employ the common notation D/Dt to indicate the material derivatives of both material and spatial vectors and tensors with no ambiguity since

$$\frac{DF_K(\mathbf{X}, t)}{Dt} = \dot{F}_K = \frac{\partial F_K}{\partial t}$$
$$\frac{Df_k(\mathbf{x}, t)}{Dt} = \dot{f}_k = \frac{\partial f_k}{\partial t} + f_{k,l}\dot{x}_l \tag{2.2.6}$$

The concept of material differentiation can be extended to two-point vectors and tensors. For example, for a two-point vector $f_{kK}(\mathbf{x}, \mathbf{X}, t)$ we have

$$\frac{Df_{kK}}{Dt} = \left.\frac{\partial f_{kK}}{\partial t}\right|_{\mathbf{x},\mathbf{X}} + \frac{\partial f_{kK}}{\partial x_l}\dot{x}_l \tag{2.2.7}$$

It is not difficult to prove that *the material derivative obeys the rules of partial differentiation involving sums and products*, that is,

$$\frac{D}{Dt}(f_k + g_k) = \frac{Df_k}{Dt} + \frac{Dg_k}{Dt}$$
$$\frac{D}{Dt}(f_k g_l) = \frac{Df_k}{Dt}g_l + f_k\frac{Dg_l}{Dt} \tag{2.2.8}$$

DEFINITION 2. *The velocity vector* v *is the material time rate of change of the position vector of a particle.*

In rectangular coordinates, the position vector p is given by

$$\mathbf{p} = \mathbf{x}_k(\mathbf{X}, t)\mathbf{i}_k$$

The velocity vector v is thus defined by

$$\mathbf{v} \equiv \frac{d\mathbf{p}}{dt} = \left.\frac{\partial x_k}{\partial t}\right|_{\mathbf{X}}\mathbf{i}_k \quad \text{or} \quad v_k = \frac{\partial x_k}{\partial t} \tag{2.2.9}$$

By using (1.4.5) we may also write

$$\mathbf{v} = \left.\frac{\partial \mathbf{u}}{\partial t}\right|_{\mathbf{X}}, \qquad V_K = \left.\frac{\partial U_K(\mathbf{X}, t)}{\partial t}\right|_{\mathbf{X}} \tag{2.2.10}$$

In (2.2.9) we may consider

$$\mathbf{v} = \mathbf{V}(\mathbf{X}, t) = V_K(\mathbf{X}, t)\mathbf{I}_K$$

so that the identity of the particle X is known. Alternatively, using (2.2.2) we have

$$\mathbf{v} = \mathbf{V}(\mathbf{X}(\mathbf{x}, t), t) = \mathbf{v}(\mathbf{x}, t) = v_k(\mathbf{x}, t)\mathbf{i}_k$$

in which case the velocity field $\mathbf{v}(\mathbf{x}, t)$ is defined at each spatial point x at time t.

DEFINITION 3. *The acceleration vector* a *is the time rate of change of the velocity vector for a given particle*, that is,

$$\mathbf{a} \equiv \frac{d\mathbf{v}}{dt} \tag{2.2.11}$$

According to (2.2.4), we therefore have

$$\mathbf{a}(\mathbf{x}, t) \equiv \frac{Dv_k}{Dt}\mathbf{i}_k \quad \text{or} \quad a_k(\mathbf{x}, t) = \frac{\partial v_k}{\partial t} + v_{k,l}v_l \tag{2.2.12}$$

Here the term $v_{k,l}v_l$ is the *convective term.* If the lagrangian variables are used, then

$$\mathbf{a}(\mathbf{X}, t) = \left.\frac{\partial V_K(\mathbf{X}, t)}{\partial t}\right|_{\mathbf{X}} \tag{2.2.13}$$

In the lagrangian representation, the material particle with a given velocity or acceleration is identifiable. This scheme is an immediate extension of that of particle mechanics. In the eulerian description, the velocity and acceleration at time t at a spatial point are known, but the particle occupying this point is not known. As each particle moves through a spatial point, it acquires the velocity and acceleration associated with that point at that time.

SOME DEFINITIONS: A spatial point where $\mathbf{v} = \mathbf{0}$ is called a *stagnation point.* If at a given place x the velocity does not change in time, we say that the motion is *steady* at that point. For a steady motion, therefore,

$$\mathbf{v} = \mathbf{v}(\mathbf{x}) \tag{2.2.14}$$

A motion is *one dimensional* (or *lineal*) if two components of the velocity vanish, and the third depends only on one space variable, for example,

$$v_1 = v_1(x_1, t), \qquad v_2 = v_3 = 0 \tag{2.2.15}$$

A motion is a *plane motion* if the velocity component perpendicular to a set of parallel planes is zero and the two velocity components in these planes are functions of two plane coordinates, for example,

$$v_1 = v_1(x_1, x_2, t), \quad v_2 = v_2(x_1, x_2, t), \quad v_3 = 0 \tag{2.2.16}$$

Other special types of motion exist and will he studied later.

2.3. PATH LINES, STREAM LINES

A *path line* is the trace of a particle X as t varies. Thus

$$x_k = x_k(X, t), \quad X = \text{fixed} \tag{2.3.1}$$

gives the path line of a particle initially located at X The path line is also obtained as the integral curve of

$$dx_k = v_k\, dt$$

that passes through X at $t = 0$.

The stream lines at time t are the curves that are tangent to the velocity field. Hence, the integral curves to

$$\mathbf{v} = k\, d\mathbf{x}$$

or

$$\frac{dx_1}{v_1} = \frac{dx_2}{v_2} = \frac{dx_3}{v_3} = \frac{1}{k} = \text{const} \tag{2.3.2}$$

are the stream lines.

Experimentally, by dropping a small visible floating particle and taking a long time exposure, we see the trace of the particle revealing its path line. A short time exposure of a fluid onto which many floating particles are dropped will show the instantaneous direction of the velocity field, thus revealing the stream lines.

The *stream sheets* and *stream tubes*, respectively, are the collection of stream lines that intersect an open and a closed curve. Since the material particles move in the direction of the velocity field, they cannot cross the stream sheets and stream tubes.

By drawing tangent curves to any vector field at a spatial point at time t, one obtains other types of lines, for example, *vortex lines*. The sheets and tubes associated with such lines are similarly defined.

A manifold consisting of a class of material particles is called a *material manifold*. Thus, a *material line* may be defined by

$$X_\alpha = X_\alpha(S), \qquad (\alpha = 1, 2, 3) \tag{2.3.3}$$

where S is a parameter. At time t its *configuration* is given by

$$\hat{\mathbf{x}}(S, t) = \mathbf{x}(\mathbf{X}(S), t) \tag{2.3.4}$$

where $\mathbf{x} = \mathbf{x}(\mathbf{X}, t)$ is the motion.

A *material surface* may be defined by

$$X_\alpha = X_\alpha(L, M) \quad \text{or} \quad F(\mathbf{X}) = 0 \tag{2.3.5}$$

where L and M are two parameters. The configuration of this material surface is obtained by substituting (2.3.5) into the equation of motion, that is,

$$\mathbf{x} = \mathbf{x}(\mathbf{X}(L, M), t) \quad \text{or} \quad F(\mathbf{X}(\mathbf{x}, t)) = 0 \tag{2.3.6}$$

A material volume is a region of material particles. We state the following two theorems without proof.[1]

THEOREM 1 (LAGRANGE'S CRITERION). *The necessary and sufficient condition for the surface $f(\mathbf{x}, t)$ to be material is that*

$$\dot{f} = \frac{\partial f}{\partial t} + f_{,k} v_k = 0 \tag{2.3.7}$$

THEOREM 2 (HELMHOLTZ-ZORAWSKI CRITERION). *The necessary and sufficient condition for lines tangential to a vector field $\mathbf{q}$ to be material is that*

$$q_k \dot{q}_l - q_l \dot{q}_k - (q_k v_{l,m} - q_l v_{k,m}) q_m = 0 \tag{2.3.8}$$

or

$$\mathbf{q} \times \left[\frac{\partial \mathbf{q}}{\partial t} + \operatorname{curl}(\mathbf{q} \times \mathbf{v}) + \mathbf{v} \operatorname{div} \mathbf{q} \right] = \mathbf{0} \tag{2.3.9}$$

When $\mathbf{q} = \mathbf{v}$, from (2.3.9) we get

$$\mathbf{v} \times \frac{\partial \mathbf{v}}{\partial t} = 0 \quad \text{or} \quad \frac{\partial \mathbf{v}}{\partial t} = K(\mathbf{x}, t)\mathbf{v} \tag{2.3.10}$$

which is the proof of the following theorem:

THEOREM 3. *The stream lines and path lines coincide if and only if* (2.3.10) *is satisfied.*

Note that when $\partial \mathbf{v}/\partial t = \mathbf{0}$, this is the case. Therefore, for steady motion, the stream lines and path lines coincide.

2.4. MATERIAL DERIVATIVE OF THE ELEMENT OF ARC, SURFACE, AND VOLUME

The time rates of elements of arc, surface, and volume are often required

[1] For proof of these theorems see Eringen [1962, Art. 18].

in the calculations of time rates of integrals taken over material lines, surfaces, and volumes. To provide the necessary apparatus we now give

FUNDAMENTAL LEMMA. *The material derivative of the displacement gradients is given by*

$$\frac{D}{Dt}(x_{k,K}) = v_{k,l}x_{l,K}, \qquad \frac{D}{Dt}(dx_k) = v_{k,l}\,dx_l \tag{2.4.1}$$

where $dx_k \equiv x_{k,K}\,dX_K$.

Proof: Consider

$$\frac{D}{Dt}(dx_k) = \frac{D}{Dt}\left(\frac{\partial x_k}{\partial X_K}\,dX_K\right) = \frac{\partial}{\partial X_K}\left(\frac{Dx_k}{Dt}\right)dX_K$$

since in the operation D/Dt we have X_K = fixed. We now write $v_k \equiv Dx_k/Dt$ so that

$$\frac{D}{Dt}(dx_k) = \frac{\partial v_k}{\partial X_K}\,dX_K = \frac{\partial v_k}{\partial x_l}\frac{\partial x_l}{\partial X_K}\,dX_K = v_{k,l}\,dx_l$$

which proves $(2.4.1)_2$. If one writes $dx_k = x_{k,K}\,dX_K$ in $(2.4.1)_2$, then $(2.4.1)_1$ follows.

A *corollary* to the fundamental lemma is

$$\frac{D}{Dt}(X_{K,k}) = -v_{l,k}X_{K,l} \tag{2.4.2}$$

To prove this, we take the material derivative of $x_{l,L}X_{L,k} = \delta_{kl}$. Hence,

$$\frac{D}{Dt}(x_{l,L})X_{L,k} + x_{l,L}\frac{D}{Dt}(X_{L,k}) = 0$$

Using $(2.4.1)_1$, we find that this reads

$$\begin{aligned} x_{l,L}\frac{D}{Dt}(X_{L,k}) &= -v_{l,m}x_{m,L}X_{L,k} \\ &= -v_{l,k} \end{aligned}$$

Multiplying both sides by $X_{K,l}$, we obtain (2.4.2).

THEOREM 1. *The material derivative of the square of the arc length is given by*

$$\frac{D}{Dt}(ds^2) = 2d_{kl}\,dx_k\,dx_l \tag{2.4.3}$$

where

$$d_{kl} \equiv v_{(k,l)} \equiv \tfrac{1}{2}(v_{k,l} + v_{l,k}) \tag{2.4.4}$$

is called the *deformation rate* tensor of Euler.

Proof:

$$\frac{D}{Dt}(ds^2) = \frac{D}{Dt}(dx_k\, dx_k) = 2\frac{D}{Dt}(dx_k)\, dx_k = 2v_{k,l}\, dx_l\, dx_k$$

where we used $(2.4.1)_2$. Since this is symmetric in indices k and l, it may also be written as

$$\frac{D}{Dt}(ds^2) = (v_{k,l} + v_{l,k})\, dx_k\, dx_l$$

which proves (2.4.3).

In the lagrangian description we express (2.4.3) as

$$\frac{D}{Dt}(ds^2) = 2d_{kl}x_{k,K}x_{l,L}\, dX_K\, dX_L \tag{2.4.5}$$

$$= 2\dot{E}_{KL}\, dX_K\, dX_L$$

where

$$\dot{E}_{KL} \equiv \frac{D}{Dt}(E_{KL}) = \tfrac{1}{2}\dot{C}_{KL} = d_{kl}x_{k,K}x_{l,L} \tag{2.4.6}$$

is the material derivative of the lagrangian strain tensor as will be shown to be the case in Section 2.7. We have therefore not introduced a new symbol.

For an arbitrary pair of points (arbitrary dx) it is clear from (2.4.3) that $D(ds^2)/Dt = 0$ if and only if $d_{kl} = 0$. But this implies that the element of arc length for any pair is not changing in time. This, of course, is the definition of rigid body motion. Thus,

THEOREM 2 (KILLING). *The necessary and sufficient condition for the motion of a body to be rigid is* $d_{kl} = 0$.

LEMMA. *The material derivative of the jacobian is given by*

$$\frac{Dj}{Dt} = jv_{k,k} \tag{2.4.7}$$

Proof: We have

$$\frac{Dj}{Dt} = \frac{D}{Dt}|x_{k,K}| = \frac{\partial j}{\partial(x_{k,K})}\frac{D(x_{k,K})}{Dt} = \frac{\partial j}{\partial(x_{k,K})}v_{k,l}x_{l,K}$$

where we used $(2.4.1)_1$. But according to (1.4.7) we have

$$\frac{\partial j}{\partial(x_{k,K})} = jX_{K,k}$$

from which, upon substituting into the extreme right of the previous expression, we get (2.4.7).

THEOREM 3. *The material derivative of the element of area is given by*

$$\frac{D}{Dt}(da_k) = v_{m,m}\, da_k - v_{m,k}\, da_m \tag{2.4 8}$$

To prove this we calculate the material derivative of $(1.11.3)_2$, that is,

$$\frac{D}{Dt}(jX_{K,k}\, dA_K) = \left[\frac{Dj}{Dt} X_{K,k} + j \frac{D}{Dt}(X_{K,k})\right] dA_K$$

Using (2.4.7) and (2.4.2) on the right, we get (2.4.8).

THEOREM 4. *The material derivative of the volume element is given by*

$$\frac{D}{Dt}(dv) = v_{k,k}\, dv = \mathrm{I}_d\, dv \tag{2.4 9}$$

where $\mathrm{I}_d \equiv v_{k,k}$ denotes the first deformation invariant.

Proof: Since $dv = j\, dV$ we have

$$\frac{D(dv)}{Dt} = \frac{D}{Dt}(j\, dV) = \frac{Dj}{Dt} dV = v_{k,k}\, dv$$

where the last step follows from (2 4.7).

Higher-order material derivatives can be carried out by repeated differentiations. Thus for example for the Mth material derivative of ds^2 we have

$$\frac{D^M}{Dt^M}(ds^2) = A_{kl}^{(M)}\, dx_k\, dx_l \tag{2.4.10}$$

where $A_{kl}^{(M)}$ are known as the *Rivlin-Ericksen* tensors of order M. A recurrence relation for $\mathrm{A}^{(M)}$ is obtained as follows

$$\frac{D^{M+1}}{Dt^{M+1}}(ds^2) = \frac{D}{Dt}\left[\frac{D^M(ds^2)}{Dt^M}\right] = \frac{D}{Dt}[A_{kl}^{(M)}\, dx_k\, dx_l]$$

$$= \frac{D}{Dt}[A_{kl}^{(M)}]\, dx_k\, dx_l + A_{kl}^{(M)} \frac{D}{Dt}(dx_k)\, dx_l + A_{kl}^{(M)}\, dx_k \frac{D}{Dt}(dx_l)$$

Using $(2.4.1)_2$ and the definition (2.4.10), we obtain

$$\begin{aligned} A_{kl}^{(M+1)} &= \frac{D}{Dt}[A_{kl}^{(M)}] + A_{km}^{(M)} v_{m,l} + A_{ml}^{(M)} v_{m,k}, \qquad M \geq 1 \\ A_{kl}^{(1)} &\equiv 2d_{kl} = v_{k,l} + v_{l,k} \end{aligned} \tag{2.4.11}$$

An explicit expression for $\mathbf{A}^{(M)}$ can be found from

$$\frac{D^M}{Dt^M}(ds^2) = \frac{D^M}{Dt^M}(dx_k\, dx_k) = \sum_{K=0}^{M} \binom{M}{K} \frac{D^{M-K}}{Dt^{M-K}}(dx_k)\frac{D^K}{Dt^K}(dx_l)$$

where $\binom{M}{K}$ is the binomial coefficient defined by

$$\binom{M}{K} \equiv \frac{M!}{(M-K)!K!}$$

As in $(2.4.1)_2$ we can show that, for $M \geq 1$,

$$\frac{D^M}{Dt^M}(dx_k) = v^{(M)}_{k,l}\, dx_l, \qquad v^{(M)}_k \equiv \frac{D^M x_k}{Dt^M}, \qquad v^{(1)}_k \equiv v_k \tag{2.4.12}$$

Using this and (2.4.10), we find that

$$A^{(M)}_{kl} = v^{(M)}_{k,l} + v^{(M)}_{l,k} + \sum_{K=1}^{M-1} \binom{M}{K} v^{(M-K)}_{n,k} v^{(K)}_{n,l}, \qquad M \geq 1 \tag{2.4.13}$$

But since

$$\frac{D^M}{Dt^M}(ds^2) = \frac{D^M}{Dt^M}(C_{KL}\, dX_K\, dX_L) = C^{(M)}_{KL}\, dX_K\, dX_L$$

$$= C^{(M)}_{KL} X_{K,k} X_{L,l}\, dx_k\, dx_l$$

we also have

$$A^{(M)}_{kl} = C^{(M)}_{KL} X_{K,k} X_{L,l}, \qquad M \geq 1 \tag{2.4.14}$$

Therefore, if $\mathbf{x}(\mathbf{X}, 0) = \mathbf{X}$, then

$$\mathbf{A}^{(M)} = \mathbf{C}^{(M)}\Big|_{t=0} \tag{2.4.15}$$

For a rigid motion, ds^2 remains unchanged, hence $\mathbf{A}^{(M)}$ must vanish.

2.5. KINEMATICS OF LINE, SURFACE, AND VOLUME INTEGRALS

LEMMA 1. *The material derivative of a line integral of any field φ over a material line $\mathscr{C}$ is calculated by*

$$\frac{D}{Dt}\int_{\mathscr{C}} \varphi\, dx_k = \int_{\mathscr{C}} (\dot{\varphi}\, dx_k + \varphi v_{k,l}\, dx_l) \tag{2.5.1}$$

Proof: A material line $\mathscr{C}$ has an equation $\mathbf{X} = \mathbf{X}(S)$. Therefore, the integral on the left of (2.5.1) has a fixed limit in the material description. Thus, the operator D/Dt may be interchanged with the integral operator.

Hence,

$$\frac{D}{Dt}\int_{\mathscr{C}} \varphi\, dx_k = \int_{\mathscr{C}} \frac{D}{Dt}(\varphi\, dx_k) = \int_{\mathscr{C}} \left[\dot{\varphi}\, dx_k + \varphi \frac{D}{Dt}(dx_k)\right]$$

using $(2.4.1)_2$ we get (2.5.1).

For an integral over a *fixed spatial line* c, the counterpart of (2.5.1) reads

$$\frac{\partial}{\partial t}\int_c \varphi\, dx_k = \int_c \frac{\partial \varphi}{\partial t}\, dx_k \tag{2.5.2}$$

Note that the difference between (2.5.1) and (2.5.2) arises from the motion of the particles constituting $\mathscr{C}$.

LEMMA 2. *The material derivative of a surface integral of any field over a material surface* $\mathscr{S}$ *is given by*

$$\frac{D}{Dt}\int_{\mathscr{S}} \varphi\, da_k = \int_{\mathscr{S}} [\dot{\varphi}\, da_k + \varphi(-v_{l,k}\, da_l + v_{l,l}\, da_k)] \tag{2.5.3}$$

Proof of this again follows by carrying D/Dt inside (since in material description $\mathscr{S}$ is fixed) and using (2.4.8).

For a *spatially fixed surface* s, the counterpart of (2.5.3) is

$$\frac{\partial}{\partial t}\int_s \varphi\, da_k = \int_s \frac{\partial \varphi}{\partial t}\, da_k \tag{2.5.4}$$

Selecting φ as a vector field $\mathbf{q}$, we may write (2.5.3) as

$$\begin{aligned}\frac{D}{Dt}\int_{\mathscr{S}} q_k\, da_k &= \int_{\mathscr{S}} (\dot{q}_k - q_l v_{k,l} + q_k v_{l,l})\, da_k \\ &= \int_{\mathscr{S}} d\mathbf{a} \cdot \left[\frac{\partial \mathbf{q}}{\partial t} + \operatorname{curl}(\mathbf{q} \times \mathbf{v}) + \mathbf{v} \operatorname{div} \mathbf{q}\right]\end{aligned} \tag{2.5.5}$$

From this there follows

ZORAWSKI'S CRITERION. *For a flux of a vector* $\mathbf{q}$ *across every material surface to remain constant, it is necessary and sufficient that*

$$\frac{\partial \mathbf{q}}{\partial t} + \operatorname{curl}(\mathbf{q} \times \mathbf{v}) + \mathbf{v} \operatorname{div} \mathbf{q} = \mathbf{0} \tag{2.5 6}$$

The foregoing result (2.5.5) is valid for an arbitrary spatial surface $s(t)$ bounded by a closed curve $c(t)$ which is moving with velocity $\mathbf{v}$, that is,

$$\frac{D}{Dt}\int_{s(t)} \mathbf{q} \cdot d\mathbf{a} = \int_{s(t)} \left[\frac{\partial \mathbf{q}}{\partial t} + \operatorname{cupl}(\mathbf{q} \times \mathbf{v}) + \mathbf{v} \operatorname{div} \mathbf{q}\right] \cdot d\mathbf{a} \tag{2.5.7}$$

Thus, this may be seen by imagining that $s(t) + c(t)$ consists of some fictitious material particles having velocity $\mathbf{v}$.

Using the Stokes theorem (A14.2) we convert the second term on the right-hand side of (2.5.8) to a line integral.

$$\frac{D}{Dt}\int_{s(t)} \mathbf{q}\cdot d\mathbf{a} = \int_{s(t)} \left(\frac{\partial \mathbf{q}}{\partial t} + \mathbf{v}\,\mathrm{div}\,\mathbf{q}\right)\cdot d\mathbf{a} + \int_{c(t)} (\mathbf{q}\times\mathbf{v})\cdot d\mathbf{p} \tag{2.5.8}$$

LEMMA 3. *The material derivative of any field over a material volume is given by*

$$\frac{D}{Dt}\int_{\mathscr{V}} \varphi\, dv = \int_{\mathscr{V}} (\dot{\varphi} + \varphi v_{k,k})\, dv = \int_{\mathscr{V}} \left[\frac{\partial \varphi}{\partial t} + (\varphi v_k)_{,k}\right] dv \tag{2.5.9}$$

The proof again follows by carrying D/Dt inside of the integral (since, in the material volume $\mathscr{V}$, the limits of the integral are fixed) and using (2.4.9).

By the Green-Gauss theorem the second term on the extreme right of (2.5.7) may be converted to a surface integral. If the field φ is continuous throughout the body, then this gives

$$\frac{D}{Dt}\int_{\mathscr{V}} \varphi\, dv = \int_{\mathscr{V}} \frac{\partial \varphi}{\partial t}\, dv + \int_{\mathscr{S}} \varphi v_k\, da_k \tag{2.5.10}$$

If we select $\mathscr{V}$ and $\mathscr{S}$ to coincide instantaneously with the fixed spatial volume v and its surface s, then (2.5.10) may also be expressed as

$$\frac{D}{Dt}\int_{\mathscr{V}} \varphi\, dv = \int_{v} \frac{\partial \varphi}{\partial t}\, dv + \int_{s} \varphi v_k\, da_k \tag{2.5.11}$$

which states that:

The rate of change of the total φ over a material volume is identical to the rate of creation of φ in a fixed volume v instantaneously coinciding with $\mathscr{V}$ and the flux φv_k through the bounding surface s of v.

By a similar argument which led to (2.5.8), we see that (2.5.11) can be extended to an arbitrary spatial volume $v(t)$ bounded by a closed surface $s(t)$ which is moving with velocity $\mathbf{v}$, that is,

$$\frac{D}{Dt}\int_{v(t)} \varphi\, dv = \int_{v(t)} \frac{\partial \varphi}{\partial t}\, dv + \int_{s(t)} \varphi\mathbf{v}\cdot d\mathbf{a} \tag{2.5.12}$$

The time rates of integrals over regions containing a discontinuity surface are common occurrences in the study of waves, shocks, and other types of discontinuities. We give below expressions modifying (2.5.5) for a surface integral containing a moving discontinuity line and (2.5.9) for a volume integral over a volume intersected by a moving discontinuity surface.

Consider first the case of a material volume $\mathscr{V}$ intersected by a discontinuity surface $\sigma(t)$ moving with a velocity $\mathbf{v}$ (Fig 2.5.1). Applying (2.5.12) to the two volumes $\mathscr{V}^+$ and $\mathscr{V}^-$ bounded by $\mathscr{S}^+ + \sigma^+$ and $\mathscr{S}^- + \sigma^-$ respectively, we write

$$\frac{D}{Dt}\int_{\mathscr{V}^+}\varphi\,dv = \int_{\mathscr{V}^+}\frac{\partial\varphi}{\partial t}\,dv + \int_{\mathscr{S}^+}\varphi\mathbf{v}\cdot d\mathbf{a} - \int_{\sigma^+}\varphi\mathbf{v}\cdot d\mathbf{a}$$

$$\frac{D}{Dt}\int_{\mathscr{V}^-}\varphi\,dv = \int_{\mathscr{V}^-}\frac{\partial\varphi}{\partial t}\,dv + \int_{\mathscr{S}^-}\varphi\mathbf{v}\cdot d\mathbf{a} + \int_{\sigma^-}\varphi\mathbf{v}\cdot d\mathbf{a}$$

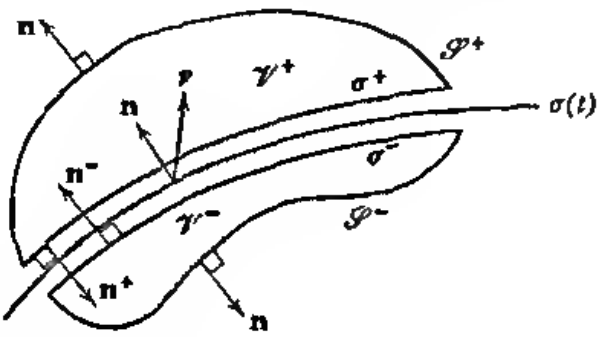


Fig. 2.5.1 Discontinuity surface.

Upon adding these two equations and letting σ^+ and σ^- approach σ, we obtain

$$\frac{D}{Dt}\int_{\mathscr{V}-\sigma}\varphi\,dv = \int_{\mathscr{V}-\sigma}\frac{\partial\varphi}{\partial t}\,dv + \int_{\mathscr{S}-\sigma}\varphi\mathbf{v}\cdot d\mathbf{a} - \int_{\sigma}[\varphi\mathbf{v}]\cdot d\mathbf{a}$$

Using the Green-Gauss theorem (A14.3) to replace the second term on the right-hand side, we obtain

(2.5.13)

$$\frac{D}{Dt}\int_{\mathscr{V}-\sigma}\varphi\,dv = \int_{\mathscr{V}-\sigma}\left[\frac{\partial\varphi}{\partial t} + \operatorname{div}(\varphi\mathbf{v})\right]dv + \int_{\sigma}[\varphi(\mathbf{v}-\mathbf{v})]\cdot d\mathbf{a}$$

where a bold-face bracket indicates the jump of the enclosed quantity across $\sigma(t)$, for example,

$$[\mathbf{A}] \equiv \mathbf{A}^+ - \mathbf{A}^-$$

Here $\mathbf{A}^+$ and $\mathbf{A}^-$ are the values of $\mathbf{A}$ from positive and negative sides of $\mathbf{n}$ of $\sigma(t)$.

A similar argument can be extended to a material surface $\mathscr{S}$ intersected by a discontinuity line $\gamma(t)$ which is moving with a velocity $\mathbf{v}$ on $\mathscr{S}$. Applying (2.5.8) to the surfaces $\mathscr{S}^+$ and $\mathscr{S}^-$ (Fig. 2.5.2), using (A14.4), this time, for the conversion of a line integral on $\mathscr{C} - \gamma$, and assuming that the normal

component of $\mathbf{v}$ to $\mathscr{S}$ is continuous, we obtain

$$\frac{D}{Dt}\int_{\mathscr{S}} \mathbf{q}\cdot d\mathbf{a} = \int_{\mathscr{S}-\gamma}\left[\frac{\partial \mathbf{q}}{\partial t} + \operatorname{curl}(\mathbf{q}\times\mathbf{v}) + \mathbf{v}\operatorname{div}\mathbf{q}\right]\cdot d\mathbf{a} + \int_{\gamma(t)}[\mathbf{q}\times(\mathbf{v}-\mathbf{v})]\cdot\mathbf{k}\,ds \tag{2.5.14}$$

This result finds important applications in the electromagnetic theory of moving media.

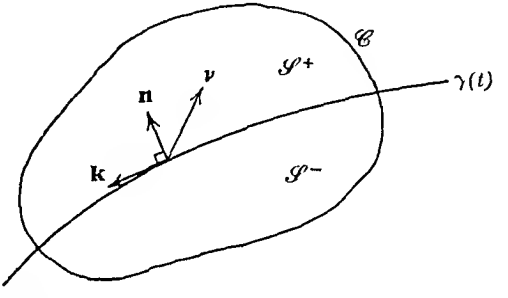


Fig. 2.5.2 Discontinuity line.

In the theory of continuous media two types of balance equations are of common occurrence:

$$\frac{D}{Dt}\int_{\mathscr{S}} \mathbf{q}\cdot d\mathbf{a} = \oint_{\mathscr{C}} \mathbf{h}\cdot d\mathbf{s} + \int_{\mathscr{S}} \mathbf{r}\cdot d\mathbf{a} \tag{2.5.15}$$

$$\frac{D}{Dt}\int_{\mathscr{V}} \varphi\, dv = \oint_{\mathscr{S}} \tau_k n_k\, da + \int_{\mathscr{V}} g\, dv \tag{2.5.16}$$

where $\mathbf{q}$, $\mathbf{h}$, and $\mathbf{r}$ are, generally, vector fields and φ, τ_k, and g are tensor fields. For future convenience we would like to give expressions of these integrals for a region $\mathscr{V}$ containing a moving discontinuity surface $\sigma(t)$ and for a surface $\mathscr{S}$ containing a moving discontinuity line $\gamma(t)$. To this end we apply (2.5.14) and the Stokes theorem (A14.4) to (2.5.15) and (2.5.13) and the Green-Gauss theorem (A14.3) to (2.5.16). The result is

$$\int_{\mathscr{S}-\gamma}\left[\frac{\partial \mathbf{q}}{\partial t} + \operatorname{curl}(\mathbf{q}\times\mathbf{v}) + \mathbf{v}\operatorname{div}\mathbf{q} - \operatorname{curl}\mathbf{h} - \mathbf{r}\right]\cdot d\mathbf{a} + \int_{\gamma(t)}[\mathbf{q}\times(\mathbf{v}-\mathbf{v}) - \mathbf{h}]\cdot\mathbf{k}\,ds = \mathbf{0} \tag{2.5.17}$$

$$\int_{\mathscr{V}-\sigma}\left[\frac{\partial \varphi}{\partial t} + \operatorname{div}(\varphi\mathbf{v}) - \tau_{k,k} - g\right] dv + \int_{\sigma(t)}[\varphi(v_k - v_k) - \tau_k]n_k\, da = 0 \tag{2.5.18}$$

These expressions find important applications in the derivation of the field equations and jump conditions in the theory of continuous media.

2.6. DEFORMATION RATE, SPIN, VORTICITY

The deformation rate tensor d_{kl} was already introduced in Section 2.4. We define this and the *spin* tensor w_{kl} below:

$$d_{kl} \equiv v_{(k,l)} \equiv \tfrac{1}{2}(v_{k,l} + v_{l,k}) \tag{2.6.1}$$

$$w_{kl} \equiv v_{[k,l]} \equiv \tfrac{1}{2}(v_{k,l} - v_{l,k}) \tag{2.6.2}$$

By adding these two equations, we get

$$v_{k,l} = d_{kl} + w_{kl} \tag{2.6.3}$$

The deformation rate is a symmetric tensor, while the spin is a skew-symmetric tensor, that is,

$$d_{kl} = d_{lk}, \qquad w_{kl} = -w_{lk} \tag{2.6.4}$$

In three dimensions we can always construct an *axial vector* w_k from a skew-symmetric tensor w_{kl}. Thus,

$$w_k \equiv e_{klm} w_{ml} = e_{klm} v_{m,l} \quad \text{or} \quad \mathbf{w} \equiv \operatorname{curl} \mathbf{v} \tag{2.6.5}$$

is an axial vector called the *vorticity vector*.

The physical significance of these tensors and vectors is understood by studying the time rate of change of material lengths and angles. In order to prepare the ground, we first introduce the physical quantity $d_{(\mathbf{n})}$ called the *stretching* in the direction of a unit vector n at x, by

$$d_{(\mathbf{n})} \equiv \frac{1}{ds}\frac{D(ds)}{Dt} = d_{kl} n_k n_l, \qquad n_k \equiv \frac{dx_k}{ds} \tag{2.6.6}$$

Therefore the normal component of the deformation rate tensor is the stretching. This point is clarified further by selecting n in the direction of one of the coordinate axes, say x_1. Then $n_1 = 1$, $n_2 = n_3 = 0$, and we get

$$d_{(1)} = d_{11} \tag{2.6.7}$$

To understand the physical meaning of the mixed components of d and those of w, we now calculate the time rate of change of the angle $\theta_{(\mathbf{n}_1\mathbf{n}_2)}$ between two unit vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ at x.

$$\begin{aligned} \frac{D}{Dt}\cos\theta_{(\mathbf{n}_1\mathbf{n}_2)} &= \frac{D}{Dt}\left(\frac{dx_{1k}}{ds_1}\frac{dx_{2k}}{ds_2}\right) \\ &= \frac{1}{ds_1 ds_2}\frac{D}{Dt}(dx_{1k}\,dx_{2k}) - [d_{(\mathbf{n}_1)} + d_{(\mathbf{n}_2)}]\cos\theta_{(\mathbf{n}_1\mathbf{n}_2)} \end{aligned} \tag{2.6.8}$$

where ds_1 and ds_2, respectively, are the lengths of the vectors $d\mathbf{x}_1$ and $d\mathbf{x}_2$ taken along $\mathbf{n}_1$ and $\mathbf{n}_2$. Consequently,

$$\mathbf{n}_1 = \frac{d\mathbf{x}_1}{ds_1}, \qquad \mathbf{n}_2 = \frac{d\mathbf{x}_2}{ds_2}$$

and $d_{(\mathbf{n}_1)}$ and $d_{(\mathbf{n}_2)}$ are the stretchings of $d\mathbf{x}_1$ and $d\mathbf{x}_2$. Using $(2.4.1)_2$, we get

$$\frac{D}{Dt}(dx_{1k}\, dx_{2k}) = 2d_{kl}\, dx_{1k}\, dx_{2l} \tag{2.6.9}$$

Substituting this into (2.6.8), we find

$$-\sin\theta_{(\mathbf{n}_1\mathbf{n}_2)}\dot{\theta}_{(\mathbf{n}_1\mathbf{n}_2)} = 2d_{kl}n_{1k}n_{2l} - [d_{(\mathbf{n}_1)} + d_{(\mathbf{n}_2)}]\cos\theta_{(\mathbf{n}_1\mathbf{n}_2)} \tag{2.6.10}$$

For $\theta_{(\mathbf{n}_1\mathbf{n}_2)} = 0$ this reduces to (2.6.6) and for $\theta_{(\mathbf{n}_1\mathbf{n}_2)} \neq 0$ it gives the *shearing* $-\dot{\theta}_{(\mathbf{n}_1\mathbf{n}_2)}$ of the directions $d\mathbf{x}_1$ and $d\mathbf{x}_2$.
Now consider the special case in which $\mathbf{n}_1$ and $\mathbf{n}_2$ are in the directions of two rectangular coordinate axes, say x_1 and x_2, respectively. Then $\theta_{(12)} = \pi/2$, $n_{1k} = \delta_{1k}$, $n_{2l} = \delta_{2l}$, and (2.6.10) reduces to

$$-\dot{\theta}_{(12)} = 2d_{12} \tag{2.6.11}$$

which shows that d_{12} is one-half of the shearing in the x_1 and x_2 directions. Hence we have proved

THEOREM 1. *The normal components of the deformation rate tensor are the stretchings, and the mixed components are half the shearings in rectangular coordinates.*

The physical significance of the spin tensor is understood if we consider the rate at which a material element $d\mathbf{x}$ rotates about a fixed direction ν. To this end we calculate the material derivative of

$$\cos\varphi \equiv \cos(d\mathbf{x}, \boldsymbol{\nu}) = \frac{dx_k}{ds}\nu_k = n_k\nu_k$$

that is,

$$-\sin\varphi\dot{\varphi} = \frac{D}{Dt}\left(\frac{dx_k}{ds}\nu_k\right) \tag{2.6.12}$$

$$= v_{l,k}n_k\nu_l - d_{(\mathbf{n})}\cos\varphi$$

When $\varphi = \pi/2$, this gives

$$-\dot{\varphi} = v_{l,k}n_k\nu_l \tag{2.6.13}$$

If ν is taken along x_1 and n along x_2, then the rate at which x_2 is rotating toward x_1 is given by

$$-\dot{\varphi}_{12} = v_{1,2}$$

Thus $v_{1,2}$ *is the rate at which a material element along the* x_2*-axis is rotating towards the* x_1*-axis in a counterclockwise direction* (Figure 2.6.1).

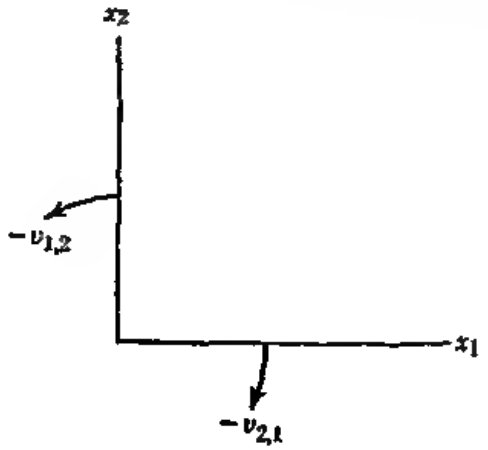


Fig. 2.6.1. Spin components

The shearing in the x_1- and x_2-directions is the sum of these spins since these axes are turning away from each other, that is,

$$-\theta_{(12)} \equiv -\dot{\phi}_{12} - \varphi_{21} = 2d_{12}$$

On the other hand, the rate of rotation in the counterclockwise direction of a material lamina lying along x_1- and x_2-axes is given by

$$\dot{\phi}_{21} - \dot{\phi}_{12} = v_{1,2} - v_{2,1} = 2w_{12}$$

Therefore *one half the difference of the relative spins* (in counterclockwise direction) *of the* x_1- and x_2*-axes gives the component* w_{12} *of the spin tensor* w.

Since $w_1 = -2w_{23}$, $w_2 = -2w_{31}$, and $w_3 = -2w_{12}$, we see that the same thing applies to $-w_k/2$.

According to Killing's theorem, the necessary and sufficient condition for a rigid motion is $d_{kl} = 0$. If this holds throughout a region of the body, we have the general solution

$$v_k = \tilde{\omega}_{kl}x_l + b_k \tag{2.6 14}$$

where b_k and $\tilde{\omega}_{kl}$ are functions of t alone and $\tilde{\omega}_{kl}$ is the rotation rate tensor and $\tilde{\omega}_k = e_{klm}\tilde{\omega}_{ml}$ is the angular velocity vector about the origin of x. From (2.6.2) for rigid motion it follows that

$$w_{kl} = \tilde{\omega}_{kl} \tag{2.6.15}$$

Therefore for a rigid motion the spin tensor is equal to the rotation rate tensor.

We close this article with the following elegant theorem.

THEOREM 2. *The spin is the angular velocity of the principal axes of the deformation-rate tensor.*

The principal directions $\mathbf{n}_\alpha$ of d_{kl} and the principal stretchings d_α are determined in a similar fashion to those of the strain tensor. Directions $\mathbf{n}_\alpha$ are the solutions of the system.

$$(d_{kl} - d\delta_{kl})n_l = 0 \tag{2.6.16}$$

for each of the roots d_α of the characteristic equation

$$|d_{kl} - d\delta_{kl}| = 0 \tag{2.6.17}$$

Since d_{kl} is a symmetric tensor, everything that has been said in connection with E_{KL} applies to this case in the eulerian representation.

Let us now calculate the material derivative of a unit vector $\mathbf{n}$.

$$\dot{n}_k = \frac{D}{Dt}\left(\frac{dx_k}{ds}\right) = \frac{1}{ds}\frac{D}{Dt}(dx_k) - \frac{1}{ds^2}\frac{D}{Dt}(ds)\,dx_k$$

Using $(2.4.1)_2$ and (2.6.6), we get

$$\dot{n}_k = (d_{kl} + w_{kl} - d\delta_{kl})n_l \tag{2.6.18}$$

When $\mathbf{n}$ is a proper direction, (2.6.16) applies and this reduces to

$$\dot{n}_k = w_{kl}n_l \tag{2.6.19}$$

This, of course, is the analytical proof of Theorem 2.

2.7. STRAIN RATE

The lagrangian and eulerian strain rates are defined by

$$\dot{E}_{KL}(\mathrm{X}, t) \equiv \frac{DE_{KL}}{Dt} = \frac{\partial E_{KL}}{\partial t} \tag{2.7.1}$$

$$\dot{e}_{kl}(\mathrm{x}, t) \equiv \frac{De_{kl}}{Dt} = \left.\frac{\partial e_{kl}}{\partial t}\right|_{\mathrm{x}} + \frac{\partial e_{kl}}{\partial x_m}v_m \tag{2.7.2}$$

THEOREM 1. *The lagrangian and eulerian strain rates are given by*

$$\dot{E}_{KL} = d_{kl}x_{k,K}x_{l,L} \tag{2.7.3}$$

$$\dot{e}_{kl} = d_{kl} - e_{mk}v_{m,l} - e_{ml}v_{m,k} \tag{2.7.4}$$

Proof: To prove (2.7.3) we calculate the material derivative of C_{KL}, that is,

$$\dot{C}_{KL} = 2\dot{E}_{KL} = \frac{D}{Dt}(x_{k,K}x_{k,L}) = \frac{Dx_{k,K}}{Dt}x_{k,L} + x_{k,K}\frac{Dx_{k\,L}}{Dt}$$

Using (2.4.1), we find that this gives (2.7.3)

To prove (2.7.4) we make use of the expression $(1.5.4)_2$ of e_{kl}

$$e_{kl} = E_{kL}X_{K,k}X_{L,l}$$

The material derivative of this expression is

$$\dot{e}_{kl} = \frac{D}{Dt}(E_{KL}X_{K,k}X_{L,l}) = \dot{E}_{KL}X_{K,k}X_{L,l} + E_{KL}\frac{DX_{K,k}}{Dt}X_{L,l} + E_{KL}X_{K,k}\frac{DX_{L,l}}{Dt}$$

Upon using (2.7.3) and (2.4.2) this may be reduced to

$$\dot{e}_{kl} = d_{kl} - v_{m,k}E_{KL}X_{K,m}X_{L,l} - v_{m,l}E_{KL}X_{K,k}X_{L,m}$$

With the help of the above expression of e_{kl}, this gives (2.7.4).

From the foregoing it is also clear that

(2.7.5) $$\dot{C}_{KL} = 2\dot{E}_{KL}, \qquad \dot{e}_{kl} = -2\dot{e}_{kl} = -c_{mk}v_{m,l} - c_{ml}v_{m,k}$$

From (2.7.1) and (2.7.2) it is clear that the two strain rates are in general not the same as the deformation rate. At time $t = 0$ the medium is unstrained. Taking $X_K = \delta_{Kk}x_k$, we get

(2.7.6) $$\dot{E}_{KL}(X, 0) = d_{kl}\delta_{kK}\delta_{lL}, \qquad \dot{e}_{kl}(x, 0) = d_{kl}$$

For infinitesimal deformations the lagrangian strain rates are approximately equal to the deformation rates, so that

(2.7.7) $$\dot{E}_{KL} \cong d_{kl}\delta_{kK}\delta_{lL}$$

where δ_{kK} is the Kronecker delta. However, for the eulerian strain rate to be equal to the deformation rate, that is,

$$\dot{e}_{kl} \cong d_{kl}$$

we need the further restriction that the spin tensor w_{kl} be of no greater order of magnitude than the deformation rate tensor, since in (2.7.4) we can write $v_{m,l} = d_{ml} + w_{ml}$.

2.8. MASS, MOMENTUM, MOMENT OF MOMENTUM, AND ENERGY

In mechanics, with each body there is associated a measure called *mass*. Mass has a dimension M independent of length L and time T. *It is non-*

negative and additive, and it is invariant under the motion. If the mass is *absolutely continuous* in the space variables, then there exists a density ρ called the *mass density.* The total mass M of the body is then determined by

$$M = \int_{\mathcal{V}} \rho \, dv, \qquad \dim \rho = M/L^3 \tag{2.8.1}$$

If the mass is not continuous throughout $\mathcal{V}$, then instead of (2.8.1) we write

$$M = \int_{\mathcal{V}_1} \rho \, dv + \sum_{\alpha} M_\alpha$$

where the summation is taken over all *discrete* masses contained in the body. We shall be dealing with a continuous mass medium in which (2.8.1) is valid, which implies that $M \to 0$ as $\mathcal{V} \to 0$. We therefore have

$$0 \leq \rho < \infty \tag{2.8.2}$$

DEFINITION 1. *The momentum (or linear momentum) $\mathcal{P}$ of a continuous mass medium contained in $\mathcal{V}$ is defined by*

$$\mathcal{P} \equiv \int_{\mathcal{V}} \rho \mathbf{v} \, dv, \qquad \mathbf{v} \equiv \dot{\mathbf{x}} \tag{2.8.3}$$

DEFINITION 2. *The moment of momentum (or angular momentum) $\underset{o}{\mathcal{H}}$, about an origin O, of a continuous mass medium contained in $\mathcal{V}$, is defined by*

$$\underset{o}{\mathcal{H}} \equiv \int_{\mathcal{V}} \mathbf{p} \times \rho \mathbf{v} \, dv \tag{2.8.4}$$

DEFINITION 3. *The kinetic energy $\mathcal{K}$ of a continuous medium contained in $\mathcal{V}$ is defined by*

$$\mathcal{K} \equiv \frac{1}{2} \int_{\mathcal{V}} \rho \mathbf{v} \cdot \mathbf{v} \, dv \tag{2.8.5}$$

In rectangular coordinates x we have the component expressions

$$\begin{aligned} \mathcal{P}_k &= \int_{\mathcal{V}} \rho v_k \, dv \\ \underset{o}{\mathcal{H}}_k &= \int_{\mathcal{V}} \rho e_{klm} x_l v_m \, dv \\ \mathcal{K} &= \frac{1}{2} \int_{\mathcal{V}} \rho v_k v_k \, dv \end{aligned} \tag{2.8.6}$$

In bivector representation $\underset{o}{\mathscr{H}}_k$ is equivalent to a skew-symmetric tensor $\underset{o}{\mathscr{H}}_{kl} = e_{klm}\underset{o}{\mathscr{H}}_m$, so that we may equivalently write

$$\underset{o}{\mathscr{H}}_{kl} = \int_v \rho(x_k v_l - x_l v_k)\, dv \tag{2.8.7}$$

in place of $\underset{o}{\mathscr{H}}_k$.

2.9. FUNDAMENTAL AXIOMS OF MECHANICS

We now enunciate four fundamental laws of mechanics.

FUNDAMENTAL AXIOM 1. *Principle of Conservation of Mass. The total mass of the body is unchanged during the motion. When this is valid for an arbitrarily small neighborhood of each material point, we say that the mass is conserved locally.*

The total or global mass conservation may be expressed by

$$\int_v \rho\, dv = \int_V \rho_0\, dV \tag{2.9.1}$$

or by the material derivative of this equation, namely

$$\frac{D}{Dt}\int_v \rho\, dv = 0 \tag{2.9.2}$$

When the volume integrals in (2.9.1) are both expressed in either the material or spatial system, we get

$$\int_V (\rho_0 - \rho J)\, dV = 0, \quad \text{or} \quad \int_v (\rho - \rho_0 J^{-1})\, dv = 0 \tag{2.9.3}$$

One may replace J here, if desired, by its expressions in the strain invariants (for example, (1.12.12)), that is,

$$\begin{aligned} J = \sqrt{\mathrm{III}_C} &= \frac{\mathrm{I}}{\sqrt{\mathrm{III}_c}} = (1 + 2\mathrm{I}_E + 4\mathrm{II}_E + 8\mathrm{III}_E)^{1/2} \\ &= (1 - 2\mathrm{I}_e + 4\mathrm{II}_e - 8\mathrm{III}_e)^{-1/2} \end{aligned} \tag{2.9.4}$$

The *local* conservation of mass requires that (2.9.3) be valid for every volume element. Hence,

$$\begin{aligned} \rho_0 = \rho J &= \rho\sqrt{\mathrm{III}_C} = \rho(1 + 2\mathrm{I}_E + 4\mathrm{II}_E + 8\mathrm{III}_E)^{1/2} \\ \rho = \rho_0 J^{-1} &= \rho_0\sqrt{\mathrm{III}_c} = \rho_0(1 - 2\mathrm{I}_e + 4\mathrm{II}_e - 8\mathrm{III}_e)^{1/2} \end{aligned} \tag{2.9.5}$$

are various expressions of the local conservation of mass

If we carry out the indicated differentiation in (2.9.2), we get

$$\int_{\mathscr{V}} \frac{D}{Dt}(\rho\, dv) = 0 \quad \text{or} \quad \int_{\mathscr{V}} \left[\frac{\partial \rho}{\partial t} + (\rho v_k)_{,k}\right] dv = 0 \tag{2.9.6}$$

From this there follows the local conservation of mass

$$\frac{D(\rho\, dv)}{Dt} = 0 \quad \text{or} \quad \frac{\partial \rho}{\partial t} + (\rho v_k)_{,k} = 0 \tag{2.9.7}$$

These are the equivalent forms of the *spatial equation* of *continuity*. Note that (2.9.5) and (2.9.7) are but different expressions of the same thing. In fact (2.9.7) is the material derivative of (2.9.5). In solid mechanics (2.9.5) and in fluid mechanics (2.9.7) are usually found suitable.

FUNDAMENTAL AXIOM 2. *Principle of Balance of Momentum. The time rate of change of momentum is equal to the resultant force* $\mathscr{F}$ *acting on the body.*

$$\frac{d\mathscr{P}}{dt} = \mathscr{F}, \quad \text{or} \quad \frac{D}{Dt}\int_{\mathscr{V}} \rho v_k\, dv = \mathscr{F}_k \tag{2.9.8}$$

FUNDAMENTAL AXIOM 3. *Principle of Balance of Moment of Momentum. The time rate of change of moment of momentum about a fixed point O is equal to the resultant moment* $\underset{O}{\mathscr{M}}$ *about O.*

$$\frac{d\underset{O}{\mathscr{H}}}{dt} = \underset{O}{\mathscr{M}} \quad \text{or} \quad \frac{D}{Dt}\int_{\mathscr{V}} \rho e_{klm} x_l v_m\, dv = \underset{O}{\mathscr{M}}_k \tag{2.9.9}$$

In bivector form, (2.9.9) reads

$$\frac{D}{Dt}\int_{\mathscr{V}} \rho(x_k v_l - x_l v_k)\, dv = \underset{O}{\mathscr{M}}_{kl} \tag{2.9.10}$$

Equations (2.9.8) and (2.9.9) are *Euler's equations of motion* for a body. They may be considered as an extension of Newton's laws of motion for a particle.

FUNDAMENTAL AXIOM 4. *Principle of Conservation of Energy. The time rate of change of the kinetic plus internal energy is equal to the sum of the rate of work* $\mathscr{W}$ *of the external forces plus all other energies that enter or leave the body per unit time.*

$$\frac{D}{Dt}(\mathscr{K} + \mathscr{E}) = \mathscr{W} + \sum_{\alpha} \mathscr{Q}_{\alpha} \tag{2.9.11}$$

where $\mathscr{E}$ and $\mathscr{W}$ are, respectively, the *internal energy* and the *work* of external forces per unit time, and $\mathscr{Q}_{\alpha}$ represents the αth kind of energy per

unit time, for example, heat energy, electrical energy, and chemical energy. This axiom implies that the energies are additive, and that if proper accounting is made of all the energies due to external effects, what is left over to *balance* is the rate of the internal energy $\mathcal{E}$. In classical thermodynamics the internal energy possesses certain important distinguishing characteristics. It is a *state function*, that is, it is independent of the *process* followed in changing the state of the body. We may state this also by saying that the internal energy is a function of the *nondissipative constitutive variables*. This latter point of view is, however, challenged by the recent developments in continuum mechanics (cf. Chapters 4 and 5).

In particle mechanics the principle of conservation of energy is obtained as the first integral of Newton's second law of motion when the forces are not explicit functions of the velocity and time. In thermodynamics it is stated as *the first principle of thermostatics*, applicable to systems in equilibrium. The principle, as stated above, is an extension of both that of classical mechanics and that of thermostatics. It is valid for every system including dissipative ones in which the energy principle of both classical mechanics and thermostatics fails to apply.

The local forms of the basic axioms 2 and 3 and the principle of conservation of mass will be obtained in Chapters 3 and 4 where the deep significance of these principles will be further appreciated. For a continuous mass medium an internal energy density per unit mass, ε, is assumed to exist such that

$$\mathcal{E} = \int_{\mathcal{V}} \rho \varepsilon \, dv \tag{2.9.12}$$

2.10. OBJECTIVE TENSORS

The physical properties of materials are not dependent on the coordinate frame selected. It is intuitively clear that whether the observer is at rest or in motion, the material properties he observes should be the same. If this viewpoint is accepted, then the measurements made in one frame of reference are sufficient to determine the material properties in all other frames that are in rigid motion with respect to one another.

In the formulation of physical laws, it is desirable to employ, as far as possible, quantities that are independent of the motion of the observer. Such quantities are called *objective*. For example, the location of a point will appear different to observers located at different places. Similarly, the velocity of a point is dependent on the velocity of the observer. Therefore, these quantities are *not* objective. On the other hand, the distance between

two points and the angles between two directions are independent of the rigid motion of the frame of reference (the observer). Newton's laws of motion have long been known to be valid only in a special frame of reference called the *galilean* frame. A galilean frame differs from a fixed reference frame by a constant translatory velocity. Attempts to free the principles of mechanics from the motion of the observer were resolved by Einstein's theory of general relativity.

We wish to stay in the domain of classical mechanics with regard to basic axioms. However, we would like to employ the *principle of objectivity* in the description of material properties (cf. Chapter 5).

Let a rectangular frame $\mathscr{F}$ be in relative rigid motion with respect to another one, $\mathscr{F}'$. A point with rectangular coordinates x_k at time t in $\mathscr{F}$ will have the rectangular coordinates x'_k at time t' in $\mathscr{F}'$. Since the frames are in rigid motion with respect to each other, we have

$$x'_k(t') = Q_{kl}(t)x_l + b_k(t), \qquad t' = t - a \tag{2.10.1}$$

where a is a constant allowing us to select the origin of time different in $\mathbf{x}'$ than in $\mathbf{x}$, and $\mathbf{Q}(t)$ and $\mathbf{b}(t)$ are functions of time alone, of which $\mathbf{Q}(t)$ is subject to

$$Q_{kl}Q_{ml} = Q_{lk}Q_{lm} = \delta_{km} \tag{2.10.2}$$

These conditions are the usual conditions satisfied by the cosine directors of $\mathbf{x}'$ with respect to $\mathbf{x}$.

It is not difficult to prove that the transformation (2.10.1) subject to (2.10.2) is the most general one in which the length and angles are preserved, for example,

$$ds'^2 = dx'_k\,dx'_k = Q_{kl}\,dx_l Q_{km}\,dx_m = \delta_{lm}\,dx_l\,dx_m = ds^2$$

A similar proof is produced in a similar fashion for the preservation of angles.

An objective vector $\mathbf{a}$ in the original frame $\mathscr{F}$ will appear in the new frame as

$$a'_k = Q_{kl}a_l \tag{2.10.3}$$

DEFINITION 1. *Two motions* $x_k(\mathrm{X}, t)$ *and* $x'_k(\mathrm{X}, t')$ *are called objectively equivalent if and only if*

$$x'_k(\mathrm{X}, t') = Q_{kl}(t)x_l(\mathrm{X}, t) + b_k(t), \qquad t' = t - a \tag{2.10.4}$$

where $Q_{kl}(t)$ is subject to (2.10.2).

Two objectively equivalent motions differ only relative to the reference frame and time. For a fixed frame and time, the two motions may be made

to coincide by the superposition of an arbitrary rigid motion of one and by a shift of time.

DEFINITION 2. *Any tensorial quantity is said to be objective if in any two objectively equivalent motions it obeys the appropriate tensor transformation law for all times*

Thus, if a vector a_k and a second-order tensor s_{kl} are objective, then they must obey the following transformation law appropriate to two objectively equivalent motions:

$$
\begin{aligned}
a'_k(\mathbf{X}, t') &= Q_{kl}(t)a_l(\mathbf{X},t) \\
s'_{kl}(\mathbf{X}, t') &= Q_{km}(t)Q_{ln}(t)s_{mn}(\mathbf{X}, t)
\end{aligned}
\tag{2.10.5}
$$

For vectors and tensors that are independent of time, objectivity readily applies For time-dependent quantities this is not always the case, as may be seen from the following example. Consider, the velocity vector $\mathbf{v} \equiv \dot{\mathbf{x}}$. From (2.10.4) we have

$$
\frac{Dx'_k}{Dt'} = Q_{kl}\dot{x}_l + \dot{Q}_{kl}x_l + b_k
$$

or

$$
v'_k = Q_{kl}v_l + \dot{Q}_{kl}x_l + b_k \tag{2.10 6}
$$

which is not of the form of $(2.10.5)_1$. Therefore, the velocity is not objective. Similarly, we can show that the acceleration is not objective either. However, we proceed to show that

THEOREM 1. *The deformation rate tensor d_{kl} is objective but the spin tensor w_{kl} is not.*

Proof· From (2.10.6) we bave

$$
v'_{k,l} = Q_{km}v_{m,n}\frac{\partial x_n}{\partial x'_l} + \dot{Q}_{km}\frac{\partial x_m}{\partial x'_l}
$$

To calculate $\partial x_n/\partial x'_l$, we first solve for x_n from (2 10.1). Hence,

$$
x_k = Q_{mk}(x'_m - b_m)
$$

thus,

$$
\frac{\partial x_n}{\partial x'_l} = Q_{ln} \tag{2.10.7}
$$

Using this expression, we have

$$
v'_{k,l} = Q_{km}Q_{ln}v_{m,n} + \dot{Q}_{km}Q_{lm} \tag{2.10 8}
$$

Similarly, we find

$$v'_{i,k} = Q_{km}Q_{ln}v_{n,m} + Q_{km}\dot{Q}_{lm} \tag{2.10.9}$$

If we add these two expressions, we get

$$d'_{kl} = Q_{km}Q_{ln}\,d_{mn} \tag{2.10.10}$$

since, according to (2.10.2), by differentiation,

$$\dot{Q}_{kl}Q_{ml} + Q_{kl}\dot{Q}_{ml} = 0, \qquad \dot{Q}_{lk}Q_{lm} + Q_{lk}\dot{Q}_{lm} = 0 \tag{2.10.11}$$

Equation (2.10.10) has the precise form $(2.10.5)_2$, thus proving the objectivity of the tensor d_{kl}.

The spin tensor w'_{kl} is calculated by subtracting (2.10.9) from (2.10.8). Hence,

$$w'_{kl} = Q_{km}Q_{ln}w_{mn} + \dot{Q}_{km}Q_{lm} \tag{2.10.12}$$

Thus, the spin tensor w_{kl} is not objective. The term

$$\Omega_{kl} \equiv \dot{Q}_{km}Q_{lm} \tag{2.10.13}$$

responsible for the non-objective character of w_{kl} is the relative angular velocity of the two frames.

By multiplying (2.10.12) by Q_{lr}, we obtain

$$\dot{Q}_{kl} = Q_{ml}w'_{km} - Q_{km}w_{ml} \tag{2.10.14}$$

Two other expressions are obtained by use of (2.10.11), (2.10.8), and (2.10.2). These are

$$\dot{Q}_{kl} = Q_{ml}v'_{k,m} - Q_{km}v_{m,l} = -Q_{ml}v'_{m,k} + Q_{km}v_{l,m} \tag{2.10.15}$$

These expressions may be used to obtain higher-order time rates of Q_{kl} by successive substitutions.

PROBLEMS

2.1. Calculate the first two material derivatives of
(*a*) The acceleration vector $a_k(\mathbf{x}, t)$;
(*b*) The second-order displacement gradient, $x_{k,KL}$.

2.2. Show that the acceleration vector, a, may be expressed in vectorial form:

$$\mathbf{a} = \frac{\partial \mathbf{v}}{\partial t} + \mathbf{w} \times \mathbf{v} + \tfrac{1}{2}\nabla v^2$$

where $\mathbf{v}(\mathbf{x}, t)$ and $\mathbf{w}(\mathbf{x}, t) \equiv \nabla \times \mathbf{v}$ are, respectively, the velocity vector and vorticity vector and $v^2 = \mathbf{v}\cdot\mathbf{v}$. Show that the above form is valid for any arbitrary curvilinear coordinates.

2.3. Let

$$\phi(\mathbf{x}, t) = -\frac{1}{4\pi}\int_{\mathscr{V}} \frac{I_4(\mathbf{y})}{r}\, dv(\mathbf{y})$$

$$\mathbf{F}(\mathbf{x}, t) = \frac{1}{4\pi}\int_{\mathscr{V}} \frac{\mathbf{w}(\mathbf{y})}{r}\, dv(\mathbf{y}), \qquad r \equiv [(x_k - y_k)(x_k - y_k)]^{1/2}$$

where $I_4 \equiv v_{k,k}$ and $\mathbf{w}$ is the vorticity vector. Prove that the velocity vector is given by

$$v_i = \phi_{,i} + e_{ijk}F_{k,j} \qquad \text{in } \mathscr{V}$$

Find the differential equation satisfied by $\mathbf{F}$

2.4. In a plane motion of a continuum, the velocity field is given by

$$\dot{x} = -V\left[1 - \frac{a^2(x^2 - y^2)}{(x^2 + y^2)^2}\right], \qquad y = -2Va^2\frac{xy}{r^4}, \qquad \dot{z} = 0$$

where V and a are constants and (x, y, z) are the spatial rectangular coordinates Determine the streamlines.

2.5. In a plane motion, streamlines, $\psi =$ constant, are given in the parametric form

$$x = \phi + e^{\psi}\cos\psi, \qquad y = \psi + e^{\psi}\sin\psi$$

Sketch the streamlines and determine the velocity field.

2.6. The velocity field of a continuum is given by

$$x = Va^2\frac{x^2 - y^2}{(x^2 + y^2)^2}, \qquad y = 2Va^2\frac{xy}{(x^2 + y^2)^2}, \qquad \dot{z} = 0$$

where V and a are constants. Determine the path lines.

2.7. Prove that at each point of a body there exists at least one material direction that is stationary instantaneously.

2.8. In a plane motion of a continuum, the velocity field is given by

$$x = -\frac{k}{4\pi}\frac{x^2}{y(x^2 + y^2)}, \qquad \dot{y} = \frac{k}{4\pi}\frac{y^2}{x(x^2 + y^2)}, \qquad \dot{z} = 0$$

where k is constant. Determine (*a*) the streamlines; (*b*) the vortex lines.

2.9. For the velocity fields given in problems 2 4, 2.6, and 2.8, determine:

(*a*) Components of the deformation rate tensor.

(*b*) Components of the spin tensor and vorticity vector.

(*c*) Invariants of the deformation rate tensor.

2.10. Calculate the first two material derivatives of the lagrangian and eulerian strain rate tensors and the spin tensor.

2.11. If in a motion of a body the material points crossing a surface σ gain or lose mass, what would be the jump condition at σ?

2.12. Across a moving surface $\sigma(t)$, momentum and linear momentum of the material points of a continuum are seen to undergo jumps because of the creation or destruction of mass. Find the jump conditions at $\sigma(t)$.

2.13. Across a moving surface, $\sigma(t)$, radiation of heat energy is causing sudden energy loss. Express the jump condition at $\sigma(t)$.

2.14. If a body is undergoing an isochoric motion ($I_d = 0$), what must be the relationship between the initial density ρ_0 and the density ρ at time t? Using the principal axes of stretch, calculate ρ up to second-order terms in velocity gradients.

2.15. The velocity field of a continuum is prescribed by

$$\dot{x} = f(y) - yg(r), \qquad \dot{y} = xg(r), \qquad \dot{z} = 0$$
$$r \equiv (x^2 + y^2)^{1/2}$$

Determine:

(*a*) The deformation rate tensor and its invariants;

(*b*) Vorticity vector.

Give a discussion of the type of motion that the body is undergoing.

2.16. Find the equations of continuity and vorticity for the following special types of motions:

(*a*) Isochoric;

(*b*) Irrotational.

Show that when the motion is isochoric and irrotational, then the velocity field is derivable from an analytic function.

2.17. Show that for a plane, isochoric, irrotational motion of a continuous medium, the velocity field is given by

$$\mathbf{v} = -\nabla\phi, \qquad \text{on } \mathscr{S}$$

where

$$\phi(\mathbf{x}, t) = -\frac{1}{2\pi}\int_{\mathscr{C}} (\phi - \phi')\frac{\partial}{\partial n}(\log r)\, da, \qquad r \equiv \sqrt{x^2 + y^2}$$

Here $\mathscr{C}$ is the boundary of the plane region $\mathscr{S}$ and ϕ' refers to the space within the inner boundary of $\mathscr{C}$. The function ϕ' is subject to the condition that $\partial\phi'/\partial n = \partial\phi/\partial n$ on $\mathscr{C}$ with $\partial/\partial n$ representing the normal derivative along $\mathscr{C}$. Under what condition is the above formula valid for the infinite plane region?

2.18. For the isochoric, irrotational motion of a body, show that the kinetic energy is given by

$$K = \tfrac{1}{2}\rho_0 \int \phi \frac{\partial\phi}{\partial n}\, da$$

where ϕ is the velocity potential, that is, $\mathbf{v} = -\nabla\phi$, and $\partial/\partial n$ represents the normal derivative along the exterior normal of the closed surface, $\mathscr{S}$.

2.19. For a simply connected region, find the integrability conditions for

$$d_{kl} = \tfrac{1}{2}(v_{k,l} + v_{l,k})$$

2.20. Investigate the objectivity of the following quantities:
(*a*) Eulerian strain rate.
(*b*) The material derivative of the deformation rate tensor.
(*c*) The material derivative of the spin tensor.
(*d*) The material derivative of the area vector.

2.21. (*Short Term Paper.*) Make a literature search and write a short paper on the stability of the Kármán vortex street.

2.22. (*Short Term Paper.*) Make a literature search and write a paper on the kinematical conditions that must be satisfied at a discontinuity surface. Discuss the conditions for velocity and acceleration waves

CHAPTER 3

Stress

3.1. SCOPE OF THE CHAPTER

This chapter deals with internal and external forces and couples and their transformations through the laws of motion. External and internal loads that cause the body to be deformed are classified, and the global laws of motion are stated in Section 3.2. Internal loads give rise to the stress and the couple stress hypotheses at any internal surface. These are discussed in Sections 3.3 and 3.4. The principles of local balance of momenta, leading to Cauchy's laws of motion, are derived in Section 3.5. These equations play a central role in the theory of continuous media. In Section 3.6 we give the jump conditions on a moving discontinuity surface for the conservation of mass and balance of momenta. These are also useful in obtaining the boundary conditions on loads.

A short account of the stress quadric of Cauchy, similar to the strain ellipsoid, is to be found in Section 3.7. For large deformations of the body, the laws of motion expressed in an eulerian system present difficulties in regard to boundary conditions which are expressed on the unknown boundary of the deforming body. Equations of motion referred to the known initial boundary of the body (Section 3.8) may be helpful in the solution of some problems. A brief account of stress flux, which finds important applications in viscoelastic theories, is given in Section 3.9.

3.2. EXTERNAL AND INTERNAL LOADS

A body undergoes deformatinn when subjected to external and internal forces. These forces may be mechanical, electrical, chemical, or of some other origin. Mechanical fnrces acting nn a particle, according to Newtonian mechanics, are functions nf the position vector x, velocity vector v, and the time t. However, in continuum mechanics the motion of a collection of many particles is analyzed so that the forces may depend on the position and velocity nf all particles in the collection at all past times. Since we do not identify particles, this means that the relative deformation of particles with respect to neighboring nnes and the history of this deformation may come into play. Thus, fnrces may depend on various order spatial gradients, their various time rates and integrals, as well as electrical and chemical variables.

As in classical mechanics, the forces are not defined. To the undefined quantities such as position, time, and mass there are added two more, namely, the force $\mathscr{F}$ and the couple $\mathscr{M}$ acting on bodies. They are vectorial quantities given by

$$\mathscr{F} = \int_{\mathscr{V}} d\mathscr{F}, \qquad \mathscr{M} = \int_{\mathscr{V}} (\mathbf{p} \times d\mathscr{F} + d\mathscr{M}) \tag{3 2 1}$$

These quantities are known *a priori*. The total force $\mathscr{F}$ consists nf the vector sum nf all forces acting on the body, and the total couple $\mathscr{M}$ consists of two parts: the total *moment*

$$\int_{\mathscr{V}} \mathbf{p} \times d\mathscr{F}$$

of the individual forces about a point (fnr example, origin O) and that nf couples $d\mathscr{M}$.

From a continuum point of view, whatever the origin may be, we divide the forces and couples into three eategories.

Extrinsic Body Loads. These are the forces and couples that arise from the external effects. They act nn the mass points of the body. A load density per unit mass is assumed tn exist. The extrinsic body loads per unit volume are called *volume* or *body loads*. Examples are the force of gravity and electrostatic forces. Extrinsic body lnads are not objective. The transformations of these loads are deduced frnm the basic Axioms 2 and 3 introduced in Section 2.9.

Extrinsic Surface Loads (Contact Loads). These loads arise from the action of one body on another through the bounding surface. The surface

density of these loads is assumed to exist. The extrinsic surface force per unit area is called the *surface traction*, and the extrinsic surface couple per unit area is called the *surface couple*. Surface tractions and surface couples depend on the orientation of the surface on which they act. The hydrostatic pressure acting on the surface of a submerged body and surface tractions produced by an external electrostatic field are examples of extrinsic surface loads.

Internal Loads (Mutual Loads). These are the result of the mutual action of pairs of particles that are located in the interior of the body. According to Newton's third law, the mutual action of a pair of particles consists of two forces acting along the line connecting the particles, equal in magnitude, and opposite in direction to one another. *Therefore the resultant internal force is zero.* Mutual loads are *objective.*

The effect of interparticle forces in a continuum appears in the form of a resultant effect of one part of the body on another part through the latter's bounding surface. This concept gives rise to the *stress hypothesis* that is studied in the following article.

Let $\mathbf{f}$ be the body force per unit mass and $\mathbf{t}_{(\mathbf{n})}$ the surface traction per unit area, acting on the surface of the body with exterior normal $\mathbf{n}$. Suppose that concentrated *forces* $\mathscr{F}_\alpha$ also act at isolated points $\mathbf{p}_\alpha$ of the body. Then the resultant external force $\mathscr{F}$ acting on the body is given by

$$\mathscr{F} = \oint_{\mathscr{S}} \mathbf{t}_{(\mathbf{n})}\, da + \int_{\mathscr{V}} \rho \mathbf{f}\, dv + \sum_\alpha \mathscr{F}_\alpha \tag{3.2.2}$$

If $\mathbf{l}$, $\mathscr{M}_{(\mathbf{n})}$, and $\mathscr{M}_\alpha$, respectively, are the body couple per unit mass, the surface couple per unit area, and the concentrated couple at the position $\mathbf{p}_\alpha$, then the resultant moment about the origin O is given by

(3.2.3)

$$\underset{O}{\mathscr{M}} = \oint_{\mathscr{S}} [\mathbf{m}_{(\mathbf{n})} + \mathbf{p} \times \mathbf{t}_{(\mathbf{n})}]\, da + \int_{\mathscr{V}} \rho(\mathbf{l} + \mathbf{p} \times \mathbf{f})\, dv + \sum_\alpha (\mathscr{M}_\alpha + \mathbf{p}_\alpha \times \mathscr{F}_\alpha)$$

Note that internal forces do not contribute to the resultant forces and couples.

To avoid difficulties in theorems concerning *local* situations, for example, infinite stress distribution in the neighborhood of a concentrated load, we shall assume that the local theorems apply to points where no concentrated loads are applied. For *global* theorems there is no need for such an assumption. With this caution, we drop the concentrated loads from our equations.

$$\mathscr{F} = \int_{\mathscr{S}} \mathbf{t}_{(\mathbf{n})}\, da + \int_{\mathscr{V}} \rho \mathbf{f}\, dv \tag{3.2.4}$$

$$\underset{o}{\mathcal{M}} = \oint_{\mathscr{S}} [\mathbf{m}_{(\mathbf{n})} + \mathbf{p} \times \mathbf{t}_{(\mathbf{n})}]\, da + \int_{\mathscr{V}} \rho(\mathbf{l} + \mathbf{p} \times \mathbf{f})\, dv \tag{3.2.5}$$

The basic Axioms 2 and 3 concerning the balance of momentum and the balance of moment of momentum now read

$$\frac{d}{dt}\int_{\mathscr{V}} \rho \mathbf{v}\, dv = \oint_{\mathscr{S}} \mathbf{t}_{(\mathbf{n})}\, da + \int_{\mathscr{V}} \rho \mathbf{f}\, dv \tag{3.2.6}$$

$$\frac{d}{dt}\int_{\mathscr{V}} \rho \mathbf{p} \times \mathbf{v}\, dv = \oint_{\mathscr{S}} [\mathbf{m}_{(\mathbf{n})} + \mathbf{p} \times \mathbf{t}_{(\mathbf{n})}]\, da + \int_{\mathscr{V}} \rho(\mathbf{l} + \mathbf{p} \times \mathbf{f})\, dv \tag{3.2.7}$$

These equations are Euler's equations of motion. Internal forces balance each other so that they do not appear in these equations.

Since the extrinsic forces are *not* objective, Euler's equations are not objective. These equations govern the global motion of the body.

3.3. STRESS HYPOTHESIS

The internal loads and their connection to the surface loads may be understood by the application of the principles of balance of global momenta on a region fully or partially contained in the body. We first consider a small region of volume v bounded by a closed surface σ such that $v + \sigma$ is fully contained in the body (Figure 3.3.1). At a point p of σ the effect of $\mathscr{V} - v$ is *equipollent* to a system of surface forces $\mathbf{t}_{(\mathbf{n})}$ called *stress vectors* and surface couples $\mathbf{m}_{(\mathbf{n})}$ called *couple stress vectors. Both stress and couple stress vectors are objective.* On surfaces passing through the same point p, but oriented differently, the stress and couple stress

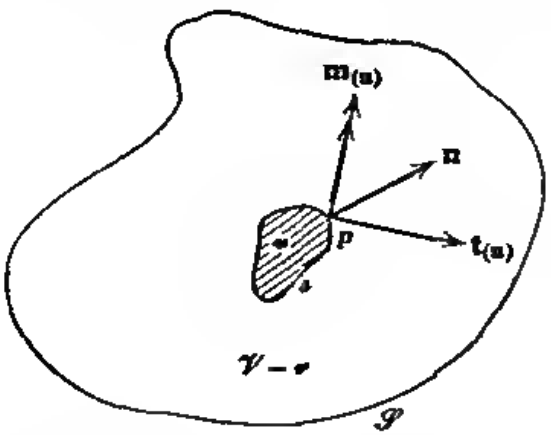


Fig. 3.3.1 Stress and couple stress vectors.

vectors at p are in general different. Thus these loads depend not only on the position vector $\mathbf{p}$ on σ but also on its exterior normal $\mathbf{n}$. This dependence is therefore indicated by the subscript $\mathbf{n}$ they carry. The body loads acting

in v are equipollent to a body force per unit volume $\rho\mathbf{f}$ and a body couple per unit volume $\rho\mathbf{l}$. The principle of balance of momenta (3.2.6) applied to body $\mathscr{V} + \mathscr{S}$ is valid also for the region $v + s$ without change except that we must now replace the limits of integrals, $\mathscr{V}$ and $\mathscr{S}$, by v and s, respectively.

To determine the dependence of the stress and couple stress vectors on the exterior normal, we next apply the principles of balance of momenta to a small tetrahedron with its vertex $\mathbf{p}$ in the interior of v and having three of its faces on the coordinate surfaces and the fourth face on s (Figure 3.3.2). The stress vector[1] on the coordinate surface $x_k = \text{const.}$ is denoted by $-\mathbf{t}_k$.

We now apply (3.2.6) to this tetrahedron. An estimate of the surface and volume integrals may be made by use of the mean value theorem. Thus

$$\frac{d}{dt}(\rho^*\mathbf{v}^*\,\Delta v) = \mathbf{t}^*_{(\mathbf{n})}\Delta a - \mathbf{t}^*_k\Delta a_k + \rho\mathbf{f}^*\Delta v$$

where ρ^*, $\mathbf{v}^*$, and $\mathbf{f}^*$ are, respectively, the values of ρ, $\mathbf{v}$, and $\mathbf{f}$ at some

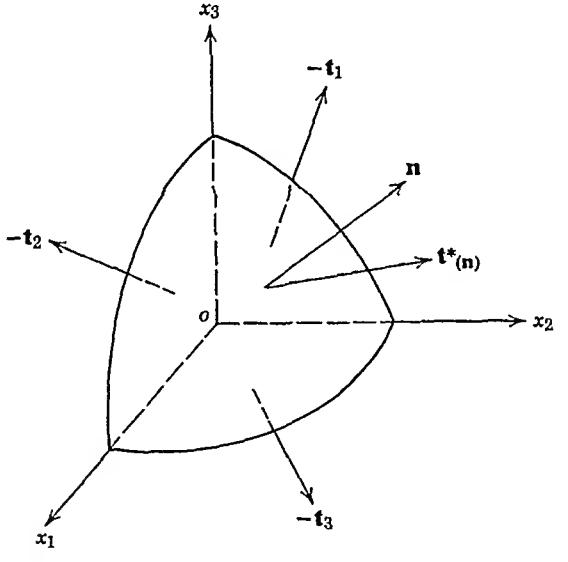


Fig. 3.3.2 Tetrahedron.

interior points of v and $\mathbf{t}^*_{(\mathbf{n})}$ and $\mathbf{t}^*_k$ are the values of $\mathbf{t}_{(\mathbf{n})}$ and $\mathbf{t}_k$ at some point of Δa and Δa_k, respectively. Using the principle of conservation of mass,

[1] Since the exterior normal of a coordinate surface $x_k = \text{const.}$ is in the direction of $-x_k$, without loss in generality, we denote the stress vector acting on this surface by $-\mathbf{t}_k$ rather than $\mathbf{t}_k$.

we may also put $d(\rho^*\mathbf{v}^*\Delta v)/dt = \rho^*\dot{\mathbf{v}}^*\Delta v$ on the left-hand side. Upon dividing both sides by Δa and letting $\Delta a \to 0$ so that point p approaches the surface a from the interior, we have $\Delta v/\Delta a \to 0$. The quantities ρ^*, $\mathbf{v}^*$, and $\mathbf{f}^*$ are bounded and this gives

(3.3 1) $$\mathbf{t}_{(n)}\, da = \mathbf{t}_k\, da_k$$

The four surfaces of the tetrahedron form a closed surface. Therefore in the limit, the vector sum of the coordinate surfaces must add up to the area vector $d\mathbf{a}$, that is,

(3 3 2) $$d\mathbf{a} = \mathbf{n}\, da = da_k \mathbf{i}_k$$

From this we get

(3 3 3) $$da_k = n_k\, da$$

Substituting this into (3.3.1), we have

(3 3 4) $$\mathbf{t}_{(n)} = \mathbf{t}_k n_k$$

We have therefore proved the following theorem:

THEOREM *The stress vector at a point p of a surface with an exterior unit normal* $\mathbf{n}$ *is a linear function of the stress vectors acting on the coordinate surfaces through the same point p, the coefficients being the cosine directors of* $\mathbf{n}$

The stress vectors $\mathbf{t}_k$ are, by definition, independent of $\mathbf{n}$. From (3.3.4) it therefore follows that

(3 3 5) $$\mathbf{t}_{(-n)} = -\mathbf{t}_{(n)}$$

hence the proof of the corollary

COROLLARY. *The stress vector acting on the opposite sides of the same surfaces at a given point are equal in magnitude and opposite in sign.* This is the counterpart of Newton's third law of motion that action is equal to reaction.

Application of a similar argument to (3.2.7) together with (3.3.4) leads to

(3 3 6) $$\mathbf{m}_{(n)} = \mathbf{m}_k n_k, \qquad \mathbf{m}_{(-n)} = -\mathbf{m}_{(n)}$$

provided of course that $\mathbf{m}_{(n)}$ and $\mathbf{m}_k$ are bounded and nonvanishing in the limit $\Delta a \to 0$, as in the case of $\mathbf{t}_{(n)}$ and $\mathbf{t}_k$.

The existence of a couple stress vector and body couples is often rejected with the following argument. A couple in classical mechanics is pictured as a pair of parallel forces having equal magnitude and opposite sense, separated by a moment arm. The forces are assumed to be *bounded*. Now

if we let the moment arm approach zero (which occurs in the above argument when $\Delta a \to 0$), and not admit unbounded forces, the moment of the couple approaches zero. Thus, in a strictly mathematical theory in which v and a (hence Δv and Δa) are allowed to vanish, while the forces remain bounded, we cannot have a couple stress vector or body couple. If, however, v and a (hence infinitesimal dv and da) are considered to represent some physical volume and surface small enough to admit such mathematical interpretations attributed to dv and da, with an admissible error, then we may admit the existence of the *couple stress* vector. In physics today, on the molecular scale, the continuity of matter is rejected. Thus mathematical infinitesimals such as dv and da are models of the smallest allowable volume and surface in which the continuum hypothesis is not violated. For example, in a small volume containing large enough particles, the vector sum of all body forces may not be passing through the mass center. This then gives rise to a body couple. Similar arguments applied to the surface forces acting on a small surface of this small volume will admit the existence of a couple stress. Be that as it may, for the sake of simplicity, in this book we shall not study the interesting consequences arising from couple stresses and body couples.[2]

3.4. STRESS TENSOR

The stress tensor t_{kl} is the lth component of the stress vector $\mathbf{t}_k$ acting on the positive side of the kth coordinate surface. The stress tensor is objective.

$$\mathbf{t}_k = t_{kl}\mathbf{i}_l \tag{3.4.1}$$

The first subscript in t_{kl} indicates the coordinate surface $x_k = \text{const.}$ on which the stress vector $\mathbf{t}_k$ acts, and the second subscript the direction of the component of $\mathbf{t}_k$. For example, t_{23} is the x_3-component of the stress vector $\mathbf{t}_2$ acting on the surface $x_2 = \text{const.}$ Now if the exterior normal of the surface $x_2 = \text{const.}$ points in the positive direction of the x_2-axis, t_{23} points in the positive direction of the x_3-axis. If the exterior normal of $x_2 = \text{const.}$ is in the negative direction of the x_2-axis, t_{23} is directed in the negative direction of the x_3-axis. The positive stress components are shown in Figure 3.4.1. Positive stress components on the surface $p_1a_1b_1c_1$ are oriented in the positive direction of the x_1-, x_2-, and x_3-axes since the exterior normal of this surface is in the direction of the $x_2 > 0$ axis. On the plane *pabc* the positive stress components are oriented in the directions of the negative coordinate axes since the exterior normal of this plane

[2] The subject matter is presently in a state of development. For an account of the subject, see Eringen and Suhubi [1964, a, b], Eringen [1964], Green and Rivlin [1964].

points in the direction of the $x_2 < 0$ axis. The above sign convention applies to the $x_1 = \text{const.}$ and the $x_3 = \text{const.}$ planes. In order not to crowd the figure, the stress components acting on the $x_1 = \text{const.}$ planes are omitted. The sign convention given above is not unique. It is, however, generally accepted in the literature.

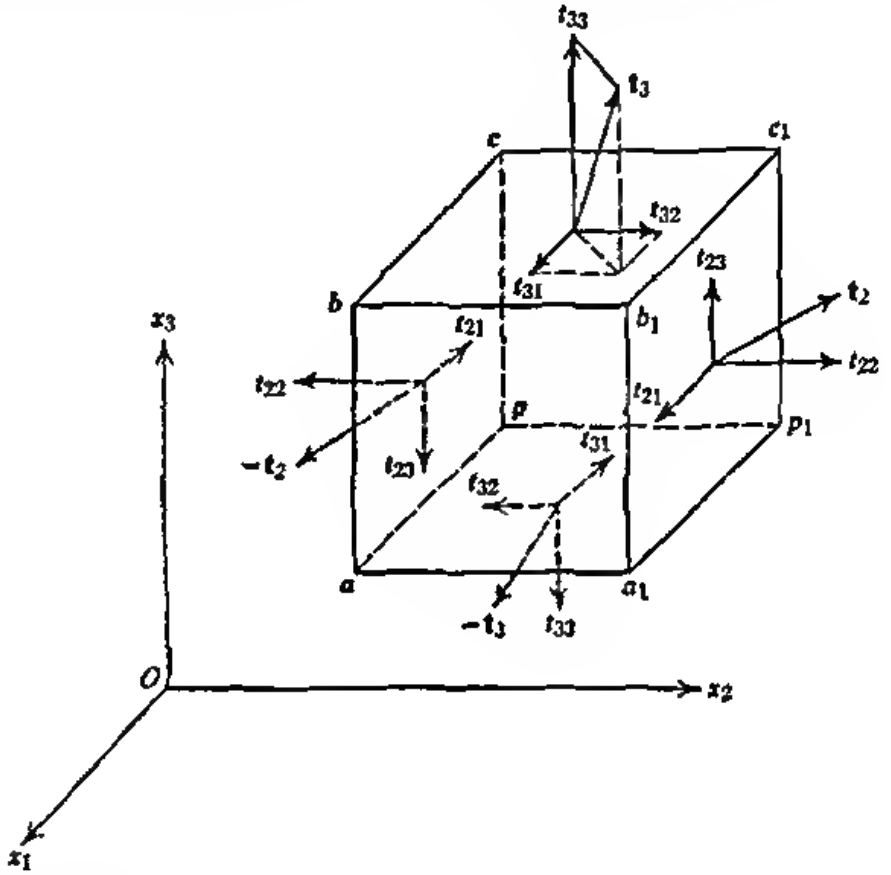


Fig. 3.4.1 Stress tensor

The components t_{11}, t_{22}, and t_{33} *are called normal stresses and the mixed components* t_{12}, t_{23}, *etc., are called shearing stresses.* Arranged in a matrix form, we may represent t_{kl} by

$$\|t_{kl}\| = \begin{bmatrix} t_{11} & t_{12} & t_{13} \\ t_{21} & t_{22} & t_{23} \\ t_{31} & t_{32} & t_{33} \end{bmatrix} \tag{3.4.2}$$

Note that in this form the normal stresses occupy the main diagonal of the matrix, and the shear components occupy the remaining spaces.

Substituting (3.4.1) into (3.3.4), we get

$$\mathbf{t}_{(n)} = t_{kl} n_k \mathbf{i}_l \quad \text{or} \quad t_{(n)l} = t_{kl} n_k \tag{3.4.3}$$

This is the proof of the following theorem:

THEOREM. *The stress vector acting on any plane through a point is fully characterized as a linear function of the stress tensor at the point.*

The tensorial character of t_{kl} may be seen as follows. Suppose that the rectangular coordinates x_k are rotated to x'_k. We have through $(3.4.3)_1$

$$\mathbf{t}_{(n)} = t_{kl}n_k\mathbf{i}_l = t'_{rs}n'_r\mathbf{i}'_s$$

By taking the scalar product with $\mathbf{i}'_m$, we obtain

$$t'_{rs}n'_r = t_{kl}n_kQ_{sl} \tag{3.4.4}$$

where

$$Q_{sl} \equiv \mathbf{i}'_s \cdot \mathbf{i}_l \tag{3.4.5}$$

Now $\mathbf{n}$ being a vector, according to $(1.8.9)_1$, it transforms as

$$n_k = Q_{rk}n'_r \tag{3.4.6}$$

Upon substituting this into (3.4.4), we get

$$t'_{rs}n'_r = t_{kl}Q_{rk}Q_{sl}n'_r$$

This must be true irrespective of the plane chosen, that is, n'_r being arbitrary. Hence

$$t'_{rs} = Q_{rk}Q_{sl}t_{kl} \tag{3.4.7}$$

In a similar fashion, we obtain

$$t_{rs} = Q_{kr}Q_{ls}t'_{kl} \tag{3.4.8}$$

Quantities t_{kl} that transform in accordance with the rules (3.4.7) and (3.4.8) are known to be *cartesian tensors*. The tensor transformation for t_{kl} remains valid in curvilinear coordinates when Q_{kr} and t_{kl} are properly defined (cf. Eringen [1962, Art. 31], also Appendix C). This establishes the tensorial character of t_{kl}. Since in the definition of t_{kl} no time rate is involved, the above transformation remains unchanged when Q_{kl} is a function of time. This is the proof of the objectivity of the stress tensor.

The analysis leading to (3.4.3) is applicable without change to any point of the surface $\mathscr{S}$ of the body whenever the tangent plane is continuous there. Thus, if the surface load per unit area is prescribed there, (3.4.3) gives the boundary condition

$$t_{(n)l} = t_{kl}n_k = s_l \qquad \text{on } \mathscr{S} \tag{3.4.9}$$

where s_l is the prescribed surface traction.

From the definitions of the body force $\mathbf{f}$, stress vector $\mathbf{t}_{(n)}$, couple stress vector $\mathbf{m}_{(n)}$, and stress tensor t_{kl}, we see that these quantities have the

physical dimensions

$$[\mathbf{f}] = \frac{L}{T^2} \approx \frac{\text{force}}{\text{mass}}, \qquad [\mathbf{t}_{(n)}] = \frac{M}{LT^2} \approx \frac{\text{force}}{\text{area}}$$

$$[\mathbf{m}_{(n)}] \approx \frac{M}{T^2} \approx \frac{\text{force} \times \text{distance}}{\text{area}} \tag{3.4.10}$$

$$[t_{kl}] \approx \frac{M}{LT^2} \approx \frac{\text{force}}{\text{area}}$$

where brackets [ ] indicate the dimension and L, M, and T respectively denote the dimension of mass, length, and time. In the CGS system, force is measured in dynes, length in centimeters, mass in grams, and time in seconds. Therefore in this system, $\mathbf{t}_{(n)}$ and t_{kl} are measured in dynes per square centimeter. In the English practical units, they are measured in pounds per square inch.

Finally, we give a short list of various notations used for the stress tensor by various writers:

Clebsch, Erngen, Truesdell	t_{11}	t_{22}	t_{33}	t_{23}	t_{31}	t_{12}
Cauchy (early work)	A	B	C	D	E	F
Cauchy (later work), St. Venant, Maxwell	p_{xx}	p_{yy}	p_{zz}	p_{yz}	p_{zx}	p_{xy}
F. Neumann, Kirchhoff, Love	X_x	Y_y	Z_z	Y_z	Z_x	X_y
K. Pearson	$\widehat{xx}$	$\widehat{yy}$	$\widehat{zz}$	$\widehat{yz}$	$\widehat{zx}$	$\widehat{xy}$
Kelvin	P	Q	R	S	T	V
Kármán, Timoshenko, engineers	σ_x	σ_y	σ_z	τ_{yz}	τ_{zx}	τ_{xy}
Russian and German writers, Green and Zerna	τ_{11}	τ_{22}	τ_{33}	τ_{23}	τ_{31}	τ_{12}
Some English and American writers and others	σ_{11}	σ_{22}	σ_{33}	σ_{23}	σ_{31}	σ_{12}
	σ_{xx}	σ_{yy}	σ_{zz}	σ_{yz}	σ_{zx}	σ_{xy}

3.5. PRINCIPLES OF BALANCE OF LOCAL MOMENTA

Global balance of momenta is expressed by (3.2.6) and (3.2.7). We now carry out the indicated differentiation on the left and use the local conservation of mass $D(\rho\, dv)/Dt = 0$. In addition we take $\mathbf{m}_{(n)} = \mathbf{l} = 0$ so that the continuum under consideration is *nonpolar*. Thus

$$\int_{\mathscr{V}} \rho \mathbf{a}\, dv = \oint_{\mathscr{S}} \mathbf{t}_{(n)}\, da + \int_{\mathscr{V}} \rho \mathbf{f}\, dv \tag{3.5.1}$$

$$\int_{\mathscr{V}} \rho \mathbf{p} \times \mathbf{a}\, dv = \oint_{\mathscr{S}} \mathbf{p} \times \mathbf{t}_{(n)}\, da + \int_{\mathscr{V}} \rho \mathbf{p} \times \mathbf{f}\, dv \tag{3.5.2}$$

where $\mathbf{a} \equiv \dot{\mathbf{v}}$ is the acceleration vector. Apply these principles to a small interior region $v + s$ of the body $\mathscr{V} + \mathscr{S}$. Using also (3.4.3), we find that these read

$$\int_v \rho \mathbf{a}\, dv = \oint_s \mathbf{t}_k n_k\, da + \int_v \rho \mathbf{f}\, dv \tag{3.5.3}$$

$$\int_v \rho \mathbf{p} \times \mathbf{a}\, dv = \oint_s \mathbf{p} \times \mathbf{t}_k n_k\, da + \int_v \rho \mathbf{p} \times \mathbf{f}\, dv \tag{3.5.4}$$

Applying the Green-Gauss theorem to convert the surface integrals into volume integrals, we get

$$\int_v [\mathbf{t}_{k,k} + \rho(\mathbf{f} - \mathbf{a})]\, dv = \mathbf{0} \tag{3.5.5}$$

$$\int_v \{\mathbf{i}_k \times \mathbf{t}_k + \mathbf{p} \times [\mathbf{t}_{k,k} + \rho(\mathbf{f} - \mathbf{a})]\}\, dv = \mathbf{0} \tag{3.5.6}$$

where we wrote $\mathbf{p}_{,k} = \mathbf{i}_k$. Equations (3.5.5) and (3.5.6) are the mathematical expressions of *the principles of balance of global momenta*, local conservation of mass being assumed. For these to be valid for any arbitrary volume v, the necessary and sufficient condition is the vanishing of the integrands. Hence

$$\mathbf{t}_{k,k} + \rho(\mathbf{f} - \mathbf{a}) = \mathbf{0} \tag{3.5.7}$$

$$\mathbf{i}_k \times \mathbf{t}_k = \mathbf{0} \tag{3.5.8}$$

Upon substituting from (3.4.1), we get

$$t_{lk,l} + \rho(f_k - a_k) = 0 \tag{3.5.9}$$

$$t_{lk} = t_{kl} \tag{3.5.10}$$

where in (3.5.8) we also used the equation $\mathbf{i}_m = e_{klm}\mathbf{i}_k \times \mathbf{i}_l$ valid for unit vectors. This then gives $e_{klm}t_{kl} = 0$, which is the same thing as (3.5.10). Equations (3.5.9) and (3.5.10) are, respectively, the *first and second laws of motion of Cauchy*, and the proofs of the following basic theorems.

THEOREM 1. *The necessary and sufficient condition for the local balance of momentum is Cauchy's first law of motion*, (3.5.7) *or* (3.5.9).

THEOREM 2. *The necessary and sufficient condition for the local balance of moment of momentum is Cauchy's second law of motion*, (3.5.8) *or* (3.5.10).

For nonpolar media, it is clear that Cauchy's second law is the statement of symmetry of the stress tensor. Thus, in this case, the stress matrix (3.4.2)

is symmetrical with respect to its main diagonal, so that instead of nine we only have six independent components for the stress tensor, that is, $t_{11}, t_{22}, t_{33}, t_{12} = t_{21}, t_{23} = t_{32}$, and $t_{31} = t_{13}$.

In expanded notation (3.5.9) and (3.5.10) usually appear in the textbooks as

$$\frac{\partial t_{xx}}{\partial x} + \frac{\partial t_{yx}}{\partial y} + \frac{\partial t_{zx}}{\partial z} + \rho(f_x - a_x) = 0$$

$$\frac{\partial t_{xy}}{\partial x} + \frac{\partial t_{yy}}{\partial y} + \frac{\partial t_{zy}}{\partial z} + \rho(f_y - a_y) = 0 \tag{3.5.11}$$

$$\frac{\partial t_{xz}}{\partial x} + \frac{\partial t_{yz}}{\partial y} + \frac{\partial t_{zz}}{\partial z} + \rho(f_z - a_z) = 0$$

$$t_{xy} = t_{yx}, \qquad t_{yz} = t_{zy}, \qquad t_{zx} = t_{xz} \tag{3.5.12}$$

Here the acceleration vector a_k is given by

$$a_k = \frac{\partial v_k}{\partial t} + v_{k,l} v_l, \qquad v_k \equiv \dot{x}_k = \dot{u}_k \tag{3.5.13}$$

where $u_k(\mathbf{x}, t)$ is the displacement vector. In expanded notation, (3.5.13) reads

$$a_x = \frac{\partial v_x}{\partial t} + \frac{\partial v_x}{\partial x} v_x + \frac{\partial v_x}{\partial y} v_y + \frac{\partial v_x}{\partial z} v_z$$

$$a_y = \frac{\partial v_y}{\partial t} + \frac{\partial v_y}{\partial x} v_x + \frac{\partial v_y}{\partial y} v_y + \frac{\partial v_y}{\partial z} v_z \tag{3.5.14}$$

$$a_z = \frac{\partial v_z}{\partial t} + \frac{\partial v_z}{\partial x} v_x + \frac{\partial v_z}{\partial y} v_y + \frac{\partial v_z}{\partial z} v_z$$

3.6. JUMP CONDITIONS AT A MOVING DISCONTINUITY SURFACE

When the region $\mathscr{V}$ contains a moving discontinuity surface $\sigma(t)$ (Figure 2.5.2), the expressions of local conservation of mass and local balance of momenta again apply at all points of $\mathscr{V} - \sigma$, that is, at the points not located on $\sigma(t)$. In fact, these equations and the jump conditions on $\sigma(t)$ may be derived in one stroke by the use of the expression (2.5.18) and the

global forms of the principles of balance:

$$\frac{d}{dt}\int_{\mathscr{V}} \rho\, dv = 0$$

$$\frac{d}{dt}\int_{\mathscr{V}} \rho \mathbf{v}\, dv = \oint_{\mathscr{S}} \mathbf{t}_k n_k\, da + \int_{\mathscr{V}} \rho \mathbf{f}\, dv \tag{3.6.1}$$

$$\frac{d}{dt}\int_{\mathscr{V}} \rho \mathbf{p} \times \mathbf{v}\, dv = \oint_{\mathscr{S}} \mathbf{p} \times \mathbf{t}_k n_k\, da + \int_{\mathscr{V}} \rho \mathbf{p} \times \mathbf{f}\, dv$$

We first use (2.5.18) in $(3.6.1)_1$ after setting $\tau_k = g = 0$ and $\varphi = \rho$.

$$\int_{\mathscr{V}-\sigma} \left[\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v})\right] dv + \int_{\sigma} [\rho(\mathbf{v} - v)] \cdot \mathbf{n}\, da = 0$$

Setting the integrand of the first integral to zero, we get the usual equation of local mass conservation. Setting the integrand of the integral over σ to zero, we get the jump condition

$$[\rho(\underset{n}{v} - \underset{n}{v})] = 0 \tag{3.6.2}$$

where $\underset{n}{v} = \mathbf{v} \cdot \mathbf{n}$ and $\underset{n}{v} = v \cdot \mathbf{n}$ are, respectively, the normal component of the material velocity and the normal component of the velocity of the moving surface $\sigma(t)$.

Similarly, by setting $\varphi = \rho \mathbf{v}$, $\tau_k = \mathbf{t}_k$, and $g = \rho \mathbf{f}$ in (2.5.18) for the left side of $(3.6.1)_2$, we get

$$\int_{\mathscr{V}-\sigma} [\mathbf{t}_{k,k} + \rho(\mathbf{f} - \mathbf{a})]\, dv + \int_{\sigma} [\mathbf{t}_k - \rho \mathbf{v}(v_k - v_k)] n_k\, da = \mathbf{0}$$

The volume integral in this expression gives Cauchy's first law and the surface integral, by setting the coefficient of da in the integrand equal to zero, gives the jump condition

$$[\mathbf{t}_k] n_k - [\rho \mathbf{v}(\underset{n}{v} - \underset{n}{v})] = \mathbf{0} \tag{3.6.3}$$

The method presented above when applied to $(3.6.1)_3$ produces no new jump conditions.

The jump conditions (3.6.2) and (3.6.3) on $\sigma(t)$ may also be written as

$$[\rho \mathbf{v}] \cdot \mathbf{n} - [\rho] v \cdot \mathbf{n} = 0 \tag{3.6.4}$$

$$[t_{kl}] n_k - [\rho v_l \mathbf{v}] \cdot \mathbf{n} + [\rho v_l] v \cdot \mathbf{n} = 0 \tag{3.6.5}$$

These results will now be applied to three special cases:

(*a*) ***The Discontinuity Surface is a Material Interface.*** In this case, $\mathbf{v} = v$, (3.6.4) is satisfied identically, and (3.6.5) reduces to

(3.6.6) $$[t_{kl}]n_k = 0$$

Hence, *on a material interface of two media the surface traction* $t_{(n)l} = t_{kl}n_k$ *is continuous.*

(*b*) ***The Discontinuity Surface Coincides with the Surface of the Body.*** In this case $\rho^+ = 0, \mathbf{v}^- = v$. Again (3.6.4) gives an identity and (3.6.5) reduces to

$$\overset{+}{t}_{kl}n_k - \overset{-}{t}_{kl}n_k = 0$$

When $\overset{+}{t}_{kl}n_k$ is interpreted as the external surface load $t_{(n)l} = \overset{+}{t}_{kl}n_k = s_l$; this gives the boundary condition on tractions (3 4.9) obtained previously in a different way.

(*c*) ***The Discontinuity Surface is a Fixed Surface.*** In this case $v = 0$ and (3.6.4) and (3.6.5) give

(3 6 7) $$[\rho v_k]n_k = 0$$

(3 6.8) $$[\rho v_l v_k - t_{kl}]n_k = 0$$

3.7. STRESS QUADRIC OF CAUCHY

The normal component N of the stress vector $\mathbf{t}_{(n)}$ acting at a point p of the surface σ with exterior normal $\mathbf{n}$ (Figure 3.7.1) is calculated by

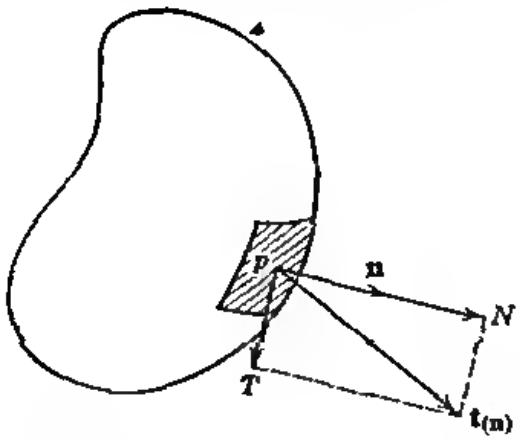


Fig. 3.7.1 Normal and tangential tractions.

(3.7.1) $$N = \mathbf{t}_{(n)} \cdot \mathbf{n}$$

Upon substituting from (3.4 3), we obtain

(3 7.2) $$N = t_{kl}n_k n_l$$

Suppose that N is fixed and the orientation $\mathbf{n}$ of the surface σ at p is variable. With this stipulation (3.7.2) represents a quadric surface called the *stress quadric of Cauchy*. The variables n_k are subject to the condition

$$t_k n_k = 1 \tag{3.7.3}$$

We can, as with the stretch, search for the extremums of N. For vanishing body couples and couple stresses, the stress tensor is symmetric. Therefore for the nonpolar case we have a complete correspondence between the stress quadric of Cauchy and the strain quadric of Cauchy. We may therefore transform the results of Sections 1.9 and 1.10 here by the mere replacement of E_{KL} by t_{kl} and N_K by n_k. With this understanding, we have the following results:

1. There exist at least three and generally three *principal directions* of stress.
2. There are three *principal stresses* $t_\alpha (\alpha = 1, 2, 3)$ that act on the principal planes.
3. On the principal planes the shearing stresses vanish and the normal stresses take the stationary values t_α.
4. In general, there exist three independent stress invariants I_t, II_t, and III_t whose expressions are identical to (1.10.8) and (1.10.9) of the strain invariants except that everywhere we replace E_{KL} and E_α by t_{kl} and t_α, respectively.

Special types of states of stress at a point have also a one-to-one correspondence with those of strain. For example, when two principal stresses vanish, we have the state of stress called *simple tension* or *uniaxial stress*. When one of the principal stresses vanishes, we have *plane* or *biaxial stress*. When none vanishes, the state of stress is *triaxial*.

A state of *simple shear* exists if one of the shear stresses is not zero while all other stress components vanish, for example,

$$\|t_{kl}\| = \begin{bmatrix} 0 & t_{xy} & 0 \\ t_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad t_{xy} \neq 0$$

In this case we have

$$\mathrm{I}_t = \mathrm{III}_t = 0, \qquad \mathrm{II}_t = -t_{xy}{}^2, \qquad t_1 = -t_2 = t_{xy}$$

so that referred to the principal axes, two principal stresses are equal in magnitude and opposite in sign, and the third principal stress is zero.

Mohr's circle constructed for plane strain is valid without change for plane stress. There exists a geometrical construction due to O. Mohr for the determination of stress tractions T and N acting in any plane when t_{kl} are given in one reference plane. This construction involves three circles, and

it is a generalization of Mohr's circle for plane stress. For this the reader is referred to books on advanced strength of materials and elasticity, cf. Sokolnikoff [1956, Art. 18].

Finally, we enunciate the following theorem regarding the stationary values of the shearing stress.

THEOREM *The maximum shearing stress acts on the plane that bisects the angle between the greatest and the smallest principal stress planes. Its value is one half the difference between the greatest and the smallest principal stresses.*

This theorem has found application in failure theories. For its proof we refer the reader to textbooks on elasticity (cf. Sokolnikoff [1956, Art. 18]).

3.8. EQUATIONS OF MOTION IN THE MATERIAL COORDINATES

Cauchy's equations of motion are expressed in the eulerian language involving stress tensor t_{kl}, body force f_k, density ρ, and acceleration vector a_k, all defined in the spatial coordinates x_k. In a deforming body, the surface of the body in the deformed state is not known at the outset, but will have to be determined through the solution of a boundary-value problem The place and the shape of the surface of the body depend on the unknown displacement vector. The problem is thus nonlinear On the other hand, the initial shape of the body is known, and therefore the boundary conditions can be expressed on the well-defined surface of the undeformed body. Some problems may be made tractable by formulating them in the material coordinates where the shape of the body is prescribed nt the outset.

In the classical theory of elasticity, the equations of motion expressed in spatial coordinates are solved under the boundary conditions referred to the original undeformed surface of the body. In this mix-up, the eulerian stresses t_{kl} are expressed as linear functions of the lagrangian strains E_{KL}. This confusion may be clarified and shown to be permissible under certain conditions For this purpose we now obtain the equations of local conservation of mass and Cauchy's equations of motion in the material coordinates X_K. For the equation of local conservation of mass from (2.9.5) we already have

$$\rho_0 = \rho J \tag{3.8.1}$$

where $J = \det x_{k,K}$.

To obtain the appropriate form of the equations of balance of momenta, we introduce the stress vector $\mathbf{T}_K$ at a spatial point x at time t occupied by a material point X referred to the undeformed area dA_K.

$$\mathbf{t}_{(n)}\, da = \mathbf{t}_k\, da_k = \mathbf{T}_K\, dA_K \tag{3.8.2}$$

According to (1.12.3) and (1.12.11) we have

$$da_k = jX_{K,k}\, dA_K, \qquad dv = j\, dV \tag{3.8.3}$$

Substituting this into (3.8.2), we obtain

$$\mathbf{t}_k = j^{-1}x_{k,K}\mathbf{T}_K, \qquad \mathbf{T}_K = jX_{K,k}\mathbf{t}_k \tag{3.8.4}$$

Clearly

$$\mathbf{t}_{k,k} = (j^{-1}x_{k,K}\mathbf{T}_K)_{,k} = j^{-1}x_{k,K}\mathbf{T}_{K,L}X_{L,k} = j^{-1}\mathbf{T}_{K,K}$$

since from (1.4.7) we can prove that $(j^{-1}x_{k,K})_{,k} = 0$.

Substituting this into (3.5.7) and using (3.8.1), we obtain

$$\mathbf{T}_{K,K} + \rho_0(\mathbf{f} - \mathbf{a}) = \mathbf{0} \tag{3.8.5}$$

This is Cauchy's equation of motion in the reference state. Following Piola,[3] we now introduce the pseudo stresses T_{Kl} and T_{KL} by

$$\mathbf{T}_K = T_{Kl}\mathbf{i}_l = T_{KL}x_{l,L}\mathbf{i}_l \tag{3.8.6}$$

so that through (3.8.4) we have

$$T_{Kl} = jX_{K,k}t_{kl}, \qquad t_{kl} = j^{-1}x_{k,K}T_{Kl} = j^{-1}x_{k,K}x_{l,L}T_{KL} \tag{3.8.7}$$

$$T_{KL} = T_{Kl}X_{L,l} = jX_{K,k}X_{L,l}t_{kl} \tag{3.8.8}$$

From $(3.8.6)_1$ and (3.8.2) it is clear that T_{Kl} is the stress at $\mathbf{x}$ measured per unit undeformed area at $\mathbf{X} = \mathbf{X}(\mathbf{x}, t)$.

Using (3.8.6) in (3.8.5), we get two different forms of the equations of motion

$$T_{Kk,K} + \rho_0(f_k - a_k) = 0 \tag{3.8.9}$$

$$(T_{KL}x_{k,L})_{,K} + \rho_0(f_k - a_k) = 0 \tag{3.8.10}$$

Cauchy's second law of motion follows from $t_{kl} = t_{lk}$ upon using $(3.8.7)_2$. Hence, either of the two forms

$$T_{Kk}x_{l,K} = T_{Kl}x_{k,K} \tag{3.8.11}$$

$$T_{KL} = T_{LK} \tag{3.8.12}$$

may be used for this purpose.

The above expressions may be used as a source for approximate theories in which deformation gradients, strains or rotations are small as compared to unity. To this end we recall the equation $(1.5.11)_2$

$$X_{K,k} = (\delta_{lk} - \tilde{e}_{lk} - \tilde{r}_{lk})\delta_{lK}$$

[3] [1833], [1848, Section 34–38]. See also Kirchhoff [1852, 763–764] and E. and F. Cosserat [1896, Section 15].

1. *Strains $\tilde{e}_{lk}$ are small as compared to unity.* In this case we also have from (2.9.4) $J \cong 1 + \tilde{e}_{kk}$. Hence

$$T_{Kl} \cong (t_{ml} - \tilde{r}_{mk} t_{kl})\delta_{mK} \tag{3.8.13}$$

$$T_{KL} \cong (t_{mk} - \tilde{r}_{ml} t_{lk} - \tilde{r}_{kl} t_{lm} + \tilde{r}_{mn}\tilde{r}_{kl} t_{nl})\delta_{mK}\delta_{kL} \tag{3.8.14}$$

2. *Both strains $\tilde{e}_{lk}$ and rotations $\tilde{r}_{lk}$ are small as compared to unity.* In this case (3.8.13) and (3.8.14) reduce to

$$T_{Kl} \cong t_{ml}\delta_{mK}, \qquad T_{KL} \cong t_{mk}\delta_{mK}\delta_{kL} \tag{3.8.15}$$

We therefore see that only when both $\tilde{\mathbf{e}}$ and $\tilde{\mathbf{r}}$ are small compared to unity (the infinitesimal deformation theory) there will be no numerical difference between the eulerian and the Piola stresses.

3. *Some of the $\tilde{r}_{kl}$ are negligibly small as compared to others.* In this case (3.8.13) and (3.8.14) are simplified by dropping the terms containing these $\tilde{r}_{kl}$. Consideration must be given, however, to the fact that the stress components multiplying these small $\tilde{r}_{kl}$ must be of the same order of magnitude as the rest. Such situations occur in the finite deformations of plates and shells. The well-known Kármán-Timoshenko theory of plates[4] represents a good example in which the component of the rotation vector perpendicular to the median plane of the plate is small as compared to other components since the plate elements are more resistant to rotation about their normals.

We conclude this article by a recapitulation of the main result for the infinitesimal theory: *When strains $\tilde{e}_{kl}$ and rotations $\tilde{r}_{kl}$ are small as compared to unity and all stress components are of the same order of magnitude, then the equations of motion and the boundary conditions of the deformed body have the same form in the material and spatial coordinates, and the Cauchy and Piola stresses are identical in value.*

3.9. STRESS FLUX

In the study of viscoelastic materials, the dependence of the dynamical behavior of these materials on the time rate of change of stress is of common occurrence. For example, the so-called simple Maxwell solid requires the use of a rate of stress in its constitutive theory (cf. Section 9.4). The stress rate, in a fashion similar to the strain rate, may be defined by

$$i_{kl} \equiv \frac{Dt_{kl}}{Dt} \tag{3.9.1}$$

[4] Cf. Eringen [1955a], Novozhilov [1948, Ch. VI].

While the stress tensor is objective, the stress-rate tensor is not. Therefore the stress rate should not occur in this form in the description of material properties. We should like to construct an objective tensor containing the rate of stress. According to Section 2.10, any objective tensor in two objectively equivalent motions will transform according to the law valid for the stress tensor, that is,

$$t'_{kl} = Q_{km}Q_{ln}t_{mn} \tag{3.9.2}$$

where

$$Q_{kl}Q_{ml} = Q_{lk}Q_{lm} = \delta_{km}, \qquad \det Q_{kl} = 1 \tag{3.9.3}$$

By taking the material derivative of both sides of (3.9.2) we get

$$\dot{t}'_{kl} = \dot{t}_{mn}Q_{km}Q_{ln} + t_{mn}\dot{Q}_{km}Q_{ln} + t_{mn}Q_{km}\dot{Q}_{ln} \tag{3.9.4}$$

From this we can eliminate $\dot{\mathbf{Q}}$ by use of (2.10.14). Hence

$$\dot{t}'_{kl} + t_{km}w'_{ml} - t'_{ml}w'_{km} = (\dot{t}_{mn} + t_{mr}w_{rn} - t_{rn}w_{mr})Q_{km}Q_{ln}$$

This shows that the quantity defined by

$$\hat{t}_{kl} \equiv \dot{t}_{kl} + t_{km}w_{ml} - t_{ml}w_{km} \tag{3.9.5}$$

is objective. The tensor $\hat{\mathbf{t}}$ so defined is called a *stress flux*. This is not the only quantity containing stress rate that is objective, that is, the definition of stress flux is *not* unique. Other stress fluxes may be obtained by adding objective tensors such as $\pm t_{km}d_{ml}$ to the right-hand side of (3.9.5). Thus for example

$$\begin{aligned}
\hat{t}_{kl} &= \dot{t}_{kl} + t_{ml}v_{m,k} + t_{km}v_{m,l} \\
\hat{t}_{kl} &= \dot{t}_{kl} - t_{ml}v_{k,m} - t_{km}v_{l,m} \\
\hat{t}_{kl} &= \dot{t}_{kl} - t_{ml}v_{k,m} + t_{km}v_{m,l} \\
\hat{t}_{kl} &= \dot{t}_{kl} + t_{ml}v_{m,k} - t_{km}v_{l,m}
\end{aligned} \tag{3.9.6}$$

All these tensors are objective. Still other expressions different from those given above can be found.[5] Other stress fluxes are obtained also through

$$\begin{aligned}
\frac{D}{Dt}T_{KL}(X, t) = \frac{\partial T_{KL}}{\partial t} &= \frac{D}{Dt}(jt_{kl}X_{K,k}X_{L,l}) \\
&= j\dot{t}_{kl}X_{K,k}X_{L,l} + jt_{kl}\frac{D}{Dt}(X_{K,k})X_{L,l} \\
&\quad + jt_{kl}X_{K,k}\frac{DX_{L,l}}{Dt} + \frac{Dj}{Dt}t_{kl}X_{K,k}X_{L,l}
\end{aligned}$$

[5] For these and for higher-order stress fluxes see Eringen [1962, Arts. 27, 72, and 93].

Now employing (2.4.2) and (2.4.6), this may be arranged to

$$\frac{\partial T_{KL}}{\partial t} = \overset{\circ}{t}_{kl} j X_{K,k} X_{L,l} \tag{3.9.7}$$

where

$$\overset{\circ}{t}_{kl} \equiv \dot{t}_{kl} - t_{km} v_{l,m} - t_{ml} v_{k,m} + t_{kl} v_{m,m} \tag{3.9.8}$$

is also a stress flux.[6] It can be verified easily that it obeys objectivity. The result (3.9.7) is also the statement of a theorem.

THEOREM. *The necessary and sufficient condition for the Piola-Kirchoff stress tensor to be stationary, in any arbitrary motion, is that the Truesdell stress rate must vanish.*[7]

Since $j \neq 0$ and $X_{K,k}$ are arbitrary, $\partial T_{KL}/\partial t = 0$ in (3.9.7) is possible if and only if $\overset{\circ}{t}_{kl} = 0$.

Oldroyd [1950] earlier introduced a stress tensor that is deduced from T_{KL}/j. If in the above derivation this is used, then we arrive at the Oldroyd stress flux

$$\overset{\circ}{t}_{kl} = \dot{t}_{kl} - t_{km} v_{l,m} - t_{ml} v_{k,m} \tag{3.9.9}$$

Many other possibilities exist.

PROBLEMS

3.1. Components of the stress tensor and the couple stress tensor are given in a rectangular coordinate frame, x_k, at a point p

Determine the surface tractions and couples at a plane through p, which intersects the segments a, $2a$, and $3a$ on the x_1-, x_2-, x_3-axis, respectively

3.2. Rectangular coordinates x_k are rotated rigidly an angle $\theta < 90°$ about the x_3-axis in a counterclockwise direction. The new coordinate axes so obtained are then rotated an angle $\phi < 90°$ about the new x_2-axes in a counterclockwise direction The stress components are given at point p in the x_k frame. Determine the components of the stress tensor in the coordinate frame x'_k obtained after these rotations.

3.3. Determine the surface tractions at a point p of a surface $x_3 = f(x_1, x_2)$ in terms of the stress components at p referred to rectangular coordinates x_k.

3.4. Determine the components of the traction acting on an area element, da_k, in the deformed body in terms of the tractions acting on the area element before deformation. The deformation is assumed to be known.

[6] Truesdell [1955a, b] derived this in another fashion.

[7] This theorem was given by Eringen [1962, p. 253]

3.5. Determine the components of stress tensor and surface tractions in cylindrical coordinates (r, θ, z), in terms of those referred to rectangular coordinates. Find the tractions on the surface. $r = \text{const}$.

3.6. Write the equations of motion (Cauchy's first law) in the following coordinate systems:

(*a*) Cylindrical coordinates.

(*b*) Spherical coordinates.

3.7. (*Short Term Paper.*) Write the equations of motion in any one of the following coordinates:

(*a*) Bi-polar coordinates.

(*b*) Ellipsoidal coordinates.

(*c*) Toroidal coordinates.

(*d*) Paraboloidal coordinates.

(*e*) Helicoidal coordinates.

3.8. (*Short Term Paper.*) Study the three-dimensional Mohr's circle and present a thorough discussion.

3.9. Prove that the maximum shearing stress acts on the plane that bisects the angle between the greatest and the smallest principal stresses.

3.10. If $\mathbf{t}_{(n)}$ and $\mathbf{t}_{(n')}$ are the tractions at a point, p, on the intersection of two surfaces having unit normals $\mathbf{n}$ and $\mathbf{n}'$, respectively, at p, prove that $\mathbf{t}_{(n')} \cdot \mathbf{n} = \mathbf{t}_{(n)} \cdot \mathbf{n}'$.

3.11. Construct the Mohr's circle for the case of the plane stress (t_{11}, t_{12}, t_{22}), $t_{13} = t_{23} = t_{33} = 0$. Find the principal stresses, maximum shearing stress, and their directions.

3.12. (*Short Term Paper.*) Study literature and discuss the concept of couple stress. Derive the equations of balance of momenta and jump conditions on a moving discontinuity surface when couple stress and body couples are present.

3.13. Obtain the traction boundary conditions on the boundary of a deforming solid. Find the equation of the deformed boundary surface when that of the undeformed surface and the motion are known.

3.14. If τ denotes the tangential surface traction at a point on a surface, show that mean square tangential stress $\langle \tau^2 \rangle$, in terms of principal stresses t_1, t_2, and t_3, is given by

$$\langle \tau^2 \rangle \equiv \lim_{s \to 0} \frac{1}{s} \oint \tau^2 \, da = \frac{1}{15}[(t_1 - t_2)^2 + (t_2 - t_3)^2 + (t_3 - t_1)^2]$$

where the integral is over the area s of a sphere.

3.15. Show that equations of equilibrium of a body for vanishing body forces are identically satisfied by

$$t_{kl} = e_{kmn} e_{lrs} \phi_{mr,ns}$$

The case of $\phi_{12} = \phi_{23} = \phi_{31} \equiv 0$ is known as the Maxwell's stress functions and $\phi_{11} = \phi_{23} = \phi_{31} \equiv 0$, Morera's stress functions. Find the two-dimensional stress function that satisfies the equations of equilibrium for the two-dimensional state of stress.

CHAPTER 4

Thermodynamics of Continuous Media

4.1. SCOPE OF THE CHAPTER

This chapter is concerned with the thermodynamics of continuous media. Thermodynamics is a study primarily centered around the principles of energy and entropy and their various applications. In the usual courses that are offered in physics or in engineering curricula, seldom is a genuinely dynamical phenomenon treated. A proper name for such studies is *thermostatics*. Most books on thermostatics and thermodynamics of irreversible processes (a proper name for thermodynamics) often cover *discrete systems*.

Thermodynamics of continuous media starts with the principle of conservation of energy presented in Section 4.2. In Section 4.3 we discuss the idea of potential energy and strain energy. Thermodynamic nondissipative stress and pressure are derived and discussed, and transition to global thermostatics is made. Section 4.4 is devoted to jump conditions at a moving discontinuity surface. In Section 4.5 we introduce the concepts of entropy, thermodynamic temperature, thermodynamic tensions, and thermodynamic paths, which are essential to the ideas of specific heat and latent heat. The principle of entropy is presented in Section 4.6. We introduce the local Clausius-Duhem inequality as a basic axiom, discuss its important consequences, and arrive at the global principle celebrated as the second law of thermodynamics. In Section 4.7 we give two examples (the elastic solid and viscous fluid) to illustrate the thermodynamic restrictions emanating from the imposition of the local Clausius-Duhem inequality. Thermodynamic and mechanical equilibria are the subjects of Section 4.8, where a variational treatment is also presented.

Thermodynamics, in spite of its age, is not a well-developed science for the treatment of dynamical and dissipative phenomena so often encountered in the discipline of continuum physics. This inadequacy, when used as a reason to avoid the treatment of the physical phenomena in which thermodynamics is essential, often eliminates the very interesting and lively aspects of the physical sciences.

4.2. PRINCIPLE OF CONSERVATION OF ENERGY

The principle of global conservation of energy has already been formulated in Section 2.9. According to this principle, the time rate of change of the kinetic energy $\mathscr{K}$ and the internal energy $\mathscr{E}$ is equal to the sum of the rate-work $\mathscr{W}$ done upon the body by the external forces and all other energy rates $\mathscr{Q}_\alpha$. Thus mathematically

$$\dot{\mathscr{K}} + \dot{\mathscr{E}} = \mathscr{W} + \sum_\alpha \mathscr{Q}_\alpha \tag{4.2.1}$$

Among the various types of energies, we may cite *heat energy*, *chemical energy*, *electromagnetic energy*, and *surface energy*. For the sake of simplicity and clear understanding, we consider the thermomechanical systems in which only the heat energy Q is present. Thus

$$\dot{\mathscr{K}} + \dot{\mathscr{E}} = \mathscr{W} + Q \tag{4.2.2}$$

Interconvertibility of heat Q into mechanical energy has been a well-known fact since the time of Carnot [1824] and a clear formulation of this equivalence was given by Joule [1843]. The thermostatic changes are considered time-independent so that $\dot{\mathscr{K}} = 0$, and writing $\dot{\mathscr{E}}\, dt \equiv d\mathscr{E}$, $\mathscr{W}\, dt \equiv \delta\mathscr{W}$, $Q\, dt \equiv \delta\mathscr{Q}$, in (4.2.2), we obtain the well-known *first* law of thermodynamics,

$$d\mathscr{E} = \delta\mathscr{W} + \delta\mathscr{Q}$$

The heat energy, Q, above is measured in mechanical units and has the dimension of the mechanical power $\mathscr{W}$, that is,

$$[Q] = [\mathscr{W}] = \frac{\text{force} \times \text{distance}}{\text{time}} = \frac{ML^2}{T^3}$$

where M, L, T, respectively, denote the units of mass, length, and time. In CGS units (Centimeter-Gram-Second) the measure of mechanical power is erg/sec = dyne cm/sec. In MKS units (Meter-Kilogram-Second) newton-m/sec is employed as the measure of power. In English units lb ft/sec is of common use.

In a continuous medium $\mathscr{V} + \mathscr{S}$, heat may enter the body through its surface $\mathscr{S}$ or it may be supplied internally, through distributed heat sources h per unit mass of the body. Let $\mathbf{q}$ denote the heat vector per unit area acting at a point $\mathbf{x}$ of the surface $\mathscr{S}$ directed outward with respect to the body (that is, it makes an acute angle with the exterior normal $\mathbf{n}$ to $\mathscr{S}$ at $\mathbf{x}$). Then for the total heat input we write

$$Q = \oint_{\mathscr{S}} \mathbf{q} \cdot \mathbf{n}\, da + \int_{\mathscr{V}} \rho h\, dv \tag{4.2.3}$$

For the kinetic and internal energies, we have

$$\mathscr{K} = \frac{1}{2}\int_{\mathscr{V}} \rho v_k v_k\, dv, \qquad \mathscr{E} = \int_{\mathscr{V}} \rho \varepsilon\, dv \tag{4.2.4}$$

where $v_k \equiv \dot{x}_k$ is the *velocity vector and* ε *is the internal energy density per* unit mass of the body.

The mechanical energy consists of the work done by the surface and body forces in unit time. For simplicity, we consider a nonpolar medium. Thus

$$\begin{aligned} \mathscr{W} &= \oint_{\mathscr{S}} \mathbf{t}_{(\mathbf{n})} \cdot \mathbf{v}\, da + \int_{\mathscr{V}} \rho \mathbf{f} \cdot \mathbf{v}\, dv \\ &= \oint_{\mathscr{S}} t_{ik} v_k n_i\, da + \int_{\mathscr{V}} \rho f_k v_k\, dv \end{aligned} \tag{4.2.5}$$

where the surface integral in the second line follows from using $(3.4.3)_1$.

We now carry out the indicated time rates for $\mathscr{K}$ and $\mathscr{E}$ and convert the surface integrals into volume integrals by use of the Green-Gauss theorem. Thus

$$\begin{aligned} \dot{\mathscr{K}} &= \int_{\mathscr{V}} \left[\rho v_k a_k\, dv + \frac{1}{2} v_k v_k \frac{D}{Dt}(\rho\, dv)\right] \\ \dot{\mathscr{E}} &= \int_{\mathscr{V}} \left[\rho \dot{\varepsilon}\, dv + \varepsilon \frac{D}{Dt}(\rho\, dv)\right] \\ \dot{\mathscr{W}} &= \int_{\mathscr{V}} (t_{ik,i} v_k + t_{ik} v_{k,i} + \rho f_k v_k)\, dv \\ Q &= \int_{\mathscr{V}} (q_{i,i} + \rho h)\, dv \end{aligned} \tag{4.2.6}$$

Upon carrying these expressions into (4.2.2) and rearranging various terms, we obtain

$$\begin{aligned} &\int_{\mathscr{V}} (\rho \dot{\varepsilon} - t_{ik} v_{k,i} - q_{i,i} - \rho h)\, dv \\ &\quad + \int_{\mathscr{V}} (\varepsilon + \tfrac{1}{2} v_k v_k) \frac{D}{Dt}(\rho\, dv) - \int_{\mathscr{V}} v_k (t_{ik,i} + \rho f_k - \rho a_k)\, dv = 0 \end{aligned} \tag{4.2.7}$$

This is the final expression of the global conservation of energy. When the energy is balanced for arbitrarily small volume elements we have the *local* conservation of energy. In this case, we also use the local conservation of mass and the local balance of momentum. Because of these latter principles the integrands of the second and the third volume integrals vanish and we obtain for the equation of the *local conservation of energy*

$$\rho\dot{\varepsilon} = t_{kl}v_{l,k} + q_{k,k} + \rho h \tag{4.2.8}$$

For the nonpolar case under consideration $t_{kl} = t_{lk}$. We also write $v_{l,k} = d_{lk} + w_{lk}$ and note that $t_{kl}w_{lk} = 0$ since $w_{lk} = -w_{kl}$. Thus (4.2.8) may also be written as

$$\rho\dot{\varepsilon} = t_{kl}\, d_{lk} + q_{k,k} + \rho h \tag{4.2.9}$$

Here the mechanical energy

$$\phi \equiv t_{kl}v_{l,k} = t_{kl}\, d_{lk} \tag{4.2.10}$$

is also called the *stress power*. Equation (4.2.8) or (4.2.9) is thus a local balance equation expressing the fact that the internal energy change per unit time is due to the stress power and heat energy.

4.3. POTENTIAL ENERGY AND STRAIN ENERGY

In classical mechanics, when the force is a function of the position vector and may be derived from a potential $U(\mathbf{x})$, Newton's second law of motion can be integrated to yield the principle of conservation of energy

$$\mathscr{K} + \mathscr{U} = \text{constant}$$

In mechanics of continua we have body forces and surface forces, the latter being connected to the stress hypothesis. The concept of potential energy needs a new interpretation. We shall confine this term to body forces $\mathbf{f}$. When the body forces are steady and derivable from a potential $U(\mathbf{x})$, we may express the global conservation of energy in a more familiar form, that is, suppose

$$f_k = -U_{,k} \tag{4.3.1}$$

then

$$\int_{\mathscr{V}} \rho f_k v_k \, dv = -\int_{\mathscr{V}} \rho U_{,k} v_k \, dv = -\dot{\mathscr{U}} + \int_{\mathscr{V}} U \frac{D}{Dt}(\rho \, dv) \tag{4.3.2}$$

where

$$\mathscr{U} \equiv \int_{\mathscr{V}} \rho U \, dv \tag{4.3.3}$$

may be called the potential energy. Upon substituting (4.3.2) into (4.2.5) and the result into (4.2.2), we get

$$\mathscr{K} + \mathscr{E} + \mathscr{U} = \oint_{\mathscr{S}} t_{kl} v_l n_k \, da + \int_{\mathscr{V}} \rho U \frac{D}{Dt}(\rho \, dv) + Q \tag{4.3.4}$$

When the right side of this is zero, we get a *theorem of conservation of mechanical energy; namely, the sum of the global kinetic, internal, and potential energies is constant,*

$$\mathscr{K} + \mathscr{E} + \mathscr{U} = \text{constant} \tag{4.3.5}$$

Note that for this to be the case we must have, in addition to the assumptions underlying (4.3.1),

1. *The body be insulated,* that is, $Q = 0$,
2. *The energy of the surface tractions be zero* (*vanishing surface integral in* 4.3.4), and
3. *The total potential carried by the created or destroyed mass be zero* (*vanishing volume integral in* 4.3.4).

All these, of course, represent very special situations.

Since we have confined the term potential energy to body forces, it is natural to inquire whether a similar situation can exist for the surface tractions. This inquiry, as we shall see later, leads to the concept of *strain energy*

For the nonpolar case under consideration it is reasonable to assume that the stress tensor can be written as the sum of two symmetric stress tensors

$$\mathbf{t} = {}_E\mathbf{t} + {}_D\mathbf{t} \quad \text{or} \quad t_{kl} = {}_Et_{kl} + {}_Dt_{kl} \tag{4.3.6}$$

of which ${}_E\mathbf{t}$ is the reversible or recoverable part of the stress that is derivable from a potential and ${}_D\mathbf{t}$ is the irreversible or dissipative part. We shall call ${}_E\mathbf{t}$ the hyperelastic stress and assume that it is derivable from a potential $\tau(x_{k,K})$ so that

$$\rho\dot{\tau} = {}_Et_{kl}\, d_{lk} = {}_Et_{kl} v_{l,k} \tag{4.3.7}$$

Substituting (4.3.6) into this and the result into the expression of $\mathscr{W}$ in (4.2.9) and integrating over the volume $\mathscr{V}$ of the deformed body, we get

$$\int_{\mathscr{V}} \rho\dot{\varepsilon} \, dv = \int_{\mathscr{V}} \rho\dot{\tau} \, dv + \int_{\mathscr{V}} {}_Dt_{lk}\, d_{kl} \, dv + \int_{\mathscr{V}} (q_{k,k} + \rho h) \, dv$$

If we recall that the equation of balance of energy (4.2.9) assumes that the

local mass is conserved and that Cauchy's first law is valid, then we may rewrite this as

$$\dot{\mathscr{E}} = \dot{\mathscr{T}} + \mathscr{D} + Q \tag{4.3.8}$$

where $\mathscr{E}$ and Q are, respectively, the total internal energy and heat energy defined earlier. New energies $\mathscr{T}$ and $\mathscr{D}$, defined below, are, respectively, called the *total strain energy* and the *total dissipative power*:

$$\mathscr{T} \equiv \int_{\mathscr{V}} \rho\tau \, dv, \qquad \mathscr{D} \equiv \int_{\mathscr{V}} {}_D t_{lk} \, d_{kl} \, dv \tag{4.3.9}$$

This may be stated as: *When local mass is conserved and Cauchy's first law is valid, then the sum of the time rate of the total strain energy, the total dissipative power, and the total heat energy is equal to the time rate of the internal energy.* Note that equation (4.3.8) is not the global form of the energy conservation when the local mass is not conserved and/or Cauchy's first law is not satisfied. In fact, the global form of the energy conservation is

$$\dot{\mathscr{E}} = \dot{\mathscr{T}} + \mathscr{D} + Q - \dot{\mathscr{K}} - \int_{\mathscr{V}} \tau \frac{D}{Dt}(\rho \, dv) + \int_{\mathscr{V}} (t_{lk,l} + \rho f_k) v_k \, dv \tag{4.3.10}$$

In a special case, when $\varepsilon = \tau$, equation (4.3.8) reduces to

$$\mathscr{D} + Q = 0 \tag{4.3.11}$$

so that in this case the *dissipation energy is totally converted into heat.*

The existence of the potential energy $\tau(x_{k,K})$ through (4.3.7) leads to determination of the hyperelastic stress. For

$$\dot{\tau} = \frac{\partial \tau}{\partial x_{k,K}} \frac{D}{Dt}(x_{k,K}) = \frac{\partial \tau}{\partial x_{k,K}} v_{k,l} x_{l,K}$$

where in the last step we used $(2.4.1)_1$. Now carry this into (4.3.7)

$$\rho \frac{\partial \tau}{\partial x_{k,K}} x_{l,K} v_{k,l} = {}_E t_{lk} v_{k,l}$$

from which, for arbitrary motions and since ${}_E t_{kl} = {}_E t_{lk}$, we get

$$\begin{aligned} {}_E t_{lk} &= \rho \frac{\partial \tau}{\partial x_{(k,K}} x_{l),K} \\ 0 &= \rho \frac{\partial \tau}{\partial x_{[k,K}} x_{l],K} \end{aligned} \tag{4.3.12}$$

It is not difficult to calculate the corresponding Piola stresses ${}_E T_{Kl}$ and

${}_E T_{KL}$, because we have

$$ {}_E T_{KI} = \frac{\rho_0}{\rho} {}_E t_{kl} X_{K,k} $$

$$ {}_E T_{KL} = \frac{\rho_0}{\rho} {}_E t_{kl} X_{K,k} X_{L,l} $$

Now, by substituting (4.3.12)$_1$ into these expressions and using (4.3.12)$_2$, we obtain

$$ {}_E T_{KI} = \rho_0 \frac{\partial \tau}{\partial x_{l,K}} $$

(4.3.13)

$$ {}_E T_{KL} = \rho_0 \frac{\partial \tau}{\partial x_{k,(K}} X_{L),k} $$

A special case for the strain energy occurs when τ depends only on $J = \det x_{k,K}$, that is,

(4.3.14) $$ \tau = \tau(j) = \tau(\rho_0/\rho) $$

For this case through (4.3.12) and using (1.4.7), we have

$$ {}_E t_{lk} = \rho \frac{\partial \tau}{\partial j} \frac{\partial j}{\partial x_{(k,K}} x_{l),K} = \rho \frac{\partial \tau}{\partial j} j X_{K,k} x_{l,K} $$

or simply

(4.3.15) $$ {}_E t_{kl} = -\pi \delta_{kl} $$

where

(4.3.16) $$ \pi(j) \equiv -\rho_0 \frac{\partial \tau}{\partial j} $$

is the elastic *hydrostatic pressure*. Conversely (4.3.15) implies (4.3.14). For this particular case we have

(4.3.17) $$ \rho t = -\pi v_{k,k} $$

so that by using $D(dv)/Dt = v_{k,k}\, dv$ we have

(4.3.18) $$ \mathscr{T} = -\int_\tau \pi \frac{D}{Dt}(dv) + \int_\tau \tau \frac{D}{Dt}(\rho\, dv) $$

We now show how we may deduce the first principle of thermostatics as a special case of the foregoing results. Classical thermostatics deals with homogeneous systems and the stress consists of a purely hydrostatic

pressure that is constant. We may more generally assume π to be a function of time alone, that is, $\pi = \pi(t)$. Under these conditions (4.3.18) gives

$$\mathscr{T} = -\pi \dot{\mathscr{V}} \tag{4.3.19}$$

Since now $\mathscr{D} = 0$ and the system is homogeneous (4.3.8) reduces to

$$d\mathscr{E} = -\pi \, d\mathscr{V} + \delta\mathscr{Q} \tag{4.3.20}$$

where we wrote

$$d\mathscr{E} \equiv \dot{\mathscr{E}} \, dt, \qquad \delta\mathscr{Q} \equiv Q \, dt \tag{4.3.21}$$

with the interpretations that all variations are considered time-independent and that they are considered to be changes between neighboring states.

Equation (4.3.20) is the well-known expression of the first law of thermostatics.

The assumptions that lead us to (4.3.20) are therefore:

1. The system is uniform and explicitly time-independent;
2. The dissipative part of the stress vanishes;
3. The nondissipative part of the stress consists of hydrostatic pressure only.[1]

The first restriction is the most critical one. If we try to incorporate the dissipative part of the stress into (4.3.20) we may be forced into considering the effect of inertia forces, since these give rise to dissipative power. A fresh approach would be to multiply (4.2.2) by dt and write $d\mathscr{K}$ for $\dot{\mathscr{K}} \, dt$ and $\delta\mathscr{W}$ for $\mathscr{W} \, dt$. Such an approach involves the term $d\mathscr{K}$ which is definitely out of the realm of classical thermostatics.

4.4. ENERGY JUMP AT A MOVING DISCONTINUITY SURFACE

At a point of a moving surface of discontinuity $\sigma(t)$, sweeping the body with a velocity $\mathbf{v}$, the jump conditions to be satisfied may be obtained by using the global balance equation (4.2.2). The equation may be written in the form (2.5.16) before the conversion of various integrals. Using the resulting form (2.5.18) of (2.5.16), we find that (4.2.2) contains the integrals appearing in (4.2.6) taken over $\mathscr{V} - \sigma$. In addition to each of the equations in (4.2.6) there are the following surface terms denoted by bold face

[1] This condition is not essential, and for a more general case of elastic stress we may replace (4.3.20) by $d\mathscr{E} = \delta\mathscr{W} + \delta\mathscr{Q}$, cf. Section 4.2. Also it is not difficult to show that this equation can be extended to a collection of homogeneous systems (heteregeneous media).

brackets respectively,

$$
\begin{aligned}
[\mathscr{K}] &= \int_\sigma ([\tfrac{1}{2}\rho v_i v_i \mathbf{v}] \cdot \mathbf{n} - [\tfrac{1}{2}\rho v_i v_i \mathbf{v}] \cdot \mathbf{n})\, da \\
[\mathscr{E}] &= \int_\sigma ([\rho \varepsilon \mathbf{v}] \cdot \mathbf{n} - [\rho \varepsilon \mathbf{v}] \cdot \mathbf{n})\, da \\
[\mathscr{W}] &= \int_\sigma [t_{ki} v_i] n_k\, da \\
[\mathscr{Q}] &= \int_\sigma [q_k] n_k\, da
\end{aligned}
\tag{4.4.1}
$$

Consequently for an arbitrary surface $\sigma(t)$ this gives

$$
[(\rho\varepsilon + \tfrac{1}{2}\rho v_i v_i)(v_k - \overset{+}{v}_k) - t_{ki}v_i - q_k]n_k = 0 \qquad \text{on } \sigma(t) \tag{4.4.2}
$$

Although some simplification is possible by resolving various vectors into normal and tangential components, it is not worth the effort since not much more clarity is obtained. Certain special cases are worthy, however, of consideration.

(*a*) ***The Discontinuity Surface is a Material Interface.*** In this case $\overset{+}{\mathbf{v}} = \mathbf{v}$ and (4.4.2) gives

$$
([t_{ki}v_i] + [q_k])n_k = 0 \tag{4.4.3}
$$

This states that *the jump on the energy of tractions across a material interface is balanced with that of the normal component of the heat vector.*

If the material velocity is continuous across $\sigma(t)$ using (3.6.5) this further reduces to

$$
[q_k]n_k = 0
$$

which states *the continuity in the normal component of the heat vector* across $\sigma(t)$.

(*b*) ***The Discontinuity Surface Coincides with the Surface of the Body.*** In this case $\overset{+}{\rho} = 0$ and $\overset{-}{\mathbf{v}} = \mathbf{v}$. We again obtain (4.4.3) which may also be expressed as

$$
(t_{ki}v_i + q_k)n_k = \mathbf{s} \cdot \overset{+}{\mathbf{v}} + q_{(n)} \tag{4.4.4}
$$

where we put

$$
\begin{aligned}
&\overset{+}{t}_{ki}n_k \equiv s_i \qquad \overset{+}{\mathbf{q}} \cdot \mathbf{n} \equiv q_{(n)} \\
&\overset{-}{t}_{ki} \equiv t_{ki} \qquad \overset{-}{q}_k \equiv q_k, \qquad \overset{-}{v}_i \equiv v_i
\end{aligned}
\tag{4.4.5}
$$

of which s_k is the surface traction as given by (3.4.9) and $q_{(n)}$ is the normal component of the heat vector. Equation (4.4.4) gives the boundary conditions on the energy flow across the surface of the body. It states that *the energy of the external forces and the normal component of heat is balanced with those taken adjacent to the surface from the interior.*

If the external forces are attached to the surface then $\bar{v}_l = \overset{+}{v}_l$ and the first terms on the left and right of (4.4.4) are equal. Hence we obtain the boundary condition

$$q_k n_k = q_{(n)} \tag{4.4.6}$$

involving the heat alone.

(c) ***The Discontinuity Surface is a Fixed Surface.*** In this case $\mathbf{v} = 0$ and (4.4.2) gives

$$[(\tfrac{1}{2}\rho v_l v_l + \rho\varepsilon)v_k - t_{kl}v_l - q_k]n_k = 0 \tag{4.4.7}$$

4.5. ENTROPY

The classical equation (4.3.20) of the first principle of thermostatics may be put into a differential form if an integrating factor $1/\bar{\theta}$ can be found so that $\delta\mathcal{Q}$ may be expressed as a total differential

$$\frac{\delta\mathcal{Q}}{\bar{\theta}} = dH \tag{4.5.1}$$

With this substitution (4.3.20) takes the form

$$d\mathcal{E} = -\pi\, d\mathcal{V} + \bar{\theta}\, dH \tag{4.5.2}$$

The new variables $\bar{\theta}$ and H so introduced are respectively called the *absolute temperature* and the *total entropy.*

The conditions under which (4.5.1) are possible were first investigated by Caratheodory [1909, 1925].[2] Caratheodory arrived at the concept of temperature and entropy through the concept of *thermal equilibrium* and the assumption that for any state there is an arbitrarily near state that the thermodynamic system cannot reach without expanding work. For non-equilibrium processes, which are more natural than the equilibrium processes, all attempts to define entropy and temperature face great difficulties. Presently an adequate theory of irreversible thermodynamics does not

[2] In this connection see also Born [1921], Ehrenfest-Afanassjewa [1925], Mimura [1931], and Waples [1952].

exist.[3] Introduction of the concepts of entropy and temperature as primitive quantities, just as in the case of mass, seems to be the safest approach at the present state of thermodynamics.

Thermodynamics States. The class of $n+1$ independent variables η, v_α ($\alpha = 1, 2, \ldots, n$) that influence the internal energy density ε and temperature θ of a material point X occupying a place x at time t is called a *thermodynamic state*. The parameters v_α have the dimensions of electrical and mechanical units, and the dimension of the *entropy density* is independent of those of v_α. The entropy density η has the dimension of *energy per unit temperature per unit mass*. Thus we write

$$\varepsilon = \hat{\varepsilon}(\eta, v_\alpha, \mathrm{X}) \tag{4.5.3}$$

$$\theta = \hat{\theta}(\eta, v_\alpha, \mathrm{X}) \tag{4.5.4}$$

Equations (4.5.3) and (4.5.4) are "thermodynamical" constitutive equations for the internal energy density and the temperature. They are subject to the restrictions of the Clausius-Duhem inequality postulated later and those of the constitutive theory studies in Chapter 5. The selection of v_α depends on the thermodynamic state of the body.[4] A certain class of variables cannot be admitted into the class of v_α. For example, *the time, place, motion, velocity, and acceleration are not admissible* on the ground that ε and θ must obey the *principle of objectivity* (cf. Chapter 5). The parameters η and v_α depend on time and place or motion, that is,

$$\eta = \eta(\mathrm{x}, t), \qquad \mathrm{v} = \mathrm{v}(\mathrm{x}, t)$$

where we wrote v, in short, to represent v_α. Since for a given motion we have $\mathrm{x} = \mathrm{x}(\mathrm{X}, t)$, we see that $\eta = \hat{\eta}(\mathrm{X}, t)$, $\mathrm{v} = \hat{\mathrm{v}}(\mathrm{X}, t)$ and therefore $\varepsilon = \hat{\varepsilon}(\mathrm{X}, t)$ and $\theta = \hat{\theta}(\mathrm{X}, t)$. However, according to (4.5.3) and (4.5.4) ε and θ are completely characterized without regard to time and place, that is, they are not explicit functions of x and t but they are dependent on the values of η and v_α at place x and time t. The choice of v_α depends on the systems under consideration. Through our experience with the physical system we select v_α to define the thermodynamic character of the system. For example, for dilute gases and some fluids, $v_1 = 1/\rho$, $v_\alpha = 0$ ($\alpha > 1$).

[3] A large number of studies in this field are confined to linear constitutive equations and processes neighboring equilibrium. Constitutive coefficients appearing in such theories are assumed to be symmetric tensors. For a treatment of mostly discrete systems, see DeGroot [1962]. For linear irreversible continuum theories, see Eringen [1960].

[4] Through the axiom of causality discussed in Chapter 5 a rule for the selection of v_α exists for the thermomechanical theories. For chemical and electromechanical systems, some approaches exist (Dixon and Eringen [1965] and Eringen and Ingram [1965]); however, they are less definitive.

Once v_α are selected, we will have the thermodynamic state of an ideal system determined. For this state to represent any physical one, we must see that it does not contradict the basic axioms of mechanics, thermodynamics, and constitution. Such restrictions provide conditions of admissibility on the thermodynamic state of an ideal system (cf. Section 4.6).

DEFINITION. *The changes that occur in* ε *and* θ *due to change of* η *and* v_α *are called a thermodynamic process.*

Equations (4.5.3) and (4.5.4) describe this process, so that they are sometimes named equations of state.[5]

DEFINITION. *A thermodynamic process is called homogeneous if the functions* ε *and* θ *are independent of* $\mathbf{X}$.

For a homogeneous state, therefore, the functional forms of the energy, density, and temperature are the same at all points of the body.

If (4.5.4) is soluble for η, then we can select θ as a new state variable to replace η. A sufficient condition for (4.5.4) to be soluble in some neighborhood of η (to be an explicit function of η) with respect to η, for fixed v_α is that $\partial\theta/\partial\eta\big|_{v,\mathbf{X}}$ is continuous and does not vanish in that neighborhood of η. In this case we have

$$\eta = \eta(\theta, v_\alpha, \mathbf{X}) \tag{4.5.5}$$

$$\varepsilon = \varepsilon[\eta(\theta, v, \mathbf{X}), v, \mathbf{X}] \equiv \hat{\varepsilon}(\theta, v, \mathbf{X}) \tag{4.5.6}$$

where both functions η and $\hat{\varepsilon}$ are given at $\mathbf{X}$. For homogeneous processes these functions do not change forms with $\mathbf{X}$.

DEFINITION. *Thermostatic temperature* $\bar{\theta}$ *and thermostatic tensions* τ_α are defined by

$$\bar{\theta} \equiv \frac{\partial\varepsilon}{\partial\eta}\bigg|_{v,\mathbf{X}}, \quad \tau_\alpha \equiv \frac{\partial\varepsilon}{\partial v_\alpha}\bigg|_{\eta,\mathbf{X},v_\beta} \qquad \beta \neq \alpha \tag{4.5.7}$$

We emphasize the distinction between the temperature θ and the thermostatic temperature $\bar{\theta}$ introduced above. In Section 4.6 it will be shown that they are the same for some thermomechanical systems.

For an arbitrary change of a thermodynamic state at a given material point $\mathbf{X}$, through (4.5.3) and (4.5.7) we have

$$d\varepsilon = \bar{\theta}\, d\eta + \tau_\alpha\, dv_\alpha \tag{4.5.8}$$

[5] The terminology is used here in a broader sense than that of classical thermodynamics where the terminology is applied only to one of the two equations (4.5.3) and (4.5.4) or $\varepsilon = \hat{\varepsilon}(\theta, v_\alpha, \mathbf{X})$, cf. (4.5.6).

where a summation over repeated indices is understood. This is the *local* form of the well-known Gibbs's equation, obtained by Gibbs [1873] for a special system. When $\nu_1 = 1/\rho$ then $-\tau_1$ is the *thermodynamic pressure*. For a mixture, if such a Gibbs's equation can be obtained, and if we take $\nu_2, \nu_3, \ldots, \nu_n$ as the concentrations of the constituents, then the corresponding $\tau_2, \tau_3, \ldots, \tau_n$ are known as the *chemical potentials*.

For a fixed X we may also write

$$\dot{\varepsilon} = \bar{\theta}\dot{\eta} + \tau_\alpha \dot{\nu}_\alpha \tag{4.5.9}$$

From (4.5.7) it is clear that

$$\bar{\theta} = \bar{\theta}(\eta, \nu, \mathrm{X}), \qquad \tau_\alpha = \tau_\alpha(\eta, \nu, \mathrm{X}) \tag{4 5 10}$$

If (4.5.10) is soluble for η, then we can select $\bar{\theta}$ as a new state variable to replace η. A sufficient condition for $(4.5.10)_1$ to be soluble with respect to η, in some neighborhood of η, is that $\partial\bar{\theta}/\partial\eta$ be continuous and not vanish in that neighborhood of η In this case we have

$$\eta = \eta(\bar{\theta}, \nu, \mathrm{X}) \tag{4.5.11}$$

By substitution into (4.5.3) and $(4.5.10)_2$, we obtain

$$\varepsilon = \hat{\varepsilon}(\bar{\theta}, \nu, \mathrm{X}) \tag{4 5 12}$$

$$\tau_\alpha = \bar{\tau}_\alpha(\bar{\theta}, \nu, \mathrm{X}) \quad \text{or} \quad \nu_\alpha = \bar{\nu}_\alpha(\bar{\theta}, \tau, \mathrm{X}) \tag{4.5 13}$$

These equations are other forms of the thermal equations of state. Equations $(4\ 5.13)_2$ are obtained by assuming that $(4.5.10)_2$ is soluble for ν. The sufficient conditions for the unique inverse of the forms $(4.5.13)_2$ to exist, in some neighborhood of ν, are the continuity of partial derivatives of $(4.5.10)_2$ with respect to ν_β and

$$\det(\partial\tau_\alpha/\partial\nu_\beta) \neq 0 \tag{4 5.14}$$

in that neighborhood of ν. The changes of variables of this type as well as interrelations obtained among various differentials of state variables occupy the subject of classical thermostatics.

DEFINITION. *Thermodynamic Path. For a given particle* X, $\eta = \eta(\lambda)$ *and* $\nu = \nu(\lambda)$, *for variable* λ, *define a thermodynamic path in the space of thermodynamic state* η *and* ν. The paths on which $\eta = \text{const.}$ are called *isentropic* and those with $\bar{\theta} = \text{const.}$ are called ***isothermal.***

Specific heat κ and *latent heats* λ_α and μ_α for a given particle X are defined by

$$\kappa \equiv \frac{\delta q}{d\bar{\theta}} = \frac{1}{\dot{\bar{\theta}}}(\dot{\varepsilon} - \tau_\beta \dot{\nu}_\beta) \tag{4.5.15}$$

$$\lambda_\alpha \equiv \frac{\delta q}{d\tau_\alpha} = \frac{1}{\dot{\tau}_\alpha}(\dot{\varepsilon} - \tau_\beta \dot{v}_\beta) \tag{4.5.16}$$

$$\mu_\alpha \equiv \frac{\delta q}{dv_\alpha} = \frac{1}{\dot{v}_\alpha}(\dot{\varepsilon} - \tau_\beta \dot{v}_\beta) \tag{4.5.17}$$

where $\delta q \equiv d\varepsilon - \tau_\beta\, dv_\beta$ is the *heat increment*. Since $\mathbf{X}$ is fixed, by taking $\lambda = \lambda(t)$ on a thermodynamic path P, we may also write these expressions as

$$\kappa = \frac{1}{d\bar{\theta}_P}(d\varepsilon_P - \tau_{\beta P}\, dv_{\beta P}) = \bar{\theta}_P \frac{d\eta_P}{d\bar{\theta}_P} \tag{4.5.18}$$

$$\lambda_\alpha = \frac{d\eta_P}{d\tau_{\alpha P}}(d\varepsilon_P - \tau_{\beta P}\, dv_{\beta P}) = \bar{\theta}_P \frac{d\eta_P}{d\tau_{\alpha P}} \tag{4.5.19}$$

$$\mu_\alpha = \frac{1}{dv_{\alpha P}}(d\varepsilon_P - \tau_{\beta P}\, dv_{\beta P}) = \bar{\theta}_P \frac{d\eta_P}{dv_{\alpha P}}, \quad (P \text{ not summed}) \tag{4.5.20}$$

where we used (4.5.8) to get the extreme rights of these expressions. For experimental measurements it is convenient to keep $\mathbf{v} = \text{const.}$ or $\boldsymbol{\tau} = \text{const.}$ Thus, for example, the specific heat at constant *substate* $\mathbf{v}$ follows from

$$\kappa_{\mathbf{v}} = \left(\frac{\partial \varepsilon}{\partial \bar{\theta}}\right)_{\mathbf{v},\mathbf{X}} \tag{4.5.21}$$

To determine the specific heat at constant tension, $\kappa_{\boldsymbol{\tau}}$, we regard ε as a function of $\bar{\theta}$ and $\mathbf{v}$. Then through (4.5.15) we write

$$\kappa = \frac{1}{d\bar{\theta}_P}\left\{\left(\frac{\partial \varepsilon}{\partial \bar{\theta}}\right)_{\mathbf{v},\mathbf{X}} d\bar{\theta}_P + \left[\left(\frac{\partial \varepsilon}{\partial v_\beta}\right)_{\bar{\theta},\mathbf{v}_\alpha,\mathbf{X}} - \tau_\beta\right] dv_{\beta P}\right\}, \qquad \alpha \neq \beta \tag{4.5.22}$$

If in (4.5.22) we hold the tensions $\tau_\beta = \text{const.}$, we get

$$\kappa_{\boldsymbol{\tau}} - \kappa_{\mathbf{v}} = \left[\left(\frac{\partial \varepsilon}{\partial v_\beta}\right)_{\bar{\theta},\mathbf{v}_\alpha,\mathbf{X}} - \tau_\beta\right]\left(\frac{\partial v_\beta}{\partial \bar{\theta}}\right)_{\boldsymbol{\tau}}, \qquad \alpha \neq \beta \tag{4.5.23}$$

where $\mathbf{v}$ is regarded as given by $(4.5.13)_2$. The ratio of specific heats

$$\gamma \equiv \frac{\kappa_{\boldsymbol{\tau}}}{\kappa_{\mathbf{v}}} \tag{4.5.24}$$

is an important quantity employed in gas dynamics, thermostatics, statistical mechanics, and other branches of physics.

4.6. PRINCIPLE OF ENTROPY

FUNDAMENTAL AXIOM 5 (*Clausius-Duhem Inequality*). *The time rate of change of the total entropy H is never less than the sum of the influx of*

entropy S *through the surface* $\mathscr{S}$ *of the body and the entropy* B *supplied by the body sources. We postulate this to be true for all parts of the body for all independent processes.*

Thus, according to the first part of this axiom, we have

$$\Gamma \equiv \frac{dH}{dt} - B - \oint_{\mathscr{S}} \mathbf{S} \cdot d\mathbf{a} \geq 0 \tag{4 6.1}$$

where Γ so defined is called *total entropy production.* For a continuous mass medium, we may express this in the form

$$\frac{d}{dt}\int_{\mathscr{V}} \rho\eta \, dv - \int_{\mathscr{V}} \rho b \, dv - \oint_{\mathscr{S}} S_k \, da_k \qquad \geq 0 \tag{4 6.2}$$

were b is the *local entropy source* per unit mass

$$B \equiv \int_{\mathscr{V}} \rho b \, dv \tag{4 6 3}$$

When (4.6.2) is valid for any part of the body we get the local production of entropy

$$\rho\gamma \equiv \rho\dot{\eta} - \rho b - \operatorname{div} \mathbf{S} \geq 0 \qquad \text{in } \mathscr{V} - \sigma(t) \tag{4.6.4}$$

where γ is the local entropy production, and the jump condition

$$[\rho\eta(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} - [\mathbf{S}] \cdot \mathbf{n} \geq 0 \qquad \text{on } \sigma(t) \tag{4.6.5}$$

where $\sigma(t)$ is a moving discontinuity surface in $\mathscr{V}$ having velocity v and positive normal n (cf Figure 2.5 2) The specific form of these equations depends on the type of process. For thermal changes, diffusion, and chemical and electrical phenomena, different types of effects make up the entropy influx S and entropy source b. It is always possible to express the entropy influx and entropy source as

$$\mathbf{S} = \frac{\mathbf{q}}{\theta} + \mathbf{S}_1, \qquad b = \frac{h}{\theta} + b_1 \tag{4 6.6}$$

where $\mathbf{q}/\theta$ is the entropy influx due to heat input, h/θ is the entropy source supplied by the energy source, and the remaining terms $\mathbf{S}_1$ and b_1, respectively, are the entropy influx and source due to all other effects.

Carrying h from (4.2.8) into (4.6.6) we have

$$\rho\theta b = \rho\theta b_1 + p\hat{\varepsilon} - t_{kl}v_{l,k} - q_{k,k}$$

Substituting this and (4.6.6) into (4.6.4) and (4.6.5) we get

$$\rho\gamma \equiv \rho\left(\dot{\eta} - \frac{\dot{\varepsilon}}{\theta}\right) + \frac{1}{\theta} t_{kl} v_{l,k} + \frac{1}{\theta} q_k (\log\theta)_{,k} - \operatorname{div} \mathbf{S}_1 - \rho b_1 \geq 0 \qquad \text{in } \mathscr{V} - \sigma(t)$$

(4.6.7)

$$[\rho\eta(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} - \left[\frac{\mathbf{q}}{\theta} + \mathbf{S}_1\right] \cdot \mathbf{n} \geq 0 \qquad \text{on } \sigma(t)$$

DEFINITION 1. *A thermodynamic process in which* $\mathbf{S}_1 = \mathbf{0}$, $b_1 = 0$, that is,

$$\mathbf{S} = \frac{\mathbf{q}}{\theta}, \qquad b = \frac{h}{\theta} \tag{4.6.8}$$

will be called a simple thermomechanical process. For a simple process (4.6.7) further reduces to[6]

$$\rho\gamma \equiv \rho\left(\dot{\eta} - \frac{\dot{\varepsilon}}{\theta}\right) + \frac{1}{\theta} t_{kl} v_{l,k} + \frac{1}{\theta} q_k (\log\theta)_{,k} \geq 0 \qquad \text{in } \mathscr{V} - \sigma(t) \tag{4.6.9}$$

$$\left[\rho\eta(\mathbf{v} - v) - \frac{\mathbf{q}}{\theta}\right] \cdot \mathbf{n} \geq 0 \qquad \text{on } \sigma(t) \tag{4.6.10}$$

For this process the global form of the Clausius-Duhem inequality is obtained by using (4.6.8) in (4.6.1).

$$\Gamma \equiv \dot{H} - \oint_{\mathscr{S}} \frac{\mathbf{q}}{\theta} da \cdot - \int_{\mathscr{V}} \frac{\rho h}{\theta} dv \geq 0 \tag{4.6.11}$$

For adiabatic processes $\mathbf{q} = \mathbf{0}$, $h = 0$, and this gives $\dot{H} \geq 0$ which is the statement that *in an adiabatic process the global entropy cannot decrease.* Henceforward we shall be dealing only with simple thermomechanical processes.[7]

The mass density ρ is nonnegative, that is, $\rho > 0$. We also impose the condition that

$$\theta > 0, \qquad \inf \theta = 0$$

that is, the temperative is absolute, or it is always positive.[8] An alternative

[6] The principle of entropy in the general form (4.6.4) was given in the second draft of this chapter (1962) and used in various publications thereafter, for example, Eringen and Ingram [1965], and Eringen [1965]. The form (4.6.9) for simple thermodynamical processes was used by Eringen [1962] and Koh and Eringen [1963] and was further developed by Coleman and Mizel [1964].

[7] For an example of nonsimple processes we cite chemically reacting media, cf. Eringen and Ingram [1965].

[8] Nuclear experiments done by Purcell and Pound [1951] on pure LiF crystal have shown that, under certain conditions, it is possible to induce a negative absolute temperature in thermally insulated systems.

form useful for the treatment of heat-conducting materials is found by introducing the free-energy function

$$\psi \equiv \varepsilon - \theta\eta \tag{4.6.12}$$

The inequality (4.6.9) now takes the form

$$\rho\gamma \equiv -\frac{\rho}{\theta}(\dot{\psi} + \eta\dot{\theta}) + \frac{1}{\theta} t_{kl} v_{l,k} + \frac{1}{\theta^2} q_k \theta_{,k} \geq 0 \tag{4.6.13}$$

and of course we have (4.6.10) across $\sigma(t)$.

DEFINITION 2. *A process will be called thermodynamically admissible if and only if it obeys the local Clausius-Duhem inequality and possesses a nonnegative finite temperature*

$$0 \leq \gamma, \qquad 0 < \theta < \infty, \qquad \inf \theta = 0 \tag{4.6.14}$$

DEFINITION 3 *A thermodynamic process is said to be mechanically admissible (or consistent) if it obeys the axioms of motion, namely, the conservation of mass, balance of momenta, and conservation of energy. It is called constitutively admissible if it is restricted by the constitutive axioms* (cf. Chapter 5).

Thus in a mechanically admissible process ρ, t_{kl}, ε, and q_k of (4.6.9) must satisfy the equations of conservation of mass, balance of momenta, and conservation of energy. For a constitutively admissible process, various axioms stated in Section 5.3 provide further restrictions. According to an axiom of constitution called the principle of equipresence, all constitutive functions must employ the same list of independent variables. Some of the variables from a chosen list may be shown not to be present in the arguments of some of the constitutive functions in view of other restrictions. Thus, for example, when ε and θ are given by functions such as (4.5.3) and (4.5.4), the constitutive equations for the stress t_{kl} and the heat q_k must also use the same dependent variables η and v_α, that is,

$$\begin{aligned} t_{kl} &= t_{kl}(\eta, v_\alpha, \mathrm{X}) \\ q_k &= q_k(\eta, v_\alpha, \mathrm{X}) \end{aligned} \tag{4.6.15}$$

The functions ε, θ given by (4.5.3) and (4.5.4) and t_{kl} and q_k given above are subject to additional constitutive requirements discussed in Chapter 5.

THEOREM 1. *In a thermodynamically admissible simple thermomechanical process characterized by a set of state variables (η, v_α) with v_α being functionally independent of η and not containing the time rates and integrals of η, the temperature θ and the thermostatic temperature $\bar{\theta}$ are the same*

Using (4.5.3) in (4.6.9) we can write the latter

$$\left(\theta - \frac{\partial \varepsilon}{\partial \eta}\right)\dot{\eta} - \rho\tau_\alpha \dot{v}_\alpha + t_{kl} v_{l,k} + q_k (\log \theta)_{,k} \geq 0 \tag{4.6.16}$$

Now, ε, θ, $\mathbf{t}$ and $\mathbf{q}$ are functions of η and v_α. This inequality, which is linear in $\dot{\eta}$, must be valid for all values of $\dot{\eta}$. The independent variable $\dot{\eta}$ appears only in the first term, on the left-hand side, of (4.6.16). To maintain (4.6.16) for all values of $\dot{\eta}$, we must set the coefficient of $\dot{\eta}$ equal to zero. Thus

$$\theta = \left.\frac{\partial \varepsilon}{\partial \eta}\right|_{v,\,\mathbf{x}} \tag{4.6.17}$$

This is the expression of the thermostatic temperature $\tilde{\theta}$ that is postulated by $(4.5.7)_1$ and the proof of the theorem.

With (4.6.17), equation (4.6.16) reduces to

$$\rho\theta\gamma \equiv -\rho\tau_\alpha \dot{v}_\alpha + t_{kl} v_{l,k} + q_k (\log \theta)_{,k} \geq 0 \tag{4.6.18}$$

Thermodynamic Forces and Fluxes. In the expression of the production of entropy (4.6.18) inner products of vectoral and tensoral quantities occur in pairs. In the thermodynamics of irreversible processes, it has become customary to refer to one set of the pairs as *thermodynamic forces* or *affinities* and the conjugate set (multiplying the forces) as the *thermodynamic fluxes.* For example, from (4.6.18) one may select

Force	*Corresponding Flux*
$\left(\frac{1}{\theta}\right)_{,k}$	q_k
$v_{l,k}$	t_{kl}
$\dot{v}_\alpha$	τ_α

Linear constitutive equations are then set up between any one of the forces and all of the fluxes with symmetric constitutive coefficients (Onsager's principle).[9] The correct selection of forces and fluxes requires ingenuity and perhaps confidence in the results to be obtained. Symmetry properties of odd-order tensors and the shaky grounds of the so-called Curie's theorem are sufficient reasons to abandon this thermodynamic approach[10] to constitutive theory.

[9] For an account on the subject see Eringen [1960].

[10] The present chapter some years ago was based on linear irreversible thermodynamics of the type criticized above. The rapid advancement of continuum mechanics now makes it possible to use the present concrete approach. While there remain large uncultivated grounds in irreversible processes, this author believes that without a sound thermodynamics the mechanics is incomplete and all mechanical ersatzes are liable to remain temporary measures.

4.7. THERMOOYNAMIC RESTRICTIONS ON ELASTIC SOLIOS AND VISCOUS FLUIOS

(*i*) *Elastic Solid.* For an elastic solid, mechanical and thermal constitutive equations may be expressed as

$$\begin{aligned} t_{kl} &= t_{kl}(\eta, x_{k,K}) \\ q_k &= q_k(\eta, x_{k,K}) \\ \varepsilon &= \varepsilon(\eta, x_{k,K}) \\ \theta &= \theta(\eta, x_{k,K}) \end{aligned} \tag{4.7.1}$$

These are the results of selecting the first nine of v_α in (4 6 15), (4.5.3), and (4.5.4) as $x_{k,K}$ and remaining $v_\alpha \equiv 0$ for $\alpha > 9$. Corresponding to this we have for the first nine thermodynamic tensions τ_{kK} and the thermodynamic temperature $\bar{\theta}$

$$\tau_{kK} \equiv \frac{\partial e}{\partial x_{k,K}}\bigg|_{\eta}, \qquad \bar{\theta} \equiv \frac{\partial \varepsilon}{\partial \eta}\bigg|_{x_{k,K}} \tag{4.7.2}$$

Now calculate

$$\dot{\varepsilon} = \frac{\partial \varepsilon}{\partial \eta}\eta + \frac{\partial \varepsilon}{\partial x_{k,K}}\frac{D}{Dt}(x_{k,K})$$

With the help of (2.4.1) this may be put into the form

$$\dot{\varepsilon} = \frac{\partial \varepsilon}{\partial \eta}\dot{\eta} + \frac{\partial \varepsilon}{\partial x_{k,K}} v_{k,l} x_{l,K}$$

Upon substituting this into (4.6.9) we have

$$\rho\gamma \equiv \rho\left(1 - \frac{1}{\theta}\frac{\partial \varepsilon}{\partial \eta}\right)\eta + \frac{1}{\theta}\left(t_{lk} - \rho\frac{\partial \varepsilon}{\partial x_{k,K}} x_{l,K}\right)v_{k,l} + \frac{1}{\theta^2} q_k \theta_{,k} \geq 0 \tag{4.7.3}$$

This inequality is linear in η, $v_{l,k}$, and $\theta_{,k}$ since according to (4.7.1) t_{kl}, q_k, and ε are independent of these quantities. Moreover these variables can be varied independently. For all independent variations of $\dot{\eta}$, $v_{l,k}$, and $\theta_{,k}$ the inequality (4.7 3) cannot be maintained unless

$$\theta = \frac{\partial \varepsilon}{\partial \eta} \tag{4 7.4}$$

$$t_{lk} = \rho\frac{\partial e}{\partial x_{(k,K}} x_{\eta),K} \tag{4.7.5}$$

$$\rho \frac{\partial \varepsilon}{\partial x_{[k,K}} x_{l],K} = 0 \tag{4.7.6}$$

$$q_k = 0 \tag{4.7.7}$$

A comparison of (4.7.4) and (4.7.5) with (4.7.2) shows that

$$t_{kl} = \rho \tau_{(kK} x_{l),K} \tag{4.7.8}$$

$$\theta = \bar{\theta} = \frac{\partial \varepsilon}{\partial \eta} \tag{4.7.9}$$

Thus the stress and the thermodynamic tensions are related to each other and the thermodynamic temperature is the same as the temperature. This latter result can be seen also from Theorem 1 of Section 4.6. Equation (4.7.4) also indicates that ε and θ as given by (4.7.1) are not independent of each other. We have thus proved

THEOREM 1. *An elastic solid having constitutive equations of the form* (4.7.1), *is thermodynamically admissible if and only if the temperature* θ *is given by* (4.7.4) *and the stress tensor by* (4.7.5) *subject to restrictions* (4.7.6). *Such a solid can respond only to adiabatic changes* (4.7.7).

We note that such severe restrictions as contained in Theorem 1 stem from the conditions that: (a) the constitutive equations (4.7.1) contain only the variables listed, (b) the Clausius-Duhem inequality must be satisfied for all independent processes and for all parts of the body. If we include, for example, the thermal gradient and/or the velocity gradient $v_{k,l}$ into the arguments of (4.7.1) (cf. Section 5.4), then the situation would be altered. Similarly, if we do not demand that (4.7.3) be satisfied for all independent processes, but for a motion and/or a thermal problem under consideration, then we do not need to impose the restrictions implied by (4.7.4) to (4.7.7).[11] *The axiom of entropy places restrictions on the material properties.* *Note*: The Clausius-Duhem inequality for an elastic solid requires that (4.7.5) and (4.7.6) be satisfied. It is interesting to note that the requirement (4.7.6) is equivalent to the statement that ε must be a function of the Green deformation tensor

$$C_{KL} = x_{k,K} x_{k,L} \tag{4.7.10}$$

This follows from the observation that (4.7.6) is a statement equivalent to requiring that ε be invariant under orthogonal transformations if it is a function of $x_{k,K}$ and some scalars (see Section 4.8 and App. B6).

[11] In this regard, see a discussion by Eringen and Grot [1965, Appendix].

(ii) Viscous Fluids. A class of viscous fluids is defined by mechanical and thermal constitutive equations of the form

$$
\begin{aligned}
t_{kl} &= t_{kl}\left(\eta, \frac{1}{\rho}, v_{k,l}\right) \\
q_k &= q_k\left(\eta, \frac{1}{\rho}, v_{k,l}\right) \\
\varepsilon &= \varepsilon\left(\eta, \frac{1}{\rho}, v_{k,l}\right) \\
\theta &= \theta\left(\eta, \frac{1}{\rho}, v_{k,l}\right)
\end{aligned}
\tag{4.7 11}
$$

In this case, the local Clausius-Duhem inequality in (4.6.9) will take the form

$$
\rho\left(1 - \frac{1}{\theta}\frac{\partial \varepsilon}{\partial \eta}\right)\dot{\eta} + \frac{1}{\rho\theta}\frac{\partial \varepsilon}{\partial \rho^{-1}}\dot{\rho} - \frac{1}{\theta}\frac{\partial \varepsilon}{\partial v_{k,l}}\dot{v}_{k,l} + \frac{1}{\theta}t_{kl}v_{l,k} + \frac{1}{\theta^2}q_k\theta_{,k} \geq 0 \tag{4.7.12}
$$

Since each of η and $\dot{v}_{k,l}$ appear linearly in this inequality, for all arbitrary variations of these quantities we cannot maintain the inequality (4.7.12) unless

$$
\theta = \frac{\partial \varepsilon}{\partial \eta}, \qquad \frac{\partial \varepsilon}{\partial v_{k,l}} = 0 \tag{4.7.13}
$$

Thus ε and consequently θ are independent of $v_{k,l}$, that is,

$$
\varepsilon = \varepsilon\left(\eta, \frac{1}{\rho}\right), \qquad \theta = \theta\left(\eta, \frac{1}{\rho}\right) = \left.\frac{\partial \varepsilon}{\partial \eta}\right|_{\rho^{-1}} = \bar{\theta} \tag{4.7.14}
$$

Carrying the expression $\dot{\rho} = -\rho v_{k,k}$ from the equation of continuity, we reduce (4.7 12) to

$$
\frac{1}{\theta}(t_{kl} + \pi\delta_{kl})v_{l,k} + \frac{1}{\theta^2}q_k\theta_{,k} \geq 0 \tag{4.7.15}
$$

where π is the *thermodynamic pressure* defined by

$$
\pi\left(\eta, \frac{1}{\rho}\right) = -\left.\frac{\partial \varepsilon}{\partial \rho^{-1}}\right|_{\eta} \tag{4.7.16}
$$

Since $\mathbf{t}$, π, and $\mathbf{q}$ are independent of $\theta_{,k}$, we must also set

$$
q_k = 0 \tag{4.7.17}
$$

thus reducing (4.7.15) to

$$
(t_{kl} + \pi\delta_{kl})v_{l,k} \geq 0 \tag{4.7.18}
$$

No further progress is possible until we know the nature of the function $t(\eta, 1/\rho, v_{l,k})$. Hence we have proved

THEOREM 2. *A viscous fluid having constitutive equations of the form* (4.7.11) *is thermodynamically admissible if and only if* ε, θ *are given by* (4.7.14) *and the stress satisfies the dissipation inequality* (4.7.18). *Such fluids can respond only to adiabatic changes.*

Again, by including $x_{k,K}$, higher rates of $v_{k,l}$, and thermal gradients into the arguments of the constitutive equations we obtain viscoelastic solids and fluids for which the conditions of thermodynamic admissibility are totally different from those of the present simple case.

4.8. THERMODYNAMIC AND MECHANICAL EQUILIBRIUM

A system is said to be in mechanical equilibrium if in any inertial system the acceleration vanishes. For local equilibrium, $\dot{v} = 0$ and Cauchy's equations reduce to

$$t_{kl,k} + \rho f_l = 0, \qquad t_{kl} = t_{lk} \tag{4.8.1}$$

A system will be called in global (local) *thermal equilibrium*[12] if the global (local) entropy production is zero, that is,

$$\Gamma = 0 \qquad \text{(global equilibrium)} \tag{4.8.2}$$

$$\gamma = 0 \qquad \text{(local equilibrium)} \tag{4.8.3}$$

This condition of local equilibrium (4.8.3) must be valid for all independent processes. Thus vanishing γ implies that the dissipation energies of independent processes vanish separately. Consider the following special cases.

(*i*) ***Elastic Solid.*** For the elastic solid treated in the previous section, we found that the conditions of thermodynamic admissibility are

$$\theta = \frac{\partial \varepsilon}{\partial \eta}, \qquad t_{kl} = \rho \frac{\partial \varepsilon}{\partial x_{(k,K}} x_{l),K}, \qquad \rho \frac{\partial \varepsilon}{\partial x_{[k,K}} x_{l],K} = 0 \tag{4.8.4}$$

$$q_k = 0 \tag{4.8.5}$$

In this case through (4.7.3) we see that $\gamma = 0$. Hence

THEOREM 1. *Thermodynamically admissible elastic solids characterized by constitutive equations of the form* (4.7.1) *are in thermal equilibrium. For such solids, the temperature and the stress are derivable from a potential* (*the internal energy function*).

[12] It should be noted that the terms global and local equilibrium are not identical to the corresponding nomenclature used in statistical mechanics where these terms refer to an absolute and local Maxwellian distribution function.

(*ii*) *Viscous Fluids.* The viscous fluids characterized by (4.7.11) will be in local thermal equilibrium if and only if

$$t_{kl} = -\pi\delta_{kl} = \frac{\partial \varepsilon}{\partial \rho^{-1}}\,\delta_{kl} \tag{4.8.6}$$

for only under this condition is $\gamma = 0$ for all $v_{1,k}$. Thus

THEOREM 2. *Thermodynamically admissible viscous fluids characterized by* (4.7.11) *are in thermal equilibrium if and only if the stress is a hydrostatic pressure. For such fluids the temperature is derivable from a potential* (eq. 4.7.13)$_1$.

For systems in mechanical and thermal equilibrium, a principle of virtual work exists:

Principle of Virtual Work. In the case of mechanical and thermodynamic equilibria, it is possible to construct a *principle of virtual work* from which Cauchy's first law of motion may be derived as a special case of a general criterion of equilibrium. To this end, consider a family of deformations $\mathbf{x} = \mathbf{x}(\mathbf{X}, \lambda)$, where λ is a parameter. Assume that for $\lambda = 0$ the deformation $\mathbf{x} = \mathbf{x}(\mathbf{X}, 0)$ is given. The first *variation*, $\delta\mathbf{x}$, of $\mathbf{x}$ is defined as

$$\delta\mathbf{x} \equiv \left.\frac{\partial \mathbf{x}}{\partial \lambda}\right|_{\lambda=0,\mathbf{X}} d\lambda \tag{4.8.7}$$

For a function $f(\mathbf{x}, \mathbf{x}_{,K}, \ldots)$ the first variation of f is defined by

$$\delta f \equiv \left.\frac{\partial f}{\partial \lambda}\right|_{\lambda=0,\mathbf{X}} d\lambda \tag{4.8.8}$$

The variation symbol δ and partial derivative $\partial/\partial X_K$ commute, that is,

$$\delta(f_{,K}) = (\delta f)_{,K} \tag{4.8.9}$$

From the above definitions it is clear that the operator $\delta/d\lambda$ is similar to material differentiation with λ replacing the time t.

We now postulate the general criterion of equilibrium as the union of the principle of virtual work and thermodynamic equilibrium

$$\delta \int_{\mathscr{V}} \rho\varepsilon \, dv = \oint_{\mathscr{S}} t_{(n)k}\delta x_k \, da + \int_{\mathscr{V}} \rho f_k \delta x_k \, dv \tag{4.8.10}$$

$$\delta \int_{\mathscr{V}} \rho\eta \, dv = 0 \tag{4.8.11}$$

subject to the condition that all variations satisfy the principle of conservation of mass

$$\delta(\rho \, dv) = 0 \tag{4.8.12}$$

The left side of (4.8.10) is the variation of total internal energy and the right is the *virtual work* of the surface and body forces, respectively. Equation (4.8.11) is the variation of the total entropy. We have $\varepsilon = \varepsilon(\eta, v_\alpha)$ so that (4.8.10) and (4.8.11) by use of (4.5.7) and (4.8.12) may be written as

$$\int_{\mathscr{V}} (\rho\bar{\theta}\delta\eta + \rho\tau_\alpha\delta v_\alpha)\, dv = \oint_{\mathscr{S}} t_{(n)k}\delta x_k\, da + \int_{\mathscr{V}} \rho f_k \delta x_k\, dv$$

(4.8.13)

$$\int_{\mathscr{V}} \rho\delta\eta\, dv = 0$$

The second condition gives $\eta = \text{const.}$ In this case, or in the case of $\bar{\theta} = \text{const.}$, the first of (4.8.13) reduces to

$$\int_{\mathscr{V}} \rho\tau_\alpha\delta v_\alpha\, dv = \oint_{\mathscr{S}} t_{(n)k}\delta x_k\, da + \int_{\mathscr{V}} \rho f_k\delta x_k\, dv \tag{4.8.14}$$

When δv_α and δx_k are treated as independent variables, this gives

$$\tau_\alpha = 0, \qquad t_{(n)k} = 0, \qquad f_k = 0 \tag{4.8.15}$$

If, however, we select v_α as $x_{k,K}$ then τ_α becomes $\partial\varepsilon/\partial x_{k,K}$ so that (4.8.14) reads

$$\int_{\mathscr{V}} \rho \frac{\partial\varepsilon}{\partial x_{k,K}}\delta x_{k,K}\, dv = \oint_{\mathscr{S}} t_{(n)k}\delta x_k\, dv + \int_{\mathscr{V}} \rho f_k\delta x_k\, dv$$

By a method similar to the one used in deriving (2.4.1) we can show that

$$\delta(x_{k,K}) = (\delta x_k)_{,l} x_{l,K}$$

We write on the left of the foregoing expression

$$\rho \frac{\partial\varepsilon}{\partial x_{k,K}}\delta x_{k,K} = \left(\rho \frac{\partial\varepsilon}{\partial x_{k,K}} x_{l,K}\delta x_k\right)_{,l} - \left(\rho \frac{\partial\varepsilon}{\partial x_{k,K}} x_{l,K}\right)_{,l}\delta x_k$$

By use of the Green-Gauss theorem we therefore get

$$\oint_{\mathscr{S}} \left(t_{(n)k} - \rho \frac{\partial\varepsilon}{\partial x_{k,K}} x_{l,K} n_l\right)\delta x_k\, da + \int_{\mathscr{V}} \left[\left(\rho \frac{\partial\varepsilon}{\partial x_{k,K}} x_{l,K}\right)_{,l} + \rho f_k\right]\delta x_k\, dv = 0$$

If there are no kinematical constraints, the local condition of equilibrium follows from this

$$\left(\rho \frac{\partial\varepsilon}{\partial x_{k,K}} x_{l,K}\right)_{,l} + \rho f_k = 0 \qquad \text{in } \mathscr{V} \tag{4.8.16}$$

$$\left(\rho \frac{\partial\varepsilon}{\partial x_{k,K}} x_{l,K}\right) n_l = t_{(n)k} \qquad \text{on } \mathscr{S} \tag{4.8.17}$$

Consistent with the principles of mechanics, we are now motivated to set

$$t_{lk} = \rho \frac{\partial \varepsilon}{\partial x_{k,K}} x_{l,K} \tag{4.8.18}$$

With this the equilibrium stresses become the same as the thermodynamic stresses (cf 4.7 5) and the equilibrium equations are

$$\begin{aligned} t_{lk,l} + \rho f_k &= 0 \qquad \text{in } \mathscr{V} \\ t_{lk} n_l &= t_{(n)k} \qquad \text{on } \mathscr{S} \end{aligned} \tag{4 8 19}$$

While the method of virtual work leads to Cauchy's first law of motion and the boundary conditions for equilibrium, Cauchy's second law (symmetry of the stress tensor in the present case) cannot be obtained by the principle as introduced above. To obtain this law, we need an additional postulate; *the internal energy density* ε *is invariant under rigid rotations of the deformed body.*

To prove this theorem we suppose ε is a function of vectors $\mathbf{x}_{,K}$ and η, that is, $\varepsilon = \varepsilon(\eta, \mathbf{x}_{,K})$ which is invariant under rotation; mathematically

$$\varepsilon(\eta, \mathbf{x}_{,K}) = \varepsilon(\eta', \mathbf{x}'_{,K})$$

where η' and $\mathbf{x}'_{,K}$ are the values of η and $\mathbf{x}_{,K}$ in the new coordinates $\mathbf{x}' = \mathbf{Q}\mathbf{x}$ with $\mathbf{Q}$ *satisfying*

$$\mathbf{Q}\mathbf{Q}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I} \quad \text{and} \quad \det \mathbf{Q} = 1$$

Under the rotation of the frame of reference we have $\eta = \eta'$ and $\mathbf{x}'_{,K} = \mathbf{Q}\mathbf{x}_{,K}$ so that

$$\varepsilon(\eta, \mathbf{x}_{,K}) = \varepsilon(\eta, \mathbf{Q}\mathbf{x}_{,K}) = \varepsilon(\eta, \mathbf{x}'_{,K}) \tag{4 8.20}$$

An infinitesimal rotation has the form $Q_{kl} = \tilde{r}_{kl} + \delta_{kl} = \varepsilon_{klm}\tilde{r}_m + \delta_{kl}$ where $\tilde{r}_{kl}$ is an arbitrary antisymmetric tensor. Thus from (4 8.20) we must have

$$\left[\frac{\partial \varepsilon}{\partial x'_{k,K}} \frac{\partial x'_{k,K}}{\partial Q_{mn}} \frac{\delta Q_{mn}}{\delta \tilde{r}_l}\right]_{\mathbf{Q}=\mathbf{I}} = 0$$

For an infinitesimal rotation this reads

$$\frac{\partial \varepsilon}{\partial x_{k,K}} x_{l,K} \varepsilon_{klm} = 0$$

This implies that

$$\frac{\partial \varepsilon}{\partial x_{[k,K}} x_{l],K} = 0 \tag{4.8.21}$$

It can be shown that this condition is also sufficient to insure that ε is invariant under *finite* rotation.[13]

Since

$$ {}_E t_{kI} = \rho \frac{\partial \varepsilon}{\partial x_{k,K}} x_{I,K} \tag{4.8.22} $$

we see that

$$ {}_E t_{kI} = {}_E t_{Ik} \tag{4.8.23} $$

which is the proof of the theorem.

In spite of all these derivations, the thermostatic and variational approaches exclude the dynamical effects. Thermostatic stresses are non-dissipative. While through some variational principles and the Euler-D'Alembert principle we may bring in the inertia forces, the inclusion of viscous stresses cannot be made without additional fundamental assumptions.

PROBLEMS

4.1. During a thermomechanical process, mass density is produced at the rate of $\hat{\rho}$ per second. Find the local conservation of energy. What form would this equation take if, instead, there were a momentum creation?

4.2. Show that if the potential energy $\tau(x_{k,K})$ is invariant under rigid rotations of the spatial frame of reference, it must be a function of the Lagrangian strain measure E_{KL}.

4.3. Find the equation of the jump of energy across a discontinuity plane (a shock wave), moving with the velocity U in the x-direction when all physical quantities are functions of x only.

4.4. For certain gases, the equation of state is described by $\varepsilon = \varepsilon(\theta, \rho^{-1})$ where ε, θ, ρ are respectively the internal energy density, temperature, and mass density. A gas is said to be ideal if $\left.\dfrac{\delta \varepsilon}{\delta \rho}\right|_{\theta} = 0$ and $\pi = R\rho\theta$, where R is a constant.

(*a*) For an ideal gas, show that $c_p - c_v = R$ when c_p and c_v are respectively specific heats at constant pressure and volume.

(*b*) If an ideal gas is expanding without any heat loss (adiabatic process), show that $\pi\rho^{-\gamma} = \text{constant}$, where $\gamma = c_p/c_v$ is a constant.

[13] This follows from the theorem of Cauchy (App. B). Since according to (4.8.20) ε is a hemitropic function of the vectors $x_{,K}$, it follows that ε must be a function of the inner products of these vectors two at a time (i.e., C_{KL}) and their scalar triple product (det C_{KL}). Hence, (4.8.21).

4.5. A gas having a constitutive equation of the form

$$\pi = \frac{R\theta}{\rho^{-1} - b} - \frac{a}{\rho^{-2}}$$

(where R, a, and b are constants) is known as the Van der Waals gas.

(*a*) Show that

$$\varepsilon = \int c_v(\theta)\, d\theta - \frac{a}{\rho^{-1}}$$

(*b*) Express ρ as a function of θ only;

(*c*) Express π as a function of θ only,

(*d*) Calculate the work for an adiabatic expansion

4.6. Calculate the entropy η as a function of θ and ρ for:

(*a*) An ideal gas;

(*b*) A Van der Waals gas

(For the definition of these gases, see problems (*4.4*) and (*4.5*) above.)

4.7. For the Van der Waals gas, undergoing an adiabatic expansion, calculate:

(*a*) The entropy as a function of temperature;

(*b*) The entropy as a function of density.

4.8. For an ideal gas, the equation of state is given by

$$\varepsilon = K\rho^{\gamma-1} \exp(\eta/c_v)$$

where K and c_v are constants Determine the pressure π and temperature θ.

4.9. For a Van der Waals gas, the equation of state is given, parametrically, by

$$\varepsilon = \int c(\lambda)\, d\lambda - a\rho$$

$$\eta = \int \frac{c(\lambda)\, d\lambda}{\lambda} - R \ln \rho$$

where a and R are constants Determine the pressure π and the temperature θ.

4.10. In a body, stress, heat, energy and entropy are assumed to depend on ρ^{-1} and $v_{k,K}$ Find the ultimate expressions of the constitutive equations Show that the free energy cannot depend on $v_{k,K}$

4.11. For a body in adiabatic process, the stress, heat energy, and entropy are known to depend on displacement gradients $x_{k,K}$, and their material time rates show that the free energy is independent of the rates of the displacement gradients. If the stress is linear in these rates, what conditions must be placed on the constitutive coefficients in order that the second law of thermodynamics is not violated for all changes?

4.12. In a body, stress, heat, energy, and entropy are known to depend on the density, ρ, and the temperature, θ, and the temperature gradient $\theta_{,k}$. If the Clausius-Duhem inequality is valid for all independent processes, what must be the form of the constitutive equations? If the above constitutive dependent variables are polynominal in $\theta_{,k}$, find the ultimate forms of the constitutive equations.

4.13. (Short Term Paper.) Make a literature search and write an exposition on the thermodynamics of porous solids.

4.14. (Short Term Paper.) Read literature and write an exposition on the thermodynamics of two fluids in binary reactions.

4.15. (Short Term Paper.) Write a critique on the linear irreversible thermodynamics and the so-called Onsager relations.

4.16. (Short Term Paper.) Write a short account of the concept of entropy based on statistical mechanical considerations.

CHAPTER 5

Constitutive Equations

5.1. SCOPE OF THE CHAPTER

The basic laws of motion, conservation of mass, balance of momenta, conservation of energy, and principle of entropy, are valid for all types of media irrespective of their internal constitutions. Material bodies having the same mass and geometry, when subjected to identical external effects, respond differently. For example, most solids under small external pressure deform slightly, whereas fluids flow and take the shape of their containers. Internal constitution of matter is responsible for these different responses. In continuum mechanics we are not concerned with the atomic structure of matter, rather, we are interested in the global behavior of matter. To this end, therefore, we need equations that reflect the gross effects in the structural differences. Since instruments measuring global phenomena automatically read the statistical averages, the continuum theory is meaningful physically as an independent discipline.

While the theory of constitutive equations goes back to the very beginning of our science, its axiomatic organization is only beginning to take shape. The universality of the axioms, verification of the old results, and the new meaningful material theories that they lead to are sufficient for a serious study of the subject matter. The foundation laid is by no means final; we believe, however, the organization presented in this chapter is solid and has promise. In any case, for the theories developed in this book the scope is more than adequate.

In Section 5.2 we discuss the need for a basic philosophy underlying the theory of constitutive equations. The axioms of constitutive theory are

presented in Section 5.3. While the eight axioms enumerated are explained with reference to thermomechanical continua, they can be adapted to media possessing electrical and chemical properties. In Section 5.4 we present in some detail a constitutive theory for a very general class of thermomechanical media. Simple and nonsimple materials having hereditary characteristics are discussed. In Sections 5.5 and 5.6 we discuss non-heat-conducting elastic materials and in Section 5.7, the Stokesian fluids. The thermal effects and heat conduction phenomena in fluids are presented in Section 5.8. The remaining two sections, 5.9 and 5.10, are devoted to the subject of thermoelasticity. The nonlinear theory is presented on par with the other sections of this chapter.

By these simple applications, the meaning and the power of various constitutive axioms are revealed. The contents of these sections are essential for the material contained in Chapters 6 through 8. We have separated Chapter 9 of this book for a more detailed discussion of viscoelastic continua in order that the beginner will not be subjected to a heavy dose before he can master the simpler situations.

5.2. THE NEED FOR CONSTITUTIVE EQUATIONS

Basic principles of continuum mechanics, namely, conservation of mass, balance of momenta, and conservation of energy, discussed in Chapters 2, 3 and 4, lead to the fundamental equations:

$$\frac{\partial \rho}{\partial t} + (\rho v_k)_{,k} = 0 \tag{5.2.1}$$

$$t_{lk,l} + \rho(f_k - \dot{v}_k) = 0 \tag{5.2.2}$$

$$t_{kl} = t_{lk} \tag{5.2.3}$$

$$\rho\dot{\varepsilon} = t_{kl}v_{l,k} + q_{k,k} + \rho h \tag{5.2.4}$$

We have a total of eight independent equations. The number of unknowns, ρ, v_k, t_{kl}, q_k and ε, are seventeen provided that f_k and h are prescribed as intended. Clearly, the foregoing basic equations are not adequate for the determination of these unknowns except for some trivial situations, for example, rigid body motions in the absence of heat conduction. The axiom of entropy

$$\rho\gamma \equiv \rho\left(\dot{\eta} - \frac{\dot{\varepsilon}}{\theta}\right) + \frac{1}{\theta}t_{kl}v_{l,k} + \frac{1}{\theta}q_k(\log\theta)_{,k} \geq 0 \tag{5.2.5}$$

introduces the additional unknowns η and θ, which does not help to reduce the number of additional equations needed for the problem. Eleven

additional equations must be supplied to make the problem determinate. When electrical and chemical variables are brought into play, the situation becomes much worse and the number of additional equations needed becomes larger.

In the derivation of the equations (5.2.1) to (5.2.5) no differentiation has been made between various types of substances. It is therefore not surprising that the foregoing equations are not sufficient to explain fully the motions of solids, fluids, viscoelastic materials, and materials having *various types of physical properties. The character of the material is brought into the formulation through appropriate constitutive equations for each material with the constitutive variables being restricted in their regions of definitions.*

A constitutive equation defines an ideal material. In the selection of constitutive variables and in the formulation of constitutive equations, certain physical and mathematical considerations will have to be satisfied. A discussion of such requirements is given in Section 5.3. At present there is no definite rule which includes all variables that must be present for a given material. No advance can be made in the description of materials without *a priori* knowledge (which may be crude) of the dependence of material behavior upon external fields and the range of phenomena to be covered. For example, if we are interested in solids which, upon the relief of applied loads, return to the original unstressed configuration, then we are dealing with elastic solids. However, from everyday observation we also know that all solids undergo a permanent deformation when the applied loads exceed certain limits. Thus the range of the elastic theory is confined by such limits of loads, or more specifically, the limits of the local state of stress. Elasticity theory is only capable of predicting the behavior of a given material in a given *range* of the state of stress. The behavior of the same material outside of this range will have to be explained by a new set of constitutive equations, for example, constitutive equations of plasticity. Similarly, such solids may also exhibit electrical phenomena upon deformation, for example, piezoelectricity. To incorporate such phenomena, we need to include other constitutive variables in our equations. For viscoelastic behavior of solids still other sets of new variables must be taken into account. Thus, a *constitutive theory is intended only to describe a limited number of physical phenomena decided at the outset for a given material.*

Since the turn of the century some writers (Jaumann [1911], Lohr [1917], Green and Rivlin [1957], Noll [1958], Jordan and Eringen [1964], Eringen [1965], and others) have given theories encompassing a large number of physical phenomena. While the importance of these theories in providing a source for approximate theories and in the explanation of some esoteric effects is appreciated from the point of view of engineering applications and

mathematical treatment of boundary and initial value problems, they often present insurmountable mathematical difficulties. We shall, therefore, apply the principles of material constitution to simpler materials for illustrative purposes and to demonstrate how they can be generalized to explain more complicated material behavior.

A constitutive theory may be built by use of the methods of *statistical mechanics*. In this approach a material is considered to be made up of atoms and molecules attracted to each other by intermolecular forces. The nature of intermolecular forces is not well known and the motions of atoms are intricate, thus making such an approach feasible only for very simple and ideal situations, for example, ideal gases. In the selection of constitutive variables, the statistical mechanical approach may provide intuitive guidance. In this regard thermodynamics and other branches of physics will also be helpful.

At present there exists no systematic rule that will point to the basic constitutive variables needed. However, certain guiding principles exist which may be used to exclude certain variables from such equations. These are discussed in Section 5.3.

5.3. AXIOMS OF CONSTITUTIVE THEORY[1]

For a constitutive theory to represent a material adequately, certain physical and mathematical requirements must be satisfied. The following axioms are basic in the formulation of the constitutive equations:

(i) Axiom of Causality
(ii) Axiom of Determinism
(iii) Axiom of Equipresence
(iv) Axiom of Objectivity
(v) Axiom of Material Invariance
(vi) Axiom of Neighborhood
(vii) Axiom of Memory
(viii) Axiom of Admissibility

(i) AXIOM OF CAUSALITY. *We consider the motion of the material points of a body and their temperatures as self-evident observable effects in every thermomechanical behavior of the body. The remaining quantities (other than those derivable from the temperature and motion) that enter the expression*

[1] Ideas underlying some of these axioms were evolved in the course of the development of continuum mechanics. Organization of these axioms together with various extensions were given by Eringen [1962, Ch. 5] and [1965]. The present chapter leans heavily on the latter work.

of entropy production[2] *are the "causes" or the dependent constitutive variables.*

The axiom of causality is intended for the selection of the independent constitutive variables of a material subject to limited changes. In a material that is subjected to some external forces, certain observable changes occur. For example, the material points of the body undergo deformations, and thermal, electromagnetic, and chemical changes occur. At the outset, one can select an independent set of these observable changes as the effects produced in the body. This set then constitutes the independent constitutive variables for a limited class of physical and chemical phenomena observable in a material. Once the independent constitutive variables are selected, the remaining quantities, in the basic equations (5.2.1) to (5.2.5), which are not derivable from the constitutive variables, make up the list of the dependent constitutive variables. Since in this book we are not concerned with electromagnetic and chemical phenomena, we state this axiom for the thermomechanical phenomena only. The limited form of this axiom can be generalized to include more complicated phenomena.[3]

Our ideas of natural phenomena are the result of some simple observations or experiences we have had in the past. Without this experience and observation it is not possible to construct a model of nature. With the help of this intuitive guidance, the axiom of causality is adequate for a decision (not necessarily unique) on the set of independent constitutive variables. However, once the effects are decided, the causes are readily obtained from the expression of entropy production or the balance laws. For example, in the treatment of thermomechanical phenomena according to this axiom, the independent constitutive variables are

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t), \qquad \theta = \theta(\mathbf{X}, t) \tag{5.3.1}$$

Now the velocity $\mathbf{v} \equiv \dot{\mathbf{x}}$ is derivable from the motion and $v_{I,k}$ from $\mathbf{v}$. The density is related to the motion through the continuity equation $\rho_0/\rho = \det(x_{k,K})$. Therefore, in the expression of the entropy production (5.2.5), the only other functions that remain to be prescribed are the stress tensor $\mathbf{t}$, the heat vector $\mathbf{q}$, the internal energy density ε, and the entropy

[2] "The expression of entropy production" may be replaced by the balance equations (equations of conservation of mass, balance of momenta, and conservation of energy) and entropy. In this case, external effects such as body force and heat source are, of course, excluded.

[3] Electromagnetic phenomena may be treated by including the electric and magnetic fields as observables Cf. Jordan and Eringen [1964a, b], Dixon and Eringen [1965a, b] Inclusion of chemical reactions requires further sophistication, cf Eringen and Ingram [1965]

density η. These then constitute the dependent constitutive variables and they are to be expressed in terms of $\mathrm{x}(\mathrm{X}, t)$ and $\theta(\mathrm{X}, t)$.

(ii) AXIOM OF DETERMINISM. *The value of the thermomechanical constitutive functions,* $(\mathrm{t}, \mathrm{q}, \varepsilon, \eta)$, *at a material point* X *of the body* $\mathscr{B}$ *at time* t *is determined by the history of the motion and temperature of all material doints of* $\mathscr{B}$.

The axiom of determinism is a principle of exclusion. It excludes the dependence of the material behavior at X on any point outside of the body and any future events. Accordingly, the future phenomena concerning the behavior of the body are fully determined and observable once all past motions of the body are known. The quantum mechanical phenomena are excluded. Following this axiom, therefore, the thermomechanical constitutive equations may be written as

$$\mathrm{t}(\mathrm{X}, t) = \mathscr{F}[\mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t] \tag{5.3.2}$$

$$\mathrm{q}(\mathrm{X}, t) = \mathscr{G}[\mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t] \tag{5.3.3}$$

$$\varepsilon(\mathrm{X}, t) = \mathscr{E}[\mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t] \tag{5.3.4}$$

$$\eta(\mathrm{X}, t) = \mathscr{N}[\mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t] \tag{5.3.5}$$

where $\mathscr{F}$ and $\mathscr{G}$ are respectively tensor valued and vector valued functionals and $\mathscr{E}$ and $\mathscr{N}$ are scalar-valued functionals defined over the field of real functions $\mathrm{x}(\mathrm{X}', t')$ and $\theta(\mathrm{X}', t')$ with

$$\mathrm{X}' \in \mathscr{B}, \qquad t' \leq t \tag{5.3.6}$$

and they are also functions of X and t. In the subsequent work, we shall place additional restrictions on these functionals and on the functions over which they are defined.

In order to provide a feeling about functionals, suppose that the body consists of only M discrete material points $\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_M$ and X being any one of these points. In this case, (5.3.2) would be replaced by

$$\mathrm{t}(\mathrm{X}, t) = \mathrm{F}[\mathrm{x}_1(t'), \mathrm{x}_2(t'), \ldots, \mathrm{x}_M(t'), \theta_1(t'), \theta_2(t'), \ldots, \theta_M(t'), \mathrm{X}, t] \tag{5.3.7}$$

where

$$\mathrm{x}_\alpha(t') \equiv \mathrm{x}(\mathrm{X}_\alpha, t'), \qquad \theta_\alpha(t') \equiv \theta(\mathrm{X}_\alpha, t'), \qquad \alpha = 1, 2, \ldots, M \tag{5.3.8}$$

Now F is a functional of the functions $\mathrm{x}_\alpha(t')$ and $\theta_\alpha(t')$ defined on the field of real functions of $t' \leq t$. Finally, if functions $\mathrm{x}_\alpha(t')$ and $\theta_\alpha(t')$ take only N discrete values $t_1, t_2, \ldots, t_N$, then we may replace the functional F by a function of MN vector variables $\mathrm{x}_{\alpha\beta}$, MN scalar variables $\theta_{\alpha\beta}$, X, and t, i.e.

$$\mathrm{t}(\mathrm{X}, t) = \mathrm{f}(\mathrm{x}_{\alpha\beta}, \theta_{\alpha\beta}, \mathrm{X}, t), \qquad \begin{matrix} \alpha = 1, 2, \ldots, M \\ \beta = 1, 2, \ldots, N \end{matrix} \tag{5.3.9}$$

Conversely, under certain conditioos of continuity and smoothness, when M and N approach infinity (the material poiots X_α and the time set t_β become dense), the function f approaches the functional $\mathscr{F}$.

A functional such as $\mathscr{F}$ is often expressed by indicating the extent of the interval in which the argument functions are defined Thus (5.3.2) may be written as

$$\text{(5.3.10)} \qquad \mathrm{t}(\mathrm{X}, t) = \overset{t'=t}{\underset{t'=-\infty}{\underset{\mathrm{X}\in\mathscr{B}}{\mathscr{F}}}} [\mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t]$$

This form is too cumbersome for printing and will not be used unless it is essential to the discussion

The constitutive equations (5 3 2) to (5 3 5) are not the most general type encountered in the characterization of the material properties. A more general and, in fact, a more natural constitutive theory begins with a set of "independent" functional equations of the form

$$\text{(5 3.11)} \qquad \mathscr{G}_\alpha[\mathrm{t}(\mathrm{X}', t'), \mathrm{q}(\mathrm{X}', t'), \varepsilon(\mathrm{X}', t'), \eta(\mathrm{X}', t'), \mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t] = 0$$

The number of such equations and conditions that they must obey are presently unknown. However, they must be of such a nature as to determine t, q, ε and η at (X, t) For example, for a nonpolar medium, we need eleven such equations ($\alpha = 1, 2, \ldots, 11$) so that the eleven quantities $t_{ki}(\mathrm{X}, t)$, $q_k(\mathrm{X}, t)$, $\varepsilon(\mathrm{X}, t)$, and $\eta(\mathrm{X}, t)$ may be soluble Thus equations of the form (5.3.2) to (5.3.5) result as a special case of (5.3 11) In a theory that makes use of (5.3.11) the history of stress, heat, internal energy, and entropy play an essential role. Accordingly, the stress at (X, t) depends on the histories of the stress, heat, internal energy, and entropy as well as on the motion and temperature at all other points of the body. Approximations of (5 3 11) exist in which the dependence on t(X', t') may be replaced by the history of various order stress rates, heat rates, etc. For example, the special case

$$\dot{\mathrm{t}}(\mathrm{X}, t) = \mathscr{F}[\mathrm{t}(\mathrm{X}, t), \mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t]$$

where $\dot{\mathrm{t}}$ is the stress rate at (X, t) More generally, we may have

$$\text{(5.3.12)} \qquad \mathrm{t}^{(p)}(\mathrm{X}, t) = \mathscr{F}[\mathrm{t}^{(p-1)}(\mathrm{X}, t), \mathrm{t}^{(p-2)}(\mathrm{X}, t), \ldots, \mathrm{t}(\mathrm{X}, t), \mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t]$$

which involves the stress rates up to the pth order at (X, t).

Still other theories of *nonsimple* media having constitutive equations of the form

$$\text{(5.3 13)} \qquad \mathrm{t}_{,\kappa_1\kappa_2\ldots\kappa_R} = \mathscr{F}[\mathrm{t}_{,\kappa_1\kappa_2\ldots\kappa_{R-1}}(t'), \ldots, \mathrm{t}_{,\kappa_1}(t'), \mathrm{t}(t'); \mathrm{x}(\mathrm{X}', t'), \theta(\mathrm{X}', t'), \mathrm{X}, t]$$

emanate as a natural consequence of (5.3.11).

Constitutive equation (5 3.11) may not be soluble for the stress or heat, etc, or it may give rise to multiple-valued stress, stress rates, heat, etc. Consquently theories of plasticity, viscoplasticity, and a large class of esoteric material behavior are included in such forms We restrain to dwell on such general theories

which at present are not fully developed. For an account of the subject see Eringen [1962, Ch. X].

(iii) AXIOM OF EQUIPRESENCE. *At the outset all constitutive functionals should be expressed in terms of the same list of independent constitutive variables until the contrary is deduced.*

This axiom is a precautionary measure. It helps us not to forget or be prejudiced against a certain class of variables and favor others in the expression of constitutive functionals. The axiom of admissibility and various approximations may eliminate the dependence on some of these variables. Until such is shown to be the case, we agree to employ the same list of independent constitutive variables in all constitutive functions. Thus, we see the presence of $\mathbf{x}(\mathbf{X}', t')$, $\theta(\mathbf{X}', t')$, $\mathbf{X}$ and t in all constitutive equations (5.3.2) to (5.3.5).

(iv) AXIOM OF OBJECTIVITY. *The constitutive equations must be form-invariant with respect to rigid motions of the spatial frame of reference.*

It is evident, from physical considerations, that the material properties cannot depend on the motion of the observer. If two frames $\bar{\mathbf{x}}$ and $\mathbf{x}$ can be made to coincide with a rigid motion and a time shift, then they must be related to each other by

$$\bar{\mathbf{x}}(\mathbf{X}, \bar{t}) = \mathbf{Q}(t)\mathbf{x}(\mathbf{X}, t) + \mathbf{b}(t) \tag{5.3.14}$$

where $\mathbf{Q}(t)$ is a proper orthogonal transformation, $\mathbf{b}(t)$ is a translation, and $\bar{t}$ is obtained from t by a constant shift of the origin of time, that is,

$$\mathbf{Q}\mathbf{Q}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}, \qquad \det \mathbf{Q} = +1 \tag{5.3.15}$$

$$\bar{t} = t - a$$

where $\mathbf{I}$ is the unit tensor. According to the axiom of objectivity, the constitutive functionals must be form-invariant when we use $\bar{\mathbf{x}}$ in place of $\mathbf{x}$ for all proper orthogonal transformations $\{\mathbf{Q}(t)\}$, translations $\{\mathbf{b}(t)\}$ of the spatial frame of reference and all shifts $\{a\}$ of the origin of time. This implies that the response functionals are *spatially hemitropic* in their tensor variables. If we include the reflection (for which $\det \mathbf{Q} = -1$) to the group $\{\mathbf{Q}(t)\}$, then we obtain *spatial isotropy*. In this case $\{\mathbf{Q}(t)\}$ is the full orthogonal group of transformations.[4] Spatially hemitropic materials

[4] The question whether we must employ the full orthogonal group or the proper orthogonal group in the formulation of the principle of objectivity is a question unsettled to this day. The use of the full group may be questioned on several grounds. For optically active fluids there exist preferred directions. It appears that the use of the full orthogonal group for the spatial objectivity may eliminate essential properties of these materials. Further support for the use of proper group is given by the recent rejection of parity in quantum mechanics.

are more general than the spatially isotropic materials, and the latter class is contained in the former. The additional requirement of form-invariance of a response function to reflection forces the hemitropic materials to be isotropic. We shall, therefore, use the proper orthogonal group as the basis of the principle of objectivity. However, we shall investigate the consequence of invariance under the reflection group, since it leads to considerable simplifications of the constitutive equations.

(v) AXIOM OF MATERIAL INVARIANCE. Crystallographic orientations of the material points in a body give rise to certain symmetries in the material properties. For example, the constitutive functionals may not change their forms when the material coordinates (X_1, X_2, X_3) are changed to $(X_1, X_2, -X_3)$. This represents a reflection of the material frame of reference with respect to the plane $X_3 = 0$ This condition, in turn, imposes restrictions on the constitutive equatioos. Let {S} be a subgroup of the full group of orthogonal transformations for the material axes and {B} be the translations of these axes. The axiom of material invariance may then be stated as:

AXIOM OF MATERIAL INVARIANCE. *Constitutive equations must be form-invariant with respect to a group of orthogonal transformations* {S} *and translations* {B} *of the material coordinates. These restrictions are the result of symmetry conditions implied by* {S} *and* {B} *in the material frame of reference* X.

Accordingly, for such materials the response functions will be form-invariant under all transformations of the form

$$\bar{\mathbf{X}} = \mathbf{S}\mathbf{X} + \mathbf{B} \tag{5.3.16}$$

where

$$\mathbf{S}\mathbf{S}^T = \mathbf{S}^T\mathbf{S} = \mathbf{I}, \qquad \det \mathbf{S} = \pm 1 \tag{5.3.17}$$

for all members of the group {S} and all translations {B} These conditions express geometrical symmetries, represented by {S} and inhomogeneities represented by {B}, at X, in the physical properties of the body. The symmetry group {S} may be a subgroup of the full orthogonal group. When {S} is the *proper orthogonal group* we say that the material is *hemitropic* If {S} is the full group, then the material is called *isotropic* A material that is not hemitropic is called *anisotropic*. When the response functions do not depend on the translations {B} of the origin of material coordinates we say that the material is *homogeneous*. When the response function changes with some translations {B} of the material axes then the material is called *inhomogeneous*.

A material may possess different types of material symmetries for its different properties. For example, an isotropic material with respect to stress and strain may not be isotropic with respect to another property, for example, electric displacement and polarization. Similarly, elastically homogeneous materials may be inhomogeneous electrically.

Materials may also possess curvilinear symmetries. In this case, the situation is similar to (5.3.16) except that X may be a curvilinear coordinate.

Materials may also be subject to other geometrical and internal constraints. For example, rubbery materials are incompressible. This is expressed by the additional condition that the local volume remains unchanged, that is, $\mathrm{III}_C = 1$. More generally, it is possible that the constitutive variables of a certain class of materials may be restricted by some internal constraints. For example, the material may contain inextensible strings or sheets. These are expressed by writing additional equations of the form

$$f_\alpha(\mathbf{E}) = 0, \qquad (\alpha = 1, 2, \ldots, n, \quad n < 6) \tag{5.3.18}$$

where E is a strain measure. These conditions on the strains restrict the class of possible deformations of the material. Since six such equations fully determine the strain field E, the number of such equations cannot exceed five. In place of (5.3.18) we may have constraints in the form of differential and integral equations. These internal constraints must be considered as part of the constitutive equations.

(vi) Axiom of Neighborhood. *The values of the independent constitutive variables at distant material points from* X *do not affect appreciably the values of the constitutive dependent variables at* X.

This axiom states that the motions $\mathbf{x}(\mathbf{X}', t')$ and the temperatures $\theta(\mathbf{X}', t')$ of material points $\mathbf{X}'$ that are not near the point X will not contribute appreciably to t, q, ε and η at $(\mathbf{X}, t)$. In other words, the values of the constitutive functionals at X and t are insensitive to the motion and temperature histories of the distant points from X. Since thermomechanical phenomena are the result of the relative motion and thermal changes of the material points near X, it is natural to expect that the thermomechanical history of the distant points will have no appreciable effect on the behavior of material at X. There is no single way of formulating this principle. Below is given a formulation useful for our purpose.[5]

Smooth Neighborhood. Suppose that the functions $\mathbf{x}(\mathbf{X}', t')$ and $\theta(\mathbf{X}', t')$ admit Taylor expansions about $\mathbf{X}' = \mathbf{X}$ for all $t' \leq t$, terminating with gradients of order P and Q respectively, that is

[5] For still another formulation, see Eringen [1965].

(5 3.19)

$$\mathbf{x}(\mathbf{X}', t') = \mathbf{x}(\mathbf{X}, t') + (X'_{K_1} - X_{K_1})\mathbf{x}_{,K_1}(\mathbf{X}, t') + \frac{1}{2!}(X'_{K_1} - X_{K_1})(X'_{K_2} - X_{K_2})\mathbf{x}_{,K_1K_2}(\mathbf{X}, t') + \cdots + \frac{1}{P!}(X'_{K_1} - X_{K_1})\cdots(X'_{K_P} - X_{K_P})\mathbf{x}_{,K_1\cdots K_P}(\mathbf{X}, t')$$

$$\theta(\mathbf{X}', t') = \theta(\mathbf{X}, t') + (X'_{K_1} - X_{K_1})\theta_{,K_1} + \frac{1}{2!}(X'_{K_1} - X_{K_1})(X'_{K_2} - X_{K_2})\mathbf{x}_{,K_1K_2}(\mathbf{X}, t') + \cdots + \frac{1}{Q!}(X'_{K_1} - X_{K_1}) \quad (X'_{K_Q} - X_{K_Q})\mathbf{x}_{,K_1 \quad K_Q}(\mathbf{X}, t')$$

If the constitutive functionals are sufficiently smooth so that they can be approximated by the functionals in the field of real functions

(5.3.20)
$$\mathbf{x}(\mathbf{X}, t'), \mathbf{x}_{,K_1}(\mathbf{X}, t'), \ldots, \mathbf{x}_{,K_1K_2 \quad K_P}(\mathbf{X}, t')$$
$$\theta(\mathbf{X}, t'), \theta_{,K_1}(\mathbf{X}, t'), \ldots, \theta_{,K_1 \quad K_Q}(\mathbf{X}, t')$$

we say that the material at X for all $t' \leq t$ satisfies the smooth neighborhood hypothesis In this case, we see that the behavior of the material point X is not affected by the histories of the distance points from X. Materials of this type are called *nonsimple materials of material gradient type.* In some regard, the range of the neighborhood is brought into play with the highest order deformation and thermal gradients involved. In the following article, we shall show that the dependence on $\mathbf{x}(\mathbf{X}, t')$ can be eliminated from the list (5.3 20) by use of the axiom of objectivity. Further in this book we shall be studying mainly the materials involving the gradients up to and including the first order so that our constitutive equations will be of the form

(5 3.21)
$$t(\mathbf{X}, t) = \mathscr{F}[\mathbf{x}_{,K}(t'), \theta(t'), \theta_{,K}(t'), \mathbf{X}, t]$$

These materials are called the *simple materials.* Since $\mathscr{F}$ is no longer a functional in the continuous range of X, we omit writing X in the arguments of $\mathbf{x}_{,K}(t')$, $\theta(t')$, and $\theta_{,K}(t')$, that is, we put for simplicity

$$\mathbf{x}_{,K}(t') \equiv \mathbf{x}_{,K}(\mathbf{X}, t')$$

In passing, we remark that when the more general constitutive equations (5 3.1 t) are used, the above interpretation of the axiom of small neighborhood provides motivation for materials characterized by equations of the type

(5.3.22)
$$t_{,K_1K_2 \quad K_P}(\mathbf{X}, t) = \mathscr{F}[t_{,K_1K_2 \quad K_{P-1}}(\mathbf{X}, t'), \ldots, t(\mathbf{X}, t'), \mathbf{x}(\mathbf{X}', t'), \theta(\mathbf{X}', t'), \mathbf{X}, t]$$

Other constitutive axioms (for example, objectivity) place restrictions on the form of these equations. The literature contains no serious study of the materials characterized by constitutive equations of the form (5.3.22).

The axiom of neighborhood places certain smoothness conditions on the constitutive functions in their dependence on the histories of neighboring points to X. There is no unique mathematical formalization of this axiom. In general terms, it can be expressed as the *functional continuity* and *functional differentiability* conditions of the constitutive functionals with respect to their argument functions $\mathbf{x}(\mathbf{X}', t')$ and $\theta(\mathbf{X}', t')$ in the neighborhood of the thermomechanical history $\mathbf{x}(\mathbf{X}, t')$, $\theta(\mathbf{X}, t')$ of the material point X for all times $t' \leq t$.[6] Of course, what has been said above applies equally well to functionals $\mathscr{G}$, $\mathscr{E}$ and $\mathscr{N}$.

(vii) AXIOM OF MEMORY. *The values of the constitutive variables at distant past from the present do not affect appreciably the values of the constitutive functions.*

This principle is the counterpart of the principle of neighborhood in the time domain. Accordingly, the memory of past motions and temperatures of any material point decays rapidly.[7] No unique mathematical formulation can be made of this axiom. The following two limited interpretations suffice for our purposes.

(*a*) *Smooth Memory.* Suppose that there exists a time $\tau_0 < t$ appropriate to each material such that the thermomechanical histories $\mathbf{x}(\mathbf{X}', t')$ and $\theta(\mathbf{X}', t')$ possess Taylor series expansions at $t' = t$ for all $\mathbf{X}' \in \mathscr{B}$ terminating with rates of order p and q, respectively, that is,

$$\mathbf{x}(\mathbf{X}', t') = \mathbf{x}(\mathbf{X}', t) + (t' - t)\dot{\mathbf{x}}(\mathbf{X}', t) + \cdots + \frac{(t' - t)^p}{p!}\mathbf{x}^{(p)}(\mathbf{X}', t)$$

(5.3.23)

$$\theta(\mathbf{X}', t') = \theta(\mathbf{X}', t) + (t' - t)\dot{\theta}(\mathbf{X}', t) + \cdots + \frac{(t' - t)^q}{q!}\theta^{(q)}(\mathbf{X}', t)$$

$$\tau_0 < t' \leq t, \qquad \mathbf{X}' \in \mathscr{B}$$

[6] Functional continuity and differentiability require the definition of distance between two neighboring argument functions or the norm in a function space of the argument functions. This norm can be so defined as to reflect the importance of the points near X by introducing a weight function. The continuity of the functionals or their functional derivatives up to a certain order in the neighborhood of the thermomechanical history of the point X in this normed function space can be used as the mathematical statement of the axiom of neighborhood, cf. Eringen [1965]. See also axiom of memory.

[7] The idea was known to Boltzmann [1874] and Volterra [1913, 1930]. Straneo [1925] formally introduced the concept of fading heredity. Recently Coleman and Noll [1961] gave another formulation. This will be presented at the end of the discussion of smooth memory.

where

$$\mathbf{x}^{(p)} = \frac{\partial^p \mathbf{x}}{\partial t^p}\bigg|_{\mathbf{X}'}, \qquad \theta^{(q)} = \frac{\partial^q \theta}{\partial t^q}\bigg|_{\mathbf{X}'}$$

If the constitutive functionals are sufficiently smooth so that the dependence on $\mathbf{x}(\mathbf{X}', t')$ and $\theta(\mathbf{X}', t')$ can be replaced by the list of real functions

$$\begin{aligned} &\mathbf{x}(\mathbf{X}', t), \quad \dot{\mathbf{x}}(\mathbf{X}', t), \ldots, \quad \mathbf{x}^{(p)}(\mathbf{X}', t) \\ &\theta(\mathbf{X}', t), \quad \dot{\theta}(\mathbf{X}', t), \ \ldots, \quad \theta^{(q)}(\mathbf{X}', t) \end{aligned} \tag{5.3.24}$$

we say that they satisfy the axiom of smooth memory. In this case, the distant histories of motion from time t of any material point $\mathbf{X}'$ do not affect the material behavior at $\mathbf{X}$ at time t. The existence of the highest order time rates $\mathbf{x}^{(p)}(\mathbf{X}', t)$ and $\theta^{(q)}(\mathbf{X}', t)$, however, displays a kind of memory dependence in a limited sense. Such materials will have constitutive equations of the form

$$\mathbf{t}(\mathbf{X}, t) = \mathscr{F}[\mathbf{x}^{(\alpha)}(\mathbf{X}', t), \theta^{(\beta)}(\mathbf{X}', t), \mathbf{X}, t] \tag{5.3.25}$$

$$\alpha = 0, 1, 2, \ldots, p, \qquad \beta = 0, 1, 2, \ldots, q$$

and they are named *materials of the rate type*. If the functional equations of the form (5.3.11) are used, then the motivation is provided for the special materials characterized by the equations of the form (5.3.12).

(*b*) *Fading Memory.* The functions $\mathbf{x}(\mathbf{X}', t')$ and $\theta(\mathbf{X}', t')$ may not be so smooth as to admit Taylor expansions of the form (5.3.23). Nevertheless, the constitutive functionals may be such as to smooth out past discontinuities in these functions and/or their time derivatives. In this sense, the principle of memory is a requirement of smoothness for the response functionals. To formalize this concept, we need to introduce an *influence function* which favors the short time dependence of the functionals to far past. Let $\mathscr{I}_p(t - t') > 0$ be such a function defined over $-\infty < t' \leq t$ such that

$$\mathscr{I}_p(0) = 1, \qquad \lim_{t' \to -\infty} (t - t')^p \mathscr{I}_p(t - t') = 0 \tag{5.3.26}$$

We then say that $\mathscr{I}_p(t - t')$ is an influence function of an order less than p. For example

$$\mathscr{I}_p(t - t') = \frac{1}{[k(\mathbf{X}')(t - t') + 1]^p} \tag{5.3.27}$$

is such a function of order less than p. Similarly

$$\mathscr{I}_\infty(t - t') = e^{-\alpha(t - t')}, \qquad \alpha > 0 \tag{5.3.28}$$

is an influence function of arbitrary order. $\mathscr{I}_p(t-t')$ may be a function of X'. The *normed distance* between $x(X', t')$ and $x(X', t)$ or the $\mathscr{I}$-*norm* of $x(X', t') - x(X', t)$ may be defined as

(5.3.29)

$$\|x(X', t') - x(X', t)\|_{\mathscr{I}} = \left\{\int_{-\infty}^{t} \mathscr{I}(t-t')|x(X', t') - x(X', t)|^2 \, dt'\right\}^{1/2}$$

where $|\ |$ indicates the magnitude of the vector enclosed.

AN AXIOM OF FADING MEMORY can now be stated as: *The response functionals are continuous from below in a neighborhood of the thermomechanical history at time t in the function space normed according to* (5.3.29) *for all* X' *in* $\mathscr{B}$.

Alternatively for every $e > 0$ there exists an $a > 0$ and $b > 0$ such that

(5.3.30) $$\|\mathscr{F}[x(X', t'), \theta(X', t'), X, t)] - \mathscr{F}[x(X', t), \theta(X', t), X, t]\| < e$$

whenever

(5.3.31) $$|x(X', t') - x(X', t)|_{\mathscr{I}} < a$$
$$|\theta(X', t') - \theta(X', t)|_{\mathscr{I}} < b$$

for all times $t' \leq t$ and X' in $\mathscr{B}$, where $\|\mathscr{F}\|$ stands for $\sqrt{\mathrm{tr}\,\mathscr{F}^2}$. Accordingly, when the thermomechanical histories $x(X', t')$ and $\theta(X', t')$ differ very little from $x(X', t)$ and $\theta(X', t)$, respectively (when inequalities (5.3.31) are satisfied for positive constants a and b), then the corresponding functionals are close together (inequality (5.3.30) is satisfied). Hence the stresses at (X, t) corresponding to these two histories differ very little. The presence of the influence function $\mathscr{I}$ makes the histories near to time t influence the stress more than the distant times.

Just as in the case of functions, we can strengthen the continuity, as defined above, by applying the latter to the functional derivatives of $\mathscr{F}$. Thus, a strong axiom of fading memory may be formulated by assuming that the so-called Fréchet derivatives of $\mathscr{F}$ up to an order n exist and are continuous in the neighborhood of histories at time t in the Hilbert space normed by an influence function of order greater than $(n + \frac{1}{2})$. Since we will be dealing with linear simple materials, no recourse will be made to such a strong axiom of fading memory.

The axiom of memory discussed above is regarded to be valid for all other constitutive functionals.

(viii) AXIOM OF ADMISSIBILITY. ***All constitutive equations must be consistent with the basic principles of continuum mechanics, that is, they are subject to the principles of conservation of mass, balance of momenta, conservation of energy, and the Clausius-Duhem inequality.***

This axiom is a statement of the fact that in the formulation of the constitutive equations we must not violate the basic laws of motion. It helps also to eliminate the dependence on certain constitutive variables whenever they are obtainable through the equations of motion. For example, we need not include the dependence on the history of mass density, since this is obtained through the equation of continuity

$$\frac{\rho_0}{\rho(X', t')} = \det \frac{\partial x_k(X', t')}{\partial X'_K} \tag{5.3.32}$$

unless the dependence on $x_{k,K}$ at a fixed time $t' = t$ is only through its determinant. For certain materials (for example, fluids) this may be the case. In this case, the dependence on $\rho(X', t')$ at $t' = t$ may be included

The Clausius-Duhem inequality can be used to eliminate the dependence on certain variables appearing in the constitutive functionals. For example, for the isothermal elastic solids we shall find that the internal energy and entropy cannot depend on the rates of deformation gradients. In Section 4 6, Theorem 1 provides an example in which the Clausius-Duhem inequality, under certain conditions, forces the temperature θ to be derivable from the internal energy density (eq. 4.6.17).

5.4. THERMOMECHANICAL MATERIALS

In this book we deal only with the deformations and motions of continuous media The study of the chemical and electromagnetic changes is not within the scope of the book. A large class of materials does not undergo chemical transition or produce appreciable electromagnetic effects when deformed. However, the deformation and motion generally produce heat. Conversely, materials subjected to thermal changes deform and flow. The effect of thermal changes on the material behavior depends on the range and severity of such changes

The thermomechanical constitutive equations are intended to describe the physical properties of materials whose behavior is influenced by the deformation and thermal changes. According to the axioms of causality, equipresence, and determinism, the coostitutive equations of such materials may be expressed in the form (5.3.2) to (5 3.5), that is,

$$\mathbf{t}(X, t) = \mathscr{F}[\mathbf{x}(X', t'), \theta(X', t'), X, t] \tag{5.4.1}$$

$$\mathbf{q}(\mathbf{X}, t) = \mathcal{G}[\mathbf{x}(\mathbf{X}', t'), \theta(\mathbf{X}', t'), \mathbf{X}, t] \tag{5.4.2}$$

$$\varepsilon(\mathbf{X}, t) = \mathcal{E}[\mathbf{x}(\mathbf{X}', t'), \theta(\mathbf{X}', t'), \mathbf{X}, t] \tag{5.4.3}$$

$$\eta(\mathbf{X}, t) = \mathcal{N}[\mathbf{x}(\mathbf{X}', t'), \theta(\mathbf{X}', t'), \mathbf{X}, t] \tag{5.4.4}$$

where the functionals $\mathcal{F}$, $\mathcal{G}$, $\mathcal{E}$ and $\mathcal{N}$ are subject to the restrictions of axioms of the constitutive theory discussed in Section 5.3. We now proceed to deduce the consequence of these axioms. Since the procedure is similar for all functionals, for the sake of brevity, we carry out the analysis only for the stress functional $\mathcal{F}$. The results for $\mathcal{G}$, $\mathcal{E}$, and $\mathcal{N}$ are then written down immediately by analogy. We start with the application of the axiom (iv).

Consequence of the Axiom of Objectivity. According to this axiom, the form of the functional $\mathcal{F}$ should be the same in any two objectively equivalent motions, that is,

$$\mathcal{F}[\mathbf{x}(\mathbf{X}', t'), \theta(\mathbf{X}', t'), \mathbf{X}, t] = \mathcal{F}[\bar{\mathbf{x}}(\mathbf{X}', \bar{t}'), \theta(\mathbf{X}', \bar{t}'), \mathbf{X}, \bar{t}'] \tag{5.4.5}$$

for all $\mathbf{Q}(t')$, $\mathbf{b}(t')$ and a such that

$$\begin{aligned} &\bar{\mathbf{x}}(\mathbf{X}', \bar{t}') = \mathbf{Q}(t')\mathbf{x}(\mathbf{X}', t') + \mathbf{b}(t') \\ &\mathbf{Q}\mathbf{Q}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}, \qquad \bar{t}' = t' - a \end{aligned} \tag{5.4.6}$$

Consider the following three special transformations of the frame $\mathbf{x}$:

(a) *Rigid Translation of the Spatial Frame.* For this case take $\mathbf{Q}(t') = \mathbf{I}$, $a = 0$, $\mathbf{b}(t') = -\mathbf{x}(\mathbf{X}, t')$. This means that the spatial frame of reference is translated so that the material point $\mathbf{X}$ at time t' remains at the origin. From $(5.4.6)_1$ it follows that

$$\bar{\mathbf{x}}(\mathbf{X}', \bar{t}') = \mathbf{x}(\mathbf{X}', t') - \mathbf{x}(\mathbf{X}, t'), \qquad \bar{t}' = t'$$

Substituting this into (5.4.5), we get

$$\mathbf{t}(\mathbf{X}, t) = \mathcal{F}[\mathbf{x}(\mathbf{X}', t') - \mathbf{x}(\mathbf{X}, t'), \theta(\mathbf{X}', t'), \mathbf{X}, t] \tag{5.4.7}$$

(b) *Shift of Time.* Selecting

$$\mathbf{Q}(t') = \mathbf{I}, \qquad \mathbf{b}(t') = \mathbf{0}, \qquad a = t$$

we obtain from equation (5.4.6)

$$\bar{\mathbf{x}}(\mathbf{X}', \bar{t}') = \mathbf{x}(\mathbf{X}', t'), \qquad \bar{t}' = t' - t, \qquad -\infty \leq \bar{t}' \leq 0$$

Upon carrying this into the right-hand side of (5.4.5) we get

$$\mathbf{t} = \mathcal{F}[\mathbf{x}(\mathbf{X}', \bar{t}' + t), \theta(\mathbf{X}', \bar{t}' + t), \mathbf{X}, 0]$$

Therefore $\mathcal{F}$ is independent of t. Introducing

$$\tau' \equiv t - t' \geq 0, \qquad 0 \leq \tau' \leq \infty \tag{5.4.8}$$

and combining the above result with (5.4.7) we have

$$\mathbf{t}(\mathbf{X}, t) = \mathscr{F}[\mathbf{x}(\mathbf{X}', t-\tau') - \mathbf{x}(\mathbf{X}, t-\tau'), \theta(\mathbf{X}', t-\tau'), \mathbf{X}] \tag{5.4.9}$$

Thus $\mathscr{F}$ is a functional in the history of the relative motions from the present and the temperatures of all particles $\mathbf{X}'$ in $\mathscr{B}$.

(c) *Rigid Rotations of the Spatial Frame.* We take $\mathbf{b}=\mathbf{0}$, $a=0$ and $\mathbf{Q}(t')$ arbitrary. This represents an arbitrary time-dependent rotation of the spatial frame of reference. In this rotation, the stress $\mathbf{t}$ transforms as

$$\bar{\mathbf{t}}(\mathbf{X}, t) = \mathbf{Q}(t)\mathbf{t}(\mathbf{X}, t)\mathbf{Q}^T(t) \tag{5.4.10}$$

Using (5.4.5) we obtain

$$\begin{aligned}\mathbf{Q}(t)\mathscr{F}[\mathbf{x}(\mathbf{X}', t-\tau') - \mathbf{x}(\mathbf{X}, t-\tau'), \theta(\mathbf{X}', t-\tau'), \mathbf{X}]\mathbf{Q}^T(t) \\ = \mathscr{F}\{\mathbf{Q}(t-\tau')[\mathbf{x}(\mathbf{X}', t-\tau') - \mathbf{x}(\mathbf{X}, t-\tau')], \theta(\mathbf{X}', t-\tau'), \mathbf{X}\}\end{aligned} \tag{5.4.11}$$

for all $\mathbf{Q}(t')$, $\tau' \geq 0$ *subject to* $(5.4.6)_2$. *This is the restriction that must be imposed on the functional* $\mathscr{F}$ *in order that it be objective. Conversely, any functional* $\mathscr{F}$ given by (5.4.1) satisfies the principle of objectivity if it has the form of (5.4.9) satisfying (5.4.11), since any general rigid motion of the spatial frame of reference and the time shift can be obtained by a succession of the above three changes. Therefore, *equation* (5.4.9) *subject to* (5.4.11) *is the most general constitutive equation for the stress in thermomechanical theories of continuous media.* Equation (5.4.9) must satisfy the remaining axioms, namely, the axiom of material invariance, the axiom of neighborhood, the axiom of memory, and the axiom of admissibility.

Consequence of the Axiom of Material Invariance. Material symmetry, if it exists, can be characterized by a group of orthogonal transformations $\{\mathbf{S}\}$ and translations $\{\mathbf{B}\}$ of the material frame of reference $\mathbf{X}$. Response functions are then restricted by imposing the conditions that they be form-invariant under the transformation

$$\bar{\mathbf{X}} = \mathbf{S}\mathbf{X} + \mathbf{B} \tag{5.4.12}$$

where $\mathbf{S}$ is subject to

$$\mathbf{S}\mathbf{S}^T = \mathbf{S}^T\mathbf{S} = \mathbf{I}, \qquad \det \mathbf{S} = \pm 1 \tag{5.4.13}$$

for all members of the group $\{\mathbf{S}\}$. The consequence of these conditions may be found by employing the same technique used in connection with the axiom of objectivity above, except that in the present case the transformations are applied to material coordinates X_K. Thus, for example, the stress response functional $\mathscr{F}$ must be subject to

$$\begin{aligned}&\mathscr{F}[\mathbf{x}(\mathbf{X}', t-\tau') - \mathbf{x}(\mathbf{X}, t-\tau'), \theta(\mathbf{X}', t-\tau'), \mathbf{X}] \\ &= \mathscr{F}[\mathbf{x}(\mathbf{S}\mathbf{X}' + \mathbf{B}, t-\tau') - \mathbf{x}(\mathbf{S}\mathbf{X} + \mathbf{B}, t-\tau'), \theta(\mathbf{S}\mathbf{X}' + \mathbf{B}, t-\tau'), \mathbf{S}\mathbf{X} + \mathbf{B}]\end{aligned} \tag{5.4.14}$$

When {S} is the proper orthogonal group, (5.4.14) gives restrictions on *hemitropic* (transversely isotropic) *materials*. When {S} is the full group, the material is called *isotropic*. When the dependence on **B** is not present, then the material is *homogeneous*. In this case, the explicit dependence of $\mathscr{F}$ on X disappears.

Symmetry restrictions of all known materials are included in (5.4.14). Out of the infinitely many members of the full orthogonal group, twelve distinct ones are adequate for the descriptions of thirty-two classes of crystalline materials. In Chapter 6 we shall give a discussion of the anisotropic elastic materials when such transformation groups are used to reduce the constitutive equations.

Consequence of the Axiom of Smooth Neighborhood. Suppose that the functions $\mathrm{x}(\mathrm{X}', t-\tau')$ and $\theta(\mathrm{X}', t-\tau')$ possess Taylor expansions of the forms (5.3.19) at $\mathrm{X}' = \mathrm{X}$ for all $\tau' \geq 0$. Then this axiom states that the response functionals are smooth enough to allow representations of the form

$$\begin{aligned}(5.4.15)\quad \mathrm{t}(\mathrm{X}, t) = \mathscr{F}[&\mathrm{x}_{,K_1}(t-\tau'), \mathrm{x}_{,K_1K_2}(t-\tau'), \ldots, \mathrm{x}_{,K_1K_2\cdots K_P}(t-\tau'),\\ &\theta(t-\tau'), \theta_{,K_1}(t-\tau'), \ldots, \theta_{,K_1K_2\cdots K_Q}(t-\tau'); \mathbf{D}_K, \mathrm{X}]\\ &(K_1, K_2, \ldots, K_P, \ldots, K_Q) = 1, 2, 3\end{aligned}$$

In the arguments of $\mathscr{F}$ we omit writing X for simplicity since $\mathscr{F}$ is no longer a functional in the continuous range of X, that is, we employ the notation

$$(5.4.16)\qquad \mathrm{x}_{,K}(t-\tau') \equiv \mathrm{x}_{,K}(\mathrm{X}, t-\tau'), \qquad \theta(t-\tau') \equiv \theta(\mathrm{X}, t-\tau')$$

In (5.4.15) the dependence of $\mathscr{F}$ on the three vectors $\mathbf{D}_K$ is indicated. Since all directional vectors $\mathrm{X}' - \mathrm{X}$ originating from X in three dimensions can be decomposed along three vectors $\mathbf{D}_K$, this dependence is the result of the presence of $X'_K - X_K$ in the expansion (5.3.19). The vectors $\mathbf{D}_K$ will be called *material descriptors*. We may elect to use the base vectors $\mathbf{I}_K$ of our coordinates X_K to replace $\mathbf{D}_K$. The indication of this dependence is therefore merely stressing the fact that the form of $\mathscr{F}$ depends on the choice of bases. Material descriptors express the directional dependence of the material properties at a material point X. When no ambiguity arises we may drop $\mathbf{D}_K$ from the arguments of the constitutive functionals.

Materials that are characterized by constitutive equations of the form (5.4.15) are called *nonsimple materials of the material gradient type*. The ordered pair of the material gradients present in the argument of $\mathscr{F}$ defines the order of the material, that is,

DEFINITION 1. *A material is said to be of mechanical grade P and thermal grade Q or G(P, Q), if and only if the constitutive functionals depend on the gradients of* $\mathrm{x}(\mathrm{X}, t-\tau')$ *up to order P and gradients of* $\theta(\mathrm{X}, t-\tau')$ *up to*

order Q. For example, the constitutive equation (5.4.15) characterizes a material $G(P, Q)$

For most materials, $\mathbf{t}$, $\mathbf{q}$, ε and η at $\mathbf{X}$ are influenced only by the immediate neighborhood of $\mathbf{X}$, that is, $|\mathbf{X}' - \mathbf{X}|$ is very small. In such cases, it may be permissible to employ only the first-order gradients in (5.4.15). Such materials are called *simple materials.*

DEFINITION 2. *Thermomechanical materials of mechanical grade one and thermal grade one* $G(1, 1)$ *are called simple materials.* For simple materials the constitutive equations (5.4 15) take the form

$$\mathbf{t} = \mathscr{F}[\mathbf{x}_{,K}(t - \tau'), \theta(t - \tau'), \theta_{,K}(t - \tau'); \mathbf{D}_K, \mathbf{X}] \tag{5.4.17}$$

It is of interest to note that simple materials contain many subclasses For example, in (5.4.17) when the dependence on $\mathbf{x}_{,K}$ is dropped we get *rigid materials*, $G(0, 1)$ Similarly, when the dependence on $\theta_{,K}$ is not present we obtain *non-heat-conducting materials*, $G(1, 0)$. Many other special materials, for example, elastic solids, Stokesian fluids, turn out to be special classes contained in (5 4 17). We also note that if we retain higher-order gradients in (5.4.15), then we obtain nonsimple materials of various classes. For example, by including $\mathbf{x}_{,KL}(t - \tau')$ into the argument of $\mathscr{F}$ we get the theory of *couple stress*. A nonsimple material with $G(0, Q)$ is a rigid material and one with $G(P, 0)$ is non-heat-conducting.

The materials described by constitutive equations of the forms (5.4 15) are memory-dependent, that is, the stress at $\mathbf{X}$ at time t depends on the history prior to time t of the deformation gradients, the temperature, and the temperature gradients. The stress also depends on the directions $\mathbf{D}_K$ at $\mathbf{X}$ at time t, that is, the material may be anisotropic.

From (5.4.15) we form a new functional $\mathscr{F}$ by

$$F_{KL} = x_{k,K}(\mathbf{X}, t)x_{l,L}(\mathbf{X}, t)\mathscr{F}_{kl} \tag{5.4.18}$$

The functional F_{KL} is a scalar invariant with respect to rigid motions of the spatial frame of reference (that is, it is objective) According to a theorem of Cauchy [App Section B6] a single-valued function F_{KL} of the argument vectors $\mathbf{x}_{,K_1}, \mathbf{x}_{,K_1K_2}, \ldots; \theta_{,K_1}, \theta_{,K_1K_2}, \ldots$ for all ($K_\alpha = 1, 2, 3$) must reduce to functions of the scalar products of these vectors taken two at a time. A list of such independent invariants is too cumbersome to produce here. For simple materials these are given by:

$$C_{KL}(t - \tau') \equiv \delta_{kl}x_{k,K}(t - \tau')x_{l,L}(t - \tau') \tag{5 4.19}$$

$$\det x_{k,K}(t - \tau') = [\det C_{KL}(t - \tau')]^{1/2} = \frac{\rho_0}{\rho(t - \tau')}$$

Thus for simple materials we have

$$F_{KL} = F_{KL}[C_{MN}(t - \tau'), \rho^{-1}(t - \tau'), \theta(t - \tau'), \theta_{,K}(t - \tau'); \mathbf{D}_K, \mathbf{X}]$$

Solving for $\mathbf{F}$ from (5.4.18), we obtain the stress constitutive equations of simple materials

$$t_{kl}(\mathbf{X}, t) = F_{KL}[C_{MN}(t-\tau'), \rho^{-1}(t-\tau'), \theta(t-\tau'), \theta_{,K}(t-\tau'); \mathbf{D}_M, \mathbf{X}]X_{K,k}X_{L,l} \tag{5.4.20}$$

Consequence of the Axiom of Smooth Memory. If the argument functions of $\mathbf{F}$ in (5.4.15) possess continuous partial derivatives with respect to τ' at $\tau' = t$, then this axiom dictates that the functionals for a class of materials with limited memory in the range $\tau_0 < t' < t$, $-\infty < \tau_0$, may be represented by functions involving various time rates of these gradients, for example,

$$\mathbf{t}(\mathbf{X}, t) = \mathbf{f}(\mathbf{x}_{,K_1}, \dot{\mathbf{x}}_{,K_1}, \ddot{\mathbf{x}}_{,K_1}, \ldots; \mathbf{x}_{,K_1K_2}, \dot{\mathbf{x}}_{,K_1K_2}, \ldots; \ldots, \theta, \dot{\theta}, \ddot{\theta}, \ldots; \theta_{,K_1}, \dot{\theta}_{,K_1}, \ldots; \theta_{,K_1K_2}, \dot{\theta}_{,K_1K_2}, \ldots; \mathbf{D}_K, \mathbf{X}) \tag{5.4.21}$$

The number of the rates for each of the gradients depends on the strength of the memory of these gradients (in some sense). We note that $\mathbf{f}$ is no longer a functional. It is a tensor-valued function of the arguments listed. These materials are called *rate type*. A simple material of the rate type involving time rates of the deformation gradients up to order p and temperature and its gradients up to order q and r has the form

$$\mathbf{t}(\mathbf{X}, t) = \mathbf{f}(\mathbf{x}_{,K}, \dot{\mathbf{x}}_{,K}, \ddot{\mathbf{x}}_{,K}, \ldots, \mathbf{x}^{(p)}_{,K}; \theta, \dot{\theta}, \ddot{\theta}, \ldots, \theta^{(q)}; \theta_{,K}, \dot{\theta}_{,K}, \ddot{\theta}_{,K}, \ldots, \theta^{(r)}_{,K}; \mathbf{D}_K, \mathbf{X}) \tag{5.4.22}$$

The principle of objectivity restricts the form of these functions. In fact, had we employed (5.4.20) and assumed that $\mathbf{C}(t-\tau')$, $\theta(t-\tau')$, etc., appearing in the argument of F_{KL} possess continuous partial derivatives of order p, q, etc., then this axiom could have been interpreted to impose sufficient conditions of smoothness on $\mathbf{F}$ to replace (5.4.20) by

$$t_{kl}(\mathbf{X}, t) = F_{KL}(C_{MN}, \dot{C}_{MN}, \ddot{C}_{MN}, \ldots, C^{(p)}_{MN}; \rho^{-1}, \dot{\rho}, \ddot{\rho}, \ldots, \rho^{(p)}; \theta, \dot{\theta}, \ddot{\theta}, \ldots, \theta^{(q)}; \theta_{,K}, \dot{\theta}_{,K}, \ddot{\theta}_{,K}, \ldots, \theta^{(r)}_{,K}; \mathbf{D}_M, \mathbf{X})X_{K,k}X_{L,l} \tag{5.4.23}$$

For a certain class of materials, and when the time rates of the gradients appearing in (5.4.15) do not exist, in place of the axiom of smooth memory, we may use the axiom of fading memory[8]. The consequence of the axiom

[8] Coleman and Noll [1960] have shown that the axiom of fading memory can be used to obtain the rate type of simple materials as an asymptotic approximation for "slow" motions.

of fading memory for linear viscoelastic materials is discussed further in Chapter 9.

Consequence of the Axiom of Admissibility. In view of the equation of continuity (5 3.32) the dependence of F_{KL} on ρ^{-1} is included in the dependence on C_{KL} Therefore, it may be excluded from the argument of F_{KL}. This is the result of the axiom of admissibility, namely that F_{KL} must be compatible with the equation of conservation of mass. Regarding this point, however, care must be exercised because of the fact that in certain situations (for example, fluids) the dependence on C_{KL} at $\tau' = t$ may disappear; however, the dependence on ρ^{-1} must be retained as explained in Section 5.3. Accordingly, we may drop the dependence on ρ^{-1} and its rates from the equations (5.4 20) and (5.4.23). Therefore, in place of (5.4.20) and (5.4.23) for simple materials we have, respectively,

$$t_{kl}(\mathrm{X}, t) = F_{KL}[C_{MN}(t-\tau'), \theta(t-\tau'), \theta_{,K}(t-\tau'), \mathrm{X}]X_{K,k}X_{L,l} \tag{5 4 24}$$

$$t_{kl}(\mathrm{X}, t) = F_{KL}[C_{MN}, \dot{C}_{MN}, C_{MN}, \ \ldots, C^{(r)}_{MN}; \theta, \dot{\theta}, \ddot{\theta}, \ldots, \theta^{(q)}, \theta_{,K}, \dot{\theta}_{,K}, \ddot{\theta}_{,K}, \ldots, \theta^{(r)}_{,K}; \mathrm{X}]X_{K,k}X_{L,l} \tag{5 4 25}$$

The Clausius-Duhem inequality places further restrictions on the form of the constitutive functionals. These restrictions will be studied in the following sections on various special classes of materials, for example, elastic materials, viscous fluids. For more general cases the reader is referred to Chapter 9 and Eringen [1965, Art. 4]

5.5. ELASTIC MATERIALS

DEFINITION *A simple material is called elastic if the stress* $\mathbf{t}$, *heat vector* $\mathbf{q}$, *internal energy density* ε, *and the entropy density* η *at* (X, t) *depend only on the deformation gradient* $\mathbf{x}_{,K}$ *and temperature* θ *at* (X, t), *not on the entire past thermomechanical history.*

Constitutive equations for elastic materials are given by

$$\mathbf{t}(\mathrm{X}, t) = \mathbf{f}(\mathbf{x}_{,K}, \theta, \mathbf{D}_K, \mathrm{X}) \tag{5.5.1}$$

$$\mathbf{q}(\mathrm{X}, t) = \mathbf{g}(\mathbf{x}_{,K}, \theta, \mathbf{D}_K, \mathrm{X}) \tag{5.5.2}$$

$$\varepsilon(\mathrm{X}, t) = e(\mathbf{x}_{,K}, \theta, \mathbf{D}_K, \mathrm{X}) \tag{5.5.3}$$

$$\eta(\mathrm{X}, t) = n(\mathbf{x}_{,K}, \theta, \mathbf{D}_K, \mathrm{X}) \tag{5.5.4}$$

Argument vectors $\mathbf{D}_K$ are the material descriptors indicating the directional dependence of the constitutive functions $\mathbf{f}$, $\mathbf{g}$, e and n. Note that

these equations are the special cases of (5.4.17) where the dependence on $\theta_{,K}(t-\tau')$ is dropped and the memory dependence on $\mathbf{x}_{,K}(t-\tau')$ and $\theta(t-\tau')$ for $\tau' \neq t$ is eliminated. They are also the special forms of (5.4.22). The change in these materials occurs only from the change of configuration at time t. Selecting the reference configuration $\mathbf{X}$ as the undeformed and unstressed initial configuration with uniform temperature, we see that the stress at time t arises from the relative change of the configuration and temperature with respect to the initial state with no regard to intermediate changes. These materials, therefore, possess perfect memory of the initial configuration—not any other.

In the same way as in (5.4.18) from (5.5.1) and (5.5.2) we form two new functions $\mathbf{F}$ and $\mathbf{Q}$ by

$$F_{KL} = x_{k,K}x_{l,L}f_{kl} \qquad Q_K = x_{k,K}g_k \tag{5.5.5}$$

The functions F_{KL}, Q_K, e and n are scalar invariants with respect to rigid motions of the spatial frame of reference, that is, they are objective. According to a theorem of Cauchy [App. Section B6] they must reduce functions of the scalar products of the three vectors $\mathbf{x}_{,K}$ and their determinant, that is, they must be functions of

$$C_{KL} = \delta_{kl}x_{k,K}x_{l,L}, \qquad \det x_{k,K} = (\det C_{KL})^{1/2} = \frac{\rho_0}{\rho} \tag{5.5.6}$$

Thus

$$\begin{aligned} F_{KL} &= F_{KL}(\mathbf{C}, \rho^{-1}, \theta, \mathbf{D}_K, \mathbf{X}) \\ Q_K &= Q_K(\mathbf{C}, \rho^{-1}, \theta, \mathbf{D}_K, \mathbf{X}) \\ \varepsilon &= e(\mathbf{C}, \rho^{-1}, \theta, \mathbf{D}_K, \mathbf{X}) \\ \eta &= n(\mathbf{C}, \rho^{-1}, \theta, \mathbf{D}_K, \mathbf{X}) \end{aligned} \tag{5.5.7}$$

Unless we are interested in some approximations of $\mathbf{F}$ and $\mathbf{Q}$, we can drop the argument ρ^{-1} on account of the axiom of admissibility, namely, given $\mathbf{C}$, ρ^{-1} is determined through the continuity equation $(5.5.6)_2$. Since F_{KL} and Q_K are referred to the coordinates X_K we need not stress the dependence on the descriptors $\mathbf{D}_K$. Using (5.5.5) we solve for $\mathbf{f}$ and $\mathbf{g}$ leading to the constitutive equations

$$t_{kl} = F_{KL}(\mathbf{C}, \theta, \mathbf{X})X_{K,k}X_{L,l} \tag{5.5.8}$$

$$q_k = Q_K(\mathbf{C}, \theta, \mathbf{X})X_{K,k} \tag{5.5.9}$$

$$\varepsilon = e(\mathbf{C}, \theta, \mathbf{X}) \tag{5.5.10}$$

$$\eta = n(\mathbf{C}, \theta, \mathbf{X}) \tag{5.5.11}$$

The axiom of admissibility requires that (5.5.8) to (5.5.11) must be co
sistent with the principles of conservation of mass, balance of moment
conservation of energy, and the local Clausius-Duhem inequality, that i

$$\frac{\rho_0}{\rho} = \sqrt{\mathrm{III}_C} \tag{5.5.12}$$

$$t_{kl,k} + \rho(f_l - \dot{v}_l) = 0 \tag{5.5.13}$$

$$t_{kl} = t_{lk} \tag{5.5.14}$$

$$\rho\varepsilon = t_{kl}\, d_{lk} + q_{k,k} + \rho h \tag{5 5.15}$$

$$\rho\left(\eta - \frac{\dot{\varepsilon}}{\theta}\right) + \frac{1}{\theta}\, t_{kl}\, d_{lk} + \frac{1}{\theta^2}\, q_k \theta_{,k} \geq 0 \tag{5 5.16}$$

Of these (5.5.12) has already been used once in eliminating ρ^{-1} from th
arguments of the response functions **F**, **Q**, e, and n. Next we employ th
Clausius-Duhem inequality. Upon substituting from (5 5.8) to (5 5.11) int
(5.5.16) we have

$$\rho\left(\frac{\partial n}{\partial C_{KL}} - \frac{1}{\theta}\frac{\partial e}{\partial C_{KL}}\right)\dot{C}_{KL} + \rho\left(\frac{\partial n}{\partial \theta} - \frac{1}{\theta}\frac{\partial e}{\partial \theta}\right)\dot{\theta} + \frac{1}{\theta} f_{kl}\, d_{lk} + \frac{1}{\theta^2}\, g_k \theta_{,k} \geq 0$$

From a combination of (2.7 3) and (2.7.5), we have

$$\dot{C}_{KL} = 2 d_{kl} x_{k,K} x_{l,L} \tag{5.5.17}$$

so that

$$\left[2\rho\left(\frac{\partial n}{\partial C_{KL}} - \frac{1}{\theta}\frac{\partial e}{\partial C_{KL}}\right) x_{k,K} x_{l,L} + \frac{1}{\theta} f_{kl}\right] d_{lk} + \rho\left(\frac{\partial n}{\partial \theta} - \frac{1}{\theta}\frac{\partial e}{\partial \theta}\right)\dot{\theta} + \frac{1}{\theta^2}\, g_k \theta_{,k} \geq \tag{5.5.18}$$

This inequality must be satisfied for all independent thermomechanica
processes. Since $\dot{\theta}$, **d**, and $\theta_{,k}$ occur only linearly with coefficients whic
are not functions of these quantities, (5.5.18) cannot be maintained for a
d, $\dot{\theta}$, and $\theta_{,k}$ unless the coefficients of these terms vanish separately. Thu

$$t_{kl} = 2\rho\left(\frac{\partial e}{\partial C_{KL}} - \theta\frac{\partial n}{\partial C_{KL}}\right) x_{(k,K} x_{l),L}$$

$$0 = 2\rho\left(\frac{\partial e}{\partial C_{KL}} - \theta\frac{\partial n}{\partial C_{KL}}\right) x_{[k,K} x_{l],L} \tag{5.5.19}$$

$$\frac{\partial n}{\partial \theta} - \frac{1}{\theta}\frac{\partial e}{\partial \theta} = 0, \qquad g_k = 0$$

We introduce the *free energy* ψ by

$$\psi \equiv \varepsilon - \theta\eta = e - \theta n = \psi(\mathbf{C}, \theta, \mathbf{X}) \tag{5.5.20}$$

Using this in the above equations, we get

$$\begin{aligned} t_{kl} &= 2\rho \frac{\partial \psi}{\partial C_{KL}} x_{(k,K} x_{l),L} \\ 0 &= 2\rho \frac{\partial \psi}{\partial C_{KL}} x_{[k,K} x_{l],L} \\ \eta &= -\frac{\partial \psi}{\partial \theta} \\ q_k &= 0 \end{aligned} \tag{5.5.21}$$

Comparing (5.5.21) with the corresponding equations of (5.5.8) to (5.5.11), we find that

$$\begin{aligned} F_{KL} &= 2\rho \frac{\partial \psi}{\partial C_{MN}} C_{MK} C_{NL} \\ Q_K &= 0 \\ \varepsilon &= \psi - \theta \frac{\partial \psi}{\partial \theta} \\ \eta &= -\frac{\partial \psi}{\partial \theta} \end{aligned} \tag{5.5.22}$$

We have therefore proved

THEOREM 1. *An elastic solid is thermodynamically admissible if and only if the stress, internal energy, and entropy are derivable from a potential* Σ *and the heat is zero* (*the solid must undergo locally adiabatic changes*), that is,

$$t_{kl} = 2\frac{\rho}{\rho_0} \frac{\partial \Sigma}{\partial C_{KL}} x_{(k,K} x_{l),L} \tag{5.5.23}$$

$$q_k = 0 \tag{5.5.24}$$

$$\varepsilon = \frac{1}{\rho_0}\left(\Sigma - \theta \frac{\partial \Sigma}{\partial \theta}\right) \tag{5.5.25}$$

$$\eta = -\frac{1}{\rho_0} \frac{\partial \Sigma}{\partial \theta} \tag{5.5.26}$$

where

$$\frac{1}{\rho_0} \Sigma(\mathbf{C}, \theta, \mathbf{X}) \equiv \psi \tag{5.5.27}$$

is the free energy subject to the conditions

$$2\frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial C_{KL}} x_{(k,K} x_{l),L} = 0 \tag{5.5.28}$$

Since $C_{KL} = C_{LK}$ equations (5.5.28) are satisfied identically. Materials characterized by the constitutive equations of the form (5.5.23) to (5.5.26) are called *Green-elastic* or *hyperelastic solids.*

Upon carrying (5.5.23) to (5.5.26) into the equation of energy (5.5.15) we get

$$\eta - \frac{h}{\theta} = 0 \tag{5.5.29}$$

Thus, *in hyperelastic solids the entropy change is solely due to heat sources* When the heat source is absent ($h = 0$), we have $\eta = 0$

We observe that the entropy production, $\rho\gamma$, for the hyperelastic solids, vanish. Hence

THEOREM 2. *Hyperelastic solids are in thermal equilibrium*

If we carry (5.5.25), (5.5.26) into equation of energy (5.5.15) and use (5.5.29) we obtain

$$\frac{\rho}{\rho_0}\dot{\Sigma} + \rho\eta\dot{\theta} = t_{kl}\, d_{lk} \tag{5.5.30}$$

Remark. We must note that the Clausius-Duhem inequality (5.5.18) may be satisfied under other special conditions. For example, we may have $\dot{\theta} = 0$. In this case, we need not satisfy $(5.5.19)_3$. This corresponds to steady temperature changes.[9] Similarly, in another special case one may have uniform temperature. In this case or in the case $\theta = \text{const.}$ (isothermal changes) we need not have $\mathbf{q} = 0$. Still a third case arises when $\mathbf{d} = 0$ (rigid motions). In this case the stress $\mathbf{t}$ is undefined since $(5.5.19)_1$ need not be satisfied. These special situations correspond to some constraints on the dependence of the constitutive functions on their variables. When such restraints exist naturally, appropriate modifications will have to be made in the final forms of the constitutive equations. An important case arises in connection with the *incompressible elastic solids.*

Incompressible Elastic Solids. In this case the mass density remains unchanged, that is,

$$\frac{\rho_0}{\rho} = \sqrt{\mathrm{III}_C} = 1 \tag{5.5.31}$$

[9] Truesdell [1952, p. 174] and Eringen [1962, Art. 45] use (5.5.30) without the second term on the left-hand side as the definition of ideally elastic solids.

This condition places a restriction on $\mathbf{C}$, namely, all components C_{KL} of $\mathbf{C}$ are not independent. Hence we must take proper caution in evaluating the partial derivatives $\partial\psi/\partial C_{KL}$ and $\partial\Sigma/\partial C_{KL}$. This is achieved by the method of Lagrange's multipliers. Thus in the calculations of these derivatives we replace ψ by

$$\psi - \frac{p}{2\rho_0}(\mathrm{III}_C - 1)$$

where p is the unknown Lagrange multiplier. Substituting this into $(5.5.22)_1$ and using the identity

$$\frac{\partial \mathrm{III}_C}{\partial C_{MN}} C_{MK} = \mathrm{III}_C \delta_{NK}$$

we obtain

$$F_{KL} = -pC_{KL} + 2\rho \frac{\partial \psi}{\partial C_{MN}} C_{MK} C_{NL} \tag{5.5.32}$$

where $\partial\psi/\partial C_{MN}$ is to be calculated without any regard to the restrictions (5.5.31). With $\mathbf{F}$ given by (5.5.32), the stress-strain relations (5.5.23) of the incompressible elastic solid take the form

$$t_{kl} = -p\delta_{kl} + 2\frac{\partial \Sigma}{\partial C_{KL}} x_{(k,K} x_{l),L} \tag{5.5.33}$$

where $\Sigma \equiv \rho_0 \psi$.

The function $p(\mathbf{x}, t)$ so introduced is an unknown of the theory. It is called *pressure*. This unknown would be determined upon integration of the differential equations and use of the boundary conditions. We caution the reader of the difference between this pressure and the thermodynamic pressure π defined by (4.7.16).

Constitutive equations (5.5.23) to (5.5.26) and (5.5.33) are valid for anisotropic elastic materials. The axiom of material invariance places restrictions on these equations. If the material has symmetry conditions characterized by a group $\{\mathbf{S}\}$ of orthogonal transformations of the material axes $\mathbf{X}$ and translations $\{\mathbf{B}\}$, that is,

$$\bar{\mathbf{X}} = \mathbf{S}\mathbf{X} + \mathbf{B} \tag{5.5.34}$$

$$\mathbf{S}\mathbf{S}^T = \mathbf{S}^T\mathbf{S} = \mathbf{I}, \qquad \det \mathbf{S} = \pm 1 \tag{5.5.35}$$

then, according to the axiom of material invariance the response function Σ must be form-invariant when any material axes $\bar{\mathbf{X}}$ related to $\mathbf{X}$ by (5.5.34) are used, that is,

$$\Sigma(\bar{\mathbf{C}}, \bar{\theta}, \bar{\mathbf{X}}) = \Sigma(\mathbf{C}, \theta, \mathbf{X}) \tag{5.5.36}$$

We have

$$\overline{\mathbf{C}} = \mathbf{S}\mathbf{C}\mathbf{S}^{\mathrm{T}}, \qquad \bar{\theta} = \theta$$

so that we must have

$$\Sigma(\mathbf{S}\mathbf{C}\mathbf{S}^{\mathrm{T}}, \theta, \mathbf{S}\mathbf{X} + \mathbf{B}) = \Sigma(\mathbf{C}, \theta, \mathbf{X}) \tag{5.5.37}$$

for all members of the group {S} and translations {B}. If the solid is such that upon taking $\mathbf{S} = \mathbf{I}$ and $\mathbf{B} = -\mathbf{X}$ for all material points in the body (5.5.37) remains valid, we see that Σ becomes independent of $\mathbf{X}$. Such solids are called *homogeneous*. In this case we have

$$\Sigma = \Sigma(\mathbf{C}, \theta) \qquad \text{(Homogeneous solids)} \tag{5.5.38}$$

$$\Sigma(\mathbf{S}\mathbf{C}\mathbf{S}^{\mathrm{T}}, \theta) = \Sigma(\mathbf{C}, \theta) \tag{5.5.39}$$

This latter condition is employed to reduce the constitutive equations of homogeneous solids having various types of directional symmetries at a point $\mathbf{X}$. Thirty-two different classes of crystalline solids are obtained by the use of eleven proper members of the group {S} and a reflection of the axes $\mathbf{X}$. In Chapter 6 we discuss various symmetry requirements for the linear theory [10]

5.6. ISOTROPIC ELASTIC MATERIALS

The simplest material symmetry is obtained when S_{KL} is the full orthogonal group, that is, when the stress does not depend on the orientation of the material axes. Such solids are called *isotropic solids*. Thus *an elastic solid is said to be isotropic if the stress potential Σ is invariant under the full orthogonal group of transformations of the material axes X_K.*[11] This means that for all possible rotations and reflections of the material axes X_K, (5.5.36) remains valid. It is clear that for this to be true Σ cannot depend on the components C_{KL} but it must depend only on the invariants of C_{KL}. Therefore, for isotropic elastic materials we have

$$\Sigma = \Sigma(\mathrm{I}_C, \mathrm{II}_C, \mathrm{III}_C, \mathbf{X}) \qquad \text{(Isotropic)} \tag{5.6.1}$$

Through the relations of I_C, II_C, and III_C to other invariants we see that:

[10] For a discussion of various crystal classes in the nonlinear theory, see Smith and Rivlin [1958]; see also Green and Adkins [1960, Art. 1.6]. For a classification in mineralogy, see Dana and Hurlbut [1959, Ch 2]

[11] Some authors refer to this as an isotropic material with a center of symmetry, or *holohedral*. If the orthogonal group is proper (reflection is not allowed), then such materials are known as isotropic materials with no center of symmetry, or *hemihedral*

THEOREM. *For isotropic hyperelastic materials* Σ *may be considered to be a function of the invariants of any one of the strain measures* $\mathbf{C}$, $\mathbf{c}$, $\overset{-1}{\mathbf{c}}$, $\mathbf{E}$, $\mathbf{e}$, that is, we may write

$$\Sigma = \Sigma(\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{X}) \tag{5.6.2}$$

where I, II, and III represent any one set of the above strain measures. For the isotropic elastic materials an important form of the constitutive equations may be found as follows. Consider

$$\Sigma = \Sigma(c_{kl}, \theta, \mathrm{X}) = \Sigma(\mathrm{I}_c, \mathrm{II}_c, \mathrm{III}_c, \theta, \mathrm{X}) \tag{5.6.3}$$

We have

$$\dot{\Sigma} = \frac{\partial \Sigma}{\partial c_{kl}} \dot{c}_{kl} + \frac{\partial \Sigma}{\partial \theta} \dot{\theta} = -2 \frac{\partial \Sigma}{\partial c_{kl}} c_{m(k} d_{ml)} - 2 \frac{\partial \Sigma}{\partial c_{kl}} c_{m(k} w_{ml)} - \rho_0 \eta \dot{\theta} \tag{5.6.4}$$

where we first used $(2.7.5)_2$ afterwards wrote $v_{r,s} = d_{rs} + w_{rs}$. Substitution of (5.6.4) into (5.5.30) gives

$$-2 \frac{\rho}{\rho_0} \frac{\partial \Sigma}{\partial c_{kl}} c_{m(k} d_{ml)} - 2 \frac{\rho}{\rho_0} \frac{\partial \Sigma}{\partial c_{kl}} c_{m(k} w_{ml)} = t_{kl} d_{lk}$$

from which for arbitrary symmetric tensor d_{kl} and antisymmetric tensor w_{kl} we obtain

$$t_{kl} = -2 \frac{\rho}{\rho_0} \frac{\partial \Sigma}{\partial c_{m(l}} c_{k)m} \tag{5.6.5}$$

$$0 = -2 \frac{\rho}{\rho_0} \frac{\partial \Sigma}{\partial c_{m[l}} c_{k]m} \tag{5.6.6}$$

These are the stress constitutive equations that are valid for an isotropic elastic solid. If we write $c_{kl} = \delta_{kl} - 2 e_{kl}$ we obtain Murnaghan's [1937] spatial form

$$t_{kl} = \frac{\rho}{\rho_0} \left(\frac{\partial \Sigma}{\partial e_{kl}} - \frac{\partial \Sigma}{\partial e_{ml}} e_{km} - \frac{\partial \Sigma}{\partial e_{km}} e_{lm} \right) \tag{5.6.7}$$

where Σ is subject to the condition

$$-\frac{\rho}{\rho_0} \left(\frac{\partial \Sigma}{\partial e_{ml}} e_{km} - \frac{\partial \Sigma}{\partial e_{mk}} e_{lm} \right) = 0 \tag{5.6.8}$$

For incompressible elastic solids (5.6.7) is modified to

$$t_{kl} = -p \delta_{kl} + \frac{\partial \Sigma}{\partial e_{kl}} - \frac{\partial \Sigma}{\partial e_{ml}} e_{km} - \frac{\partial \Sigma}{\partial e_{km}} e_{lm} \qquad \text{(incompressible)} \tag{5.6.9}$$

where now Σ is a function of two invariants I_e, II_e, θ, and $\mathbf{X}$.

$$\Sigma = \Sigma(I_e, II_e, \theta, \mathbf{X}) \tag{5.6 10}$$

since in this case $III_e = 1$, hence through $(I.10.24)_3$ we have

$$III_e = -\tfrac{1}{2}I_e + \tfrac{1}{2}II_e \tag{5.6.11}$$

An alternative derivation may be made by expressing the stress as a function of the Finger deformation tensor $\overset{-1}{\mathbf{e}}$ as follows. For isotropic materials, Σ is a function of the invariants

(5.6.12)

$$I_1 = \operatorname{tr} \mathbf{C} = \operatorname{tr} \overset{-1}{\mathbf{e}}, \qquad I_2 = \operatorname{tr} \mathbf{C}^2 = \operatorname{tr} (\overset{-1}{\mathbf{e}})^2, \qquad I_3 = \operatorname{tr} \mathbf{C}^3 = \operatorname{tr} (\overset{-1}{\mathbf{e}})^3$$

Using (5.5 23) we write

$$t_{kl} = 2\frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial I_a}\frac{\partial I_a}{\partial C_{KL}} x_{(k,K}x_{l),L} \tag{5.6 13}$$

Now

$$\frac{\partial I_1}{\partial C_{KL}} = \delta_{LK}, \qquad \frac{\partial I_2}{\partial C_{KL}} = 2C_{LK}$$

$$\frac{\partial I_3}{\partial C_{KL}} = 3C_{LM}C_{MK} \tag{5.6.14}$$

Substituting these into (5 6.13) and using the expressions

$$C_{KL} \equiv x_{k,K}x_{k,L}, \qquad \overset{-1}{c}_{kl} \equiv x_{k,K}x_{l,K} \tag{5 6.15}$$

we get

$$t = 2\frac{\rho}{\rho_0}\left[\frac{\partial \Sigma}{\partial I_1}\overset{-1}{\mathbf{c}} + 2\frac{\partial \Sigma}{\partial I_2}(\overset{-1}{\mathbf{c}})^2 + 3\frac{\partial \Sigma}{\partial I_3}(\overset{-1}{\mathbf{c}})^3\right] \tag{5.6.16}$$

Now according to the Cayley-Hamilton theorem for a matrix $\mathbf{a}$ we have

$$\begin{aligned}\mathbf{a}^3 = (\operatorname{tr} \mathbf{a})\mathbf{a}^2 - \tfrac{1}{2}[(\operatorname{tr} \mathbf{a})^2 - \operatorname{tr} \mathbf{a}^2]\mathbf{a} \\ + [\tfrac{1}{3}\operatorname{tr} \mathbf{a}^3 + \tfrac{1}{6}(\operatorname{tr} \mathbf{a})^3 - \tfrac{1}{2}\operatorname{tr} \mathbf{a} \operatorname{tr} \mathbf{a}^2]\mathbf{I}\end{aligned} \tag{5 6 17}$$

By use of (5.6.17) we can replace $(\overset{-1}{\mathbf{e}})^3$ in (5 6.16) by its equivalent, thus obtaining

$$\begin{aligned}t = \frac{\rho}{\rho_0}\Bigg\{ &(I_1^3 - 3I_1I_2 + 2I_3)\frac{\partial \Sigma}{\partial I_3}\mathbf{I} \\ &+ \left[2\frac{\partial \Sigma}{\partial I_1} - 3(I_1^2 - I_2)\frac{\partial \Sigma}{\partial I_3}\right]\overset{-1}{\mathbf{c}} \\ &+ \left(4\frac{\partial \Sigma}{\partial I_2} + 6I_1\frac{\partial \Sigma}{\partial I_3}\right)(\overset{-1}{\mathbf{c}})^2\Bigg\}\end{aligned} \tag{5.6 18}$$

Here, for $\underset{-1}{\mathbf{I}}_a$, we may of course, use their equivalent forms expressed in terms of $\underset{-1}{\mathbf{c}}$ given by (5.6.12). Other forms for the stress are possible. For these, the reader is referred to Eringen [1962, Arts. 45 to 48].

5.7. STOKESIAN FLUIDS

A fluid may be recognized by the vague notion that in any of its distorted shapes, leaving the mass density unchanged, it has no memory of its past states. We may formalize this idea as follows:

DEFINITION 1. *A body is called a fluid if every configuration of the body leaving the density unchanged can be taken as the reference configuration.*

For simplicity, consider the simple thermomechanical materials described by constitutive equations of the form (5.4.25) not including terms beyond the first order rates, that is,

$$t_{kl}(\mathbf{X}, t) = F_{KL}(\mathbf{C}, \dot{\mathbf{C}}, \rho^{-1}, \dot{\rho}, \theta, \dot{\theta}, \theta_{,K}, \dot{\theta}_{,K}; \mathbf{X})\, X_{K,k}\, X_{L,l} \tag{5.7.1}$$

If every configuration is to be a reference configuration, then we can write $\mathbf{X} = \mathbf{x}$ with ρ unchanged. With this

$$\begin{aligned} x_{k,L} &\to \delta_{kl}, \qquad C_{MN} \to \delta_{mn} \\ \dot{C}_{KL} &= 2\, d_{kl} x_{k,K} x_{l,L} \to 2\, d_{kl} \\ \theta_{,K} &\to \theta_{,k}, \qquad \dot{\theta}_{,K} \to \dot{\theta}_{,k} \end{aligned}$$

By using the equation of continuity $\dot{\rho} = -\rho\, d_{kk}$ we eliminate the dependence on $\dot{\rho}$ so that (5.7.1), for fluids, may be written as

$$t_{kl} = f_{kl}(\mathbf{d}, \rho^{-1}, \theta, \dot{\theta}, \theta_{,k}, \dot{\theta}_{,k}; \mathbf{x}) \tag{5.7.2}$$

For simplicity we consider fluids that are independent of $\theta_{,k}$ and $\dot{\theta}_{,k}$.

DEFINITION 2. *A stokesian (non-heat-conducting) fluid is a thermomechanical medium having constitutive equations of the form*

$$t_{kl} = f_{kl}(\mathbf{d}, \rho^{-1}, \theta, \mathbf{x}) \tag{5.7.3}$$

$$q_k = g_k(\mathbf{d}, \rho^{-1}, \theta, \mathbf{x}) \tag{5.7.4}$$

$$\varepsilon = e(\mathbf{d}, \rho^{-1}, \theta, \mathbf{x}) \tag{5.7.5}$$

$$\eta = n(\mathbf{d}, \rho^{-1}, \theta, \mathbf{x}) \tag{5.7.6}$$

subject to the axioms of admissibility, objectivity, and material invariance.

Note that in the above passage from (5.7.1) to (5.7.2) the objective character of t_{kl} given by (5.7.1) is destroyed. To restore it, the principles of objectivity

must be applied. As in Section 5.4, by use of the axiom of objectivity we can show that the response functions $\mathbf{f}$, $\mathbf{g}$, e, and n cannot depend on $\mathbf{x}$ explicitly. Moreover, they must be form-invariant under arbitrary time-dependent rotations of the spatial frame of reference $\mathbf{x}$. Let $\bar{\mathbf{x}}(\mathbf{X}, t)$ be a motion objectively equivalent to $\mathbf{x}(\mathbf{X}, t)$, that is,

$$\begin{aligned} \bar{\mathbf{x}} &= \mathbf{Q}(t)\mathbf{x} + \mathbf{b}(t) \\ \mathbf{Q}\mathbf{Q}^T &= \mathbf{Q}^T\mathbf{Q} = \mathbf{I} \end{aligned} \tag{5.7.7}$$

The axiom of objectivity requires that for all $\mathbf{Q}(t)$ and $\mathbf{b}(t)$ we must have

$$\bar{\mathbf{t}} = \mathbf{f}(\bar{\mathbf{d}}, \rho^{-1}, \theta, \bar{\mathbf{x}}) = \mathbf{f}(\mathbf{d}, \rho^{-1}, \theta, \mathbf{x}) \tag{5.7.8}$$

and similar conditions written for $\mathbf{g}$, e, and n.

If we select $\mathbf{Q} = \mathbf{I}$, $\mathbf{b} = -\mathbf{x}$ (5.7.8) gives

$$\mathbf{f}(\mathbf{d}, \rho^{-1}, \theta, \mathbf{0}) = \mathbf{f}(\mathbf{d}, \rho^{-1}, \theta, \mathbf{x}) = \mathbf{t}$$

Therefore $\mathbf{f}$ cannot depend on $\mathbf{x}$.

Next we select $\mathbf{b} = \mathbf{0}$ and $\mathbf{Q}(t)$ arbitrary. Since we have

$$\bar{\mathbf{t}} = \mathbf{Q}\mathbf{t}\mathbf{Q}^T, \qquad \bar{\mathbf{d}} = \mathbf{Q}\mathbf{d}\mathbf{Q}^T$$

equation (5.7.8) reduces to

$$\mathbf{Q}\mathbf{f}(\mathbf{d}, \rho^{-1}, \theta)\mathbf{Q}^T = \mathbf{f}(\mathbf{Q}\mathbf{d}\mathbf{Q}^T, \rho^{-1}, \theta) \tag{5.7.9}$$

The axiom of objectivity when applied to $\mathbf{g}$, e, and n similarly leads to the restrictions

$$\begin{aligned} \mathbf{Q}\mathbf{g}(\mathbf{d}, \rho^{-1}, \theta) &= \mathbf{g}(\mathbf{Q}\mathbf{d}\mathbf{Q}^T, \rho^{-1}, \theta) \\ e(\mathbf{d}, \rho^{-1}, \theta) &= e(\mathbf{Q}\mathbf{d}\mathbf{Q}^T, \rho^{-1}, \theta) \\ n(\mathbf{d}, \rho^{-1}, \theta) &= n(\mathbf{Q}\mathbf{d}\mathbf{Q}^T, \rho^{-1}, \theta) \end{aligned} \tag{5.7.10}$$

The tensor functions $\mathbf{f}$, $\mathbf{g}$, e, and n satisfying the conditions (5.7.9) and (5.7.10) for all proper orthogonal transformations $\{\mathbf{Q}(t)\}$ are called *hemitropic*. If the reflection of axes is adjoined to the group $\{\mathbf{Q}(t)\}$ they are called *isotropic*.[12] This completes the restrictions emanating from the axiom of objectivity.

Next we consider the restrictions due to the axiom of admissibility. According to this axiom $\mathbf{t}$, $\mathbf{q}$, ε, and η must satisfy the equations of conservation of mass, balance of momenta, conservation of energy, and the Clausius-Duhem inequality, that is,

[12] Note that this is spatial isotropy, not material isotropy discussed in Section 5.6 for elastic materials.

$$\frac{\partial \rho}{\partial t} + (\rho v_k)_{,k} = 0 \tag{5.7.11}$$

$$t_{kl,k} + \rho(f_l - \dot{v}_l) = 0 \tag{5.7.12}$$

$$t_{kl} = t_{lk} \tag{5.7.13}$$

$$\rho\dot{\varepsilon} = t_{kl}\, d_{lk} + q_{k,k} + \rho h \tag{5.7.14}$$

$$\rho\left(\dot{\eta} - \frac{\dot{\varepsilon}}{\theta}\right) + \frac{1}{\theta} t_{kl}\, d_{lk} + \frac{1}{\theta^2} q_k \theta_{,k} \geq 0 \tag{5.7.15}$$

Of these the inequality (5.7.15) must be satisfied for all independent processes. When constitutive equations (5.7.3) to (5.7.6) are brought into (5.7.15) we have

$$\rho\left(\frac{\partial n}{\partial d_{kl}} - \frac{1}{\theta}\frac{\partial e}{\partial d_{kl}}\right)\dot{d}_{kl} - \left(\frac{\partial n}{\partial \rho^{-1}} - \frac{1}{\theta}\frac{\partial e}{\partial \rho^{-1}}\right)\frac{\dot{\rho}}{\rho}$$

$$+ \rho\left(\frac{\partial n}{\partial \theta} - \frac{1}{\theta}\frac{\partial e}{\partial \theta}\right)\dot{\theta} + \frac{1}{\theta} f_{kl}\, d_{lk} + \frac{1}{\theta^2} g_k \theta_{,k} \geq 0$$

Here the coefficients of $\dot{\mathbf{d}}$, $\dot{\theta}$, and $\theta_{,k}$ are independent of these quantities since n, e, $\mathbf{f}$, and $\mathbf{g}$ are only functions of $\mathbf{d}$, ρ^{-1} and θ. The above inequality cannot be maintained for all variations of the independent thermomechanical processes described by $\dot{\mathbf{d}}$, $\dot{\theta}$ and $\theta_{,k}$ since these are linearly involved in the above inequality. Hence the coefficients of these variables must vanish, that is,

$$\frac{\partial n}{\partial d_{kl}} - \frac{1}{\theta}\frac{\partial e}{\partial d_{kl}} = 0 \tag{5.7.16}$$

$$\frac{\partial n}{\partial \theta} - \frac{1}{\theta}\frac{\partial e}{\partial \theta} = 0 \tag{5.7.17}$$

$$g_k = 0 \tag{5.7.18}$$

so that

$$\left(\frac{\partial n}{\partial \rho^{-1}} - \frac{1}{\theta}\frac{\partial e}{\partial \rho^{-1}}\right) d_{kk} + \frac{1}{\theta} f_{kl}\, d_{lk} \geq 0 \tag{5.7.19}$$

To obtain (5.7.19) we also used $\dot{\rho} = -\rho\, d_{kk}$, which is another form of (5.7.11). Equation (5.7.16) indicates that the function $e - n\theta$ is independent of $\mathbf{d}$. We now introduce the *free energy* function $\psi(\rho^{-1}, \theta)$ by

$$\psi(\rho^{-1}, \theta) \equiv e - n\theta \tag{5.7.20}$$

Using this in (5.7.17) we get

$$\eta = -\frac{\partial \psi}{\partial \theta} \tag{5.7.21}$$

Carrying (5.7.20) into (5.7.19), we can write this inequality in the form

$$\frac{1}{\theta}(f_{kl} + \pi\delta_{kl})\, d_{lk} \geq 0 \tag{5.7.22}$$

where

$$\pi(\rho^{-1}, \theta) \equiv -\frac{\partial \psi}{\partial \rho^{-1}} \tag{5.7.23}$$

is the *thermodynamic pressure*. We recognize

$${}_D t_{kl} \equiv t_{kl} + \pi\delta_{kl} \tag{5.7.24}$$

as the *dissipative* part of the stress tensor. Since $\theta > 0$ we see that (5.7.22) is reduced to the compact form

$${}_D t_{kl}\, d_{lk} \geq 0 \tag{5.7.25}$$

Hence the Clausius-Duhem inequality imposes the condition that *the dissipative energy* must be nonnegative.

Upon using (5.7.20), (5.7.21), and (5.7.23) in the equation of energy (5.7.14), we obtain the expression of the entropy rate

$$\rho\theta\eta = {}_D t_{kl}\, d_{lk} + \rho h \tag{5.7.26}$$

which states that *the dissipative energy and the internal heat sources are the sole contributors to the entropy change*. Thermodynamic equilibrium is established when the dissipative energy vanishes. This can occur if one of the following conditions is satisfied:

$$t_{kl} = -\pi\delta_{kl}, \qquad d_{kl} = 0, \qquad {}_D\mathbf{t} \perp \mathbf{d} \tag{5.7.27}$$

The first of these gives us nonviscous fluids, the second expresses rigid body motion, and the third is a motion in which the principal stretchings are perpendicular to the principal stresses (probably a hypothetical fluid motion). We again remark that these are the results of special internal constraints for which proper attention is required. The case of incompressible fluids is discussed later in this section.

Collecting our findings, we see that the constitutive equations of these fluids can be written in the form

$$t_{kl} = -\pi(\rho^{-1}, \theta)\delta_{kl} + {}_D f_{kl}(\mathbf{d}, \rho^{-1}, \theta) \tag{5.7.28}$$

$$q_k = 0 \tag{5.7.29}$$

$$\varepsilon = \psi(\rho^{-1}, \theta) - \theta\frac{\partial \psi}{\partial \theta} \tag{5.7.30}$$

$$\eta = -\frac{\partial \psi}{\partial \theta} \tag{5.7.31}$$

where

$$\pi(\rho^{-1}, \theta) \equiv -\frac{\partial \psi}{\partial \rho^{-1}} \tag{5.7.32}$$

and ${}_D\mathbf{f}$ is subject to the conditions

$$\mathbf{Q}(t)\,{}_D\mathbf{f}(\mathbf{d}, \rho^{-1}, \theta)\mathbf{Q}^T(t) = {}_D\mathbf{f}(\mathbf{Q}\mathbf{d}\mathbf{Q}^T, \rho^{-1}, \theta) \tag{5.7.33}$$

$${}_Df_{kl}\, d_{lk} \geq 0 \tag{5.7.34}$$

If we assume that ${}_D\mathbf{f}$ is continuous in $\mathbf{d}$ the inequality (5.7.34) implies that

$${}_D\mathbf{f}(\mathbf{0}, \rho^{-1}, \theta) = \mathbf{0} \tag{5.7.35}$$

To see this, let all $d_{kl} = 0$, except d_{11}; then by (5.7.34) we must have $f_{11}d_{11} \geq 0$. If $d_{11} > 0$, then ${}_Df_{11} > 0$; if $d_{11} < 0$, then ${}_Df_{11} < 0$. Thus, ${}_Df_{11} = 0$ when $d_{11} = 0$. This argument when applied to other components of d_{kl} leads to the proof of (5.7.35). Hence: *The dissipative stress vanishes with the deformation rate.*

Equations (5.7.28) to (5.7.34) must be employed along with equations (5.7.11) to (5.7.14) in dealing with any problems of Stokesian fluids. These fluids are nonthermal in the sense that there is no heat conduction ($q_k = 0$). However, the fluid properties change with temperature.

It is clear that in (5.7.28) no reference is made to any initial state, moreover ${}_D\mathbf{f}$ satisfies (5.7.34). Since we took $\mathbf{X} = \mathbf{x}$ with $\rho \equiv$ fixed, the constitutive equations are valid in all frames of reference $\mathbf{x}$ leaving the density unchanged. Since $\mathbf{Q}(t)$ is a proper orthogonal group, we see that *all Stokesian fluids are hemitropic.* If we include the reflection of the spatial axes in the group $\mathbf{Q}$, the fluids become *isotropic.*

Incompressible Fluids. For the incompressible fluids we have the internal constraints that $\rho = \text{const.} = \rho_0$. Hence $\partial\psi/\partial\rho^{-1}$ in (5.7.23) is undefined. Introducing an unknown Lagrange multiplier $p(\mathbf{x}, t)$ we must replace $\psi(\rho^{-1}, \theta)$ in this case by

$$\psi(\theta) + p\left(\frac{1}{\rho_0} - \frac{1}{\rho}\right)$$

Equation (5.6.23) now reads

$$\pi(\theta) = p \tag{5.7.36}$$

With this the dissipative part of the stress tensor is now defined by

$${}_Dt_{kl} = t_{kl} + p\delta_{kl} \tag{5.7.37}$$

The constitutive equations have the same form as (5.7.28) to (5.7.31) except that π is replaced by p and all functions are now independent of ρ^{-1}. The pressure p introduced above is an unknown of the theory and it would be determined through solutions of the field equations under a given set of boundary conditions.

Further simplification of the constitutive equations is obtained by assuming that ${}^{4}_{D}\mathbf{f}$ is a polynomial in $\mathbf{d}$. In this case ${}_{D}\mathbf{f}$ can be expressed in finite form involving only terms quadratic in $\mathbf{d}$ with coefficients that are polynomial functions in the invariants of $\mathbf{d}$. For an account on this and for the linear theory we refer the reader to Chapter 7 of this book. For the nonlinear theory see also Eringen [1962, Chapter 7] Other examples of fluids, solids, and heat-conducting and viscoelastic materials are to be found in Sections 5.8 to 5 10 and Chapters 8 and 9 of this book.

5.8. THERMOVISCOUS STOKESIAN FLUIDS

The effect of heat conduction is brought into play by including the temperature gradient into the constitutive variables of Stokesian fluids. Thus

DEFINITION 1. *A heat-conducting Stokesian fluid is a thermomechanical medium having constitutive equations of the form*

$$t_{kl} = f_{kl}(\rho^{-1}, \mathbf{d}, \theta_{,k}, \theta) \tag{5.8.1}$$

$$q_k = g_k(\rho^{-1}, \mathbf{d}, \theta_{,k}, \theta) \tag{5.8.2}$$

$$\varepsilon = e(\rho^{-1}, \mathbf{d}, \theta_{,k}, \theta) \tag{5.8.3}$$

$$\eta = n(\rho^{-1}, \mathbf{d}, \theta_{,k}, \theta) \tag{5.8.4}$$

subject to the axioms of admissibility and objectivity.

In the same way as in (5.7.10), we can easily show that the axiom of objectivity places the restrictions on $\mathbf{f}$, $\mathbf{g}$, e, and n,

$$\begin{aligned} \mathbf{Q}\mathbf{f}(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta)\mathbf{Q}^T &= \mathbf{f}(\rho^{-1}, \mathbf{Q}\mathbf{d}\mathbf{Q}^T, \mathbf{Q}\nabla\theta, \theta) \\ \mathbf{Q}\mathbf{g}(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta) &= \mathbf{g}(\rho^{-1}, \mathbf{Q}\mathbf{d}\mathbf{Q}^T, \mathbf{Q}\nabla\theta, \theta) \\ e(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta) &= e(\rho^{-1}, \mathbf{Q}\mathbf{d}\mathbf{Q}^T, \mathbf{Q}\nabla\theta, \theta) \\ n(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta) &= n(\rho^{-1}, \mathbf{Q}\mathbf{d}\mathbf{Q}^T, \mathbf{Q}\nabla\theta, \theta) \end{aligned} \tag{5.8.5}$$

for all proper orthogonal transformations $\mathbf{Q}(t)$. Thus the tensor functions $\mathbf{f}$, $\mathbf{g}$, e, and n are *hemitropic*.

Next we consider the restrictions emanating from the local Clausius-Duhem inequality (5.7.15). To this end we first introduce the free energy function ψ by

$$\psi = \varepsilon - \theta\eta \tag{5.8.6}$$

Inequality (5.7.15) can then be written as

$$-\frac{\rho}{\theta}(\dot{\psi} + \dot{\theta}\eta) + \frac{1}{\theta} t_{kl}\, d_{lk} + \frac{1}{\theta^2} q_k \theta_{,k} \geq 0 \tag{5.8.7}$$

or since

$$\psi = \psi(\rho^{-1}, \mathbf{d}, \theta_{,k}, \theta)$$

this reads

$$-\frac{\rho}{\theta}\left(\frac{\partial\psi}{\partial d_{kl}}\frac{D}{Dt} d_{kl} + \frac{\partial\psi}{\partial\theta_{,k}}\frac{D}{Dt}\theta_{,k}\right) - \frac{\rho}{\theta}\left(\frac{\partial\psi}{\partial\theta} + n\right)\dot{\theta}$$

$$+\frac{1}{\theta}\left(f_{kl} - \frac{\partial\psi}{\partial\rho^{-1}}\delta_{kl}\right) d_{lk} + \frac{1}{\theta^2} g_k\theta_{,k} \geq 0$$

where we used the equation of continuity in computing $\dot{\rho}$. This inequality is to be satisfied for all independent variations of $D\, d_{kl}/Dt$, $D\theta_{,k}/Dt$, $\dot{\theta}$, d_{kl} and $\theta_{,k}$. It is linear in the first three of these variables. Hence we must have

$$\frac{\partial\psi}{\partial d_{kl}} = 0, \qquad \frac{\partial\psi}{\partial\theta_{,k}} = 0$$

$$n = -\frac{\partial\psi}{\partial\theta} \tag{5.8.8}$$

$$\frac{1}{\theta}\left(f_{kl} - \frac{\partial\psi}{\partial\rho^{-1}}\delta_{kl}\right) d_{lk} + \frac{1}{\theta^2} g_k\theta_{,k} \geq 0$$

The first two of these equations indicate that ψ cannot depend on $\mathbf{d}$ and $\theta_{,k}$, that is,

$$\psi = \psi(\rho^{-1}, \theta) \tag{5.8.9}$$

If we introduce the thermodynamic pressure

$$\pi(\rho^{-1}, \theta) \equiv -\frac{\partial\psi}{\partial\rho^{-1}} \tag{5.8.10}$$

the inequality in $(5.8.8)_4$ reads

$$\frac{1}{\theta} D f_{kl}\, d_{lk} + \frac{1}{\theta^2} g_k\theta_{,k} \geq 0 \tag{5.8.11}$$

where

$$ {}_D f_{kl} \equiv f_{kl} + \pi \delta_{kl} \tag{5.8.12} $$

is the dissipative part of the stress Since ${}_D\mathbf{f}$ and $\mathbf{g}$ are functions of $\mathbf{d}$ and $\theta_{,k}$ no other simplification is possible in (5.8.11). Thus

THEOREM 1. *Stokesian thermoviscous fluids are thermodynamically admissible if and only if they possess constitutive equations of the form*

$$ t_{kl} = -\pi(\rho^{-1}, \theta)\delta_{kl} + {}_D f_{kl}(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta) \tag{5.8.13} $$

$$ q_k = g_k(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta) \tag{5.8.14} $$

$$ \varepsilon = \psi(\rho^{-1}, \theta) - \theta \frac{\partial \psi}{\partial \theta} \tag{5.8.15} $$

$$ \eta = -\frac{\partial \psi}{\partial \theta} \tag{5.8.16} $$

where

$$ \pi(\rho^{-1}, \theta) = -\frac{\partial \psi}{\partial \rho^{-1}} \tag{5.8.17} $$

and ${}_D\mathbf{f}$ *and* $\mathbf{g}$ *are subject to the conditions*

$$ \mathbf{Q}(t)\,{}_D\mathbf{f}(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta)\mathbf{Q}^T(t) = {}_D\mathbf{f}(\rho^{-1}, \mathbf{Q}\mathbf{d}\mathbf{Q}^T, \mathbf{Q}\nabla\theta, \theta) \tag{5.8.18} $$

$$ \mathbf{Q}(t)\mathbf{g}(\rho^{-1}, \mathbf{d}, \nabla\theta, \theta) = \mathbf{g}(\rho^{-1}, \mathbf{Q}\mathbf{d}\mathbf{Q}^T, \mathbf{Q}\nabla\theta, \theta) \tag{5.8.19} $$

$$ \frac{1}{\theta}\,{}_D f_{kl}\, d_{lk} + \frac{1}{\theta^2} g_k \theta_{,k} \geq 0 \tag{5.8.20} $$

Assuming that ${}_D\mathbf{f}$ and $\mathbf{g}$ are continuous in $\mathbf{d}$ and $\nabla\theta$, in the same way as in (5.7.35), we can prove that

$$ \begin{aligned} {}_D\mathbf{f}(\rho^{-1}, \mathbf{0}, \mathbf{0}, \theta) &= \mathbf{0} \\ \mathbf{g}(\rho^{-1}, \mathbf{0}, \mathbf{0}, \theta) &= \mathbf{0} \end{aligned} \tag{5.8.21} $$

Hence: *The viscous dissipation and heat vanish with* $\mathbf{d}$ *and* $\nabla\theta$. The conditions (5.8.18) and (5.8.19) are equivalent to

THEOREM 2. *Stokesian thermoviscous fluids are hemitropic.* If reflection is included in the group of transformations, $\{\mathbf{Q}(t)\}$, then this is equivalent to the statement that *Stokesian thermoviscous fluids are isotropic.*

Equations (5.8.13) to (5.8.20) must be employed along with the usual equations of continuity, balance of momenta, and energy given by (5.7.11) to (5.7.14) to deal with the problems of heat-conducting Stokesian fluids.

The constitutive equations (5.8.13) to (5.8.16) are valid for nonlinear phenomena. An example of constitutively linear thermoviscous fluids is dealt with in Chapter 7.

The equation of heat conduction follows from combining (5.8.13) to (5.8.16) with the equation of energy (5.7.14), that is,

$$\rho\theta \frac{\partial^2 \psi}{\partial \theta^2} \dot{\theta} + \theta \frac{\partial^2 \psi}{\partial \theta \partial \rho^{-1}} d_{kk} + {}_D f_{kl}\, d_{lk} + q_{k,k} + \rho h = 0 \tag{5.8.22}$$

Incompressible Fluids. The constitutive equations for incompressible fluids are obtained from (5.8.13) to (5.8.20) by setting $\rho = \text{constant} = \rho_0$, hence $d_{kk} = 0$ and replacing π by an unknown function $p(\mathrm{x}, t)$. Thus

$$t_{kl} = -p\delta_{kl} + {}_D f_{kl}(\mathrm{d}, \nabla\theta, \theta) \tag{5.8.23}$$

$$q_k = g_k(\mathrm{d}, \nabla\theta, \theta) \tag{5.8.24}$$

(incompressible)

$$\varepsilon = \psi(\theta) - \theta \frac{\partial \psi}{\partial \theta} \tag{5.8.25}$$

$$\eta = -\frac{\partial \psi}{\partial \theta} \tag{5.8.26}$$

where ${}_D\mathrm{f}$ and g are subject to (5.8.18) to (5.8.21) and

$$d_{kk} = 0 \tag{5.8.27}$$

The heat conduction equation (5.8.22) reduces to

$$\rho_0 \theta \frac{\partial^2 \psi}{\partial \theta^2} \dot{\theta} + {}_D f_{kl}\, d_{lk} + q_{k,k} + \rho h = 0 \tag{5.8.28}$$

because of (5.8.27).

5.9. THERMOELASTIC SOLIDS

The effect of heat conduction can be incorporated in the behavior of elastic materials by including temperature gradients among the constitutive variables. From (5.4.25) we deduce that the constitutive equations of thermoelastic solids, in the simplest case, will have the form,

$$t_{kl} = F_{KL}(\mathbf{C}, \theta_{,K}, \theta, \mathrm{X}) X_{K,k} X_{L,l} \tag{5.9.1}$$

$$q_k = Q_K(\mathbf{C}, \theta_{,K}, \theta, \mathrm{X}) X_{K,k} \tag{5.9.2}$$

$$\varepsilon = e(\mathbf{C}, \theta_{,K}, \theta, \mathrm{X}) \tag{5.9.3}$$

$$\eta = n(\mathbf{C}, \theta_{,K}, \theta, \mathrm{X}) \tag{5.9.4}$$

These equations satisfy the axiom of objectivity. They are valid for nonlinear, anisotropic, and inhomogeneous thermoelastic solids. The Clausius-Duhem inequality (5.8.7), in this case, has the specific form

$$\left(-\frac{\rho}{\theta}\frac{\partial\psi}{\partial C_{KL}}+\frac{1}{2\theta}t_{kl}X_{K,k}X_{L,l}\right)\dot{C}_{KL}-\frac{\rho}{\theta}\frac{\partial\psi}{\partial\theta_{,K}}\dot{\theta}_{,K}-\frac{\rho}{\theta}\left(\frac{\partial\psi}{\partial\theta}+\eta\right)\dot{\theta}+\frac{1}{\theta^2}q_k\theta_{,k}\geq 0$$

where we have used (5.5.17) to replace d_{kl} in (5.8.7). This inequality is linear in $\dot{\mathbf{C}}$, $\dot{\theta}_{,k}$ and $\dot{\theta}$. To maintain it for all values of these variables it is necessary and sufficient that their coefficients vanish. Hence,

$$t_{kl}=2\rho\frac{\partial\psi}{\partial C_{KL}}x_{(k,K}x_{l),L}$$

$$0=\frac{\partial\psi}{\partial C_{KL}}x_{[k,K}x_{l],L}$$

$$\frac{\partial\psi}{\partial\theta_{,K}}=0 \tag{5.9.5}$$

$$\eta=-\frac{\partial\psi}{\partial\theta}$$

$$q_k\theta_{,k}\geq 0$$

We have therefore proved

THEOREM 1. *A thermoelastic solid characterized by* (5.9.1) *to* (5.9.4) *is thermodynamically admissible if the stress, internal energy, and entropy are derivable from a potential* $\Sigma\equiv\rho_0\psi$ (*where* Σ *or the free energy* ψ *is a function of* $\mathbf{C}$, θ, *and* $\mathbf{X}$ *only*) *and the heat satisfies the inequality* $(5.9.5)_5$. *More specifically, the constitutive equations of thermoelastic solids must have the forms*

$$t_{kl}=2\frac{\rho}{\rho_0}\frac{\partial\Sigma}{\partial C_{KL}}x_{(k,K}x_{l),L} \tag{5.9.6}$$

$$q_k=Q_K(\mathbf{C},\theta_{,K},\theta,\mathbf{X})X_{K,k} \tag{5.9.7}$$

$$\varepsilon=\frac{1}{\rho_0}\left(\Sigma-\theta\frac{\partial\Sigma}{\partial\theta}\right) \tag{5.9.8}$$

$$\eta=-\frac{1}{\rho_0}\frac{\partial\Sigma}{\partial\theta} \tag{5.9.9}$$

where

$$\frac{1}{\rho_0}\Sigma(\mathbf{C},\theta,\mathbf{X})\equiv\psi \tag{5.9.10}$$

is the free energy subject to the conditions

$$2\frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial C_{KL}} x_{[k,K}x_{l],L} = 0 \tag{5.9.11}$$

and q_k is subject to

$$q_k \theta_{,k} \geq 0 \tag{5.9.12}$$

The condition (5.9.11) is a statement that Σ is invariant to rigid rotations of the spatial frame of reference. It is an identity since $C_{KL} = C_{LK}$.

Upon comparing (5.9.6) with (5.9.1), we find that

$$F_{KL} = 2\frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial C_{MN}} C_{MK}C_{NL} \tag{5.9.13}$$

If we introduce a new heat function G_K by

$$Q_K = G_M(\mathbf{C}, \theta_{,K}, \theta, \mathbf{X})C_{MK} \tag{5.9.14}$$

the inequality (5.9.12) can be put into the form

$$G_K(\mathbf{C}, \theta_{,K}, \theta, \mathbf{X})\theta_{,K} \geq 0 \tag{5.9.15}$$

In terms of G_M the thermal constitutive equation (5.9.7) takes the form

$$q_k = G_K(\mathbf{C}, \theta_{,K}, \theta, \mathbf{X})x_{k,K} \tag{5.9.16}$$

which is valid for anisotropic solids.

If we assume that G_M is continuous in $\theta_{,K}$, then the inequality (5.9.15) implies that

$$G_M(\mathbf{C}, \mathbf{0}, \theta, \mathbf{X}) = \mathbf{0} \quad \text{or} \quad Q_M(\mathbf{C}, \mathbf{0}, \theta, \mathbf{X}) = \mathbf{0} \tag{5.9.17}$$

Hence, *the heat vanishes with the gradient of θ.*

The equation of heat conduction is obtained through the equation of conservation of energy (5.5.15) and using (5.9.6) to (5.9.9). Hence,

$$\frac{\rho\theta}{\rho_0}\left(\frac{\partial^2 \Sigma}{\partial \theta^2}\dot{\theta} + \frac{\partial^2 \Sigma}{\partial\theta\partial C_{KL}}\dot{C}_{KL}\right) + q_{k,k} + \rho h = 0 \tag{5.9.18}$$

Incompressible Solids. For incompressible solids $\rho = \rho_0 =$ constant, and the stress constitutive equations are modified in the same way as in previous sections. Thus

$$t_{kl} = -p\delta_{kl} + 2\frac{\partial \Sigma}{\partial C_{KL}} x_{(k,K}x_{l),L} \tag{5.9.19}$$

where the pressure $p(\mathbf{x}, t)$ is an unknown of the theory.

Material symmetry requirements are provided by use of the axiom of

material invariance. The situation here is similar to the case of the elastic solid (Section 5.5 and Chapter 6). The following section presents only the case of isotropic materials.

5.10. ISOTROPIC THERMOELASTIC MATERIALS

For isotropic materials, the functions Σ and Q_K must be form-invariant under the full orthogonal group of transformations of the material frame, that is,

$$\begin{aligned} \Sigma(\mathbf{C}, \theta, \mathbf{X}) &= \Sigma(\mathbf{S}\mathbf{C}\mathbf{S}^T, \theta, \mathbf{S}\mathbf{X}) \\ \mathbf{S}\mathbf{Q}(\mathbf{C}, \nabla\theta, \theta, \mathbf{X}) &= \mathbf{Q}(\mathbf{S}\mathbf{C}\mathbf{S}^T, \mathbf{S}\nabla\theta, \theta, \mathbf{S}\mathbf{X}) \end{aligned} \tag{5.10.1}$$

where {S} is the full orthogonal group of transformations.[13] The conditions (5.10.1) express the fact that Σ and Q_K must be isotropic functions. This means that Σ shall be a function of the invariants of C, (5 6.12), that is,

$$\Sigma = \Sigma(I_1, I_2, I_3, X) \tag{5.10.2}$$

and Q_K has the form (B9.8), namely,

$$\begin{aligned} \mathbf{P} = K_1\mathbf{a} + K_2(\mathbf{ab} + \mathbf{ba}) + K_3(\mathbf{ab}^2 + \mathbf{b}^2\mathbf{a}) + K_4(\mathbf{a}^2\mathbf{b} - \mathbf{ba}^2) \\ + K_5(\mathbf{a}^2\mathbf{b}^2 - \mathbf{b}^2\mathbf{a}^2) + K_6(\mathbf{ba}^2\mathbf{b}^2 - \mathbf{b}^2\mathbf{a}^2\mathbf{b}) \end{aligned}$$

where K_1 to K_6 are functions of the invariants listed in (B8.3) and

$$P_{KL} \equiv e_{KLM}Q_M, \qquad a_{KL} \equiv e_{KLM}\theta_{,M}, \qquad b_{KL} \equiv C_{KL}$$

Upon multiplying both sides of the above expression of P_{KL} by e_{KLR} and using the identity

$$e_{KLM}e_{KRS} = \delta_{LR}\delta_{MS} - \delta_{LS}\delta_{MR}$$

after some manipulations, we obtain

(5.10.3)

$$\begin{aligned} Q_K = [(K_1 + K_2C_{MM} + K_3C_{NM}C_{MN})\delta_{KL} - K_2C_{KL} - K_3C_{KM}C_{ML}]\theta_{,L} \\ + e_{KLM}(K_4C_{MR} + K_5C_{RS}C_{SM})\theta_{,L}\theta_{,R} \\ + K_6e_{KLM}C_{MN}C_{NP}C_{RL}\theta_{,R}\theta_{,P} \end{aligned}$$

[13] Equation (5.10.1) is valid for anisotropic materials also when {S} is taken as a subgroup of the full orthogonal group of transformations The forms {S} depend on the material symmetry.

This is valid for hemitropic materials. For isotropic materials the requirement of form-invariance of Q_K under the reflection of any one of the material axes X_K imposes the additional conditions that

$$K_4 = K_5 = K_6 = 0$$

Hence for isotropic materials we may write

$$Q_K = (\kappa_0 \delta_{KL} + \kappa_1 C_{KL} + \kappa_2 C_{KM} C_{ML}) \theta_{,L} \tag{5.10.4}$$

where κ_0, κ_1, and κ_2, in accordance with (B8.3) and the above definitions of a and b, are polynomials in the following list of joint invariants of C_{KL} and $\theta_{,K}$.

$$\begin{aligned} &I_1 \equiv C_{KK}, \quad I_2 \equiv C_{KL} C_{LK}, \quad I_3 \equiv C_{KL} C_{LM} C_{MK} \\ &I_4 \equiv \theta_{,K} \theta_{,K}, \quad I_5 \equiv C_{KL} \theta_{,K} \theta_{,L}, \quad I_6 \equiv C_{KM} C_{ML} \theta_{,K} \theta_{,L} \\ &I_7 \equiv C_{KM} C_{MN} C_{NL} \theta_{,K} \theta_{,L} \end{aligned} \tag{5.10.5}$$

Upon writing

$$C_{KL} \equiv x_{k,K} x_{k,L}, \qquad \overset{-1}{c}_{kl} \equiv x_{k,K} x_{l,K} \qquad \theta_{,K} \equiv \theta_{,k} x_{k,K} \tag{5.10.6}$$

we find that the invariants (5.10.5) take the forms

$$\begin{aligned} &I_1 = \overset{-1}{c}_{kk}, \quad I_2 = \overset{-1}{c}_{kl} \overset{-1}{c}_{lk}, \quad I_3 = \overset{-1}{c}_{kl} \overset{-1}{c}_{lm} \overset{-1}{c}_{mk} \\ &I_4 = \overset{-1}{c}_{kl} \theta_{,k} \theta_{,l}, \quad I_5 = \overset{-1}{c}_{kl} \overset{-1}{c}_{lm} \theta_{,k} \theta_{,m}, \quad I_6 = \overset{-1}{c}_{kl} \overset{-1}{c}_{lm} \overset{-1}{c}_{kn} \theta_{,m} \theta_{,n} \\ &I_0 \equiv \theta_{,k} \theta_{,k} \end{aligned} \tag{5.10.7}$$

Here I_1 to I_6 are the expressions of the same listed in (5.10.5) in terms of $\overset{-1}{c}$ and $\theta_{,k}$ and I_0 results from expressing I_7 in terms of the I_1 to I_6 and I_0 through the use of the Cayley-Hamilton theorem. In the same way as in Section 5.6 we find for the expression of stress (5.6.18)

$$\begin{aligned} t_{kl} = \frac{\rho}{\rho_0} \Bigg\{ &(I_1^3 - 3I_1 I_2 + 2I_3) \frac{\partial \Sigma}{\partial I_3} \delta_{kl} \\ &+ \left[2 \frac{\partial \Sigma}{\partial I_1} - 3(I_1^2 - I_2) \frac{\partial \Sigma}{\partial I_3} \right] \overset{-1}{c}_{kl} + \left(4 \frac{\partial \Sigma}{\partial I_2} + 6 I_1 \frac{\partial \Sigma}{\partial I_3} \right) \overset{-1}{c}_{km} \overset{-1}{c}_{ml} \Bigg\} \end{aligned} \tag{5.10.8}$$

The constitutive equation for heat is obtained by carrying (5.10.4) into (5.9.7) and using (5.10.6). Hence,

$$q_k = (\kappa_0 \delta_{kl} + \kappa_1 \overset{-1}{c}_{kl} + \kappa_2 \overset{-1}{c}_{km} \overset{-1}{c}_{ml}) \theta_{,l} \tag{5.10.9}$$

where κ_0, κ_1, and κ_2 are functions of the invariants listed in (5.10.7).

Equation $(5.9.5)_2$ expresses the fact that Σ is invariant under rigid motions of the frame of reference and of course, $t_{kl} = t_{lk}$ and $(5.9.5)_5$ restricts the coefficients κ_α by

$$\kappa_0 I_0 + \kappa_1 I_4 + \kappa_2 I_5 \geq 0 \tag{5.10.10}$$

For the incompressible solid we have $III_C = 1, \rho_0 = \rho$ and $\Sigma = \Sigma(I_1, I_2, X)$. In this case, we add an arbitrary pressure term $-p(x, t)$ on the right of (5.10.8). This results in

$$t_{kl} = -p\delta_{kl} + 2\frac{\partial \Sigma}{\partial I_1} c^{-1}_{kl} + 4\frac{\partial \Sigma}{\partial I_2} c^{-1}_{km} c^{-1}_{ml} \tag{5.10.11}$$

The expression of the heat (5.10.9) and the inequality (5.10.10) does not change form except that κ_1, κ_2, and κ_3 are now independent of I_3.

PROBLEMS

5.1 At a material point, X, of a body, the stress rate, heat, energy, and entropy are assumed to depend on the stress and the density at that point Apply the axioms of constitution to determine the final forms of the constitutive equations

5.2. At a material point of a body, the constitutive dependent variables (stress, heat, energy and entropy) are assumed to be linear functions in the strain **E** and linear functionals in the rate of strain, $\dot{\mathbf{E}}$. By use of the axioms of constitution, find the forms of the constitutive equations For the case of isotropic solids, carry out all reductions to get the final forms of these equations

5.3. A purely elastic solid is known to possess an axis of symmetry Find the stress constitutive equations.

5.4. A rigid body is known to possess an axis of symmetry for heat. Find the constitutive equation of heat. For the linear theory, how many independent constitutive coefficients are needed? Obtain the heat conduction equation for this case.

5.5. For isotropic, hyperelastic solids, prove that the principal axes of stress and strain coincide.

5.6. For an isotropic, hyperelastic solid, find the stress constitutive equations up to and including the second-order terms in the eulerian strains.

5.7. For a Stokesian fluid, find the second-degree constitutive equations in terms of the deformation rate tensor.

5.8. Prove that the second-degree Stokesian fluids are linear in the components of deformation-rate tensor, **d**, the coefficients of which being nonlinear in the invariants of **d**.

5.9. If a fluid is described by a set of constitutive equations in which the stress,

heat, energy, and entropy are assumed to depend on the density and the first- and second-order gradients of temperature, what form must the constitutive equations have? Obtain the equation of heat conduction.

5.10. Find the constitutive equations of an isotropic thermoelastic solid which includes all second-degree terms in the strain and the temperature gradient.

5.11. A constitutively linear, thermoviscous Stokesian fluid is known to possess an equation of state of the Van der Waals type (*cf.* Problem 4.5). Find the equation of heat conduction.

5.12. A viscoelastic solid is characterized by constitutive equations in which the stress, heat, energy, and entropy depend on the temperature, strain, and temperature gradient and their first two time rates. The solid is isotropic and linear. Determine the constitutive equations and find the restrictions emanating from the Clausius-Duhem inequality.

5.13. (Short Term Paper) Study literature and present a discussion of the classification of all crystal classes.

5.14. (Short Term Paper) Make a literature survey on the invariants of a vector and two symmetric tensors of the second order.

5.15. (Short Term Paper) Study literature and derive the constitutive equations of a solid in which the constitutive dependent variables (t, q, ε, and η) are linear functionals in the lagrangian strain, strain rate, and temperature gradient for all times prior to t.

5.16. (Short Term Paper) Study literature and give a discussion of the constitutive equations of a porous, elastic solid (elastic solid containing ideal fluids).

5.17. (Short Term Paper) Give a derivation of the constitutive equation of diffusion when two different fluids are mixed.

5.18. (Short Term Paper) A material is called polar if the couple stress does not vanish. Study literature and present a discussion of the polar, elastic solids.

5.19. (Short Term Paper) Give a survey of polar fluids.

CHAPTER 6

Theory of Elasticity

6.1. SCOPE OF THE CHAPTER

This chapter is concerned with the development of the basic equations of the linear theory of elasticity and some illustrative solutions of these equations As in the case of the remaining chapters, our primary interest is the derivation of these equations rather than an exhaustive exposition of the methods of solution. In Section 6.2 we present an account of the linear constitutive equations for anisotropic and isotropic elastic solids. Various types of material symmetry restrictions are studied. Section 6.3 is concerned with the conditions that must be imposed on the elastic coefficients in order that they represent thermally stable materials. Section 6 4 is devoted to a discussion of the experimental determination of elastic constants. In Section 6.5 a résumé of the basic equations of balance and jump as well as of constitutive equations is presented. The field equation of Navier is given together with boundary and initial conditions Section 6.6 is an account of curvilinear coordinates. Here we give the forms of the basic equations in orthogonal curvilinear coordinates. The results for cylindrical and spherical coordinates are presented at the end of this Section. Sections 6.7 to 6 12 are devoted to illustrative solutions of static and dynamic problems. Among the static solutions given we cite the half-space subject to uniform pressure (Section 6 7), the cylindrical tube subject to pressures (Section 6.8), and the spherical shell subject to pressures (Section 6 9) Section 6 10 discusses elastic waves in isotropic solids Both plane and spherical waves are presented. In Section 6 11 we return to the problem of a half space under a dynamical load, and in Section 6.12, to that of a spherical cavity

under dynamical pressure. The static and dynamic versions of the same type of problems are therefore exhibited for the benefit of the student. This scheme is, in fact, continued throughout the remaining chapters whenever possible. For instance, in Chapter 8 we take up some of these problems again for thermoelastic solids, and in Chapter 9, for some viscoelastic materials.

6.2. LINEAR CONSTITUTIVE EQUATIONS

In Section (5.5) we found that the most general constitutive equations of an elastic solid have the forms expressed by equations (5.5.23) to (5.5.26). For our purpose we write these equations in terms of the lagrangian strain measure $\mathbf{E}$ related to $\mathbf{C}$ by

$$E_{KL} = \tfrac{1}{2}(C_{KL} - \delta_{KL})$$

Thus the potential Σ is now to be considered a function of $\mathbf{E}$, θ and $\mathbf{X}$, that is,

$$\Sigma = \Sigma(\mathbf{E}, \theta, \mathbf{X}) = \rho_0 \psi \tag{6.2.1}$$

where ψ is the free energy introduced in Section 5.5. The constitutive equations (5.5.23) to (5.5.26) now take the forms

$$t_{kl} = \frac{\rho}{\rho_0} \frac{\partial \Sigma}{\partial E_{KL}} x_{(k,K} x_{l),L} \tag{6.2.2}$$

$$q_k = 0 \tag{6.2.3}$$

$$\varepsilon = \frac{1}{\rho_0}\left(\Sigma - \theta \frac{\partial \Sigma}{\partial \theta}\right) \tag{6.2.4}$$

$$\eta = -\frac{1}{\rho_0} \frac{\partial \Sigma}{\partial \theta} \tag{6.2.5}$$

when Σ is subject to the conditions

$$\frac{\rho}{\rho_0} \frac{\partial \Sigma}{\partial E_{KL}} x_{[k,K} x_{l],L} = 0 \tag{6.2.6}$$

which are satisfied identically since $E_{KL} = E_{LK}$.

By using the master equation (6.2.2) of nonlinear elastic solids, we can derive various approximate theories. A polynomial approximation in the strain components E_{KL} is based upon an expansion of the form

$$\Sigma = \Sigma_0 + \Sigma_{KL} E_{KL} + \tfrac{1}{2}\Sigma_{KLMN} E_{KL} E_{MN} + \cdots \tag{6.2.7}$$

where Σ_0, Σ_{KL}, Σ_{KLMN} are, in general, functions of X, for inhomogeneous solids, and θ. For the homogeneous solid they are functions of θ only. Without loss in generality we take

$$\Sigma_{KL} = \Sigma_{LK}$$
$$\Sigma_{KLMN} = \Sigma_{MNKL} = \Sigma_{LKMN} = \Sigma_{KLNM} \tag{6.2 8}$$

since the term containing this latter tensor is symmetric in E_{KL} and E_{MN} and $E_{KL} = E_{LK}$.

We are interested only in the linear theory so that we need not retain terms beyond the quadratic terms in E If we also substitute

$$\frac{\rho}{\rho_0} \simeq 1 - \mathrm{I}_E, \qquad x_{k,K} = (\delta_{MK} + \bar{E}_{MK} + \bar{R}_{MK})\,\delta_{Mk}$$

into (6.2 2) and recall that for the linear theory $E_{KL} \simeq \bar{E}_{KL}$ and δ_{Mk} is the Kronecker delta, we get

$$t_{kl} = [(1 - \mathrm{I}_E)\Sigma_{KL} + \Sigma_{ML}(\bar{E}_{KM} + \bar{R}_{KM}) + \Sigma_{KM}(\bar{E}_{LM} + \bar{R}_{LM}) + \Sigma_{KLMN}\bar{E}_{MN}]\,\delta_{Kk}\,\delta_{Ll} \tag{6.2 9}$$

It can be seen that (6.2 6) is satisfied automatically. When the natural state is stress free, t_{kl} must vanish, that is, whenever $x_k = X_K\,\delta_{Kk}$, $t_{kl} = 0$ In this case $\bar{E} = \bar{R} = 0$ and we get $\Sigma_{KL} = 0$. Therefore for vanishing initial stress, we have

$$t_{kl} = \Sigma_{KLMN}\bar{E}_{MN}\,\delta_{Kk}\,\delta_{Ll} \tag{6 2 10}$$

where the coefficients Σ_{KLMN} are subject to the restrictions given by (6.2.8) These symmetry conditions show that the total number of the elastic coefficients Σ_{KLMN} is 21 If we remember that for small strains

$$\bar{E}_{MN} \simeq \bar{e}_{mn}\,\delta_{Mm}\,\delta_{Nn} \tag{6.2 11}$$

we may express (6 2.10) in spatial form

$$t_{kl} = \sigma_{klmn}\bar{e}_{mn} \tag{6 2 12}$$

where

$$\sigma_{klmn} \equiv \Sigma_{KLMN}\,\delta_{Kk}\,\delta_{Ll}\,\delta_{Mm}\,\delta_{Nn} \tag{6 2.13}$$

are the spatial elastic moduli that are subject to the symmetry conditions

$$\sigma_{klmn} = \sigma_{mnkl} = \sigma_{lkmn} = \sigma_{klnm} \tag{6.2.14}$$

The stress potential can now be expressed as

$$\Sigma = \tfrac{1}{2}\sigma_{klmn}\bar{e}_{kl}\bar{e}_{mn} \tag{6.2.15}$$

Material Symmetry. Material symmetry (when it exists) imposes restrictions on these coefficients, thus further reducing the number of independent coefficients. Material symmetry is exhibited by the form-invariance of Σ under a group of transformations of the material axes. Accordingly, for a homogeneous solid, if there is a symmetry group $\{S\}$ for which under the transformation of axes X_K

$$X'_K = S_{KL}X_L \tag{6.2.16}$$

$$S_{KL}S_{ML} = S_{LK}S_{LM} = \delta_{KM}, \qquad \det S_{KL} = \pm 1$$

the expression of Σ remains invariant, we have

$$\Sigma = \Sigma_0 + \Sigma_{KL}E'_{KL} + \tfrac{1}{2}\Sigma_{KLMN}E'_{KL}E'_{MN}$$

Upon substitution of

$$E'_{KL} = E_{MN}S_{KM}S_{LN}$$

into this expression and comparing the result with (6.2.7) we find that

$$\Sigma_{KL} = \Sigma_{MN}S_{MK}S_{NL}$$
$$\Sigma_{KLMN} = \Sigma_{PQRS}S_{PK}S_{QL}S_{RM}S_{SN} \tag{6.2.17}$$

Under the conditions of symmetry (6.2.8) we may organize these material constants into the following symmetric square matrices with 6 and 21 independent members.

$$\|\Sigma_{KL}\| = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{12} \\ & \Sigma_{22} & \Sigma_{23} \\ & & \Sigma_{33} \end{bmatrix}$$

$$\|\Sigma_{KLMN}\| = \begin{bmatrix} \Sigma_{1111} & \Sigma_{1122} & \Sigma_{1133} & \Sigma_{1123} & \Sigma_{1113} & \Sigma_{1112} \\ & \Sigma_{2222} & \Sigma_{2233} & \Sigma_{2223} & \Sigma_{2213} & \Sigma_{2212} \\ & & \Sigma_{3333} & \Sigma_{3323} & \Sigma_{3313} & \Sigma_{3312} \\ & & & \Sigma_{2323} & \Sigma_{2313} & \Sigma_{2312} \\ & & & & \Sigma_{1313} & \Sigma_{1312} \\ & & & & & \Sigma_{1212} \end{bmatrix} \tag{6.2.18}$$

where the unwritten members are identical to those located symmetrically with respect to the main diagonal of these matrices. We now proceed to discuss the consequences of various material symmetries.[1]

(i) *Symmetry with Respect to a Plane.* If the material is symmetric with respect to the plane $X_3 = 0$, upon the transformation of axes

$$X'_1 = X_1, \qquad X'_2 = X_2, \qquad X'_3 = -X_3$$

[1] For discussions of anisotropic elasticity, see Love [1944], Lekhnitskii [1950], and Hearman [1961].

Σ should remain invariant. This is a reflection about the plane $X_3 = 0$. In this case, according to (6 2.16) we have

$$S_{11} = 1, \qquad S_{22} = 1, \qquad S_{33} = -1$$

all other $S_{KL} = 0, \qquad \det S_{KL} = -1$

Upon substituting this into $(6.12.17)_1$ we find that $\Sigma_{13} = -\Sigma_{13}$, $\Sigma_{23} = -\Sigma_{23}$, so that $\Sigma_{13} = \Sigma_{23} = 0$. An examination of $(6.2.17)_2$ reveals that all Σ_{KLMN} in which the index 3 occurs an odd number of times are zero. Thus the matrices (6 2.18), respectively, reduce to the forms:

$$\|\Sigma_{KL}\| = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & 0 \\ & \Sigma_{22} & 0 \\ & & \Sigma_{33} \end{bmatrix}$$

(6 2.19)

$$\|\Sigma_{KLMN}\| = \begin{bmatrix} \Sigma_{1111} & \Sigma_{1122} & \Sigma_{1133} & 0 & 0 & \Sigma_{1112} \\ & \Sigma_{2222} & \Sigma_{2233} & 0 & 0 & \Sigma_{2212} \\ & & \Sigma_{3333} & 0 & 0 & \Sigma_{3312} \\ & & & \Sigma_{2323} & \Sigma_{2313} & 0 \\ & & & & \Sigma_{1313} & 0 \\ & & & & & \Sigma_{1212} \end{bmatrix}$$

The number of independent Σ_{KL} is now 4 and that of Σ_{KLMN} is 13.

(ii) *Symmetry with Respect to Two Orthogonal Planes (Orthotropy).* If the material properties are symmetrical with respect to two orthogonal planes $X_3 = 0$ and $X_1 = 0$, then upon the transformation of axes

$$X'_1 = -X_1, \qquad X'_2 = X_2, \qquad X'_3 = -X_3$$

and

$$X'_1 = -X_1, \qquad X'_2 = X_2, \qquad X'_3 = X_3$$

Σ remains invariant. By use of (6.2.17), we find that the nonzero elastic coefficients are further reduced to

$$\|\Sigma_{KL}\| = \begin{bmatrix} \Sigma_{11} & 0 & 0 \\ & \Sigma_{22} & 0 \\ & & \Sigma_{33} \end{bmatrix}$$

(6.2.20)

$$\|\Sigma_{KLMN}\| = \begin{bmatrix} \Sigma_{1111} & \Sigma_{1122} & \Sigma_{1133} & 0 & 0 & 0 \\ & \Sigma_{2222} & \Sigma_{2233} & 0 & 0 & 0 \\ & & \Sigma_{3333} & 0 & 0 & 0 \\ & & & \Sigma_{2323} & 0 & 0 \\ & & & & \Sigma_{1313} & 0 \\ & & & & & \Sigma_{1212} \end{bmatrix}$$

The number of Σ_{KL} is now 3, and that of Σ_{KLMN} is 9. It is obvious from this result that the body is also symmetric about the plane $X_2 = 0$. Such a material therefore is called an *orthotropic* material. The elastic properties of certain woods are approximated fairly well with the orthotropic elastic moduli.

(iii) *Hexagonal System.* A material is said to have hexagonal symmetry if upon an arbitrary rotation of axes X_K about one of the axes X_L, for example, X_3, Σ remains unaltered. In this case

$$X'_1 = X_1 \cos\alpha + X_2 \sin\alpha, \qquad X'_2 = -X_1 \sin\alpha + X_2 \cos\alpha, \qquad X'_3 = X_3$$

so that

$$\begin{aligned} S_{11} &= \cos\alpha & S_{12} &= \sin\alpha & S_{13} &= 0 \\ S_{21} &= -\sin\alpha & S_{22} &= \cos\alpha & S_{23} &= 0 \\ S_{31} &= 0 & S_{32} &= 0 & S_{33} &= 1 \end{aligned}$$

Now through $(6.2.17)_1$ we find that

$$\begin{aligned} \Sigma_{11} &= \Sigma_{11}\cos^2\alpha + \Sigma_{22}\sin^2\alpha - 2\Sigma_{12}\sin\alpha\cos\alpha \\ \Sigma_{22} &= \Sigma_{11}\sin^2\alpha + \Sigma_{22}\cos^2\alpha + 2\Sigma_{12}\sin\alpha\cos\alpha \\ \Sigma_{12} &= (\Sigma_{11} - \Sigma_{22})\sin\alpha\cos\alpha + \Sigma_{12}(\cos^2\alpha - \sin^2\alpha) \\ \Sigma_{13} &= \Sigma_{13}\cos\alpha - \Sigma_{23}\sin\alpha \\ \Sigma_{23} &= \Sigma_{13}\sin\alpha + \Sigma_{23}\cos\alpha, \qquad \Sigma_{33} = \Sigma_{33} \end{aligned}$$

If these equations are to be valid for any arbitrary α we must have

$$\Sigma_{11} = \Sigma_{22}, \qquad \Sigma_{12} = \Sigma_{13} = \Sigma_{23} = 0$$

Similarly we can apply the above transformation to $(6.2.17)_2$. After some algebraic manipulations, we find various relations among the components Σ_{KLMN}. The matrices of Σ_{KL} and Σ_{KLMN} for hexagonal symmetry thus reduce to

$$\|\Sigma_{KL}\| = \begin{bmatrix} \Sigma_{11} & 0 & 0 \\ & \Sigma_{11} & 0 \\ & & \Sigma_{33} \end{bmatrix}$$

(6.2.21)

$$\|\Sigma_{KLMN}\| = \begin{bmatrix} \Sigma_{1111} & \Sigma_{1122} & \Sigma_{1133} & 0 & 0 & 0 \\ & \Sigma_{1111} & \Sigma_{1133} & 0 & 0 & 0 \\ & & \Sigma_{3333} & 0 & 0 & 0 \\ & & & \Sigma_{1313} & 0 & 0 \\ & & & & \Sigma_{1313} & 0 \\ & & & & & \frac{1}{2}(\Sigma_{1111} - \Sigma_{1122}) \end{bmatrix}$$

The number of independent constants for Σ_{KL} is now 2 and for Σ_{KLMN} is 5.

(iv) *Isotropic Solids.* An isotropic solid possesses no preferred direction with respect to its elastic properties. This means that (6.2.17) must be valid for the full orthogonal group of transformations of X_K. This may be obtained by applying arbitrary rotations about two orthogonal axes. By repeating the arguments of (iii) we obtain further conditions on Σ_{KL} and Σ_{KLMN}.

$$\Sigma_{KL} \equiv \alpha_E \delta_{KL}, \qquad \Sigma_{1313} = \tfrac{1}{2}(\Sigma_{1111} - \Sigma_{1122}) \equiv \mu_E$$

$$\Sigma_{3333} = \Sigma_{1111}, \qquad \Sigma_{1122} = \Sigma_{1133} \equiv \lambda_E$$

so that

$$\|\Sigma_{KL}\| = \begin{bmatrix} \alpha_E & 0 & 0 \\ & \alpha_E & 0 \\ & & \alpha_E \end{bmatrix}$$

(6.2.22)

$$\|\Sigma_{KLMN}\| = \begin{bmatrix} \lambda_E + 2\mu_E & \lambda_E & \lambda_E & 0 & 0 & 0 \\ & \lambda_E + 2\mu_E & \lambda_E & 0 & 0 & 0 \\ & & \lambda_E + 2\mu_E & 0 & 0 & 0 \\ & & & \mu_E & 0 & 0 \\ & & & & \mu_E & 0 \\ & & & & & \mu_E \end{bmatrix}$$

where λ_E and μ_E are known as the *Lamé constants*. For this case, we may also express (6.2.22) in index notation

(6.2.23)

$$\Sigma_{KL} = \alpha_E \delta_{KL}$$

$$\Sigma_{KLMN} = \lambda_E \delta_{KL} \delta_{MN} + \mu_E(\delta_{KM} \delta_{LN} + \delta_{KN} \delta_{LM})$$

Substitution of these into (6.2.7) gives the stress potential

(6.2.24)

$$\Sigma = \Sigma_0 + \alpha_E \mathrm{I}_E + \tfrac{1}{2}\lambda_E \mathrm{I}_E^2 + \mu_E E_{KL} E_{LK}$$

$$= \Sigma_0 + \alpha_E \mathrm{I}_E + \tfrac{1}{2}(\lambda_E + 2\mu_E)\mathrm{I}_E^2 - 2\mu_E \mathrm{II}_E$$

where we employed the relation

$$E_{KL} E_{LK} = \mathrm{I}_E^2 - 2\mathrm{II}_E$$

to obtain the last form of Σ. The stress-strain relations follow from carrying (6.2.23) into (6.2.9)

(6.2.25) $$t_{kl} = [\alpha_E \delta_{KL} + (\lambda_E - \alpha_E)\mathrm{I}_E \delta_{KL} + 2(\mu_E + \alpha_E)\tilde{E}_{KL}]\, \delta_{Kk}\, \delta_{Ll}$$

In the linear theory we have

(6.2.26) $$\tilde{e}_{kl} = \tilde{E}_{KL}\, \delta_{Kk}\, \delta_{Ll}, \qquad \mathrm{I}_E \simeq \mathrm{I}_e$$

so that (6.2.25) may also be written as

$$t_{kl} = \alpha_e \delta_{kl} + (\lambda_e - \alpha_e)\mathrm{I}_{\tilde{e}} \delta_{kl} + 2(\mu_e - \alpha_e)\tilde{e}_{kl} \tag{6.2.27}$$

where

$$\alpha_e \equiv \alpha_E, \qquad \lambda_e \equiv \lambda_E, \qquad \mu_e \equiv \mu_E + 2\alpha_E \tag{6.2.28}$$

In this case, the stress potential Σ may also be expressed in terms of the invariants of $\tilde{\mathbf{e}}$. To this end, through (1.10.23), (1.10.24), and (1.10.27), we find that

$$\mathrm{I}_E = \frac{\mathrm{I}_e - 4\mathrm{II}_e + 12\mathrm{III}_e}{1 - 2\mathrm{I}_e + 4\mathrm{II}_e - 8\mathrm{III}_e}, \qquad \mathrm{II}_E = \frac{\mathrm{II}_e - 6\mathrm{III}_e}{1 - 2\mathrm{I}_e + 4\mathrm{II}_e - 8\mathrm{III}_e}$$

$$\mathrm{III}_E = \frac{\mathrm{III}_e}{1 - 2\mathrm{I}_e + 4\mathrm{II}_e - 8\mathrm{III}_e} \tag{6.2.29}$$

Substituting the first two of these into (6.2.24) and retaining only the quadratic terms in $\mathbf{e}$, we find that

$$\Sigma = \Sigma_0 + \alpha_e \mathrm{I}_{\tilde{e}} + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}}^2 - 2\mu_e \mathrm{II}_{\tilde{e}} \tag{6.2.30}$$

The fact that the stress potential, for an isotropic solid, is a function of the invariants I_e, II_e, and III_e was shown, for the general case, by a theorem in Section 5.6. Indeed, the stress constitutive equation (6.2.27) may also be derived from (5.6.18) and a polynomial approximation of Σ in terms of the invariants I_e, II_e and III_e. The result so obtained is identical to (6.2.27). Such a method when applied to a general function $\Sigma = \Sigma(\mathrm{I}_e, \mathrm{II}_e, \mathrm{III}_e, \theta)$ leads to nonlinear constitutive equations of the theory of elasticity.[2]

From the structure of (6.2.27) is is clear that:

THEOREM 1. *In a linear isotropic elastic solid the only possible initial stress is a state of hydrostatic stress.*

This theorem is valid also for the nonlinear theory, as can be seen from (5.6.8) with $\overset{-1}{\mathbf{c}} = \mathbf{I}$.

For an initial stress-free state we have the *Hooke-Cauchy* law of the classical linear theory of isotropic solids.

$$t_{kl} = \lambda_e \tilde{e}_{mm} \delta_{kl} + 2\mu_e \tilde{e}_{kl} \tag{6.2.31}$$

and the corresponding stress potential,

$$\Sigma = \rho_0(\varepsilon - \theta\eta) = \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}}^2 - 2\mu_e \mathrm{II}_{\tilde{e}} \tag{6.2.32}$$

[2] See Eq. (5.6.18). For other forms of nonlinear constitutive equations and various approximate theories, see Eringen [1962, Arts. 47, 60].

In this expression we omitted the unessential initial energy Σ_0 since the origin of Σ can be so chosen as to make $\Sigma_0 = 0$.

The expression of entropy for the general linear anisotropic solid is obtained by using (6.2.7) in (6.2.5). Hence

$$\eta = -\frac{1}{\rho_0}\left(\frac{\partial \Sigma_0}{\partial \theta} + \frac{\partial \Sigma_{KL}}{\partial \theta} E_{KL} + \frac{1}{2}\frac{\partial \Sigma_{KLMN}}{\partial \theta} E_{KL}E_{MN}\right) \tag{6.2.33}$$

For the isotropic solid (and with $\Sigma_0 = 0$) this reduces to

$$\eta = -\frac{1}{\rho_0}\left[\frac{1}{2}\left(\frac{\partial \lambda_e}{\partial \theta} + 2\frac{\partial \mu_e}{\partial \theta}\right) I_E^2 - 2\frac{\partial \mu_e}{\partial \theta} II_E\right] \tag{6.2.34}$$

Elastic solids are in thermal equilibrium [cf. eq (5.5.29)] $q = 0$ and ε and η are not coupled. Therefore, we need not make any reference to q, ε, and η any further, and the constitutive theory of elasticity will be governed fully with the stress constitutive equations

Incompressible Materials. In this case we have

$$\frac{\rho_0}{\rho} = \sqrt{III_C} = (1 + 2I_E + 4II_E + 8III_E)^{1/2} = 1 \tag{6.2.35}$$

For small strains through the binomial expansion, we get

$$\frac{\rho_0}{\rho} = 1 + I_E + 0(|\tilde{E}|^2) = 1$$

This means that in the linear theory of incompressible materials we have

$$I_E = 0 \quad \text{or} \quad I_e = 0 \tag{6.2.36}$$

For the linear theory, (6.2.7) is now subject to the conditions (6.2.36). Using the same argument that led to the general equation (5.5.33) of incompressible elastic solids, we find for the linear isotropic solid

$$t_{kl} = -p\delta_{kl} + 2\mu_e \tilde{e}_{kl} \tag{6.2.37}$$

$$\Sigma = \rho_0(\varepsilon - \theta\eta) = -2\mu_e II_E \tag{6.2.38}$$

where $p(\mathbf{x}, t)$ is an unknown of the theory.

6.3. RESTRICTIONS ON ELASTIC COEFFICIENTS

Physical observations confirm that the volume of a material subjected to hydrostatic pressure becomes smaller. Similarly, in any state of strain, the direction of the shear in any plane is the same as the direction of the

shear stress in that plane. More abstractly, the conditions of stability of thermal equilibrium impose restrictions upon the elastic moduli λ_e and μ_e. In this article, we discuss the consequence of such restrictions that must be placed upon the elastic moduli in order that they may represent a real material adequately.

We have found that the stress constitutive equations of isotropic linear elastic materials and the corresponding stress potential are given, respectively, by

$$t_{kl} = \lambda_e \tilde{e}_{mm}\delta_{kl} + 2\mu_e \tilde{e}_{kl} \tag{6.3.1}$$

$$\Sigma = \rho_0(\varepsilon - \theta\eta) = \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}}^2 - 2\mu_e \mathrm{II}_{\tilde{e}} \tag{6.3.2}$$

We assume that (6.3.1) is soluble for $\tilde{e}_{kl}$. In fact, setting $k = l$, this equation gives

$$t_{kk} \equiv \mathrm{I}_t = (3\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}} \tag{6.3.3}$$

Elimination of $\mathrm{I}_{\tilde{e}}$ from (6.3.1) by use of this equation gives

$$\tilde{e}_{kl} = -\frac{\lambda_e}{2\mu_e(3\lambda_e + 2\mu_e)} t_{mm}\delta_{kl} + \frac{1}{2\mu_e} t_{kl} \tag{6.3.4}$$

We observe that for $\tilde{e}_{kl}$ *to be uniquely determined by* t_{kl} *we must have*

$$\mu_e \neq 0, \qquad 3\lambda_e + 2\mu_e \neq 0 \tag{6.3.5}$$

and in order not to have zero strain for a finite stress we must have

$$|\mu_e| < \infty, \qquad |3\lambda_e + 2\mu_e| < \infty \tag{6.3.6}$$

(*i*) *Hydrostatic Pressure.* Experimental observations indicate that under hydrostatic pressure the volume of an elastic solid diminishes. The state of stress at a point of the body is said to be hydrostatic if the stress tensor has the form

$$t_{kl} = -\bar{p}\delta_{kl}, \qquad \bar{p} > 0 \tag{6.3.7}$$

From (6.3.3) we find that

$$\bar{p} = -k\mathrm{I}_{\tilde{e}}, \qquad k \equiv \lambda_e + \tfrac{2}{3}\mu_e \tag{6.3.8}$$

where k is called the *modulus of compression* or the *bulk modulus*. For infinitesimal deformation, the volume change is given by $\mathrm{I}_{\tilde{e}}$. The state of stress in a small test body subjected to a constant normal pressure $\bar{p}$ on its surface is approximated by (6.3.7). If the hydrostatic pressure is to diminish volume of the body, we must have $k \geq 0$. Thus

$$0 \leq 3\lambda_e + 2\mu_e$$

(*ii*) *Simple Shear.* Consider a simple shear in which

$$t_{12} \neq 0 \qquad t_{kl} = 0, \qquad (k, l \neq 1, 2)$$

In this case (6.3.4) gives

$$t_{12} = 2\mu_e \tilde{e}_{12} \tag{6.3.9}$$

Experimental observation of small deformations of elastic solids subjected to simple shear indicates that t_{12} and $\tilde{e}_{12}$ have the same direction. Consequently $\mu_e > 0$.

Collecting the results of (i) and (ii) we have

$$0 < 3\lambda_e + 2\mu_e < \infty, \qquad 0 < \mu_e < \infty \tag{6.3.10}$$

The restrictions (6.3.10) on the moduli of isotropic materials are universally agreed. They are also confirmed from various other considerations, such as thermal stability and uniqueness theorems We now present one more argument based on the stability of thermal equilibrium.

(*iii*) *Stability of Thermal Equilibrium.*

THEOREM. *The inequalities* (6 3.10) *are necessary and sufficient for the stress potential* (*or free energy*) *to be nonnegative in any nonrigid deformation for a fixed temperature.*

Sufficiency of (6.3.10) for Σ to be nonnegative can be shown by writing (6.3.2) in the form

$$6\Sigma = (3\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}}^2 + 2\mu_e J_2 \tag{6.3.11}$$

where J_2 is a new invariant given by

$$J_2 \equiv 2\mathrm{I}_{\tilde{e}}^2 - 6\mathrm{II}_{\tilde{e}} = (\tilde{e}_1 - \tilde{e}_2)^2 + (\tilde{e}_2 - \tilde{e}_3)^2 + (\tilde{e}_3 - \tilde{e}_1)^2 \tag{6.3.12}$$

Here $\tilde{e}_\alpha$ are the principal strains. Clearly $J_2 \geq 0$, so that when (6 3.10) is satisfied, then $\Sigma \geq 0$.

To prove that (6.3.10) is necessary, we express Σ in a quadratic form in terms of the principal strains $\tilde{e}_\alpha$, that is,

$$\Sigma = \tfrac{1}{2} K_{\alpha\beta}\, \tilde{e}_\alpha \tilde{e}_\beta \tag{6.3.13}$$

where

$$\|K_{\alpha\beta}\| = \begin{bmatrix} \lambda_e + 2\mu_e & \lambda_e & \lambda_e \\ \lambda_e & \lambda_e + 2\mu_e & \lambda_e \\ \lambda_e & \lambda_e & \lambda_e + 2\mu_e \end{bmatrix}$$

If Σ is to be a nonnegative function of $\tilde{e}_\alpha$, then the principal values of $K_{\alpha\beta}$ must be nonnegative so that (6 3.13) is an ellipsoid. The principal values

K_α of $K_{\alpha\beta}$ are the roots of the equation

$$\det (K_{\alpha\beta} - K\delta_{\alpha\beta}) = 0$$

This gives

$$(\lambda_e + 2\mu_e - K)^3 - 3\lambda_e{}^2(\lambda_e + 2\mu_e - K) + 2\lambda_e{}^3 = 0$$

whose roots are

$$K_1 = K_2 = 2\mu_e, \qquad K_3 = 3\lambda_e + 2\mu_e$$

Thus the quadratic form (6.3.13) is an ellipsoid of revolution when (6.3.10) is satisfied. By definition we have $\Sigma \equiv \rho_0(\varepsilon - \theta\eta) = \rho_0\psi$, hence the proof of the theorem.

Thus we have proved that under the conditions (6.3.10) the *free energy* is a convex function of $\tilde{e}_{kl}$ for fixed θ. In this case, thermal equilibrium is said to be stable. *The conditions* (6.3.10) *are therefore necessary and sufficient for stable thermal equilibrium.*

The conditions resulting from nonnegativity of the stress potential for an anisotropic solid are more complicated. However, the methods of approach just used can be applied to various types of solids discussed in Section 6.2. We shall not pursue this point further.

6.4. EXPERIMENTAL DETERMINATION OF ELASTIC CONSTANTS

The linear theory presented above has found extensive application in the deformation of metallic structures. Steel, aluminum, copper, zinc, and many other materials possess linear stress-strain curves in fairly large regions of stress close to a failure point. For isotropic elastic materials, the stress-strain relations contain only two constants, λ_e and μ_e. Therefore, we only need to perform two experiments to determine these constants. In the engineering literature, often two other constants (*Young's modulus* E and *Poisson's ratio* ν) are encountered. These are connected to λ_e and μ_e by

$$\lambda_e \equiv \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu_e \equiv G = \frac{E}{2(1+\nu)} \tag{6.4.1}$$

These constants are appropriate for the *tension* test and/or the *simple shear* test, respectively.

(*a*) *Uniaxial Tension Test.* To determine E and ν, a circular cylinder is subjected to a uniform axial tension, $t_{11} = T$ and all other $t_{kl} = 0$ (Figure 6.4.1). Through (6.3.4) for this case we find that

$$\tilde{e}_{11} = \frac{T}{E} \equiv e \qquad \tilde{e}_{22} = \tilde{e}_{33} = -\frac{\nu}{E}T = -\nu e$$
$$\tilde{e}_{23} = \tilde{e}_{31} = \tilde{e}_{12} = 0 \tag{6.4.2}$$

Thus for $T > 0$ ($T < 0$), cylinder of unit length and of unit diameter will elongate (shorten) an amount T/E, and its diameter will contract (expand) an amount $\nu T/E$. By measuring this elongation and contraction one can determine E and ν, and through (6.4.1), λ_e and μ_e. If T is plotted against e, we obtain a straight line through the origin, the slope of which gives E.

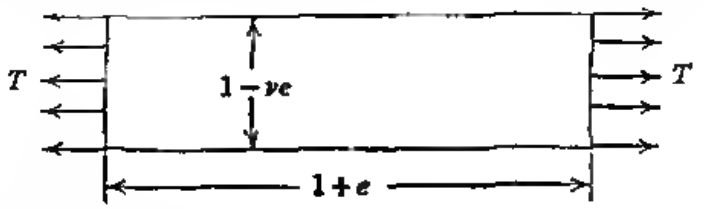


Fig. 6.4.1 Uniaxial tension test.

This test was enunciated as a law in 1676 by Robert Hooke in an anagram "ceiinossttuu" which was translated by him two years later as *Ut Tensio sic vis*, "the extension is proportional to the force." Thus it is known as Hooke's law.

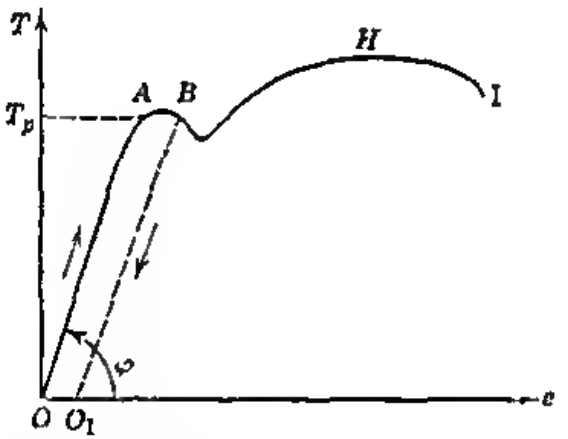


Fig. 6.4.2 Tension test (linear materials for $0 < T < T_p$)

Experiments conducted in the laboratory indicate that the result predicted is valid for soft and *ductile* materials, such as steel and aluminum, up to a limit marked A on the diagram (Figure 6.4.2). In many materials, upon the relief of load, a small permanent elongation of the order of magnitude 10^{-5} in/in is left. However, this is negligible for all engineering purposes. The greatest tension T_A applied without producing a permanent deformation is called the *elastic limit*. When T exceeds T_A, a material exhibits both elastic and *plastic* behavior. In some neighborhood marked by B, *yielding* occurs; this is noticed by the fact that upon relief of the tension a permanent strain is observed. After this point, with a small increase in load T, the rod suddenly stretches a large amount. Finally, the

maximum (ultimate) load is reached at the point H; then, the bar breaks at I. When the ultimate stress is reached, brittle materials, such as carbon steel, break; but ductile materials begin to "neck" (that is, the cross section of the bar is reduced a great deal). For low-carbon steel, $T_A =$ 35,000 psi; the ultimate stress $T_H = 60{,}000$ psi. For high-carbon steel, the ultimate stress may exceed 200,000 psi.

If the yield point is reached, upon the reduction of load, the strain reduces on a straight line parallel to OA leaving a permanent strain OO_1, for $T = 0$. Thus the theory of elasticity applies only in the region $O \leq T \leq T_A$. Beyond that region, a new theory is needed which deals with elastic-plastic materials. For the nonlinear elastic materials the straight line OA is curved as in Figure 6.4.3. Upon the relief of load, however, the strain reduces to zero following the same curve OA. For experiments on nonlinear elasticity see Eringen [1962, Art 61].

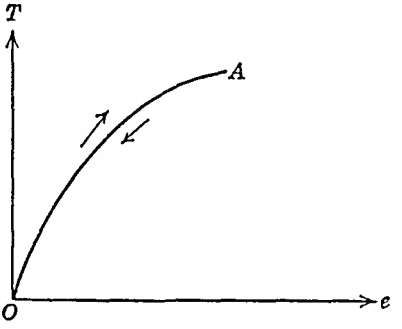


Fig. 6.4.3 Tension test (nonlinear materials).

(*b*) ***Simple Shear Test.*** In the case of a square block subjected to simple shear (Figure 6.4.4), we have

$$t_{12} = S, \qquad t_{11} = t_{22} = t_{33} = t_{23} = t_{31} = 0 \tag{6.4.3}$$

It follows from (6.3.9) that the shear modulus

$$\mu_e \equiv G = \frac{S}{2\tilde{e}_{12}} = \frac{S}{\alpha_{12}} \tag{6.4.4}$$

is obtained by measuring the angle change α_{12}. The modulus G, which is the same as μ_e, is for this reason called the *modulus of rigidity* or the *shear modulus.*

(*c*) ***Hydrostatic Pressure Test.*** In a small piece of material (a cube or a sphere) subjected to a hydrostatic pressure $\bar{p}$ on its surface, the state of stress may be approximated by Figure 6.4.5,

$$t_{kl} = -\bar{p}\delta_{kl} \tag{6.4.5}$$

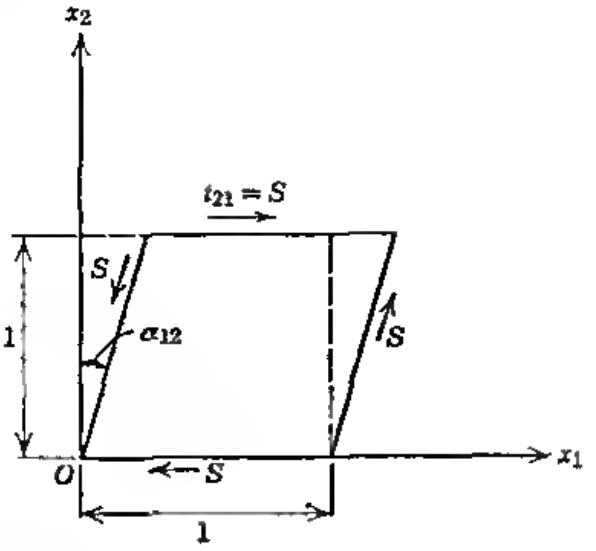


Fig. 6.4.4 Simple shear test

Equation (6.3.4) gives the strain

$$\bar{e}_{kl} = -\frac{\bar{p}}{3\lambda_e + 2\mu_e}\delta_{kl} = -\frac{1-2\nu}{E}\bar{p}\delta_{kl} \tag{6.4.6}$$

This may be written as

$$\bar{p} = -k\mathrm{I}_{\bar{e}}, \qquad k \equiv \lambda_e + \tfrac{2}{3}\mu_e = \frac{E}{3(1-2\nu)} \tag{6.4.7}$$

where k is the modulus of compression or bulk modulus. Since hydrostatic pressure tends to diminish the volume, it is clear that k must be positive This means that $\nu \leq \frac{1}{2}$ In most metals, ν is in the neighborhood of 1/3 For incompressible solids $\nu = \frac{1}{2}$ In Table 6.4.1 there are listed values of E and ν for several isotropic substances

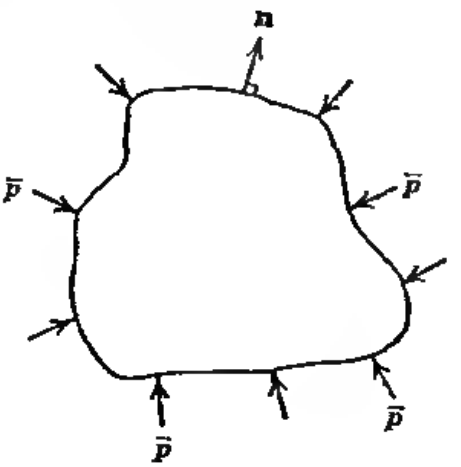


Fig. 6 4.5 Hydrostatic pressure.

For vulcanized rubber the simple tension test curve is given on Figure 6.4.6 which exhibits the nonlinear dependence on strain. For other experiments see Eringen [1962, Art. 61]. For various types of anisotropic elastic solids the elastic moduli may be found in Cady [1946], Mason [1950], Dana and Hurlbut [1959], and Hearman [1961]. The reader is referred to these books for various interesting studies in anisotropic elasticity.

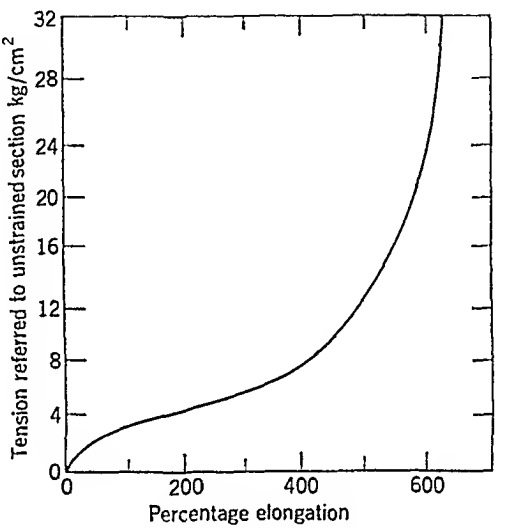


Fig. 6.4.6 Typical force-extension curve for vulcanized rubber (after Treloar [1958]).

Table 6.4.1. Elastic Constants

Substance	E	ν
Wrought Iron	27 to 29 $\times$ 10^6 psi	0.28
Mild Steel	29 to 31 $\times$ 10^6 psi	0.28
Cast Iron	14 to 23 $\times$ 10^6 psi	0.28
Aluminum (Cast)	12.5 $\times$ 10^6 psi	0.34
Duraluminum	10 $\times$ 10^6 psi	0.25
Brass	12 $\times$ 10^6 psi	0.33
Copper	15 $\times$ 10^6 psi	0.35
Glass	8 to 10.6 $\times$ 10^6 psi	0.23
Rubber	250 to 350 psi	0.50
Spruce (15% moisture)	1.3 $\times$ 10^6 psi	0.08
Nitrocellulose	0.355 $\times$ 10^6 psi	0.41

6.5. RÉSUMÉ OF BASIC EQUATIONS OF ELASTICITY

The linear theory of isotropic elastic solids is based on the following equations

Equations of Balance

$$\frac{\rho_0}{\rho} = 1 + I_\varepsilon \tag{6.5.1}$$

$$t_{kl,k} + \rho(f_l - \dot{v}_l) = 0 \quad \text{in } \mathscr{V} \tag{6.5.2}$$

$$t_{kl} = t_{lk} \tag{6.5.3}$$

$$\dot{\eta} - \frac{h}{\theta} = 0 \tag{6.5.4}$$

The Clausius-Duhem inequality is satisfied automatically The energy equation (6.5.4) is the result of using the constitutive theory peculiar to elasticity cf (5.5.29).

Equations of Jump. If there is a moving discontinuity surface $\sigma(t)$ sweeping the body with a velocity $\mathbf{v}$ in the direction of the normal $\mathbf{n}$ the following jump conditions must be satisfied on the surface $\sigma(t)$

$$[\rho(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} = 0 \tag{6.5.5}$$

$$-[t_{kl}]n_k + [\rho v_l(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} = 0 \tag{6.5.6}$$

$$[(\varepsilon + \tfrac{1}{2}v_l v_l)\rho(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} - [t_{kl}v_l + q_k]n_k = 0 \tag{6.5.7}$$

$$\left[\rho\eta(\mathbf{v} - \mathbf{v}) - \frac{\mathbf{q}}{\theta}\right] \cdot \mathbf{n} \geq 0 \quad \text{on } \sigma(t) \tag{6.5.8}$$

Constitutive Equations (Compressible Solids)

$$t_{kl} = \lambda_e \mathrm{I}_e \delta_{kl} + 2\mu_e \bar{e}_{kl} \tag{6.5.9}$$

$$\Sigma = \rho_0(\varepsilon - \theta\eta) = \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_e^2 - 2\mu_e \mathrm{II}_e \tag{6.5.10}$$

$$q_k = 0 \tag{6.5.11}$$

$$\eta = -\frac{1}{\rho_0}\frac{\partial \Sigma}{\partial \theta}\bigg|_e \tag{6.5.12}$$

Constitutive Equations (Incompressible Solids)

$$t_{kl} = -p\delta_{kl} + 2\mu_e \bar{e}_{kl} \tag{6.5.13}$$

$$\Sigma = \rho_0(\varepsilon - \theta\eta) = -2\mu_e \mathrm{II}_e \tag{6.5.14}$$

$$q_k = 0 \tag{6.5.15}$$

$$\eta = -\frac{1}{\rho_0}\frac{\partial \Sigma}{\partial \theta}\bigg|_e \tag{6.5.16}$$

Kinematical Relations

$$\dot{v}_l \simeq \frac{\partial^2 u_l}{\partial t^2} \tag{6.5.17}$$

$$\tilde{e}_{kl} \equiv \tfrac{1}{2}(u_{k,l} + u_{l,k}) \tag{6.5.18}$$

$$I_{\tilde{e}} \equiv \tilde{e}_{kk}, \qquad \mathrm{II}_{\tilde{e}} \equiv \tfrac{1}{2}(\tilde{e}_{kk}\tilde{e}_{ll} - \tilde{e}_{kl}\tilde{e}_{lk}) \tag{6.5.19}$$

where u_l is the displacement vector.

Field Equations. The field equations are obtained by combining (6.5.9) or (6.5.13) with (6.5.2) and using (6.5.17) and (6.5.18). This leads to the celebrated field equations of Navier.

$$(\lambda_e + \mu_e)u_{l,lk} + \mu_e u_{k,ll} + \rho(f_k - \ddot{u}_k) = 0 \qquad \text{in } \mathscr{V} \tag{6.5.20}$$

(compressible)

$$-p_{,k} + \mu_e u_{k,ll} + \rho_0(f_k - \ddot{u}_k) = 0 \qquad \text{in } \mathscr{V} \tag{6.5.21}$$

$$u_{k,k} = 0 \quad \text{(incompressible)}$$

Boundary Conditions

$$t_{lk}n_l = \bar{t}_k \qquad \text{on } \mathscr{S}_t \tag{6.5.22}$$

$$u_k = \bar{u}_k \qquad \text{on } \mathscr{S}_u \tag{6.5.23}$$

where $\bar{t}_k$ is the surface traction prescribed on a part $\mathscr{S}_t$ of the surface of the body $\mathscr{S} = \mathscr{S}_t + \mathscr{S}_u$, and $\bar{u}_k$ is the displacement field prescribed on the remaining part $\mathscr{S}_u$ not overlapping $\mathscr{S}_t$. Other mixed combinations of the boundary conditions are encountered in the literature.[3] However, a *well-posed* theory requires that the boundary conditions together with the initial conditions must not violate the uniqueness theorem of the theory or, in other words they must yield a unique resultant motion.[4]

Initial Conditions

$$u_k(\mathrm{x}, 0) = u_k^0(\mathrm{x}) \qquad \text{in } \mathscr{V} \tag{6.5.24}$$

$$\dot{u}_k(\mathrm{x}, 0) = v_k^0(\mathrm{x}) \qquad \text{in } \mathscr{V} \tag{6.5.25}$$

[3] For example $\bar{t}_1$, $\bar{t}_2$, and $\bar{u}_3$ may be prescribed, cf. Eringen [1955b].

[4] For the uniqueness theorems in elasticity, see Sokolnikoff [1956, Art. 27], Gurtin and Sternberg [1961], Hill [1961], Bramble and Payne [1962], and references in these works.

It could be said that the entire theory of linear isotropic elastic solids is based on solutions of Navier's equations (6.5.20) or (6 5.21) subject to some boundary and initial conditions of the type (6.5.22) to (6.5.25).

For anisotropic solids the Navier equations must be replaced by appropriate equations, since, in this case, the stress constitutive equations do not have the forms (6.5.9) or (6.5.13). In this case, using (6.2.12) in (6.5.2) we get

$$\sigma_{klmn}\, u_{m\,nk} + \rho(f_l - \dot{v}_l) = 0 \qquad \text{in } \mathscr{V} \tag{6.5.26}$$

where σ_{klmn} is related to Σ_{KLMN} by (6.2 13) and it may possess various symmetry restrictions studied in Section 6.2.

6.6. CURVILINEAR COORDINATES

If a set of curvilinear coordinates x^k is used to express the basic equations of the theory of elasticity in place of rectangular coordinates, then all we need is to make some simple interpretations to find the forms of the equations collected in Section 6 5 Let $g_{kl}(\mathbf{x})$ be the metric tensor in the curvilinear coordinates x^k. The square of the element of arc length is now given by (cf. App. C2 11)

$$ds^2 = g_{kl}\, dx^k\, dx^l \tag{6.6.1}$$

The passage from rectangular coordinates to curvilinear coordinates may be made by observing the following two simple rules: (a) The partial differentiation symbol (,) must be replaced by the covariant differentiation symbol (;), (b) The repeated indices must be on diagonal positions.

Thus, for example, the Cauchy equations of motion (6 5 2) in curvilinear coordinates will have the form

$$t^k{}_{l;k} + \rho(f_l - \dot{v}_l) = 0 \tag{6 6 2}$$

where $t^k{}_l$ are the mixed components of the stress tensor and an index following a semi-colon indicates the covariant partial differentiation, that is,

$$t^k{}_{l;r} \equiv t^k{}_{l,r} + \begin{Bmatrix} k \\ mr \end{Bmatrix} t^m{}_l - \begin{Bmatrix} m \\ rl \end{Bmatrix} t^k{}_m \tag{6.6.3}$$

where $\left\{ {k \atop mr} \right\}$ are the Christoffel symbols of the second kind (cf. App. C5)

The infinitesimal strains and rotations are given by

$$\tilde{e}_{kl} = \tfrac{1}{2}(u_{k,l} + u_{l,k}), \qquad \tilde{r}_{kl} = \tfrac{1}{2}(u_{k,l} - u_{l,k}) \tag{6.6.4}$$

where

$$u_{k;l} \equiv u_{k,l} - \begin{Bmatrix} m \\ kl \end{Bmatrix} u_m \tag{6 6.5}$$

Often these equations are expressed in terms of the physical components of the vectors and tensors involved. The physical components $t^{(k)}{}_{(l)}$ and $u^{(k)}$ of $t^k{}_l$ and u^k are related to each other by (App. C4).

$$t^k{}_l = t^{(k)}{}_{(l)}\sqrt{g_{ll}/g_{kk}}, \qquad u^k = u^{(k)}/\sqrt{g_{kk}} \tag{6.6.6}$$

We now give the expressions of (6.6.2) and (6.6.4) in *orthogonal curvilinear* coordinates only. In this case we have $g_{kl} = 0$ when $k \neq l$, and

$$\begin{aligned} &ds^2 = g_{11}(dx^1)^2 + g_{22}(dx^2)^2 + g_{33}(dx^3)^2 \\ &g^{kk} = 1/g_{kk}, \qquad g \equiv \det g_{kl} = g_{11}g_{22}g_{33} \\ &\begin{Bmatrix} k \\ lm \end{Bmatrix} = \frac{1}{2g_{kk}}\left(\frac{\partial g_{kk}}{\partial x^m}\delta_{kl} + \frac{\partial g_{mm}}{\partial x^l}\delta_{km} - \frac{\partial g_{ll}}{\partial x_k}\delta_{lm}\right) \end{aligned} \tag{6.6.7}$$

Using (6.6.6) and (6.6.7) in (6.6.2), we get

$$\sum_{k=1}^{3}\left\{\frac{1}{\sqrt{g}}\frac{\partial}{\partial x^k}\left[t^{(k)}{}_{(l)}\frac{\sqrt{g}}{\sqrt{g_{kk}}}\right] + \frac{1}{\sqrt{g_{kk}g_{ll}}}\frac{\partial\sqrt{g_{ll}}}{\partial x^k}t^{(k)}{}_{(l)} - \frac{1}{\sqrt{g_{kk}g_{ll}}}\frac{\partial\sqrt{g_{kk}}}{\partial x^l}t^{(k)}{}_{(k)}\right\} + \rho(f^{(l)} - \dot{v}^{(l)}) = 0 \tag{6.6.8}$$

where $f^{(l)} = f^l/\sqrt{g_{ll}}$ are the physical components of the body force.

To express (6.6.4) in physical components of the displacement vector we first express it in the form

$$\tilde{e}^k{}_l = \tfrac{1}{2}(u^k{}_{;l} + g_{nl}g^{km}u^n{}_{;m}) \tag{6.6.9}$$

where

$$u^k = g^{kl}u_l, \qquad u^k{}_{;l} \equiv u^k{}_{,l} + \begin{Bmatrix} k \\ ml \end{Bmatrix}u^m \tag{6.6.10}$$

The forms (6.6.6) can be employed to replace $\tilde{\mathbf{e}}$ and $\mathbf{u}$ by their physical components $\tilde{e}^{(k)}{}_{(l)}$ and $u^{(k)}$. Upon using (6.6.7) in (6.6.9) we obtain

$$\tilde{e}^{(k)}{}_{(l)} = \tilde{e}^k{}_l\frac{\sqrt{g_{kk}}}{\sqrt{g_{ll}}} = \frac{1}{2}\left[\frac{\sqrt{g_{kk}}}{\sqrt{g_{ll}}}\frac{\partial}{\partial x^l}\left(\frac{u^{(k)}}{\sqrt{g_{kk}}}\right) + \frac{\sqrt{g_{ll}}}{\sqrt{g_{kk}}}\frac{\partial}{\partial x^k}\left(\frac{u^{(l)}}{\sqrt{g_{ll}}}\right) + \frac{1}{\sqrt{g_{kk}g_{ll}}}\sum_{m=1}^{3}\frac{\partial g_{kk}}{\partial x^m}\frac{u^{(m)}}{\sqrt{g_{mm}}}\delta_{kl}\right] \tag{6.6.11}$$

Similarly

$$\tilde{r}^{(k)}{}_{(l)} = \frac{1}{2\sqrt{g_{\underline{kk}}g_{\underline{ll}}}}\left[\frac{\partial}{\partial x^l}(\sqrt{g_{\underline{kk}}}\,u^{(k)}) - \frac{\partial}{\partial x^k}(\sqrt{g_{\underline{ll}}}\,u^{(l)})\right] \tag{6.6.12}$$

$$2\tilde{r}^{(m)} = \sum_{k,l} e^{mlk}\tilde{r}^{(k)}{}_{(l)} \quad \text{or} \quad 2\vec{r} = \nabla \times \mathbf{u}$$

The stress constitutive equations, as usual, are

$$t^{(k)}{}_{(l)} = \lambda_e \tilde{e}^{(m)}{}_{(m)}\delta^k{}_l + 2\mu_e \tilde{e}^{(k)}{}_{(l)} \tag{6.6.13}$$

Equations (6.6 8), (6.6.11) and (6.6.12) are all that are needed for the treatment of problems in any set of orthogonal curvilinear coordinates. The passage from physical components of vectors and tensors to tensor components is made through such equations as (6.6 6)

Navier's equations in curvilinear coordinates are obtained by combining (6.6.11) and (6.6.13) with (6 6.8). This is rather cumbersome. The tensor expression of these equations is written immediately by use of the rules stated above. In fact (6 5.2) in any curvilinear coordinates may be expressed as

$$(\lambda_e + \mu_e)u^l{}_{,lk} + \mu_e u_k{}^{,l}{}_{,l} + \rho(f_k - \ddot{v}_k) = 0 \tag{6 6.14}$$

A vectorial form of this equation is found if we remember the vector identities

$$u_k{}^{,l}{}_{,l} = -(\nabla \times \nabla \times \mathbf{u})_k + (\nabla\nabla \cdot \mathbf{u})_k$$

$$u^l{}_{,kl} = (\nabla\nabla \cdot \mathbf{u})_k$$

where ∇ is the gradient operator. Upon using these identities in (6.6.14) we get

$$(\lambda_e + 2\mu_e)\nabla\nabla \cdot \mathbf{u} - \mu_e \nabla \times \nabla \times \mathbf{u} + \rho(\mathbf{f} - \ddot{\mathbf{u}}) = 0 \tag{6 6.15}$$

By substituting the expressions of the gradient, divergence, and curl operators (cf. Appendix C5 26), this equation can be expressed in any coordinate system. We now give the forms of these equations in cylindrical and spherical coordinates

(*a*) *Cylindrical Coordinates.* Cylindrical coordinates (r, θ, z) are related to rectangular coordinates x, y, z by (Figure 6.6.1)

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z \tag{6.6 16}$$

The square of arc length, ds^2, and metric tensor, g_{kl}, are given by

$$ds^2 = dr^2 + r^2\,d\theta^2 + dz^2$$

$$g_{11} = 1, \qquad g_{22} = r^2, \qquad g_{33} = 1, \qquad g_{kl} = 0 \quad (k \neq l) \tag{6.6.17}$$

The gradient, divergence, curl, and laplacian operators are given in Appendix C (eq. C5.26).

Cauchy's equations of motion, strains, and rotations are obtained by

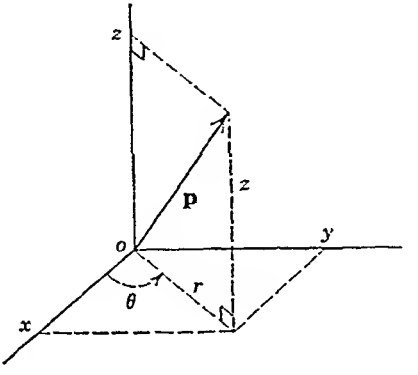


Fig. 6.6.1 Cylindrical coordinates.

substituting g_{kl}, given above, into (6.6.8), (6.6.11), and $(6.6.12)_2$, respectively. Thus

$$\frac{\partial t_{rr}}{\partial r}+\frac{1}{r}\frac{\partial t_{r\theta}}{\partial \theta}+\frac{\partial t_{rz}}{\partial z}+\frac{1}{r}(t_{rr}-t_{\theta\theta})+\rho(f_r-\ddot{u}_r)=0$$

$$\frac{\partial t_{r\theta}}{\partial r}+\frac{1}{r}\frac{\partial t_{\theta\theta}}{\partial \theta}+\frac{\partial t_{\theta z}}{\partial z}+\frac{2}{r}t_{\theta r}+\rho(f_\theta-\ddot{u}_\theta)=0 \tag{6.6.18}$$

$$\frac{\partial t_{rz}}{\partial r}+\frac{1}{r}\frac{\partial t_{\theta z}}{\partial \theta}+\frac{\partial t_{zz}}{\partial z}+\frac{1}{r}t_{rz}+\rho(f_z-\ddot{u}_z)=0$$

$$\tilde{e}_{rr}=\frac{\partial u_r}{\partial r},\quad \tilde{e}_{\theta\theta}=\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r},\quad \tilde{e}_{zz}=\frac{\partial u_z}{\partial z}$$

(6.6.19)

$$2\tilde{e}_{\theta z}=\frac{\partial u_\theta}{\partial z}+\frac{1}{r}\frac{\partial u_z}{\partial \theta},\qquad 2\tilde{e}_{zr}=\frac{\partial u_r}{\partial z}+\frac{\partial u_z}{\partial r},\qquad 2\tilde{e}_{r\theta}=\frac{1}{r}\frac{\partial u_r}{\partial \theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}$$

$$2\tilde{r}_r=\frac{1}{r}\frac{\partial u_z}{\partial \theta}-\frac{\partial u_\theta}{\partial z},\qquad 2\tilde{r}_\theta=\frac{\partial u_r}{\partial z}-\frac{\partial u_z}{\partial r},\qquad 2\tilde{r}_z=\frac{1}{r}\left(\frac{\partial}{\partial r}(ru_\theta)-\frac{\partial u_r}{\partial \theta}\right)$$

where $(t_{rr}, t_{r\theta}, \ldots, t_{zz})$, $(\tilde{e}_{rr}, \tilde{e}_{r\theta}, \ldots, \tilde{e}_{zz})$, (u_r, u_θ, u_z) and $(\tilde{r}_r, \tilde{r}_\theta, \tilde{r}_z)$ denote, respectively, the physical components of stress, strain, displacement, and rotations in cylindrical coordinates.

Navier's equations (6.6.15) in component form read

$$(\lambda_e + 2\mu_e)\frac{\partial \mathrm{I}_\varepsilon}{\partial r} - 2\mu_e\left(\frac{1}{r}\frac{\partial \tilde{r}_z}{\partial \theta} - \frac{\partial \tilde{r}_\theta}{\partial z}\right) + \rho(f_r - \ddot{u}_r) = 0$$

$$(\lambda_e + 2\mu_e)\frac{\partial \mathrm{I}_\varepsilon}{r\partial \theta} - 2\mu_e\left(\frac{\partial \tilde{r}_r}{\partial z} - \frac{\partial \tilde{r}_z}{\partial r}\right) + \rho(f_\theta - \ddot{u}_\theta) = 0 \tag{6.6.20}$$

$$(\lambda_e + 2\mu_e)\frac{\partial \mathrm{I}_\varepsilon}{\partial z} - \frac{2\mu_e}{r}\left[\frac{\partial}{\partial r}(r\tilde{r}_\theta) - \frac{\partial \tilde{r}_r}{\partial \theta}\right] + \rho(f_z - \ddot{u}_z) = 0$$

where

$$\mathrm{I}_\varepsilon \equiv \operatorname{div} \mathbf{u} = \frac{1}{r}\frac{(\partial r u_r)}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z}$$

(*b*) ***Spherical Coordinates.*** Spherical coordinates (r, θ, φ) are related to rectangular coordinates (x, y, z) by (Figure 6.6.2)

$$x = r\sin\theta\cos\varphi, \qquad y = r\sin\theta\sin\varphi, \qquad z = r\cos\theta \tag{6.6.21}$$

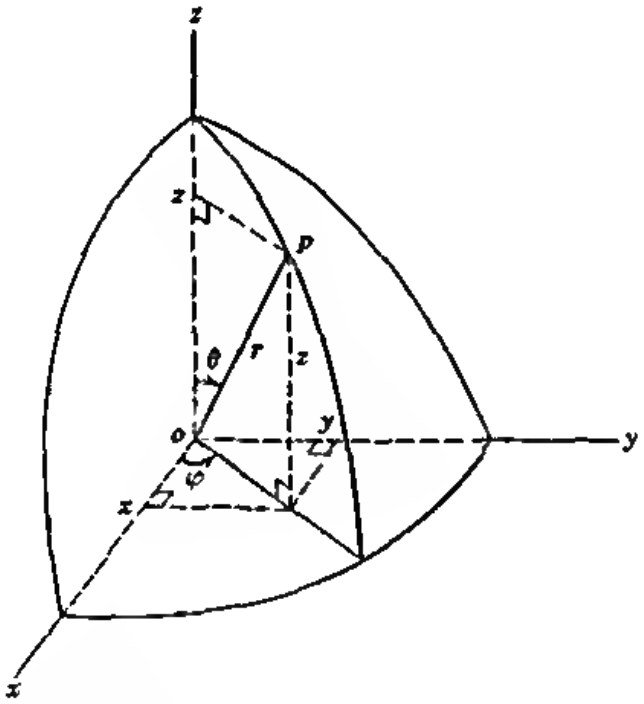


Fig. 6.6.2 Spherical coordinates.

The square of the arc length, ds^2, and the metric tensor, g_{kl}, are given by

$$ds^2 = dr^2 + r^2\,d\theta^2 + r^2\sin^2\theta\,d\varphi^2$$

$$g_{11} = 1, \quad g_{22} = r^2, \quad g_{33} = r^2\sin^2\theta, \quad g_{kl} = 0 \qquad (k \neq l) \tag{6.6.22}$$

The gradient, divergence, curl, and laplacian operators are given in

Appendix C (eq. C27). Cauchy's equations of motion in terms of physical components of stress are

$$\frac{\partial t_{rr}}{\partial r}+\frac{1}{r}\frac{\partial t_{r\theta}}{\partial \theta}+\frac{1}{r\sin\theta}\frac{\partial t_{r\varphi}}{\partial \varphi}+\frac{1}{r}(2t_{rr}-t_{\theta\theta}-t_{\varphi\varphi}+t_{r\theta}\cot\theta)+\rho(f_r-\ddot{u}_r)=0$$

$$\frac{\partial t_{r\theta}}{\partial r}+\frac{1}{r}\frac{\partial t_{\theta\theta}}{\partial \theta}+\frac{1}{r\sin\theta}\frac{\partial t_{\theta\varphi}}{\partial \varphi}+\frac{1}{r}[3t_{r\theta}+(t_{\theta\theta}-t_{\varphi\varphi})\cot\theta]+\rho(f_\theta-\ddot{u}_\theta)=0 \tag{6.6.23}$$

$$\frac{\partial t_{r\varphi}}{\partial r}+\frac{1}{r}\frac{\partial t_{\theta\varphi}}{\partial \theta}+\frac{1}{r\sin\theta}\frac{\partial t_{\varphi\varphi}}{\partial \varphi}+\frac{1}{r}[3t_{r\varphi}+2t_{\theta\varphi}\cot\theta]+\rho(f_\varphi-\ddot{u}_\varphi)=0$$

Physical components of strains ($\tilde{e}_{rr}$, $\tilde{e}_{r\theta}$, ..., $\tilde{e}_{\varphi\varphi}$) and rotations ($\tilde{r}_r$, $\tilde{r}_\theta$, $\tilde{r}_\varphi$) are

$$\tilde{e}_{rr}=\frac{\partial u_r}{\partial r},\quad \tilde{e}_{\theta\theta}=\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r},\quad \tilde{e}_{\varphi\varphi}=\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta$$

$$2\tilde{e}_{\theta\varphi}=\frac{1}{r}\frac{\partial u_\varphi}{\partial \theta}+\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial \varphi}-\frac{u_\varphi\cot\theta}{r}$$

$$2\tilde{e}_{\varphi r}=\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi}+\frac{\partial u_\varphi}{\partial r}-\frac{u_\varphi}{r} \tag{6.6.24}$$

$$2\tilde{e}_{r\theta}=\frac{1}{r}\frac{\partial u_r}{\partial \theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}$$

$$2\tilde{r}_r=\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(u_\varphi\sin\theta)-\frac{\partial u_\theta}{\partial \varphi}\right]$$

$$2\tilde{r}_\theta=\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi}-\frac{1}{r}\frac{\partial}{\partial r}(ru_\varphi)$$

$$2\tilde{r}_\varphi=\frac{1}{r}\frac{\partial}{\partial r}(ru_\theta)-\frac{1}{r}\frac{\partial u_r}{\partial \theta}$$

Navier's equations have the component form

$$(\lambda_e+2\mu_e)\frac{\partial \mathrm{I}_{\tilde{e}}}{\partial r}-\frac{2\mu_e}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\tilde{r}_\varphi\sin\theta)-\frac{\partial \tilde{r}_\theta}{\partial \varphi}\right]+\rho(f_r-\ddot{u}_r)=0$$

$$(\lambda_e+2\mu_e)\frac{\partial \mathrm{I}_{\tilde{e}}}{r\partial \theta}-\frac{2\mu_e}{r}\left[\frac{1}{\sin\theta}\frac{\partial \tilde{r}_r}{\partial \varphi}-\frac{\partial}{\partial r}(r\tilde{r}_\varphi)\right]+\rho(f_\theta-\ddot{u}_\theta)=0 \tag{6.6.25}$$

$$\frac{(\lambda_e+2\mu_e)}{r\sin\theta}\frac{\partial \mathrm{I}_{\tilde{e}}}{\partial \phi}-\frac{2\mu_e}{r}\left[\frac{\partial}{\partial r}(r\tilde{r}_\theta)-\frac{\partial \tilde{r}_r}{\partial \theta}\right]+\rho(f_\varphi-\ddot{u}_\varphi)=0$$

where

$$I_2 \equiv \operatorname{div} \mathbf{u} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 u_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(u_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial u_\varphi}{\partial \varphi}$$

6.7. HALF SPACE SUBJECT TO UNIFORM PRESSURE

An isotropic elastic solid occupying the half space $x \geq 0$ is subjected to a uniform static pressure p at the bounding plane $x = 0$ (Figure 6.7.1) We

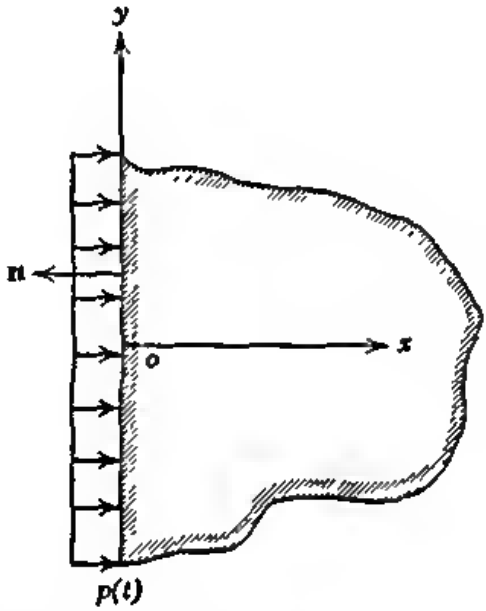


Fig. 6.7.1 Half space subject to uniform pressure.

would like to determine the displacement and stress fields. A suitable coordinate system is a set of rectangular coordinates (x, y, z). Displacements on any plane $x = \text{const.}$ are independent of y and z and also displacement components $u_y = u_z = 0$. The problem is a static one so that the displacement and stress fields are independent of time. We also assume that the body force vanishes. The Navier's equations (6.5 20) reduce to a single equation

$$\frac{d^2 u_x}{dx^2} = 0 \tag{6 7.1}$$

Since the exterior normal $\mathbf{n}$ to the plane $x = 0$ has the components $(-1, 0, 0)$, for the boundary conditions (6.5.22) on tractions we have

$$-t_{xx} = -(\lambda_e + 2\mu_e)\frac{du_x}{dx} = \bar{t}_x = p, \qquad t_{xy} = t_{xz} = 0 \quad \text{on } x = 0 \tag{6.7.2}$$

We leave the conditions $x = \infty$ undecided for the time being.

The general solution of (6.7.1) subject to $(6.7.2)_1$ is

$$u_x = -\frac{p}{\lambda_e + 2\mu_e} x + C_1, \qquad u_y = u_z = 0 \tag{6.7.3}$$

The stress field now follows from (6.5.9) and (6.5.18)

$$t_{xx} = -p, \qquad t_{yy} = t_{zz} = -\frac{\lambda_e}{\lambda_e + 2\mu_e} p, \qquad t_{xz} = t_{yz} = t_{xy} = 0 \tag{6.7.4}$$

The stress component t_{xx} at every plane $x = \text{const.}$ is therefore constant and is equal to $-p$. Consequently, at $x = \infty$, it is also $-p$. Unless we apply an equal and opposite pressure at $x = \infty$, the body is not in global equilibrium. For $x = \infty$ we get $u_x = -\infty$. This situation is typical for media (extending to infinity) that are subject to nonequilibrating external forces on its boundaries. The undetermined constant C_1 represents a rigid body translation that is unessential to the stress field.

6.8. INFINITE CYLINDRICAL TUBE SUBJECT TO INTERNAL AND EXTERNAL PRESSURES

An infinite elastic circular cylindrical tube is subject to an internal pressure p_0 and an external pressure p_1. We would like to determine the displacement and stress fields. The appropriate coordinate system for this case is cylindrical coordinates (r, θ, z) with the z taken along the axis of the cylinder (Figure 6.8.1). We ignore body forces. Since any point on

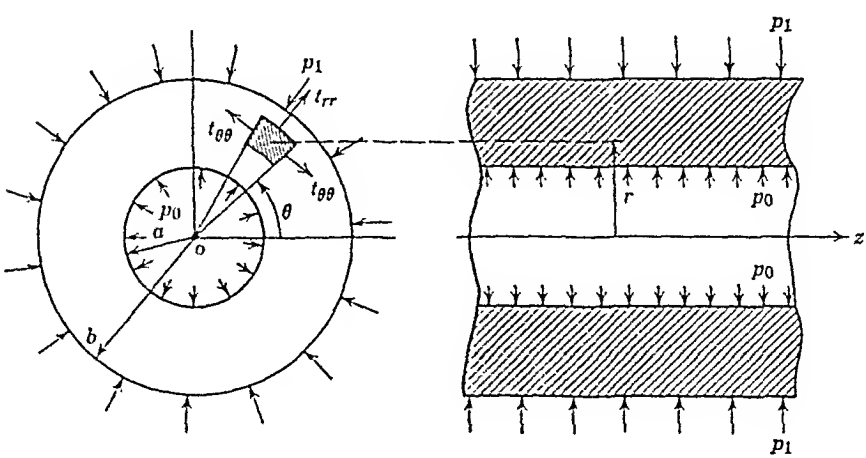


Fig. 6.8.1 Infinite circular cylindrical tube subject to internal and external pressure.

a cylindrical surface $r = \text{const.}$ will not move in either the θ or z direction, we have $u_\theta = u_z = 0$. The displacement component u_r, on the other hand, will not depend on θ and z as dictated by the symmetry in loading and geometry, that is, $u_r = u_r(r)$. With these simplifications, Navier's equation (6 6 20) in cylindrical coordinates reduces to a single equation

$$\frac{d}{dr}\left[\frac{1}{r}\frac{d}{dr}(ru_r)\right] = 0 \tag{6.8.1}$$

For the stress components through (6.6.19) and (6.6.13) we have

$$\begin{aligned} t_{rr} &= \frac{\lambda_e}{r}\frac{d(ru_r)}{dr} + 2\mu_e\frac{du_r}{dr} \\ t_{\theta\theta} &= \frac{\lambda_e}{r}\frac{d(ru_r)}{dr} + 2\mu_e\frac{u_r}{r} \\ t_{zz} &= \frac{\lambda_e}{r}\frac{d(ru_r)}{dr} \\ t_{rz} &= t_{\theta z} = t_{r\theta} = 0 \end{aligned} \tag{6 8 2}$$

The components of the exterior normal to the inner and outer surfaces $r = a$ and $r = b$, respectively, are $(-1, 0, 0)$ and $(1, 0, 0)$. Thus the boundary conditions are

$$\begin{aligned} -\left[\frac{\lambda_e}{r}\frac{d(ru_r)}{dr} + 2\mu_e\frac{du_r}{dr}\right]_{r=a} &= \bar{t}_r(a) = p_0 \\ \left[\frac{\lambda_e}{r}\frac{d(ru_r)}{dr} + 2\mu_e\frac{du_r}{dr}\right]_{r=b} &= \bar{t}_r(b) = -p_1 \end{aligned} \tag{6.8.3}$$

The general solution of (6.8.1) is

$$u_r = C_1\frac{r}{2} + \frac{C_2}{r} \tag{6 8.4}$$

where C_1 and C_2 are two constants that are determined through (6.8.3)

$$C_1 = \frac{a^2p_0 - b^2p_1}{(\lambda_e + \mu_e)(b^2 - a^2)}, \qquad C_2 = \frac{a^2b^2(p_0 - p_1)}{2\mu_e(b^2 - a^2)} \tag{6.8.5}$$

The displacement and stress fields are therefore obtained by carrying these into (6.8 4) and (6.8.2)

$$\begin{aligned} u_r &= \frac{a^2p_0}{2(b^2 - a^2)}\left(\frac{r}{\lambda_e + \mu_e} + \frac{b^2}{\mu_e r}\right) - \frac{b^2p_1}{2(b^2 - a^2)}\left(\frac{r}{\lambda_e + \mu_e} + \frac{a^2}{\mu_e r}\right) \\ u_\theta &= u_z = 0 \end{aligned} \tag{6.8.6}$$

$$t_{rr} = \frac{a^2 p_0}{b^2 - a^2}\left(1 - \frac{b^2}{r^2}\right) - \frac{b^2 p_1}{b^2 - a^2}\left(1 - \frac{a^2}{r^2}\right),$$

(6.8.7)
$$t_{\theta\theta} = \frac{a^2 p_0}{b^2 - a^2}\left(1 + \frac{b^2}{r^2}\right) - \frac{b^2 p_1}{b^2 - a^2}\left(1 + \frac{a^2}{r^2}\right),$$

$$t_{zz} = \frac{\lambda_e}{\lambda_e + \mu_e}\frac{a^2 p_0 - b^2 p_1}{b^2 - a^2}, \qquad t_{rz} = t_{\theta z} = t_{r\theta} = 0$$

(*a*) ***Cylindrical Cavity.*** If we let $b \to \infty$ the tube extends to infinity in the r-direction becoming a solid with a cavity. The displacement and stress fields for this case follow from (6.8.6) and (6.8.7) by setting $b = \infty$.

(6.8.8)
$$u_r = \frac{a^2 p_0}{2\mu_e r} - \frac{p_1}{2}\left(\frac{r}{\lambda_e + \mu_e} + \frac{a^2}{\mu_e r}\right), \qquad u_\theta = u_z = 0$$

$$t_{rr} = -p_0\frac{a^2}{r} - p_1\left(1 - \frac{a^2}{r^2}\right)$$

(6.8.9)
$$t_{\theta\theta} = p_0\frac{a^2}{r^2} - p_1\left(1 - \frac{a^2}{r^2}\right)$$

$$t_{zz} = -\frac{\lambda_e}{\lambda_e + \mu_e}p_1, \qquad t_{rz} = t_{\theta z} = t_{r\theta} = 0$$

When $p_1 = 0$ we see that as $r \to \infty$, $u_r \to 0$. In this case, the tractions on the interior surface have zero resultants so that we do not encounter infinite displacements at $r = \infty$.

(*b*) ***Cylinder with No Hole.*** In this case $a = 0$. If p_0 is bounded, then (6.8.7) gives

(6.8.10)
$$u_r = -\frac{p_1 r}{2(\lambda_e + \mu_e)}, \qquad u_\theta = u_z = 0$$

(6.8.11)
$$t_{rr} = t_{\theta\theta} = -p_1, \qquad t_{zz} = -\frac{\lambda_e}{\lambda_e + \mu_e}p_1, \qquad t_{rz} = t_{\theta z} = t_{r\theta} = 0$$

6.9. SPHERICAL SHELL SUBJECT TO INTERNAL AND EXTERNAL PRESSURES

A spherical shell, of inner and outer radii a and b, is subject to an internal pressure p_0 and an external pressure p_1. We would like to determine the displacement and stress fields. The appropriate coordinate system is a set of spherical coordinates (r, θ, φ), as in Figure 6.9.1. Any point A on

a concentric spherical surface $r = \text{const.}$ will move only along the radius connecting A to the center O of the shell. Therefore $u_\theta = u_\varphi = 0$ and $u_r = u_r(r)$. Navier's equations with vanishing body forces are

$$\frac{d}{dr}\left(\frac{du_r}{dr} + 2\frac{u_r}{r}\right) = 0 \tag{6.9.1}$$

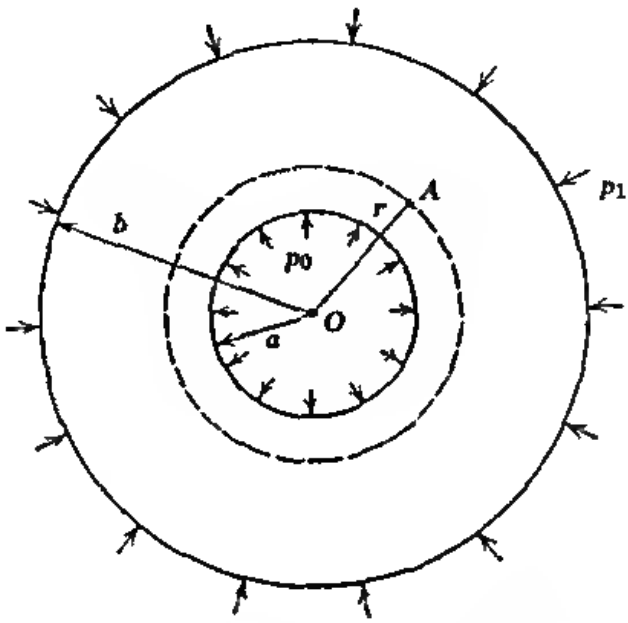


Fig. 6.9.1 Spherical shell subject to internal and external pressures

For the components of the stress, through (6.6.19) and (6.6 13) we have

$$\begin{aligned} t_{rr} &= \lambda_e\left(\frac{du_r}{dr} + 2\frac{u_r}{r}\right) + 2\mu_e\frac{du_r}{dr} \\ t_{\theta\theta} &= t_{\varphi\varphi} = \lambda_e\left(\frac{du_r}{dr} + 2\frac{u_r}{r}\right) + 2\mu_e\frac{u_r}{r} \\ t_{r\theta} &= t_{r\varphi} = t_{\theta\varphi} = 0 \end{aligned} \tag{6.9.2}$$

The boundary conditions on the inner and outer surfaces $r = a$ and $r = b$ are

$$\begin{aligned} -\left[\lambda_e\left(\frac{du_r}{dr} + 2\frac{u_r}{r}\right) + 2\mu_e\frac{du_r}{dr}\right]_{r=a} &= p_0 \\ \left[\lambda_e\left(\frac{du_r}{dr} + 2\frac{u_r}{r}\right) + 2\mu_e\frac{du_r}{dr}\right]_{r=b} &= -p_1 \end{aligned} \tag{6.9.3}$$

The problem is thus reduced to finding the solution of (6.9.1) subject to

(6.9.3). Equations (6.9.1) intcgrate to

$$u_r = \tfrac{1}{3}C_1 r + \frac{C_2}{r^2} \tag{6.9.4}$$

The constants of integrations C_1 and C_2 are determined through the boundary conditions (6.9.3)

$$C_1 = \left(p_0 \frac{a^3}{b^3} - p_1\right)(\lambda_e + \tfrac{2}{3}\mu_e)^{-1}\left(1 - \frac{a^3}{b^3}\right)^{-1}$$

$$C_2 = (p_0 - p_1)a^3\left[4\mu_e\left(1 - \frac{a^3}{b^3}\right)\right]^{-1}$$

The displacement and stress fields follow from (6.9.4) and (6.9.2)

$$u_r = \frac{p_0 \dfrac{a^3}{b^3} - p_1}{(3\lambda_e + 2\mu_e)\left(1 - \dfrac{a^3}{b^3}\right)} r + \frac{(p_0 - p_1)a^3}{4\mu_e\left(1 - \dfrac{a^3}{b^3}\right)} \frac{1}{r^2} \tag{6.9.5}$$

$$u_\theta = u_\varphi = 0$$

$$t_{rr} = \left[p_0\left(\frac{a^3}{b^3} - \frac{a^3}{r^3}\right) - p_1\left(1 - \frac{a^3}{r^3}\right)\right]\left(1 - \frac{a^3}{b^3}\right)^{-1}$$

$$t_{\theta\theta} = t_{\varphi\varphi} = \left[p_0\left(\frac{a^3}{b^3} + \frac{a^3}{2r^3}\right) - p_1\left(1 + \frac{a^3}{2r^3}\right)\right]\left(1 - \frac{a^3}{b^3}\right)^{-1} \tag{6.9.6}$$

$$t_{rz} = t_{\theta z} = t_{r\theta} = 0$$

(*a*) ***Spherical Cavity.*** If we set $b = \infty$ we obtain a solid of infinite extent having a spherical cavity of radius a. In this case, (6.9.5) and (6.9.6) reduce to

$$u_r = -\frac{p_1 r}{3\lambda_e + 2\mu_e} + \frac{(p_0 - p_1)a^3}{4\mu_e r^2} \tag{6.9.7}$$

$$u_\theta = u_\varphi = 0$$

$$t_{rr} = -p_0 \frac{a^3}{r^3} - p_1\left(1 - \frac{a^3}{r^3}\right)$$

$$t_{\theta\theta} = t_{\varphi\varphi} = p_0 \frac{a^3}{2r^3} - p_1\left(1 + \frac{a^3}{2r^3}\right) \tag{6.9.8}$$

$$t_{r\theta} = t_{\theta\varphi} = t_{r\varphi} = 0$$

At $r = \infty$ we have $t_{rr} = t_{\theta\theta} = t_{\varphi\varphi} = -p_1$. For $p_1 = 0$ we see that $u_r \to 0$

as $1/r^2$ with $r \to \infty$. For $p_0 = 0$ we find that the hoop stresses $t_{\theta\theta}$ and $t_{\varphi\varphi}$ at $r = a$ are $1\frac{1}{2}$ times as large as the applied internal pressure.

(b) Elastic Sphere. If p_0 is bounded and $a = 0$, we obtain a sphere of radius b subject to a uniform external pressure p_1. In this case, (6.9.5) and (6.9.6) reduce to

$$u_r = -\frac{p_1 r}{3\lambda_e + 2\mu_e}, \quad u_\theta = u_\varphi = 0 \tag{6.9.9}$$

$$t_{rr} = t_{\theta\theta} = t_{\varphi\varphi} = -p_1, \qquad t_{r\theta} = t_{\theta\varphi} = t_{r\varphi} = 0 \tag{6.9.10}$$

The state of stress is therefore a hydrostatic pressure.

6.10. PROPAGATION OF ELASTIC WAVES IN ISOTROPIC SOLIDS

In this Section and in the following two we give examples of dynamical motions of linear isotropic elastic solids. For simplicity, we consider propagation of waves in solids that extend to infinity.

The dynamical theory of isotropic elastic solids is governed by Navier's equations whose vector form was given in (6.6.15), that is,

$$(\lambda_e + 2\mu_e)\nabla\nabla \cdot \mathbf{u} - \mu_e \nabla \times \nabla \times \mathbf{u} + \rho(\mathbf{f} - \ddot{\mathbf{u}}) = 0 \tag{6.10.1}$$

We introduce a *scalar potential* ϕ and *vector potential* ψ for the displacement field u and a similar pair for the body force f

$$\mathbf{u} = \nabla\phi + \nabla \times \boldsymbol{\psi}, \qquad \nabla \cdot \boldsymbol{\psi} = 0 \tag{6.10.2}$$

$$\mathbf{f} = -\nabla g - \nabla \times \mathbf{h}, \qquad \nabla \cdot \mathbf{h} = 0$$

Upon substitution of (6.10.2) into (6.10.1) we get

$$\nabla(c_1^2 \nabla^2 \phi - g - \ddot{\phi}) - \nabla \times (c_2^2 \nabla \times \nabla \times \boldsymbol{\psi} + \mathbf{h} + \ddot{\boldsymbol{\psi}}) = \mathbf{0} \tag{6.10.3}$$

where

$$c_1 \equiv \left(\frac{\lambda_e + 2\mu_e}{\rho}\right)^{1/2}, \qquad c_2 \equiv \left(\frac{\mu_e}{\rho}\right)^{1/2} \tag{6.10.4}$$

Equation (6.10.3) is satisfied if

$$c_1^2 \nabla^2 \phi - g = \ddot{\phi} \tag{6.10.5}$$

$$-c_2^2 \nabla \times \nabla \times \boldsymbol{\psi} - \mathbf{h} = \ddot{\boldsymbol{\psi}} \tag{6.10.6}$$

These are the scalar and vector wave equations, which are valid also in

curvilinear coordinates. In rectangular coordinates (6.10.6) has the component representation

$$c_2^2\nabla^2\psi_k - h_k = \ddot{\psi}_k, \qquad (k = 1, 2, 3) \tag{6.10.7}$$

This last expression is *not*, however, valid in curvilinear coordinates.

When the motion is isochoric (no volume change) we know that div $\mathbf{u}$ vanishes. Also for irrotational motion $\tilde{r}_{kl} \equiv u_{[k,l]} = 0$ or curl $\mathbf{u} = 0$. These are accomplished, respectively, if ψ and ϕ vanish. Therefore equations (6.10.5) and (6.10.6) are, respectively, the equations of *irrotational* and *equivoluminal* waves. The positive constants c_1 and c_2 have the dimension of velocity and thus are called the velocities of irrotational and equivoluminal waves. Since Young's modulus $E > 0$, we see that c_1 and c_2 are real and nonnegative in the region of Poisson's ratio

$$-1 \leq \nu \leq \tfrac{1}{2} \tag{6.10.8}$$

The decomposition of a continuous vector field into (6.10.2) is always possible, under very general conditions (cf. Morse and Feshbach [1953, p. 53], Courant and Hilbert [1956, p. 246], so that the solution of (6.10.5) and (6.10.6), under the appropriate boundary and initial conditions, for example, (6.5.22) to (6.5.25), will describe uniquely the motion of a homogeneous, isotropic elastic solid.

(i) Plane Waves in Infinite Solid. We now give the solutions of Navier's equation in the form of plane waves advancing in an infinite solid. The boundary and initial conditions are unspecified, the body force vanishes and we are only interested in the existence of such waves.

A plane wave solution of (6.10.5) with $g = 0$ is a solution of the form

$$\phi = \phi(\nu_k x_k \mp ct) \tag{6.10.9}$$

where ν_k and c are constants. Substituting (6.10.9) into (6.10.5) we see that it is satisfied if $c = c_1$ and

$$\nu_k\nu_k = 1 \tag{6.10.10}$$

Thus in (6.10.9) ν_k are the cosine directors of a line L perpendicular to a plane P having the equation

$$\nu_k x_k \mp c_1 t = \text{const.}$$

for a fixed time (Figure 6.10.1). The plane moves with a speed $\pm c_1$ along L remaining perpendicular to it.

Next we try a vector plane wave solution of Navier's equation (6.10.7) of the form

$$\psi = \psi(\nu_k x_k \mp ct), \qquad \nabla \cdot \psi = 0 \tag{6.10.11}$$

By substituting (6.10.11) into (6.10.7) with $\mathbf{h} = 0$, we find that this is a solution if $c = c_2$, and v_k satisfies (6.10.10).

The displacement field corresponding to (6.10.9) and (6.10.11) follows from (6.10.2), that is,

$$\mathbf{u} = \phi'\mathbf{v} + \mathbf{v} \times \boldsymbol{\psi}', \qquad \mathbf{v} \cdot \boldsymbol{\psi} = 0 \tag{6.10.12}$$

where a prime indicates differentiation with respect to the arguments of ϕ and $\boldsymbol{\psi}$. Remembering $(6.10.11)_2$, we observe that

$$\begin{aligned} \mathbf{u} \cdot \mathbf{v} &= \phi' \\ \mathbf{u} \times \mathbf{v} &= \boldsymbol{\psi}' \end{aligned} \tag{6.10.13}$$

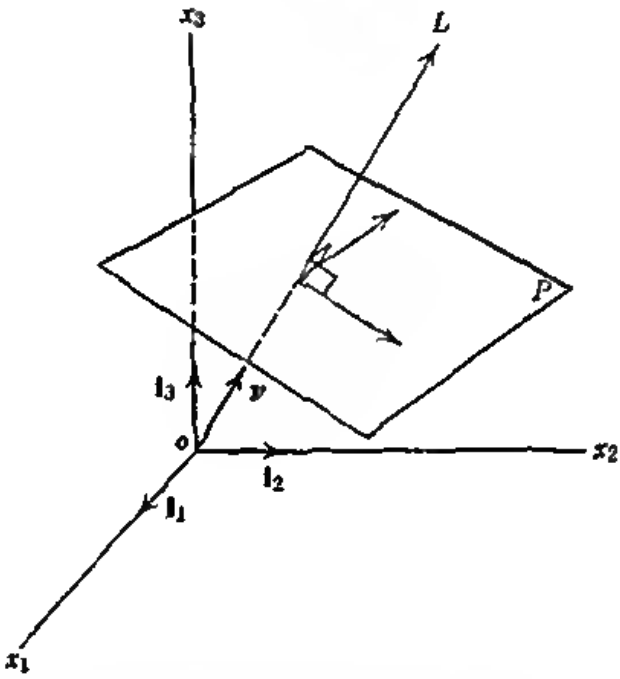


Fig. 6.10.1 Plane elastic wave propagating in the direction of L.

Thus the scalar plane waves $\phi = \phi(\mathbf{v} \cdot \mathbf{x} \mp c_1 t)$ propagate along the direction $\mathbf{v}$, and are associated with the component of the displacement in this direction The vector waves $\boldsymbol{\psi} = \boldsymbol{\psi}(\mathbf{v} \cdot \mathbf{x} \mp c_2 t)$ propagate in a plane perpendicular to the direction $\mathbf{v}$, and are associated with the components of the displacement field in this plane. A plane wave propagating along a direction $\mathbf{v}$, therefore, consists of three independent components corresponding to an irrotational wave and two plane polarized equivoluminal waves.[5]

[5] Irrotational waves are sometimes called *dilatational*, or *compressional*, or *P-waves* (push waves); and the *equivoluminal waves* are called *shear waves*, or *SV* (vertically polarized shear waves) and SH (horizontally polarized shear waves) when there is a preferred horizontal plane as in seismic waves.

(ii) Spherical Waves. In spherical coordinates (r, θ, φ) the Laplacian ∇^2 has the expression (App. C5.27)

$$\nabla^2\phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\varphi^2} \tag{6.10.14}$$

Here we study only the waves that have central symmetry so that there is no dependence on the angles θ and φ. These are waves advancing with spherical wave fronts. In this case $\phi = \phi(r, t)$, and when $g \equiv 0$ (6.10.5) reduces to

$$\frac{\partial^2\phi}{\partial r^2} + \frac{2}{r}\frac{\partial\phi}{\partial r} = \frac{1}{c_1^2}\frac{\partial^2\phi}{\partial t^2} \tag{6.10.15}$$

Upon the substitution of

$$\phi = \frac{1}{r}F(r, t) \tag{6.10.16}$$

this equation reduces to

$$\frac{\partial^2 F}{\partial r^2} = \frac{1}{c_1^2}\frac{\partial^2 F}{\partial t^2} \tag{6.10.17}$$

The general solution of this equation is

$$F = F_1(r - c_1 t) + F_2(r + c_1 t) \tag{6.10.18}$$

where F_1 and F_2 are two arbitrary functions. Thus the general solution of (6.10.15) is

$$\phi = \frac{1}{r}F_1(r - c_1 t) + \frac{1}{r}F_2(r + c_1 t) \tag{6.10.19}$$

Of the two terms F_1 represents a spherical wave front $r - c_1 t = \text{const.}$ advancing in radial direction from the origin $r = 0$. The second term F_2 represents a system of spherical waves advancing towards the origin. The velocity of these waves is the velocity of irrotational waves.

Similarly a solution of (6.10.6) with $\mathbf{h} \equiv \mathbf{0}$ is given by

$$\psi = \frac{1}{r}\mathbf{G}_1(r - c_2 t) + \frac{1}{r}\mathbf{G}_2(r + c_2 t) \tag{6.10.20}$$

The general solution for the displacement field $\mathbf{u}$ is then found by carrying these into $(6.10.2)_1$. The functions F_1, F_2, $\mathbf{G}_1$, and $\mathbf{G}_2$ are to be determined from the initial conditions.

For an explosion at the origin $r = 0$ the waves will travel in the $r > 0$ direction so that in this case one sets $F_2 = 0$ and $\mathbf{G}_2 = \mathbf{0}$. This condition

is related to the problem of uniqueness and, in the case of steady-state motion, is known as the *Sommerfeld radiation* condition. This condition is a mathematical expression of the physical intuition that the steady-state waves of any type generated at a point must travel towards $r = \infty$ and must not radiate back.[6]

6.11. HALF SPACE SUBJECT TO SUDDENLY APPLIED UNIFORM PRESSURE

In this Section we investigate the nature of elastic waves generated by the sudden application of a time dependent uniform pressure $p(t)$ on the plane boundary $x = 0$ of an isotropic elastic solid occupying the half space $x \geq 0$ (Figure 6 7.1) Initially, the body is at rest. At time $t = 0+$ a uniform time-dependent pressure $p(t)$ is applied to the plane boundary $x = 0$. We would like to determine the displacement and stress fields in the body for all times $t > 0$. The problem is one-dimensional The boundary and initial conditions are expressed as

$$t_{xx} = -p(t), \qquad x = 0, \qquad t > 0 \tag{6.11.1}$$

$$u_x(x, t) = \dot{u}_x(x, t) = 0, \qquad x \geq 0, \qquad t = 0 \tag{6 11.2}$$

Since the pressure is uniform, we have a plane wave solution of the wave equation (6 10.5) so that the displacement field may be expressed as

$$u_x \equiv u = F_1(x - \tau) + F_2(x + \tau), \qquad u_y = u_z = 0 \tag{6.11.3}$$

where

$$\tau \equiv c_1 t \tag{6.11.4}$$

and F_1 and F_2 are two arbitrary functions In Section 6 10 we found that (6.11 3) is a solution of Navier's equation. The initial conditions (6 11.2) require that

$$F_1(x) + F_2(x) = 0$$

$$-F'_1(x) + F'_2(x) = 0 \qquad x \geq 0$$

whose solution is

$$F_1(x) = -F_2(x) = A \qquad \text{for } x \geq 0$$

where A is a constant. Since $x + \tau > 0$ for $x, \tau > 0$ we must have

$$u = 0 \quad \text{for} \quad x - \tau \geq 0$$

$$u = F_1(x - \tau) - A \quad \text{for} \quad x - \tau < 0$$

[6] For a further discussion of this and related subjects we refer the reader to Sommerfeld [1949, Art. 28], Stoker [1954], Ewing, Jardetzki and Press [1957], and Stratton [1959].

The constant A can be absorbed into F_1. By setting $F(x) = F_1(x) - A$ with $F(0) = 0$, we have, in fact, a solution

$$u = F(x - \tau) \qquad x - \tau < 0$$
$$u = 0 \qquad x - \tau \geq 0 \tag{6.11.5}$$

To determine the form of the functions F we employ the boundary condition (6.11.1). The stress field is given by

$$t_{xx} = (\lambda_e + 2\mu_e)\frac{\partial u}{\partial x}, \qquad t_{yy} = t_{zz} = \lambda_e \frac{\partial u}{\partial x}$$
$$t_{xz} = t_{yz} = t_{xy} = 0 \tag{6.11.6}$$

The boundary condition (6.11.1) now gives

$$(\lambda_e + 2\mu_e)F'(-\tau) = -p(\tau/c_1)$$

whose integral for $F(0) = 0$ is

$$F(-\tau) = \frac{1}{\lambda_e + 2\mu_e}\int_0^\tau p(s/c_1)\, ds$$

Upon substituting $x - c_1 t$ for τ we get

$$u(x, t) = \frac{c_1}{\lambda_e + 2\mu_e}\int_0^{t-(x/c_1)} p(\sigma)\, d\sigma \quad \text{for} \quad t > \frac{x}{c_1}$$
$$u(x, t) = 0 \quad \text{for} \quad t \leq \frac{x}{c_1} \tag{6.11.7}$$

The stresses as determined from (6.11.6) are

$$t_{xx} = -p\left(t - \frac{x}{c_1}\right), \qquad t_{xz} = t_{yz} = t_{xy} = 0$$
$$t_{yy} = t_{zz} = -\frac{\lambda_e}{\lambda_e + 2\mu_e}\, p\left(t - \frac{x}{c_1}\right) \quad \text{for} \quad t > x/c_1 \tag{6.11.8}$$
$$t_{kl} = 0 \quad \text{for} \quad t \leq x/c_1$$

The stress is zero at points where $x \geq c_1 t$. At time $t = x/c_1$ the wave front arrives. The normal stress t_{xx} at a place $x_0 < c_1 t$ is compressive and has the value of the pressure at $t - x_0/c_1$. According to (6.11.5) the displacement at a point $x < c_1 t$ is proportional to the area under the pressure curve from 0 to the point $t - x/c_1$.

6.12. SPHERICAL CAVITY SUBJECT TO SUDDENLY APPLIED UNIFORM PRESSURE

A second example of elastic waves generated by the boundary loading is encountered in the case of an infinite elastic solid containing a spherical cavity subjected to a uniform dynamical pressure field (Figure 6.12.1).

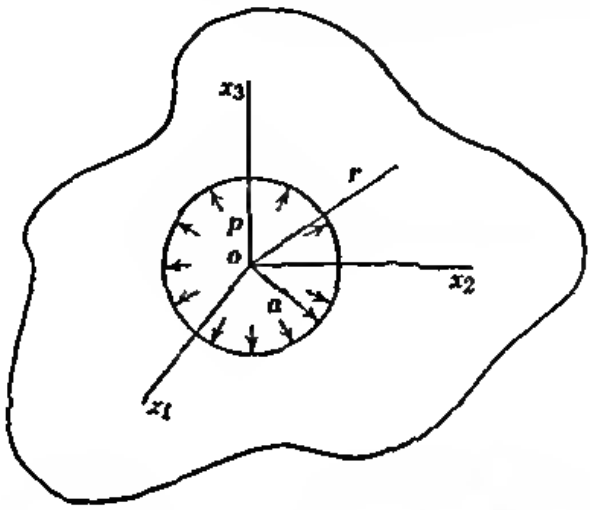


Fig. 6.12.1 Spherical waves generated by the suddenly applied pressure in a spherical cavity.

Upon the sudden application of a uniform pressure $p(t)$ into the cavity, spherical waves are generated propagating away from the cavity. The problem is to determine the displacement and stress fields.

An appropriate coordinate system, in this case, is the set of spherical coordinates (r, θ, φ). The body force is assumed to vanish. Because of the central symmetry of the geometry and the loading we have

$$u \equiv u_r = u(r, t), \qquad u_\theta = u_\varphi = 0$$

The boundary conditions on the surface of the cavity $r = a$ are

$$\begin{aligned} &t_{rr} = -p(t), \qquad r = a, \qquad t > 0 \\ &t_{r\theta} = t_{r\varphi} = 0 \end{aligned} \tag{6.12.1}$$

Initially, the body is at rest so that

$$u(r, 0) = \dot{u}(r, 0) = 0, \qquad r \geq a, \qquad t = 0 \tag{6.12.2}$$

The problem is thus reduced to the determination of a single function satisfying the conditions (6.12.1) and (6.12.2). The general spherical wave solution (6.10.19) of Navier's equation may be employed in the form

$$\phi = \frac{1}{r} F_1\left(t - \frac{r-a}{c_1}\right) + \frac{1}{r} F_2\left(t + \frac{r-a}{c_1}\right) \tag{6.12.3}$$

The initial conditions (6.12.2) imply that

$$F_1(-\xi) + F_2(\xi) = 0$$

$$-F'_1(-\xi) + F'_2(\xi) = 0, \qquad \xi \equiv \frac{r-a}{c_1} \geq 0$$

The solution of these equations is

$$F_1(-\xi) = -F_2(\xi) = A \quad \text{for} \quad \xi \geq 0$$

where A is a constant. Since $t + (r-a)/c_1 > 0$ for $(r-a)/c$, $t > 0$ we must have

$$\phi = 0 \quad \text{for} \quad t - \frac{r-a}{c_1} \leq 0$$

$$\phi = \frac{1}{r} F\left(t - \frac{r-a}{c_1}\right) \quad \text{for} \quad t - \frac{r-a}{c_1} > 0$$

where the constant A is absorbed into F so that $F(0) = 0$.

The displacement field is now given by

$$u_r = \frac{\partial \phi}{\partial r} = \frac{\partial}{\partial r}\left[\frac{1}{r} F(\tau)\right] = -\frac{1}{r^2} F(\tau) - \frac{1}{c_1 r} F'(\tau) \qquad \text{when } \tau > 0$$

(6.12.4) $$u_r = 0 \qquad \text{when } \tau \leq 0$$

$$u_\theta = u_\varphi = 0 \qquad \text{for all } \tau$$

where

(6.12.5) $$\tau \equiv t - \frac{r-a}{c_1}$$

The nonvanishing physical components of the stress tensor in spherical coordinates are given by (6.9.2)

(6.12.6) $$t_{rr} = (\lambda_e + 2\mu_e)\frac{\partial u_r}{\partial r} + 2\lambda_e \frac{u_r}{r}$$

$$t_{\theta\theta} = t_{\varphi\varphi} = \lambda_e \frac{\partial u_r}{\partial r} + 2(\lambda_e + \mu_e)\frac{u_r}{r}$$

Upon substituting (6.12.4) into this we obtain

(6.12.7) $$t_{rr} = (\lambda_e + 2\mu_e)\frac{F''}{c_1^2 r} + 4\mu_e\left(\frac{F'}{c_1 r^2} + \frac{F}{r^3}\right)$$

$$t_{\theta\theta} = t_{\varphi\varphi} = \lambda_e \frac{F''}{c_1^2 r} - 2\mu_e\left(\frac{F'}{c_1 r^2} + \frac{F}{r^3}\right), \qquad \tau > 0$$

The boundary condition (6.12.1) now leads to the following linear differential equation for $F(t)$

$$F'' + \frac{4\mu_e}{\lambda_e + 2\mu_e}\left(\frac{c_1}{a}F' + \frac{c_1^2}{a^2}F\right) = -\frac{c_1{}^2 a}{\lambda_e + 2\mu_e}p(t) \tag{6.12.8}$$

The solution of this equation is subject to the condition $F = F' = 0$ for $\tau \leq 0$, since u_r must be continuous. Thus

$$F(t) = -\frac{c_1 a^2}{2\mu_e}(1-2\nu)^{1/2}\int_0^t e^{-\alpha(t-s)}\sin\beta(t-s)p(s)\,ds \tag{6.12.9}$$

where ν is Poisson's ratio and α and β are defined by

$$\alpha \equiv \frac{1-2\nu}{1-\nu}\frac{c_1}{a}, \qquad \beta \equiv \frac{\sqrt{1-2\nu}}{1-\nu}\frac{c_1}{a} \tag{6.12.10}$$

The function $F(\tau)$ is obtained from (6.12.9) by mere replacement of t by τ.

The case of constant pressure $p(t) = p_0$ applied at $t = 0$ and retained for all time $t > 0$ is of particular interest. In this case (6.12.9) leads to

$$\begin{aligned} F(\tau) &= -\frac{p_0 a^3}{4\mu_e}[1 - (2-2\nu)^{1/2}e^{-\alpha\tau}\sin(\beta\tau + \gamma)] \quad \text{for} \quad \tau > 0 \\ F(\tau) &= 0 \quad \text{for} \quad \tau \leq 0 \end{aligned} \tag{6.12.11}$$

where γ is defined by

$$\cot\gamma \equiv \sqrt{1-2\nu}, \qquad \frac{\pi}{4} \leq \gamma \leq \frac{\pi}{2} \tag{6.12.12}$$

The displacement and stress fields now follow from (6.12.4) and (6.12.6), respectively.

As t becomes larger, the second term in the bracket of (6.12.11) gets smaller. For $t \to \infty$ we obtain the *static* solution. In this limit we have

$$\begin{aligned} u_r &= \frac{p_0 a^3}{4\mu_e}\frac{1}{r^2}, \qquad u_\theta = u_\varphi = 0 \\ t_{rr} &= -2t_{\theta\theta} = -2t_{\varphi\varphi} = -p_0\frac{a^3}{r^3} \end{aligned} \tag{6.12.13}$$

In the theory of plasticity, based on the Tresca yield criterion, when the maximum shear stress $t_{\theta\theta} - t_{rr} = t_{\varphi\varphi} - t_{rr}$ exceeds a limit, yielding occurs. Following Hunter (1954), the plot of this quantity versus r/a for different times is shown in Figure 6.12.2 for $\nu = 0.29$, representative of a typical

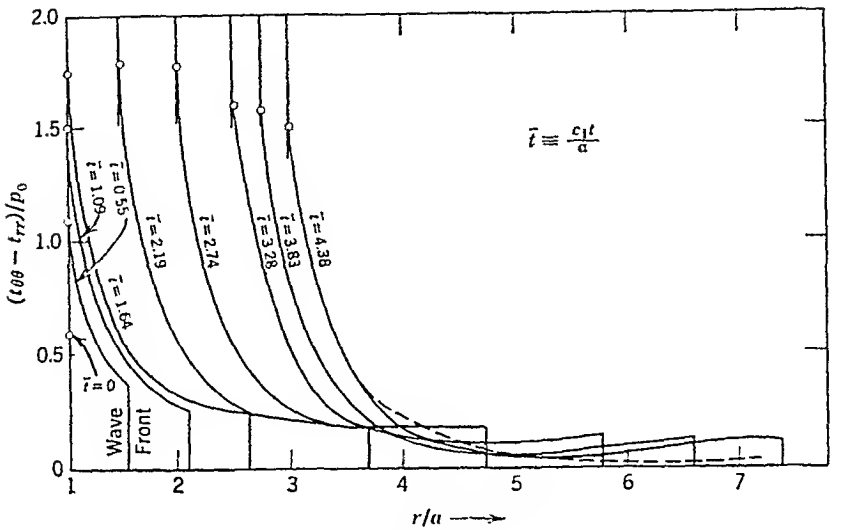


Fig. 6.12.2 Variation of shear stress with distance for various times (after Hunter [1954]).

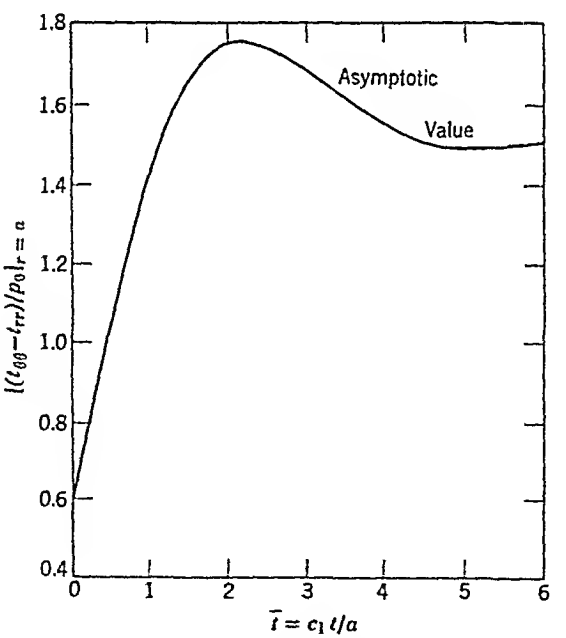


Fig. 6.12.3 Variation of shear stress at cavity surface with time (after Hunter [1954]).

steel. This figure shows that after $\bar{t} = c_1 t/a$ exceeds 4, the static situation is very well approximated in the vicinity of the cavity. The quantity $(t_{\theta\theta} - t_{rr})/p_0$ is always largest at the cavity surface. The time variation of this quantity is shown on Figure 6.12.3. Here, $(t_{\theta\theta} - t_{rr})_{r=a}/p_0$ starts at 0 592 when $\bar{t} = 0$, rises to a maximum 1.75 when $\bar{t} = 2.19$, and falls ultimately to 1.5 as $\bar{t} \to \infty$. Thus the minimum pressure to cause yielding at the cavity is 0.571 t_y ($t_y \equiv$ yield stress) for the dynamic loading and 0.667 t_y for the static loading. It is interesting to note that instantaneously the hoop stresses, $t_{\theta\theta}$ and $t_{\varphi\varphi}$, are compressive and become tensile after a passage of time.

The general case of arbitrary dynamic loading at the surface of a spherical cavity is much more difficult to treat. In this case, polar and azimuthal variations of the surface tractions must be considered, so that we need to determine not only the scalar wave solution but also the vector wave solutions of Navier's equation [7]

PROBLEMS

6.1. Prove that in a nonlinear, isotropic, elastic solid the only possible initial stress is a state of hydrostatic stress. In this regard, what can be said about an anisotropic elastic solid?

6.2. Find the expression of energy for an isotropic, elastic solid that possesses stress-constitutive equations of the second degree in the strains.

6.3. What must be the restrictions on the elastic constants of an elastic solid that possesses stress-constitutive equations of the second degree in the strains, in order that the internal energy density be nonnegative for all states of the strain.

6.4. Suggest a test program to determine all elastic coefficients of a second-degree elastic solid.

6.5. Devise a test program to determine the constitutive functions of an incompressible, nonlinear, hyperelastic solid.

6.6. In a linear, isotropic, elastic solid, show that the stress, t, satisfies the Beltrami-Michell equations

$$t_{rs,kk} + \frac{1}{1+\nu} t_{kk,rs} + \frac{\nu}{1-\nu} F_{k,k}\delta_{rs} + F_{r,s} + F_{s,r} = 0$$

where $\mathbf{F} = \rho(\mathbf{f} - \mathbf{a})$.

6.7. In a linear, isotropic, elastic solid, $f_{r,s} = a_{r,s} = 0$, show that the stress, t_{kl} and strain e_{kl} satisfy the equations

$$t_{rs,kkll} = 0, \qquad e_{rs,kkll} = 0.$$

[7] The general solution was given by Eringen [1957].

6.8. When the body force and the acceleration vanishes, show that the following constitutes special solutions of the Navier's equations for the linear, isotropic, elastic solids:

(*a*) $2\mu_e \mathbf{u} = \nabla F, \qquad \nabla^2 F = 0$

(*b*) $2\mu_e \mathbf{u}_\alpha = A \nabla G_\alpha - (3 - 4\nu) G_\alpha \nabla A, \qquad \nabla^2 G_\alpha = 0$

$A(\mathbf{x}) \equiv a_\alpha x_\alpha, \qquad (\alpha = 1, 2, 3)$

where a_α are constants.

(*c*) $\mu_e \mathbf{u}_\alpha = \nabla H_\alpha \times \nabla A, \qquad \nabla^2 H_\alpha = 0, \qquad (\alpha = 1, 2, 3)$

and A is the same as in (*b*).

6.9. For each of the special solutions of the Navier's equations listed in Problem 6.8, find:

(*a*) Dilatation.

(*b*) Infinitesimal rotation vector.

(*c*) Components of the stress tensor.

6.10. By superposition of the special solutions listed in Problem 6.8, find a solution of the Navier's equation which gives zero shearing stresses at a plane boundary $x_3 = 0$.

6.11. A linear, isotropic, elastic solid of infinite extent is subject to body force, **f**, only. If the acceleration vanishes, show that the displacement field is given by

$$u_l = \frac{1}{4\pi E} \frac{1+\nu}{1-\nu} \left\{ (2 - 2\nu) \int \frac{\rho f_l(\xi)}{r(\mathbf{x}, \xi)} \, dv(\xi) - \frac{1}{2} \int \frac{\partial}{\partial x_k} \left[\frac{x_l - \xi_l}{r(\mathbf{x}, \xi)} \right] \rho f_k(\xi) \, dv(\xi) \right\}$$

where

$$r(\mathbf{x}, \xi) \equiv [(x_k - \xi_k)(x_k - \xi_k)]^{1/2}$$

and the integral is over the three-dimensional space.

6.12. If $(\mathbf{t}^{(1)}, \tilde{\mathbf{e}}^{(1)})$ and $(\mathbf{t}^{(2)}, \tilde{\mathbf{e}}^{(2)})$ are two sets of stress and strain tensors in a medium $\mathscr{V} + \mathscr{S}$, prove that

$$\int_{\mathscr{V}} t_{kl}^{(1)} e_{kl}^{(2)} \, dv = \int_{\mathscr{V}} t_{kl}^{(2)} \tilde{e}_{kl}^{(1)} \, dv$$

6.13. By use of the result given in Problem 6.12, prove the *reciprocal theorem* of Betti and Rayleigh:

$$\int_{\mathscr{S}} t_k^{(1)} u_k^{(2)} \, da + \int_{\mathscr{V}} \rho f_k^{(1)} u_k^{(2)} \, dv = \int_{\mathscr{S}} t_k^{(2)} u_k^{(1)} \, da + \int_{\mathscr{V}} \rho f_k^{(2)} u_k^{(1)} \, dv$$

where $t_k^{(1)}$, $t_k^{(2)}$ are respectively stress resultants; $f_k^{(1)}$, $f_k^{(2)}$ are body forces; and $u_k^{(1)}$ and $u_k^{(2)}$ are the displacement vectors corresponding to each set of state of strain and stress given in Problem 6.12.

6.14. An isotropic, elastic solid of infinite extent is subject to a concentrated load of magnitude, F, in x-direction at the origin of coordinates. If the state of stress is two-dimensional, show that

$$t_{xx} = \frac{Fx}{4\pi(1-\nu)r^2}\left[(1-2\nu)+\frac{2x^2}{r^2}\right]$$

$$t_{yy} = \frac{Fx}{4\pi(1-\nu)r^2}\left[(1+2\nu)-\frac{2x^2}{r^2}\right]$$

$$t_{xy} = \frac{Fy}{4\pi(1-\nu)r^2}\left[(1-2\nu)+\frac{2x^2}{r^2}\right], \quad r \equiv \sqrt{x^2+y^2}$$

Discuss the nature of singularity at $x = y = 0$.

6.15. A half plane is subject to a concentrated load, P, perpendicular to the surface, $x = 0$, of the medium at the origin, $x = y = 0$. Show that the stress field in the medium is given by

$$t_{xx} = -\frac{2Px^3}{\pi r^4}, \quad t_{yy} = -\frac{2Pxy^2}{\pi r^4}, \quad t_{xy} = -\frac{2Px^2y}{\pi r^4}$$

Discuss the nature of singularity at $x = y = 0$.

6.16. Using the result given in Problem 6.15, determine the state of stress in a half plane when (*a*) the load is a distributed load of constant magnitude, p, over $|y| < a$, and (*b*) it is a variable magnitude, $p(y)$, over the $x = 0$ line.

6.17. In a cylindrical bar, the displacement field is prescribed by

$$u_1 = -\theta x_2 x_3, \quad u_2 = \theta x_1 x_3, \quad u_3 = \phi(x_1, x_2)$$

Determine the stresses and the equation that ϕ must satisfy if the bar is in equilibrium. Neglect the body forces. If the bar is subject only to torques at its ends, set up the boundary-value problem for ϕ.

6.18. Using the displacement field given in Problem 6.17, show that in an elliptic cylinder

$$\phi = -\frac{a^2-b^2}{a^2+b^2}\theta x_1 x_2$$

where a and b are the major and minor axes of the cylinder. Show that the torque, T, at any section is given by

$$T = \frac{\pi a^3 b^3 \mu \theta}{a^2+b^2}.$$

6.19. An equivoluminal or irrotational plane wave is reflected from a flat stress-free boundary of a half space. Show that two plane waves are reflected from the boundary.

(*a*) Find the angles that their directions of propagation make with the normal of the plane boundary.

(*b*) Determine the amplitude ratios of the reflected waves to that of the incident wave.

(*c*) Determine the conditions for which one or both reflected waves may disappear.

6.20. In an elastic half space, surface waves occur. These waves are usually most damaging in earthquakes and other explosions. The amplitude of these waves exponentially decays away from the surface. Find their velocity of propagations.

6.21. (Short Term Paper) Write the expressions of strain deformation relations and Cauchy's equations of motion in one of the following coordinate systems.

(*a*) Ellipsoidal coordinates.
(*b*) Conical coordinates.
(*c*) Paraboloidal coordinates.
(*d*) Bi-polar coordinates.
(*e*) Toroidal coordinates.

6.22. (Short Term Paper) Read literature and present a complex function technique for solving two-dimensional problems in elastostatics.

6.23. (Short Term Paper) An infinite sheet containing a slit (crack) of finite length is subject to a uniform axial tension perpendicular to the slit at infinity. The crack is beginning to spread under the tension. Determine the stress field.

6.24. (Short Term Paper) Study literature and present a mathematical technique for the solution of mixed boundary-value problems in which stress is prescribed along a part of the boundary and displacements on the rest (for example, punch problems).

CHAPTER 7

Fluid Dynamics

7.1. SCOPE OF THE CHAPTER

This chapter is intended for the derivation of the basic equations of linear Stokesian fluids and some illustrative solutions. The linear constitutive theory for compressible and incompressible viscous fluids is presented in Section 7.2. An account of the experimental origin of viscosities and a method of their determination are given in Section 7.3. A résumé of basic equations (of the Navier-Stokes theory) and the boundary and initial conditions are collected in Section 7.4. In Section 7.5 we give the basic equations in orthogonal curvilinear coordinates. The remaining sections, 7.6 to 7.15, are devoted to illustrative solutions. Section 7.6 discusses some simple fluid motions, such as irrotational motion, isochoric motion, and plane motion. Slow motions of the Navier-Stokes fluids in a circular tube are discussed in Section 7.7. In Section 7.8 some nonsteady one-dimensional problems are discussed. These are illustrative examples of some simple boundary and initial value problems. Motions of Taylor, Section 7.9, is another illustrative example of an exact solution. In Section 7.10 we present plane stagnation flow where the importance of the nonlinear character of the fluid is exhibited. Here we see the origin of the ideas leading to the boundary layer theory so prominent in classical fluid dynamics. Up to now, the fluid was assumed to be incompressible. With Section 7.11 we return to compressible fluids. For the inviscid fluid, the acoustic approximation is presented, followed by two illustrative problems—one on radiation from a vibrating sphere (Section 7.12) and the other a diffraction problem (Section 7.13). In Section 7.14 we delve into the problem

of propagation of shock discontinuities. Here, of course, we see the first genuine use of jump conditions across a moving discontinuity surface (shock wave). The last section, 7.15, is on spherical blast waves. The special difficulties of this problem that arise from its nonlinear character are particularly noticed when we recall the solution of a corresponding problem (Section 6.12) in the linear theory of elasticity.

The problems discussed in this chapter are but a few of the exact solutions in the field. They are presented for the purpose of making the equations derived meaningful for the student. Needless to say, even a representative class of solutions or an exposition of various exact and approximate methods of solutions is beyond the scope of this chapter.

7.2. LINEAR CONSTITUTIVE EQUATIONS

In Section 5.7 we obtained the constitutive equation, of Stokesian fluids. The general form of these equations is given by (5.7.28) to (5.7.34), that is,

$$t_{kl} = -\pi(\rho^{-1}, \theta)\,\delta_{kl} + {}_D f_{kl}(\mathbf{d}, \rho^{-1}, \theta) \tag{7.2.1}$$

$$q_k = 0 \tag{7.2.2}$$

$$\varepsilon = \psi(\rho^{-1}, \theta) - \theta \frac{\partial \psi}{\partial \theta} \tag{7.2.3}$$

$$\eta = -\frac{\partial \psi}{\partial \theta} \tag{7.2.4}$$

where π is the thermodynamic pressure defined by

$$\pi(\rho^{-1}, \theta) \equiv -\frac{\partial \psi}{\partial \rho^{-1}} \tag{7.2.5}$$

and ${}_D\mathbf{f}$ is subject to the restrictions arising from the axioms of objectivity and the Clausius-Duhem inequality

$$\mathbf{Q}(t)\,{}_D\mathbf{f}(\mathbf{d}, \rho^{-1}, \theta)\mathbf{Q}^{\mathrm{T}}(t) = {}_D\mathbf{f}(\mathbf{Q}\mathbf{d}\mathbf{Q}^{\mathrm{T}}, \rho^{-1}, \theta) \tag{7.2.6}$$

$${}_D f_{kl}\, d_{lk} \geq 0 \tag{7.2.7}$$

Here $\{\mathbf{Q}(t)\}$ is the full group of all orthogonal transformations. These fluids are therefore *isotropic*. Simplifications can be achieved when an isotropic tensor function ${}_D\mathbf{f}$ of the symmetric tensors $\mathbf{d}$ is a polynomial in $\mathbf{d}$. In this case, ${}_D\mathbf{f}$ can be expressed as

$${}_D\mathbf{f} = \sum_{\alpha=0}^{N} a_\alpha(\rho^{-1}, \theta)\,\mathbf{d}^\alpha \tag{7.2.8}$$

Upon using the Cayley-Hamilton theorem (Appendix B4.8) in three dimensions,

$$\mathbf{d}^3 - \mathrm{I}_d\,\mathbf{d}^2 + \mathrm{II}_d\,\mathbf{d} - \mathrm{III}_d = 0 \tag{7.2.9}$$

we can replace all terms of (7.2.8) having powers of $\mathbf{d}$ greater than 2, so that (7.2.8) can be written in the form (cf. Appendix B9.3)

$$ {}_{\rho}\mathbf{f} = \alpha_0\mathbf{I} + \alpha_1\,\mathbf{d} + \alpha_2\,\mathbf{d}^2 \tag{7.2.10}$$

where α_κ are polynomials in the invariants of $\mathbf{d}$ with coefficients that are functions of ρ^{-1} and θ

$$\alpha_\kappa = \alpha_\kappa(\rho^{-1}, \theta; I_d, \mathrm{II}_d, \mathrm{III}_d), \qquad (\kappa = 0, 1, 2) \tag{7.2.11}$$

Inequality (7.2.7) is valid for all $\mathbf{d}$. If ${}_{\rho}\mathbf{f}(\rho^{-1}, \theta; \mathbf{d})$ is continuous in $\mathbf{d}$, we must have ${}_{\rho}\mathbf{f}(\rho^{-1}, \theta, 0) = 0$. Consequently

$$\alpha_0(\rho^{-1}, \theta, 0, 0, 0) = 0 \tag{7.2.12}$$

This therefore gives, for the general nonlinear stress, constitutive equations

$$t_{kl} = (-\pi + \alpha_0)\delta_{kl} + \alpha_1\, d_{kl} + \alpha_2\, d_{km}\, d_{ml} \tag{7.2.13}$$

We state this as

THEOREM. *The stress constitutive equations of Stokesian fluids* (7.2.1), *with* ${}_{\rho}\mathbf{f}$ *being a polynomial in* $\mathbf{d}$, *are equivalent to* (7.2.13) *with* α_κ *subject to* (7.2.12) *and*

$$\alpha_0\,\mathrm{tr}\,\mathbf{d} + \alpha_1\,\mathrm{tr}\,\mathbf{d}^2 + \alpha_2\,\mathrm{tr}\,\mathbf{d}^3 \geq 0 \tag{7.2.14}$$

The above inequality follows from using ${}_{\rho}\mathbf{f}$, given by (7.2.10), in (7.2.7). Using the identities

$$\begin{aligned} \mathrm{tr}\,\mathbf{d} &= \mathrm{I}_d \\ \mathrm{tr}\,\mathbf{d}^2 &= \mathrm{I}_d^2 - 2\mathrm{II}_d \\ \mathrm{tr}\,\mathbf{d}^3 &= \mathrm{I}_d^3 - 3\mathrm{I}_d\mathrm{II}_d + 3\mathrm{III}_d \end{aligned} \tag{7.2.15}$$

we can also express inequality (7.2.14) as

$$\alpha_0\mathrm{I}_d + \alpha_1(\mathrm{I}_d^2 - 2\mathrm{II}_d) + \alpha_2(\mathrm{I}_d^3 - 3\mathrm{I}_d\mathrm{II}_d + 3\mathrm{III}_d) \geq 0 \tag{7.2.16}$$

Linear Theory. The master equations (7.2.13) can be used to derive various approximate theories.[1] Here we are concerned only with the linear theory. In this case we have

$$\alpha_0 = \lambda_v(\rho^{-1}, \theta)\mathrm{I}_d, \qquad \alpha_1 = 2\mu_v(\rho^{-1}, \theta), \qquad \alpha_2 = 0 \tag{7.2.17}$$

[1] Cf. Eringen [1962, Art. 49 and Ch. 7]

where λ_v and μ_v are called, respectively, the *dilatational* and *shear* viscosities. Consequently, for the stress constitutive equations we have

$$t_{kl} = (-\pi + \lambda_v \mathrm{I}_d)\delta_{kl} + 2\mu_v\, d_{kl} \tag{7.2.18}$$

Using this result in (7.2.16) we find the restrictions that must be imposed on these fluids

$$(\lambda_v + 2\mu_v)\mathrm{I}_d^{\,2} - 4\mu_v \mathrm{II}_d \geq 0 \tag{7.2.19}$$

This inequality is postulated to be valid for all motions. The consequence of this is found in the same way as in Section 6.3 (theorem). Hence

$$0 \leq 3\lambda_v + 2\mu_v < \infty, \qquad 0 \leq \mu_v < \infty \tag{7.2.20}$$

THEOREM. *Linear Stokesian fluids have the stress constitutive equations of the form* (7.2.18) *with the viscosity coefficients* λ_v *and* μ_v *subject to* (7.2.20), *and the thermodynamic pressure* π *given by* (7.2.5).

The entropy changes in these fluids because of the dissipative power and internal heat sources. According to (5.7.26) this is calculated by

$$\rho\theta\dot{\eta} = {}_D f_{kl}\, d_{lk} + \rho h \tag{7.2.21}$$

In the case of the linear theory, by using (7.2.18) we reduce it to

$$\rho\theta\dot{\eta} = (\lambda_v + 2\mu_v)\mathrm{I}_d^{\,2} - 4\mu_v \mathrm{II}_d + \rho h \tag{7.2.22}$$

Incompressible Fluids. The incompressibility restriction is made by setting $\mathrm{I}_d = 0$ and replacing π by the unknown pressure function $p(\mathbf{x}, t)$. Hence, for the linear theory, we have

$$t_{kl} = -p\delta_{kl} + 2\mu_v\, d_{kl} \tag{7.2.23}$$

$$d_{kk} = 0 \tag{7.2.24}$$

$$0 \leq \mu_v(\theta) < \infty \tag{7.2.25}$$

$$\rho_0\theta\dot{\eta} = -4\mu_v \mathrm{II}_d + \rho_0 h \tag{7.2.26}$$

The pressure $p(\mathbf{x}, t)$ is an unknown of the theory, and is to be determined from the solution of each specific problem under given boundary conditions. In this case $\rho =$ constant $= \rho_0$, so that μ_v is a function of the temperature θ only.

7.3. EXPERIMENTAL ORIGIN OF VISCOSITIES

The linear constitutive equations obtained in the previous section have found applications in the treatment of motions of a large class of fluids.

Applications to the motion of water, air, and a variety of lubrication oils have been so successful that most hydrodynamicists today consider these equations to be the laws of nature! We must point out that various important physical phenomena known as the Poynting effect, or normal stress effect, and Kelvin effect, cannot be derived from these linear equations. In highly viscous fluids, and in motions having large deformation rates, certain fluids, exhibit these effects. Thus, for example, sweetened milk, between two coaxial cylinders with the inner cylinder rotating and the external one fixed, has been observed to climb up to the surface of the inner cylinder, contrary to the behavior of Navier-Stokes fluids Similarly, the swelling that occurs in a pipe flow, upon its exit from the pipe, cannot be explained by use of linear stress constitutive equations For these and other interesting nonlinear phenomena, we refer the reader to Eringen [1962, Chapter 7].

Within the context of the linear theory we have two viscosity coefficients, λ_v and μ_v, which are adequate for the characterization of Navier-Stokes fluids. Of these, λ_v affects the dissipation in a dilatational motion, whereas μ_v is responsible for the equivoluminal motion. Thus, we consider the following two simple experiments.

(*i*) ***Hydrostatic Pressure.*** The state of stress in a body is said to be a hydrostatic pressure if

$$t_{kl} = -\bar{p}\delta_{kl} \tag{7 3.1}$$

Upon substituting this into (7.2.18) and setting $k = l$, we find

$$\pi - \bar{p} = (\lambda_v + \tfrac{2}{3}\mu_v)\mathrm{I}_d \tag{7.3.2}$$

and

$$d_{kl} = 0 \quad \text{when} \quad k \neq l$$

And for an incompressible fluid $\mathrm{I}_d = 0$ we have

$$\bar{p} = \pi \tag{7.3.3}$$

Note that this would also be the case when the fluid undergoes a rigid motion, that is, $\mathbf{d} = 0$. In this case, the term "hydrostatic pressure" for $\bar{p}$ is certainly appropriate. For the compressible fluid, it has been argued that $\bar{p}$ must also be the same as π. Hence, nearly all works in classical hydrodynamics are based upon the condition

$$\lambda_v + \tfrac{2}{3}\mu_v = 0 \qquad \text{(compressible)} \tag{7.3.4}$$

This condition is known as the ***Stokes condition*** Stokes himself did not place great faith in this condition. Kinetic theory is in support of (7 3.4), however, for dense gases and liquids there is no evidence that it is valid

Some experimental investigations are found to contradict this condition.[2] The use of the Stokes condition does not lead to great discrepancies in the calculations for most problems, unless the problem is so chosen as to eliminate the effect of μ_v. For example, if the distortion of fluid consists of pulsating expansions and contractions that are the same in all directions then, under the condition (7.3.4), the dissipative power will be zero. The conclusion that there will be no dissipation in this case appears physically unreasonable, no matter how close λ_v may be to $-2\mu_v/3$. Moreover, for some fluids, experiments indicate that $\lambda_v > 0$. Accurate determination of λ_v is of some difficulty, since measurements of amplitude decay in spherically symmetric waves are difficult.

(ii) Rectilinear Shearing Flow. The steady flow of a fluid between two parallel rigid walls, one stationary and the other moving with constant velocity u_0, is important in explaining the experimental observation of the effect of the shear viscosity μ_v. Selecting rectangular coordinates (x, y, z) with x taken along the fixed wall and y and z perpendicular to it (Figure

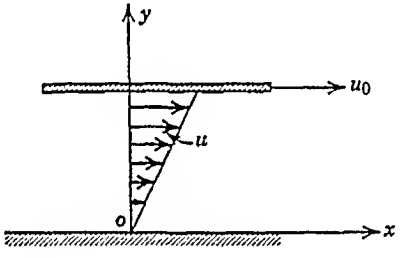


Fig. 7.3.1 Rectilinear Flow.

7.3.1), we expect the following one-dimensional velocity field to be the solution of the problem

$$\dot{x} = u(y), \qquad \dot{y} = \dot{z} = 0 \tag{7.3.5}$$

Through (7.2.18) and (7.3.5) for the stress we get

$$\begin{aligned} t_{xx} &= t_{yy} = t_{zz} = -\pi \\ t_{xy} &= \mu_v \frac{du}{dy}, \qquad t_{xz} = t_{yz} = 0 \end{aligned} \tag{7.3.6}$$

Cauchy's equations of motion for vanishing body force give

$$-\frac{\partial \pi}{\partial x} + \mu_v \frac{d^2u}{dy^2} = 0, \qquad \frac{\partial \pi}{\partial y} = \frac{\partial \pi}{\partial z} = 0$$

[2] For a discussion of various experimental results, see Truesdell [1952, p.228].

Thus $\pi = \pi(x)$ and

$$\frac{d\pi}{dx} = \mu_v \frac{d^2u}{dy^2} = A$$

Since u is a function of y only, A is constant. Integration gives

$$\mu_v u = A\frac{y^2}{2} + By + C, \qquad \pi = Ax + D$$

Assuming that the fluid adheres to the walls for the boundary conditions, we write

$$u = 0 \qquad \text{for} \quad y = 0$$

$$u = u_0 \qquad \text{for} \quad y = h$$

This determines the constants B and C, and results in the solution

$$\mu_v u = \frac{A}{2}(y^2 - hy) + \mu_v \frac{u_0}{h} y$$

$$t_{xy} = \frac{A}{2}(2y - h) + \mu_v \frac{u_0}{h}, \qquad \pi = Ax + D \tag{7.3.7}$$

$$t_{xx} = t_{yy} = t_{zz} = -\pi, \qquad t_{xz} = t_{yz} = 0$$

When the pressure gradient $A = 0$, we get

$$u = \frac{u_0}{h} y, \qquad t_{xy} = \mu_v \frac{u_0}{h} y \tag{7.3.8}$$

Therefore, the horizontal component of the velocity varies linearly across the channel and take its maxima at the moving wall $y = h$ (Figure 7.3.1) Consequently, the resistance (the horizontal force) encountered by the upper plate gives a measure of the viscosity, that is,

$$\mu_v = t_{xy}/u_0$$

The channel flow described above can be used for the measurement of the shear viscosity μ_v. Because of the difficulty in maintaining the conditions (7.3.5) for this type of flow in real channels having finite depths, instead the flow of a viscous fluid in a circular tube is used as a viscometer. The solution of this problem is given in Section 7.7. By the use of capillary tubes (in diameters varying from 0 001 to 0 014 cm), as initiated by Poiseuille [1840],[3] it is possible to obtain a very high accuracy. The

[3] A physician interested in circulation of blood experimentally established the law of discharge of fluids through capillary tubes, and carried out the first series of significant experiments in this field

viscosity of water, for example, at atmospheric pressure is probably known within 1/10th of 1% accuracy.

There exist other types of viscometers based on the Couette flow (torsional flow), parallel plate viscometers, etc. For a discussion of some of these cases, see Michell [1946, Chapter III].

There also exist empirical formulas for the temperature- and pressure-dependence of viscosity. For instance, Lamb [1952, Art. 326], gives the formula, attributed to Helmholtz.

$$\mu_v = \frac{0.01779}{1 + 0.03368\theta + 0.0002209\theta^2} \tag{7.3.9}$$

for the viscosity of water in C.G.S. units, when the temperature is measured in the Centigrade scale. For air, he quotes the equation

$$\mu_v = 0.0001702\ (1 + 00.0329\theta + 0.0000070\theta^2) \tag{7.3.10}$$

The pressure dependence is understood through the fact that $\pi = \pi(\rho^{-1}, \theta)$, implying $\rho^{-1} = f(\pi, \theta)$. Consequently

$$\mu_v = \mu_v[f(\pi, \theta), \theta] = \bar{\mu}_v(\pi, \theta) \tag{7.3.11}$$

It is found in fluids and in air that μ_v increases with pressure. In air, $\nu_v \equiv \mu_v/\rho$, known as the *kinematic viscosity*, varies inversely as the pressure. In Table 7.3.1 we quote the viscosities of a few fluids at atmospheric pressure.[4]

7.4. RÉSUMÉ OF BASIC EQUATIONS OF FLUID DYNAMICS

Fluid dynamics, linear in stress constitutive equations, are based on the following equations.

Equations of Balance

$$\frac{\partial \rho}{\partial t} + (\rho v_k)_{,k} = 0 \tag{7.4.1}$$

$$t_{kl,k} + \rho(f_l - \dot{v}_l) = 0 \tag{7.4.2}$$

$$t_{kl} = t_{lk} \tag{7.4.3}$$

$$\rho\theta\dot{\eta} = (\lambda_v + 2\mu_v)\mathrm{I}_d^2 - 4\mu_v \mathrm{II}_d + \rho h \tag{7.4.4}$$

$$0 \leq 3\lambda_v + 2\mu_v < \infty, \qquad 0 \leq \mu_v < \infty \tag{7.4.5}$$

The last conditions are necessary and sufficient for the satisfaction of the Clausius-Duhem inequality.

[4] The values of μ_v are taken from Michell [1946, p. 118] except for water and air (marked by asterisks) which are taken from Goldstein [1938, pp. 5–7].

Table 7.3.1. Viscosity of fluids in C G.S. units (gram/cm sec) (at Atmospheric Pressure)

	0°C	20°C	40°C	60°C	80°C	100°C
Water*	0.01792	0.01005	0 00656	0.00469	0 00356	0 00284
Mercury	0.0170	0.0157	—	—	0.00122	—
Carbon bisulfide	0.0044	0 0038	0.0032	—	—	—
Glycerine	46.	8.7	—	—	—	—
Sperm Oil	1.13	0.335	0.173	0 009	0.040	—
Castor Oil	—	7.24	2 23	0.68	0 28	0.12
Mobil Oil BB	—	9.47	1.30	0.54	0.26	0.12
Air*	1.709×10^{-4}	1.808×10^{-4}	1.904×10^{-4}	1.997×10^{-4}	$2\,088 \times 10^{-4}$	2.175×10^{-4}

For viscosities of other substances the reader is referred to *The International Critical Tables.*

Equations of Jump. If a discontinuity surface $\sigma(t)$ is sweeping the body with a velocity v, in the direction of its positive normal $\mathbf{n}$, the following jump conditions must be satisfied at $\sigma(t)$

$$[\rho(\mathbf{v}-v)]\cdot\mathbf{n}=0 \tag{7.4.6}$$

$$-[t_{kl}]n_k+[\rho v_l(\mathbf{v}-v)]\cdot\mathbf{n}=0 \tag{7.4.7}$$

$$[(\varepsilon+\tfrac{1}{2}v^2)\rho(\mathbf{v}-v)]\cdot\mathbf{n}-[t_{kl}v_l+q_k]n_k=0 \tag{7.4.8}$$

$$\left[\rho\eta(\mathbf{v}-v)-\frac{\mathbf{q}}{\theta}\right]\cdot\mathbf{n}\geq 0 \tag{7.4.9}$$

Constitutive Equations (Compressible Fluids)

$$t_{kl}=(-\pi+\lambda_v I_d)\delta_{kl}+2\mu_v\, d_{kl} \tag{7.4.10}$$

$$q_k=0 \tag{7.4.11}$$

$$\varepsilon=\psi(\rho^{-1},\theta)-\theta\frac{\partial\psi}{\partial\theta} \tag{7.4.12}$$

$$\eta=-\frac{\partial\psi}{\partial\theta},\qquad \pi=-\frac{\partial\psi}{\partial\rho^{-1}} \tag{7.4.13}$$

where the free energy $\psi(\rho^{-1},\theta)$ characterizes the thermodynamic state of the system.

Constitutive Equations (Incompressible Fluids)

$$t_{kl}=-p\delta_{kl}+2\mu_v\, d_{kl} \tag{7.4.14}$$

$$q_k=0 \tag{7.4.15}$$

$$\varepsilon=\psi(\theta)-\theta\frac{\partial\psi}{\partial\theta} \tag{7.4.16}$$

$$\eta=-\frac{\partial\psi}{\partial\theta} \tag{7.4.17}$$

In this case, $\rho=\rho_0=\text{const.}$, so that we must also satisfy

$$v_{k,k}=0 \tag{7.4.18}$$

which follows from (7.4.1). The function $p(\mathbf{x},t)$ is an unknown of the theory, to be determined upon integrating the field equations and using the boundary conditions. In this case, ε and η are uncoupled from the mechanical state of the system, so that we need not prescribe ψ in order to determine the motion of incompressible fluids.

Kinematical Relations

$$a_k = \dot{v}_k = \frac{\partial v_k}{\partial t} + v_{k,l} v_l \qquad \text{(acceleration)} \tag{7.4 19}$$

$$d_{kl} = \tfrac{1}{2}(v_{k,l} + v_{l,k}) \qquad \text{(deformation rate)} \tag{7.4.20}$$

$$w_{kl} = \tfrac{1}{2}(v_{k,l} - v_{l,k}) \qquad \text{(spin)} \tag{7.4.21}$$

$$w_k = e_{klm} w_{ml} \qquad \text{(vorticity)} \tag{7.4.22}$$

$$\mathrm{I}_d = v_{k,k}, \qquad \mathrm{II}_d = \tfrac{1}{2}(d_{kk} d_{ll} - d_{kl} d_{lk}) \tag{7.4.23}$$

Navier-Stokes Equations. Upon substituting t given by (7 4 10) and (7.4.14) into (7.4.2), we obtain the celebrated equations of *Navier-Stokes.*

$$(\lambda_v + \mu_v) v_{k,kl} + \mu_v v_{l,kk} - \pi_{,l} + \rho(f_l - \dot{v}_l) = 0 \qquad \text{(compressible)} \tag{7.4.24}$$

$$\mu_v v_{l,kk} - p_{,l} + \rho(f_l - \dot{v}_l) = 0 \qquad \text{(incompressible)} \tag{7.4 25}$$

The Navier-Stokes equations (7.4.24) for compressible fluids must be supplemented with the equation of continuity (7 4.1), the equation of energy (7.4.4), the equation of state $\psi(\rho^{-1}, \theta)$, or (7.4 13), thus providing five partial differential equations for the determination of five unknowns v_k, ρ, and θ. For incompressible fluids, no need arises for the equations of energy and state. We need to supplement (7.4 25) with the incompressibility condition (7 4 18), thus obtaining four partial differential equations to determine the four unknowns v_k and p. Thus the problem is determinate in both cases when the boundary and the initial conditions are "well posed."

We emphasize the difference between the Navier-Stokes equations and Navier's equations (6 5.20) and (6.5 21) In the former, we observe not only an extra pressure term but also that these equations are nonlinear; this can be seen from the expression of the acceleration (7.4.19) and from the products of ρ and $\dot{v}$ present in (7 4.24) and in the continuity equation (7.4.1) In the linear theory of elasticity, this situation does not occur, on account of the approximation (6.5.17) being made for the acceleration, and the density ρ being taken constant in the treatment of (6.5 20)

Boundary Conditions. If on a part of the surface $\mathscr{S}_t$ of the boundary $\mathscr{S}$ the traction, and on the remainder $\mathscr{S}_v \equiv \mathscr{S} - \mathscr{S}_t$ the velocity is prescribed, then we write

$$\begin{aligned} t_{kl} n_k &= \bar{t}_l \qquad \text{on } \mathscr{S}_t \\ v_k &= \bar{v}_k \qquad \text{on } \mathscr{S}_v \end{aligned} \tag{7.4.26}$$

Here $\mathscr{S}_t$ and $\mathscr{S}_v$ are not overlapping. According to Lamb, at a free surface or at the contacting surface of two dissimilar fluids, the tractions $\bar{t}_l$ should

be continuous. Whether or not the continuity of velocity across the surface or the interface follows from this is not known, although special examples exhibit the continuity.

The conditions to be satisfied at a solid boundary contacting the fluid medium are more controversial. According to Stokes, the fluid must adhere to the solid, that is, $(7.4.26)_2$ must be satisfied at the contacting surface with $\bar{v}_k$ being the surface velocity of the solid, or

$$\Delta \mathbf{v} = \mathbf{0} \qquad \text{on } \mathscr{S} \text{ of solid} \tag{7.4.27}$$

where $\Delta \mathbf{v}$ denotes the difference between the fluid and solid boundary velocities. For high-altitude aerodynamics, the condition of adherence is known to be invalid. Among the various slip conditions proposed, the most important one is

$$\Delta v_n = 0, \qquad \Delta \mathbf{v}_t = k\mathbf{t}_t \tag{7.4.28}$$

where the subscripts n and t denote the normal and tangential components, respectively, at the boundary, and k is a coefficient that may depend on ρ^{-1} and θ. These conditions have found some support from kinetic theory. The coefficient k is nearly zero, except for very low pressures.

Initial Conditions. Initial conditions are statements of the velocity and density fields throughout the body at an initial time

$$\begin{aligned} \mathbf{v}(\mathbf{x}, 0) &= \mathbf{v}_0(\mathbf{x}) \\ \rho(\mathbf{x}, 0) &= \rho_0(\mathbf{x}) \end{aligned} \qquad \text{in } \mathscr{V} \text{ at } t = 0 \tag{7.4.29}$$

The boundary and initial conditions must reasonably represent the physical situation of the fluid. In a well-posed theory, these conditions are admissible on the grounds of existence and uniqueness theorems.[5] The ultimate success of the theory, however, lies in the experimental verifications of the predicted phenomena. The success of the Navier-Stokes theory to date has surely been beyond the expectations of its originators.

7.5. CURVILINEAR COORDINATES

To obtain the expressions of various equations in curvilinear coordinates, we employ the procedure introduced in Section 6.6. In *orthogonal curvilinear coordinates*, the square of the arc length, ds^2, is given by

$$ds^2 = g_{11}(dx^1)^2 + g_{22}(dx^2)^2 + g_{33}(dx^3)^2$$

[5] For existence and uniqueness theorems, see Serrin [1959, Art. 72] and the references therein.

The metric tensor g_{kl} has only three nonvanishing components g_{11}, g_{22}, and g_{33}. The form of Cauchy's equation of motion is already given by (6.6.8). The expressions for the physical components of the deformation rate, spin, and vorticity are readily obtained by the mere replacement of **u** by **v** in (6.6.11) and (6 6 12). Thus

$$(7.5.1)\qquad d^{(k)}_{(l)} = d^k{}_l \frac{\sqrt{g_{kk}}}{\sqrt{g_{ll}}} = \frac{1}{2}\left[\frac{\sqrt{g_{kk}}}{\sqrt{g_{ll}}}\frac{\partial}{\partial x^l}\left(\frac{v^{(k)}}{\sqrt{g_{kk}}}\right) + \frac{\sqrt{g_{ll}}}{\sqrt{g_{kk}}}\frac{\partial}{\partial x^k} \times \left(\frac{v^{(l)}}{\sqrt{g_{ll}}}\right) + \frac{1}{\sqrt{g_{kk}g_{ll}}}\sum_{m=1}^{3}\frac{\partial g_{kk}}{\partial x^m}\frac{v^{(m)}}{\sqrt{g_{mm}}}\delta_{kl}\right]$$

$$(7.5.2)\qquad w^{(k)}_{(l)} = \frac{1}{2\sqrt{g_{kk}g_{ll}}}\left[\frac{\partial}{\partial x^l}(\sqrt{g_{kk}}v^{(k)}) - \frac{\partial}{\partial x^k}(\sqrt{g_{ll}}v^{(l)})\right]$$

$$(7.5.3)\qquad w^{(m)} = \sum_{k,l} e^{mlk} w^{(k)}_{(l)} \quad \text{or} \quad \mathbf{w} = \nabla \times \mathbf{v}$$

The stress constitutive equations are,

$$(7.5.4)\qquad t^{(k)}_{(l)} = (-\pi + \lambda_v d^{(m)}_{(m)})\delta^k_l + 2\mu_v d^{(k)}_{(l)}$$

The Navier-Stokes equations may be obtained in the same manner as in Section 6.6. In vectorial form they can be expressed as

$$(7.5.5)\qquad (\lambda_v + 2\mu_v)\nabla\nabla\cdot\mathbf{v} - \mu_v \nabla \times \nabla \times \mathbf{v} - \nabla\pi + \rho(\mathbf{f} - \dot{\mathbf{v}}) = 0$$

By mere substitution into (7.5.5) of the expressions for the gradient, divergence, and curl operators in curvilinear coordinates, we obtain the curvilinear expression of this equation.

The expression of the acceleration vector $\dot{\mathbf{v}}$ is obtained as follows. First write

$$\dot{v}_k = \frac{\partial v_k}{\partial t} + v_{k,l}v^l = \frac{\partial v_k}{\partial t} + (v_{k,l} - v_{l,k})v^l + v_{l,k}v^l$$

$$= \frac{\partial v_k}{\partial t} + 2w_{kl}v^l + \tfrac{1}{2}(v_l v^l)_{,k}$$

Each one of the quantities appearing in this last expression can be replaced by their physical components

$$\dot{v}_k = \sqrt{g_{kk}}\,\dot{v}^{(k)}, \qquad v_k = \sqrt{g_{kk}}\,v^{(k)}, \qquad v^l = v^{(l)}/\sqrt{g_{ll}}$$

$$w_{kl} = w^{(k)}_{(l)}\sqrt{g_{kk}g_{ll}}$$

Hence

$$\dot{v}^{(k)} = \frac{\partial v^{(k)}}{\partial t} + 2w^{(k)}{}_{(l)}v^{(l)} + \frac{1}{2\sqrt{g_{kk}}}(\mathbf{v}\cdot\mathbf{v})_{,k} \tag{7.5.6}$$

Since the term $v_l v^l$ is an absolute scalar, its covariant derivative is the same as its partial derivative. Expression (7.5.6) can also be written as

$$\dot{\mathbf{v}} = \frac{\partial \mathbf{v}}{\partial t} - \mathbf{v}\times\mathbf{w} + \tfrac{1}{2}\nabla(v^2) \tag{7.5.7}$$

where w is the vorticity vector whose physical components are given by (7.5.3). A more explicit form of (7.5.7) is

$$\dot{\mathbf{v}} = \frac{\partial \mathbf{v}}{\partial t} - \mathbf{v}\times(\nabla\times\mathbf{v}) + \tfrac{1}{2}\nabla(v^2) \tag{7.5.8}$$

For the vector form of the Navier-Stokes equation we therefore have

$$(\lambda_v + 2\mu_v)\nabla\nabla\cdot\mathbf{v} - \mu_v\nabla\times\nabla\times\mathbf{v} - \nabla\pi + \rho\mathbf{f} = \rho\left[\frac{\partial \mathbf{v}}{\partial t} - \mathbf{v}\times(\nabla\times\mathbf{v}) + \tfrac{1}{2}\nabla(v^2)\right] \quad \text{(compressible)} \tag{7.5.9}$$

For the incompressible fluid $\nabla\cdot\mathbf{v} = 0$ and we replace π by p so that

$$-\mu_v\nabla\times\nabla\times\mathbf{v} - \nabla p + \rho\mathbf{f} = \rho_0\left[\frac{\partial \mathbf{v}}{\partial t} - \mathbf{v}\times(\nabla\times\mathbf{v}) + \tfrac{1}{2}\nabla(v^2)\right] \tag{7.5.10}$$

(incompressible)

The expression of the continuity equation in vector form, suitable for curvilinear coordinates, is

$$\frac{\partial \rho}{\partial t} + \mathbf{v}\cdot\nabla\rho + \rho\nabla\cdot\mathbf{v} = 0 \tag{7.5.11}$$

For an incompressible fluid $\rho = \text{const.} = \rho_0$ and (7.5.11) reduces to

$$\nabla\cdot\mathbf{v} = 0 \quad \text{(incompressible)} \tag{7.5.12}$$

The boundary conditions in vector form can be expressed by replacing the tensor components of vectors and tensor by their physical components in (7.4.26). This gives

$$\begin{aligned} t^{(k)}{}_{(l)}n_{(k)} &= \bar{t}_{(l)} \quad \text{on } \mathscr{S}_t \\ v^{(k)} &= \bar{v}^{(k)} \quad \text{on } \mathscr{S}_v \end{aligned} \tag{7.5.13}$$

If we substitute from (7.5.4) into (7.5.13) these could be arranged into

$$-\pi\mathbf{n} + \lambda_v(\nabla \cdot \mathbf{v})\mathbf{n} + 2\mu_v(\mathbf{n} \cdot \nabla)\mathbf{v} + \mu_v\mathbf{n} \times (\nabla \times \mathbf{v})$$

$$= \bar{\mathbf{f}}(\mathbf{x}, t) \quad \text{on } \mathscr{S}_t, t \geq 0 \tag{7.5.14}$$

$$\mathbf{v} = \bar{\mathbf{v}}(\mathbf{x}, t) \quad \text{on } \mathscr{S}_v \quad \text{(compressible)}$$

$$-p\mathbf{n} + 2\mu_v(\mathbf{n} \cdot \nabla)\mathbf{v} + \mu_v\mathbf{n} \times (\nabla \times \mathbf{v}) = \bar{\mathbf{f}}(\mathbf{x}, t) \quad \text{on } \mathscr{S}_t, t \geq 0 \tag{7.5.15}$$

$$\mathbf{v} = \bar{\mathbf{v}}(\mathbf{x}, t) \quad \text{on } \mathscr{S}_v \text{ (incompressible)}$$

The vector form of the initial conditions is

$$\begin{aligned} \mathbf{v}(\mathbf{x}, 0) &= \mathbf{v}_0(\mathbf{x}) \\ \rho(\mathbf{x}, 0) &= \rho_0(\mathbf{x}) \end{aligned} \qquad \text{in } \mathscr{V}, t = 0 \tag{7.5.16}$$

Using the expressions of the gradient, divergence, and curl operators in orthogonal curvilinear coordinates, (7.5.9) to (7.5.12) and (7.5.14) to (7.5.16), we get the necessary equations for the treatment of problems in hydrodynamics.

7.6. SPECIAL CLASSES OF FLUID MOTION

In this section we shall derive certain reduced forms of the field equations (Navier-Stokes equations and the equation of continuity)

$$(\lambda_v + 2\mu_v)\nabla\nabla \cdot \mathbf{v} - \mu_v\nabla \times \nabla \times \mathbf{v} - \nabla\pi + \rho\mathbf{f}$$

$$= \rho\left[\frac{\partial \mathbf{v}}{\partial t} - \mathbf{v} \times (\nabla \times \mathbf{v}) + \nabla\left(\frac{v^2}{2}\right)\right] \tag{7.6.1}$$

$$\frac{\partial \rho}{\partial t} + \mathbf{v} \cdot \nabla\rho + \rho\nabla \cdot \mathbf{v} = 0 \tag{7.6.2}$$

which are useful for the treatment of special classes of motion.

(*i*) ***Irrotational Motion.*** If the velocity is derivable from a potential, that is,

$$\mathbf{v} = \nabla\phi \tag{7.6.3}$$

then we have $\mathbf{w} = \nabla \times \mathbf{v} = 0$ and the motion is called irrotational. We also decompose the body force as

$$\mathbf{f} = -\nabla g - \nabla \times \mathbf{h}, \qquad \nabla \cdot \mathbf{h} = 0 \tag{7.6.4}$$

where g and $\mathbf{h}$ are scalar and vector potentials. Substituting (7.6.3) and (7.6.4) into (7.6.1) and (7.6.2), we obtain

$$\frac{\lambda_v + 2\mu_v}{\rho} \nabla\nabla^2\phi - \nabla \times \mathbf{h} - \nabla\left[\frac{\partial\phi}{\partial t} + \tfrac{1}{2}(\nabla\phi)^2 + \int \frac{d\pi}{\rho} + g\right] = 0 \tag{7.6.5}$$

$$\frac{\partial\rho}{\partial t} + \nabla\phi \cdot \nabla\rho + \rho\nabla^2\phi = 0 \tag{7.6.6}$$

Further significant progress is possible when $(\lambda_v + 2\mu_v)/\rho$ is constant and $\nabla \times \mathbf{h} = 0$. The former represents a very special situation, placing a condition on $\lambda_v + 2\mu_v$ since in this case we must write $\lambda_v + 2\mu_v = k\rho$ in (7.6.6) where k is a constant. For this case, (7.6.5) integrates to

$$-\frac{\lambda_v + 2\mu_v}{\rho} \nabla^2\phi + \frac{\partial\phi}{\partial t} + \tfrac{1}{2}(\nabla\phi)^2 + \int \frac{d\pi}{\rho} + g = K_1(t) \tag{7.6.7}$$

Another interesting case is that of an inviscid fluids, $\lambda_v = \mu_v = 0$. In this case, the integral of (7.6.5) is

$$\frac{\partial\phi}{\partial t} + \tfrac{1}{2}(\nabla\phi)^2 + \int \frac{d\pi}{\rho} + g = K_1(t) \tag{7.6.8}$$

where $K_1(t)$ is an arbitrary function of time. At a fixed time at all spatial points of the body, the left side of (7.6.8) is a constant. By evaluating the left side of (7.6.8) at a material point we can eliminate $K_1(t)$. Equation (7.6.8) is the celebrated *Bernoulli's equation* of inviscid, irrotational, compressible fluids.

(ii) Isochoric Motions. For the motions in which volume does not change we have $\nabla \cdot \mathbf{v} = 0$. Substituting this and (7.6.4) into (7.6.1) and (7.6.2) we obtain

$$-\nabla\left(\int \frac{d\pi}{\rho} + g + \frac{v^2}{2}\right) - \frac{\mu_v}{\rho} \nabla \times \nabla \times \mathbf{v} - \nabla \times \mathbf{h} = \frac{\partial\mathbf{v}}{\partial t} - \mathbf{v} \times (\nabla \times \mathbf{v}) \tag{7.6.9}$$

$$\frac{d\rho}{dt} = 0 \tag{7.6.10}$$

From (7.6.10) it follows that

$$\rho = \text{const.} \qquad \text{(along a material line)} \tag{7.6.11}$$

By taking the curl and divergence of both sides of (7.6 9), for the case of $\rho = \text{const.} = \rho_0$ throughout the body, we get

$$\nabla \times (\nu_0 \nabla \times \mathbf{w}) + \nabla \times \nabla \times \mathbf{h} + \frac{\partial \mathbf{w}}{\partial t} + \nabla \times (\mathbf{w} \times \mathbf{v}) = 0 \tag{7.6.12}$$

$$\nabla^2 \left(\frac{\pi}{\rho_0} + g + \frac{v^2}{2} \right) = 0 \tag{7.6 13}$$

where we wrote

$$\mathbf{w} = \nabla \times \mathbf{v}, \qquad \nu_0 \equiv \frac{\mu_0}{\rho} \tag{7.6 14}$$

Now consider the inviscid case $\nu_0 = 0$ and take $\mathbf{h} = 0$. In this instance we have

$$\frac{\partial \mathbf{w}}{\partial t} + \nabla \times (\mathbf{w} \times \mathbf{v}) = 0 \tag{7.6 15}$$

According to Zorawski's criterion (2 5.6) we see that in this case the flux of the vorticity w across every material surface remains constant. Thus *the vorticity is convected with the fluid.* When ν_0 is not zero and $\mathbf{h}$ does not vanish, we have vorticity created by viscous friction and body forces.

Introducing a vector velocity potential

$$\mathbf{v} = \nabla \times \boldsymbol{\psi} \tag{7.6.16}$$

we see that $\nabla \cdot \mathbf{v} = 0$ is satisfied automatically and (7.6.12) and (7.6 13) can be written as

$$\nabla \times \left[\nu_0 \nabla \times \nabla \times \nabla \times \boldsymbol{\psi} + \nabla \times \mathbf{h} + \frac{\partial}{\partial t} (\nabla \times \boldsymbol{\psi}) + (\nabla \times \nabla \times \boldsymbol{\psi}) \times (\nabla \times \boldsymbol{\psi}) \right] = 0 \tag{7.6.17}$$

$$\nabla^2 \left[\frac{\pi}{\rho_0} + \tfrac{1}{2} (\nabla \times \boldsymbol{\psi})^2 + g \right] = 0 \tag{7.6 18}$$

Contrary to the case of linear elasticity, unfortunately, the nonlinear character of equations (7.6.1) and (7 6.2) does not allow superposition of cases (i) and (ii) above to obtain the solution of the general fluid motions.

(*iii*) ***Incompressible Fluids.*** In this case, (7.6.17) and (7.6.18) apply with π replaced by p.

(*iv*) ***Plane Motions of Incompressible Fluids.*** For a plane motion in the

plane $x^3 \equiv z = 0$, the z-component of the velocity vanish and the two other components of the velocity are independent of z. Thus

$$\mathbf{v}(x^1, x^2, t) = \nabla \times \psi(x^1, x^2, t) = (v^{(1)}, v^{(2)}, 0)$$

$$\mathbf{w}(x^1, x^2, t) = \nabla \times \nabla \times \psi = (0, 0, w^{(3)}) \tag{7.6.19}$$

Consequently we may take

$$\psi = \psi(x^1, x^2, t)\mathbf{e}_z \tag{7.6.20}$$

where $\mathbf{e}_z$ is the unit vector in the z-direction. For this case the equations (7.6.17) and (7.6.18) simplify a great deal. We now give the expressions of these equations for $\nu_v = \text{const.}$ and $\mathbf{h} = \mathbf{0}$ for the case of incompressible fluids, referred to rectangular and plane polar coordinates.

(*a*) *Rectangular Coordinates* (x, y)

$$\mathbf{v} = \nabla \times \psi = \frac{\partial \psi}{\partial y}\mathbf{i}_1 - \frac{\partial \psi}{\partial x}\mathbf{i}_2$$

$$\mathbf{w} = \nabla \times \mathbf{v} = -\nabla^2 \psi \mathbf{i}_3 \tag{7.6.21}$$

$$-\frac{\partial}{\partial t}(\nabla^2 \psi) - \left(\frac{\partial \psi}{\partial y}\frac{\partial}{\partial x} - \frac{\partial \psi}{\partial x}\frac{\partial}{\partial y}\right)\nabla^2 \psi + \nu_v \nabla^4 \psi = 0 \tag{7.6.22}$$

$$\nabla^2\left[\frac{p}{\rho_0} + \frac{1}{2}\left(\frac{\partial \psi}{\partial x}\right)^2 + \frac{1}{2}\left(\frac{\partial \psi}{\partial y}\right)^2 + g\right] = 0 \tag{7.6.23}$$

where

$$\nabla^2 \psi \equiv \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \tag{7.6.24}$$

(*b*) *Plane Polar Coordinates* (r, θ)

$$-\frac{\partial}{\partial t}(\nabla^2 \psi) - \frac{1}{r}\left(\frac{\partial \psi}{\partial \theta}\frac{\partial}{\partial r} - \frac{\partial \psi}{\partial r}\frac{\partial}{\partial \theta}\right)\nabla^2 \psi + \nu_v \nabla^4 \psi = 0 \tag{7.6.25}$$

$$\nabla^2\left[\frac{p}{\rho_0} + \frac{1}{2}\left(\frac{\partial \psi}{\partial r}\right)^2 + \frac{1}{2}\left(\frac{1}{r}\frac{\partial \psi}{\partial \theta}\right)^2 + g\right] = 0 \tag{7.6.26}$$

where

$$\nabla^2 \psi \equiv \frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r}\frac{\partial \psi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \psi}{\partial \theta^2} \tag{7.6.27}$$

Upon solving (7.6.22) or (7.6.25) we can determine ψ. Using ψ in (7.6.23) or (7.6.26) together with the boundary conditions we can determine p.

7.7. POISEUILLE FLOW

The steady motion of a viscous fluid in a straight pipe of uniform circular cross section of radius a is known as the Poiseuille flow. In cylindrical coordinates (r, θ, z) with z taken along the axis of the cylinder, we consider the one-dimensional motion given by

$$\dot{r} = \dot{\theta} = 0, \qquad \dot{z} = w(r), \qquad 0 \leq r \leq a \tag{7.7.1}$$

As the boundary condition, we assume that the fluid adheres to the wall $r = a$ of the cylinder, that is,

$$w(a) = 0 \tag{7.7.2}$$

The differential equations of motion follow from (7 5 10). For this motion through (7.7.1) and (Appendix C5.26) we have

$$\nabla \cdot \mathbf{v} = 0, \qquad \nabla \times \nabla \times \mathbf{v} = -\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right)\mathbf{e}_z$$

$$-\mathbf{v} \times (\nabla \times \mathbf{v}) + \tfrac{1}{2}\nabla(v^2) = 0, \qquad \frac{\partial \mathbf{v}}{\partial t} = 0$$

The motion is isochoric. Compatible with this, we also consider the fluid to be incompressible. Hence

$$\mu_v \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right)\mathbf{e}_z - \nabla p + \rho \mathbf{f} = 0$$

Neglecting the body force $\mathbf{f}$, we see that $\partial p/\partial r = \partial p/\partial \theta = 0$ and $p = p(z)$ Consequently

$$\frac{\mu_v}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right) = \frac{\partial p}{\partial z} \tag{7.7.3}$$

The integral of this equation is

$$w = \frac{r^2}{4\mu_v}\frac{\partial p}{\partial z} + A \log r + B \tag{7.7 4}$$

At the axis of the cylinder $r = 0$ the velocity must be finite. This implies that $A = 0$. The condition (7.7.2) is adequate for the determination of B Hence

$$w = \frac{r^2 - a^2}{4\mu_v}\frac{\partial p}{\partial z} \tag{7.7.5}$$

Thus the axial velocity distribution is parabolic along the r-axis and has its maxima at the axis $r = 0$ of the cylinder (Figure 7.7.1) where we took

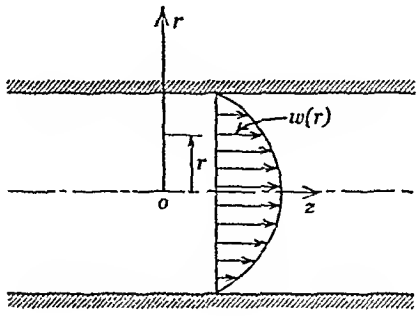


Fig. 7.7.1 Poiseuille flow.

$\frac{\partial p}{\partial z} < 0$ so that the flow is toward the positive z-axis.

$$w_{\max} = -\frac{a}{4\mu_v}\frac{\partial p}{\partial z} \tag{7.7.6}$$

The flux of fluid across any cross section is calculated from

$$\mathcal{M} = \int_0^a \rho_0 w 2\pi r \, dr = -\rho_0 \frac{\pi}{8}\frac{a^4}{\mu_v}\frac{\partial p}{\partial z} \tag{7.7.7}$$

where ρ_0 is the constant density of the fluid. The physical components of the stress tensor are given by

$$\begin{aligned} t_{rr} &= t_{\theta\theta} = t_{zz} = -p \\ t_{rz} &= \mu_v \frac{\partial w}{\partial r} = \frac{r}{2}\frac{\partial p}{\partial z}, \qquad t_{r\theta} = t_{\theta z} = 0 \end{aligned} \tag{7.7.8}$$

Equation (7.7.7) agrees exactly with the findings of Poiseuille [1840–1] and others. These results are valid for slow motions of the fluid. For high velocities, the flow becomes unstable and the adherence condition at the wall is violated. The separation of the flow from the wall and the succeeding turbulent flow are both mathematically and conceptually too difficult to analyze here.

A different type of physical phenomenon is encountered for certain types of highly viscous fluids known as non-Newtonian fluids. Experiments of Merrington [1943] on pipe flow have shown that the fluid at the exit section of a viscometer swell out. This phenomenon cannot be explained by the use of the constitutively linear theory of viscous fluids. The

nonlinear terms in the constitutive equations must be taken into account. These additional terms can be shown to account for the above phenomenon, known as the Merrington effect or normal stress effect. An account of the subject may be found in Eringen [1962, Chapter 7].

7.8. MOTION OF INCOMPRESSIBLE VISCOUS FLUIDS IN ONE-SPACE DIMENSION

Here we consider the motion of an incompressible, viscous fluid along the x-axis of rectangular coordinates (x, y, z). We will study the class of motions having velocity components of the form:

$$v_1 = u(x, y, t), \qquad v_2 = v_3 = 0 \tag{7.8.1}$$

The equation of continuity (7.4.1) reduces to

$$\frac{\partial u}{\partial x} = 0$$

Hence $u = u(y, t)$. The equations of motion (7.4.25), with vanishing body force $\mathbf{f} = 0$, read

$$\mu_v \frac{\partial^2 u}{\partial y^2} - \frac{\partial p}{\partial x} - \rho_0 \frac{\partial u}{\partial t} = 0$$

$$\frac{\partial p}{\partial y} = 0, \qquad \frac{\partial p}{\partial z} = 0$$

From the last two equations, we have $p = p(x, t)$. Since u is not a function of x through the first equation we see that

$$\nu_v \frac{\partial^2 u}{\partial y^2} - \frac{\partial u}{\partial t} = K(t)$$

$$\frac{1}{\rho_0} \frac{\partial p}{\partial x} = K(t)$$

where $\nu_v \equiv \mu_v/\rho_0$. The second of these equations gives

$$\frac{p}{\rho_0} = K(t)x + K_1(t)$$

If the motion considered extends to $x = \infty$, then $p \to \infty$ as $x \to \infty$. To forbid such large pressures, we must set $K(t) = 0$, so that the final equations to be dealt with are

$$\nu_v \frac{\partial^2 u}{\partial y^2} = \frac{\partial u}{\partial t}, \qquad \nu_v \equiv \frac{\mu_v}{\rho_0} \tag{7.8.2}$$

$$p = \rho_0 K_1(t) \tag{7.8.3}$$

Here $K_1(t)$ will have to be determined from the boundary conditions. The velocity profile of such motions is the integral of $(7.8.2)_1$ subject to boundary and initial conditions. Equation $(7.8.2)_1$ is identical with that of heat conduction in rigid solids. We now give a few solutions for illustrative purposes.

A "similarity solution" of $(7.8.2)_1$ is obtained by exploring a solution of the form,

$$u = u(\eta), \qquad \eta \equiv y^2/4\nu_v t \tag{7.8.4}$$

Substitution of (7.8.4) into $(7.8.2)_1$ gives

$$\eta u'' + (\eta + \tfrac{1}{2})u' = 0$$

The solution of this equation is

$$u(\eta) = A_1 \int_0^\eta e^{-\zeta} \zeta^{-1/2}\, d\zeta + B_1$$

where A_1 and B_1 are constants. By a change of variable

$$\eta \equiv \xi^2 = y^2/4\nu_v t \tag{7.8.5}$$

we can write this result as

$$u(y, t) = A \operatorname{erf} \xi + B \tag{7.8.6}$$

where the error function, erf ξ, is defined by

$$\operatorname{erf} \xi = \frac{2}{\sqrt{\pi}} \int_0^\xi e^{-x^2}\, dx \tag{7.8.7}$$

This function is of frequent occurrence in heat-conduction and diffusion problems. A few of its important properties are

$$\begin{gathered} \operatorname{erf} 0 = 0, \qquad \operatorname{erf} \infty = 1 \\ \operatorname{erf} (-\xi) = -\operatorname{erf} \xi \end{gathered} \tag{7.8.8}$$

The constants A and B can be determined to satisfy certain boundary conditions. An example is given below.

(i) *Motion with Constant Velocity at $y = 0$.* Consider a fluid occupying the half space $y \geq 0$ which is at rest when $t \leq 0$. For all times $t \geq 0$ the boundary $y = 0$ of this fluid is set to motion with a constant velocity u_0 along the x-axis, that is,

$$\begin{aligned} u(0, t) &= u_0, \qquad y = 0, \qquad t \geq 0 \\ u(y, 0) &= 0, \qquad y > 0, \qquad t = 0 \end{aligned} \tag{7.8.9}$$

Substituting from (7.8.6) into (7.8.9) we determine A and B, thus obtaining the solution

$$u = u_0 - u_0 \operatorname{erf}(y/2\sqrt{\nu_v t}) \tag{7.8.10}$$

(*ii*) ***Motion Due to Initial Velocity Distribution.*** Consider a two-dimensional motion of a fluid occupying an infinite region with the initial conditions

$$u(y, 0) = u_0(y), \qquad -\infty < y < \infty, \qquad t = 0 \tag{7.8.11}$$

This problem may easily be solved by use of the Fourier integral

$$u(y, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \bar{u}(\eta, t) e^{-i\eta y}\, d\eta \tag{7.8.12}$$

which has as its inverse the Fourier transform defined by

$$\bar{u}(\eta, t) = \int_{-\infty}^{\infty} u(y, t) e^{i\eta y}\, dy \tag{7.8.13}$$

Substitution of (7.8.12) into equation $(7.8.2)_1$ gives

$$\int_{-\infty}^{\infty} [-\dot{\bar{u}}(\eta, t) - \eta^2 \nu_v \bar{u}(\eta, t)] e^{-i\eta y}\, d\eta = 0$$

which is satisfied if

$$\dot{\bar{u}} + \eta^2 \nu_v \bar{u} = 0$$

The solution of this equation subject to (7.8.11) is

$$\bar{u}(\eta, t) = \bar{u}_0(\eta) e^{-\nu_v \eta^2 t}$$

where $\bar{u}_0(\eta)$ is the Fourier transform of $u_0(y)$ defined by (7.8.13) at $t = 0$. Thus (7.8.12) becomes

$$u(y, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \bar{u}_0(\eta) e^{-\nu_v \eta^2 t - i\eta y}\, d\eta \tag{7.8.14}$$

If $\bar{f}$ and $\bar{g}$ are the Fourier transforms of f and g, then according to the Faltung theorem[6]

$$\int_{-\infty}^{\infty} f(\eta) g(y - \eta)\, d\eta = \frac{1}{2\pi} \int_{-\infty}^{\infty} \bar{f}(\eta) \bar{g}(\eta) e^{-i\eta y}\, d\eta \tag{7.8.15}$$

Thus selecting $\bar{f}(\eta) = \bar{u}_0(\eta)$ and $\bar{g}(\eta) = \exp(-\nu_v \eta^2 t)$ we have the inverse transforms

$$f(y) = u_0(y), \qquad g(y) = \frac{1}{\sqrt{4\pi \nu_v t}} e^{-y^2/4\nu_v t}$$

[6] Cf Sneddon [1951, p 24].

so that using (7.8.15) we get

$$u(y, t) = \frac{1}{\sqrt{4\pi\nu_v t}} \int_{-\infty}^{\infty} u_0(\eta) \exp\left[-(y-\eta)^2/4\nu_v t\right] d\eta \tag{7.8.16}$$

which is the solution of our problem. By a change of variable $\xi = (\eta - y)/2\sqrt{\nu_v t}$ this result may also be expressed as

$$u(y, t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} u_0(y + 2\sqrt{\nu_v t}\xi)\, e^{-\xi^2}\, d\xi \tag{7.8.17}$$

This is also the solution of the corresponding heat-conduction problem in an infinite solid (cf. Carslaw and Jaeger [1959, Art. 2.3]).

7.9. MOTIONS OF TAYLOR

A large class of special plane motions is obtained whenever the nonlinear terms in (7.6.22) vanish separately, that is, when

$$-\frac{\partial}{\partial t}\nabla^2\psi + \nu_v\nabla^4\psi = 0 \tag{7.9.1}$$

$$\frac{\partial\psi}{\partial y}\frac{\partial}{\partial x}\nabla^2\psi - \frac{\partial\psi}{\partial x}\frac{\partial}{\partial y}\nabla^2\psi = 0 \tag{7.9.2}$$

Among such motions are[7]

$$\nabla^2\psi = T(t)F(x, y) \quad \text{and} \quad \nabla^2\psi = K \tag{7.9.3}$$

where $T(t)$ is a function of time alone and K is a constant. The first of these equations, when substituted into (7.9.1), gives by separation of variables

$$\frac{\dot{T}}{\nu_v T} = \frac{\nabla^2 F}{F} = -k^2$$

so that

$$T = Ce^{-\nu_v k^2 t}, \qquad \nabla^2 F + k^2 F = 0$$

Hence

$$\begin{aligned} \nabla^2\psi &= Ce^{-\nu_v k^2 t}F \\ \nabla^2 F + k^2 F &= 0 \end{aligned} \tag{7.9.4}$$

[7] All solutions satisfying (7.9.1) and (7.9.2) were given by Kampé de Fériet [1932].

From these it also follows that

$$\psi = -\frac{C}{k^2} e^{-\nu_e k^2 t} F + G \tag{7.9.5}$$

$$\nabla^2 F + k^2 F = 0, \qquad \nabla^2 G = 0$$

Taylor [1923] discovered a special case of these motions, namely $G = 0$, so that the motions

$$\psi(x, y, t) = -\frac{C}{k^2} e^{-\nu_e k^2 t} F(x, y) \tag{7.9.6}$$

$$\nabla^2 F + k^2 F = 0$$

are known as Taylor motions.

The celebrated Helmholtz equation $(7.9.4)_2$ has many interesting solutions. For example, Taylor gave a solution of the form

$$\psi = A \exp(-2\pi^2 \nu_e t/a^2) \cos\frac{\pi x}{a} \cos\frac{\pi y}{a} \tag{7.9.7}$$

which is a model of a system of vortices rotating in opposite directions in squares of side a (Figure 7.9.1). The magnitude of the vortices decays in

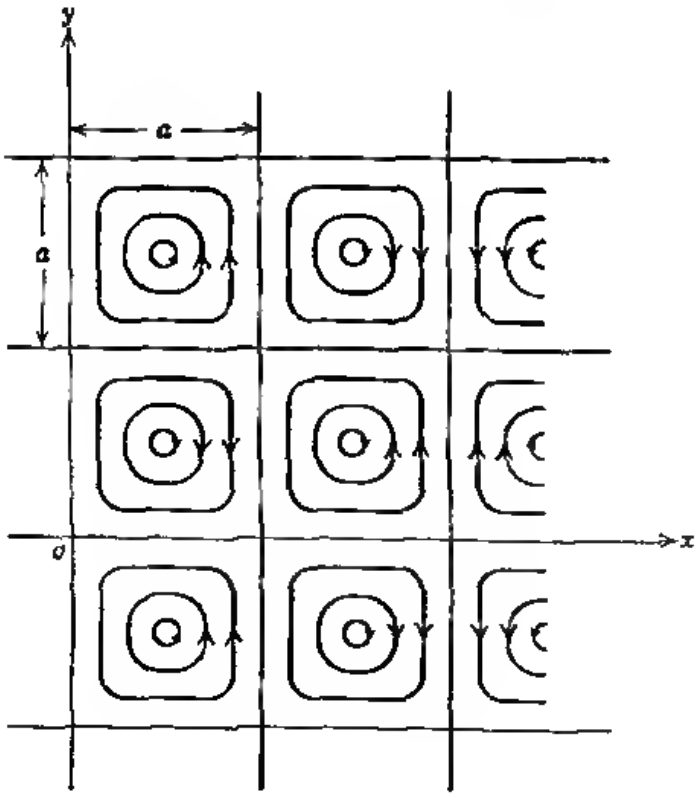


Fig. 7.9.1 Taylor vortices.

time. In a characteristic time $t_1 = a^2/2\pi^2\nu_v$ the reduction in magnitude is $1/e$. For air at 20°C, $\nu_v = 0.15$ cm^2/sec; the characteristic time for a vortex having the diameter $a = 1$ cm is $t_1 = 1/3$ second.

Many other problems of interest can be solved by use of the solution of (7.9.4). We give another example concerned with vortex motion.[8]

Motion Due to an Initial Vortex. If we are interested in the motion due to a vortex $w_z(r, t)$ of constant magnitude acting in $0 \leq r \leq a$ we have the initial conditions

$$\begin{aligned} 2w_z &= \Omega(r, 0) = K \quad \text{for} \quad 0 \leq r \leq a \\ 2w_z &= \Omega(r, 0) = 0 \quad \text{for} \quad r > a \end{aligned} \tag{7.9.8}$$

where r is the radial distance from the origin. In this case we solve $(7.9.4)_2$ in cylindrical coordinates. The equation

$$F'' + \frac{1}{r} F' + k^2 F = 0$$

has the solution

$$F(r) = AJ_0(kr) + BY_0(kr)$$

where J_0 and Y_0 are Bessel functions of the first and second kinds of order zero. Of these the functions $Y_0(kr) \to -\infty$ for $r \to 0$ so that we must set $B = 0$. Hence, according to $(7.6.21)_1$ and $(7.9.4)_1$ we have

$$\Omega(r, t) = -\nabla^2\psi(r, t) = -Ae^{-\nu_v k^2 t} J_0(kr) \tag{7.9.9}$$

According to $(7.6.21)_2$ we have $w_z = -\nabla^2\psi$ which at $t = 0$ does not give a constant, so that we need the general solution

$$\Omega(r, t) = -\int_0^\infty A(k) e^{-\nu_v k^2 t} J_0(kr)\, dk \tag{7.9.10}$$

This gives for the boundary conditions

$$-\int_0^\infty A(k) J_0(kr)\, dk = \begin{cases} K & \text{when} \quad 0 \leq r \leq a \\ 0 & \text{when} \quad r > a \end{cases} \tag{7.9.11}$$

The function $A(k)$ may be solved from here by use of the well-known Hankel's formula, namely, if

$$\bar{f}(k) = \int_0^\infty f(r) J_0(kr) r\, dr \tag{7.9.12}$$

then

$$f(r) = \int_0^\infty \bar{f}(k) J_0(kr) k\, dk \tag{7.9.13}$$

[8] For a collection of problems of this type see Berker [1963, Articles 33–39, 41].

Using this we find that

$$A(k) = -Kk\int_0^a J_0(kr)r\,dr = -aKJ_1(ka) \tag{7.9.14}$$

Thus (7.9.10) now reads

$$\Omega(r, t) = aK\int_0^\infty J_1(ka)e^{-\nu_0 k^2 t}J_0(kr)\,dk \tag{7.9.15}$$

The velocity field is given by

$$\begin{aligned} v_r &= 0 \\ v_\theta &= aK\int_0^\infty J_1(ka)e^{-\nu_0 k^2 t}J_1(kr)\,dk \end{aligned} \tag{7.9.16}$$

A different expression for $\Omega(r, t)$ may be obtained if we substitute (7 9 14) in integral form into (7 9.10) and use the expression[9]

$$\int_0^\infty e^{-p^2 t^2}J_0(a\,)J_0(bt)t\,dt = \frac{1}{2p^2}\exp\left(-\frac{a^2+b^2}{4p^2}\right)I_0\left(\frac{ab}{2p^2}\right) \tag{7.9.17}$$

where I_0 is the modified Bessel function of zero order. Thus

$$\Omega(r, t) = \frac{K}{2\nu_0 t}\exp\left(-\frac{r^2}{4\nu_0 t}\right)\int_0^a \exp\left(-\frac{\rho^2}{4\nu_0 t}\right)I_0\left(\frac{r\rho}{2\nu_0 t}\right)\rho\,d\rho \tag{7.9.18}$$

The change in the value of the vortex in time at $r = 0$ is

$$\Omega(0, t) = K\left[1 - \exp\left(-\frac{a^2}{4\nu_0 t}\right)\right] \tag{7.9.19}$$

7.10. PLANE STAGNATION FLOW

To illustrate the effect of nonlinearity, we present here a flow problem which is amenable to exact treatment. This is the flow of a steady incompressible fluid approaching perpendicularly a flat wall $y = 0$. The flow is separated by the y-axis into two symmetrical regions. On the $x > 0$ side it moves along the wall to the right and on $x < 0$ to the left (Figure 7.10 1). The stream lines may, of course, be reversed when the flow direction is reversed.

The solution of this problem falls into a class of solutions of (7.6.22) of the form

$$\psi(x, y) = F(y)x + G(y) \tag{7.10.1}$$

[9] See Watson [1958, p 395]

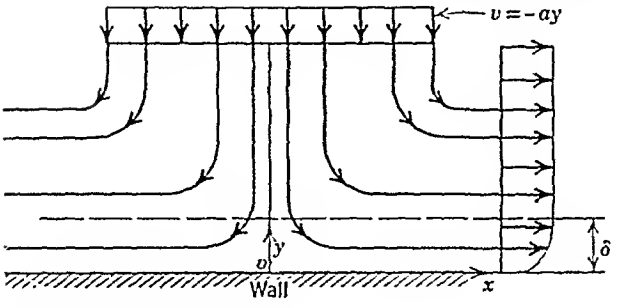


Fig. 7.10.1 Plane stagnation flow.

attributed to Riabouchinsky [1924], where $F(y)$ and $G(y)$ are to be determined from (7.6.22) and the boundary conditions. For our purpose it is adequate to take $G = 0$. Through (7.6.21) to (7.6.23) we get

$$u = xF'(y), \qquad v = -F(y), \qquad w = 0 \tag{7.10.2}$$

$$w_x = w_y = 0, \qquad w_z = -xF''(y)$$

and

$$\frac{p}{\rho_0} + \tfrac{1}{2}(F^2 + x^2F'^2) = K_1 = \text{const.} \tag{7.10.3}$$

$$\nu_v F^{IV} + FF''' - F'F'' = 0$$

This last equation is integrated once immediately to

$$\nu_v F''' - F'^2 + FF'' + K^2 = 0 \tag{7.10.4}$$

where K^2 is a constant. In order to solve this equation numerically it is convenient to remove the constants by setting

$$\eta = \alpha y, \qquad F(y) = \beta f(\eta) \tag{7.10.5}$$

Upon substitution of this into (7.10.4) and setting

$$\alpha = (-K/\nu_v)^{1/2}, \qquad \beta = (-K\nu_v)^{1/2} \tag{7.10.6}$$

we get

$$f''' + ff'' - f'^2 + 1 = 0 \tag{7.10.7}$$

The boundary conditions require vanishing velocity components at the wall $y = 0$, and $u = ax$ (where a is a constant) at a large distance ($y \to \infty$) from the wall. Hence

$$f(0) = f'(0) = 0 \qquad \text{at } y = 0 \tag{7.10.8}$$

$$f'(\infty) = 1 \qquad \text{at } y = \infty$$

where without loss in generality we set

$$K = -a \tag{7.10.9}$$

The velocity and vorticity components in terms of $f(\eta)$ now read

$$u = axf'(\eta), \qquad v = -\sqrt{av_v}\, f(\eta)$$
$$w_x = w_y = 0, \qquad w_z = -\sqrt{a/v_v}\, axf'' \tag{7.10.10}$$

The pressure p is determined through $(7.10.3)_1$. If we let p_0 denote the *stagnation pressure* at $x = y = 0$, then $K_1 = p_0/\rho_0$ and we have for the pressure

$$\frac{p - p_0}{\rho_0} + \tfrac{1}{2}a^2\left(\frac{v_v}{a}f^2 + x^2f'^2\right) = 0 \tag{7.10.11}$$

Thus once $f(\eta)$ is determined by solving (7.10.7) through (7.10.10) and (7.10.11) we obtain the velocity, the vorticity, and the pressure fields, respectively.

The solution of (7.10.7) was first given by Hiemenz [1911] and later improved by Howarth [1935]. The results for f, f', f'' are plotted in Figure 7.10.2. We observe that at approximately $\eta = 2.4$ we have $f' = 0.99$ so that

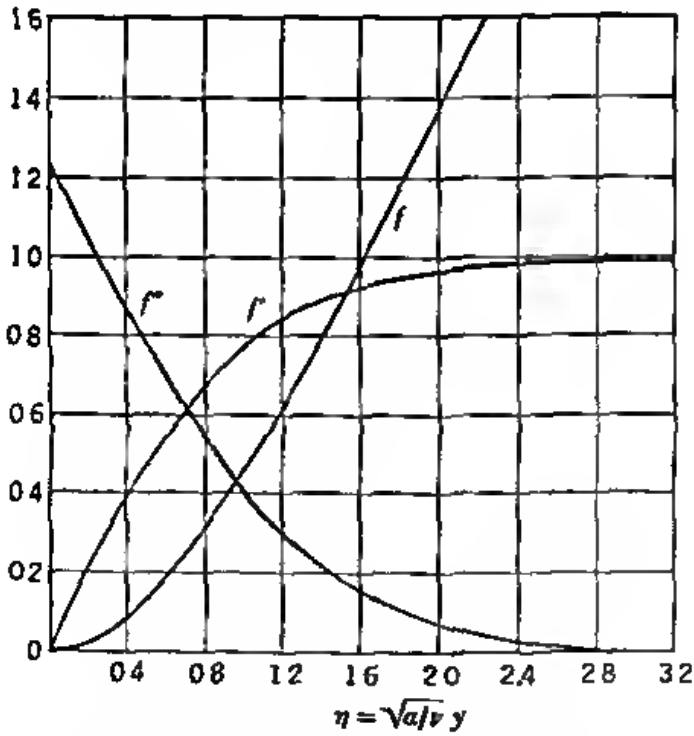


Fig. 7.10.2 Velocity distribution of plane stagnation flow.

the velocity $u = ax$ at $y = \infty$ is reached within 1 per cent at a distance

$$y = \delta = 2.4\sqrt{v_v/a} \tag{7.10.12}$$

This indicates that for low values of v_v, δ is small, so that the influence of

the viscosity is confined to a small layer near the wall. The pressure gradient $\partial p/\partial y$ is also proportional to $\rho_0 a\sqrt{\nu_v a}$ which is also small for small ν_v. These considerations are the basis of our intuitive feeling that the Navier-Stokes equations, may be replaced by a set of simpler equations which can properly take into account the effect of viscosity near an object, while at far distances the much simpler *ideal flow theory*, obtained by setting $\nu_v = 0$, may be used to solve the flow problems. Such a theory, known as the *boundary layer theory*, was initiated by Prandtl [1904], and has been developed into a powerful mathematical tool for the solutions of problems not only in hydrodynamics but also in many other branches of continuum physics. The developments of the last sixty years are large in the field and, of course, are beyond the scope of the present book. For a few representative accounts we refer the reader to Dryden, Murnaghan, and Bateman [1932]; Goldstein [1938], Howarth [1953], Schlichting [1955], Serrin [1959], and Berker [1963].

7.11. COMPRESSIBLE INVISCID FLOW, ACOUSTICS

When the viscosity of a fluid is so small as to allow setting $\lambda_v = \mu_v = 0$, the basic equations of Stokesian fluids reduce to those of compressible, inviscid fluids. These are, respectively, the equations of balance of momentum and conservation of mass, that is,

$$\rho\left(\frac{\partial v_k}{\partial t} + v_{k,l}v_l\right) = \rho f_k - \pi_{,k} \tag{7.11.1}$$

$$\frac{\partial \rho}{\partial t} + (\rho v_k)_{,k} = 0 \tag{7.11.2}$$

where the thermodynamic pressure π is given by an equation of state of the form

$$\pi = \pi(\rho, \theta) \quad \text{or} \quad \pi = \pi(\rho, \eta) \tag{7.11.3}$$

For a heat-conducting gas, to these equations we must adjoin the equation of heat conduction, cf. Section 5.8. Here we shall only discuss adiabatic cases with no heat sources so that $\mathbf{q} = 0$ and $\dot{\eta} = 0$ or

$$\frac{\partial \eta}{\partial t} + \eta_{,k}v_k = 0 \tag{7.11.4}$$

We may use as an equation of state

$$\begin{aligned} &\pi = \pi(\rho) && \text{Barotropic gas} \\ &\pi = p_0\rho^\gamma && \text{Polytropic gas} \\ &\pi = R\rho\theta && \text{Ideal gas } (\theta = \text{const.}) \end{aligned} \tag{7.11.5}$$

The boundary conditions require that we specify only the normal component of the velocity on the surface of the solid and/or the pressure

$$(7\ 11.6) \qquad \begin{aligned} v_k n_k &= v_{(n)} \quad \text{on } \mathscr{S}_v \\ \pi &= \pi_{(n)} \quad \text{on } \mathscr{S}_t \equiv \mathscr{S} - \mathscr{S}_v \end{aligned}$$

This is because the order of the partial differential equations is reduced by one. The boundary-layer thickness now goes to zero and the fluid does not stick on the boundary of the solid However, the normal component of the velocity must be the same as the normal velocity of the solid at any point of the surface of the solid. Sometimes the pressure may be given on the surface of the fluid.

The initial conditions are

$$(7.11.7) \qquad \begin{aligned} \mathbf{v}(\mathbf{x}, 0) &= \mathbf{v}^0(\mathbf{x}) \\ \rho(\mathbf{x}, 0) &= \rho_0(\mathbf{x}), \qquad \mathbf{x} \text{ in } \mathscr{V}, \qquad t = 0 \end{aligned}$$

The complete behavior of an inviscid, compressible gas, in adiabatic process, is therefore predicted by the solution of (7.11.1), (7.11.2), (7.11.4), and one of (7.11.5) subject to the boundary conditions (7.11.6) and the initial conditions (7.11.7)

Acoustic Approximation. When a small acoustical disturbance occurs in a gas at rest, we may use a scheme of perturbation to linearize the above equations of motion. In this case we write

$$(7.11.8) \qquad \begin{aligned} \rho &= \rho_0 + \varepsilon\rho_a + 0(\varepsilon^2), \qquad \mathbf{v} = \varepsilon\mathbf{u} + 0(\varepsilon^2), \\ \pi &= \pi_0 + \varepsilon\pi_a + 0(\varepsilon^2), \qquad \mathbf{f} = \varepsilon\bar{\mathbf{f}} + 0(\varepsilon^2) \end{aligned}$$

where $\varepsilon \ll 1$ is a small perturbation parameter and ρ_0 and π_0 are, respectively, the ambient constant density and the pressure. Substituting (7.11.8) into (7.11 1) and (7.11.2) and setting the coefficient of ε equal to zero, we get

$$(7.11.9) \qquad \rho_0 \frac{\partial u_k}{\partial t} = -c_0^2 \frac{\partial \rho_a}{\partial x_k} + \rho_0 \bar{f}_k$$

$$(7.11.10) \qquad \frac{\partial \rho_a}{\partial t} + \rho_0 \frac{\partial u_k}{\partial x_k} = 0$$

where

$$(7.11.11) \qquad c_0^2 \equiv \left.\frac{\partial \pi}{\partial \rho}\right|_{\rho=\rho_0}$$

is the square of the *velocity of sound.* Eliminating u_k from these equations, we get

$$\frac{\partial^2 \rho_a}{\partial t^2} - c_0^2 \nabla^2 \rho_a = -\rho_0 \bar{f}_{k,k} \tag{7.11.12}$$

This is the well-known equation of sound waves.

An alternative but equivalent equation of sound waves involves the use of the velocity and body-force potentials

$$u_k = -\phi_{,k}, \qquad \bar{f}_k = -g_{,k} \tag{7.11.13}$$

Substituting these into (7.11.9), and integrating, we get

$$c_0^2 \rho_a + \rho_0(g - \dot{\phi}) = K(t)$$

By redefining ϕ we can eliminate $K(t)$ from this equation so that

$$c_0^2 \rho_a + \rho_0(g - \dot{\phi}) = 0 \tag{7.11.14}$$

Using this equation and (7.11.13) in (7.11.10), we obtain

$$\frac{\partial^2 \phi}{\partial t^2} - c_0^2 \nabla^2 \phi = \dot{g} \tag{7.11.15}$$

This equation is more suitable for boundary and initial-value problems involving the velocity, while (7.11.12) is better suited for those involving the density or pressure changes.

The boundary and initial conditions appropriate to (7.11.15) are

$$A\phi(\mathbf{x}', t) + B\phi_{,k} n_k = F(\mathbf{x}', t), \qquad \mathbf{x}' \text{ on } \mathscr{S}, \qquad t > 0 \tag{7.11.16}$$

$$\begin{aligned} &\phi(\mathbf{x}, 0) = \phi_0(\mathbf{x}), \\ &\dot{\phi}(\mathbf{x}, 0) = \phi_1(\mathbf{x}), \end{aligned} \qquad \mathbf{x} \text{ in } \mathscr{V}, \qquad t = 0 \tag{7.11.17}$$

where $A = A(\mathbf{x}')$ and $B = B(\mathbf{x}')$ may be any functions of $\mathbf{x}'$. By including discontinuous functions, such as the Heaviside function, in the selection of A and B in (7.11.16), we may treat mixed boundary conditions of various varieties contained in (7.11.6). Once ϕ is determined, through (7.11.13) and (7.11.14) we obtain the velocity and density fields. Thus, it is clear that the solution of compressible flow problems in the limit of small amplitudes will give the solution of the corresponding problems in acoustics.

Finally, we remark briefly on the velocities of sound waves in various gases

$$\begin{aligned} c_0 &= (R\theta)^{1/2} = (\pi_0/\rho_0)^{1/2} \qquad \text{ideal gas} \\ c_0 &= (p_0 \gamma \rho_0^{\gamma - 1})^{1/2} = (\pi_0 \gamma / \rho_0)^{1/2} \qquad \text{polytropic gas} \end{aligned} \tag{7.11.18}$$

where $\gamma \equiv c_p/c_v$ and R are gas constants.

The velocity of sound in an ideal gas was first obtained by Newton, It gives a value for air 20 per cent less than the experimentally observed one. The value for polytropic gases, first suggested by Laplace, agrees well with experimental results at low frequencies, so that it is used for a precise determination of γ.

For liquids, the situation is much more complicated and requires a more elaborate relation $\pi = \pi(\rho, \theta)$. For van der Waals gas, we have

$$\pi = \frac{R\rho\theta}{1 - b\rho} - a\rho^2 \tag{7.11.19}$$

Using this and considering a and b as functions of ρ and θ, Schaaffs [1940] gave a semi-empirical expression for the van der Waals gas

$$c_0^2 \simeq \frac{\gamma R\theta}{1 - b}\left[\frac{1}{3(1 - b\rho)} - 2\right] \tag{7.11 20}$$

where

$$b \simeq \frac{b_1}{1 + (b_1 - b_0)\rho} \tag{7.11 21}$$

and b_0 and b_1 are constants (the values of b at $\theta = \pi = 0$ and $\rho = \infty$, respectively). For certain organic liquids, such as methyl alcohol and benzene, this gives a fairly good agreement with experiment at 20°C. For further results, see Schaaffs [1940] and Bergmann [1954]

7.12. RADIATION FROM A VIBRATING SPHERE

As an illustrative example, here we shall treat the problem of radiation of sound from a vibrating sphere. This problem constitutes a mathematical model of a sonar device. The mathematical problem requires the solution of

$$\frac{\partial^2\phi}{\partial t^2} - c_0{}^2\nabla^2\phi = 0 \tag{7.12.1}$$

subject to

$$\phi_{,k}n_k = -U \quad \text{when} \quad r = a, \qquad t > 0 \tag{7.12.2}$$

$$\phi(\mathbf{x}, 0) = \dot{\phi}(\mathbf{x}, 0) = 0 \quad \text{when} \quad r > a, \qquad t = 0 \tag{7.12.3}$$

where r is the radial distance and $r = a$ is the surface of the sphere. Note that U is none other than the velocity of a point on the sphere normal to the surface of the sphere.

The appropriate solution of (7.12.1) for diverging sound waves in the direction $r > 0$ is given by

$$\phi(r, t) = \frac{1}{r} F(\tau), \qquad \tau \equiv t - \frac{r-a}{c_0} > 0 \tag{7.12.4}$$

$$F(0) = 0$$

See, for instance, Section 6.12. To determine the specific form of $F(\tau)$ we substitute this into (7.12.2). Hence

$$-\frac{1}{a^2} F(\tau) - \frac{1}{c_0 a} F'(\tau) = -U(\tau), \qquad r = a, \qquad \tau > 0$$

The solution of this equation for $F = 0$ at $t = 0$ must be found, since, for $t \leq \frac{r-a}{c_0}$, ϕ and $\dot{\phi}$ must vanish for all r. Thus

$$F(\tau) = c_0 a \int_0^\tau U(s) \exp\left[\frac{c_0}{a}(s-\tau)\right] ds, \qquad \tau \geq 0 \tag{7.12.5}$$

The solution of our problem is therefore

$$\phi(r, t) = \frac{c_0 a}{r} \int_0^\tau U(s) \exp\left[\frac{c_0}{a}(s-\tau)\right] ds, \qquad \tau \geq 0$$
$$\phi(r, t) = 0, \qquad \tau < 0 \tag{7.12.6}$$

where τ is given by $(7.12.4)_2$. This completes the solution of the problem. Note that this solution is also the solution of the analogous problem of radiation of elastic waves from a spherical cavity with a given surface displacement $u_r = U$ at the surface of the cavity $r = a$.

We may investigate, as an example, a sinusoidal, radial oscillation of the sphere. In this case

$$U = U_0 \sin \lambda t \quad \text{at} \quad r = a, \qquad t > 0 \tag{7.12.7}$$

where U_0 and λ are constants. Substitution of this into (7.12.6) gives

$$\phi(r, t) = \frac{U_0 c_0 a}{r\sqrt{(c_0/a)^2 + \lambda^2}} \left\{ \sin\left[\lambda\left(t - \frac{r-a}{c_0}\right) - \delta\right] + \frac{\lambda}{\sqrt{(c_0/a)^2 + \lambda^2}} \exp\left[-\frac{c_0}{a}\left(t - \frac{r-a}{c_0}\right)\right] \right\}$$
$$\text{for } t \geq \frac{r-a}{c_0} \tag{7.12.8}$$

$$\phi(r, t) = 0 \qquad \text{for } t < \frac{r-a}{c_0}$$

where

$$\delta = \arcsin\left[\lambda/\sqrt{(c_0/a)^2 + \lambda^2}\right] \tag{7.12.9}$$

After the passage of a long time at a point $r < c_0 t + a$ we see that the second term in the brace of $(7.12.8)_1$ quickly dies out and the velocity potential is governed by the first sinusoidal term in the brace. The waves will have a phase angle δ.

The displacement $u_r(r, t)$ and the density change ρ_a are easily obtained by using (7.12.6) or (7.12.8) in (7.11.13) and (7.11.14), respectively.

7.13. DIFFRACTION OF A PLANE SOUND WAVE FROM A CIRCULAR CYLINDER

As another example of a solution of the acoustical wave equation, we consider the diffraction (scattering) of a plane, harmonic sound wave, propagating normally to the axis of a circular cylinder, from the cylinder. Since the velocity field is independent of the axial direction, we have a two-dimensional problem of diffraction of a plane wave from a circular disk (Figure 7.13.1).

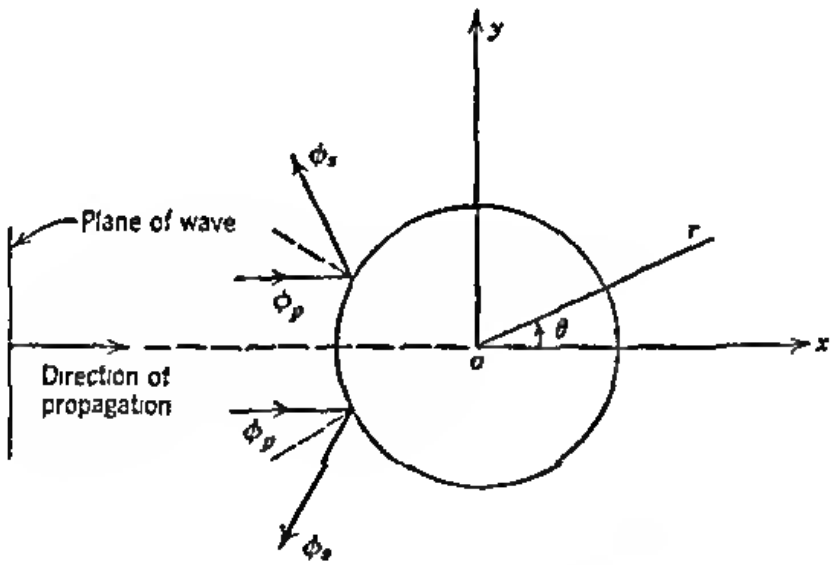


Fig. 7.13.1 Diffraction of a plane wave from a circular cylinder.

A harmonic plane wave propagating along the x-axis may be represented by a velocity potential

$$\phi_p = A_p e^{ikx - i\omega t} \tag{7.13.1}$$

When this wave hits the cylinder it gives rise to a scattered wave for which the velocity potential ϕ_s must satisfy the wave equation

$$\frac{\partial^2 \phi_s}{\partial t^2} - c_0{}^2 \nabla^2 \phi_s = 0 \tag{7.13.2}$$

subject to the boundary conditions that the normal velocity at the surface of the cylinder $r = a$ is zero, that is,

$$\frac{\partial}{\partial r}(\phi_p + \phi_s) = 0 \tag{7.13.3}$$

and, of course, that the scattered waves must radiate in the direction of $r > 0$. Since we are dealing with harmonic waves, and since the solution must be periodic in θ with a period of 2π, we try a solution of the form

$$\phi_s = R(r)e^{im\theta - i\omega t} \tag{7.13.4}$$

where m is an integer. Substitution of this into (7.13.2), and using the expression for the laplacian in plane polar coordinates (r, θ), we get

$$R'' + \frac{1}{r}R' + \left(k^2 - \frac{m^2}{r^2}\right)R = 0 \tag{7.13.5}$$

where

$$k \equiv \omega/c_0 \tag{7.13.6}$$

The differential equation (7.13.5) is Bessel's equation, whose general solution is a linear combination of Bessel functions of the first and second kinds, $J_m(kr)$ and $Y_m(kr)$, of order m, that is,

$$R(r) = AJ_m(kr) + BY_m(kr) \tag{7.13.7}$$

where A and B are arbitrary constants. These functions have the power series representation

$$\begin{aligned} J_m(x) &= \sum_{j=0}^{\infty} \frac{(-)^j x^{m+2j}}{2^{m+2j} j!(m+j)!} \\ Y_m(x) &= J_m(x)\log x - \frac{x^{-m}}{2^{1-m}} \sum_{j=0}^{\infty} \frac{(m-j-1)!\,x^{2j}}{2^{2j}\cdot j!} \\ &\quad - \frac{x^m}{2^{1+m}} \sum_{j=0}^{\infty} \frac{(-)^j}{(m+j)!\,j!}\left[\Omega_j + \Omega_{j+1}\left(\frac{x}{2}\right)^{2j}\right] \end{aligned} \tag{7.13.8}$$

where

$$\Omega_j \equiv 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{j} \tag{7.13.9}$$

For large values of x, these functions have asymptotic values

$$J_m(x) \simeq \left(\frac{2}{\pi x}\right)^{1/2} \cos\left(x - \frac{2m+1}{4}\pi\right)$$
$$Y_m(x) \simeq \left(\frac{2}{\pi x}\right)^{1/2} \sin\left(x - \frac{2m+1}{4}\pi\right) \tag{7.13.10}$$

If the waves are to be outgoing in the direction $r > 0$, then we cannot employ $J_m(kr)$ or $Y_m(kr)$ to represent the solution of our problem, because, according to the Sommerfeld radiation condition in two dimensions, we must have

$$\lim_{r\to\infty} \sqrt{r}\left(\frac{\partial\phi}{\partial r} - ik\phi\right) = 0 \tag{7.13.11}$$

Such a solution is obtained by a combination of $J_m(x)$ and $Y_m(x)$, the Haokel functions defined by

$$H_m^{(1)}(x) = J_m(x) + iY_m(x)$$
$$H_m^{(2)}(x) = J_m(x) - iY_m(x) \tag{7.13.12}$$

The functions $H_m^{(1)}$ satisfy (7.13.11) as can be seen by using (7.13.10) in (7.13.12) and the result in (7.13.11). Thus, by superposition, we write for the general solution

$$\phi_s(r, \theta, t) = e^{-i\omega t} \sum_{m=0}^{\infty} B_m H_m^{(1)}(kr) \cos m\theta \tag{7.13.13}$$

since we also have $\phi_s(r, \theta, t) = \phi_s(r, -\theta, t)$. This solution dies out as $1/\sqrt{r}$ for $r \to \infty$ and satisfies the radiation condition. The constants B_m must be determined from the boundary condition (7.13.3). In order to achieve this, ϕ_p must be expressed as a trigonometric series. To this end, we write

$$e^{ikr\cos\theta} = \sum_{m=0}^{\infty} C_m(r) \cos m\theta$$

Now $C_m(r)$ are evaluated by multiplying both sides by $\cos n\theta$ and integrating over θ from 0 to 2π. Hence

$$C_m(r) = \frac{1}{\pi}\int_0^{2\pi} e^{ikr\cos\theta} \cos m\theta \, d\theta$$

$$C_m(r) = 2i^m J_m(kr) \qquad \text{for } m \geq 1$$

$$C_m(r) = J_0(kr) \qquad \text{for } m = 0$$

Thus a plane wave is represented in terms of cylindrical waves by the series

$$e^{ikr\cos\theta} = J_0(kr) + 2\sum_{m=1}^{\infty} i^m J_m(kr)\cos m\theta \tag{7.13.14}$$

Substituting this into (7.13.1) and using the result together with (7.13.13) in (7.13.3), we get

$$kA_p\left[J'_0(ka) + 2\sum_{m=1}^{\infty} i^m J'_m(ka)\cos m\theta\right] + k\left[B_0 H'^{(1)}_0(ka) + \sum_{m=1}^{\infty} B_m H^{(1)}_m(ka)\cos m\theta\right] = 0$$

where a prime indicates differentiation with respect to the argument of the function. Since this equation must be valid for all θ, we obtain

$$\begin{aligned} B_0 &= -A_p \frac{J'_0(ka)}{H'^{(1)}_0(ka)} \\ B_m &= -2A_p \frac{i^m J'_m(ka)}{H'^{(1)}_m(ka)}, \qquad m > 0 \end{aligned} \tag{7.13.15}$$

This, in principle, completes the solution of our problem, since in (7.13.13) we have B_m completely determined in terms of the amplitude A_p of the incoming plane wave.

Using (7.13.12) in (7.13.15), we may equivalently write the latter as

$$\begin{aligned} B_0 &= -iA_p e^{-i\gamma_0}\sin\gamma_0 \\ B_m &= -2i^{m+1}A_p e^{-i\gamma_m}\sin\gamma_m, \qquad m > 0 \end{aligned}$$

where

$$\tan\gamma_m \equiv -J'_m(ka)/Y'_m(ka) \tag{7.13.16}$$

The solution for ϕ_s now takes the form

$$\phi_s(r,\theta,t) = A_p e^{-i\omega t}\left[-ie^{-i\gamma_0}\sin\gamma_0 H^{(1)}_0(kr) - 2\sum_{m=1}^{\infty} i^{m+1}e^{-i\gamma_m}\sin\gamma_m H^{(1)}_m(kr)\cos m\theta\right] \tag{7.13.17}$$

For large values of kr, that is, $kr \gg 1$, by using the asymptotic formula (7.13.10), we see that

$$\phi_s(r,\theta,t) \simeq A_p\left(\frac{2}{\pi kr}\right)^{1/2}\psi_s(\theta)e^{ik(r-c_0t)-3i\pi/4} \tag{7.13.18}$$

where

$$\psi_s(\theta) \equiv \sin \gamma_0 e^{-i\gamma_0} + 2\sum_{m=1}^{\infty} \sin \gamma_m e^{-i\gamma_m} \cos m\theta \tag{7.13.19}$$

The magnitude of ψ_s, $|\psi_s(\theta)|$, represents the angular distribution of the intensity of the sound wave on some scale For a few values of ka, a polar plot of $|\psi_s(\theta)|^2$ is shown in Figure 7.13 2.

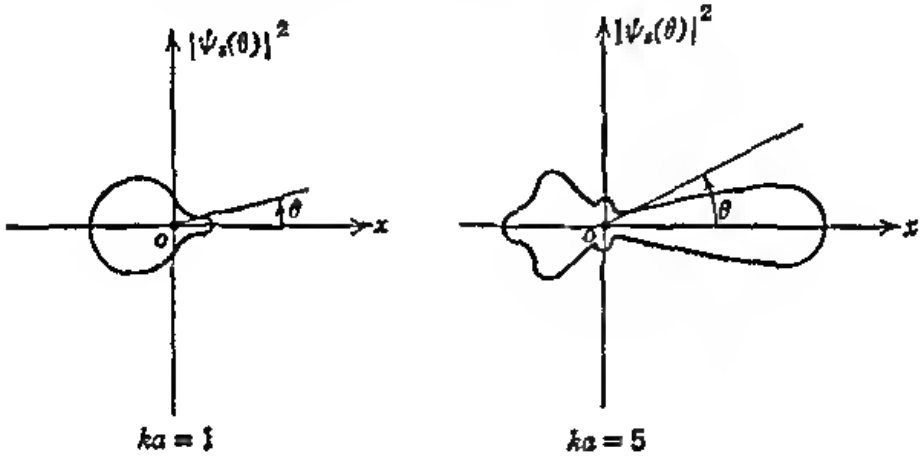


Fig. 7.13.2 Angular distribution in scattering from a circular cylinder

The integral

$$P_s = \int_0^{2\pi} \frac{1}{\rho_0 c_0} |\phi_s|^2 r \, d\theta = \frac{A_p^2}{\pi \rho_0 \omega} \int_0^{2\pi} |\psi_s|^2 \, d\theta$$

is a measure of the total average rate at which acoustic energy is scattered For small and large ka, we find that

$$P_s \simeq \frac{6\pi^5 a^4}{\lambda^3} P_0, \qquad ka \ll 1 \tag{7.13.20}$$

$$P_s \simeq 4aP_0, \qquad ka \gg 1$$

where $\lambda \equiv 2\pi/k$ is the wavelength and

$$P_0 \equiv A_p^2/2c_0\rho_0 \tag{7.13.21}$$

is the intensity of the plane wave. A plot of $P_s/4aP_0$ as a function of $ka = 2\pi a/\lambda$ is shown in Figure 7.13.3. From Figure 7 13.3 we observe that the total scattered power increases with decreasing wavelength. Figure 7.13.2 indicates that for large wavelengths ($ka = 1$) the scattering is mainly in the direction of the incident wave. As the wavelength decreases (frequency increases) the forward scattering begins to increase, and complicated peaks appear. For a very large wavelength, the scattered intensity goes to zero and the wave pays no attention to the obstacle. On the other hand

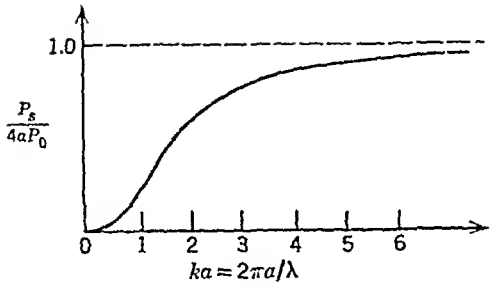


Fig. 7.13.3 Total scattered power.

for a very small wavelength, we may use the ray reflection principle (geometrical acoustics) to follow the wave propagation. Indeed, for such cases, there exist asymptotic theories by which sufficiently accurate results may be obtained.[10]

7.14. PROPAGATION OF SHOCK WAVES

The jump conditions across a surface $\sigma(t)$ moving with a velocity $\mathbf{v}$ were collected in Section 7.4. They are given below in the order of the local conservation of mass, the local balance of momentum, the local conservation of energy, and the Clausius-Duhem inequality.

(7.14.1) $[\rho(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} = 0$ (conservation of mass)

(7.14.2) $[-t_{kl}]n_k + [\rho v_l(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} = 0$ (balance of momentum)

(7.14.3) $[(\varepsilon + \frac{1}{2}v_l v_l)\rho(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n}$
$- [t_{kl}v_l + q_k]n_k = 0$ (conservation of energy)

(7.14.4) $\left[\rho\eta(\mathbf{v} - \mathbf{v}) - \frac{\mathbf{q}}{\theta}\right] \cdot \mathbf{n} \geq 0$ (Clausius-Duhem inequality)

Here $\mathbf{n}$ is the unit normal to $\sigma(t)$ making an acute angle with its velocity $\mathbf{v}$ at P (Figure 7.14.1). A bold-face bracket indicates the difference in the enclosed quantity in the positive and negative directions of $\mathbf{n}$ at any point on $\sigma(t)$, that is,

$$[A] = A_1 - A_0$$

where A_0 is the value of A in $\mathcal{V}_0$ near $\sigma(t)$ and A_1 is that in $\mathcal{V}_1$. The normal $\mathbf{n}$ is continuous across $\sigma(t)$ so that it may be carried inside the bracket if desired. We are dealing with compressible, inviscid fluids so that

(7.14.5) $$t_{kl} = -\pi\delta_{kl}$$

[10] Cf. Kline [1951], [1954], Keller [1953], [1954], [1962], Keller, *et al.* [1956]. For the case of an elastic solid, see Karal and Keller [1959].

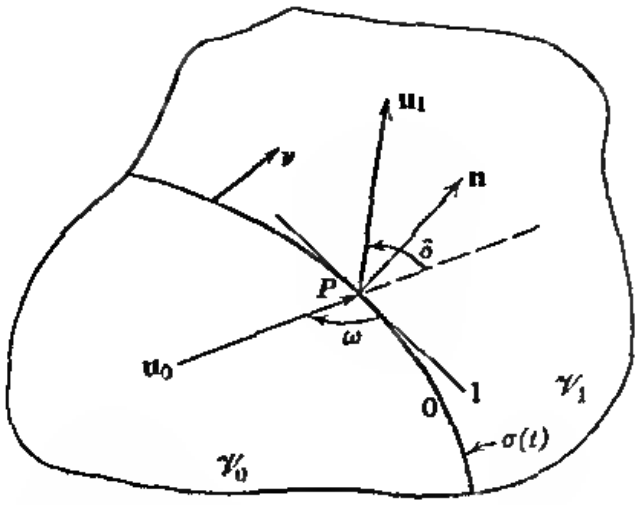


Fig. 7.14.1 Moving discontinuity surface

If we further write for the *mass flux* m across $\sigma(t)$

$$m \equiv \rho(\mathbf{v} - \mathbf{v}) \cdot \mathbf{n} \tag{7.14 6}$$

and use the relative velocity $\mathbf{u}$ with respect to the surface

$$\mathbf{u} \equiv \mathbf{v} - \mathbf{v} \tag{7.14.7}$$

we find that the above jump conditions simplify to

$$[\rho \mathbf{u} \cdot \mathbf{n}] = 0 \quad \text{or} \quad [m] = 0 \tag{7 14 8}$$

$$[\pi \mathbf{n} + m\mathbf{u}] = \mathbf{0} \tag{7.14 9}$$

$$\left[m\left(\varepsilon + \frac{u^2}{2} + \frac{\pi}{\rho}\right) - \mathbf{q} \cdot \mathbf{n} \right] = 0 \tag{7.14.10}$$

$$\left[\rho\eta\mathbf{u} - \frac{\mathbf{q}}{\theta} \right] \cdot \mathbf{n} \geq 0 \tag{7.14 11}$$

These relations are valid across any discontinuity surface If the surface of discontinuity is a *contact surface*, then the mass flux $m = 0$. If, however, it is not a contact surface, so that $m \neq 0$, then the surface of discontinuity is called a *shock front*.

To simplify the situation we only study the case in which there is no flux of heat energy, that is $\mathbf{q} = 0$. Further, we give the solution for a perfect gas when the specific heats are independent of temperature. The equation of state is thus given by

$$\pi = (c_p - c_v)\rho\theta, \qquad \varepsilon = c_v\theta \tag{7.14.12}$$

For an ideal gas that is polytropic, the pressure has the form

$$\pi = S(\eta)\rho^\gamma, \qquad \gamma \equiv c_p/c_v \tag{7.14.13}$$

where S is a function of the entropy η, and γ is a constant between 1 and 5/3 for most gases. For air at moderate temperatures, $\gamma = 1.4$. When c_p and c_v are constants, it is not difficult to show that

$$S(\eta) = (\gamma - 1) \exp [(\eta - \eta_0)/c_v] \tag{7.14.14}$$

where η_0 is a constant. Using (7.14.12) and $(7.14.13)_2$, we see that

$$i \equiv \varepsilon + \frac{\pi}{\rho} = \frac{\gamma}{\gamma - 1}\frac{\pi}{\rho} = \frac{c^2}{\gamma - 1} \tag{7.14.15}$$

where i is called the *specific enthalpy* and c is the velocity of sound for polytropic gases (cf. (7.11.18)). Denoting by subscripts 0 and 1 the values of the quantities in front and behind the shock front, and by ω the angle that $\mathbf{u}_0$ makes with the tangent of $\sigma(t)$ at a point P, and by δ the deflection angle of $\mathbf{u}_1$ from $\mathbf{u}_0$ at P, all taken in the plane of $\mathbf{u}_0$ and n, the explicit forms of (7.14.8) to (7.14.11) read

$$\begin{aligned}
&\rho_0 u_0 \sin \omega = \rho_1 u_1 \sin (\omega - \delta) = m \\
&u_0 \cos \omega = u_1 \cos (\omega - \delta) \\
&\pi_0 - \pi_1 = m[u_1 \sin (\omega - \delta) - u_0 \sin \omega] \\
&i_0 + \tfrac{1}{2}u_0^2 \sin^2 \omega = i_1 + \tfrac{1}{2}u_1^2 \sin^2 (\omega - \delta) \\
&\eta_0 \leq \eta_1
\end{aligned} \tag{7.14.16}$$

For i_0 and i_1 we made use of (7.14.15). Note that the second and third of (7.14.16) are the results of taking the tangential and normal projections of (7.14.9) on the tangent, and on the normal of $\sigma(t)$ at P.

By use of (7.14.12) to (7.14.15) we can determine the ratio of various quantities in front of and behind the shock, the entropy change, and the deflection angle δ. Thus

$$\begin{aligned}
&\frac{\rho_1}{\rho_0} = \frac{(\gamma + 1)\sigma}{2 + (\gamma - 1)\sigma} \\
&\frac{\pi_1}{\pi_0} = \frac{2\gamma\sigma - \gamma + 1}{\gamma + 1} \\
&\frac{\theta_1}{\theta_0} = \frac{(2\gamma\sigma - \gamma + 1)[2 + (\gamma - 1)\sigma]}{(\gamma + 1)^2\sigma} \\
&\left(\frac{u_1}{u_0}\right)^2 = \cos^2 \omega + \left[\frac{2 + (\gamma - 1)\sigma}{(\gamma + 1)\sigma}\right]^2 \sin^2 \omega \\
&\eta_1 - \eta_0 = c_v \log (\pi_1 \rho_0^\gamma / \pi_0 \rho_1^\gamma) \\
&\cot \delta = \left[\frac{(\gamma + 1)\sigma \operatorname{cosec}^2 \omega}{2(\sigma - 1)} - 1\right] \tan \omega
\end{aligned} \tag{7.14.17}$$

where

$$\sigma \equiv \left(\frac{u_0}{c_0}\right)^2 \sin^2 \omega = M_0^2 \sin^2 \omega \tag{7.14.18}$$

The quantity $M \equiv v/c$ is called the *Mach number*. The Clausius-Duhem inequality $(7.14\ 16)_5$ can be shown to imply that the only admissible values of σ are those for which

$$\sigma > 1 \tag{7.14.19}$$

Otherwise the entropy decreases across the shock This, of course, is not admissible on physical grounds To show (7.14.19) from $(7.14\ 17)_5$ we calculate

$$\frac{d}{d\sigma}(\eta_1 - \eta_0) = c_p \frac{2(\gamma - 1)(\sigma - 1)^2}{\sigma(2\gamma\sigma - \gamma + 1)[2 + (\gamma - 1)\sigma]}$$

This cannot be negative, since from $(7\ 14.17)_2$ $\pi_1/\pi_0 \geq 0$ implies $\sigma \geq (\gamma - 1)/2\gamma$. Hence $\eta_1 - \eta_0$, which vanishes when $\sigma = 1$, increases with σ and therefore is positive only if $\sigma > 1$. This proves $\pi_1 \geq \pi_0$, hence the discontinuities are compression waves. The values of π_1/π_0, θ_1/θ_0, and $\eta_1 - \eta_0$ have no upper bound. However, ρ_1/ρ_0 is restricted to the range

$$1 < \frac{\rho_1}{\rho_0} < \frac{\gamma + 1}{\gamma - 1}$$

For $\gamma = 1.4$ we have $(\rho_1/\rho_0)_{max} = 6$.

The condition $\sigma \geq 1$ implies that the flow ahead of the shock wave is supersonic. The flow behind the shock may by supersonic or subsonic depending on M_0 and σ as may be seen from

$$M_1^2 \equiv (u_1/c_1)^2 = [2 + (\gamma - 1)M_0^2] \times \left\{\frac{1}{2\gamma\sigma - \gamma + 1} + \frac{2}{(\gamma - 1)[2 + (\gamma - 1)\sigma]}\right\} - \frac{2}{\gamma - 1} \tag{7.14.20}$$

For the smallest value of $\sigma = 1$, we have $M_1 = M_0 \geq 1$, and the value of $\omega = \mu_0 =$ the Mach angle. For a given M_0 the greatest value of $\sigma = M_0^2$, is the condition of a *normal shock* wave In this case

$$M_1^2 = \frac{2 + (\gamma - 1)M_0^2}{2\gamma M_0^2 - \gamma + 1} \leq 1, \qquad \omega = \frac{\pi}{2}$$

If for a given M_0, ω increases, then σ increases and M_1 decreases Thus with the increase of ω from μ_0 to $\pi/2$, the type of flow behind the shock wave changes monotonically from supersonic with the Mach number M_0 to subsonic with Mach number

$$M_1 = \left[\frac{2 + (\gamma - 1)M_0^2}{2\gamma M_0^2 - \gamma + 1}\right]^{1/2}$$

This also shows that the flow behind a normal shock wave is subsonic and the Mach number of the flow behind any shock cannot be less than $[(\gamma - 1)/2\gamma]^{1/2}$. For $\gamma = 1.4$, this minimum is $1/\sqrt{7}$.

The ratios ρ_1/ρ_0, π_1/π_0, and θ_1/θ_0 for $\gamma = 1.4$ are given as functions of σ in Table 7.14.1 and the values of the deflection angle δ as a function of ω with M_0 the parameter are plotted in Figure 7.14.2, (Howarth [1953]).

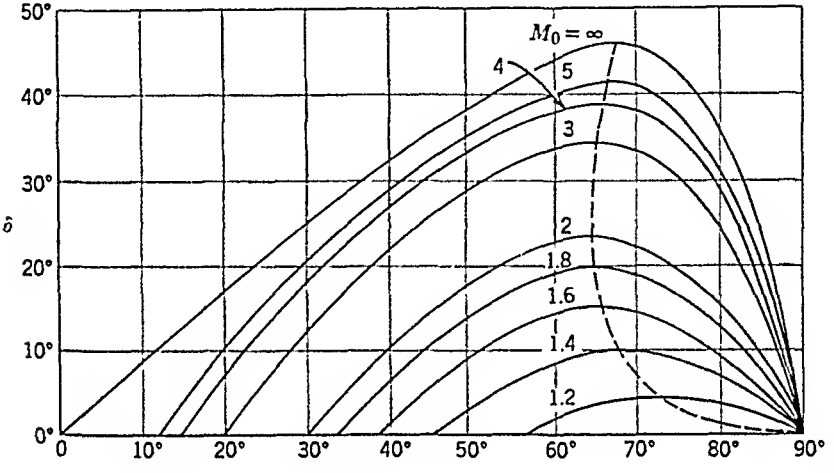


Fig. 7.14.2 Deflection angle δ for $\gamma = 1.4$.

Table 7.14.1. Conditions behind the shock wave for $\gamma = 1.4$

$M_0 \sin\omega$	π_1/π_0	ρ_1/ρ_0	θ_1/θ_0	$M_0 \sin\omega$	π_1/π_0	ρ_1/ρ_0	θ_1/θ_0
1.0	1	1	1	3.0	10.33	3.857	2.679
1.2	1.51	1.342	1.128	3.2	11.78	4.031	2.922
1.4	2.12	1.690	1.255	3.4	13.32	4.188	3.180
1.6	2.82	2.032	1.388	3.6	14.95	4.330	3.454
1.8	3.61	2.359	1.532	3.8	16.68	4.457	3.743
2.0	4.5	2.667	1.688	4.0	18.5	4.571	4.047
2.2	5.48	2.951	1.857	4.2	20.41	4.675	4.366
2.4	6.55	3.212	2.040	4.4	22.42	4.768	4.702
2.6	7.72	3.449	2.238	4.6	24.52	4.853	5.052
2.8	8.98	3.664	2.451	4.8	26.71	4.930	5.418
3.0	10.33	3.857	2.679	5.0	29	5	5.8
				∞	∞	6	∞

There exist geometrical methods, for example, the shock polar diagram, by which, for a given u_0 and ω, one can determine u_1 and the angle of deflection δ. For this, and for other details of shock wave theory, we refer

readers to the standard literature on compressible flow theory; see, for instance, Courant and *Friedrichs [1948]*, Howarth *[1953, I]*, Shapiro *[1953, I]*, and Stanyukovich **[1960]**.

Structure of Shock Waves. It is instructive to approach the one-dimensional, steady, normal shock wave from the point of view of the solution of the equations of compressible viscous flow theory. For a one-dimensional motion in the x-direction the equations of continuity, balance of momentum, and conservation of energy (for $\lambda_v = -\frac{2}{3}\mu_v$) are, respectively, given by

$$\frac{d}{dx}(\rho u) = 0$$

$$\rho u \frac{du}{dx} = -\frac{d}{dx}\left(\pi - \tfrac{4}{3}\mu_v \frac{du}{dx}\right) \tag{7.14.21}$$

$$\rho u \frac{di}{dx} - u\frac{d\pi}{dx} = \frac{d}{dx}\left(\kappa \frac{d\theta}{dx}\right) + \tfrac{4}{3}\mu_v\left(\frac{du}{dx}\right)^2$$

By combining the last two equations, we may replace the energy equation by

$$\rho u \frac{d}{dx}(i + \tfrac{1}{2}u^2) = \frac{d}{dx}\left[\mu_v\left(\frac{1}{\sigma_p}\frac{di}{dx} + \tfrac{4}{3}u\frac{du}{dx}\right)\right]$$

where $\sigma_p \equiv \mu_v c_p/\kappa$ is the *Prandtl number*, not necessarily a constant. We may now integrate the first two of (7.14.21) and this last equation to obtain

$$\rho u = \rho_0 u_0 = \rho_1 u_1 = m$$

$$mu + \pi - \tfrac{4}{3}\mu_v \frac{du}{dx} = mu_0 + \pi_0 = mu_1 + \pi_1 \equiv mV \tag{7.14.22}$$

$$m(i + \tfrac{1}{2}u^2) - \mu_v\left(\frac{1}{\sigma_p}\frac{di}{dx} + \tfrac{4}{3}u\frac{du}{dx}\right) = m(i_0 + \tfrac{1}{2}u_0^2) = m(i_1 + \tfrac{1}{2}u_1^2) \equiv mI$$

From the last two equations, with the use of the equation of state

$$\pi = (c_p - c_v)\rho\theta = (m/u)(i - \varepsilon)$$

we get

$$\tfrac{4}{3}\mu_v u \frac{du}{dx} = m[i - \varepsilon + u(u - V)]$$

$$\frac{\mu_v}{\sigma_p}\frac{di}{dx} = m(\varepsilon + Vu - \tfrac{1}{2}u^2 - I) \tag{7.14.23}$$

These equations have been integrated for the ideal gas in which c_p and c_v are constants so that $i = c_p\theta$ and $\varepsilon = c_v\theta$. Taylor **[1910]** gave a solution of (7.14.23) for a weak shock in which μ_v, κ, and σ_p are constants. Here we give Becker's [1921] solution modified by Thomas **[1944]** for a temperature-dependent μ_v. The value $\sigma_p = 3/4$ very closely approximates diatomic gases. For this case, addition of the two equations of (7.14.23) gives

$$\tfrac{4}{3}\mu_v \frac{d}{dx}(i + \tfrac{1}{2}u^2) = m(i + \tfrac{1}{2}u^2 - I)$$

This shows that

$$i + \tfrac{1}{2}u^2 = \mathrm{I} \tag{7.14.24}$$

since the solution involving $\exp[(3m/4)\int dx/\mu_v]$ is not permissible. Thus, the energy is constant throughout the wave. By use of (7.14.24), $(7.14.23)_1$ may be written as

$$\frac{D}{M_0}\frac{\mu_v}{\mu_0}u\frac{du}{dx} = -(u_0 - u)(u - u_1) \tag{7.14.25}$$

where

$$D \equiv \frac{8\gamma}{3(\gamma+1)}\frac{u_1}{c_1} \tag{7.14.26}$$

According to kinetic theory, μ_v is proportional to $\sqrt{\theta}$, so that $\mu_v/\mu_0 = \sqrt{i/i_1}$. Using this, we integrate (7.14.25) to get

$$x = \frac{D}{M_0}\left(1 + \frac{\gamma-1}{2}M_0\right)^{1/2}[(U_0 + U_1)\arccos U + (1 - U^2)^{1/2} - U_0F(U_0, U) + U_1F(U_1, U)] + \text{const.} \tag{7.14.27}$$

where

$$U \equiv \frac{u}{\sqrt{2\mathrm{I}}}\left[\frac{(\gamma-1)M^2}{2 + (\gamma-1)M^2}\right]^{1/2}$$

$$U_i \equiv \left[\frac{(\gamma-1)M_i^2}{2 + (\gamma-1)M_i^2}\right]^{1/2}, \qquad (i = 0, 1) \tag{7.14.28}$$

$$F(w, v) \equiv \frac{2(1 - w^2)^{1/2}}{v_0 - v_1}\operatorname{arctanh}\left[\frac{(1+w)(1-v)}{(1-w)(1+v)}\right]^{1/2}$$

The width W of the shock wave is often defined by (Figure 7.14.3)

$$W \equiv -\frac{u_0 - u_1}{|du/dx|_{\max}}$$

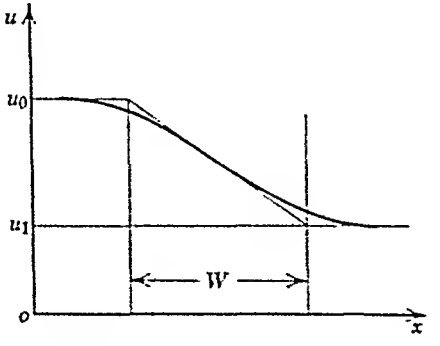


Fig. 7.14.3 Width of shock wave.

A calculation gives the result shown in Table 7.14.2. It is then clear from this result that the width is extremely small when the Mach number of the incident flow exceeds 2 The consideration of the shock wave as a moving discontinuity surface is thus justified.

Table 7.14.2. Structure of shock waves (after Howarth [1953])

Mach number of incident flow	Width of shock wave in air at 300°K ($\gamma = 1.4$)	
	mm.	Number of mean free paths
1	∞	∞
2	17.38×10^{-5}	3.98
3	10.66×10^{-5}	2.69
4	8.76×10^{-5}	2.28
5	7.95×10^{-5}	2.09
∞	6.58×10^{-5}	1.74

The Navier-Stokes equations predict small values for the shock wave thickness which are comparable to the molecular mean free path. On this ground, and on the ground that the Navier-Stokes equations emerge as a low-order approximation in a perturbation scheme applied to the Boltzmann equations of kinetic theory, the validity of these equations in calculating the shock wave thickness

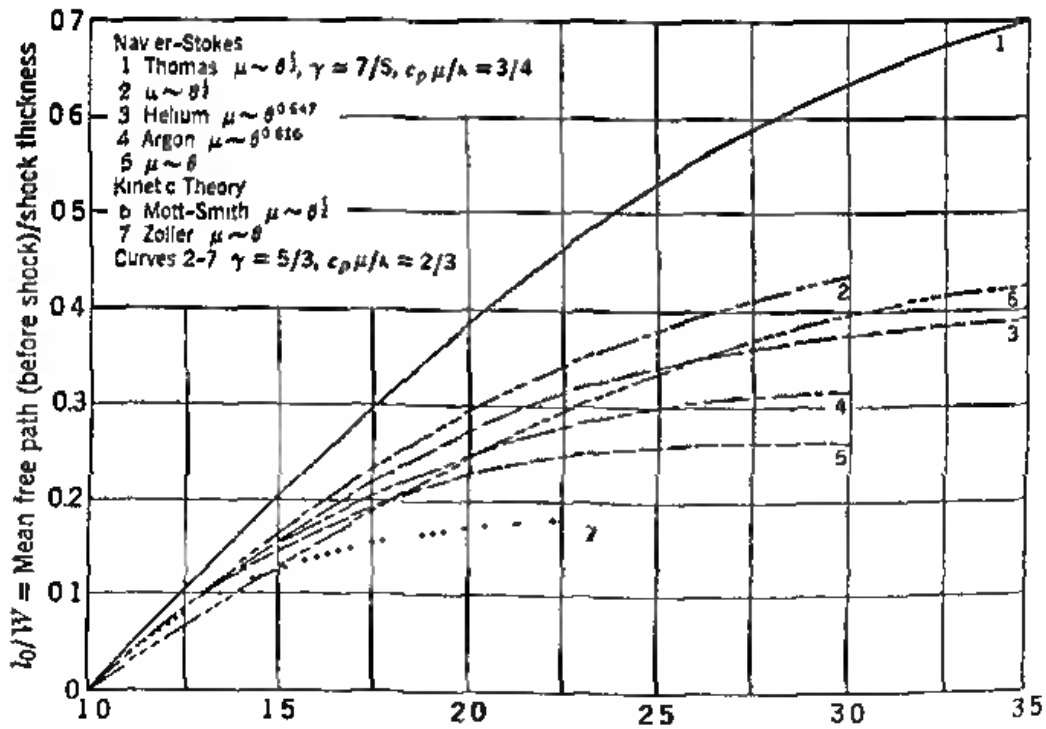


Fig. 7.14.4 Comparison of Navier-Stokes shock thickness with the kinetic theory values (after Gilbarg and Paolucci [1953]).

is questioned. Several works based on the kinetic theory exist, cf. Wang Chang [1948], Mott-Smith [1951], Zoller [1951], Grad [1952], and others. Counter arguments in favor of the Navier-Stokes and continuum theory are also given, cf. Gilbarg and Paolucci [1953]. These authors also gave numerical solutions for a temperature-dependent viscosity μ, in the more general form $\mu/\mu_0 = (\theta/\theta_0)^s$, where s is an exponent that is 0.647 for helium and 0.816 for argon. Based on these results and that of Thomas ($s = \frac{1}{2}$) given above, the ratio, l_0/W, of the Maxwell mean free path l_0 before the shock to the width of the shock wave W, defined above, is tabulated in Table 7.14.3. For l_0 we have

$$l_0 = \frac{16}{5} \frac{\mu_0}{\rho_0 \sqrt{2\pi R\theta_0}}$$

In Figure 7.14.4 there are plotted the ratios of mean free path before the shock to the shock thickness as a function of the Mach number M_0 for various exponents s. As can be seen, these results are not conclusive enough to support the thesis that the Navier-Stokes equations are inadequate for the treatment of shock wave phenomena.

Table 7.14.3. Here l_0/W for $\mu \propto \theta^s$ (after Gilbarg and Paolucci [1953])

M_0/s	0.500	0.647 (Helium)	0.816 (Argon)	1.000
1.2	0.070	0.068	0.067	0.067
1.4	0.136	0.132	0.128	0.122
1.7	0.222	0.210	0.199	0.187
2.0	0.292	0.270	0.248	0.224
2.5	0.381	0.344	0.303	0.264
3.0	0.437	0.377	0.314	0.261
4.0	—	0.410	—	—

7.15. SPHERICAL BLAST WAVES

In this section we give the solution of the basic equations of a compressible, inviscid fluid for the problem of a spherical explosion. Suppose that at a point in a gas a finite energy is liberated instantaneously. A strong spherical shock wave will start to propagate, since the initial pressure and temperature are infinitely large, theoretically. We would like to determine the pressure, density, and velocity at a point inside of the shock front as a

function of time. The basic equations of motion in spherical coordinates with central symmetry are

$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial r} + \frac{1}{\rho}\frac{\partial \pi}{\partial r} = 0$$

$$\frac{\partial \rho}{\partial t} + v\frac{\partial \rho}{\partial r} + \rho\frac{\partial v}{\partial r} + 2\rho\frac{v}{r} = 0 \tag{7.15.1}$$

$$\frac{\partial \eta}{\partial t} + v\frac{\partial \eta}{\partial r} = 0$$

of which the first is the equation of balance of momentum, the second is the continuity equation, and the third is the equation of isentropic flow. The equation of state for an ideal polytropic gas is

$$\pi = S(\eta)\rho^{\gamma} \tag{7.15.2}$$

$$S(\eta) \equiv (\gamma - 1)\exp\left[(\eta - \eta_0)/c_v\right]$$

From (7.15.2) we find that

$$d\log S = d\eta/c_v = d\log c^2 - (\gamma - 1)\,d\log\rho$$

$$\frac{d\pi}{\rho} = \frac{1}{\gamma}(dc^2 + c^2\,d\log\rho) \tag{7.15.3}$$

$$c^2 \equiv \gamma S\rho^{\gamma-1}$$

By use of these equations, we can express (7.15.1) in the form

$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial r} + \frac{1}{\gamma}\left(\frac{\partial c^2}{\partial r} + c^2\frac{\partial}{\partial r}\log\rho\right) = 0$$

$$\frac{\partial}{\partial t}(\log\rho) + v\frac{\partial}{\partial r}(\log\rho) + \frac{\partial v}{\partial r} + \frac{2v}{r} = 0 \tag{7.15.4}$$

$$\frac{\partial c^2}{\partial t} + v\frac{\partial c^2}{\partial r} + (\gamma - 1)c^2\left(\frac{\partial v}{\partial r} + \frac{2v}{r}\right) = 0$$

We seek a solution, called *self-similar motion*, of the system in the form

$$v = x(z)\frac{r}{t}, \qquad c^2 = y(z)\frac{r^2}{t^2} \tag{7.15.5}$$

where x and y are functions of a single parameter

$$z \equiv r/t^{\alpha} \tag{7.15.6}$$

where α is a constant. The use of (7.15.5) in (7.15.4) shows that for these equations not to contain r and t explicitly, we must have

$$\rho = t^{\beta}\xi(z) \tag{7.15.7}$$

Substitution of (7.15.5) to (7.15.7) into (7.15.4) now gives

$$x'(x-\alpha) + x^2 - x + \frac{1}{\gamma}[2y + y' + y(\log \xi)'] = 0$$

$$(\log \xi)'(x-\alpha) + \beta + x' + 3x = 0 \tag{7.15.8}$$

$$\frac{y'}{y}(x-\alpha) + (\gamma-1)x' + (3\gamma-1)x - 2 = 0$$

where a prime indicates differentiation with respect to $\log z$, for example,

$$x' \equiv \frac{dx}{d\log z} \tag{7.15.9}$$

From $(7.15.8)_2$ we solve for $(\log \xi)'$ and use this in $(7.15.8)_3$ so that the system (7.15.8) is now reduced to

$$(\log \xi)' = -\frac{3x + \beta + x'}{x-\alpha}$$

$$\frac{d\log z}{dx} = \frac{(\alpha - x)(d\log y/dx) - (\gamma - 1)}{(3\gamma-1)x - 2} \tag{7.15.10}$$

$$\frac{d\log z}{dx} = \frac{(\alpha-x)^2 - y}{y\{[2(\alpha-1)+\beta]\gamma^{-1} + 3x\} - x(1-x)(\alpha-x)}$$

Accordingly, the problem is reduced to solving a first-order differential equation of the form

$$\frac{dy}{dx} = y\frac{F_1(x,y)}{F_2(x,y)}$$

and two quadratures. The final solutions will therefore contain additional constants C_1, C_2, and C_3. Since we can always replace t by $t + \tau$ we will have, in all, six constants of integration α, β, τ, C_1, C_2, and C_3 in the general solution of the system. Determination of these constants must be made so as to have π, ρ, and v take the value they have at the front of the shock wave and the total energy be the same as the initial energy.

We now determine the velocity, pressure, and density inside the shock sphere. For a strong normal ($\omega = \pi/2$) shock wave from $(7.14.17)_4$ we have

$$\frac{u_1}{u_0} = \frac{2 + (\gamma-1)(u_0/c_0)^2}{(\gamma+1)(u_0/c_0)^2} \tag{7.15.11}$$

According to (7.14.7) $u_1 = v_1 - U$, $u_0 = v_0 - U$ where $U = v \cdot n$ is the normal velocity of the shock front and in the present case v_1 and v_0 are, respectively, the radial flow velocities inside and outside of the spherical shock front. Thus, $v_0 = 0$ and from the above equation we can solve for the shock front velocity

$$U/c_0 = \frac{\gamma+1}{4}\frac{v_1}{c_0} + \left[\left(\frac{\gamma+1}{4}\frac{v_1}{c_0}\right)^2 + 1\right]^{1/2} \tag{7.15.12}$$

For very large $v_1/c_0 \gg 1$ this gives

$$U \simeq \frac{\gamma+1}{2} v_1$$

Therefore, through (7.14.17) for this value of U and $v_0 = 0$, we will have

$$\pi_1 \simeq \frac{2}{\gamma+1}\rho_0 U^2, \qquad \frac{\rho_1}{\rho_0} \simeq \frac{\gamma+1}{\gamma-1}, \qquad v_1 \simeq \frac{2}{\gamma+1} U \tag{7.15.13}$$

Returning to the original problem, we see through (7.15.7) that for $\rho_1 =$ const. we must have $\beta = 0$. The conditions on the front of the shock wave according to (7.15 5) and (7.15 13) are

$$x_1 = v_1 \frac{t}{r} = \frac{2}{\gamma+1} U \frac{t}{r}$$

$$y_1 = c_1^2 \frac{t^2}{r^2} = \frac{2\gamma(\gamma-1)}{(\gamma+1)^2} U^2 \frac{t^2}{r^2}$$

Upon substituting

$$U = \frac{dr}{dt} = \alpha \frac{r}{t}$$

these reduce to

$$x_1 = \frac{2\alpha}{\gamma+1}, \qquad y_1 = \frac{2\gamma(\gamma-1)}{(\gamma+1)^2}\alpha^2 \tag{7.15.14}$$

Thus, the problem is reduced to finding the solution of $(7\ 15\ 10)_{2,3}$ passing through the points (7.15.14). In addition, the parameter α will have to be determined. A physically reasonable condition is the conservation of total energy inside the shock front $r = r_1(t)$.

$$E = 4\pi \int_0^{r_1} \left(\frac{\pi}{\gamma-1} + \rho\frac{v^2}{2}\right) r^2\, dr = \text{const} \tag{7.15.15}$$

A simple solution of $(7.15.10)_{2,3}$ subject to (7.15.14) was given by Sedov [1946a, b][11] for $\alpha = 2/5$ which has the form

$$y = \frac{\gamma - 1}{2} x^2 \frac{\alpha - x}{x - (\alpha/\gamma)}, \qquad \alpha = \frac{2}{5} \tag{7.15.16}$$

By a quadrature from $(7.15.10)_1$ and $(7.15.10)_2$ we obtain ρ and z, and through (7.15.5) v and c^2; hence from $c^2 = \gamma\pi/\rho$ the pressure, and from the equation of state the temperature. Thus

$$\begin{aligned}
z &= C_1 x^{-a}\left(x - \frac{\alpha}{\gamma}\right)^{(\gamma-1)a/2[(\gamma-2)a+1]} \\
&\quad \times \left[x + \frac{\alpha}{\alpha(\gamma-2)+1-\gamma}\right]^{a} \equiv C_1 \varphi_1(x) \\
\rho &= C_2(\alpha - x)^{2/(\gamma-2)}\left(x - \frac{\alpha}{\gamma}\right)^{(1-a)/[(\gamma-2)a+1]} \\
&\quad \times \left[x + \frac{\alpha}{\alpha(\gamma-2)+1-\gamma}\right]^{2a/(\gamma-2)a} \equiv C_2 \varphi_2(x) \\
v &= C_1 t^{a-1} x^{1-a}\left(x - \frac{\alpha}{\gamma}\right)^{(\gamma-1)a/2[(\gamma-2)a+1]} \\
&\quad \times \left[x + \frac{\alpha}{\alpha(\gamma-2)+1-\gamma}\right]^{a} \equiv C_1 (x/t)^{1-a} \varphi_3(x) \\
\pi &= \frac{\gamma - 1}{2\gamma} C_1^2 C_2 t^{2a-2} x^{2-2a}(\alpha - x)^{\gamma/(\gamma-2)} \\
&\quad \times \left[x + \frac{\alpha}{\alpha(\gamma-2)+1-\gamma}\right]^{b} \equiv \frac{\gamma - 1}{2\gamma} C_1^2 C_2 (x/t)^{2a-2} \varphi_4(x) \\
\theta &= \frac{\pi}{c_v(\gamma - 1)\rho}
\end{aligned} \tag{7.15.17}$$

where

$$\begin{aligned}
a &\equiv \frac{\alpha}{\alpha(\gamma-2)+1-\gamma}\left[\frac{\gamma-1}{2}\frac{\gamma}{\alpha(\gamma-2)+1} + 1\right] + \alpha \\
b &\equiv 2a\left[1 + \frac{1}{(\gamma-2)a}\right]
\end{aligned} \tag{7.15.18}$$

[11] Taylor [1950] gave a similar solution in lagrangian coordinates. For other discussions of this and related problems see Sedov [1959] and Hayes and Probstein [1959].

and C_1 and C_2 are constants yet to be determined. By mere substitution of π, u, and ρ into (7.15.15) we find that this integral does not depend on t explicitly, since $\alpha = 2/5$.

Now conditions at $x = x_1$, $y = y_1$ determine the constants C_1 and C_2 immediately, through (7.15.17), that is,

$$C_1 = r_1^{1-\epsilon} v_1^{\epsilon}/\varphi_3(x_1), \qquad C_2 = \rho_1/\varphi_2(x_1) \tag{7.15.19}$$

and the solution of our problem is complete.

The pressure, velocity, and density ratio's are plotted in Figure 7.15.1 for $\gamma = 7/5$ as a function of r/r_1 where r_1 is the radius of the shock wave.

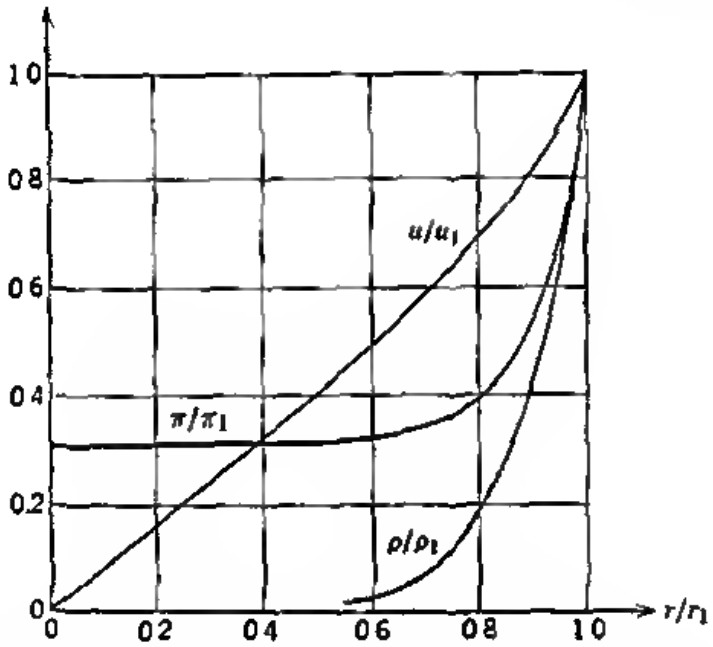


Fig. 7.15.1 Spherical blast wave accompanied by a shock wave $\gamma = 7/5$.

The self-similar solution is known to fail in time Therefore, objections have been raised to this type of solution. We note for example from (7 15.17) that the pressure behind the shock front approaches zero as $t \to \infty$ for $\alpha = 2/5$. Consequently, the assumption that the shock is strong is violated in time. Furthermore, near the center of explosion the experimentally observed pressure does not agree with the above result. Nevertheless, the Taylor-Sedov solution gives a good description of the blast wave at sufficiently distant places not affected too strongly by the explosion process provided, of course, that the shock wave can still be considered strong Other special solutions for special values of γ exist. For example, for $\gamma = 7$, which approximates water, a simple solution was found by Primakoff[12]

[12] For this and other simple solutions see Courant and Friedrichs [1948, Arts. 159 to 166] For other elaborate solutions see Stanyukovich [1960]

PROBLEMS

7.1. For an incompressible Stokesian fluid, second degree in the deformation rate tensor, show that the stress is given by

$$t_{kl} = -p\,\delta_{kl} + 2\mu_v\,d_{kl} + 4\nu_v\,d_{kr}\,d_{rl}$$

where $p(x,t)$ is arbitrary and μ_v and ν_v are functions of temperature only. Find the consequency of thermodynamics restrictions on such fluids.

7.2. Suggest a test program to determine the constitutive coefficients of compressible second-degree Stokesian fluids.

7.3. The space between two long coaxial, circular cylinders is occupied with a Navier-Stokes fluid. The cylinders are rotating with the angular velocities ω_1 and ω_2, and the fluid is assumed to stick to the cylinders. Determine the velocity and stress field in the fluid.

7.4. An incompressible Navier-Stokes fluid contained between two inclined parallel planes is flowing down under the force of gravity. If the upper plate is moving with a velocity $u = U(t)$ and the lower plate is stationary, determine the velocity field in the fluid. Assume the motion is planar.

7.5. An incompressible, inviscid fluid occupying the half space, $y > 0$, is free to flow through an aperture of length $2a$ made at a rigid plate at $y = 0$. The motion is irrotational and planar. Determine the velocity field.

7.6. In a two-dimensional motion of an incompressible, inviscid fluid with zero body loads, show that the resultant force (F_x, F_y) and resultant couple (M_x, M_y), acting on a cylindrical body, are given by the *Blasius Theorem*:

$$F_x - iF_y = \frac{i}{2}\rho\oint\left(\frac{d\Phi}{dz}\right)^2 dz, \qquad z \equiv x + iy$$

$$M_x + iM_y = -\frac{\rho}{2}\oint z\left(\frac{d\Phi}{dz}\right)^2 dz, \qquad i \equiv \sqrt{-1}$$

where the integrals are taken over the periphery of the cylindrical body and $\Phi(z)$ is the complex potential related to the velocity components, v_x and v_y by

$$v_x - iv_y = \frac{d\Phi}{dz}$$

7.7. Using the Blasius Theorem (Problem 7.6), show that the lift F_y on a circular cylinder exerted by an ideal flow in x-direction is given by

$$F_y = \rho U\Gamma$$

where U is the x-component of the velocity at $x = -\infty$ and Γ is the circulation about the center of cylinder (also the origin of the coordinates). Prove also that the drag $F_x = 0$ and there are no couples acting on the cylinder.

7.8. An incompressible, viscous fluid occupying the half space, $y > 0$, is set into an oscillatory motion by a thin plate (at $y = 0$) which is vibrating harmonically in the x-direction. Assuming that the fluid sticks to the plate, determine the velocity.

7.9. An infinite lamina of incompressible, viscous fluid occupying the half space, $z > 0$, over a plate at $z = 0$, is set into a rotatory motion. The constant angular velocity of the plate about the z-axis is Ω. The fluid is assumed to stick to the plate at $z = 0$ and has a vanishing velocity at $z = \infty$. Determine the velocity field. Assume a velocity field in the form

$$u = rf(z),\ v = rg(z),\ w = h(z),\ r \equiv \sqrt{x^2 + y^2}$$

where f, g, and h are unknown functions to be determined.

7.10. If $\phi(t)$ denotes the value of an acoustical wave $\phi(\mathbf{x},t)$ at time, t, on the boundary surface $\mathscr{S}$ and $f(t)$ denotes the normal derivative, $\partial\phi/\partial n$ on $\mathscr{S}$ at time, t, show that the wave, ϕ_P, at any point of the region $\mathscr{V}$ within $\mathscr{S}$ at time, t, is given by [Kirchhoff],

$$\phi_P(t) = -\frac{1}{4\pi}\int_{\mathscr{S}} \left\{\frac{1}{r} f\left(t - \frac{r}{c}\right) + \frac{\partial}{\partial n}\left[\frac{1}{r}\phi\left(t - \frac{r}{c}\right)\right]\right\} da$$

where c is the velocity of sound and $\dfrac{\partial}{\partial n}$ applied only to r as it appears explicitly.

7.11. An acoustic source, $f(t)$, applied to the origin, $x = y = 0$, produces diverging, symmetric waves about the origin in a two-dimensional, infinite plane. Show that the velocity potential, $\phi(r,t)$, is given by

$$\phi(r, t) = \frac{1}{2\pi}\int_0^{\infty} f\left(t - \frac{r}{c}\cosh u\right) du, \qquad r \equiv \sqrt{x^2 + y^2}$$

where c is the velocity of sound.

7.12. A small fly at time, $t = 0$, touches the center of a circular lake. Determine the subsequent motion.

7.13. Discuss the motion of a steady, irrotational, two-dimensional, compressible, inviscid fluid in which the velocity field depends only on the angle, θ, of a cylindrical coordinate.

7.14. Find the pressure rise upon the reflection of a plane shock wave from a flat rigid wall.

7.15. Show that in a steady, irrotational motion of an inviscid, compressible fluid, the velocity potential, ϕ, satisfies the equation

$$(c^2 - \phi_{,x}^2)\,\phi_{,xx} - 2\phi_{,x}\phi_{,y}\phi_{,xy} + (c^2 - \phi_{,y}^2)\,\phi_{,yy} = 0$$

where c is the velocity of sound related to ϕ. Determine the expression of c^2.

7.16. The equations of the incompressible fluids given in Problem 7.15 can be

transformed into a linear equation by the *hodograph* transformation

$$x = x(u, v), \qquad y = y(u, v)$$

where (x, y) are the rectangular coordinates in two dimensions and (u, v) are the velocity components in these coordinates. Determine the linear equations replacing the equations given in Problem 7.15.

7.17. (Short Term Paper) Linearize the Navier-Stokes equations of an incompressible fluid about a constant free stream velocity, $u_x = U$ at $x = -\infty$. For a steady flow about a sphere, determine the velocity field and the drag when .

(*a*) $U \equiv 0$ (Stokes flow);

(*b*) $U \neq 0$ (Oseen flow).

The fluid is assumed to stick to the surface of the sphere.

7.18. (Short Term Paper) Give a discussion of the boundary layer approximations. Obtain the equations of viscous incompressible fluids for a boundary layer near a flat plate. Give a solution for a fluid past a flat plate.

7.19. (Short Term Paper) Study the literature and discuss the method of characteristics for the solution of two-dimensional problems in compressible fluids.

7.20. (Short Term Paper) Derive the equations of one-dimensional non-steady incompressible inviscid fluids. Present the methods of characteristics and the concept of simple waves for the solution of the fluid-flow problem resulting from the motion of an accelerating cylinder in a straight tube.

7.21 (Short Term Paper) Give a short discussion of homogeneous, isotropic turbulence.

CHAPTER 8

Thermoelasticity

8.1. SCOPE OF THE CHAPTER

This chapter gives an account of the theory of linear thermoelasticity. The constitutive equations of the theory are derived in Section 8.2 for the linear theory of anisotropic solids. These include the constitutive equations for stress, heat conduction, internal energy, and entropy. The equation of heat conduction is obtained. In Section 8.3 the equations of the linear isotropic solids are presented. Section 8.4 collects in one place the fundamental equations, jump conditions, and boundary and initial conditions necessary in the solution of problems in this field. The Navier-Duhamel field equations are derived. The remaining sections, 8.5 to 8.8, are devoted to some illustrative solutions In Section 8.5 we give Duhamel's theorem, which is of central importance in the theory of heat conduction in solids Some heat-conduction problems concerned with the temperature distribution in a half space and outside of a spherical cavity are solved in Section 8.6 and Section 8.7 discusses thermoelastic waves. In Section 8 8 we return to the problem of a spherical cavity subject to dynamical pressure and temperature. Corresponding problems were treated previously for isothermal elasticity, spherical blast waves, and heat conduction. Once again we are prepared to observe the different physical phenomena brought into play through the physical constitution of the medium.

This chapter is not intended to be exhaustive either in the presentation of problems or in the method of solutions.

8.2. LINEAR CONSTITUTIVE EQUATIONS

In Section 5.9 we obtain the constitutive equations of a thermoelastic solid. These equations were expressed in terms of the Green deformation tensor C by (5.9.6) to (5.9.9), subject to restrictions (5.9.11) to (5.9.12). In terms of the lagrangian strain measure

$$E_{KL} = \tfrac{1}{2}(C_{KL} - \delta_{KL})$$

they may be expressed as

$$t_{kl} = \frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial E_{KL}} x_{(k,K}x_{l),L} \tag{8.2.1}$$

$$q_k = Q_K(\mathbf{E}, \theta_{,K}\,; \theta, \mathbf{X})X_{K,k} \tag{8.2.2}$$

$$\varepsilon = \frac{1}{\rho_0}\left(\Sigma - \theta\frac{\partial \Sigma}{\partial \theta}\right) \tag{8.2.3}$$

$$\eta = -\frac{1}{\rho_0}\frac{\partial \Sigma}{\partial \theta} \tag{8.2.4}$$

where

$$\frac{1}{\rho_0}\Sigma(\mathbf{E}, \theta, \mathbf{X}) \equiv \psi \tag{8.2.5}$$

is the free energy. The function Q_K is subject to the restrictions expressed by the Clausius-Duhem inequality

$$q_k\theta_{,k} \geq 0 \tag{8.2.6}$$

which expresses the fact that the heat flows from hot to cold. If we assume that $\mathbf{q}(\mathbf{E}, \theta_{,K}, \theta, \mathbf{X})$ is continuous in the temperature gradient $\theta_{,K}$, from (8.2.6), then it follows that

$$\mathbf{q}(\mathbf{E}, 0, \theta, \mathbf{X}) = 0 \quad \text{or} \quad Q_K(\mathbf{E}, 0, \theta, \mathbf{X}) = 0 \tag{8.2.7}$$

The above expressions are valid for the anisotropic inhomogeneous solids.

This chapter is concerned with the linear theory, hence we approximate the strain measure E by the infinitesimal strain $\tilde{\mathbf{E}}$ and employ the infinitesimal rotation $\tilde{\mathbf{R}}$

$$\tilde{E}_{KL} \equiv \tfrac{1}{2}(U_{K,L} + U_{L,K}) \tag{8.2.8}$$

$$\tilde{R}_{KL} \equiv \tfrac{1}{2}(U_{K,L} - U_{L,K}) \tag{8.2.9}$$

The linearization of (8.2.1) to (8.2.4) in terms of $\tilde{\mathbf{E}}$, $\tilde{\mathbf{R}}$, and $\theta_{,K}$ may be made in the same fashion as given in Section 6.2. Thus, we write

$$\Sigma = \Sigma_0 + \Sigma_{KL}\bar{E}_{KL} + \tfrac{1}{2}\Sigma_{KLMN}\bar{E}_{KL}\bar{E}_{MN} \tag{8.2 10}$$

$$Q_K = B_K + B_{KM}\theta_{,M} + B_{KLM}\bar{E}_{LM} \tag{8.2.11}$$

where Σ_0, Σ_{KL}, Σ_{KLMN}, B_K, B_{KM}, and B_{KLM} are functions of θ and $\mathbf{X}$, and satisfy the symmetry conditions

$$\Sigma_{KL} = \Sigma_{LK}, \qquad \Sigma_{KLMN} = \Sigma_{MNKL} = \Sigma_{LKMN} = \Sigma_{KLNM} \tag{8.2.12}$$

$$B_{KLM} = B_{KML}$$

We substitute (8 2 10) and (8.2.11) into (8.2 1) and (8.2.2.) and use the expressions

$$\begin{aligned} x_{k,K} &= (\delta_{MK} + \bar{E}_{MK} + \bar{R}_{MK})\,\delta_{Mk} \\ X_{K,k} &= (\delta_{mk} - \tilde{e}_{mk} - \tilde{r}_{mk})\,\delta_{mK} \end{aligned} \tag{8.2.13}$$

Hence

$$\begin{aligned} t_{kl} &= \frac{\rho}{\rho_0}\left[\Sigma_{KL} + \Sigma_{ML}(\bar{E}_{KM} + \bar{R}_{KM}) + \Sigma_{KM}(\bar{E}_{LM} + \bar{R}_{LM})\right. \\ &\quad \left. + \Sigma_{KLMN}\bar{E}_{MN}\right]\delta_{Kk}\,\delta_{Ll} \\ q_k &= B_K(\delta_{mk} - \tilde{e}_{mk} - \tilde{r}_{mk})\,\delta_{mK} + (B_{KM}\theta_{,M} + B_{KLM}\bar{E}_{LM})\,\delta_{kK} \end{aligned} \tag{8 2.14}$$

In the linear theory we have

$$\begin{aligned} \bar{E}_{KL} &\cong \tilde{e}_{kl}\,\delta_{Kk}\,\delta_{Ll}, \qquad \bar{R}_{KL} \cong \tilde{r}_{kl}\,\delta_{Kk}\,\delta_{Ll} \\ \frac{\rho}{\rho_0} &= 1 - \bar{E}_{KK} \cong 1 - \tilde{e}_{kk} \end{aligned} \tag{8 2.15}$$

Consequently (8.2.14) can be expressed in the spatial forms

$$\begin{aligned} t_{kl} &= (1 - \tilde{e}_{mm})\sigma_{kl} + \sigma_{ml}(\tilde{e}_{km} + \tilde{r}_{km}) \\ &\quad + \sigma_{km}(\tilde{e}_{lm} + \tilde{r}_{lm}) + \sigma_{klmn}\tilde{e}_{mn} \end{aligned} \tag{8.2.16}$$

$$q_k = b_m(\delta_{mk} - \tilde{e}_{mk} - \tilde{r}_{mk}) + b_{km}\theta_{,m} + b_{klm}\tilde{e}_{lm} \tag{8.2.17}$$

where

$$\begin{aligned} \sigma_{kl} &= \Sigma_{KL}\,\delta_{kK}\,\delta_{lL} \\ \sigma_{klmn} &= \Sigma_{KLMN}\,\delta_{kK}\,\delta_{lL}\,\delta_{mM}\,\delta_{nN} \\ b_k &= B_K\,\delta_{kK} \\ b_{kl} &= B_{KL}\,\delta_{kK}\,\delta_{lL} \\ b_{klm} &= B_{KLM}\,\delta_{kK}\,\delta_{lL}\,\delta_{mM} \end{aligned} \tag{8.2.18}$$

are functions of θ and X, and satisfy the symmetry relations

$$\begin{aligned} &\sigma_{kl} = \sigma_{lk}, \qquad \sigma_{klmn} = \sigma_{mnkl} = \sigma_{lkmn} = \sigma_{klnm} \\ &b_{kl} = b_{lk}, \qquad b_{klm} = b_{kml} \end{aligned} \tag{8.2.19}$$

The restrictions (8.2.7) imply that

$$b_m = 0, \qquad b_{klm} = 0 \tag{8.2.20}$$

With this the inequality (8.2.6) reads

$$b_{km}\theta_{,k}\theta_{,m} \geq 0 \tag{8.2.21}$$

We have therefore proved

THEOREM 1. *The natural state of a linear thermoelastic solid is adiabatic heat,* q, *vanishes with* $\nabla\theta$ *and* $\tilde{\mathbf{e}}$, *and the matrix of heat-conduction coefficients* b_{km} *is a nonnegative definite form.*

The constitutive equation of heat (8.2.17) is therefore reduced to

$$q_k = b_{km}(\theta, \mathrm{X})\theta_{,m} \tag{8.2.22}$$

where b_{km} is a symmetric, nonnegative, definite form.

The stress potential Σ has the form

$$\Sigma = \Sigma_0 + \sigma_{kl}\tilde{e}_{kl} + \tfrac{1}{2}\sigma_{klmn}\tilde{e}_{kl}\tilde{e}_{mn} \tag{8.2.23}$$

The material symmetry conditions further restrict the number of independent material properties σ_{kl}, σ_{klmn}, b_{kl}. We shall not delve into these details. The situation here is similar to the analyses presented in Section 6.2. In the following sections we shall study in detail only the case of the isotropic solid.

The equation of heat conduction for the anisotropic solid is obtained either from (5.9.18) or, equivalently, from

$$\rho\theta\dot{\eta} = q_{k,k} + \rho h \tag{8.2.24}$$

where η is given by (8.2.4).

Through (8.2.23) and (8.2.4) we have

$$\eta = -\frac{1}{\rho_0}\left(\frac{\partial\Sigma_0}{\partial\theta} + \frac{\partial\sigma_{kl}}{\partial\theta}\tilde{e}_{kl} + \frac{1}{2}\frac{\partial\sigma_{klmn}}{\partial\theta}\tilde{e}_{kl}\tilde{e}_{mn}\right) \tag{8.2.25}$$

Dropping the quadratic terms in strain, and substituting it into (8.2.24), we obtain the equation of heat conduction

$$\frac{\rho\theta}{\rho_0}\left(\frac{\partial^2\Sigma_0}{\partial\theta^2}\dot{\theta} + \frac{\partial^2\sigma_{kl}}{\partial\theta^2}\dot{\theta}\tilde{e}_{kl} + \frac{\partial\sigma_{kl}}{\partial\theta}\dot{\tilde{e}}_{kl}\right) + (b_{kl}\theta_{,l})_{,k} + \rho h = 0 \tag{8.2.26}$$

Further linearization is possible by using $(8.2.15)_3$ for ρ/ρ_0.

We note that in all the equations given here, the constitutive coefficients Σ_0, σ_{kl}, σ_{klmn} and b_{kl} are functions of θ and X. Therefore, they are valid for any temperature large or small. The linearization has been affected only with respect to temperature gradients and strains. In the linear theory of thermoelasticity, further linearization is carried out by assuming that the temperature change from the temperature T_0 of the natural state is small, that is, we write

$$\theta = T_0 + T, \qquad |T| \ll T_0, \qquad T_0 > 0 \tag{8.2.27}$$

In this case we may write

$$\begin{aligned} \Sigma_0 &= S_0 - \rho_0 \eta_0 T - \frac{\rho_0 \gamma}{2T_0} T^2 \\ \sigma_{kl} &= \alpha_{kl} - \beta_{kl} T \end{aligned} \tag{8.2.28}$$

so that

$$\Sigma = S_0 - \rho_0 \eta_0 T - \frac{\rho_0 \gamma}{2T_0} T^2 + \alpha_{kl} \tilde{e}_{kl} - \beta_{kl} \tilde{e}_{kl} T + \tfrac{1}{2} \sigma_{klmn} \tilde{e}_{kl} \tilde{e}_{mn} \tag{8.2.29}$$

where S_0, η_0, γ, α_{kl}, β_{kl}, and σ_{klmn} are functions of X only. Upon using (8.2.27) to (8.2.29) in (8.2.16), (8.2.22), (8.2.3), and (8.2.25), we obtain the following constitutive equations of the linear theory of anisotropic thermoelastic solids

$$\begin{aligned} t_{kl} = {} & \alpha_{kl}(1 - \tilde{e}_{mm}) - \beta_{kl} T + \alpha_{ml}(\tilde{e}_{km} + \tilde{r}_{km}) \\ & + \alpha_{km}(\tilde{e}_{lm} + \tilde{r}_{lm}) + \sigma_{klmn} \tilde{e}_{mn} \end{aligned} \tag{8.2.30}$$

$$q_k = b_{kl} \theta_{,l} \tag{8.2.31}$$

$$\begin{aligned} \rho_0 \varepsilon = {} & S_0 + \rho_0 \eta_0 T_0 + \rho_0 \gamma \left(T + \frac{T^2}{2T_0} \right) + (\alpha_{kl} + T_0 \beta_{kl}) \tilde{e}_{kl} \\ & + \tfrac{1}{2} \sigma_{klmn} \tilde{e}_{kl} \tilde{e}_{mn} \end{aligned} \tag{8.2.32}$$

$$\eta = \eta_0 + \frac{\gamma}{T_0} T + \frac{1}{\rho_0} \beta_{kl} \tilde{e}_{kl} \tag{8.2.33}$$

The linear equation of heat conduction follows from (8.2.24)

$$-\rho_0 \gamma \dot{T} - T_0 \beta_{kl} \dot{\tilde{e}}_{kl} + (b_{kl} T_{,l})_{,k} + \rho h = 0 \tag{8.2.34}$$

When the natural state is stress free, we have $\alpha_{kl} = 0$ and the stress constitutive equation (8.2.30) for this case further reduces to

$$t_{kl} = -\beta_{kl} T + \sigma_{klmn} \tilde{e}_{mn} \tag{8.2.35}$$

with the corresponding ε obtained from (8.2.32) by setting $\alpha_{kl} = 0$.

8.3. ISOTROPIC SOLIDS

The isotropic solid possesses no preferred direction with respect to its material properties. As shown in Section 6.2, the material tensors σ_{kl}, σ_{klmn}, α_{kl}, and b_{kl} are isotropic tensors. Hence, they may be expressed as

$$\sigma_{kl} = \alpha_e\,\delta_{kl}, \qquad \sigma_{klmn} = \lambda_e\,\delta_{kl}\,\delta_{mn} + \mu_e(\delta_{km}\,\delta_{ln} + \delta_{kn}\,\delta_{lm}),$$
$$b_{kl} = \kappa\,\delta_{kl} \tag{8.3.1}$$

The stress and heat constitutive equations (8.2.16) and (8.2.22) now take the forms

$$t_{kl} = \alpha_e\,\delta_{kl} + (\lambda_e - \alpha_e)\tilde{e}_{rr}\,\delta_{kl} + 2(\mu_e + \alpha_e)\tilde{e}_{kl} \tag{8.3.2}$$

$$q_k = \kappa\theta_{,k} \tag{8.3.3}$$

of which the second is the celebrated *heat conduction law of Fourier* for isotropic solids. The inequality (8.2.21) now gives

$$\kappa \geq 0 \tag{8.3.4}$$

The stress potential (8.2.23) takes the form

$$\Sigma = \Sigma_0 + \alpha_e \mathrm{I}_{\tilde{e}} + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}}^2 - 2\mu_e \mathrm{II}_{\tilde{e}} \tag{8.3.5}$$

where $\mathrm{I}_{\tilde{e}}$ and $\mathrm{II}_{\tilde{e}}$ are the invariants of the strain tensor $\tilde{\mathbf{e}}$.

The entropy, η, given by (8.2.25), reduces to

$$\eta = -\frac{1}{\rho_0}\left[\frac{\partial \Sigma_0}{\partial \theta} + \frac{\partial \alpha_e}{\partial \theta}\mathrm{I}_{\tilde{e}} + \frac{1}{2}\left(\frac{\partial \lambda_e}{\partial \theta} + 2\frac{\partial \mu_e}{\partial \theta}\right)\mathrm{I}_{\tilde{e}}^2 - 2\frac{\partial \mu_e}{\partial \theta}\mathrm{II}_{\tilde{e}}\right] \tag{8.3.6}$$

The equation of heat conduction (8.2.26) becomes

$$\frac{\rho\theta}{\rho_0}\left(\frac{\partial^2 \Sigma_0}{\partial \theta^2}\dot{\theta} + \frac{\partial^2 \alpha_e}{\partial \theta^2}\dot{\theta}\mathrm{I}_{\tilde{e}} + \frac{\partial \alpha_e}{\partial \theta}\dot{\tilde{e}}_{rr}\right) + (\kappa\theta_{,k})_{,k} + \rho h = 0 \tag{8.3.7}$$

The above equations are valid for any temperature, that is, they are nonlinear in θ. The linearization is affected as in the previous section. In fact, by putting

$$\theta = T_0 + T, \qquad |T| \ll T_0, \qquad T_0 > 0 \tag{8.3.8}$$

where T_0 is the temperature of the natural state and writing the isotropic values

$$\alpha_{kl} = \alpha_e\,\delta_{kl}, \qquad \beta_{kl} = \beta\,\delta_{kl}$$
$$b_{kl} = \kappa\,\delta_{kl}, \qquad \sigma_{klmn} = \lambda_e\,\delta_{kl}\,\delta_{mn} + \mu_e(\delta_{km}\,\delta_{ln} + \delta_{kn}\,\delta_{lm}) \tag{8.3.9}$$

for the material tensors appearing in (8.2.30) to (8.2.33) we obtain the constitutive equation of linear isotropic solids

$$t_{kl} = (1 - \tilde{e}_{mm})\alpha_e \delta_{kl} - \beta T \delta_{kl} + \lambda_e \tilde{e}_{mn} \delta_{kl} + 2(\mu_e + \alpha_e)\tilde{e}_{kl} \tag{8.3.10}$$

$$q_k = \kappa T_{,k} \tag{8.3.11}$$

$$\rho_0 \varepsilon = S_0 + \rho_0 \eta_0 T_0 + \rho_0 \gamma \left(T + \frac{T^2}{2T_0} \right) + (\alpha_e + \beta T_0) \mathrm{I}_{\tilde{e}} + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}}^2 - 2\mu_e \mathrm{II}_{\tilde{e}} \tag{8.3.12}$$

$$\eta = \eta_0 + \frac{\gamma}{T_0} T + \frac{\beta}{\rho_0} \mathrm{I}_{\tilde{e}} \tag{8.3.13}$$

The equation of heat conduction linearizes to

$$-\rho_0 \gamma \dot{T} - \beta T_0 \dot{u}_{k,k} + (\kappa T_{,k})_{,k} + \rho h = 0 \tag{8.3.14}$$

From (8.3.10) it is clear that when $\tilde{\mathbf{e}} = 0$ and $T = 0$ we have

$$t_{kl} = \alpha_e \delta_{kl} \tag{8.3.15}$$

Hence

THEOREM: *The isotrapic, thermaelastic solid in its natural state can support anly a hydrastatic state af stress.*

This theorem can be shown to be valid for the nonlinear solid by observing that the nonlinear constitutive equations (5.10.8) would reduce to the form (8.3.15) when $c_{kl}^{-1} = \delta_{kl}$.

If the solid is stress-free in its natural state (that is, $\mathbf{t} = 0$ when $T = 0$ aod $\tilde{\mathbf{e}} = 0$), we will have $\alpha_e = 0$. The stress constitutive equations (8 3.10) now take the form

$$t_{kl} = (-\beta T + \lambda_e \tilde{e}_{rr}) \delta_{kl} + 2\mu_e \tilde{e}_{kl} \tag{8.3 16}$$

The stress potential compatible with this approximation is then given by

$$\Sigma = S_0 - \rho_0 \eta_0 T - \frac{\rho_0 \gamma}{2T_0} T^2 - \beta T \mathrm{I}_{\tilde{e}} + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_{\tilde{e}}^2 - 2\mu_e \mathrm{II}_{\tilde{e}} \tag{8.3.17}$$

For vanishing β and γ these equations reduce to those of the purely elastic theory presented in Section 6.2.

Incompressible Materials. In this case $\tilde{e}_{rr} = 0$. The equations (8.3.10) to (8.3.13) are respectively modified to

$$t_{kl} = -(\beta T + p) \delta_{kl} + 2\mu_e \tilde{e}_{kl} \tag{8.3.18}$$

$$q_k = \kappa T_{,k} \tag{8.3.19}$$

$$\rho_0 \varepsilon = S_0 + \rho_0 \eta_0 T_0 + \rho_0 \gamma \left(T + \frac{T^2}{2T_0} \right) - 2\mu_e \mathrm{II}_{\tilde{e}} \tag{8.3.20}$$

$$\eta = \eta_0 + \frac{\gamma}{T_0} T \tag{8.3.21}$$

and the equation of heat conduction (8.3.14) reads

$$-\rho_0 \gamma \dot{T} + (\kappa T_{,k})_{,k} + \rho_0 h = 0 \tag{8.3.22}$$

We note that (8.3.22) is also the equation of heat transfer in a rigid body. It is clear from Killing's theorem that for a rigid motion we have

$$\dot{\tilde{e}}_{kk} \equiv \dot{u}_{k,k} = 0$$

so that (8.3.14) reduces to (8.3.22).

8.4. RÉSUMÉ OF BASIC EQUATIONS OF THERMOELASTICITY

The linear theory of isotropic thermoelastic solids is based on the following equations.

Equations of Balance

$$\frac{\rho_0}{\rho} = 1 + \mathrm{I}_{\tilde{e}} \tag{8.4.1}$$

$$t_{kl,k} + \rho(f_l - \dot{v}_l) = 0 \tag{8.4.2}$$

$$t_{kl} = t_{lk} \tag{8.4.3}$$

$$\rho \theta \dot{\eta} = q_{k,k} + \rho h \tag{8.4.4}$$

Equations of Jump. If there is a moving discontinuity surface $\sigma(t)$ sweeping the body with a velocity $\mathbf{v}$ in the direction of its normal $\mathbf{n}$ we must satisfy the following jump conditions on the surface $\sigma(t)$:

$$[\rho(\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} = 0 \tag{8.4.5}$$

$$-[t_{kl}] n_k + [\rho v_l (\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} = 0 \tag{8.4.6}$$

$$[(\varepsilon + \tfrac{1}{2} v_l v_l) \rho (\mathbf{v} - \mathbf{v})] \cdot \mathbf{n} - [t_{kl} v_l + q_k] n_k = 0 \tag{8.4.7}$$

$$\left[\rho \eta (\mathbf{v} - \mathbf{v}) - \frac{\mathbf{q}}{\theta} \right] \cdot \mathbf{n} \geq 0 \quad \text{on} \quad \sigma(t) \tag{8.4.8}$$

Constitutive Equations (Compressible Solids)

$$t_{kl} = (-\beta T + \lambda_e \mathrm{I}_{\tilde{e}})\, \delta_{kl} + 2\mu_e \tilde{e}_{kl} \tag{8.4.9}$$

$$q_k = \kappa T_{,k} \tag{8.4.10}$$

$$\rho_0\varepsilon = S_0 + \rho_0\eta_0 T_0 + \rho_0\gamma\left(T + \frac{T^2}{2T_0}\right) + \beta T_0 \mathrm{I}_e + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_e^2 - 2\mu_e \mathrm{II}_e \tag{8.4.11}$$

$$\eta = \eta_0 + \frac{\gamma}{T_0} T + \frac{\beta}{\rho_0} \mathrm{I}_e \tag{8.4.12}$$

Constitutive Equations (Incompressible Solids)

$$t_{kl} = -(\beta T + p)\,\delta_{kl} + 2\mu_e \tilde{e}_{kl} \tag{8.4.13}$$

$$q_k = \kappa T_{,k} \tag{8.4.14}$$

$$\rho_0\varepsilon = S_0 + \rho_0\eta_0 T_0 + \rho_0\gamma\left(T + \frac{T^2}{2T_0}\right) - 2\mu_e \mathrm{II}_e \tag{8.4.15}$$

$$\eta = \eta_0 + \frac{\gamma}{T_0} T \tag{8.4.16}$$

Kinematical Relations

$$\dot{v}_l \cong \frac{\partial^2 u_l}{\partial t^2} \tag{8.4.17}$$

$$\tilde{e}_{kl} \cong \tfrac{1}{2}(u_{k,l} + u_{l,k}) \tag{8.4.18}$$

$$\mathrm{I}_e \equiv \tilde{e}_{kk}, \qquad \mathrm{II}_e \cong \tfrac{1}{2}(\tilde{e}_{kk}\,\tilde{e}_{ll} - \tilde{e}_{kl}\,\tilde{e}_{lk}) \tag{8.4.19}$$

where u_l is the displacement vector.

Field Equations. The field equations of thermoelasticity consist of the equations of motion and the equation of heat transfer. The former is obtained by substituting (8.4.9) (for compressible solids) or (8.4.13) (for incompressible solids) into (8.4.2). The latter was already given by (8.3.14) for compressible solids and (8.3.22) for incompressible solids. Hence,

$$(\lambda_e + \mu_e)u_{l,lk} + \mu_e u_{k,ll} - (\beta T)_{,k} + \rho(f_k - \dot{u}_k) = 0 \qquad \text{in } \mathscr{V} \tag{8.4.20}$$

$$-\rho_0\gamma\dot{T} - \beta T_0 \dot{u}_{k,k} + (\kappa T_{,k})_{,k} + \rho h = 0 \qquad \text{(compressible)} \tag{8.4.21}$$

$$\mu_e u_{k,ll} - (\beta T + p)_{,k} + \rho_0(f_k - \dot{u}_k) = 0 \tag{8.4.22}$$

$$u_{k,k} = 0 \qquad \text{in } \mathscr{V}$$

$$-\rho_0\gamma\dot{T} + (\kappa T_{,k})_{,k} + \rho_0 h = 0 \qquad \text{(incompressible)} \tag{8.4.23}$$

Boundary Conditions

(8.4.24) $$t_{kl}n_l = \bar{t}_k \quad \text{on} \quad \mathscr{S}_t$$

(8.4.25) $$u_k = \bar{u}_k \quad \text{on} \quad \mathscr{S}_u = \mathscr{S} - \mathscr{S}_t$$

(8.4.26) $$\mathbf{q} \cdot \mathbf{n} = q_{(n)} \quad \text{on} \quad \mathscr{S}_q$$

(8.4.27) $$T = T_s \quad \text{on} \quad \mathscr{S}_T = \mathscr{S} - \mathscr{S}_q$$

where $\bar{t}_k$ is the surface traction prescribed on $\mathscr{S}_t$ of the surface $\mathscr{S} \equiv \mathscr{S}_t + \mathscr{S}_u$ and $\bar{u}_k$ is the displacement field prescribed on the remaining part $\mathscr{S}_u$ not overlapping $\mathscr{S}_t$.

Similarly, $q_{(n)}$ is prescribed on a part $\mathscr{S}_q$ of the total surface $\mathscr{S}$, and T_s is prescribed on the remainder $\mathscr{S}_T$ of the surface not overlapping $\mathscr{S}_q$. Other mixed combinations of the boundary conditions are possible. These must, of course, represent the physical conditions realistically, and together with the initial conditions must not violate the uniqueness theorem of the theory.

Another important condition, replacing any one of (8.4.26) and (8.4.27), is concerned with heat radiation from a part $\mathscr{S}_r$ of the surface $\mathscr{S}$. In this case, we have

(8.4.28) $$\mathbf{q} \cdot \mathbf{n} + a(T - T_1) = 0$$

where $a \geq 0$ is an appropriate function of the surface coordinates and T_1 is the temperature outside of the body near $\mathscr{S}_r$.

Initial Conditions

(8.4.29) $$u_k(\mathbf{x}, 0) = u_k^0(\mathbf{x})$$

(8.4.30) $$\dot{u}_k(\mathbf{x}, 0) = v_k^0(\mathbf{x})$$

(8.4.31) $$T(\mathbf{x}, 0) = T^0(\mathbf{x})$$

where $\mathbf{u}^0$, $\mathbf{v}^0$ and T^0 are prescribed functions.

It could be said that the entire theory of linear isotropic thermoelastic solids is based on the solutions of the *Navier-Duhamel* equations (8.4.20) and (8.4.21), for compressible solids, or (8.4.22) and (8.4.23), for incompressible solids, subject to some boundary and initial conditions of the types given above.

For anisotropic solids, the field equations (8.4.20) and (8.4.21) are replaced, respectively, by

(8.4.32) $$\sigma_{klmn}u_{m,nk} - (\beta_{kl}T)_{,k} + \rho(f_l - \ddot{u}_l) = 0$$

(8.4.33) $$-\rho_0\gamma\dot{T} - \beta_{kl}T_0\dot{u}_{k,l} + (b_{kl}T_{,l})_{,k} + \rho h = 0$$

For various material symmetry conditions the number of constitutive coefficients σ_{klmn}, β_{kl} and b_{kl} are reduced from their maximum number $21 + 6 + 6 = 33$ to the smallest number (5 constants) valid for isotropic bodies. These symmetry restrictions are applied in the manner presented in Section 6.2.

The form of the above equations in curvilinear coordinates is obtained by mere transfer of various expressions from Section 6.6.

8.5. HEAT CONDUCTION IN SOLIDS, DUHAMEL'S THEOREM

According to (8.3.22) the temperature $T(\mathbf{x}, t)$ is determined at a point $\mathbf{x}$ of a rigid isotropic solid $\mathscr{V} + \mathscr{S}$ at time t by the solution of the partial differential equation

$$\dot{T} = \frac{1}{\rho_0 \gamma} (\kappa T_{,k})_{,k} + \frac{h}{\gamma} \tag{8.5.1}$$

subject to some boundary conditions on $\mathscr{S}$

$$T = T_s(\mathbf{x}, t) \quad \text{on} \quad \mathscr{S}_T \tag{8.5.2}$$

$$q_{(n)} = \kappa T_{,k} n_k = q_s(\mathbf{x}, t) \quad \text{on} \quad \mathscr{S}_q \equiv \mathscr{S} - \mathscr{S}_T \tag{8.5.3}$$

for all $t > 0$ and the initial condition

$$T(\mathbf{x}, 0) = T^0(\mathbf{x}) \quad \text{in} \quad \mathscr{V} \tag{8.5.4}$$

An important mixed boundary value problem is encountered when radiation takes place in a medium having temperature T_1 near the surface. In this case the boundary conditions may be expressed as (8.4.28)

$$\kappa T_{,k} n_k + a(T - T_1) = 0 \quad \text{on} \quad \mathscr{S}_r \tag{8.5.5}$$

where $a \geq 0$ and $\mathscr{S}_r$ is that part of the total surface on the remainder of which either (8.5.2) or (8.5.3), or both, are specified. In this case, of course, $\mathscr{S}_T + \mathscr{S}_q + \mathscr{S}_r = \mathscr{S}$. We note that if we consider only a linear dependence on temperature for h, that is,

$$\frac{h}{\gamma} = H(\mathbf{x})T + F(\mathbf{x}, t) \tag{8.5.6}$$

then all of the boundary and initial value problems expressed above can be put into the compact form

$$\begin{aligned} &\dot{T}(\mathbf{x}, t) = L\{T\} + F(\mathbf{x}, t), \qquad \mathbf{x} \text{ in } \mathscr{V}, \qquad t > 0 \\ &l\{T(\mathbf{x}', t)\} = G(\mathbf{x}', t), \qquad \mathbf{x}' \text{ on } \mathscr{S}, \qquad t > 0 \\ &T(\mathbf{x}, 0) = T^0(\mathbf{x}), \qquad \mathbf{x} \text{ in } \mathscr{V}, \qquad t = 0 \end{aligned} \tag{8.5.7}$$

where the linear operators L and l are defined by

$$
\begin{aligned}
L\{T\} &\equiv \frac{1}{\rho_0 \gamma}(\kappa T_{,k})_{,k} + H(\mathbf{x})T \\
l\{T\} &\equiv c_0(\mathbf{x})T + c_k(\mathbf{x})T_{,k} \qquad (k = 1, 2, 3)
\end{aligned} \tag{8.5.8}
$$

where c_0, c_1, c_2, c_3 are selected to suit any of the foregoing problems, and $G(\mathbf{x}', t)$ is a given function.

Once the temperature is found, the heat vector $\mathbf{q}$ is then determined from Fourier's law of heat conduction

$$\mathbf{q} = \kappa \nabla T \tag{8.5.9}$$

Before giving some specific solutions, we present a theorem of great value to the solution of boundary value problems in heat conduction. In fact we will reduce the solution of the boundary value problem (8.5.7) with variable source and boundary conditions to the following problem in which the source and the surface conditions are constant.

$$
\begin{aligned}
\left(\frac{\partial}{\partial t} - L\right)\{\phi(\mathbf{x}, t, t')\} &= F(\mathbf{x}, t') \\
l\{\phi(\mathbf{x}, t, t')\} &= G(\mathbf{x}', t') \\
\phi(\mathbf{x}, 0, t') &= T^0(\mathbf{x})
\end{aligned} \tag{8.5.10}
$$

THEOREM (Duhamel). *The solution $T(\mathbf{x}, t)$ of the problem* (8.5.7) *with variable source and boundary and initial conditions is expressed in terms of the solution $\phi(\mathbf{x}, t, t')$ of the boundary value problem* (8.5.10) *with constant source and boundary conditions by*

$$T(\mathbf{x}, t) = \frac{\partial}{\partial t}\int_0^t \phi(\mathbf{x}, t - t', t')\, dt' \tag{8.5.11}$$

The proof of this theorem may be made by use of the laplace transform[1] defined by

$$\bar{\theta}(\mathbf{x}, s) = \int_0^\infty e^{-st}\theta(\mathbf{x}, t)\, dt \tag{8.5.12}$$

Multiplying both sides of (8.5.7) by e^{-st} and integrating with respect to t from 0 to ∞, we get

$$
\begin{aligned}
(s - L)\{\bar{T}(\mathbf{x}, s)\} &= T^0(\mathbf{x}) + \bar{F}(\mathbf{x}, s) \\
l\{\bar{T}(\mathbf{x}', s)\} &= \bar{G}(\mathbf{x}', s)
\end{aligned} \tag{8.5.13}
$$

[1] Bartels and Churchill [1942, p. 276]; see also Sneddon [1951, p. 163]. For an alternative approach, see Carslaw and Jaeger [1959, p. 70].

The same method applied to (8.5.10) gives

$$(s - L)\{\bar{\phi}(\mathbf{x}, s, t')\} = T^0(\mathbf{x}) + F(\mathbf{x}, t')s^{-1}$$
$$l\{\bar{\phi}(\mathbf{x}, s, t')\} = \bar{G}(\mathbf{x}', t')s^{-1} \tag{8 5 14}$$

Now multiply (8.5.14) with $e^{-st'}$ and integrate over t' from 0 to ∞. This gives

$$(s - L)\{\Phi(\mathbf{x}, s)\} = s^{-1}T^0(\mathbf{x}) + s^{-1}\bar{F}(\mathbf{x}, s)$$
$$l\{\Phi(\mathbf{x}', s)\} = s^{-1}\bar{G}(\mathbf{x}', s) \tag{8.5.15}$$

where

$$\Phi(\mathbf{x}, s) = \int_0^\infty e^{-st'}\bar{\phi}(\mathbf{x}, s, t')\,dt'$$
$$= \int_0^\infty \int_0^\infty e^{-s(t+t')}\phi(\mathbf{x}, t, t')\,dt\,dt' \tag{8.5.16}$$
$$= \int_0^\infty e^{-st}\left[\int_0^t \phi(\mathbf{x}, t - t', t')\,dt'\right]dt$$

Comparing (8 5.13) and (8.5.15) we see that these two sets are equivalent provided

$$\bar{T}(\mathbf{x}, s) = s\Phi(\mathbf{x}, s) \tag{8 5.17}$$

and the boundary value problem has a unique solution. It is clear from (8 5.16) that $\Phi(\mathbf{x}, s)$ is the laplace transform of the integral inside the bracket, so that $s\Phi(\mathbf{x}, s)$ is the laplace transform of the function

$$\frac{\partial}{\partial t}\int_0^t \phi(\mathbf{x}, t - t', t')\,dt'$$

From (8 5.17) and this there follows (8 5.11), hence the proof of the theorem. In the next section, we illustrate the use of this theorem.

8.6. TEMPERATURE DISTRIBUTIONS IN A HALF SPACE AND OUTSIDE A CAVITY

In this section we consider three simple problems of heat conduction

(*i*) ***Half Space with Constant Boundary Temperature.*** In this case we have a uniformly applied constant temperature on the boundary $x = 0$ of a half space, $x > 0$ (Figure 8.6 I). The boundary and initial conditions are

$$T(x, t) = T_0, \qquad x = 0, \qquad t \geq 0$$
$$T(x, t) = 0, \qquad t = 0, \qquad x > 0 \tag{8.6.1}$$

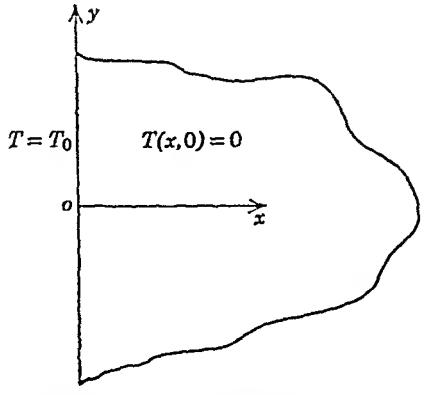


Fig. 8.6.1 Half space subject to constant temperature at $x = 0$.

Since everything is uniform on the parallel planes $x =$ constant, we have $T = T(x, t)$. Thus, the heat conduction equation (8.5.1) for an homogeneous medium with no heat source reduces to

$$\frac{\partial T}{\partial t} = \nu \frac{\partial^2 T}{\partial x^2}, \qquad \nu \equiv \frac{\kappa}{\rho_0 \gamma} \tag{8.6.2}$$

A solution of this equation can be found by considering

$$T = f(\zeta), \qquad \zeta \equiv \frac{x^2}{4\nu t} \tag{8.6.3}$$

Substituting this in (8.6.2) we get

$$\zeta f'' + (\zeta + \tfrac{1}{2}) f' = 0$$

whose solution is:

$$f(\zeta) = A_1 \int_0^{\zeta} \frac{e^{-\eta}}{\eta^{1/2}}\, d\eta + B_1$$

where A_1 and B_1 arc constant. By a change of variables

$$\zeta \equiv \xi^2 \equiv \frac{x^2}{4\nu t} \tag{8.6.4}$$

this results as

$$T(x, t) = A \operatorname{erf} \xi + B \tag{8.6.5}$$

where the error function, erf ξ, is defined by

$$\operatorname{erf} \xi = \frac{2}{\sqrt{\pi}} \int_0^{\xi} e^{-u^2}\, du \tag{8.6.6}$$

This function occurs frequently in heat-conduction problems. A few of its important properties are:

$$\text{(8.6.7)} \qquad \operatorname{erf} 0 = 0, \qquad \operatorname{erf} \infty = 1, \qquad \operatorname{erf}(-\xi) = -\operatorname{erf} \xi$$

With these properties we can find A and B such that equations (8.6.1) are satisfied

$$T(0, t) = 0 + B = T_0$$

$$T(x, 0) = A + B = 0$$

Thus,

$$\text{(8.6.8)} \qquad T(x, t) = T_0(1 - \operatorname{erf} \xi), \qquad \xi \equiv \frac{x}{2\sqrt{\nu t}}$$

The variation of temperature with ξ is shown in Figure 8.6 2.

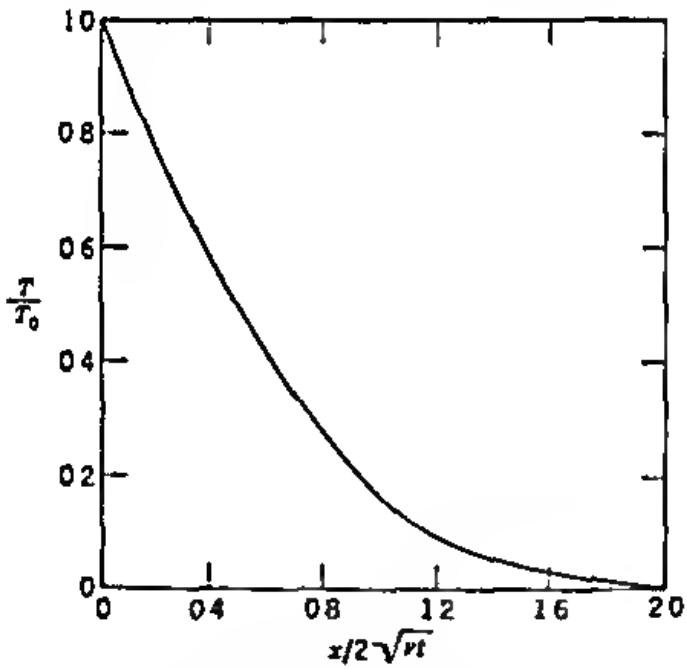


Fig. 8.6.2 Temperature distribution in half space.

(ii) Spherical Cavity Subject to a Uniform Temperature. The temperature distribution outside of a spherical cavity $r > a$ subject to a uniform temperature T_0 at $x = a$ and zero initial temperature throughout $r > a$ is determined by the solution of the following boundary value problem.

$$\text{(8.6 9)} \qquad \begin{aligned} T(r, t) &= T_0, \qquad r = a, \quad t \geq 0 \\ T(r, t) &= 0, \qquad t = 0, \quad r > a \end{aligned}$$

The heat conduction equation (8.5.1) for a homogeneous medium with

spherical symmetry and vanishing heat source reduces to

$$\frac{\partial T}{\partial t} = \nu \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial T}{\partial r}\right) \tag{8.6.10}$$

Let $T = \phi/r$; ϕ now satisfies

$$\frac{\partial \phi}{\partial t} = \nu \frac{\partial^2 \phi}{\partial r^2}$$

which is identical to (8.6.2). Hence it has a solution

$$T(r, t) = \frac{1}{r}(A \operatorname{erf} \xi + B) \tag{8.6.11}$$

where this time $\xi = (r - a)/2\sqrt{\nu t}$. The condition (8.6.9) is satisfied by choosing $-A = B = aT_0$. Thus

$$T(r, t) = \frac{aT_0}{r}(1 - \operatorname{erf} \xi), \qquad \xi \equiv \frac{r - a}{2\sqrt{\nu t}} \tag{8.6.12}$$

(iii) Half Space or Spherical Cavity Subject to Time-Dependent Temperature. The foregoing results may be employed to give solutions of those problems in which the boundary of the half space or the spherical cavity is subject to a time-dependent temperature, while the medium is initially under zero temperature, that is,

$$\begin{aligned} T(x, t) &= \phi(t), \quad x = 0, \quad t \geq 0 \\ T(x, t) &= 0, \qquad t = 0, \quad x > 0 \end{aligned} \tag{8.6.13}$$

for the half space. For the spherical cavity x is replaced by $r - a$.

If t' is a fixed parameter, then from (8.6.7) it is clear that the solution of (8.5.1) which satisfies the boundary conditions

$$\begin{aligned} T &= \phi(t'), \qquad x = 0, \qquad t > 0 \\ T(x, 0) &= 0, \qquad x > 0 \end{aligned}$$

is

$$T = \frac{2}{\sqrt{\pi}} \phi(t') \int_{x/2\sqrt{\nu t}}^{\infty} e^{-u^2}\, du$$

From Duhamel's theorem it follows that the solution of the heat equation satisfying (8.6.13) is

$$\begin{aligned} T(x, t) &= \frac{2}{\sqrt{\pi}} \frac{\partial}{\partial t} \int_0^t \phi(t')\, dt' \int_{x/2\sqrt{\nu t - \nu t'}}^{\infty} e^{-u^2}\, du \\ &= \frac{x}{2\sqrt{\pi \nu}} \int_0^t \phi(t') \frac{\exp[-x^2/4\nu(t - t')]}{(t - t')^{3/2}}\, dt' \end{aligned}$$

By the change of variable

$$t' = t - \frac{x^2}{4v\xi^2}$$

we can convert this result to

$$T(x,t) = \frac{2}{\sqrt{\pi}} \int_{x/2\sqrt{vt}}^{\infty} \phi\left(t - \frac{x^2}{4v\xi^2}\right) e^{-\xi^2}\, d\xi \tag{8.6.14}$$

For a case of the spherical cavity, we replace x by $r - a$ in this equation and multiply it by a/r.

If we take $\phi(t') = T_0 H(t')$ where $H(t')$ is the Heaviside function[2] in (8.6.14), we get back the solution (8 6.8) which shows that a *suddenly applied* temperature T_0 on the boundary is felt immediately thereafter at all points x in the interior. Thus, the heat conduction *does not have a finite propagation velocity* as in the case of elastic waves All material points receive the applied heat signal immediately. *Heat expands with infinite velocity*, a point of view that must be challenged from the viewpoint of relativity since no velocity may exceed the velocity of light c. Some argument in favor of this contradiction may be provided by the physical reason that heat conduction is of statistical, not dynamical, origin.[3]

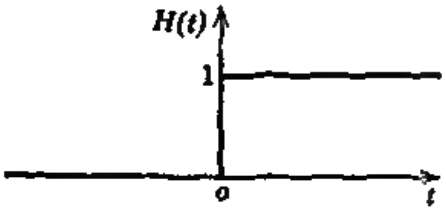


Fig. 8.6.3 Heaviside unit step function.

8.7. THERMOELASTIC WAVES

The motion and temperature of a linear, isotropic, homogeneous thermoelastic solid are governed by the Navier-Duhamel equations (8.4.20) and (8.4.21), namely

$$(\lambda_e + \mu_e)u_{l,lk} + \mu_e u_{k,ll} - \beta T_{,k} + \rho(f_k - \ddot{u}_k) = 0 \tag{8.7.1}$$

$$-\kappa T_{,kk} - \rho h + T_0 \beta \ddot{u}_{k,k} + \rho_0 \gamma \dot{T} = 0 \tag{8.7.2}$$

where the constitutive constants λ_e, μ_e, β, κ, and γ are subject to

$$0 < 3\lambda_e + 2\mu_e < \infty, \qquad 0 < \mu_e < \infty, \qquad 0 \leq \kappa, \beta, \gamma < \infty \tag{8.7.3}$$

[2] The Heaviside function, $H(t)$, is defined as 1 for $t > 0$ and 0, $t < 0$, and has a jump 1 at $t = 0$. See Figure 8 6 3.

[3] Sommerfeld [1949, p. 60].

For an incompressible solid, these equations are replaced by (8.4.22) and (8.4.23). The boundary conditions may be of the following type

$$t_{ik}n_i = \bar{t}_k \quad \text{on} \quad \mathscr{S}_t \tag{8.7.4}$$

$$u_k = \bar{u}_k \quad \text{on} \quad \mathscr{S}_u \equiv \mathscr{S} - \mathscr{S}_t$$

$$T = T_s \quad \text{on} \quad \mathscr{S}_T \tag{8.7.5}$$

$$\mathbf{q}\cdot\mathbf{n} = q_{(n)} \quad \text{on} \quad \mathscr{S}_q \equiv \mathscr{S} - \mathscr{S}_T$$

for all $t > 0$ and the initial conditions

$$u_k(\mathrm{x}, 0) = u_k{}^0(\mathrm{x})$$

$$\dot{u}_k(\mathrm{x}, 0) = v_k{}^0(\mathrm{x}) \quad \text{in} \quad \mathscr{V} \tag{8.7.6}$$

$$T(\mathrm{x}, 0) = T^0(\mathrm{x})$$

for $t = 0$. Of course, other more complicated boundary conditions are possible. The stress tensor and the heat vector are related to the displacement and temperature by the Hooke-Fourier constitutive equations

$$t_{kl} = -\beta T\,\delta_{kl} + \lambda_c u_{m,m}\,\delta_{kl} + \mu_c(u_{k,l} + u_{l,k}) \tag{8.7.7}$$

$$q_k = \kappa T_{,k} \tag{8.7.8}$$

For some treatments, the vectorial forms of (8.7.1) and (8.7.2) are convenient and are obtained by using identities following (6.6.14) in the above equations. Thus

$$(\lambda_c + 2\mu_c)\nabla(\nabla\cdot\mathbf{u}) - \mu_c\,\nabla\times(\nabla\times\mathbf{u}) - \beta\,\nabla T + \rho(\mathbf{f} - \ddot{\mathbf{u}}) = 0 \tag{8.7.9}$$

$$-\kappa\nabla^2 T - \rho h + T_0\beta\,\nabla\cdot\dot{\mathbf{u}} + \rho_0\gamma\dot{T} = 0 \tag{8.7.10}$$

where ∇ is the gradient operator and ∇^2 is the laplacian operator.

We may decompose the displacement field and body-force field in the same way as in Section 6.10 into

$$\mathbf{u} = \nabla\phi + \nabla\times\psi, \qquad \mathbf{f} = -\nabla g - \nabla\times\mathbf{h}; \qquad \nabla\cdot\psi = \nabla\cdot\mathbf{h} = 0 \tag{8.7.11}$$

Upon substituting these into (8.7.9) and (8.7.10) we see that they are satisfied if

$$c_1^2\,\nabla^2\phi - \frac{\beta}{\rho_0}T - g - \ddot{\phi} = 0 \tag{8.7.12}$$

$$c_2^2\,\nabla\times\nabla\times\psi + \mathbf{h} + \ddot{\psi} = 0 \tag{8.7.13}$$

$$-\kappa\,\nabla^2 T - \rho_0 h + T_0\beta\,\nabla^2\dot{\phi} + \rho_0\gamma\dot{T} = 0 \tag{8.7.14}$$

where for the linear theory $\rho = \rho_0$ and

$$c_1 \equiv \left(\frac{\lambda_c + 2\mu_c}{\rho_0}\right)^{1/2}, \qquad c_2 \equiv (\mu_c/\rho_0)^{1/2} \tag{8.7.15}$$

are, respectively, the velocities of irrotational and equivoluminal isothermal elastic waves. The above equations are valid in curvilinear coordinates In Section 6 10 we learned that the scalar and vector potential ϕ and ψ represent, respectively, the irrotational and equivoluminal isothermal elastic waves Thus, for example, when $\phi = 0$ the waves are distortional (shear waves) and the volume does not change, and when $\psi = 0$ the waves are irrotational, changing the volume From (8.7.13) it is clear that thermal changes do not affect ψ so that *the shear waves are uncoupled from the thermal effects.* On the other hand, coupling occurs between ϕ and T so that *heat conduction not only affects the volume change but also the volume change produces heat.*

In a thermally coupled elastic solid ($\beta \neq 0$) we see that the equivoluminal wave velocity c_2 is unaffected. However, in general, there is no wave propagating with velocity c_1 when $\beta \neq 0$

Upon combining (8.7.12) and (8 7 14) we see that ϕ and T independently satisfy the following fourth order partial differential equations

$$\left(\frac{\partial}{\partial t} - \frac{\kappa}{\rho_0 \gamma} \nabla^2\right)(\dot{\phi} - c_1^2 \nabla^2 \phi + g) - \frac{\beta^2 T_0}{\rho_0^2 \gamma} \nabla^2 \dot{\phi} + \frac{\beta}{\rho_0 \gamma} h = 0 \tag{8.7.16}$$

$$\left(\frac{\partial}{\partial t} - \frac{\kappa}{\rho_0 \gamma} \nabla^2\right)(\dot{T} - c_1^2 \nabla^2 T) - \frac{1}{\gamma}(\ddot{h} - c_1^2 \nabla^2 h) - \frac{T_0 \beta^2}{\rho_0^2 \gamma} \nabla^2\left(\dot{T} + \frac{\rho_0}{\beta} \dot{g}\right) = 0 \tag{8.7.17}$$

For vanishing body force and heat source both displacement potential ϕ and temperature field T satisfy the same field equation

$$\left(\frac{\partial}{\partial t} - \frac{\kappa}{\rho_0 \gamma} \nabla^2\right)\left(\frac{\partial^2}{\partial t^2} - c_1^2 \nabla^2\right)\{\phi, T\} - \frac{\beta^2 T_0}{\rho_0^2 \gamma} \frac{\partial}{\partial t} \nabla^2 \{\phi, T\} = 0 \tag{8.7.18}$$

The coupling of equivoluminal waves with the temperature field will in general occur through the boundary conditions. However, the field equations for ψ are uncoupled and, therefore, in the study of wave propagation in an infinite medium there is no need to give any solutions of (8.7.13) again, for elastic waves in an infinite medium satisfy the same equation We therefore turn our attention to the coupled equations (8.7.12) and (8 7 14) for ϕ and T.

A plane harmonic wave has the general form

$$\{\phi, T\} = \{\phi_0, \theta_0\} \exp[i(k \nu_m x_m - \omega t)], \qquad i \equiv \sqrt{-1} \tag{8.7.19}$$

where ϕ_0, θ_0, k and ω are, in general, complex quantities The wavelength λ and frequency f are, in general, given by

$$\lambda = 2\pi / Re(k), \qquad f = Re(\omega / 2\pi) \tag{8.7.20}$$

By selecting λ or ω real, we may study the propagation of waves of a given wavelength or a given frequency. As we saw in Section 6.10, ν_m is a unit vector in the direction of propagation of the plane wave, that is,

$$\nu_m \nu_m = 1 \tag{8.7.21}$$

By selecting the direction of the x-axis in the direction of propagation (that is, $\nu_1 = 1$, $\nu_2 = \nu_3 = 0$) without loss in generality we try a solution of the form[4]

$$\{\phi, T\} = \{\phi_0, \theta_0\} \exp[i(kx - \omega t)] \tag{8.7.22}$$

where we wrote $x = x_1$. Upon substituting this into (8.7.12) and (8.7.14) with $g = h = 0$, we obtain two equations

$$\begin{aligned} (\omega^2 - c_1^2 k^2)\phi_0 - \frac{\beta}{\rho_0}\theta_0 &= 0 \\ iT_0\beta k^2 \omega \phi_0 + (\kappa k^2 - i\rho_0 \gamma \omega)\theta_0 &= 0 \end{aligned} \tag{8.7.23}$$

The necessary condition for a nontrivial solution of this set is the vanishing of the coefficient determinant

$$(K^2 - \Omega^2)(\Omega + iK^2) + \varepsilon \Omega K^2 = 0 \tag{8.7.24}$$

where we set

$$\begin{aligned} K &\equiv \frac{\kappa}{\rho_0 \gamma c_1} k, \qquad \Omega \equiv \frac{\kappa}{\rho_0 \gamma c_1^2} \omega \\ \varepsilon &\equiv \frac{\beta^2 T_0}{\rho_0^2 \gamma c_1^2} \end{aligned} \tag{8.7.25}$$

The roots of (8.7.24) for K are $\pm K_1$ and $\pm K_2$, where

$$\begin{aligned} K_1 &= \tfrac{1}{2}\sqrt{\Omega}\,\{[\Omega + (1+i)\sqrt{2\Omega} + i(1+\varepsilon)]^{1/2} \\ &\quad + [\Omega - (1+i)\sqrt{2\Omega} + i(1+\varepsilon)]^{1/2}\} \\ K_2 &= \tfrac{1}{2}\sqrt{\Omega}\,\{[\Omega + (1+i)\sqrt{2\Omega} + i(1+\varepsilon)]^{1/2} \\ &\quad - [\Omega - (1+i)\sqrt{2\Omega} + i(1+\varepsilon)]^{1/2}\} \end{aligned} \tag{8.7.26}$$

We distinguish two cases:

(i) *Uncoupled Waves.* If we take $\varepsilon = 0$ either in (8.7.26) or in (8.7.24), we

[4] Propagation of plane thermoelastic waves in infinite media was studied by Deresiewicz [1957] and by Chadwick and Sneddon [1958]. For other expositions see Boley and Weiner [1960], Nowacki [1962, Ch. V], and Sneddon and Hill [1960, Ch. VI].

find that for real Ω the phase velocities of thermoelastic waves are given by

$$c = \frac{\omega}{Rek_1} = \pm c_1, \qquad c_T = \frac{\omega}{Rek_2} = \pm(2\kappa\omega/\rho_0\gamma)^{1/2} \tag{8.7.27}$$

In this case, the temperature field is uncoupled from the displacement field The usual irrotational elastic waves propagate in positive and negative directions of the x-axis and the phase velocity is c_1. In addition we have a new wave, called the *thermal wave*, which propagates in the same direction with a phase velocity c_T given by $(8.7.27)_2$. Since the phase velocity of this wave depends on the frequency, thermal waves are said to be *dispersive*. The displacement potential and the temperature field, for the case of real ω, are given by

$$\begin{aligned}
\phi &= \phi_0^+ \exp\left[-i\omega\left(t - \frac{x}{c_1}\right)\right] \\
&\quad + \phi_0^- \exp\left[-i\omega\left(t + \frac{x}{c_1}\right)\right] \\
T &= \theta_0^+ \exp\left[-\frac{\omega x}{c_T} - i\omega\left(t - \frac{x}{c_T}\right)\right] \\
&\quad + \theta_0^- \exp\left[\frac{\omega x}{c_T} - i\omega\left(t + \frac{x}{c_T}\right)\right]
\end{aligned} \tag{8.7.28}$$

We see here also that the amplitude of thermal waves changes with distance. For any ω, the solutions for the uncoupled waves are

$$\begin{aligned}
\phi &= \phi_0^+ \exp\left[ik(x - c_1 t)\right] + \phi_0^- \exp\left[ik(x + c_1 t)\right] \\
T &= \theta_0 \exp\left[-\frac{\kappa}{\rho_0\gamma} k^2 t + ikx\right]
\end{aligned} \tag{8.7.29}$$

In this case the elastic waves have the same character, however, the temperature distribution now takes the form of a standing wave, its amplitude decaying in time.

(ii) Coupled Waves. In this case $\varepsilon \neq 0$, and we must employ the results (8.7.26) in our calculations. If, for an assigned real ω, we let

$$K_j = c_1\left(\frac{\Omega}{v_j} + i\frac{q_j}{\omega^*}\right), \qquad j = 1, 2 \tag{8.7.30}$$

where $c_1\Omega/v_j$ and $c_1 q_j/\omega^*$ are the real and imaginary parts of K_1 and K_2 and

$$\omega^* \equiv \frac{\rho_0\gamma c_1^2}{\kappa} \tag{8.7.31}$$

then the plane wave solution of the thermoelastic waves may be expressed as[5]

$$\begin{aligned}\phi = {} & \phi_0^+ \exp\left[-q_1 x - i\omega^*\Omega\left(t - \frac{x}{v_1}\right)\right] \\ & + \phi_0^- \exp\left[q_1 x - i\omega^*\Omega\left(t + \frac{x}{v_1}\right)\right] \\ & + \frac{\beta}{\rho_0}\frac{v_2^2}{(\omega^*\Omega v_2)^2 - c_1^2(\Omega\omega^* + iq_2 v_2)^2} \\ & \times \left\{\theta_0^+ \exp\left[-q_2 x - i\omega^*\Omega\left(t - \frac{x}{v_2}\right)\right]\right. \\ & \left. + \theta_0^- \exp\left[q_2 x - i\omega^*\Omega\left(t + \frac{x}{v_2}\right)\right]\right\}\end{aligned}$$

(8.7.32)

$$\begin{aligned}T = {} & \frac{\rho_0}{\beta v_1^2}\left[(v_1\omega^*\Omega)^2 - c_1^2(\Omega\omega^* + iq_1 v_1)^2\right] \\ & \times \left\{\phi_0^+ \exp\left[-q_1 x - i\omega^*\Omega\left(t - \frac{x}{v_1}\right)\right]\right. \\ & \left. + \phi_0^- \exp\left[q_1 x - i\omega^*\Omega\left(t + \frac{x}{v_1}\right)\right]\right\} \\ & + \theta_0^+ \exp\left[-q_2 x - i\omega^*\Omega\left(t - \frac{x}{v_1}\right)\right] \\ & + \theta_0^- \exp\left[q_2 x - i\omega^*\Omega\left(t + \frac{x}{v_2}\right)\right]\end{aligned}$$

From (8.7.32), there follows the displacement field by taking the partial derivative of ϕ with respect to x. These equations indicate that there exist two types of waves propagating with velocities v_1 and v_2. These waves are dispersive since v_1 and v_2 are functions of ω; and they attenuate according to the values of q_1 and q_2. For small values of ε the effect of thermal coupling can be studied by expanding v_1, v_2, q_1, and q_2 into a power series of ε. For these results we refer the reader to Chadwick's article in Sneddon and Hill [1960].

Waves of an assigned wavelength may be studied in a similar fashion by considering K real and determining the roots (8.7.24). In this case the equation is a cubic in Ω. While the solution of the cubic is tedious the roots

[5] Chadwick [1960].

of this equation can be determined easily for small ε by the usual perturbation technique. We do not wish to pursue this subject further except that we wish to give some idea of the magnitude of the effect of thermal coupling upon the wave velocity and upon the attenuation of the wave amplitude. To this end, define

$$q_\infty = \lim_{\Omega \to \infty} q_1 = \varepsilon\omega^*/2c_1 \qquad \text{for real } \Omega \tag{8 7.33}$$

then the plots of q_1/q_∞ and v_1/c_1 as functions of Ω are shown in Figures 8.7.1 and 8.7.2 and tabulated in Table 8.7.1

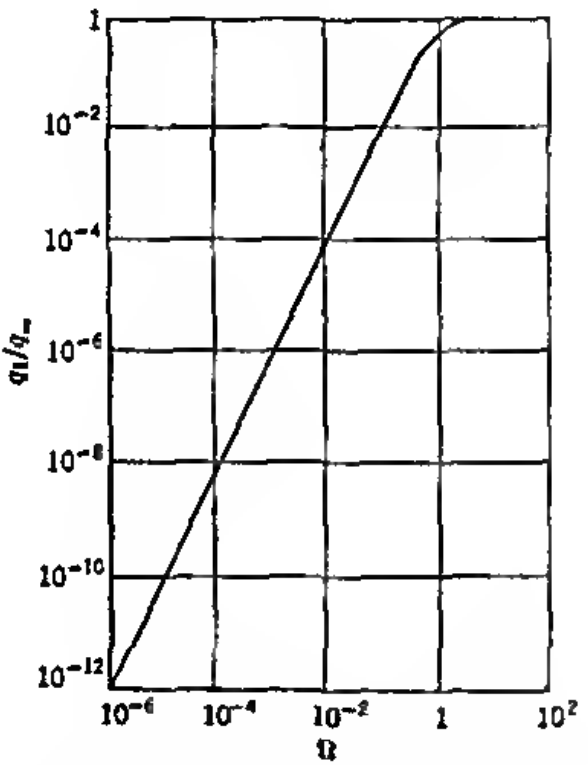


Fig. 8.7.1 Attenuation coefficient as a function of Ω *in copper at 20°C (after Chadwick* [1960]).

The range of frequencies that can be attained in an elastic solid is limited above by the Debye cutoff frequency [Brillouin 1938, p. 324],

$$\omega_c \equiv 2\pi c_2(3\rho_0/4\pi M)^{1/3} \tag{8.7.34}$$

where M is the mass of a constituent atom. In Table 8.7.2 we give for various metals c_1, ε, ω^*, q_∞ and ω_c. It is clear from this table that ω^* is a hundred times smaller than ω_c. Thus the ceiling attainable in an elastic solid is imposed not by the atomic nature of the materials but the thermal damping This table also shows that the coefficient of thermoelastic coupling is extremely small so that for most mechanical problems the

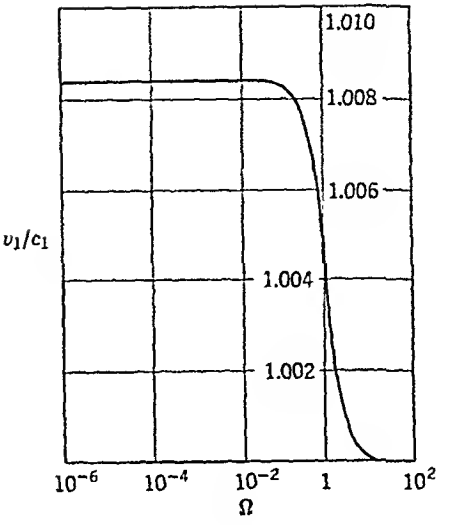


Fig. 8.7.2 Wave velocity as a function of Ω *in copper at* 20°C *(after Chadwick* [1960]).

Table 8.7.1. Variation of wave velocity and attenuation coefficient with frequency in copper at 20°C (after Chadwick [1960])

Ω	q_1/q_∞	v_1/c_1	Ω	q_1/q_∞	v_1/c_1
10^{-2}	0.0001	1.0084	3	0.9033	1.0009
10^{-1}	0.0095	1.0083	7	0.9822	1.0002
0.3	0.0800	1.0077	10	0.9921	1.0001
0.7	0.3250	1.0057	10^{2}	1.0000	1.0000
1	0.4987	1.0043			

Table 8.7.2. Thermoelastic coefficients for metals at 20°C (after Chadwick and Sneddon [1958])

Quantity	Units	Aluminum	Copper	Iron	Lead
c_1	cm/sec	6.32×10^{5}	4.36×10^{5}	5.80×10^{5}	2.14×10^{5}
ε	—	3.56×10^{-2}	1.68×10^{-2}	2.97×10^{-4}	7.33×10^{-2}
ω^*	sec^{-1}	4.66×10^{11}	1.73×10^{12}	1.75×10^{12}	1.91×10^{11}
q_∞	cm^{-1}	1.31×10^{4}	3.29×10^{3}	4.48×10^{2}	3.27×10^{4}
ω_c	sec^{-1}	9.80×10^{13}	7.55×10^{13}	9.95×10^{13}	3.69×10^{13}

uncoupled thermoelastic equations or a perturbation in ε of these equations is adequate for the treatment of thermal stress problems. The full coupling may, however, need to be considered in electronic equipment.

8.8. SPHERICAL CAVITY SUBJECT TO DYNAMICAL PRESSURE AND TEMPERATURE

Here we determine the displacement and temperature fields u and T produced in a thermoelastic infinite solid occupying the space outside a spherical cavity that is subjected to a dynamical pressure and temperature at its surface. The appropriate coordinate system is spherical coordinates r, θ, ϕ, with $r = 0$ being the center of the cavity and $r = a$ being its surface Since the disturbance is radially symmetric and g, h, and h are zero, we see that u_θ and u_ϕ are zero and u_r is determined through

$$u_r = \frac{\partial \phi}{\partial r}, \qquad u_\theta = u_\phi = 0 \tag{8.8 1}$$

This also shows that $\psi \equiv 0$ and we only need to solve (8.7.12) and (8.7.14) to determine u_r and T. By setting

$$\phi = \psi/r, \qquad T = \theta/r$$
$$u_r = \frac{\partial}{\partial r}\left(\frac{\psi}{r}\right) \tag{8.8.2}$$

further simplification is achieved so that

$$c_1^2 \frac{\partial^2 \psi}{\partial r^2} - \frac{\beta}{\rho_0}\theta - \frac{\partial^2 \psi}{\partial t^2} = 0$$
$$-\kappa \frac{\partial^2 \theta}{\partial r^2} + T_0 \beta \frac{\partial^3 \psi}{\partial t \partial r^2} + \rho_0 \gamma \frac{\partial \theta}{\partial t} = 0 \tag{8.8.3}$$

The boundary conditions are

$$t_{rr}(a, t) = \left[(\lambda_e + 2\mu_e)\left(\frac{\psi}{r}\right)_{,rr} + 2\frac{\lambda_e}{r}\left(\frac{\psi}{r}\right)_{,r} - \frac{\beta\theta}{r}\right]_{r=a} = -p(t)$$
$$T(a, t) = \frac{\theta(a, t)}{a} = f(t) \tag{8.8.4}$$

and, of course, at $r = \infty$, $T = t_{rr} = 0$ and the waves must be advancing towards $r > a$.

The initial conditions are

$$\psi = \frac{\partial \psi}{\partial t} = \theta = 0 \quad \text{for} \quad r > a, \quad t = 0 \tag{8.8.5}$$

The problem is best attacked by the use of the laplace transform

$$\{\bar{\psi}, \bar{\theta}\} \equiv \int_0^\infty \{\psi, \theta\} e^{-st}\, dt \tag{8.8.6}$$

Upon the application of the laplace transform to (8.8.3), and using (8.8.5), we get

$$\begin{aligned} c_1^2 \frac{d^2\bar{\psi}}{dr^2} - s^2\bar{\psi} - \frac{\beta}{\rho_0}\bar{\theta} &= 0 \\ -\kappa \frac{d^2\bar{\theta}}{dr^2} + \rho_0 \gamma s \bar{\theta} + T_0 \beta s \frac{d^2\bar{\psi}}{dr^2} &= 0 \end{aligned} \tag{8.8.7}$$

This is a set of linear differential equations whose general solution is

$$\begin{aligned} \bar{\psi} &= A \exp(-\zeta_1 r) + B \exp(-\zeta_2 r) \\ \bar{\theta} &= \frac{\rho_0}{\beta} c_1^2 \left[A\left(\zeta_1^2 - \frac{s^2}{c_1^2}\right) \exp(-\zeta_1 r) + B\left(\zeta_2^2 - \frac{s^2}{c_1^2}\right) \exp(-\zeta_2 r) \right] \end{aligned} \tag{8.8.8}$$

where ζ_1 and ζ_2 are the roots of the following biquadratic equation having positive real parts.

$$\zeta^4 - \left[\frac{s^2}{c_1^2} + \frac{\rho_0 \gamma}{\kappa}(1 + \varepsilon)s \right] \zeta^2 + \frac{\rho_0 \gamma}{\kappa c_1^2} s^3 = 0, \quad \varepsilon \equiv \frac{T_0 \beta^2}{\gamma \rho_0^2 c_1^2} \geq 0 \tag{8.8.9}$$

This is the same frequency equation obtained before (8.7.24), if ζ and s are replaced by ik and $-i\omega$ respectively.

Upon carrying (8.8.8) into the boundary conditions (8.8.4) we obtain two linear equations for A and B whose solution is found to be

$$\begin{aligned} A &= \frac{\exp(\zeta_1 a)}{\Delta(s)} \left\{ \frac{a^3}{\rho_0} \left(\zeta_2^2 - \frac{s^2}{c_1^2} \right) \bar{p}(s) \right. \\ &\qquad \left. + \frac{a\beta}{\rho_0 c_1^2} [s^2 a^2 + 4c_2^2(a\zeta_2 + 1)] \bar{f}(s) \right\} \\ B &= -\frac{\exp(\zeta_2 a)}{\Delta(s)} \left\{ \frac{a^3}{\rho_0} \left(\zeta_1^2 - \frac{s^2}{c_1^2} \right) \bar{p}(s) \right. \\ &\qquad \left. + \frac{a\beta}{\rho_0 c_1^2} [s^2 a^2 + 4c_2^2(a\zeta_1 + 1)] \bar{f}(s) \right\} \end{aligned} \tag{8.8.10}$$

where

$$\Delta(s) \equiv (\zeta_1 - \zeta_2)\left[(\zeta_1 + \zeta_2)(s^2a^2 + 4c_2^2) + 4ac_2^2\left(\zeta_1\zeta_2 + \frac{s^2}{c_1^2}\right)\right] \tag{8.8.11}$$

Equations (8.8.8) to (8.8.11) give the complete operational solution of the problem under consideration. The inversions of $\bar{\psi}$ and $\bar{\theta}$ are given by the integral

$$\{\psi(r, t), \theta(r, t)\} = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \{\bar{\psi}(r, s), \bar{\theta}(r, s)\} \exp(st)\, ds \tag{8.8.12}$$

where the contour $\operatorname{Re} s = c$ lies to the right of all singularities of the integrand in the complex s-plane The function $\bar{\psi}$ and $\bar{\theta}$ possess branch points at $s/\omega^* = 1 - \varepsilon \pm 2i\sqrt{\varepsilon}$, a characteristic which is fundamental to the operational solution of all thermoelastic boundary value problems Evaluation of the integral in (8 8 12) is a formidable task and has not been achieved to date Approximate solutions, however, exist,[6] based on a perturbation expansion about $\varepsilon = 0$ and for short-time behavior. Thus, in the first instance, by expanding $\bar{\psi}$ and $\bar{\theta}$ into a series

$$\begin{aligned} \bar{\psi} &= \bar{\psi}_0 + \varepsilon\bar{\psi}_1 + O(\varepsilon^2) \\ \bar{\theta} &= \bar{\theta}_0 + \varepsilon\bar{\theta}_1 + O(\varepsilon^2) \end{aligned} \tag{8 8.13}$$

where $\bar{\psi}_0$ and $\bar{\psi}_1$, etc , are independent of ε, one can easily see that ψ_0 is the solution of the problem in which the thermal field uncoupled (that is, the second term in $(8\ 8.3)_2$ has been neglected) and that ψ_1 is the first-order correction due to this coupling. For the ease of $f(t) = 0$, ψ_0 is the solution of isothermal elastic waves emanating from the dynamical pressure applied on the spherical cavity. This problem has already been treated in Section 6.12. Hence, we do not repeat this result. Chadwick in the reference cited above gives an approximate result for $\psi_1(a, t)$ at the cavity surface, for a pressure of the type $p(t) = p_0H(t)$ where $H(t)$ is the Heaviside unit function We refrain from reproducing this result here. In the same reference there is also given an asymptotic solution for the short time behavior of ψ.

For the case of $\varepsilon = 0$ but $f(t) \neq 0$ the solution set (ψ_0, θ_0) of the uncoupled dynamical problem may be obtained by inverting the integrals in (8.8.12). This solution is tantamount to neglecting the heat produced by the local volume change The fact that this effect is minutely small has been known for some time.[7] The problem with this simplification may be treated

[6] Cf. the article by Chadwick in Sneddon and Hill [1960]

[7] For the estimate of this effect see Boley and Weiner [1960, Art 2 2]

for the case of

$$p(t) = 0, \qquad T(a, t) = f(t) = T_a H(t) \tag{8.8.14}$$

In this case, the temperature field $T(r, t)$ can be obtained immediately. Indeed this has already been done for the corresponding problem in heat conduction as can be seen from the following: For $\varepsilon = 0$ from (8.8.9) we have the required roots

$$\zeta_1 = \frac{s}{c_1}, \qquad \zeta_2 = \left(\frac{\rho_0 \gamma}{\kappa} s\right)^{1/2} \tag{8.8.15}$$

with these (8.8.8) takes the form

$$\bar{\psi} = \frac{a\beta}{\rho_0 c_1^2 s} \frac{T_a}{(s/c_1)^2 - (\rho_0\gamma/\kappa)s} \left\{ \frac{a^2 s^2 + 4c_2^2 \left[a\left(\frac{\rho_0\gamma}{\kappa} s\right)^{1/2} + 1 \right]}{a^2 s^2 + 4c_2^2 a(s/c_1) + 4c_2^2} \right.$$

$$\left. \cdot \exp\left[-\frac{s}{c_1}(r-a) \right] - \exp\left[-\left(\frac{\rho_0\gamma s}{\kappa}\right)^{1/2} (r-a) \right] \right\} \tag{8.8.16}$$

$$\bar{\theta} = \frac{aT_a}{s} \exp\left[-\left(\frac{\rho_0\gamma}{\kappa} s\right)^{1/2} (r-a) \right]$$

The inverse of $\bar{\theta}$ is immediate and it gives the result already obtained in Section 8.6, that is,

$$\frac{T}{T_a} = \frac{a}{r} \operatorname{erfc}\left(\frac{r-a}{\sqrt{4vt}} \right) \tag{8.8.17}$$

where

$$v \equiv \frac{\kappa}{\rho_0 \gamma}, \qquad \operatorname{erfc}(x) \equiv 1 - \operatorname{erf}(x) \tag{8.8.18}$$

The inverse of ψ is somewhat more elaborate. The problem for this case was treated by Katasanov [1957] and later by Sternberg and Chakravorty [1959]. The resulting displacement field, by using (8.8.2), is given by

$$u_r(R, t) = \frac{\beta}{\rho_0 c_1^2} a T_a \{ V(R, \tau) + W(R, \tau) H[\tau - m(R-1)] \} \tag{8.8.19}$$

where

$$V(R, \tau) \equiv \frac{1}{2R^2} \left[(R^2 - 2\tau - 1 - 2m^2) \operatorname{erfc}(\xi) - 2(R+1) \right.$$

$$\sqrt{\frac{\tau}{\pi}}\exp(-\xi^2)+m^2\exp\left(\frac{\tau}{m^2}+\frac{R-1}{m}\right)\operatorname{erfc}\left(\xi+\frac{\sqrt{\tau}}{m}\right)$$

$$+m^2\exp\left(\frac{\tau}{m^2}-\frac{R-1}{m}\right)\operatorname{erfc}\left(\xi-\frac{\sqrt{\tau}}{m}\right)$$

$$-mR\exp\left(\frac{\tau}{m^2}+\frac{R-1}{m}\right)\operatorname{erfc}\left(\xi+\frac{\sqrt{\tau}}{m}\right)$$

$$+mR\exp\left(\frac{\tau}{m^2}-\frac{R-1}{m}\right)\operatorname{erfc}\left(\xi-\frac{\sqrt{\tau}}{m}\right)\Bigg]$$

(8.8.20)

$$W(R,t)=-\frac{m}{R^2}(m+R)\exp\left(\frac{\omega}{m^2}\right)-\frac{A}{K^2}\,Re\Big\{F(R)$$

$$\cdot\left[m^3\exp\left(\frac{\omega}{m^2}\right)\operatorname{erfc}\left(\frac{\sqrt{\omega}}{m}\right)-K^{-\frac{1}{2}}\exp(K\omega)\right.$$

$$\left.\cdot\operatorname{erfc}(\sqrt{K\omega})+B\exp(K\omega)+C\left(\omega+2\sqrt{\frac{\omega}{\pi}}\right)+D\right]\Big\}$$

$$R=r/a,\qquad \tau=v\frac{t}{a^2},\qquad \xi=\frac{R-1}{2\sqrt{\tau}}$$

$$m=\frac{v}{ac_1},\qquad \omega=\tau-m(R-1)$$

$$K=\frac{1}{m}(-p+iq),\qquad p=\frac{1-2\nu_e}{1-\nu_e},\qquad q=\frac{\sqrt{1-2\nu_e}}{1-\nu_e}$$

$$A=\frac{2p}{mq(1+2mp+2m^2p)},\qquad B=\frac{m^2}{2p}+K^{-2}+K^{-\frac{1}{2}}$$

$$C=m^2-K^{-1},\qquad D=m^4-K^{-2}$$

$$F(R)=-q(m+R)+i[1+p(m-2mR-R)]$$

and ν_e is Poisson's ratio. For $\sqrt{K}$ the principal value is to be taken. The stresses are now determined through

$$t_{rr}=-\beta T+(\lambda_e+2\mu_e)\frac{\partial u}{\partial r}+2\lambda_e\frac{u}{r}$$

(8.8.21)

$$t_{\theta\theta}=t_{\varphi\varphi}=-\beta T+\lambda_e\frac{\partial u}{\partial r}+2(\lambda_e+\mu_e)\frac{u}{r}$$

$$t_{r\theta}=t_{r\varphi}=t_{\theta\varphi}=0$$

We do not reproduce the expressions for the stresses here.

According to (8.8.19) the displacement field contains a step loading for $m > 0$, represented by the term containing the Heaviside step function H. These terms for $m > 0$ indicate that a spherical shock wave starts at the boundary of the cavity at $t = 0+$, and propagates outward with the velocity c_1 of irrotational waves. The remaining terms are diffusive in character and, as in all heat-conduction problems, are received instantaneously throughout the medium. The displacement field is continuous in R and τ for $1 < R < \infty$, $0 < \tau < \infty$ as it should be since the axiom of continuity of the material must be satisfied. Time and space derivatives of the displacement field and stresses suffer discontinuities, however, at the *wave front* $\tau = \tau^* = m(R - 1)$. At the wave front the jump in the stresses can be shown to be

$$t_{rr}(R, \tau^*+) - t_{rr}(R, \tau^*-) = \frac{\beta T_a}{R}$$

(8.8.22)

$$t_{\varphi\varphi}(R, \tau^*+) - t_{\varphi\varphi}(R, \tau^*-) = \frac{\nu_e}{1 - \nu_e} \frac{\beta T_a}{R}$$

which are independent of m. The jump discontinuity decreases with distance.

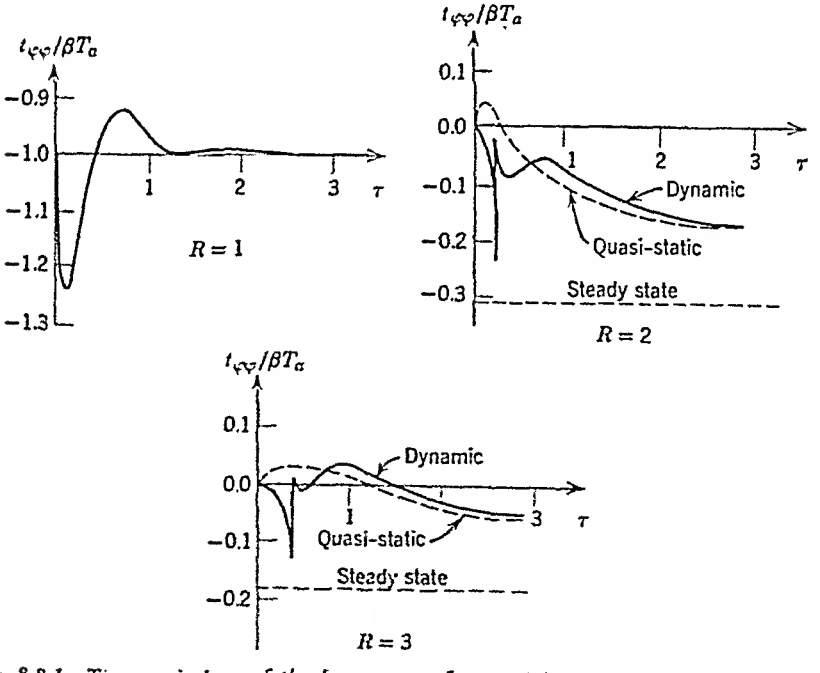


Fig. 8.8.1 Time variations of the hoop stress for $\nu_e = 0.25$, $m = 0.20$ (after Sternberg and Chakravorty [1959]).

By letting $m \to 0$, we obtain the solution of the quasistatic problem in which the inertia terms are neglected. On the other hand, by letting $\tau \to \infty$, one obtains the solution for the stationary problem. The stress component $t_{\varphi\varphi}$ for a Poisson material $\nu_e = 0.25$ and for $m = 0.20$ have beeen plotted against τ for three values of $R = 1, 2, 3$ (Figure 8.8 1).

PROBLEMS

8.1. Find he constitutive equations of isotropic thermoelastic solids which include second-degree terms in the strain and temperature gradients Give the equation of heat conduction

8.2. Find the constitutive equations of linear orthotropic thermoelastic solids

8.3. Show that

$$T(x,t) = \frac{1}{8(\pi Kt)^{3/2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y_1,y_2,y_3) \exp\left[-(x_k - y_k)(x_k - y_k)/4Kt\right] dy_1 dy_2 dy_3$$

is the solution of the heat equation for an infinite rigid solid. Find the heat energy. What is the value of this energy at time, $t = 0$?

8.4. Determine the stress and displacement fields in an infinitely long, thick, circular tube that is free of surface tractions. The cylinder is subject to a steady temperature distribution, $T = T(r)$

8.5. Find the stress and displacement fields in a thick, spherical shell free of tractions. The shell is subject to a centrally symmetric (radial) temperature distribution.

8.6. Prove that in a thermoelastic solid free of surface tractions, when the temperature is steady, the total volume change is given by

$$\Delta V = \int_{\mathcal{V}} \frac{3\beta}{3\lambda_e + 2\mu_e} T \, dv$$

where β is the coefficient of thermal expansion (see Eq 8.4.9). For a hollow sphere having concentric surfaces, $r = a, b$, calculate the volume of the cavity when the temperature rise is $T(x, y, z)$.

8.7. An elastic, isotropic, cylindrical shell of outer radius, r_0, is inserted inside of another one having the same inner radius. The cylinders are made of different materials. The temperatures of the inner and outer shells are respectively $T_0(r)$ and $T_i(r)$. The contacting surface of the two media is assumed to keep contact and not to slide on each other. Determine the stress and displacement fields

8.8. *In a rigid slab, the temperature at time, $t = 0$, is prescribed as a function of x.* Both ends of the bar are insulated and the heat conduction is one-dimensional and in the x-direction. Determine the temperature distribution for all times, $t > 0$

8.9. A rigid, infinite, circular cylinder is subjected to a constant temperature, T_0, at its outer surface. Initially the cylinder is at zero temperature. Determine the temperature distribution.

8.10. A rigid sphere, initially, is at zero temperature. The sphere is subjected to a constant temperature jump at its surface. Determine the temperature distribution.

8.11. Two ends of an elastic bar are placed on immovable supports. The bar is suddenly heated to a constant uniform temperature. Determine the stresses.

8.12. (Short Term Paper) Formulate the problem of thermoelastic flexure for bars.

8.13. (Short Term Paper) Formulate the problem of thermoelastic flexure for thin plates.

8.14. (Short Term Paper) Study literature and give an estimate of the order of magnitude of the dilatational heat production term, $\beta T_0 \dot{u}_{k,k}$ (see Eq. 8.3.14).

8.15. (Short Term Paper) A plane wave is reflected from the flat boundary of isotropic, thermoelastic half space. Determine the angle of reflection and the amplitudes of various types of reflected waves.

8.16. (Short Term Paper) Discuss Rayleigh surface waves for an isotropic, thermoelastic half space.

CHAPTER 9

Viscoelasticity

9.1. SCOPE OF THE CHAPTER

This chapter takes up the study of more complicated types of media, namely, viscoelastic materials The distinction between fluids and solids is arbitrary and often not very sharp All solids under the influence of external loads not only deform but also flow to a certain extent and dissipate energy Fluent bodies also possess a certain amount of rigidity Moreover, deformation and flow produce heat, and conversely, the addition of heat to a body causes deformation and flow. Thus physical bodies possess the characteristics of both solids and fluids.

The theories of viscoelasticity initiated by Maxwell [1867], Meyer [1874a,b, 1875], and Boltzmann [1874], and studied by Voigt [1889], [1892a,b, 1910], Kelvin [1875], Duhem [1903a,b,c,d,e,f, 1904], Natanson [1901a,b], Zaremba [1903a,b,c 1937], Volterra [1930], and others at the turn of the century remained dormant for nearly fifty years With advances in technology and demands from the solid propellant industry the interests of theoretical workers have been stimulated once again The result has been the recent increase in literature with works at all levels of sophistication.

The status of nonlinear theories up to 1961 was presented by Eringen [1962 Ch. 10]. An assessment of the fundamental advances since then was also made recently by Eringen and Grot [1965]. On the linear theories we may cite expositions by Hunter [1960], Gurtin and Sternberg [1962], and Sternberg [1964] Various volumes edited by Eirich [1956, 1958], Staverman and Schwarzl [1956], and Eringen, Liebowitz, Crowley, and

Koh [1967] contain comprehensive accounts of recent theoretical and experimental research.

With apologies to many authors whose important works could not be quoted here, we shall proceed to give an elementary exposition of the theory.

The first eleven sections of this chapter are devoted to the derivation and discussion of the constitutive equations of various types of viscoelastic materials, and the other six sections, to illustrative problems. The linear theory is best assessed by first understanding the more exact nonlinear theory. For this reason, we decided to give an elementary exposition of the nonlinear theory (in Section 9.2) in the simplest instance—Kelvin-Voigt solids. Section 9.3 is devoted to the linear theory of Kelvin-Voigt solids, both anisotropic and isotropic. In Section 9.4 we present a corresponding material for Maxwell solids. With the foundation so exposed, we are ready for the theory of general strain-rate dependent materials as presented in Section 9.5. Section 9.6 returns to the theory of functional constitutive equations, discussed in Chapter 5, and proceeds with a linear functional theory of viscoelasticity. This theory ultimately leads to the so-called Boltzmann-Volterra theory of linear viscoelasticity. The relationship between the rate and the functional theories is established in Section 9.7.

Experiments indicate that thermal effects are important in the physical behavior of viscoelastic materials. Therefore, we must ultimately bring the effect of temperature gradient into play. This is accomplished in Section 9.8 in the theory of thermoviscoelasticity. In Section 9.9 there is to be found a discussion of thermoviscoelastic fluids. In Section 9.10 we sketch the methods of determination of the memory functions characterizing isotropic viscoelastic materials. The fundamental equations of linear viscoelasticity are collected in Section 9.11.

Exact solutions of problems in viscoelasticity are handicapped with mathematical difficulties. The theorem of correspondence, presented in Section 9.12, often finds important applications in the solution of certain problems in linear viscoelasticity when the solution of the corresponding problem in elasticity is known. In Section 9.13 some quasi-static solutions are presented. These deal with the problems of half space and cylindrical tube subject to pressures. The dynamical problems of viscoelasticity are much more difficult to deal with. A discussion of the propagation of waves is presented in Section 9.14. In Section 9.15, we study in detail one-dimensional waves in Kelvin-Voigt and Maxwell solids. While these models are found not to be realistic, nevertheless they cast light onto the nature of waves in viscoelastic solids. Section 9.16 contains a discussion of a half space, and in Section 9.17, a spherical cavity subject to impulsive pressures

is treated. These discussions are a continuation of similar problems dealt with in previous chapters.

The theory of viscoelasticity is presently an evolving subject. The scope of this chapter is limited, in that it is basically intended for derivations of the basic equations of the theory. Some illustrative solutions are included to make the theory balanced and meaningful.

9.2. NONLINEAR KELVIN-VOIGT SOLID

One of the simplest rate-dependent solids is the Kelvin-Voigt solid The fundamental idea underlying this material is that the stress is dependent on the strain and deformation-rate tensors. For a one-dimensional linear theory

$$t = E\tilde{e} + E_v\dot{\tilde{e}}$$

where t is the tension in a bar, $\tilde{e}$ is the strain, and E and E_v are material constants. We see that this model is a combination of an elastic solid with a viscous fluid. It is often represented as a parallel spring-dashpot model as shown in Figure 9 2.1 with E being the spring constant and E_v the viscosity of the dashpot.

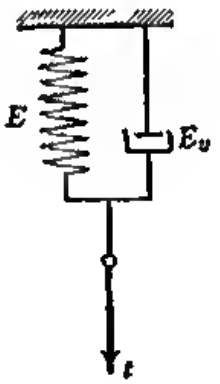


Fig. 9.2.1 Mechanical model for linear Kelvin-Voigt solid

A proper generalization of this idea to any nonlinear, three-dimensional solid begins with a constitutive equation of the form

$$t_{kl} = f_{kl}(\mathbf{x}_{,K}, \dot{\mathbf{x}}_{,K}, \theta, \mathbf{X})$$

and similar expressions for the heat vector $\mathbf{q}$, internal energy ε, and entropy η These are special forms of equations (5.4 22) which have been shown to

lead (5.4.23). In the present case, they have the explicit forms

$$t_{kl} = F_{KL}(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X})X_{K,k}X_{L,l} \tag{9.2.1}$$

$$q_k = Q_K(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X})X_{K,k} \tag{9.2.2}$$

$$\varepsilon = e(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X}) \tag{9.2.3}$$

$$\eta = n(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X}) \tag{9.2.4}$$

We have dropped the dependence on ρ^{-1} since this dependence is implied through the argument $\mathbf{C}$. Here $\mathbf{F}$ and $\mathbf{Q}$ are tensor and vector functions, respectively, and e and n are scalar functions of the tensor variables $\mathbf{C}$, $\dot{\mathbf{C}}$, the scalar θ, and the material position $\mathbf{X}$. The independent variables $\mathbf{C}$ and $\dot{\mathbf{C}}$ are the Green deformation tensor and its rate given by

$$\begin{aligned} C_{KL} &\equiv x_{k,K}x_{k,L} \\ \dot{C}_{KL} &\equiv \frac{DC_{KL}}{Dt} = 2\, d_{kl}x_{k,K}x_{l,L} \end{aligned} \tag{9.2.5}$$

where $\mathbf{d}$ is the deformation-rate tensor

$$d_{kl} \equiv \tfrac{1}{2}(v_{k,l} + v_{l,k}) \tag{9.2.6}$$

More convenient forms of (9.2.1) to (9.2.4) are obtained if we write

$$\begin{aligned} F_{KL} &= j^{-1}T_{MN}C_{MK}C_{NL} \\ Q_K &= j^{-1}G_L C_{LK}, \qquad j^{-1} = [\det(C_{KL})]^{-1/2} = \rho/\rho_0 > 0 \end{aligned} \tag{9.2.7}$$

Upon using (9.2.5) and (9.2.7) in (9.2.1) to (9.2.4) we get

$$t_{kl} = j^{-1}T_{KL}(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X})x_{k,K}x_{l,L} \tag{9.2.8}$$

$$q_k = j^{-1}G_K(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X})x_{k,K} \tag{9.2.9}$$

$$\varepsilon = e(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X}) \tag{9.2.10}$$

$$\eta = n(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X}) \tag{9.2.11}$$

where T_{KL} is a tensor function and G_K is a vector function of the tensor variables $\mathbf{C}$, $\dot{\mathbf{C}}$, and scalars θ and $\mathbf{X}$. We may now use these equations as the equations defining a Kelvin-Voigt solid.

DEFINITION 1. *A Kelvin-Voigt solid is a thermomechanical medium having constitutive equations of the form* (9.2.8) *to* (9.2.11) *subject to the axioms of admissibility, objectivity, and material invariance.*

Instead of this, we could have started with the more general form (5.4.22) if we had wished. Using the axiom of objectivity we, of course, end up with either the set (9.2.1)–(9.2.4) or, equivalently, (9.2.8)–(9.2.11).

The response functions **T**, **G**, e, and n are further restricted by the axioms of admissibility (requiring the thermodynamic admissibility), objectivity, and material invariance.

(*i*) ***Consequence of the Clausius–Duhem Inequality.*** According to the axiom of entropy, **t**, **q**, ε, and η must satisfy the inequality

$$-\frac{\rho}{\theta}(\dot{\psi}+\eta\dot{\theta})+\frac{1}{\theta}t_{kl}v_{l,k}+\frac{1}{\theta^2}q_k\theta_{,k}\geq 0 \tag{9.2.12}$$

for all independent processes. Here ψ is the Helmholtz free-energy function defined by

$$\psi \equiv \varepsilon - \theta\eta \tag{9.2.13}$$

Upon substituting (9.2.8) to (9.2.11) into (9.2.12) we get

$$-\frac{\rho_0}{\theta}\left(n+\frac{\partial\psi}{\partial\theta}\right)\dot{\theta}+\frac{1}{2\theta}\left(T_{KL}-2\rho_0\frac{\partial\psi}{\partial C_{KL}}\right)\dot{C}_{KL} \tag{9.2.14}$$
$$-\frac{\rho_0}{\theta}\frac{\partial\psi}{\partial\dot{C}_{KL}}\ddot{C}_{KL}+\frac{1}{\theta^2}G_K\theta_{,K}\geq 0$$

where we employed (9.2.5) and the conservation of mass $\rho_0 = \rho J$ ($J>0$), as given by $(9.2.7)_3$. The inequality (9.2.14) is postulated to be valid for all independent variations of $\dot{\theta}$, $\dot{C}_{KL}$, $\ddot{C}_{KL}$, and $\theta_{,K}$. Since the coefficients of $\dot{\theta}$, $\ddot{C}_{KL}$, $\theta_{,K}$, and other terms in (9.2.14) are independent of $\dot{\theta}$, $\ddot{C}_{KL}$, and $\theta_{,K}$, we deduce that the necessary and sufficient conditions for (9.2.14) to be valid for all possible values of these variables are

$$n=-\frac{\partial\psi}{\partial\theta},\qquad \frac{\partial\psi}{\partial\dot{C}_{KL}}=0,\qquad G_K=0$$

Consequently $\psi=\psi(\mathbf{C},\theta,\mathbf{X})$ and (9.2.14) reduces to

$$\frac{1}{2\theta}\left(T_{KL}-2\rho_0\frac{\partial\psi}{\partial C_{KL}}\right)\dot{C}_{KL}\geq 0$$

It is now suggested that T_{KL} may be expressed in two parts as

$$\mathbf{T}={}_E\mathbf{T}(\mathbf{C},\theta,\mathbf{X})+{}_D\mathbf{T}(\mathbf{C},\dot{\mathbf{C}},\theta,\mathbf{X})$$

where

$${}_ET_{KL}\equiv 2\rho_0\frac{\partial\psi}{\partial C_{KL}}$$

With this the above inequality becomes

$$\frac{1}{2\theta}\,{}_DT_{KL}\dot{C}_{KL}\geq 0$$

Since $C_{KL} = C_{LK}$, both ${}_E\mathbf{T}$ and ${}_D\mathbf{T}$ are symmetric tensors. Thus we have the result

THEOREM. *A Kelvin-Voigt solid is thermodynamically admissible if and only if it can be characterized by a set of constitutive equations of the form*

$$t_{kl} = 2\frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial C_{KL}} x_{k,K} x_{l,L} + \frac{\rho}{\rho_0} {}_D T_{KL}(\mathbf{C}, \dot{\mathbf{C}}, \theta, \mathbf{X}) x_{k,K} x_{l,L} \tag{9.2.15}$$

$$q_k = 0 \tag{9.2.16}$$

$$\varepsilon = \frac{1}{\rho_0}\left[\Sigma(\mathbf{C}, \theta, \mathbf{X}) - \theta\frac{\partial \Sigma}{\partial \theta}\right] \tag{9.2.17}$$

$$\eta = -\frac{1}{\rho_0}\frac{\partial \Sigma}{\partial \theta} \tag{9.2.18}$$

where

$$\frac{1}{\rho_0}\Sigma(\mathbf{C}, \theta, \mathbf{X}) \equiv \psi \tag{9.2.19}$$

is the Helmholtz free energy subject to[1]

$$2\frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial C_{KL}} x_{(k,K} x_{l),L} = 0 \tag{9.2.20}$$

and ${}_D\mathbf{T}$ is subject to

$$\frac{1}{2\theta} {}_D T_{KL} \dot{C}_{KL} \geq 0, \qquad {}_D T_{KL} = {}_D T_{LK} \tag{9.2.21}$$

for all independent variations of $\dot{\mathbf{C}}$.

The inequality (9.2.21), in particular, implies that

$${}_D\mathbf{T}(\mathbf{C}, \mathbf{0}, \theta, \mathbf{X}) = \mathbf{0} \tag{9.2.22}$$

provided ${}_D\mathbf{T}$ is continuous in $\dot{\mathbf{C}}$. To see this, let us set all $\dot{C}_{KL} = 0$ except $\dot{C}_{11}$, then by (9.2.21) we must have ${}_D T_{11}\dot{C}_{11}/2\theta \geq 0$. If $\dot{C}_{11} > 0$, then ${}_D T_{11} > 0$; if $\dot{C}_{11} < 0$, then ${}_D T_{11} < 0$. Thus ${}_D T_{11} = 0$ when $\dot{C}_{11} = 0$. This argument when applied to other components of $\dot{C}_{KL}$ leads to the proof of (9.2.22).

Several observations may be made:

(a) The stress tensor consists of a *purely elastic* part ${}_E\mathbf{T}$ and a *dissipative* part ${}_D\mathbf{T}$. The dissipative energy corresponding to ${}_D\mathbf{T}$, in accordance with $(9.2.21)_1$ must be nonnegative.

[1] Condition (9.2.20), which results from the symmetry of T_{KL} and t_{kl}, is an identity since $C_{KL} = C_{LK}$.

(b) The dissipative stress, according to (9.2.22) contains no purely elastic effect.

(c) The heat vector must vanish (9.2.16) This, of course, is the consequence of not including $\theta_{,K}$ into the arguments of the constitutive functions. When it is included, we obtain a thermoviscoelastic solid.

(d) The internal energy and entropy are independent of the rate variable $\dot{\mathbf{C}}$. Moreover, they are derivable from a potential (the Helmholtz free energy) This result is in accordance with the concepts in classical thermodynamics. Nevertheless, more general cases exist in which Σ may depend on rate variables,[2] cf. Section 9.5.

(ii) Material Symmetry Restrictions. According to the axiom of material invariance, the forms of ψ and ${}_D\mathbf{T}$ further depend on material symmetry The material symmetry is expressed in the form of the invariance of response functions under a group of transformations {S} of the material axes X. The group {S} is constructed to characterize the material symmetry present in a given material. Out of an infinite number of such transformation groups, twelve particular transformations turn out to be adequate for the description of material symmetries of thirty-two known crystal classes. An account of the subject of elastic materials is to be found in Green and Adkins [1960] For viscoelastic materials the literature does not seem to contain a similar reduction.[3] For a homogeneous material the dependence of ψ and ${}_D\mathbf{T}$ on X disappears, and for isotropic solids the forms of ψ and ${}_D\mathbf{T}$ are known from invariant theory (cf. Appendix B). For isotropic solids, ψ and ${}_D\mathbf{T}$ are form-invariant under the full group of orthogonal transformations of the material frame X, that is, they are, respectively, isotropic, scalar, and tensor functions of C. From App. B7 we have the result that ψ depends on the invariants of C only, or, equivalently, the invariants of $\overset{-1}{\mathbf{c}}$, that is,

$$\psi = \psi(\mathbf{I}_1, \mathbf{I}_2, \mathbf{I}_3, \theta) \tag{9.2.23}$$

where

$$\mathbf{I}_1 \equiv \operatorname{tr} \overset{-1}{\mathbf{c}}, \qquad \mathbf{I}_2 \equiv \operatorname{tr} (\overset{-1}{\mathbf{c}})^2, \qquad \mathbf{I}_3 \equiv \operatorname{tr} (\overset{-1}{\mathbf{c}})^3 \tag{9.2.24}$$

$$\overset{-1}{c}_{kl} \equiv x_{k,K} x_{l,K} \tag{9.2.25}$$

[2] For a discussion of this point, see Eringen and Grot [1965, Art 5 and Appendix]

[3] By use of known results in the invariant theory (Smith and Rivlin [1964] and Spencer [1966]), it is possible to construct the constitutive equations under all symmetry regulations.

Substituting (9.2.23) into (9.2.15) and using the chain rule of differentiation, we obtain

$$\mathbf{t} = \alpha_0 \mathbf{I} + \alpha_1 \overset{-1}{\mathbf{c}} + \alpha_2 (\overset{-1}{\mathbf{c}})^2 + {}_D\mathbf{t}(\mathbf{c}, \mathbf{d}, \theta) \tag{9.2.26}$$

where

$$\begin{aligned} \alpha_0 &\equiv \rho\ (I_1^3 + 2I_3 - 3I_1 I_2) \frac{\partial \psi}{\partial I_3} \\ \alpha_1 &\equiv 2\rho \frac{\partial \psi}{\partial I_1} - 3\rho(I_1^2 - I_2) \frac{\partial \psi}{\partial I_3} \\ \alpha_2 &\equiv 4\rho \frac{\partial \psi}{\partial I_2} + 6\rho I_1 \frac{\partial \psi}{\partial I_3} \end{aligned} \tag{9.2.27}$$

Further simplification is achieved if we assume that ${}_D\mathbf{T}$ is a polynomial in the matrices $\overset{-1}{\mathbf{c}}$ and $\mathbf{d}$. In this case according to (App. B9.7), we can write

$$\begin{aligned} {}_D\mathbf{t} = {} & \kappa_0 \mathbf{I} + \kappa_1 \overset{-1}{\mathbf{c}} + \kappa_2 \overset{-2}{\mathbf{c}} + \kappa_3 \mathbf{d} + \kappa_4 \mathbf{d}^2 \\ & + \kappa_5 (\overset{-1}{\mathbf{c}}\,\mathbf{d} + \mathbf{d}\,\overset{-1}{\mathbf{c}}) + \kappa_6 (\overset{-2}{\mathbf{c}}\,\mathbf{d} + \mathbf{d}\,\overset{-2}{\mathbf{c}}) \\ & + \kappa_7 (\overset{-1}{\mathbf{c}}\,\mathbf{d}^2 + \mathbf{d}^2\,\overset{-1}{\mathbf{c}}) + \kappa_8 (\overset{-2}{\mathbf{c}}\,\mathbf{d}^2 + \mathbf{d}^2\,\overset{-2}{\mathbf{c}}) \end{aligned} \tag{9.2.28}$$

where κ_0 to κ_8 are polynomials in the joint invariants of $\overset{-1}{\mathbf{c}}$ and $\mathbf{d}$, that is,

$$\begin{aligned} & I_1, I_2, I_3, \operatorname{tr} \mathbf{d}, \operatorname{tr} \mathbf{d}^2, \operatorname{tr} \mathbf{d}^3 \\ & \operatorname{tr} \overset{-1}{\mathbf{c}}\mathbf{d}, \operatorname{tr} \overset{-1}{\mathbf{c}}\mathbf{d}^2, \operatorname{tr} \overset{-2}{\mathbf{c}}\mathbf{d}, \operatorname{tr} \overset{-2}{\mathbf{c}}\mathbf{d}^2 \end{aligned} \tag{9.2.29}$$

and

$$\kappa_0 = \kappa_1 = \kappa_2 = 0 \quad \text{when} \quad \mathbf{d} = \mathbf{0} \tag{9.2.30}$$

In these equations we have written $\overset{-2}{\mathbf{c}}$ for $(\overset{-1}{\mathbf{c}})^2$.

9.3. LINEAR KELVIN-VOIGT SOLID

The theory given in Section 9.2 is too complicated for most applications. It does, however, provide a source from which various other theories may be constructed. The most common, practical theories are obtained by polynomial expansions of the coefficient functions in (9.2.15) or (9.2.26) and retaining terms up to a certain degree in whatever strain and deformation rate measures are used.

For the general anisotropic solid it is appropriate to use the Lagrangian strain measure E defined by

$$2E_{KL} \equiv C_{KL} - \delta_{KL} \tag{9.3.1}$$

For the linear theory it is appropriate to employ the infinitesimal strain tensor $\tilde{E}$ in place of E so that

$$2E_{KL} \cong 2\tilde{E}_{KL} = U_{K,L} + U_{L,K} \tag{9.3.2}$$

where U_K is the displacement vector. We also need the expression

$$\begin{aligned} x_{k,K} &= (\delta_{MK} + U_{M,K})\,\delta_{Mk} \\ &= (\delta_{MK} + \tilde{E}_{MK} + \tilde{R}_{MK})\,\delta_{Mk} \end{aligned} \tag{9.3.3}$$

A linear theory for the stress requires a second-order expansion of Σ in E. Hence, writing

$$\Sigma(\mathrm{E}, \theta) \equiv \rho_0\psi(\mathrm{E}, \theta) = \Sigma_0(\theta) + \Sigma_{KL}(\theta)E_{KL} + \tfrac{1}{2}\Sigma_{KLMN}(\theta)E_{KL}E_{MN} \tag{9.3.4}$$

we obtain

$${}_E T_{KL}(\mathrm{E}, \theta) = \frac{\partial \Sigma}{\partial E_{KL}} = \Sigma_{KL}(\theta) + \Sigma_{KLMN}(\theta)E_{MN} \tag{9.3.5}$$

The linear theory for ${}_D\mathrm{T}$ requires an expansion of the form

$${}_D T_{KL}(\dot{\mathrm{E}}, \theta) = \Gamma_{KLMN}(\theta)\dot{E}_{MN} \tag{9.3.6}$$

In these equations, the constitutive coefficients Σ_{KL}, Σ_{KLMN}, and Γ_{KLMN} may depend on the temperature θ. Because of the symmetry of the stress and strain tensors we also have

$$\begin{aligned} \Sigma_{KL} &= \Sigma_{LK} \\ \Sigma_{KLMN} &= \Sigma_{MNKL} = \Sigma_{LKMN} = \Sigma_{KLNM} \\ \Gamma_{KLMN} &= \Gamma_{LKMN} = \Gamma_{KLNM} \end{aligned} \tag{9.3.7}$$

so that the number of independent coefficients for Σ_{KL} is 6, for Σ_{KLMN} 21, and for Γ_{KLMN} is 36.[3]

If the reference state is unstressed, then we also have

$$\Sigma_{KL} = 0 \tag{9.3.8}$$

[3] If it is also assumed that the part of stress ${}_D T_{KL}$ that does not contribute to dissipation is zero, then $\Gamma_{KLMN} = \Gamma_{MNKL}$. This further reduces the number of Γ_{KLMN} to 21. This assumption is tantamount to the use of the Onsager principles of linear irreversible thermodynamics.

In the spirit of the linear theory, upon writing $\tilde{\mathbf{E}}$ for $\mathbf{E}$ and using (9.3.3), we find that the stress constitutive equations (9.2.15) take the form

$$t_{kl} = \frac{\rho}{\rho_0}[\Sigma_{KL} + \Sigma_{ML}(\tilde{E}_{KM} + \tilde{R}_{KM}) + \Sigma_{KM}(\tilde{E}_{LM} + \tilde{R}_{LM}) + \Sigma_{KLMN}\tilde{E}_{MN} + \Gamma_{KLMN}\dot{\tilde{E}}_{MN}]\,\delta_{Kk}\,\delta_{Ll} \tag{9.3.9}$$

In the linear theory, we have the approximations (8.2.15) so that (9.3.9) may be written (in the Eulerian representation) as

$$t_{kl} = (1 - \tilde{e}_{mm})\sigma_{kl} + \sigma_{ml}(\tilde{e}_{km} + \tilde{r}_{km}) + \sigma_{km}(\tilde{e}_{lm} + \tilde{r}_{lm}) + \sigma_{klmn}\tilde{e}_{mn} + \gamma_{klmn}\dot{\tilde{e}}_{mn} \tag{9.3.10}$$

where

$$\begin{aligned} \sigma_{kl} &\equiv \Sigma_{KL}\,\delta_{Kk}\,\delta_{Ll} \\ \sigma_{klmn} &\equiv \Sigma_{KLMN}\,\delta_{Kk}\,\delta_{Ll}\,\delta_{Mm}\,\delta_{Nn} \\ \gamma_{klmn} &\equiv \Gamma_{KLMN}\,\delta_{Kk}\,\delta_{Ll}\,\delta_{Mm}\,\delta_{Nn} \end{aligned} \tag{9.3.11}$$

are functions of θ and $\mathbf{X}$ and satisfy the symmetry conditions

$$\begin{aligned} &\sigma_{kl} = \sigma_{lk}, \qquad \sigma_{klmn} = \sigma_{mnkl} = \sigma_{lkmn} = \sigma_{klnm} \\ &\gamma_{klmn} = \gamma_{lkmn} = \gamma_{klnm} \end{aligned} \tag{9.3.12}$$

Further, the inequality (9.2.21) requires the dissipative energy to be non-negative, that is,

$$\gamma_{klmn}\dot{\tilde{e}}_{kl}\dot{\tilde{e}}_{mn} \geq 0 \tag{9.3.13}$$

This then places restrictions on the viscosities γ_{klmn}, namely that they *are positive semi-definite*. The expressions for ε and η can be found easily in the same fashion. Since there will be no need for these in the treatment of nonthermal viscoelasticity we do not give their expressions.

Isotropic Solids. Material symmetry places further restrictions on the constitutive coefficients σ_{kl}, σ_{klmn}, and γ_{klmn} of the anisotropic solid. These restrictions are studied in the same manner as in Section 6.2. Here we give the results for isotropic solids. In this case the constitutive coefficients are isotropic tensors, so that we may write

$$\begin{aligned} \Sigma_{KL} &= \alpha_E\,\delta_{KL} \\ \Sigma_{KLMN} &= \lambda_E\,\delta_{KL}\,\delta_{MN} + \mu_E(\delta_{KM}\,\delta_{LN} + \delta_{KN}\,\delta_{LM}) \\ \Gamma_{KLMN} &= \lambda_v\,\delta_{KL}\,\delta_{MN} + \mu_v(\delta_{KM}\,\delta_{LN} + \delta_{KN}\,\delta_{LM}) \end{aligned} \tag{9.3.14}$$

where α_E, λ_E, μ_E, λ_v, and μ_v are functions of θ and $\mathbf{X}$ in general. With this,

the stress potential (9.3.4) and the stress constitutive equation (9.3.10) take, respectively, the forms

$$\Sigma \equiv \rho_0\psi = \Sigma_0(\theta) + \alpha_e \mathrm{I}_\varepsilon + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_\varepsilon^2 - 2\mu_e \mathrm{II}_\varepsilon \tag{9.3.15}$$

$$t_{kl} = \alpha_e\,\delta_{kl} + (\lambda_e - \alpha_e)\mathrm{I}_\varepsilon\,\delta_{kl} + 2(\mu_e - \alpha_e)\tilde{e}_{kl} + \lambda_v \dot{\mathrm{I}}_\varepsilon\,\delta_{kl} + 2\mu_v \dot{\tilde{e}}_{kl} \tag{9.3.16}$$

where we also wrote

$$\alpha_e \equiv \alpha_E, \qquad \lambda_e \equiv \lambda_E, \qquad \mu_e \equiv \mu_E + 2\alpha_E \tag{9.3.17}$$

We note that the expression for Σ (9 3.15) is identical to (6.2 30) obtained in Section 6.2 while that for the stress to (6.2.27) except for the rate terms involving λ_v and μ_v.

When the natural state is *stress free* we have $\alpha_e = 0$.

For the *incompressible solid*, we set $\mathrm{I}_\varepsilon = 0$ and add an unknown pressure term $-p\,\delta_{kl}$ to the right of t_{kl}. Hence

$$\Sigma = \rho_0\psi = -2\mu_e \mathrm{II}_\varepsilon \tag{9.3.18}$$

(incompressible)

$$t_{kl} = -p\,\delta_{kl} + 2\mu_e\tilde{e}_{kl} + 2\mu_v\dot{\tilde{e}}_{kl} \tag{9.3.19}$$

9.4. MAXWELL SOLID

A different model for a class of viscoelastic solids was introduced by Maxwell [1867], and involves the material time rate of the stress. For the one-dimensional linear theory the stress constitutive equation of such a solid is expressed as

$$\dot{t} + kt = E\dot{e}$$

Such a solid is represented by a linear spring and dashpot attached in series (Figure 9.4 1).

Various nonlinear generalizations of the above equation exist, and an account of the subject together with various historical references is to be found in Eringen [1962, Ch 10] Here we discuss one of the simplest cases. The theory of Maxwell solids is obtained as a special case of the general theory expressed by (5 3.12) for $p = 1$ in the form

$$\dot{\mathbf{t}}(\mathbf{X}, t) = \mathbf{f}(\mathbf{t}, \dot{\mathbf{x}}_{,\mathbf{x}}, \theta, \mathbf{X}) \tag{9 4.1}$$

The axioms of constitution place restrictions on (9.4.1). In particular, according to the axiom of objectivity, in a frame of reference $\bar{\mathbf{x}}$ differing

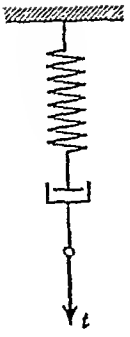


Fig. 9.4.1 Mechanical model for linear Maxwell solid.

from x by a rigid motion, the form of (9.4.1) should remain the same, that is,

$$\dot{\bar{\mathbf{t}}} = \mathbf{f}(\bar{\mathbf{t}}, \dot{\bar{\mathbf{x}}}_{,k}, \theta, \mathbf{X}) \tag{9.4.2}$$

where

$$\bar{\mathbf{t}} = \mathbf{Q}\mathbf{t}\mathbf{Q}^T, \qquad \bar{\mathbf{x}} = \mathbf{Q}\mathbf{x} + \mathbf{b} \tag{9.4.3}$$

with $\mathbf{Q}(t)$ being an arbitrary time-dependent rotation of the spatial frame x satisfying (5.3.15) and $\mathbf{b}(t)$ is a translation. Substituting (9.4.3) into (9.4.2) we get the relation

(9.4.4)

$$\mathbf{Q}\mathbf{f}(\mathbf{t}, \dot{\mathbf{x}}_{,k}, \theta, \mathbf{X})\mathbf{Q}^T + \dot{\mathbf{Q}}\mathbf{f}\mathbf{Q}^T + \mathbf{Q}\mathbf{f}\dot{\mathbf{Q}}^T = \mathbf{f}(\mathbf{Q}\mathbf{t}\mathbf{Q}^T, \mathbf{Q}\nabla\dot{\mathbf{x}}\mathbf{Q}^T + \dot{\mathbf{Q}}\mathbf{Q}^T, \theta, \mathbf{X})$$

which must hold for all $\mathbf{Q}(t)$ subject to (5.3.15). Here $\nabla\dot{\mathbf{x}} \equiv \dot{\mathbf{x}}_{,k}\mathbf{i}_k$. In particular, choosing at any arbitrary point x and instant t a rigidly rotating frame such that $\mathbf{Q} = \mathbf{I}$, $\dot{\mathbf{Q}} = -\mathbf{w}$, where w is the vorticity tensor,

$$w_{kl} \equiv \tfrac{1}{2}(v_{k,l} - v_{l,k}) \tag{9.4.5}$$

we see that (9.4.4) can be reduced to

$$\hat{\mathbf{t}} = \mathbf{f}(\mathbf{t}, \mathbf{d}, \theta, \mathbf{x}) \tag{9.4.6}$$

where $\hat{\mathbf{t}}$ is the stress flux defined by

$$\hat{\mathbf{t}} = \dot{\mathbf{t}} + \mathbf{t}\mathbf{w} - \mathbf{w}\mathbf{t} \tag{9.4.7}$$

In place of $\hat{\mathbf{t}}$ so defined, of course, other forms of stress flux listed in (3.9.6) may be used. This point was fully discussed in Section 3.9. We thus see that the theory of Maxwell solids requires writing constitutive equations for the stress flux $\hat{\mathbf{t}}$ as a tensor function of stress t, deformation rate tensor d, θ and X. This appears to be equivalent to replacing t by $\hat{\mathbf{t}}$ on

the left, and **C** by **t** and $\dot{\mathbf{C}}$ by **d** on the right of (9.2 8) of the stress constitutive equations for Kelvin-Voigt solids. This formal equivalence is unfortunately a superficial one. For presently no thermodynamic treatment of Maxwell solids exists, and this is true for all theories involving stress rates. However, if we abandon the condition of thermodynamic admissibility because of our ignorance, we can write constitutive equations for such solids. Since $\hat{\mathbf{t}}$ is objective, from (9.4 6) we see that

$$\mathbf{Q}\mathbf{f}(\mathbf{t}, \mathbf{d}, \theta, \mathbf{x})\mathbf{Q}^{\mathrm{T}} = \mathbf{f}(\mathbf{Q}\mathbf{t}\mathbf{Q}^{\mathrm{T}}, \mathbf{Q}\mathbf{d}\mathbf{Q}^{\mathrm{T}}, \theta, \mathbf{X}) \tag{9.4.8}$$

for arbitrary rigid rotations $\mathbf{Q}(t)$ satisfying (5 3.15). This implies that **f** is an hemitropic tensor function of **t** and **d**. If **f** is also a polynomial in these matrices then further reduction is possible. We assume that **f** is expressible as a polynomial in **t** and **d**. Moreover, for simplicity we also include reflection of axes in the group $\{\mathbf{Q}\}$, that is, $\det \mathbf{Q} = \pm 1$. In this case **f** is an isotropic matrix polynomial and can be reduced to the following form (App. B9.7):

$$\begin{aligned}\hat{\mathbf{t}} = {} & \kappa_0\mathbf{I} + \kappa_1\mathbf{t} + \kappa_2\mathbf{t}^2 + \kappa_3\mathbf{d} + \kappa_4\mathbf{d}^2 \\ & + \kappa_5(\mathbf{t}\mathbf{d} + \mathbf{d}\mathbf{t}) + \kappa_6(\mathbf{d}\mathbf{t}^2 + \mathbf{t}^2\mathbf{d}) \\ & + \kappa_7(\mathbf{d}^2\mathbf{t} + \mathbf{t}\mathbf{d}^2) + \kappa_8(\mathbf{d}^2\mathbf{t}^2 + \mathbf{t}^2\mathbf{d}^2)\end{aligned} \tag{9.4.9}$$

where κ_0 to κ_8 are polynomials in the joint invariants

$$\begin{array}{llll} \operatorname{tr}\mathbf{t}, & \operatorname{tr}\mathbf{t}^2, & \operatorname{tr}\mathbf{t}^3 & \\ \operatorname{tr}\mathbf{d}, & \operatorname{tr}\mathbf{d}^2, & \operatorname{tr}\mathbf{d}^3 & \\ \operatorname{tr}\mathbf{d}\mathbf{t}, & \operatorname{tr}\mathbf{d}\mathbf{t}^2, & \operatorname{tr}\mathbf{d}^2\mathbf{t}, & \operatorname{tr}\mathbf{d}^2\mathbf{t}^2 \end{array} \tag{9.4.10}$$

with coefficients that are functions of θ and **X**.

Hypoelasticity.[4] A Maxwell solid is called hypoelastic if the stress constitutive equations are linear in the deformation rate **d**. The explicit forms of the constitutive equations are obtained by linearizing (9.4 9) with respect to **d**. It is clear that in this case $\kappa_4 = \kappa_7 = \kappa_8 = 0$ and the others are linear in **d**. Hence the following general form is obtained

$$\begin{aligned}\hat{\mathbf{s}} = {} & \mathrm{I}_d\alpha_0\mathbf{I} + \alpha_1\mathbf{d} + \mathrm{I}_d\alpha_2\mathbf{s} + M\alpha_3\mathbf{I} + \tfrac{1}{2}\alpha_4(\mathbf{d}\mathbf{s} + \mathbf{s}\mathbf{d}) \\ & + \mathrm{I}_d\alpha_5\mathbf{s}^2 + M\alpha_6\mathbf{s} + N\alpha_7\mathbf{I} + \tfrac{1}{2}\alpha_8(\mathbf{d}\mathbf{s}^2 + \mathbf{s}^2\mathbf{d}) \\ & + M\alpha_9\mathbf{s}^2 + N\alpha_{10}\mathbf{s} + N\alpha_{11}\mathbf{s}^2\end{aligned} \tag{9.4.11}$$

[4] The theory was first introduced by Truesdell [1955 a, b]. An account of the subject is to be found in Eringen [1962, Ch. 8] *Note added at proof.* In this connection we refer the reader also to Truesdell and Noll [1965, D IV].

where we have taken $\hat{\mathbf{s}} = \mathbf{0}$ for $\mathbf{d} = \mathbf{0}$ and

$$\mathbf{s} \equiv \mathbf{t}/2\mu$$

$$M \equiv \operatorname{tr} \mathbf{sd} \qquad N \equiv \operatorname{tr} \mathbf{s}^2\mathbf{d} \tag{9.4.12}$$

$$\alpha_\kappa = \alpha_\kappa(\operatorname{tr} \mathbf{s}, \operatorname{tr} \mathbf{s}^2, \operatorname{tr} \mathbf{s}^3), (\kappa = 0, 1, 2, \ldots, 11)$$

and μ is the Lamé shear modulus nondimensionalizing the stress.

For the *incompressible* body $I_d = 0$ and $\mathbf{s}$ is replaced by $\mathbf{s} + (p/2\mu)\mathbf{I}$ where p is the unknown pressure.

The theory of hypoelasticity possesses certain potential applications. For example, the large deformations of solids that possess no time constant (such as relaxation time) can be studied. Certain classes of initially stressed bodies can be treated. The integration of the differential equations of motion and (9.4.11) in some cases lead to stress-strain curves that exhibit *yield-like* phenomena. To date, no solids are known to be representable with the hypoelastic models. Thus the potential of theory remains dormant.

Linear Theory of Maxwell Solids. The equations (9.4.9) may be linearized to give

$$t_{kl} = (\alpha_0 + \alpha t_{rr} + \lambda d_{rr})\delta_{kl} + 2\beta t_{kl} + 2\mu d_{kl} \tag{9.4.13}$$

where α_0, α, λ, μ, and β are, in general, functions of θ and X. For homogeneous solids at uniform temperature, they are constants. Here the coefficients λ and μ are selected to be the Lamé constants of linear elasticity, so that in the case of $\alpha_0 = \alpha = \beta = 0$ the integral of (9.4.13) gives the theory of elasticity.

The case of *incompressible solids* is obtained by setting $d_{rr} = 0$ and replacing $\mathbf{t}$ by $\mathbf{t} + p\mathbf{I}$ where $p(\mathbf{x}, t)$ is the unknown pressure.

9.5. GENERAL STRAIN-RATE-DEPENDENT MATERIALS

Kelvin-Voigt and Maxwell materials possess a certain simplicity that is attractive to the practitioners of viscoelastic theory. At the same time, these theories have limited applicability in physical situations, involving rapid changes in the strain rate. This situation can be remedied by assuming that the stress depends on higher-order strain rates. The theory of general strain-rate-dependent materials is highly complicated in its full generality. We therefore sketch the starting point and the procedure,[5] thereafter confining our attention to the linear theory.

[5] For an account and a review of such theories see Eringen [1962, Ch. 10] and Eringen and Grot [1965].

According to the general theory of rate-dependent thermomechanical materials, the stress constitutive equations have the form (5.4.25). Since we are neglecting heat conduction, we express the constitutive equations in the forms

$$t_{kl} = \overset{-t}{J} x_{k,K} x_{l,L} T_{KL}(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots, \overset{(p)}{\mathbf{C}}; 0, \mathbf{X}) \tag{9.5.1}$$

$$q_k = \overset{-1}{J} x_{k,K} G_K(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots, \overset{(p)}{\mathbf{C}}; 0, \mathbf{X}) \tag{9.5.2}$$

$$\varepsilon = e(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots, \overset{(p)}{\mathbf{C}}; 0, \mathbf{X}) \tag{9.5.3}$$

$$\eta = n(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots, \overset{(p)}{\mathbf{C}}, 0, \mathbf{X}) \tag{9.5.4}$$

where $\overset{(p)}{\mathbf{C}} \equiv D^p\mathbf{C}/Dt^p$. By carrying these into the Clausius-Duhem inequality (9.2.12) one can, as in Section 9.2, show that

$$t_{kl} = 2\rho \frac{\partial \psi}{C_{KL}} x_{k,K} x_{l,L} + \frac{\rho}{\rho_0} {}_D T_{KL} x_{k,K} x_{l,L} \tag{9.5.5}$$

$$q_k = 0 \tag{9.5.6}$$

$$\varepsilon = \psi - 0 \frac{\partial \psi}{\partial 0} \tag{9.5.7}$$

$$\eta = -\frac{\partial \psi}{\partial 0} \tag{9.5.8}$$

where ψ and ${}_D\mathbf{T}$ are subject to

$$\frac{\partial \psi}{\overset{(p)}{\partial C_{KL}}} = 0 \tag{9.5.9}$$

$$ {}_D T_{KL} \dot{C}_{KL} - 2\rho_0 \sum_{s=1}^{p-1} \frac{\partial \psi}{\overset{(s)}{\partial C_{KL}}} \overset{(s+1)}{C_{KL}} \geq 0 \tag{9.5.10}$$

Here ${}_D\mathbf{T}$ is a tensor function of $\mathbf{C}, \dot{\mathbf{C}}, \ldots, \overset{(p)}{\mathbf{C}}, 0$, and $\mathbf{X}$, that is,

$$ {}_D\mathbf{T} = {}_D\mathbf{T}(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots, \overset{(p)}{\mathbf{C}}; 0, \mathbf{X}) \tag{9.5.11}$$

and ψ is a scalar function of $\mathbf{C}, \dot{\mathbf{C}}, \ldots, \overset{(p-1)}{\mathbf{C}}, 0$, and $\mathbf{X}$

$$\psi = \psi(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots, \overset{(p-1)}{\mathbf{C}}; 0, \mathbf{X}) \tag{9.5.12}$$

Thus we see for the first time that a thermodynamic state function ψ can

be rate dependent. From (9.5.10), under the assumption of continuity of the functions ${}_DT$ and ψ, we also deduce that

$$(9.5.13)\qquad {}_DT_{KL} = 0, \frac{\partial \psi}{\partial \overset{(n)}{C}_{KL}} = 0 \quad \text{when} \quad \dot{C}_{KL} = \ddot{C}_{KL} = \cdots = \overset{(p)}{C}_{KL} = 0 \qquad (n = 1, 2, \ldots, p-1)$$

If ${}_DT$ and ψ are polynomials in the strain rates, (9.5.13) implies that ${}_DT$ may be linear in the strain rates while ψ has no linear terms in the strain rates. This provides a useful guidance in constructing approximate theories. Further advance can be made if ψ and ${}_DT$ are polynomials in the strain rates. In order to avoid writing tedious expressions we present here only the linear theory.

Linear Theory. We give the linear theory in terms of the Lagrangian strain

$$(9.5.14)\qquad 2E_{KL} \equiv C_{KL} - \delta_{KL}$$

The stress potential is defined, as usual, by

$$(9.5.15)\qquad \Sigma \equiv \rho_0 \psi$$

Hence write

$$(9.5.16)\qquad \Sigma = \Sigma_0(0) + \Sigma_{KL}(0)E_{KL} + \tfrac{1}{2}\Sigma_{KLMN}(0)E_{KL}E_{MN} + \frac{1}{2}\sum_{\alpha,\beta=1}^{p-1} F^{\alpha\beta}_{KLMN}E^{(\alpha)}_{KL}E^{(\beta)}_{MN}$$

$$(9.5.17)\qquad {}_ET_{KL} \equiv \frac{\partial \Sigma}{\partial E_{KL}} = \Sigma_{KL}(0) + \Sigma_{KLMN}(0)E_{MN}$$

$$(9.5.18)\qquad {}_DT_{KL} = \sum_{\alpha=1}^{p} \Gamma^{\alpha}_{KLMN}(0)\overset{(\alpha)}{E}_{MN}$$

Upon carrying these into (9.5.5) and using $\tilde{\mathbf{E}}$ in place of $\mathbf{E}$ and (9.3.3), we obtain the stress constitutive equations of the anisotropic solid.

$$(9.5.19)\qquad t_{kl} = (1 - \tilde{e}_{mm})\sigma_{kl} + \sigma_{ml}(\tilde{e}_{km} + \tilde{r}_{km}) + \sigma_{km}(\tilde{e}_{lm} + \tilde{r}_{lm}) + \sigma_{klmn}\tilde{e}_{mn} + \sum_{\alpha=1}^{p} \gamma^{\alpha}_{klmn}\overset{(\alpha)}{\tilde{e}}_{mn}$$

where σ_{kl}, σ_{klmn}, and γ^{α}_{klmn} are defined as in (9.3.11) and possess the symmetry conditions as expressed by (9.3.12), that is,

$$(9.5.20)\qquad \sigma_{kl} = \sigma_{lk}, \qquad \sigma_{klmn} = \sigma_{mnkl} = \sigma_{lkmn} = \sigma_{klnm}$$

$$\gamma^{\alpha}_{klmn} = \gamma^{\alpha}_{lkmn} = \gamma^{\alpha}_{klnm} \qquad (\alpha = 1, 2, \ldots, p)$$

Further restrictions on these coefficients are obtained through (9.5.10). Hence

$$\sum_{\alpha=1}^{p} \gamma^{\alpha}_{klmn} \overset{(\alpha)}{\tilde{e}}_{mn} \overset{(\alpha)}{\tilde{e}}_{kl} - \sum_{\alpha,\beta=1}^{p-1} f^{\alpha\beta}_{klmn} \overset{(\alpha+1)\,(\beta)}{\tilde{e}_{kl}\ \tilde{e}_{mn}} \geq 0 \tag{9.5.21}$$

where

$$f^{\alpha\beta}_{klmn} \equiv F^{\alpha\beta}_{KLMN}\, \delta_{Kk}\, \delta_{Ll}\, \delta_{Mm}\, \delta_{Nn} \tag{9.5.22}$$

Isotropic Solids. In the case of isotropic solids the elastic and viscous moduli are isotropic tensors. Using expressions of the form (9.3.14) for these tensors in (9.5.16) and (9.5 19) we obtain

$$\Sigma \equiv \rho_0 \psi = \Sigma_0(\theta) + \alpha_e \mathrm{I}_{\tilde{e}} + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}^2_{\tilde{e}} - 2\mu_e \mathrm{II}_{\tilde{e}} + \frac{1}{2}\sum_{\alpha,\beta=1}^{p-1} \left[f_{\alpha\beta}\, \mathrm{tr}\, \overset{(\alpha)}{\tilde{\mathbf{e}}}\, \mathrm{tr}\, \overset{(\beta)}{\tilde{\mathbf{e}}} + 2h_{\alpha\beta}\, \mathrm{tr}\, \overset{(\alpha)(\beta)}{(\tilde{\mathbf{e}}\ \tilde{\mathbf{e}})}\right] \tag{9.5.23}$$

$$t_{kl} = \alpha_e\, \delta_{kl} + (\lambda_e - \alpha_e)\tilde{e}_{rr}\, \delta_{kl} + 2(\mu_e - \alpha_e)\tilde{e}_{kl} + \sum_{\alpha=1}^{p} \left[\lambda_\alpha \overset{(\alpha)}{\tilde{e}}_{rr}\, \delta_{kl} + 2\mu_\alpha \overset{(\alpha)}{\tilde{e}}_{kl}\right] \tag{9.5.24}$$

where α_e, λ_e, μ_e, $f_{\alpha\beta} = f_{\beta\alpha}$, $h_{\alpha\beta} = h_{\beta\alpha}$, λ_α, and μ_α are functions of θ and $\mathbf{X}$.

When the natural state is stress free, that is, $\mathrm{t} = 0$ when $\tilde{\mathbf{e}} = \overset{(\alpha)}{\tilde{\mathbf{e}}} = 0$, these equations reduce to

$$\Sigma = \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}^2_{\tilde{e}} - 2\mu_e \mathrm{II}_{\tilde{e}} + \frac{1}{2}\sum_{\alpha,\beta=1}^{p-1} \left[f_{\alpha\beta}\, \mathrm{tr}\, \overset{(\alpha)}{\tilde{\mathbf{e}}}\, \mathrm{tr}\, \overset{(\beta)}{\tilde{\mathbf{e}}} + 2h_{\alpha\beta}\, \mathrm{tr}\, \overset{(\alpha)(\beta)}{(\tilde{\mathbf{e}}\ \tilde{\mathbf{e}})}\right] \tag{9.5.25}$$

$$t_{kl} = \lambda_e \tilde{e}_{rr}\, \delta_{kl} + 2\mu_e \tilde{e}_{kl} + \sum_{\alpha=1}^{p} \left[\lambda_\alpha \overset{(\alpha)}{\tilde{e}}_{rr}\, \delta_{kl} + 2\mu_\alpha \overset{(\alpha)}{\tilde{e}}_{kl}\right] \tag{9.5.26}$$

The restriction on the viscosity coefficients λ_α, μ_α, $f_{\alpha\beta}$, and $h_{\alpha\beta}$ given by (9.5.21) now reduces to

$$\sum_{\alpha=1}^{p} \left[\lambda_\alpha\, \mathrm{tr}\, \overset{(\alpha)}{\tilde{\mathbf{e}}}\, \mathrm{tr}\, \overset{(\alpha)}{\dot{\tilde{\mathbf{e}}}} + 2\mu_\alpha\, \mathrm{tr}\, \overset{(\alpha)}{(\tilde{\mathbf{e}}\, \dot{\tilde{\mathbf{e}}})}\right] - \sum_{\alpha,\beta=1}^{p-1} \left[f_{\alpha\beta}\, \mathrm{tr}\, \overset{(\alpha+1)}{\tilde{\mathbf{e}}}\, \mathrm{tr}\, \overset{(\beta)}{\tilde{\mathbf{e}}} + 2h_{\alpha\beta}\, \mathrm{tr}\, \overset{(\alpha+1)(\beta)}{(\tilde{\mathbf{e}}\ \ \tilde{\mathbf{e}})}\right] \geq 0 \tag{9 5.27}$$

which must be satisfied for all $\overset{(\alpha)}{\tilde{\mathbf{e}}}$ $(\alpha = 1, 2, \ldots, p)$.

In a theory of this type we see that the elastic part of the stress constitutive equations is superimposed with a series of viscous terms containing the time rates of the strain $\tilde{e}_{kl}$ up to order p. In the literature, the constitutive equation (9.5 26) is often expressed in a differential operator form

$$t_{kl} = P\tilde{e}_{rr}\, \delta_{kl} + 2Q\tilde{e}_{kl} \tag{9.5.28}$$

where

$$P \equiv \lambda_c + \sum_{\alpha=1}^{p} \lambda_\alpha \frac{\partial^\alpha}{\partial t^\alpha}$$

(9.5.29)

$$Q \equiv \mu_c + \sum_{\alpha=1}^{p} \mu_\alpha \frac{\partial^\alpha}{\partial t^\alpha}$$

Theories Involving Stress Rates. If the starting point for the rate theories is made with constitutive equations involving stress fluxes and strain rates of various orders, then the constitutive equations also involve various time rates of stress. As in the simplest case of Maxwell solids, no thermodynamic condition of admissibility is known for such solids. Nevertheless, the literature of linear viscoelasticity has many such equations. We give here only the linear stress constitutive equations of such solids, involving stress rates up to an order r. From the result (9.5.28) and that of Maxwell solids (9.4.13), it can be deduced that the linear stress constitutive equations for isotropic solids will have the form

(9.5.30) $$Rt_{rr}\,\delta_{kl} + 2St_{kl} = P\tilde{e}_{rr}\,\delta_{kl} + 2Q\tilde{e}_{kl}$$

where P and Q are the linear operators given by (9.5.29) and R and S are linear differential operators of the forms

$$R \equiv -\alpha + \sum_{\kappa=1}^{r} \alpha_\kappa \frac{\partial^\kappa}{\partial t^\kappa}$$

(9.5.31)

$$S \equiv -\beta + \sum_{\kappa=1}^{r} \beta_\kappa \frac{\partial^\kappa}{\partial t^\kappa}$$

where α, β, α_k, β_k ($\kappa = 1, 2, \ldots, r$) are in general functions of θ and X also. Clearly if $\mu_1 = \mu$, $\lambda_1 = \lambda$, $\beta_1 = 1/2$, α and β are the only non-zero coefficients, we obtain the Maxwell solid (9.4.13). Also, when $\alpha = \alpha_k = \beta_k = 0$, $\beta = -1/2$, (9.5.30) goes into (9.5.28). Nevertheless, as mentioned earlier, the foundation of the theory underlying (9.5.28) is presently stronger than the more general theory (9.5.30). Through the relation of (9.5.30) to the functional theories (cf. Eringen [1955a]), there is ample reason to believe that a theory of the form (9.5.30) is, in general, acceptable. However, in certain cases the stress, when solved from the differential equations (9.5.30), contains some strong delta function singularities that are intuitively disturbing.

9.6. FUNCTIONAL CONSTITUTIVE EQUATIONS

Viscoelastic materials may be characterized by functional constitutive equations as given by the master equations (5.4.24) of thermomechanical

materials In the special case of non-heat-conducting materials we drop the dependence on the temperature gradients so that the constitutive equations of these materials may be expressed as

$$t_{kl}(\mathbf{X}, t) = j^{-1}x_{k,K}x_{l,L}T_{KL}[\mathbf{C}(t-\tau'), \theta(t-\tau'), \mathbf{X}] \tag{9.6.1}$$

$$q_k(\mathbf{X}, t) = j^{-1}x_{k,K}G_K[\mathbf{C}(t-\tau'), \theta(t-\tau'), \mathbf{X}] \tag{9.6.2}$$

$$\varepsilon(\mathbf{X}, t) = e[\mathbf{C}(t-\tau'), \theta(t-\tau'), \mathbf{X}] \tag{9.6.3}$$

$$\eta(\mathbf{X}, t) = n[\mathbf{C}(t-\tau'), \theta(t-\tau'), \mathbf{X}] \tag{9.6.4}$$

where primed quantities indicate the dependence on all values of the functions in the interval $0 \leq \tau' \leq \infty$. Thus for example, T_{KL} is a tensor functional of all values of the functions $\mathbf{C}(t-\tau')$ and $\theta(t-\tau')$ in the interval $0 \leq \tau' \leq \infty$.

To satisfy the conditions of thermodynamic admissibility we substitute (9.6.1) to (9.6.4) into the Clausius-Duhem inequality (9.2.12). This gives

$$-\frac{\rho_0}{\theta}(\dot{\psi} + \eta\dot{\theta}) + \frac{1}{2\theta}T_{KL}C_{KL} + \frac{1}{\theta^2}G_K\theta_{,K} \geq 0 \tag{9.6.5}$$

Since ψ, η, $\mathbf{T}$, and $\mathbf{G}$ are independent of $\theta_{,K}$, from this it is immediate that

$$G_K = 0 \tag{9.6.6}$$

$$-\frac{\rho_0}{\theta}(\dot{\psi} + \eta\dot{\theta}) + \frac{1}{2\theta}T_{KL}\dot{C}_{KL} \geq 0 \tag{9.6.7}$$

For simplicity, we further assume that (9.6.1) to (9.6.4) possess no memory of temperature. This means replacing $\theta(t-\tau')$ by $\theta(t)$ in these equations.[6] In order to find the implications of (9.6.7) we introduce the difference histories

$$\mathbf{C}^t(\tau') \equiv \mathbf{C}(t-\tau') - \mathbf{C}(t) \tag{9.6.8}$$

so that (9.6.1), (9.6.3) and (9.6.4) and ψ may be expressed as

$$t_{kl} = j^{-1}x_{k,K}x_{l,L}T_{KL}[\mathbf{C}^t(\tau'); \mathbf{C}, \theta] \tag{9.6.9}$$

$$\varepsilon = e[\mathbf{C}^t(\tau'); \mathbf{C}, \theta] \tag{9.6.10}$$

$$\eta = n[\mathbf{C}^t(\tau'); \mathbf{C}, \theta] \tag{9.6.11}$$

$$\psi = \varepsilon - \theta\eta = \psi[\mathbf{C}^t(\tau'); \mathbf{C}, \theta] \tag{9.6.12}$$

Under proper smoothness conditions the response functionals may be expanded in a power series. Such series are in the forms of multiple integrals

[6] There is no difficulty in relaxing this assumption (see Coleman [1964]).

cf. Volterra [1930, Art. 3]. Since we are concerned only with the linear theory, we consider here an expansion of the free energy up to a second-order approximation.

$$\rho_0\psi = \Sigma(\mathbf{C}, \theta) + \int_0^\infty F_{KL}(s)C^t_{KL}(s)\,ds + \int_0^\infty\int_0^\infty F_{KLMN}(s_1, s_2)C^t_{KL}(s_1)C^t_{MN}(s_2)\,ds_1\,ds_2 \tag{9.6.13}$$

where the kernels $F_{KL}(s)$ and $F_{KLMN}(s_1, s_2)$ satisfy the symmetry conditions

$$\begin{aligned} &F_{KL}(s) = F_{LK}(s) \\ &F_{KLMN}(s_1, s_2) = F_{MNKL}(s_2, s_1) \\ &F_{KLMN}(s_1, s_2) = F_{LKMN}(s_1, s_2) = F_{KLNM}(s_1, s_2) \end{aligned} \tag{9.6.14}$$

Assuming the existence of a class of materials possessing a free energy of the form (9.6.13) the problem can be treated exactly. From (9.6.13) we can calculate $\dot\psi$ as

$$\begin{aligned} \rho_0\dot\psi = {} & \frac{\partial\Sigma}{\partial\theta}\dot\theta + \left[\frac{\partial\Sigma}{\partial C_{KL}} - \int_0^\infty F_{KL}(s)\,ds \right. \\ & \left. - 2\int_0^\infty\int_0^\infty F_{KLMN}(s_2, s_1)\,C^t_{MN}(s_1)\,ds_1 ds_2\right]\dot C_{KL} \\ & - \int_0^\infty F_{KL}(s)\frac{d}{ds}C_{KL}(t-s)\,ds - 2\int_0^\infty\int_0^\infty F_{KLMN}(s_1, s_2)C^t_{KL}(s_1) \\ & \times \frac{d}{ds_2}C_{MN}(t-s_2)\,ds_1\,ds_2 \end{aligned}$$

Substituting this into (9.6.7) we get

$$\begin{aligned} & -\frac{\rho_0}{\theta}\left(\frac{1}{\rho_0}\frac{\partial\Sigma}{\partial\theta} + \eta\right)\dot\theta + \frac{1}{2\theta}\left[T_{KL} - 2\frac{\partial\Sigma}{\partial C_{KL}} + 2\int_0^\infty F_{KL}(s)\,ds \right. \\ & \left. + 4\int_0^\infty\int_0^\infty F_{KLMN}(s_2, s_1)\,C^t_{MN}(s_1)\,ds_1\,ds_2\right]\dot C_{KL} \\ & + \frac{1}{\theta}\int_0^\infty F_{KL}(s)\frac{d}{ds}C_{KL}(t-s)\,ds + \frac{2}{\theta}\int_0^\infty\int_0^\infty F_{KLMN}(s_1, s_2) \\ & \times C^t_{KL}(s_1)\frac{d}{ds_2}C_{MN}(t-s_2)\,ds_1\,ds_2 \geq 0 \end{aligned} \tag{9.6.15}$$

This is linear in $\dot{\theta}$. Hence for arbitrary variations of $\dot{\theta}$ the inequality cannot be maintained unless the coefficient of $\dot{\theta}$ vanishes. Also it can be shown that for any history $C_{KL}(\tau)$, there exist many sufficiently near histories such that $\dot{C}_{KL}(t)$ is arbitrary. This implies that the coefficient of $\dot{C}_{KL}$ must vanish. Thus

$$\eta = -\frac{1}{\rho_0}\frac{\partial \Sigma}{\partial \theta} \tag{9.6.16}$$

(9.6 17)

$$T_{KL} = 2\frac{\partial \Sigma}{\partial C_{KL}} - 2\int_0^\infty F_{KL}(s)\,ds - 4\int_0^\infty\int_0^\infty F_{KLMN}(s_1, s_2)C^t_{MN}(s_2)\,ds_1\,ds_2,$$

$$\frac{1}{\theta}\int_0^\infty F_{KL}(s)\frac{d}{ds}C_{KL}(t-s)\,ds + \frac{2}{\theta}\int_0^\infty\int_0^\infty F_{KLMN}(s_1, s_2)C^t_{KL}(s_1)\frac{d}{ds_2}C_{MN}(t-s_2)\,ds_1\,ds_2 \geq 0 \tag{9.6.18}$$

This inequality implies that

$$F_{KL}(s) = 0 \tag{9.6.19}$$

$$\frac{1}{\theta}\int_0^\infty\int_0^\infty \phi_{KLMN}(s_1, s_2)\dot{C}^t_{KL}(s_1)\dot{C}^t_{MN}(s_2)\,ds_1\,ds_2 \geq 0 \tag{9.6.20}$$

where we have used (9.6.8) and the definitions

$$\phi_{KLMN}(s_1, s_2) \equiv \int_{s_1}^\infty F_{KLMN}(s_3, s_2)\,ds_3$$
$$\dot{C}^t_{MN}(s) \equiv \frac{d}{d\tau}C_{MN}(\tau)\Big|_{\tau=t-s} \tag{9.6.21}$$

Upon introducing the notation

$$\Phi_{KLMN}(s) \equiv \int_s^\infty \phi_{KLMN}(0, s_2)\,ds_2 \tag{9.6.22}$$

the constitutive equations for the stress becomes

$$T_{KL} = 2\frac{\partial \Sigma}{\partial C_{KL}} + 4\int_0^\infty \Phi_{KLMN}(s)\dot{C}^t_{MN}(s)\,ds \tag{9.6.23}$$

The constitutive equations (9.6 1) now take the form

$$t_{kl} = 2\frac{\rho}{\rho_0}x_{k,K}x_{l,L}\frac{\partial \Sigma}{\partial C_{KL}} + 4\frac{\rho}{\rho_0}x_{k,K}x_{l,L}\int_0^\infty \Phi_{KLMN}(s)\dot{C}^t_{MN}(s)\,ds \tag{9.6.24}$$

We note that this expression is valid for large strains but slow motions. Equations (9.6.24) are nonlinear in the strains but linear in the strain rate. For this reason, it is sometimes called *finite linear viscoelasticity* (Coleman and Noll [1961]).

Boltzmann-Volterra Theory. Further linearization of (9.6.24) with respect to strains E_{KL} is possible. Upon using an expansion of the form (9.3.4) for Σ, writing $\tilde{\mathbf{E}}$ for $\mathbf{E}$, and using (9.3.3) we can linearize (9.6.24) in the form

$$t_{kl} = (1 - \tilde{e}_{mm})\sigma_{kl} + \sigma_{ml}(\tilde{e}_{km} + \tilde{r}_{km}) + \sigma_{km}(\tilde{e}_{lm} + \tilde{r}_{lm}) + \sigma_{klmn}\tilde{e}_{mn} + \int_{-\infty}^{t} \gamma_{klmn}(t-s)\frac{\partial \tilde{e}_{mn}(s)}{\partial s}\,ds \tag{9.6.25}$$

where the material moduli σ_{kl} and σ_{klmn} are defined in the same manner as in (9.3.11) and possess the symmetry properties given by (9.3.12) and

$$\gamma_{klmn}(t) \equiv 8\Phi_{KLMN}(t)\,\delta_{Kk}\,\delta_{Ll}\,\delta_{Mm}\,\delta_{Nn} \tag{9.6.26}$$

also satisfy the symmetry conditions (9.3.12), that is,

$$\gamma_{klmn} = \gamma_{lkmn} = \gamma_{klnm} \tag{9.6.27}$$

In addition these moduli are restricted through the thermodynamical conditions (9.6.20). The knowledge of $\gamma_{klmn}(t)$ is equivalent to that of $\Phi_{KLMN}(s)$. This through (9.6.22) by differentiation gives $\phi_{KLMN}(0, s) = -\partial\Phi_{KLMN}/\partial s$.

This is not, however, adequate for the full knowledge of the function $\phi_{KLMN}(s_1, s_2)$. Therefore, the conditions (9.6.22) do not lead to definite results for γ_{klmn}. This point is clarified if we remember that the full structure of $\phi_{KLMN}(s_1, s_2)$ is needed for the characterization of the free energy but only $\phi_{KLMN}(0, s)$ needed for the stress. In terms of the strains $\mathbf{E}$, (9.6.20) reads

$$\int_{-\infty}^{t}\int_{-\infty}^{t} \phi_{KLMN}(t-s_1, t-s_2)\frac{dE_{KL}(s_1)}{ds_1}\frac{dE_{MN}(s_2)}{ds_2}\,ds_1\,ds_2 \geq 0 \tag{9.6.28}$$

If we define similar to (9.6.26),

$$\Gamma_{klmn}(s_1, s_2) \equiv 8\phi_{KLMN}(s_1, s_2)\,\delta_{Kk}\,\delta_{Ll}\,\delta_{Mm}\,\delta_{Nn} \tag{9.6.29}$$

The inequality (9.6.28) can be approximated, in terms of small eulerian strains, by

$$\int_{-\infty}^{t}\int_{-\infty}^{t} \Gamma_{klmn}(t-s_1, t-s_2)\frac{d\tilde{e}_{kl}(s_1)}{ds_1}\frac{d\tilde{e}_{mn}(s_2)}{ds_2}\,ds_1\,ds_2 \geq 0 \tag{9.6.30}$$

The *memory functions* $\gamma_{klmn}(t)$ are then related to $\Gamma_{klmn}(s_1, s_2)$ through (9.6.22) and (9.6.26)

$$\gamma_{klmn}(s) = \int_s^\infty \Gamma_{klmn}(0, s_2)\, ds_2 \tag{9.6.31}$$

It is now clear that we need the full description of Γ_{klmn} before we can satisfy the thermodynamic inequality (9.6.30) and that knowledge of $\gamma_{klmn}(s)$ does not lead to a unique determination of $\Gamma_{klmn}(s_1, s_2)$.

Isotropic Solids. The constitutive equations of the Boltzmann-Volterra theory for isotropic solids are obtained by replacing the material moduli by their isotropic expressions. Using expressions of the form (9.3.14) for second- and fourth-order isotropic tensors, we see that stress constitutive equation (9.6.25) reduces to

$$\begin{aligned} t_{kl} = {} & \alpha_e \delta_{kl} + (\lambda_e - \alpha_e)\tilde{e}_{rr}\, \delta_{kl} + 2(\mu_e - \alpha_e)\tilde{e}_{kl} \\ & + \int_{-\infty}^{t} \left[\lambda_v(t-s) \frac{\partial \tilde{e}_{rr}(s)}{\partial s} \delta_{kl} + 2\mu_v(t-s) \frac{\partial \tilde{e}_{kl}(s)}{\partial s} \right] ds \end{aligned} \tag{9.6.32}$$

where α_e, λ_e, μ_e are elastic moduli and $\lambda_v(t)$ and $\mu_v(t)$ are memory functions, all of which may also be functions of θ and X. These moduli are to be determined experimentally. For the initially unstressed isotropic solid, we also set $\alpha_e = 0$

Axiom of Fading Memory. So far, nothing has been said in regard to the character of memory functions, such as $\lambda_v(t)$ and $\mu_v(t)$, of isotropic solids, and $\gamma_{klmn}(t)$ of anisotropic solids. According to the axiom of fading memory, presented in Section 5.3, these functions are further restricted. The physical reason for this is that the values of the strains at a distant past should not affect the stress appreciably. This is usually achieved by considering decaying exponential types of memory functions. In general we may take a Prony series for these functions, for example, take

$$\lambda_v(t) = \sum_{r=1}^{N} a_r e^{-k_r t}, \qquad k_r > 0 \tag{9.6.33}$$

where a_r, k_r, and N are determined to fit experimental curves for $\lambda_v(t)$ to any desired degree of approximation. Of course, many other such expansions providing a fading memory are possible. Since the memory of the far past is lost it is also permissible to restrict the interval of time $(-\infty, t)$ to $(-T, t)$, $t < T < \infty$. When an initial state $t = 0$ exists for which the body is stress free, the integral in (9.6.32) will have the limits (0 to t).

9.7. RELATION OF RATE THEORY TO FUNCTIONAL THEORY

The passage can be made from the rate theory to the functional theory, and vice versa, under suitable smoothness conditions placed upon the response functions. We already know from Section 5.3 that the rate-type constitutive equations are obtained from the functional equations whenever the smooth memory hypothesis is in effect. We recall that according to these smoothness conditions, we may substitute the power series expansion

$$\mathbf{C}(t-\tau') = \mathbf{C}(t) - \tau'\dot{\mathbf{C}}(t) + \frac{\tau'^2}{2!}\ddot{\mathbf{C}}(t) + \cdots \tag{9.7.1}$$

in the functional $\mathbf{T}$ so that

$$\mathbf{T}[\mathbf{C}(t-\tau')] = \mathbf{F}(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots) \tag{9.7.2}$$

If the functional is smooth enough to allow (9.7.2), then we say that the memory possesses a smooth neighborhood. The conditions for a functional $\mathbf{T}$ to satisfy this are a subject in the theory of functionals. Similarly, the passage from $\mathbf{F}(\mathbf{C}, \dot{\mathbf{C}}, \ddot{\mathbf{C}}, \ldots)$ to a functional $\mathbf{T}[\mathbf{C}(t-\tau')]$ requires certain representation theorems and conditions of convergence. We shall not delve into these mathematical questions[7] here. Instead we proceed to illustrate such a passage by actually performing it for the linear theory; cf. Eringen [1955a].

Consider the linear isotropic solid characterized by the equations (9.5.30), that is,

$$Rt_{rr}\,\delta_{kl} + 2St_{kl} = P\tilde{e}_{rr}\,\delta_{kl} + 2Q\tilde{e}_{kl} \tag{9.7.3}$$

where

$$\begin{aligned} R &\equiv -\alpha + \sum_{\kappa=1}^{r} \alpha_\kappa \frac{\partial^\kappa}{\partial t^\kappa}, \qquad S \equiv -\beta + \sum_{\kappa=1}^{r} \beta_\kappa \frac{\partial^\kappa}{\partial t^\kappa} \\ P &\equiv \lambda_e + \sum_{\kappa=1}^{p} \lambda_\kappa \frac{\partial^\kappa}{\partial t^\kappa}, \qquad Q \equiv \mu_e + \sum_{\kappa=1}^{p} \mu_\kappa \frac{\partial^\kappa}{\partial t^\kappa} \end{aligned} \tag{9.7.4}$$

We assume that at time $t = 0$ the stress, strain and their time rates vanish. We proceed to solve for the stress $t_{kl}(t)$ from these differential equations. To

[7] Interested readers are referred to Volterra [1930], Coleman and Noll [1960], and Chacon and Rivlin [1964].

this end we apply the Laplace transform defined by

$$\bar{t}_{kl}(\zeta) \equiv \int_0^\infty e^{-\zeta t} t_{kl}(t)\, dt$$

$$\bar{e}_{kl}(\zeta) \equiv \int_0^\infty e^{-\zeta t} e_{kl}(t)\, dt \tag{9.7.5}$$

to (9.7.3), that is, we multiply both sides of (9.7.3) by $\exp(-\zeta t)$ and integrate over t from 0 to ∞. Since t, ē, and their time rates are zero for $t = 0$, this gives

$$\bar{R}(\zeta)\bar{t}_{rr}\,\delta_{kl} + 2\bar{S}(\zeta)\bar{t}_{kl} = \bar{P}(\zeta)\bar{e}_{rr}\,\delta_{kl} + 2\bar{Q}(\zeta)\bar{e}_{kl} \tag{9.7.6}$$

$$\bar{R}(\zeta) \equiv -\alpha + \sum_{\kappa=1}^{r} \alpha_\kappa \zeta^\kappa, \qquad \bar{S}(\zeta) \equiv -\beta + \sum_{\kappa=1}^{r} \beta_\kappa \zeta^\kappa$$

$$\bar{P} \equiv \lambda_e + \sum_{\kappa=1}^{p} \lambda_\kappa \zeta^\kappa, \qquad \bar{Q}(\zeta) \equiv \mu_e + \sum_{\kappa=1}^{p} \mu_\kappa \zeta^\kappa \tag{9.7.7}$$

To solve $\bar{t}_{kl}$ from (9.7.6) we contract on the index k and solve for $\bar{t}_{kk}$. Hence

$$\bar{t}_{kk} = \frac{(3\bar{P} + 2\bar{Q})}{(3\bar{R} + 2\bar{S})}\,\bar{e}_{kk} \tag{9.7.8}$$

Using this in (9.7.6) we can solve for $\bar{t}_{kl}$

$$\bar{t}_{kl} = \lambda_e \bar{e}_{rr}\,\delta_{kl} + 2\mu_e \bar{e}_{kl} + \bar{\lambda}_v \zeta \bar{e}_{rr}\,\delta_{kl} + 2\bar{\mu}_v \zeta \bar{e}_{kl} \tag{9.7.9}$$

where λ_e and μ_e are Lamé elastic constants and

$$\bar{\lambda}_v(\zeta) \equiv \frac{1}{\zeta \bar{S}} \frac{\bar{P}\bar{S} - \bar{R}\bar{Q}}{3\bar{R} + 2\bar{S}} - \frac{\lambda_e}{\zeta}$$

$$\bar{\mu}_v(\zeta) \equiv \frac{\bar{Q}}{2\zeta \bar{S}} - \frac{\mu_e}{\zeta} \tag{9.7.10}$$

The inverse transform of (9.7.10)—if it exists—can, with the use of the convolution theorem, be given by

$$t_{kl} = \lambda_e \bar{e}_{rr}\,\delta_{kl} + 2\mu_e \bar{e}_{kl} + \int_0^t \left[\lambda_v(t-s) \frac{\partial \bar{e}_{rr}(s)}{\partial s}\,\delta_{kl} + 2\mu_v(t-s) \frac{\partial \bar{e}_{kl}(s)}{\partial s} \right] ds \tag{9.7.11}$$

where $\lambda_v(t)$ and $\mu_v(t)$ are the inverse Laplace transforms of $\bar{\lambda}_v(\zeta)$ and $\bar{\mu}_v(\zeta)$, respectively. This result is, of course, identical to the functional form (9.6.32) under the condition that the initial state at $t = 0$ is stress free. This formalism gives rise to an immediate question, namely, under what

conditions do the inverse transforms of $\bar{\lambda}_v(\zeta)$ and $\bar{\mu}_v(\zeta)$ exist? Clearly, these conditions place restrictions on the functions P, Q, R, and S.

A theorem providing the sufficient condition for the existence of the inverse transform is known from the theory of Laplace transforms (Churchill [1958, Art. 63]). According to this theorem, $\bar{\lambda}_v(\zeta)$ and $\bar{\mu}_v(\zeta)$ must be of order $O(\zeta^{-k})$ $k > 1$ for all ζ on the right half plane $\mathrm{Re}\ \zeta \geq t_0$, and $\bar{\lambda}_v(\zeta)$ and $\bar{\mu}_v(\zeta)$ must be real on $\mathrm{Re}\,\zeta \geq t_0$. An examination of (9.7.10) shows that a sufficient condition for this is

$$p \leq r \tag{9.7.12}$$

This condition is not, however, necessary and it can be relaxed in many instances, as in the case of the all strain rate dependent materials.

9.8. THEORY OF THERMOVISCOELASTICITY

In the previous sections on the formulation of viscoelastic materials the effects of heat conduction were ignored. For slow deformations and motions, or for very rapid motions of materials, heat conduction can be neglected on the grounds that in the former the temperature rises too slowly and in the latter there is no time for heat conduction to take place. The effect of heat conduction cannot be ignored for intermediary situations where appreciable thermal changes occur during the motion or where the system is subjected to thermal gradients. We have seen that when the thermal gradient is not included in the constitutive equations as an independent variable the heat conduction is not thermodynamically admissible even though the material coefficients are affected by the local temperature. To take care of heat conduction phenomena, in addition, the equation of conservation of energy must be used.

Here we proceed to show how a thermoelastic Kelvin-Voigt solid may be formulated in a similar fashion as in Section 9.2. Afterwards we sketch the scheme for more general rate and functional types of materials.

(*i*) *Thermoelastic Kelvin–Voigt Solids.*[8] The thermoelastic Kelvin-Voigt solid is defined by a set of constitutive equations of the form

$$t_{kl} = j^{-1}T_{KL}(\mathbf{C}, \dot{\mathbf{C}}, \theta_{,K}; \theta, \mathrm{X})x_{k,K}x_{l,L} \tag{9.8.1}$$

$$q_k = j^{-1}G_K(\mathbf{C}, \dot{\mathbf{C}}, \theta_{,K}; \theta, \mathrm{X})x_{k,K} \tag{9.8.2}$$

$$\varepsilon = e(\mathbf{C}, \dot{\mathbf{C}}, \theta_{,K}; \theta, \mathrm{X}) \tag{9.8.3}$$

$$\eta = n(\mathbf{C}, \dot{\mathbf{C}}, \theta_{,K}; \theta, \mathrm{X}) \tag{9.8.4}$$

[8] The nonlinear theory was given first by Koh and Eringen [1963] and later by Coleman and Mizel [1964].

which falls into a special form of the general rate-dependent materials characterized by (5.4 25). These equations are subject to the constitutive axioms. Upon substituting these expressions into (9 2.12) we obtain

$$
\begin{aligned}
&-\frac{\rho_0}{\theta}\left(\frac{\partial\psi}{\partial\theta}+\eta\right)\dot{\theta}+\frac{1}{2\theta}\left(T_{KL}-2\rho_0\frac{\partial\psi}{\partial C_{KL}}\right)\dot{C}_{KL}\\
&-\frac{\rho_0}{\theta}\frac{\partial\psi}{\partial\dot{C}_{KL}}\ddot{C}_{KL}-\frac{\rho_0}{\theta}\frac{\partial\psi}{\partial\theta_{,K}}\dot{\theta}_{,K}+\frac{1}{\theta^2}G_K\theta_{,K}\geq 0
\end{aligned}
\tag{9.8.5}
$$

This inequality is to be satisfied for all independent variations of $\dot{\theta}$, $\dot{\mathbf{C}}$, $\mathbf{C}$, $\theta_{,K}$ and $\dot{\theta}_{,K}$. Since it is linear in $\dot{\theta}$, $\mathbf{C}$, and $\dot{\theta}_{,K}$, we see that this is possible if and only if

$$
\eta = -\frac{\partial\psi}{\partial\theta}, \qquad \frac{\partial\psi}{\partial\dot{C}_{KL}}=\frac{\partial\psi}{\partial\theta_{,K}}=0 \tag{9.8.6}
$$

and

$$
\frac{1}{2\theta}\left(T_{KL}-2\rho_0\frac{\partial\psi}{\partial C_{KL}}\right)\dot{C}_{KL}+\frac{1}{\theta^2}G_K\theta_{,K}\geq 0 \tag{9.8 7}
$$

The results (9.8.6) imply that

$$
\psi \equiv \varepsilon-\theta\eta=\psi(\mathbf{C};\theta,\mathbf{X}) \tag{9.8.8}
$$

The composition of (9.8.7) suggests that we decompose T_{KL} into two parts as

$$
T_{KL}={}_{E}T_{KL}(\mathbf{C},\theta,\mathbf{X})+{}_{D}T_{KL}(\mathbf{C},\dot{\mathbf{C}},\theta_{,K};\theta,\mathbf{X}) \tag{9.8.9}
$$

From (9 8.7), it then follows that

$$
{}_{E}T_{KL}=2\rho_0\frac{\partial\psi}{\partial C_{KL}} \tag{9.8.10}
$$

$$
\frac{1}{2\theta}\,{}_{D}T_{KL}\dot{C}_{KL}+\frac{1}{\theta^2}G_K\theta_{,K}\geq 0 \tag{9.8.11}
$$

This inequality, on account of the continuity conditions on ${}_{D}\mathbf{T}$ and G_K, implies that

$$
{}_{D}T_{KL}(\mathbf{C},\mathbf{0},\mathbf{0};\theta,\mathbf{X})=0 \tag{9.8.12}
$$

$$
G_K(\mathbf{C},\mathbf{0},\mathbf{0};\theta,\mathbf{X})=0 \tag{9.8.13}
$$

Collecting the results, we have found that the constitutive equations (9 8.1) to (9.8.4) are reducible to

$$t_{kl} = 2\frac{\rho}{\rho_0}\frac{\partial \Sigma}{\partial C_{KL}} x_{k,K} x_{l,L} + \frac{\rho}{\rho_0}\, {}_D T_{KL} x_{k,K} x_{l,L} \tag{9.8.14}$$

$$q_k = \frac{\rho}{\rho_0} G_K x_{k,K} \tag{9.8.15}$$

$$\varepsilon = \frac{1}{\rho_0}\left(\Sigma - \theta \frac{\partial \Sigma}{\partial \theta}\right) \tag{9.8.16}$$

$$\eta = -\frac{1}{\rho_0}\frac{\partial \Sigma}{\partial \theta} \tag{9.8.17}$$

where $\Sigma(\mathbf{C}, \theta, \mathbf{X})/\rho_0 \equiv \varepsilon - \theta\eta$, and ${}_D T(\mathbf{C}, \dot{\mathbf{C}}, \theta_{,K}; \theta, \mathbf{X})$ and $G_K(\mathbf{C}, \dot{\mathbf{C}}, \theta_{,K}; \theta, \mathbf{X})$ are subject to

$$\frac{1}{2\theta}\, {}_D T_{KL} \dot{C}_{KL} + \frac{1}{\theta^2} G_K \theta_{,K} \geq 0 \tag{9.8.18}$$

$${}_D T_{KL}(\mathbf{C}, 0, 0; \theta, \mathbf{X}) = 0 \tag{9.8.19}$$

$$G_K(\mathbf{C}, 0, 0; \theta, \mathbf{X}) = 0 \tag{9.8.20}$$

The material symmetry restrictions can now be applied in a fashion as before. However, these are too long to be discussed here. We give below only the linear theory in the rates $\dot{C}_{KL}$. For this case

$${}_D T_{KL} = G_{KLMN}\dot{C}_{MN} + H_{KLM}\theta_{,M} \tag{9.8.21}$$

$$G_K = K_{KL}\theta_{,L} + J_{KLM}\dot{C}_{LM} \tag{9.8.22}$$

where G_{KLMN}, H_{KLM}, K_{KL}, and J_{KLM} are functions of $\mathbf{C}$, θ, and $\mathbf{X}$. For homogeneous materials the dependence on $\mathbf{X}$ also disappears. The elastic part of the stress involving Σ can be expanded into a series $\mathbf{C}$ for various order theories. The procedure here is similar to that given before.

For *linear isotropic materials*, by using (9.8.21) and (9.8.22) in (9.8.14) and (9.8.15) and employing isotropic tensors for the coefficients, we arrive at the results

$$t_{kl} = \alpha_0\, \delta_{kl} + \left(\lambda + \lambda_v \frac{\partial}{\partial t}\right)\tilde{e}_{rr}\, \delta_{kl} + 2\left(\mu + \mu_v \frac{\partial}{\partial t}\right)\tilde{e}_{kl} \tag{9.8.23}$$

$$q_k = \kappa\theta_{,k} \tag{9.8.24}$$

where α_0, λ, λ_v, μ, μ_v and κ are functions of θ and $\mathbf{X}$ From (9.8.11) we can also deduce that these equations are thermodynamically admissible if and only if

$$3\lambda_v + 2\mu_v \geq 0, \qquad \mu_v \geq 0, \qquad \kappa \geq 0 \tag{9.8.25}$$

We thus see that *in the case of linear isotropic materials the stress is independent of the temperature gradient and the heat is independent of the strain.*

Since ε and η are functions of $\bar{e}$ and θ only, the method of Section 8.3 for small temperature variations leads to the following final forms of the constitutive equations for zero initial stress

$$t_{kl} = -\beta T\,\delta_{kl} + \left(\lambda_e + \lambda_v \frac{\partial}{\partial t}\right)\bar{e}_{rr}\,\delta_{kl} + 2\left(\mu_e + \mu_v \frac{\partial}{\partial t}\right)\bar{e}_{kl} \tag{9.8.26}$$

$$q_k = \kappa\theta_{,k} \tag{9.8.27}$$

$$\Sigma = S_0 - \rho_0\eta_0 T - \frac{\rho_0\gamma}{2T_0}T^2 - \beta T\mathrm{I}_{\bar{e}} + \tfrac{1}{2}(\lambda_e + 2\mu_e)\mathrm{I}_{\bar{e}}^2 - 2\mu_e\mathrm{II}_{\bar{e}} \tag{9.8.28}$$

$$\eta = \eta_0 + \frac{\gamma}{T_0}T + \frac{\beta}{\rho_0}\mathrm{I}_{\bar{e}} \tag{9.8.29}$$

where T is the temperature change from that of the natural state T_0, that is,

$$\theta = T_0 + T \tag{9.8.30}$$

and β, λ_e, λ_v, μ_e, μ_v, κ, S_0, ρ_0, η_0, and γ are material constants all of which may also depend on T and X. For homogeneous solids, of course, the dependence on X disappears. The equation of heat conduction follows from that of energy in the same way as in Section 8.3.

$$-\rho_0\gamma\dot{T} - \beta T_0\dot{u}_{k,k} + (\kappa T_{,k})_{,k} + \rho h = 0 \tag{9.8.31}$$

The theory being linear, the effect of viscous dissipation has been neglected in (9.8.31). We note that for viscoelastic fluids this viscous dissipation may become important and cannot, in general, be neglected

Equations (9.8.26), (9.8.27), and (9.8.31) together with the equations of balance of momenta constitute the field equations of this theory. We note that the only difference between this theory and linear isotropic thermoelasticity is in the stress constitutive equations (9.8.26). For the anisotropic solid, however, major differences occur in that the stress and heat constitutive equations involve both the strain rates and the temperature gradients. This is apparent from (9.8.21) and (9.8.22)

(ii) The Linear Rate Theory. In this theory the response functions T_{KL}, G_K, e, and n depend also on the higher-order rates $\dot{\bar{C}}$, $\ddot{\bar{C}}$, $\ldots$, $\overset{(p)}{\bar{C}}$. The axiom of thermodynamic admissibility now leads to the result that ψ is independent of $\theta_{,K}$ and the highest-order rate $\overset{(p)}{\bar{C}}$. The stress and heat constitutive equations under zero initial stress for the linear isotropic theory are found to be

$$t_{kl} = -\beta T\,\delta_{kl} + \lambda_e \tilde{e}_{rr}\,\delta_{kl} + 2\mu_e \tilde{e}_{kl} + \sum_{\alpha=1}^{p} \left(\lambda_\alpha \frac{\partial^\alpha \tilde{e}_{rr}}{\partial t^\alpha}\,\delta_{kl} + 2\mu_\alpha \frac{\partial^\alpha \tilde{e}_{kl}}{\partial t^\alpha} \right) \tag{9.8.32}$$

$$q_k = \kappa T_{,k} \tag{9.8.33}$$

The expression of Σ is obtained by adding the rate terms contained in the summation on the right-hand side of (9.5.23) to the right-hand side of Σ in (9.8.28). The expression of η is identical to (9.8.29) and for the linear theory, the equation of heat conduction is the same as (9.8.31).

Again in the linear theory of isotropic solids the difference between thermoelasticity and thermo-viscoelasticity occurs only in the stress constitutive equations. The situation for the anisotropic theory is different and both the stress and heat constitutive equation depend on the thermal gradient and all strain rates (cf. Eringen and Grot [1965, Art. 11]).

(iii) Functional Constitutive Equations. We begin with the functional equations

$$t_{kl} = j^{-1} T_{KL}[\mathbf{C}(t-\tau), \theta_{,K}(t-\tau), \theta(t-\tau); \theta, \mathbf{X}] x_{k,K} x_{l,L} \tag{9.8.34}$$

$$q_k = j^{-1} G_K[\mathbf{C}(t-\tau), \theta_{,K}(t-\tau), \theta(t-\tau); \theta, \mathbf{X}] x_{k,K} \tag{9.8.35}$$

$$\varepsilon = e[\mathbf{C}(t-\tau), \theta_{,K}(t-\tau), \theta(t-\tau); \theta, \mathbf{X}] \tag{9.8.36}$$

$$\eta = n[\mathbf{C}(t-\tau), \theta_{,K}(t-\tau), \theta(t-\tau); \theta, \mathbf{X}] \tag{9.8.37}$$

In this generality the theory of heat conduction in a material with continuous memory has not been fully developed.[9] For the materials having no memory of temperature gradient (that is, no dependence on $\theta_{,K}(t-\tau)$) the theory is simpler. In this case, the axioms of thermodynamic admissibility lead to the conclusions arrived at in Section 9.6 for the material with linear memory. For the linear isotropic solid, the stress and heat constitutive equations take the forms

$$t_{kl} = -\beta T\,\delta_{kl} + \lambda_e \tilde{e}_{rr}\,\delta_{kl} + 2\mu_e \tilde{e}_{kl} + \int_{-\infty}^{t} \left[\lambda_s(t-s) \frac{\partial \tilde{e}_{rr}(s)}{\partial s}\,\delta_{kl} + 2\mu_s(t-s) \frac{\partial \tilde{e}_{kl}(s)}{\partial s} \right] ds \tag{9.8.38}$$

$$q_k = \kappa \theta_{,k} \tag{9.8.39}$$

[9] Eringen [1960] gave a linear theory within the framework of linear irreversible processes. A general scheme for the nonlinear theory is included in Eringen [1965]. However, the thermodynamic treatment has not been carried out for the fully nonlinear theory. Coleman [1964a, 1964b] has treated a theory in which material has no memory of temperature gradient.

Within the framework of linear isotropic solids again, we find that the only difference between thermoelasticity and thermoviscoelasticity is in the stress constitutive equations. For the anisotropic case, t and q both depend on the history of strain (cf Eringen and Grot [1965]). The material moduli $\beta(T)$, $\lambda_e(T)$, $\mu_e(T)$, $\lambda_v(t, T)$, $\mu_v(t, T)$, and $\kappa(T)$ are functions of temperature T. For the inhomogeneous materials, they may also depend on X In addition, as in the isothermal case, the memory functions $\lambda_v(t, T)$ and $\mu_v(t, T)$ must satisfy the axiom of fading memory as explained in Section 9.6.

9.9. THERMOVISCOELASTIC FLUIDS

The constitutive equations of thermoviscoelastic fluids and the equation of heat conduction can be obtained from those of thermoelastic solids by certain observations. For fluids, all configurations keeping the density fixed can be reference configurations (cf. Section 5 7) Accordingly, the spatial frame of reference x can be taken as the reference frame with ρ fixed. With this

$$
\begin{aligned}
&C_{KL} = x_{k,K}x_{k,L} \to \delta_{kl} \\
&\dot{C}_{KL} = 2\,d_{kl}x_{k,K}x_{l,L} \to 2\,d_{kl} \equiv A^{(1)}_{kl} \\
&C^{(p)}_{KL} \to A^{(p)}_{KL}, \qquad \theta_{,K} \to \theta_{,k}
\end{aligned} \tag{9.9.1}
$$

where $A^{(p)}$ are called Rivlin-Ericksen tensors; they are obtained by calculating $C^{(p)}_{KL}$ and then setting X = x, or by the recurrence formula (see also (2.4 11))

$$
\begin{aligned}
&A^{(1)}_{kl} = 2\,d_{km} = v_{k,m} + v_{m,k} \\
&A^{(p)}_{km} = \dot{A}^{(p-1)}_{km} + A^{(p-1)}_{kl}v_{l,m} + A^{(p-1)}_{lm}v_{l,k}
\end{aligned} \tag{9.9.2}
$$

In addition, the constitutive functions may depend on ρ^{-1}. The general nonlinear theory is too complicated to present here (cf. Eringen [1962, Ch 10]). Here we present only the linear theory For the linear theory we have

$$
\mathbf{A}^{(p+1)} \to 2\,\mathbf{d}^{(p)} = 2D^p\mathbf{d}/Dt^p \simeq 2D^{p+1}\mathbf{e}/Dt^{p+1} \tag{9.9.3}
$$

As a result, we see that if in the constitutive equations derived in Section 9.5, we replace the elastic part of the stress by a pressure, and the strain rates $\mathbf{e}^{(\alpha)}$ by $\mathbf{d}^{(\alpha-1)}$ then we obtain the corresponding constitutive equations of thermoviscoelastic fluids. Thus

$$
t_{kl} = -\pi(\rho^{-1}, \theta)\delta_{kl} + \sum_{\alpha=0}^{p-1}(\lambda_\alpha d^{(\alpha)}_{rr}\,\delta_{kl} + 2\mu_\alpha d^{(\alpha)}_{kl}) \tag{9.9.4}
$$

$$
q_k = \kappa(\rho^{-1}, \theta)\theta_{,k} \tag{9.9.5}
$$

where $\mathbf{d}^{(0)} \equiv \mathbf{d}$ and the pressure $\pi(\rho^{-1}, \theta)$ is defined by

$$\pi \equiv -\left.\frac{\partial \psi_0(\rho^{-1}, \theta)}{\partial \rho^{-1}}\right|_{\theta} \tag{9.9.6}$$

The Helmholtz free energy (9.5.23) is replaced by

$$\psi = \psi_0(\rho^{-1}, \theta) + \frac{1}{2}\sum_{\alpha,\beta=0}^{p-2}[f_{\alpha\beta}\,\mathrm{tr}\overset{(\alpha)}{\mathbf{d}}\,\mathrm{tr}\overset{(\beta)}{\mathbf{d}} + 2h_{\alpha\beta}\,\mathrm{tr}(\overset{(\alpha)}{\mathbf{d}}\,\overset{(\beta)}{\mathbf{d}})] \tag{9.9.7}$$

and the energy and entropy are, respectively, given by

$$\varepsilon = \psi - \theta\frac{\partial \psi}{\partial \theta} \tag{9.9.8}$$

$$\eta = -\frac{\partial \psi}{\partial \theta} \tag{9.9.9}$$

The viscosities $\lambda_\alpha(\rho^{-1}, \theta)$, $\mu_\alpha(\rho^{-1}, \theta)$, $f_{\alpha\beta}(\rho^{-1}, \theta) = f_{\beta\alpha}$ and $h_{\alpha\beta}(\rho^{-1}, \theta) = h_{\beta\alpha}$ are subject to restrictions of the form (9.5.27) where $\overset{(\alpha)}{\mathbf{e}}$ is replaced by $\overset{(\alpha-1)}{\mathbf{d}}$.

The equation of heat conduction follows from the equation of energy

$$\rho\dot{\varepsilon} = t_{kl}\,d_{lk} + q_{k,k} + \rho h \tag{9.9.10}$$

With the help of (9.9.4) to (9.9.8) this may be expressed as

$$-\rho\gamma\dot{\theta} - \beta\theta v_{k,k} + (\kappa\theta_{,k})_{,k} + \rho h = \Phi(\theta, \overset{(\nu)}{\mathbf{d}}),\quad \nu = 0, 1, \ldots, p-1 \tag{9.9.11}$$

where

$$\gamma \equiv -\theta\frac{\partial^2\psi_0}{\partial\theta^2} - \frac{\theta}{2}\sum_{\alpha,\beta=0}^{p-2}\left[\frac{\partial^2 f_{\alpha\beta}}{\partial\theta^2}\,\mathrm{tr}\overset{(\alpha)}{\mathbf{d}}\,\mathrm{tr}\overset{(\beta)}{\mathbf{d}} + 2\frac{\partial^2 h_{\alpha\beta}}{\partial\theta^2}\,\mathrm{tr}(\overset{(\alpha)}{\mathbf{d}}\,\overset{(\beta)}{\mathbf{d}})\right] \tag{9.9.12}$$

$$\beta \equiv -\frac{\partial^2\psi_0}{\partial\theta\partial\rho^{-1}} - \frac{1}{2}\sum_{\alpha,\beta=0}^{p-2}\left[\left(\frac{\partial^2 f_{\alpha\beta}}{\partial\theta\partial\rho^{-1}} - \frac{1}{\theta}\frac{\partial f_{\alpha\beta}}{\partial\rho^{-1}}\right)\mathrm{tr}\overset{(\alpha)}{\mathbf{d}}\,\mathrm{tr}\overset{(\beta)}{\mathbf{d}}\right. \tag{9.9.13}$$
$$\left. + 2\left(\frac{\partial^2 h_{\alpha\beta}}{\partial\theta\partial\rho^{-1}} - \frac{1}{\theta}\frac{\partial h_{\alpha\beta}}{\partial\rho^{-1}}\right)\mathrm{tr}(\overset{(\alpha)}{\mathbf{d}}\,\overset{(\beta)}{\mathbf{d}})\right]$$

$$\Phi(\theta, \overset{(\nu)}{\mathbf{d}}) \equiv \rho\sum_{\alpha,\beta=0}^{p-2}\left\{\left(f_{\alpha\beta} - \theta\frac{\partial f_{\alpha\beta}}{\partial\theta}\right)\mathrm{tr}\overset{(\alpha+1)}{\mathbf{d}}\,\mathrm{tr}\overset{(\beta)}{\mathbf{d}}\right. \tag{9.9.14}$$
$$\left. + 2\left(h_{\alpha\beta} - \theta\frac{\partial h_{\alpha\beta}}{\partial\theta}\right)\mathrm{tr}(\overset{(\alpha+1)}{\mathbf{d}}\,\overset{(\beta)}{\mathbf{d}})\right\}$$
$$-\sum_{\alpha=0}^{p-1}[\lambda_\alpha\,\mathrm{tr}\overset{(\alpha)}{\mathbf{d}}\,\mathrm{tr}\,\mathbf{d} + 2\mu_\alpha\,\mathrm{tr}(\overset{(\alpha)}{\mathbf{d}}\,\mathbf{d})]$$

The equation of heat conduction (9.9.11) is too complicated for analytical work. Further approximation requires certain assumptions concerning the magnitudes of θ and $\overset{(s)}{\mathbf{d}}$. In general, $f_{\alpha\beta}$ and $h_{\alpha\beta}$ are taken as zero, thus simplifying these equations. However, the second summation in the expression of Φ is the dissipative viscous energy ${}_D t_{kl} d_{lk}$ and, in general, cannot be ignored as in the case of solids

For the functional theory, similar modifications lead to the constitutive equations for the stress and heat

$$(9.9.15)\quad t_{kl} = -\pi(\rho^{-1}, \theta)\delta_{kl} + \int_{-\infty}^{t} [\lambda_v(t-s)\, d_{rr}(s)\delta_{kl} + 2\mu_v(t-s)\, d_{kl}(s)]\, ds$$

$$(9.9.16)\quad q_k = \kappa(\rho^{-1}, \theta)\theta_{,k}$$

where π is the pressure defined by (9.9.6).

The expression for ψ is obtained by the methods sketched in Section 9.6. If we ignore the rate terms (equivalent to setting $f_{\alpha\beta} = h_{\alpha\beta} = 0$ in the above rate theory) we can write approximately

$$(9.9.17)\qquad \psi \cong \psi_0(\rho^{-1}, \theta)$$

$$(9.9.18)\qquad \varepsilon \cong \psi_0 - \theta \frac{\partial \psi_0}{\partial \theta}$$

$$(9.9.19)\qquad \eta \cong -\frac{\partial \psi_0}{\partial \theta}$$

This then yields an approximate heat conduction equation

$$(9.9.20)\qquad -\rho\gamma\dot{\theta} - \beta\theta v_{k,k} + (\kappa\theta_{,k})_{,k} + \rho h = \Phi$$

where

$$(9.9.21)\qquad \gamma \cong -\theta \frac{\partial^2 \psi_0}{\partial \theta^2}, \qquad \beta \cong -\frac{\partial^2 \psi_0}{\partial \rho^{-1} \partial \theta}$$

(9.9.22)

$$\Phi \cong {}_D t_{kl}\, d_{lk} = \int_{-\infty}^{t} [\lambda_v(t-s)\, \mathrm{tr}\, \mathbf{d}(s)\, \mathrm{tr}\, \mathbf{d}(t) + 2\mu_v(t-s)\, \mathrm{tr}\, \mathbf{d}(s)\, \mathbf{d}(t)]\, ds$$

The memory functions $\lambda_v(t, 0)$ and $\mu_v(t, 0)$ are in general restricted by the hypothesis of fading memory.

9.10. EXPERIMENTAL DETERMINATION OF VISCOELASTIC MODULI

Viscoelastic moduli α_κ, β_κ, λ_κ, μ_κ appearing in the rate theories (9.7.3) or the memory functions $\lambda_v(t)$ and $\mu_v(t)$ of the linear theory of isotropic

viscoelastic materials are determined either through experiments or through statistical mechanical considerations. The latter method has found some success in connection with one-dimensional models, otherwise it has been limited in scope. Because of the urgent problems existing in the solid-propellant industry the experimental work in the field is extensive. However, a large amount of this work is classified and unpublished. Those works that are published are usually limited in scope, for example, they are quasi-static, one- or two-dimensional, and the temperature and time ranges are limited. For the thermal problems, the work is searce and generally confined to the so-called *thermorheologically simple* materials. No rational experiments on the anisotropie viscoelastic materials exist. A complete list of references on the topie is beyond the scope of this seetion. For various reviews and detailed discussion of the methods, the reader is referred to Zener [1948], Nolle [1948, 1949], Fleteher and Gent [1957], Marvin [1952, 1954], Kolsky [1956], Sherby and Dorn [1958], Payne [1958], Hunter [1960], Britton [1965], and Fitzgerald [1967]. A deseription of test proeedure and findings is also recorded in ICRPG, *solid-propellant mechanical behavior* manual.

Early experiments have shown that simple models, such as Kelvin-Voigt and Maxwell, have a very limited range of applieability and that a large number of terms in (9.7.3) is required for the rate theories to be of signifieant use in engineering problems. For this reason, reeent experiments are mostly confined to the determination of memory functionals $\lambda_e(t)$ and $\mu_e(t)$. The experiments can be placed into two eategories: (a) quasi-statie tests and (b) dynamie tests. We now give a brief aeeount of these two types of experiments.

First we give two other equivalent expressions to (9.6.32) with $\alpha_e = 0$, of the constitutive equations of the Boltzmann-Volterra theory. These are

$$t_{kl} = \delta_{kl} \int_{-\infty}^{t} K(t-\tau)\dot{e}_{rr}(\tau)\,d\tau + 2\int_{-\infty}^{t} G(t-\tau)\dot{e}'_{kl}(\tau)\,d\tau \tag{9.10.1}$$

$$e_{kl} = \tfrac{1}{9}\delta_{kl} \int_{-\infty}^{t} E(t-\tau)\dot{t}_{rr}(\tau)\,d\tau + \tfrac{1}{2}\int_{-\infty}^{t} J(t-\tau)\dot{t}'_{kl}(\tau)\,d\tau \tag{9.10.2}$$

where

$$e'_{kl} \equiv e_{kl} - \tfrac{1}{3}e_{rr}\,\delta_{kl}, \qquad t'_{kl} \equiv t_{kl} - \tfrac{1}{3}t_{rr}\,\delta_{kl} \tag{9.10.3}$$

are the deviatorie parts of the strain and stress, respectively. Clearly we have

$$e'_{kk} = 0, \qquad t'_{kk} = 0 \tag{9.10.4}$$

By comparing (9.10.1) with (9.6 32) we see that the new memory functions $K(t)$, $G(t)$ are related to λ_e, μ_e, λ_v and μ_v by

$$
\begin{aligned}
K(t) &\equiv \lambda_e + \tfrac{2}{3}\mu_e + \lambda_v(t) + \tfrac{2}{3}\mu_v(t) \\
G(t) &\equiv \mu_e + \mu_v(t)
\end{aligned}
\tag{9.10.5}
$$

In general these functions also depend on the temperature T The form (9.10 1) is found useful in the ***relaxation tests*** where the strain components are fixed over the time interval $(0, t)$ and zero for $\tau < 0$. For the *creep* test where the stress is fixed, the form (9.10.2) of the constitutive equations is found to be more useful. Hence $K(t)$ and $G(t)$ are, respectively, named *relaxation bulk modulus* and ***shear modulus***; $B(t)$, *creep bulk compliance*; and $J(t)$, ***creep shear compliance***. The Fourier transforms of these quantities and their various combinations take other names [10]

Often, in the tests performed, the material is unstressed before the initial time $t = 0$, that is, both $t_{kl}(t) = \bar{e}_{kl}(t) = 0$ for $t < 0$ Henceforth we shall assume that this is the case. Thus the lower limits of integrals in (9.10 1) and (9.10 2) are to be taken as zero. Since the strain must be soluble from (9.10.1), it is clear that the function pairs K, G and B, J must be related to each other This relationship can be illustrated best by taking the Laplace transforms (or Fourier transforms) (9.10.1) and (9 10 2) and equating the transforms of t_{kl} obtained from these equations Hence

$$
\bar{K}(\zeta)\bar{B}(\zeta) = 1/\zeta^2, \qquad \bar{G}(\zeta)\bar{J}(\zeta) = 1/\zeta^2
\tag{9.10 6}
$$

where ζ is the transform variable and a superposed bar indicates the Laplace transforms, that is,

$$
\bar{K}(\zeta) \equiv \int_0^\infty K(t)e^{-\zeta t}\, dt
\tag{9.10.7}
$$

Thus, given $K(t)$ and $G(t)$ through (9.10.6), we can calculate $\bar{B}(\zeta)$ and $\bar{J}(\zeta)$ and by inversion $B(t)$ and $J(t)$.

We mentioned earlier that $\lambda_v(t)$ and $\mu_v(t)$ must obey the axiom of memory and that this can be taken care of by using a Prony series for their representation, for example, we may take

$$
\mu_v(t) = \sum_{\alpha=0}^{\infty} \mu_\alpha e^{-k_\alpha t}, \; k_\alpha > 0
\tag{9.10 8}
$$

where μ_α and k_α may be functions of temperature also and $\mu_v(t)$ is a monotonically increasing function. A similar expression is valid for $\lambda_v(t)$ Instead

[10] For a collection of the varied nomenclature used commonly, see Leaderman [1957]

of (9.10.8) a continuous relaxation spectrum may be employed so that

$$\mu_v(t) = \mu_e \int_0^\infty \phi(\alpha)(e^{-t/\alpha} - 1)\, d\alpha \tag{9.10.9}$$

where the relaxation spectrum $\phi(\alpha)$ satisfies the conditions

$$0 \leq \phi(\alpha), \qquad \phi(\infty) \leq 1 \tag{9.10.10}$$

These characterizations of the memory functions are in accordance with the current statistical mechanical models of elastomers and linear polymers (cf. Rouse [1953], Bueche [1954], and Marvin and Oser [1962]). In terms of $G(t)$, (9.10.9) may be written as

$$G(t) = \mu_e - \mu_e \int_0^\infty \phi(\alpha)(1 - e^{-t/\alpha})\, d\alpha \tag{9.10.11}$$

where $G(t)$ also satisfies the inequality

$$0 \leq G(\infty)/\mu_e$$

Similar representations are valid for $K(t)$. The character of the creep moduli $B(t)$ and $J(t)$ can be obtained from the properties of the inversion of (9.10.6) (cf. Hunter [1960], Gross [1953], Alfrey [1948]).

For uniaxial tension and compression tests, the following memory functions are found to be useful

$$\begin{aligned} \bar{E}(\zeta) &\equiv \frac{9\bar{G}(\zeta)\bar{K}(\zeta)}{3\bar{K}(\zeta) + \bar{G}(\zeta)} \\ \bar{\nu}(\zeta) &\equiv \frac{3\bar{K}(\zeta) - 2\bar{G}(\zeta)}{6\bar{K}(\zeta) + 2\bar{G}(\zeta)} \end{aligned} \tag{9.10.12}$$

Relaxation Young's modulus $E(t)$ and *Poisson's ratio* $\nu(t)$ correspond to Young's modulus and Poisson's ratio of classical elasticity and are often employed in the application of the principle of correspondence and in passage to elastic limits, etc.

We now sketch the experimental procedure for the determination of viscoelastic moduli.

(*a*) *Quasi-Static Test.* The material moduli $E(t)$ and $G(t)$—sometimes named relaxation moduli—can be determined from either a constant strain-rate test or a constant strain test. The constant strain test is called *stress relaxation.* For the determination of $E(t)$, a uniaxial test is often performed, and for $G(t)$, a shear or torsion test. Consider now a test in which all stress components vanish except $t_{11}(t)$ and

$$\epsilon_{11} = e(t), \tag{9.10.13}$$

which also vanish for $t < 0$. The Laplace transform of (9.10.1) with the use of (9.10.12) now leads to $\bar{t}_{11} = \bar{E}\bar{\zeta}\bar{e}$, or upon inversion

$$t_{11} = \int_0^t E(t-\tau)e(\tau)\,d\tau, \qquad \text{all other } t_{kl} = 0 \tag{9.10.14}$$

If we now take the axial strain $e(t)$ constant for all $t > 0$ and zero for all $t < 0$, that is,

$$e(t) = eH(t)$$

(where $H(t)$ is the Heaviside unit function defined as 1 for $t > 0$ and zero for $t < 0$), then (9.10.14) yields

$$t_{11} = E(t)e \tag{9.10.15}$$

The state of strain in this case corresponds to a uniaxial state of stress. Therefore by applying constant axial strain to a tension bar and measuring the time behavior of the axial tension one can determine $E(t)$. In practice, the step function $H(t)$ will have to be approximated.

It is equally possible to use the constant strain-rate test. In this case, in (9.10.14) $\dot{e}$ is constant, and we obtain

$$E(t) = \frac{1}{\dot{e}}\frac{dt_{11}}{dt} = \frac{dt_{11}}{de} \tag{9.10.16}$$

Therefore $E(t)$ is determined from the slope of the stress-strain curve at a time $t = e/\dot{e}$.

For the determination of the shear modulus $G(t)$ the relaxation test may be used in connection with a state of simple shear. Thus by taking

$$\tilde{e}_{12} = \gamma H(t), \qquad \tilde{e}_{kl} = 0 \qquad (k, l \neq 1, 2) \tag{9.10.17}$$

where γ is constant, we find from (9.10.1) that

$$t_{12} = \gamma G(t), \qquad t_{kl} = 0 \qquad (k, l \neq 1, 2) \tag{9.10.18}$$

Hence in a simple-shear relaxation test where the shearing strain is constant, by measuring the time behavior of t_{12}, one can determine $G(t)$. Of course, one may also employ the constant-shear strain-rate test. Sometimes a torsion test is used to produce a state of simple shear. This is accomplished by applying a constant twist to a circular cylindrical bar and measuring the torque as a function of time.

Once $E(t)$ and $G(t)$ are known, (9.10.12) may be inverted to give $K(t)$ and with this (9.10.5) gives $\lambda_v(t)$ and $\mu_v(t)$ since the Lamé elastic constants λ_e and μ_e are already known from elastic tests. Relaxation Young's modulus $E(t)$ for a propellant material, at several different temperatures, is plotted on Figure 9.10.1 for illustrative purposes. Relaxation bulk modulus $K(t)$ and shear modulus $G(t)$ of polyisobutylene are shown in Figure 9.10.2.

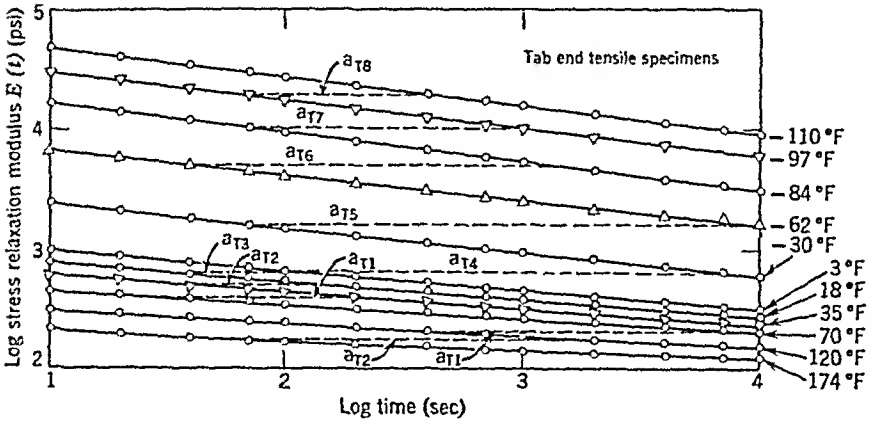


Fig. 9.10.1 Stress relaxation data, polycarbutene R propellant (after Jones [1963]).

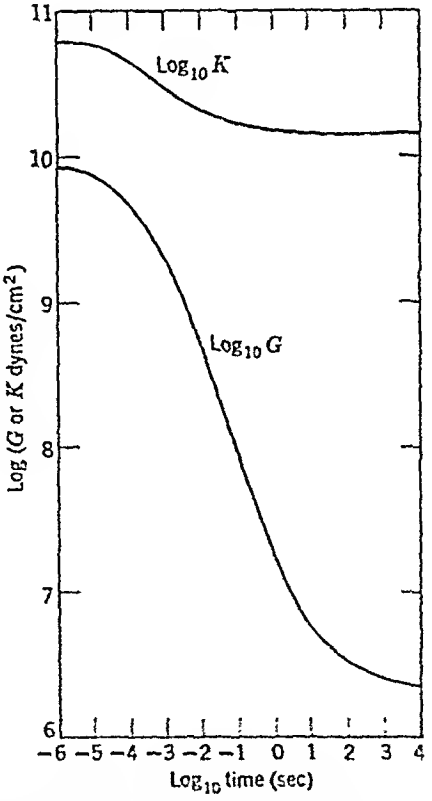


Fig. 9.10.2 Relaxation moduli of polyisobutylene in dilatation (K) and shear (G) (after Huang, Lee, and Rogers [1963]).

In principle, the creep moduli $B(t)$ and $J(t)$ could be determined from (9.10.6) by inversion. However, this process is complicated and leads to inaccuracies. Instead, creep tests are preferred. The procedure in such cases is to apply constant-stress or stress-rate tests in uniaxial tension and shear and use (9.10.2) to determine $B(t)$ and $J(t)$ The method is analogous to the relaxation test discussed earlier. Sometimes $\nu(t)$ is determined by measuring the volume change in time upon the application of uniaxial strain.

It is clear that all of these quasi-static tests are suggested by the methods known in classical elasticity. In general, several different tests are performed in order to obtain reliable data at different time ranges. For example, the stress-relaxation test provides reliable information for short-time behavior, and the creep test provides similar information for long-time behavior The viscoelastic moduli are determined in a temperature-controlled chamber for fixed uniform temperatures. Test procedure and the difficulties involved in such measurements are discussed by Britton [1965] where references are also made to other industrial work

(*b*) ***Dynamic Tests.*** The dynamic tests may be carried out to determine the viscoelastic moduli $\lambda_v(t, T)$, $\mu_v(t, T)$ or $E(t, T)$ and $G(t, T)$ Either a vibration test or a wave-propagation test may be used The dynamic tests are suitable for the determination of complex moduli

$$\bar{\lambda}_v(\omega, T), \bar{\mu}_v(\omega, T), \qquad \bar{E}(\omega, T), \qquad \bar{G}(\omega, T)$$

which are the Fourier transforms of the corresponding viscoelastic moduli Here ω represents the circular frequency. For a reasonably accurate characterization of these functions usually a wide range of frequencies is needed (for example, for rubber 0.1 to 10^5 cps).

Below we give briefly some results obtained by Hillier [1949], Kolsky [1956], and others. These experiments consist of measurement of amplitude and phase velocities at various points along prestretched filaments of polyethene, neopene, and nylon that are set into steady-state extensional vibrations at one end.

Propagation of waves in a bar based on the Fourier transform of the one-dimensional equation of motion

$$\frac{\partial t_{11}}{\partial x} = \rho \frac{\partial^2 u}{\partial t^2} \tag{9.10.19}$$

which is obtained by setting $t_{kl} = 0$ for all k, l except $k = l = 1$ in the equations of balance of momenta. In this case, the Fourier transform of the constitutive equation (9 10 14) gives

$$\bar{t}_{11} = E^*(\omega) \frac{\partial \bar{u}(x, \omega)}{\partial x} \tag{9.10.20}$$

where we used $e = u_{,x}H(t)$ and introduced

$$E^*(\omega) \equiv -i\omega\bar{E}(\omega) \tag{9.10.21}$$

which is sometimes called *complex* or *dynamic Young's modulus*. Here quantities carrying a bar represent the Fourier transform, for example,

$$\bar{u}(x, \omega) \equiv \int_{-\infty}^{\infty} e^{i\omega t}u(x, t)\, dt, \qquad i \equiv \sqrt{-1}$$

Upon substituting (9.10.20) into (9.10.19), we get

$$\frac{\partial^2 \bar{u}}{\partial x^2} + \lambda^2(\omega)\bar{u} = 0 \tag{9.10.22}$$

where

$$\lambda(\omega) = \lambda_1(\omega) + i\lambda_2(\omega) = \omega[\rho/E^*(\omega)]^{1/2}, \qquad \lambda_1 > 0 \tag{9.10.23}$$

The general solution of (9.10.22) has the form

$$\bar{u} = A(\omega)e^{\lambda x} + B(\omega)e^{-\lambda x} \tag{9.10.24}$$

where A and B are determined from the boundary conditions. In the experimental work, we are interested only in the wave velocity and attenuation. Thus only the steady-state solutions are needed. Defining two positive even functions

$$\alpha(\omega) \equiv \lambda_1(\omega), \qquad c(\omega) \equiv \omega/\lambda_2(\omega) \tag{9.10.25}$$

we see that exponentially damped harmonic waves are expressible in terms of the eigenfunctions

$$e^{-\alpha(\omega)x}\begin{Bmatrix}\cos \\ \sin\end{Bmatrix}\omega\left[t \mp \frac{x}{c(\omega)}\right] \tag{9.10.26}$$

where the sign is $-$ for waves propagating in the direction of $x > 0$ and $+$ for $x < 0$. We now set

$$E^*(\omega) = E_1(\omega) + iE_2(\omega) = |E^*(\omega)|e^{i\Delta(\omega)} \tag{9.10.27}$$

where $E_1(\omega)$, $E_2(\omega)$, and $\tan \Delta$ are real, and sometimes called *storage modulus*, *loss modulus*, and *loss tangent*, respectively. This gives for the attenuation constant $\alpha(\omega)$ and phase velocity $c(\omega)$

$$\frac{\alpha c}{\omega} = \tan\frac{\Delta}{2}, \qquad c = \left\{\frac{|E^*(i\omega)|}{\rho}\right\}^{1/2}\sec\frac{\Delta}{2} \tag{9.10.28}$$

By determining $c(\omega)$ and $\alpha(\omega)$ experimentally, through (9.10.28) and (9.10.27) we can calculate E_1 and E_2, and hence $\bar{E}(\omega)$.

The vibration frequencies 1 to 16 kc/sec are used in a temperature range 0 to 40°C. The phase velocity and attenuation in polythene (I.C.I Alkathene—grade 20) at 10°C are given in Figure 9.10.3 (Kolsky [1956]). For polyethene at 45°C, $E_1(\omega)$ and $E_2(\omega)$ are shown in Figure 9.10.4. This curve is taken from Hunter [1960] where a detailed discussion is given of such experiments

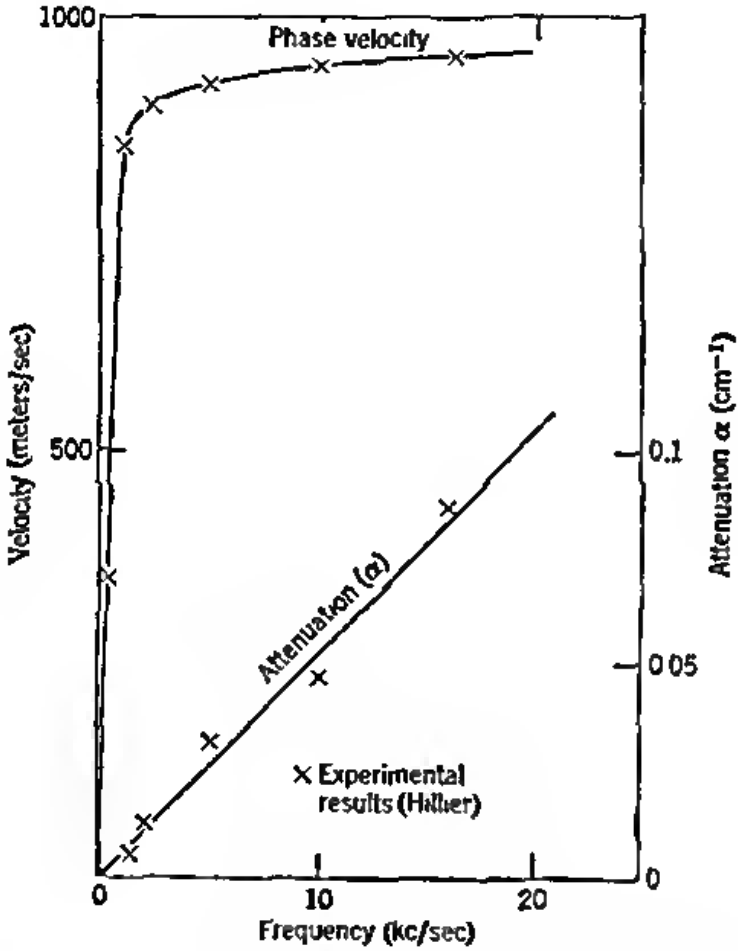


Fig. 9.10.3 Experimental measurements of attenuation and velocity in polythene filaments at 10°C (after Kolsky [1956]).

Instead of extensional ultrasonic wave propagations one may use shear or torsional oscillations to determine $\bar{G}(\omega)$. Such experiments were conducted by Gottenberg and Christensen [1964] for torsional oscillations and the results are shown on Figures 9.10.5 and 9 10.6 for a solid rocket-propellant material consisting of 20 percent polyurethane matrix and 80 percent salt crystals and aluminum powder. Based on these curves, these

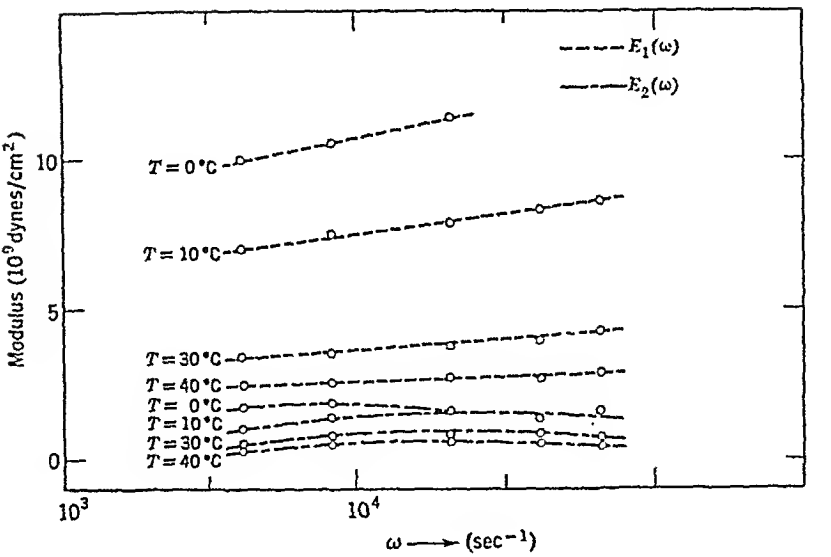


Fig. 9.10.4 Complex modulus of polythene.

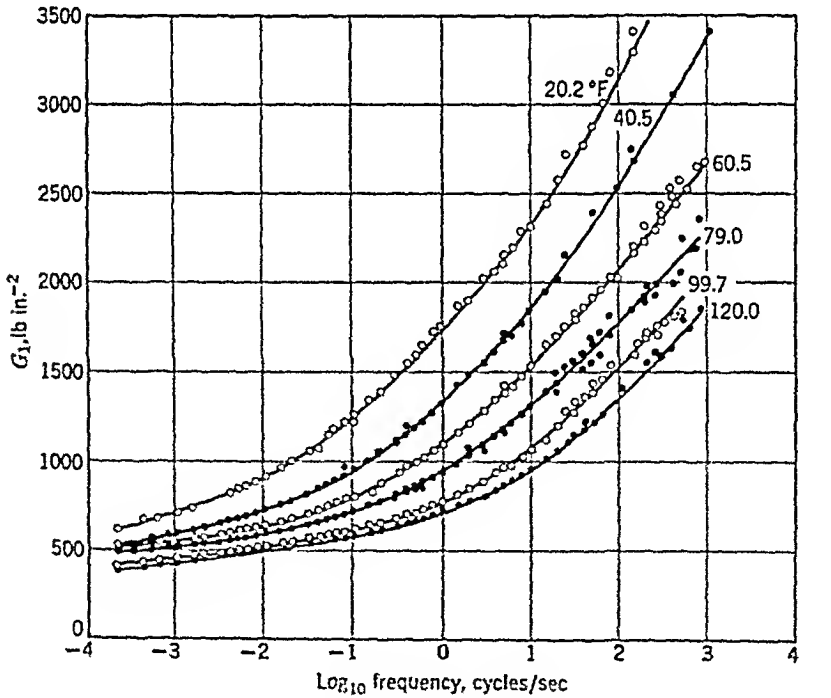


Fig. 9.10.5 Real part of complex shear modulus (after Gottenberg and Christensen [1964]).

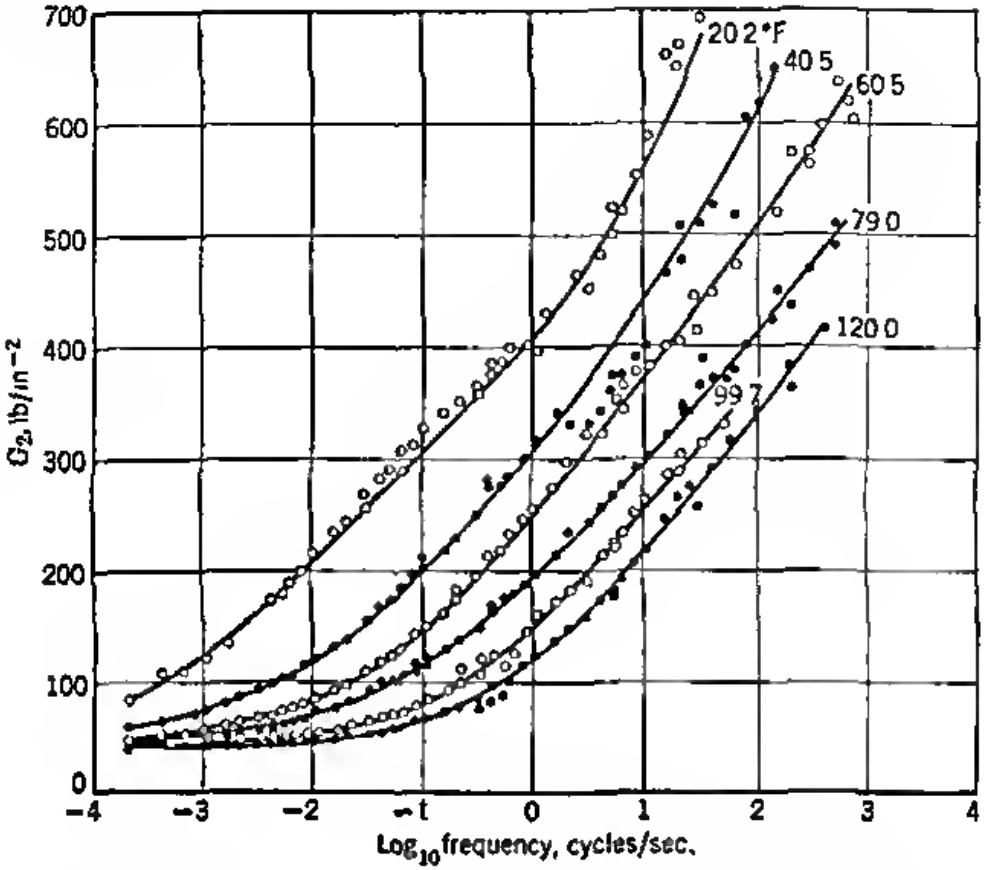


Fig. 9.10.6 Imaginary part of complex shear modulus (after Gottenberg and Christensen [1964])

authors have also calculated the relaxation modulus through the inversion of

$$G^*(\omega) = G_1(\omega) + iG_2(\omega) \equiv i\omega \int_0^\infty e^{-i\omega t} G(t)\, dt \tag{9.10.29}$$

$$= i\omega \bar{G}(-\omega)$$

where $\bar{G}(\omega)$ is the Fourier transform of $G(t)$ with $G(t) = 0$ for $t < 0$ This is shown in Figure 9.10 7.

The ICRPG manual, Section 4.6, contains another method for the determination of $G^*(\omega)$ by use of the vibrating circular plates clamped at the outer edge, subject to sinusoidal displacements along an inner circle (Section 4 6.2) and sinusoidal shear applied to a specimen (Section 4 6.2 3) In Figures 9 10 8 and 9.10.9 is presented dynamic *shear compliance* $J^*(\omega)$ defined by

$$J^*(\omega) = J_1(\omega) - iJ_2(\omega) \equiv i\omega \bar{J}(-\omega) \tag{9.10.30}$$

where $J_1(\omega)$ and $J_2(\omega)$ are real and they are named *storage* and *loss compliance*, respectively. Here a superposed bar indicates Fourier transform.

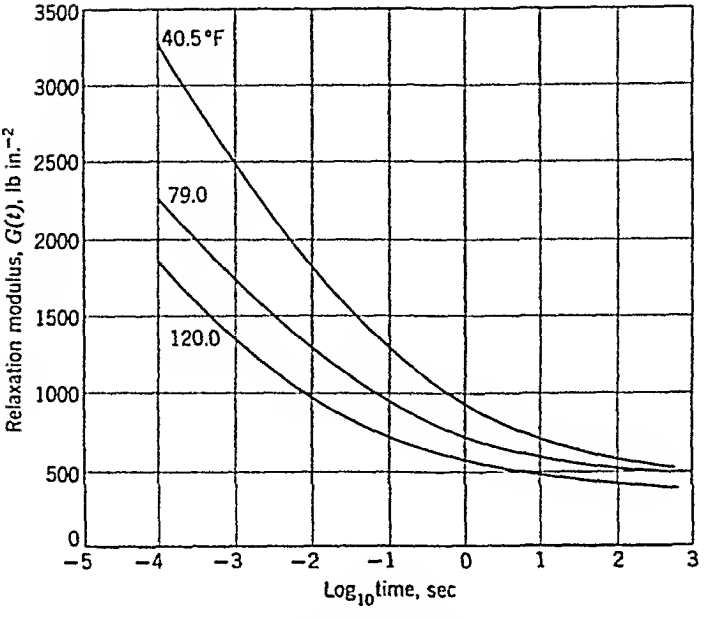


Fig. 9.10.7 Calculated relaxation modulus in shear (after Gottenberg and Christensen [1964]).

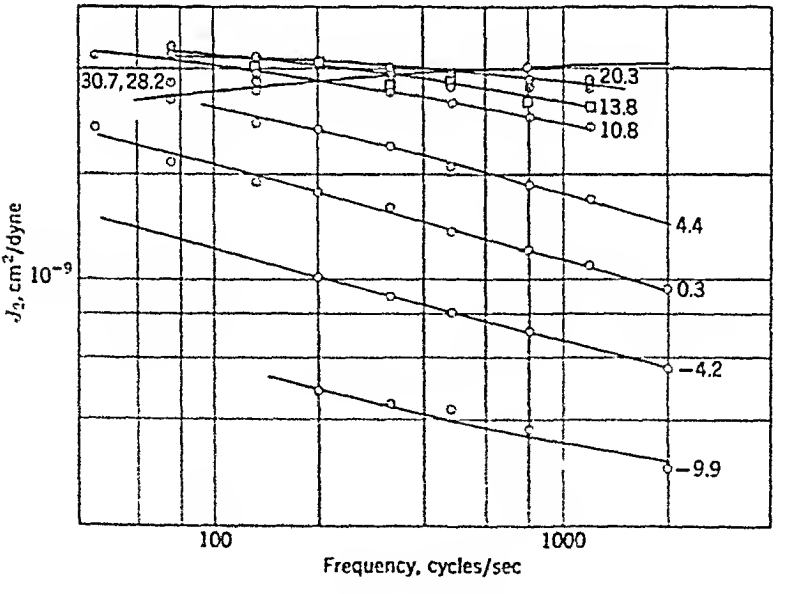


Fig. 9.10.8 Loss shear compliance of a polyurethane propellant at various frequencies and temperatures.

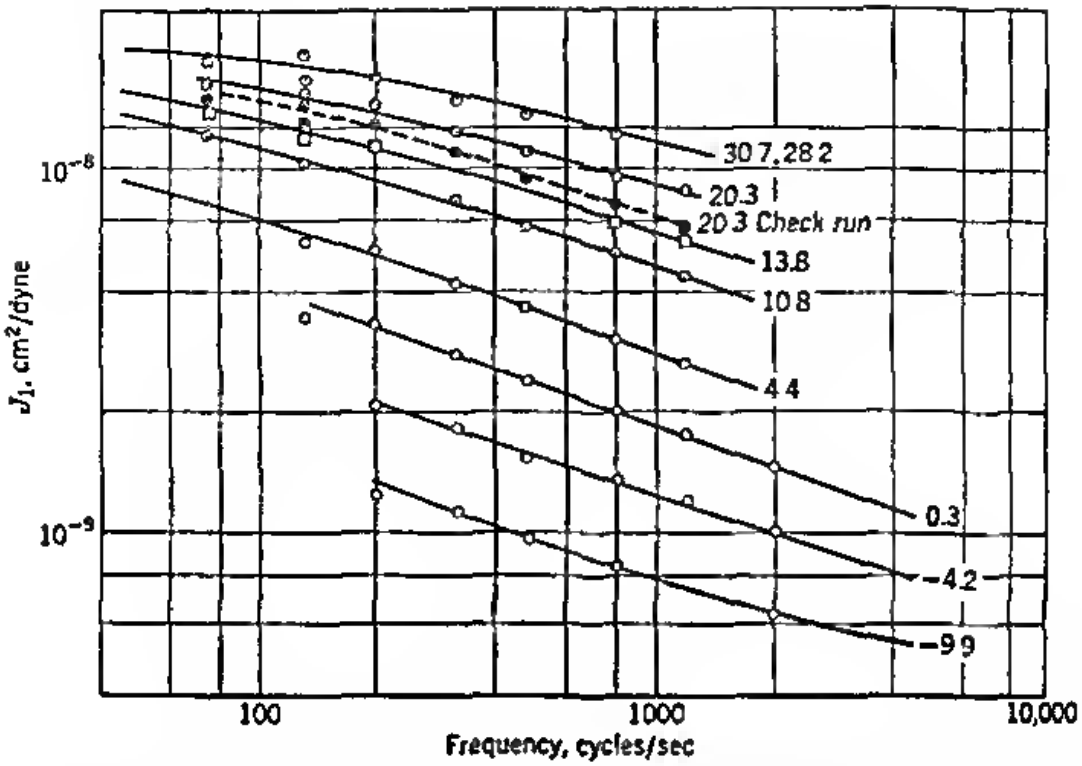


Fig. 9.10.9 Storage shear compliance of a polyurethane propellant at various frequencies and temperatures (after Landel [1963]).

By use of $(9.10.6)_2$ with $\zeta = -i\omega$ from these data one can calculate $\bar{G}(\omega)$ and hence $G(t)$ (Figure 9.10.10).

Thermal Effect. Viscoelastic moduli are highly sensitive to thermal changes. Ferry [1950] verified extensively that, for most polymeric materials, by a change in time scale we can reproduce any viscoelastic modulus at a given temperature T from its value at a fixed temperature T_s. Thus if we set

$$\xi \equiv t/a_T \tag{9.10.31}$$

where t is time and a_T is a function of $T - T_s$, then

$$E(\xi, T) = E(t, T_s) \tag{9.10.32}$$

similar expressions being valid for other moduli. Viscoelastic materials obeying this rule are sometimes called *thermorheologically simple materials* For a large class of elastomers it has been estimated that

$$\log_{10} a_T = \frac{-8.86(T - T_s)}{101.6 + T - T_s} \tag{9.10.33}$$

is valid for a temperature range $|T - T_s| = 50°C$. The best fit appears to be $T_s = 45°C$. In view of an empirical relation

$$T_s = T_g + 50°C \tag{9.10.34}$$

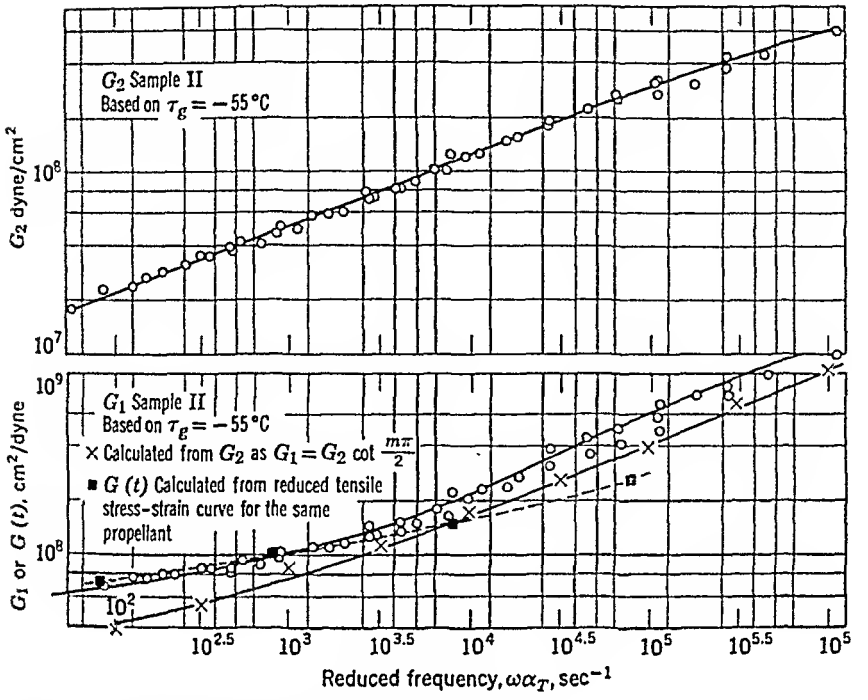


Fig. 9.10.10 Dependence of storage and loss compliance on reduced frequency for a polyurethane propellant (after Landel [1963]).

observed by Williams, Landell, and Ferry [1955] relating T_s to a *glass transition temperature* T_g, it turns out that when $T_0 - T_g$ varies significantly from 45°C, the constants 8.86 and 101.6 appearing in (9.10.33) must therefore be altered significantly.

In Figure 9.10.11 we give $E_1(\omega)$ obtained from Figure 9.10.4 by this process of shift. Similarly, Figure 9.10.12 was obtained from (9.10.1) by

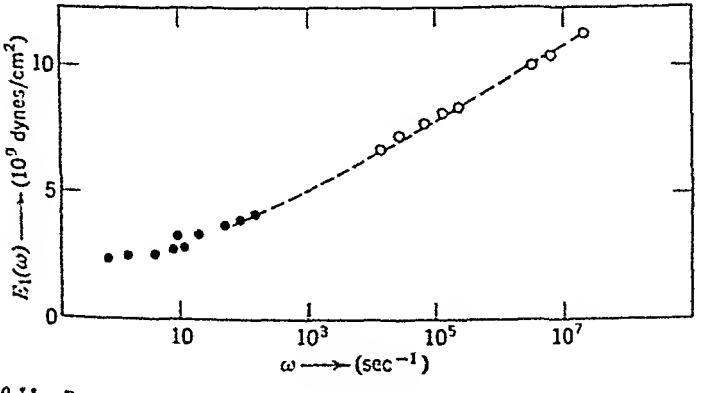


Fig. 9.10.11 Dynamic modulus of polythene at reference temperature $T_s = 45°C$.

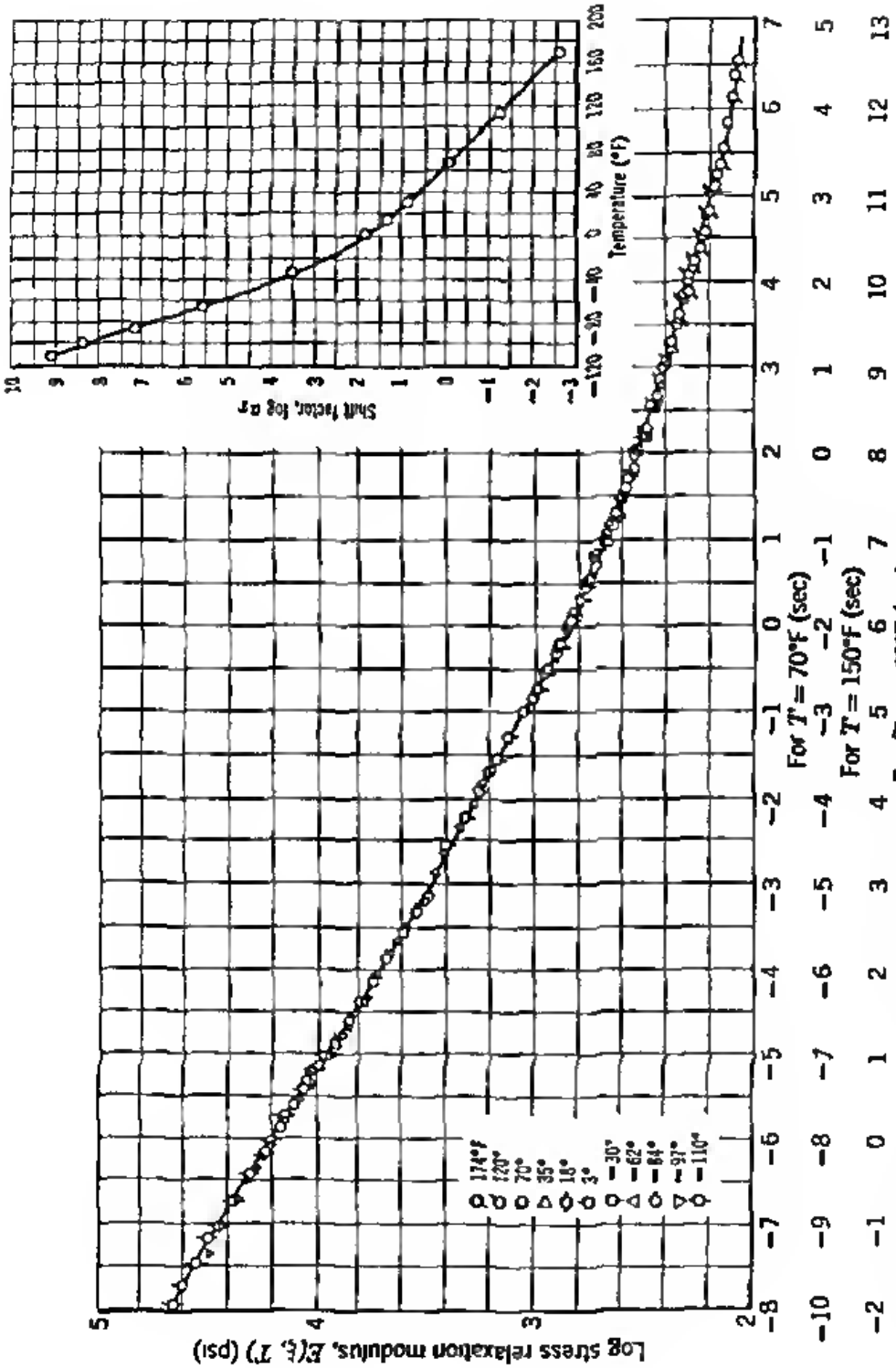


Fig. 9.10.12 Master stress relaxation modulus at $T = 70°$, $150°$, and $-68°F$, polycarbutene R propellant (after Jones [1963]).

this process. For other data on viscoelastic properties of polymers, see Ferry [1961] and Eirich [1956, 1958]. See also Landel [1958] for the properties of polyisobutylene-glass beads; Schwarzl, Bree, and Nederveen [1963] for highly filled elastomers; and Takayanagi [1963] for crystalline polymers. A recent review of theoretical and experimental work is also to be found in Eringen, Liebowitz, Crowley, and Koh [1967].

9.11. RESUMÉ OF BASIC EQUATIONS

The motions of non-heat-conducting linear isotropic viscoelastic materials are governed by the following set of equations:

Equations of Balance

$$\rho_0 = \rho(1 + u_{k,k}) \qquad \text{(mass)} \tag{9.11.1}$$

$$t_{lk,l} + \rho(f_k - \ddot{u}_k) = 0 \qquad \text{(momentum)} \tag{9.11.2}$$

$$t_{kl} = t_{lk} \qquad \text{(moment of momentum)} \tag{9.11.3}$$

Constitutive Equations

$$Rt_{rr}\,\delta_{kl} + 2St_{kl} = P\tilde{e}_{rr}\,\delta_{kl} + 2Q\tilde{e}_{kl} \qquad \text{(differential form)} \tag{9.11.4}$$

$$t_{kl} = \lambda_e \tilde{e}_{rr}\,\delta_{kl} + 2\mu_e \tilde{e}_{kl} + \int_{-\infty}^{t} \left[\lambda_v(t-s) \frac{\partial \tilde{e}_{rr}(s)}{\partial s}\,\delta_{kl} + 2\mu_v(t-s) \frac{\partial \tilde{e}_{kl}}{\partial s} \right] ds \qquad \text{(functional form)} \tag{9.11.5}$$

where R, S, P, and Q are differential operators of the form (9.7.4), and $\lambda_v(t)$ and $\mu_v(t)$ are memory functions all of which are to be determined by experimental investigations (cf. Section 9.10). The memory functions must satisfy the axiom of memory. Two special forms of (9.11.4) providing some simplicity for mathematical work are

$$t_{kl} = \left(\lambda_e + \lambda_v \frac{\partial}{\partial t}\right) \tilde{e}_{rr}\,\delta_{kl} + 2\left(\mu_e + \mu_v \frac{\partial}{\partial t}\right) \tilde{e}_{kl} \qquad \text{(Kelvin-Voigt solid)} \tag{9.11.6}$$

$$\dot{t}_{kl} + \alpha_m t_{rr}\,\delta_{kl} + 2\beta_m t_{kl} = \lambda_m \dot{\tilde{e}}_{rr}\,\delta_{kl} + 2\mu_m \dot{\tilde{e}}_{kl} \qquad \text{(Maxwell solid)} \tag{9.11.7}$$

where the material moduli $\lambda_e, \lambda_v, \mu_e, \mu_v, \alpha_m, \beta_m, \lambda_m$, and μ_m are constants for homogeneous, isothermal solids, functions of X for nonhomogeneous solids, and temperature T for nonisothermal solids.

Other special forms of (9.11.4) may be obtained by retaining various terms in the operators R, S, P, and Q.

In nontbermal problems $\mathbf{q} = \mathbf{0}$, and no need arises for the description of ε and η.

The above constitutive equations are for compressible solids. For *incompressible solids*, we set $\tilde{e}_{rr} = 0$ and replace $\lambda\tilde{e}_{rr}$ and $\lambda_m\tilde{e}_{rr}$ in these equations by an unknown pressure term $-p(\mathbf{x}, t)$.

Strain-Displacement Relations. The infinitesimal strains $\tilde{e}_{kl}$ are related to the displacements u_k by

$$2\tilde{e}_{kl} = u_{k,l} + u_{l,k} \tag{9.11.8}$$

Field Equations. The displacement field $\mathbf{u}$ is governed by the field equations that result from combining (9.11 2), (9.11.8), and one of (9.11.4) to (9.11.7). We have seen that (cf. Section 9.7) we can pass from (9.11.4) to the functional form (9 11.5) and vice versa. Therefore, we give the field equations resulting from the use of (9 11.5).

$$(\lambda_e + \mu_e)u_{l,lk} + \mu_e u_{k,ll} + \int_{-\infty}^{t} \left\{ [\lambda_v(t-s) + \mu_v(t-s)] \frac{\partial u_{l,lk}(s)}{\partial s} + \mu_v(t-s) \frac{\partial u_{k,ll}(s)}{\partial s} \right\} ds + \rho(f_k - \ddot{u}_k) = 0 \tag{9.11.9}$$

The vector form of (9.11 9), useful in the ease of curvilinear coordinates, is obtained in the same manner as in Section 6 6.

$$(\lambda_e + 2\mu_e)\nabla\nabla \cdot \mathbf{u} - \mu_e \nabla \times \nabla \times \mathbf{u} + \int_{-\infty}^{t} \left\{ [\lambda_v(t-s) + 2\mu_v(t-s)]\nabla\nabla \cdot \frac{\partial \mathbf{u}(s)}{\partial s} - \mu_v(t-s)\nabla \times \nabla \times \frac{\partial \mathbf{u}(s)}{\partial s} \right\} ds + \rho(\mathbf{f} - \ddot{\mathbf{u}}) = \mathbf{0} \tag{9.11.10}$$

For problems in which the state of body is undeformed and stress free, for $t < 0$ lower limits of these integrals are replaced by zero.

Boundary Conditions. A set of usual boundary conditions is

$$t_{lk}n_l = s_k \quad \text{on } \mathscr{S}_t \tag{9.11.11}$$

$$u_k = w_k \quad \text{on } \mathscr{S}_u \tag{9.11.12}$$

for all t

where s_k is the surface traction prescribed on a part $\mathscr{S}_t$ of the surface of the body $\mathscr{S} = \mathscr{S}_t + \mathscr{S}_u$ and w_k is the displacement field prescribed over the remainder of the surface $\mathscr{S}_u$. Other "mixed-mixed" types of conditions are possible (cf. footnote, Section 6.5). All possible boundary conditions must be admissible on the ground of existence and uniqueness theorems. Such

theorems were given recently by Breuer and Onat [1962] and Gurtin and Sternberg [1962].

Initial Conditions. The usual set of initial conditions is

$$u_k(\mathrm{x}, 0) = u_k^0(\mathrm{x}) \qquad \text{in } \mathscr{V} \tag{9.11.13}$$

$$\dot{u}_k(\mathrm{x}, 0) = v_k^0(\mathrm{x}) \qquad \text{in } \mathscr{V} \tag{9.11.14}$$

The determination of the displacement field in a linear isotropic viscoelastic solid is therefore reduced to the solution of three partial differential equations (9.11.9) under requisite boundary and initial conditions. Once u is so determined we can calculate the stress through (9.11.5) and (9.11.8).

Anisotropic Solids. For the initial stress-free ($\sigma_{kl} = 0$) anisotropic solids the stress constitutive equations (9.6.25) reduce to

$$t_{kl} = \sigma_{klmn}\tilde{e}_{mn} + \int_{-\infty}^{t} \gamma_{klmn}(t - s)\frac{\partial \tilde{e}_{mn}(\mathrm{s})}{\partial s}\, ds \tag{9.11.15}$$

so that the field equations in this case are

$$\sigma_{klmn}u_{m,nk} + \int_{-\infty}^{t} \gamma_{klmn}(t - s)\frac{\partial}{\partial s}\, u_{m,nk}(s)\, ds + \rho(f_l - \ddot{u}_l) = 0 \tag{9.11.16}$$

Thermoviscoelastic Solids. For isotropic solids the only change is in the stress and heat constitutive equations. They have the forms

$$t_{kl} = -\beta T\, \delta_{kl} + \lambda_e \tilde{e}_{rr}\, \delta_{kl} + 2\mu_e \tilde{e}_{kl} + \int_{-\infty}^{t} \left[\lambda_v(t - s, T)\frac{\partial \tilde{e}_{rr}(s)}{\partial s}\, \delta_{kl} + 2\mu_v(t - s, T)\frac{\partial \tilde{e}_{kl}(s)}{\partial s}\right] ds \tag{9.11.17}$$

$$q_k = \kappa T_{,k} \tag{9.11.18}$$

where $\beta(T)$, $\lambda_e(T)$, $\mu_e(T)$, and $\kappa(T)$ are functions of temperature T and $\lambda_v(t, T)$ and $\mu_v(t, T)$ are functions of t and T. For thermorheologically simple materials, by a shift in time scale, (9.10.31), one can reproduce any viscoelastic modulus at a given temperature T from its value at a fixed temperature (cf. 9.10.32). Such a change of time variable may be used, as convenience demands, in the analytical treatment of problems of thermoviscoelasticity.

For the Kelvin-Voigt solid, equation (9.11.17) takes the special form

$$t_{kl} = -\beta T\delta_{kl} + \left(\lambda_e + \lambda_v \frac{\partial}{\partial t}\right)\tilde{e}_{rr}\delta_{kl} + 2\left(\mu_e + \mu_v \frac{\partial}{\partial t}\right)\tilde{e}_{kl} \tag{9.11.19}$$

where β, λ_e, μ_e, λ_v, and μ_v are functions of temperature only.

For higher order rate-dependent materials we have the form (9.8.32). The equations of balance are now supplemented by the equation of heat conduction (9.8.31). The field equations of the theory of linear thermoviscoelasticity for isotropic solids are therefore

$$(9.11.20)\qquad -\beta T_{,k} + (\lambda_e + \mu_e)u_{l,lk} + \mu_e u_{k,ll} + \int_{-\infty}^{t} \left\{ [\lambda_v(t-s, T) + \mu_v(t-s, T)] \frac{\partial u_{l,lk}(s)}{\partial s} + \mu_v(t-s, T) \frac{\partial u_{k,ll}(s)}{\partial s} \right\} ds + \rho(f_k - \ddot{u}_k) = 0,$$

$$(9.11.21)\qquad -\rho_0 \gamma \dot{T} - \beta T_0 \dot{u}_{k,k} + (\kappa T_{,k})_{,k} + \rho h = 0$$

The boundary and initial conditions (9.11.11) to (9.11.14) are supplemented by conditions of the form

$$(9.11.22)\qquad \begin{aligned} \mathbf{q}\cdot\mathbf{n} &= q_{(n)} \qquad \text{on } \mathscr{S}_q \\ T &= T_s \qquad \text{on } \mathscr{S}_T = \mathscr{S} - \mathscr{S}_q \end{aligned}$$

and

$$(9.11.23)\qquad T(\mathbf{x}, 0) = T^0(\mathbf{x}) \qquad \text{in } \mathscr{V}$$

where $q_{(n)}$, T_s, and T^0 are prescribed. Other mixed conditions of the type (8.4.28) are possible.

Thermoviscoelastic Fluids. For the constitutive equations and the equation of heat conduction for thermoviscoelastic fluids, see Section 9.9. For nonlinear Stokesian thermofluids, see Section 5.7.

9.12. THEOREM OF CORRESPONDENCE

The solutions of all boundary-value problems concerning linear viscoelastic solids with time independent boundaries can be obtained from the solutions of corresponding problems in elasticity. This is known as the *theorem of correspondence*. This theorem has many applications. We give this theorem for non-heat-conducting solids. Extension to thermoviscoelasticity is obvious.

We assume that the solid is undisturbed prior to time $t = 0$ and take the Laplace transform of (9.11.10) subject to (9.11.13) and (9.11.14). Thus

(9.12.1)

$$[\lambda(\zeta) + 2\mu(\zeta)]\nabla\nabla\cdot\bar{\mathbf{u}} - \mu(\zeta)\nabla\times\nabla\times\bar{\mathbf{u}} + \rho(\mathbf{F}^* - \zeta^2\bar{\mathbf{u}} + \zeta\mathbf{u}^0 + \mathbf{v}^0) = 0$$

where

$$\lambda(\zeta) \equiv \lambda_e + \zeta\bar{\lambda}_v(\zeta), \qquad \mu(\zeta) \equiv \mu_e + \zeta\bar{\mu}_v(\zeta) \tag{9.12.2}$$

$$\rho F^* \equiv \rho\bar{\mathbf{f}} - [\bar{\lambda}_v(\zeta) + 2\bar{\mu}_v(\zeta)]\nabla\nabla \cdot \mathbf{u}^0 + \bar{\mu}_v(\zeta)\nabla \times \nabla \times \mathbf{u}^0 \tag{9.12.3}$$

and a superposed bar indicates the Laplace transform, for example,

$$\bar{\mathbf{u}}(\mathbf{x}, \zeta) = \int_0^\infty \mathbf{u}(\mathbf{x}, t)e^{-\zeta t}\, dt \tag{9.12.4}$$

Assuming $\mathscr{S}$ is not time dependent, the Laplace transforms of the boundary conditions (9.11.11) and (9.11.12) with t_{kl} given by (9.11.5) are

$$[\lambda(\zeta)\bar{u}_{r,r}\,\delta_{kl} + \mu(\zeta)(\bar{u}_{k,l} + \bar{u}_{l,k})]n_k = T_l^* \qquad \text{on } \mathscr{S}_t \tag{9.12.5}$$

$$\bar{u}_k = \bar{w}_k \qquad \text{on } \mathscr{S}_u \equiv \mathscr{S} - \mathscr{S}_t \tag{9.12.6}$$

where

$$T_l^* \equiv \bar{s}_l + [\bar{\lambda}_v(\zeta)u^0_{r,r}\,\delta_{kl} + \bar{\mu}_v(\zeta)(u^0_{k,l} + u^0_{l,k})]n_k \tag{9.12.7}$$

A comparison of (9.12.1) through (9.12.6) with the Laplace transforms of the corresponding equations of the theory of elasticity reveals the

THEOREM OF CORRESPONDENCE.[11] *The solution of the Laplace transform of any mixed boundary-value problem of linear isotropic viscoelasticity (for boundary conditions on time independent boundary surfaces) is identical with that of linear isotropic elasticity in which* λ_e, μ_e, $\bar{\mathbf{f}}$, *and* $\bar{\mathbf{s}}$ *are replaced by* $\lambda(\zeta)$, $\mu(\zeta)$, $\mathbf{T}^*$, *and* $\mathbf{F}^*$, respectively.

To obtain the solution in the time domain we must, of course, invert the transform so obtained. It is now clear that by use of the known solutions of various problems of elasticity theory, we can construct the solutions of corresponding problems in viscoelasticity. Some illustrative examples are given in the following sections.

It is important to note here that the theorem of correspondence does not apply to solids having time-dependent boundaries. This is because of the impossibility of obtaining the Laplace transforms of the boundary conditions. Thus, for example, in an indentation problem such as that of a rigid sphere impacting a viscoelastic solid, the boundary line of the contact surface changes in time. In this case, the history of loading and/or the deformation is needed on $\mathscr{S}_t$, which is changing in time. The theorem of

[11] This analogy has its origin in a paper by Alfrey [1944]. This and most of its later improvements disregard the initial conditions, cf. Lee [1955] and Sternberg [1964]. The last reference gives an account of the evolution of the analogy. See also Biot [1965, p. 349].

correspondence may not be used in such a problem, but may be used in all other boundary-value problems. Nevertheless, in general, it is very difficult to invert the transformed solutions of the problems of elasticity with the above-mentioned replacement. The inversion of integrals requires contour integrations in the complex ζ-plane and often contain complicated singularities Thus, in spite of the tool provided by the theorem of correspondence, the inversion of the solution is a problem of great computational difficulty.

It must be noted that instead of the Laplace transform, to construct a theorem of correspondence we may employ the Fourier transform, for example,

$$\bar{u}(x, \omega) = \int_{-\infty}^{\infty} u(x, t) e^{i\omega t}\, dt \tag{9.12 8}$$

In this case, the initial conditions cannot be taken into account, and we are dealing with so-called *steady-state* problems. Other than that, the above theorem remains valid except, of course, that in place of the Laplace transform variable ζ, we must substitute $-i\omega$ in all functions, for example, we write

$$\bar{\lambda}_v(\omega) = \bar{\lambda}_v(\zeta)\Big|_{\zeta = -i\omega}, \qquad i \equiv \sqrt{-1} \tag{9.12.9}$$

The inverse Fourier transform is given by

$$\lambda_v(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \bar{\lambda}_v(\omega) e^{-i\omega t}\, d\omega \tag{9.12.10}$$

Through the Laplace transforms of (9.11.6) and (9 11.7) we find the special forms of $\lambda(\zeta)$ and $\mu(\zeta)$ for Kelvin-Voigt and Maxwell models:

$$\begin{aligned} &\lambda(\zeta) \equiv \lambda_e + \lambda_v \zeta \\ &\mu(\zeta) \equiv \mu_e + \mu_v \zeta \qquad \text{(Kelvin-Voigt)} \end{aligned} \tag{9.12.11}$$

$$\begin{aligned} &\lambda(\zeta) \equiv \frac{-2\alpha_m \mu_m + \lambda_m(\zeta + 2\beta_m)}{\zeta + 2\beta_m + 3\alpha_m} \cdot \frac{\zeta}{\zeta + 2\beta_m} \\ &\mu(\zeta) \equiv \frac{\mu_m \zeta}{\zeta + 2\beta_m} \qquad \text{(Maxwell)} \end{aligned} \tag{9.12.12}$$

where λ_e, μ_e, λ_v, μ_v, α_m, β_m, λ_m, and μ_m are constants.

9.13. SOME QUASI-STATIC SOLUTIONS

In this section we give two simple solutions for illustrative purposes. These problems are concerned with deformations of viscoelastic solids in which inertia effects are negligible. Such solutions are called *quasistatic*. In viscoelasticity there is no truly static problem, since the time dependence is present in the constitutive equations as well as in the balance laws. However, for solids undergoing slow changes the inertia terms may be neglected, thus achieving a great deal of simplicity. These solutions also find applications in *creep* and *relaxation* experiments.

(i) Half Space Subject to Uniform Pressure. Suppose that at time $t = 0$ a uniform pressure $p(t)$ is applied to the plane surface $x = 0$ of the half space $x \geq 0$ and maintained. We wish to determine the stress and displacement fields at any place x at time t. Because of the symmetry present in the problem the state of deformation is uniform at any point on any plane $x =$ constant. Thus for the displacement components we have $u_x = u_x(x, t)$, $u_y = u_z = 0$ and the problem is one-dimensional. The inertia terms being neglected together with the body force, equations (9.11.9) for a solid (unstressed, and undeformed prior to $t = 0$ reduce to

$$(9.13.1) \qquad (\lambda_e + 2\mu_e)\frac{\partial^2 u_x}{\partial x^2} + \int_0^t [\lambda_v(t-s) + 2\mu_v(t-s)]\frac{\partial^3 u_x(x,s)}{\partial s \partial x^2}\,ds = 0$$

subject to boundary condition

$$(9.13.2) \qquad t_{xx} = (\lambda_e + 2\mu_e)\frac{\partial u_x}{\partial x} + \int_0^t [\lambda_v(t-s) + 2\mu_v(t-s)]\frac{\partial^2 u_x(x,s)}{\partial s \partial x}\,ds$$

$$= -p(t) \qquad \text{at } x = 0$$

We leave the condition at $x = \infty$ undecided. The Laplace transforms of (9.13.1) and (9.13.2), respectively, are

$$(9.13.3) \qquad \frac{d^2\bar{u}_x(x,\zeta)}{dx^2} = 0$$

$$(9.13.4) \qquad [\lambda(\zeta) + 2\mu(\zeta)]\frac{d\bar{u}_x(x,s)}{dx}\bigg|_{x=0} = -\bar{p}(\zeta)$$

where a bar placed on quantities indicates the Laplace transform, where ζ is the transform variable, and where

$$(9.13.5) \qquad \lambda(\zeta) \equiv \lambda_e + \zeta\bar{\lambda}_v(\zeta), \qquad \mu(\zeta) \equiv \mu_e + \zeta\bar{\mu}_v(\zeta)$$

The solution of (9.13.3) under (9.13.4) is

$$\bar{u}_x = -\frac{\bar{p}(\zeta)x}{\lambda(\zeta) + 2\mu(\zeta)} + \bar{C}_1(\zeta) \tag{9.13.6}$$

where $\bar{C}_1(\zeta)$ is an arbitrary function of ζ. The transform of the stress field is obtained through (9.11.5), (9.11.8), and (9.13.5) as

$$\begin{aligned} &\bar{t}_{xx} = -\bar{p}(\zeta), \qquad \bar{t}_{yy} = \bar{t}_{zz} = -\frac{\lambda(\zeta)}{\lambda(\zeta) + 2\mu(\zeta)}\bar{p}(\zeta) \\ &\bar{t}_{xy} = \bar{t}_{yz} = \bar{t}_{zx} = 0 \end{aligned} \tag{9.13.7}$$

The results (9.13.6) and (9.13.7) are identical in forms, respectively, to (6.7.3) and (6.7.4) of the corresponding problem in elasticity. This is, of course, as it should be in accordance with the theorem of correspondence. The displacement and stress fields are obtained by finding the inverse Laplace transform of (9.13.6) and (9.13.7). Formally, through the convolution theorem we can write

$$\begin{aligned} u_x(x, t) &= -x\int_0^t U(t-s)p(s)\,ds + C_1(t) \\ t_{yy} = t_{zz} &= -\int_0^t T(t-s)p(s)\,ds \\ t_{xx} &= -p(t), \qquad t_{xy} = t_{yz} = t_{zx} = 0 \end{aligned} \tag{9.13.8}$$

where

$$\begin{aligned} U(t) &\equiv L^{-1}\left\{\frac{1}{\lambda(\zeta) + 2\mu(\zeta)}\right\} \\ T(t) &\equiv L^{-1}\left\{\frac{\lambda(\zeta)}{\lambda(\zeta) + 2\mu(\zeta)}\right\} \end{aligned} \tag{9.13.9}$$

Here $L^{-1}\{\ \}$ indicates the inverse Laplace transform of the quantity enclosed within $\{\ \}$. The problem is complete formally. However, the inversions of (9.13.9) present a task that cannot be carried out without the knowledge of the special forms of $\lambda_v(t)$ and $\mu_v(t)$. We present the following special solution for the Kelvin-Voigt model as an illustration.

Kelvin-Voigt Model For this case through (9.12.11) we have

$$\lambda(\zeta) = \lambda_e + \zeta\lambda_v, \qquad \mu(\zeta) = \mu_e + \zeta\mu_v \tag{9.13.10}$$

where λ_v and μ_v are now constants. From (9.13.9)

$$U(t) = L^{-1}\left\{\frac{1}{\lambda_e + 2\mu_e + (\lambda_v + 2\mu_v)\zeta}\right\} = \frac{1}{\lambda_v + 2\mu_v}\, e^{-t/\tau}$$

(9.13.11)
$$T(t) = L^{-1}\left\{\frac{\lambda_e + \zeta\lambda_v}{\lambda_e + 2\mu_e + (\lambda_v + 2\mu_v)\zeta}\right\}$$
$$= \frac{2(\lambda_e\mu_v - \lambda_v\mu_e)}{(\lambda_v + 2\mu_v)^2}\, e^{-t/\tau} + \frac{\lambda_v}{\lambda_v + 2\mu_v}\,\delta(t)$$

where τ is a *time constant* defined by

(9.13.12)
$$\tau \equiv \frac{\lambda_v + 2\mu_v}{\lambda_e + 2\mu_e}$$

and $\delta(t)$ is the *Dirac delta* function with the properties that

(9.13.13)
$$\delta(t) = \begin{cases} \infty & \text{when } t = 0 \\ 0 & \text{when } t \neq 0 \end{cases}$$
$$\int_{-\infty}^{\infty} F(t)\,\delta(t) = F(0)$$

Here $F(t)$ is an arbitrary function. We thus see that inversion can involve such discontinuous "functions" as the delta function and sometimes even its "derivatives." The operational use of the delta function is justifiable, cf. Schwartz [1950].

Substitution of (9.13.11) into (9.13.8), together with the interpretation that the lower limits of these integrals are 0^-, leads to

$$u_x(x, t) = -\frac{x}{\lambda_v + 2\mu_v}\int_0^t p(s)\exp\left[-(t-s)/\tau\right] ds$$

(9.13.14)
$$t_{yy} = t_{zz} = -\frac{\lambda_v}{\lambda_v + 2\mu_v}\,p(t) - \frac{2(\lambda_e\mu_v - \lambda_v\mu_e)}{(\lambda_v + 2\mu_v)^2}$$
$$\times \int_0^t p(s)\exp\left[-(t-s)/\tau\right] ds$$

This completes the solution.

(*ii*) ***Cylindrical Tube Subject to Internal Pressure.*** The solution of the corresponding elastic problem was given in (6.8.6). Here the external pressure is absent ($p_1 = 0$), and we write $p_0 = p(t)$. Through (6.8.6), and the theorem of correspondence, we have the transforms of the displacement field for the viscoelastic problem

$$\bar{u}_r(r,\zeta) = \frac{a^2\bar{p}(\zeta)}{2(b^2-a^2)}\left[\frac{r}{\lambda(\zeta)+\mu(\zeta)} + \frac{b^2}{\mu(\zeta)r}\right] \tag{9.13.15}$$

$$u_\theta = u_z = 0$$

The stress components t_{rr} and $t_{\theta\theta}$, given by (6.8.7), remain valid for this viscoelastic problem since no material moduli are present in the expressions of these components. For t_{zz}, however, we have the corresponding transform of the viscoelastic problem

$$\bar{t}_{zz} = \frac{\lambda(\zeta)}{\lambda(\zeta)+\mu(\zeta)}\frac{a^2\bar{p}(\zeta)}{b^2-a^2} \tag{9.13.16}$$

The inversion of (9.13.15) and (9.13.16) must be obtained to find $u_r(r,t)$ and $t_{zz}(r,t)$ of the viscoelastic problem. These can be inverted for the Kelvin-Voigt model in a similar fashion to problem (*i*) above. The result is

$$u_r(r,t) = \frac{a^2}{2(b^2-a^2)} \times \int_0^t \left[\frac{r}{\lambda_v+\mu_v}e^{-(t-s)/\tau_1} + \frac{b^2}{r\mu_v}e^{-(t-s)/\tau_2}\right]p(s)\,ds \tag{9.13.17}$$

$$t_{zz}(r,t) = \frac{a^2}{b^2-a^2}\left[\frac{\lambda_v}{\lambda_v+\mu_v}p(t) + \frac{\lambda_e\mu_v-\lambda_v\mu_e}{(\lambda_v+\mu_v)^2}\int_0^t e^{-(t-s)/\tau_1}p(s)\,ds\right]$$

where τ_1 and τ_2 are two time constants defined by

$$\tau_1 \equiv \frac{\lambda_v+\mu_v}{\lambda_e+\mu_e}, \qquad \tau_2 \equiv \frac{\mu_v}{\mu_e} \tag{9.13.18}$$

It is now clear how we may proceed from the solutions of problems in elasticity to the corresponding ones in viscoelasticity.

9.14. PROPAGATION OF WAVES

In the same way as in Section 6.10 we may decompose the displacement and body force fields into scalar and vector potentials

$$\mathbf{u} = \nabla\phi + \nabla\times\psi, \qquad \mathbf{f} = -\nabla g - \nabla\times\mathbf{h}, \qquad \nabla\cdot\psi = \nabla\cdot\mathbf{h} = 0 \tag{9.14.1}$$

The vector field equations (9.12.1) are satisfied if ϕ and ψ satisfy the equations

$$c_1^2(\zeta)\nabla^2\bar{\phi} - \bar{G} = \zeta^2\bar{\phi} - \zeta\phi(\mathbf{x},0) - \dot{\phi}(\mathbf{x},0) \tag{9.14.2}$$

$$-c_2^2(\zeta)\nabla\times\nabla\times\bar{\psi} - \bar{\mathbf{H}} = \zeta^2\bar{\psi} - \zeta\psi(\mathbf{x},0) - \dot{\psi}(\mathbf{x},0) \tag{9.14.3}$$

where

$$
\begin{aligned}
c_1^2(\zeta) &\equiv [\lambda(\zeta) + 2\mu(\zeta)]/\rho \\
c_2^2(\zeta) &\equiv \mu(\zeta)/\rho \\
\bar{G}(\zeta) &\equiv \bar{g}(\mathbf{x}, \zeta) + \rho^{-1}[\bar{\lambda}_e(\zeta) + 2\bar{\mu}_e(\zeta)]\nabla^2\phi(\mathbf{x}, 0) \\
\bar{\mathbf{H}}(\zeta) &\equiv \bar{\mathbf{h}}(\mathbf{x}, \zeta) - \rho^{-1}\bar{\mu}_e(\zeta)\nabla \times \nabla \times \psi(\mathbf{x}, 0)
\end{aligned} \tag{9.14.4}
$$

Equations (9.14.2) and (9.14.3) are identical with the Laplace transforms of the corresponding equations (6.10.5) and (6.10.6), provided that we replace c_1^2, c_2^2, g, and $\mathbf{h}$ by $c_1^2(\zeta)$, $c_2^2(\zeta)$, $\bar{G}$, and $\bar{\mathbf{H}}$, respectively. This is simply one more confirmation of the theorem of correspondence.

We note, however, that unless the initial conditions $\phi(\mathbf{x}, 0)$ and $\psi(\mathbf{x}, 0)$ are prescribed $\bar{G}(\zeta)$ and $\bar{\mathbf{H}}(\zeta)$ will remain unknown, since the knowledge of $\mathbf{u}(\mathbf{x}, 0)$ does not readily lend itself to the determination of these functions.

For steady-state problems, the initial conditions are immaterial, and it may be more convenient to employ Fourier integrals. In this case, the equations corresponding to (9.14.2) to (9.14.4) are obtained by merely *replacing* ζ by $-i\omega$ and setting $\phi(\mathbf{x}, 0) = 0$, $\psi(\mathbf{x}, 0) = 0$.

Alternatively, for periodic solutions with vanishing body forces we write

$$\phi = \phi_1(\mathbf{x})e^{i\omega t}, \qquad \psi = \psi_1(\mathbf{x})e^{i\omega t} \tag{9.14.5}$$

so that in this special case (9.11.10) is satisfied if

$$c_1^2(\omega)\nabla^2\phi_1 + \omega^2\phi_1 = 0 \tag{9.14.6}$$

$$-c_2^2(\omega)\nabla \times \nabla \times \psi_1 + \omega^2\psi_1 = 0; \qquad \nabla \cdot \psi_1 = 0 \tag{9.14.7}$$

where

$$
\begin{aligned}
c_1^2(\omega) &\equiv [\lambda(\omega) + 2\mu(\omega)]/\rho \\
c_2^2(\omega) &\equiv \mu(\omega)/\rho \\
\lambda(\omega) &\equiv \lambda_e + i\omega\int_0^\infty \lambda_v(s)e^{-i\omega s}\,ds \\
\mu(\omega) &\equiv \mu_e + i\omega\int_0^\infty \mu_v(s)e^{-i\omega s}\,ds
\end{aligned} \tag{9.14.8}
$$

We now proceed to give a few special solutions

(*i*) *Irrotational Plane Waves.* In this case, $\psi_1 = 0$ and we are to determine a solution of (9.14.6). The fundamental solution of this equation may be written as

$$\phi_1 = \phi_0 \exp[-iP_k(\omega)x_k], \qquad i \equiv \sqrt{-1} \tag{9.14.9}$$

where ϕ_0 is a complex constant and $P_k(\omega)$ is a complex vector. Substituting (9.14 9) into (9.14.6) we find that $P_k(\omega)$ must satisfy

$$-c_t^{\,2}(\omega)P_k(\omega)P_k(\omega) + \omega^2 = 0 \tag{9.14 10}$$

We express $\mathrm{P}(\omega)$, $\lambda(\omega)$ and $\mu(\omega)$ in terms of their real and imaginary parts as

$$\begin{aligned} \mathrm{P}(\omega) &= \mathrm{P}_1(\omega) - i\mathrm{P}_2(\omega) \\ \lambda(\omega) &= \lambda_1(\omega) + i\lambda_2(\omega) \\ \mu(\omega) &= \mu_1(\omega) + i\mu_2(\omega) \end{aligned} \tag{9.14.11}$$

where $\mathrm{P}_1(\omega)$, $\mathrm{P}_2(\omega)$, $\lambda_1(\omega)$, $\lambda_2(\omega)$, $\mu_1(\omega)$, and $\mu_2(\omega)$ are real

Substituting (9 14 11) into (9 14 10), using $(9.14\ 8)_{1,}$ and equating the real and imaginary parts of both sides we get

$$\begin{aligned} &[\lambda_1(\omega) + 2\mu_1(\omega)](P_1^{\,2} - P_2^{\,2}) + 2[\lambda_2(\omega) + 2\mu_2(\omega)] \\ &\qquad \times \mathrm{P}_1 \cdot \mathrm{P}_2 - \rho\omega^2 = 0 \\ &[\lambda_2(\omega) + 2\mu_2(\omega)](P_1^{\,2} - P_2^{\,2}) - 2[\lambda_1(\omega) + 2\mu_1(\omega)] \\ &\qquad \times \mathrm{P}_1 \cdot \mathrm{P}_2 = 0 \end{aligned} \tag{9.14.12}$$

These equations are sufficient to determine the magnitudes P_1 and P_2 if the angle β between P_1 and P_2 is given Suppose that e_1 and e_2 are arbitrary unit vectors lying along the directions of P_1 and P_2, respectively, and

$$\mathrm{e}_1 \cdot \mathrm{e}_2 = \cos\beta \tag{9 14.13}$$

We may also define the phase velocity c_d and the *attenuation coefficient* α_d for the irrotational waves by

$$\mathrm{P}_1 = [\omega/c_d(\omega)]\mathrm{e}_1, \qquad \mathrm{P}_2 = \alpha_d(\omega)\mathrm{e}_2 \tag{9.14.14}$$

Through (9.14.12) we solve for c_d and α_d

$$\begin{aligned} c_d^{\,2}(\omega) &= \{2|\lambda(\omega) + 2\mu(\omega)| \cos\beta/\rho \sin\Delta_d\} B(\Delta_d, \beta) \\ \alpha_d(\omega) &= [\omega/c_d(\omega)]B(\Delta_d, \beta) \end{aligned} \tag{9.14.15}$$

where

$$B(\Delta_d, \beta) \equiv (1 + \cot^2\Delta_d \cos^2\beta)^{1/2} - \cot\Delta_d \cos\beta \tag{9.14.16}$$

Here Δ_d is known as the *loss angle*

$$\tan[\Delta_d(\omega)] = [\lambda_2(\omega) + 2\mu_2(\omega)]/[\lambda_1(\omega) + 2\mu_1(\omega)] \tag{9.14.17}$$

For negative values of ω, $c_d(\omega)$ and $\alpha_d(\omega)$ can be shown to be

$$c_d(\omega) = c_d(-\omega) > 0, \qquad \alpha_d(\omega) = \alpha_d(-\omega) > 0$$

Since c_d is real, β is restricted by

$$|\beta| \leq \frac{\pi}{2}$$

With $c_d(\omega)$ and $\alpha_d(\omega)$ given by (9.14.15) the fundamental solution $\mathbf{u}$ can be written as

$$\mathbf{u} = -\left(\frac{i\omega}{c_d}\mathbf{e}_1 + \alpha_d \mathbf{e}_2\right)\phi_0 \exp\left[i\omega\left(t - \frac{\mathbf{e}_1 \cdot \mathbf{x}}{c_d}\right) - \alpha_d \mathbf{e}_2 \cdot \mathbf{x}\right] \tag{9.14.18}$$

where ϕ_0 is an arbitrary constant, and $\mathbf{e}_1$ and $\mathbf{e}_2$ are unit vectors subject to the restriction

$$\mathbf{e}_1 \cdot \mathbf{e}_2 = \cos\beta > 0$$

In the particular limiting case, $\beta = 0$, $\mathbf{e}_1 = \mathbf{e}_2$, (9.14.18) simplifies to

$$u_x = -\left(\frac{i\omega}{c_d} + \alpha_d\right)\phi_0 \exp\left[i\omega\left(t - \frac{x}{c_d}\right) - \alpha_d x\right] \tag{9.14.19}$$

$$u_y = u_z = 0$$

where $\mathbf{e}_1$ is along the x-direction. Also for $\beta = 0$ the expressions for c_d and α_d simplify considerably

$$c_d = \{|\lambda(\omega) + 2\mu(\omega)|/\rho\}^{1/2} \sec(\Delta_d/2) \tag{9.14.20}$$

$$\alpha_d = (\omega/c_d)\tan(\Delta_d/2)$$

Waves for which $\beta = 0$ are sometimes called "simple irrotational waves." It is clear that in this case the directions of propagation and attenuation are the same. However, when $\beta \neq 0$ these directions do not coincide.[12]

(ii) Equiroluminal Waves. For these waves $\phi = 0$ and ψ_1 satisfies (9.14.7). The fundamental solution is of the form

$$\psi_1 = \psi_0 \exp\left[-iQ_k(\omega)x_k\right] \tag{9.14.21}$$

where ψ_0 and $\mathbf{Q}(\omega)$ are complex vectors satisfying

$$-c_2^2(\omega)Q_k(\omega)Q_k(\omega) + \omega^2 = 0; \qquad \psi_0 \cdot \mathbf{Q} = 0 \tag{9.14.22}$$

If we let

$$\mathbf{Q} = \mathbf{Q}_1(\omega) - i\mathbf{Q}_2(\omega) \tag{9.14.23}$$

and use $(9.14.11)_{2,3}$ we obtain two equations similar to (9.14.12) in order to solve for $|\mathbf{Q}_1|$ and $|\mathbf{Q}_2|$ when the angle between these two vectors is

[12] These results are due to Lockett [1962]. Lockett also studied scattering problems from a plane boundary in which the incident waves are simple, that is, $\mathbf{e}_1 = \mathbf{e}_2$.

known. The solution follows the same method of approach as in irrotational waves. Here we give only the final result

$$(9\ 14.24)\quad \mathbf{u} = -\left[\frac{i\omega}{c_s(\omega)}\mathbf{e}_1 + \alpha_s(\omega)\mathbf{e}_2\right] \times \psi_0 \exp\left[i\omega\left(t - \frac{\mathbf{e}_1 \cdot \mathbf{x}}{c_s(\omega)}\right) - \alpha_s \mathbf{e}_2 \cdot \mathbf{x}\right]$$

where ψ_0 is an arbitrary vector subject to $(9.14.22)_1$, and $\mathbf{e}_1$ and $\mathbf{e}_2$ are subject to

$$(9.14\ 25)\qquad \mathbf{e}_1 \cdot \mathbf{e}_2 = \cos\beta > 0$$

and $c_s(\omega)$ and $\alpha_s(\omega)$ are given by equations similar to (9.14.15) where $\lambda(\omega) + 2\mu(\omega)$ is replaced by $\mu(\omega)$, and Δ_d and $B(\Delta_d, \beta)$ by Δ_s and $B(\Delta_s, \beta)$, respectively, where

$$(9\ 14\ 26)\qquad \begin{aligned} &\tan\Delta_s(\omega) \equiv \mu_2(\omega)/\mu_1(\omega) \\ &B(\Delta_s, \beta) \equiv (1 + \cot^2\Delta_s \cos^2\beta)^{1/2} - \cot\Delta_s \cos\beta \end{aligned}$$

Again when $\beta = 0$ we obtain the special case of "simple equivoluminal waves". In this case $\mathbf{e}_1 = \mathbf{e}_2$ and (9 14 24) reduces to

$$(9.14.27)\quad (u_x, u_y, u_z) = (0, \psi_{0z}, -\psi_{0y}) \times \left[i\frac{\omega}{c_s(\omega)} + \alpha_s(\omega)\right] \exp\left[i\omega\left(t - \frac{x}{c_s(\omega)}\right) - \alpha_s(\omega)x\right]$$

where we have assumed that $\mathbf{e}_1$ is along the positive direction of the x-axis. In this case c_s and α_s simplify to

$$(9.14.28)\qquad \begin{aligned} c_s &= |\mu(\omega)/\rho|^{1/2} \sec(\Delta_s/2) \\ \alpha_s &= (\omega/c_s)\tan(\Delta_s/2) \end{aligned}$$

Again, for this case, the direction of propagation and that of attenuation coincide. However, when $\beta \neq 0$ this is not the case.

The foregoing solutions of the two problems indicate that the propagation of plane waves in a viscoelastic medium is distinguished by the existence of separate directions for attenuation and propagation Except for the case of the so-called "simple waves" ($\mathbf{e}_1 = \mathbf{e}_2$), this is the general situation.

9.15. PROPAGATION OF ONE-DIMENSIONAL WAVES IN KELVIN-VOIGT AND MAXWELL SOLIDS

Special viscoelastic models such as the Kelvin-Voigt and Maxwell solids have limited applicability. Nevertheless, they possess a certain amount of mathematical simplicity, and in the solution of wave propagation problems

a discussion of waves in these solids has qualitative and illustrative values. Here we give an account on waves in Kelvin-Voigt and Maxwell solids.

For one dimensional waves we have only a single nonvanishing displacement component, that is,

$$u_1 = u(x, t), \qquad u_2 = u_3 = 0 \tag{9.15.1}$$

(i) Kelvin–Voigt Solid. The field equations (9.11.2), (9.11.6), and (9.11.8) in this case reduce to

$$c_1^2 \frac{\partial^2 u}{\partial x^2} + 2k \frac{\partial^3 u}{\partial x^2 \partial t} - \frac{\partial^2 u}{\partial t^2} = 0 \tag{9.15.2}$$

where

$$c_1 \equiv \left(\frac{\lambda_e + 2\mu_e}{\rho}\right)^{1/2}, \qquad k \equiv \frac{\lambda_v + 2\mu_v}{2\rho} \tag{9.15.3}$$

λ_v and μ_v are constant. The stress components in this case are given by

$$\begin{aligned} t_{xx} &= \left[\lambda_e + 2\mu_e + (\lambda_v + 2\mu_v)\frac{\partial}{\partial t}\right]\frac{\partial u}{\partial x} \\ t_{yy} &= t_{zz} = \left(\lambda_e + \lambda_v \frac{\partial}{\partial t}\right)\frac{\partial u}{\partial x} \\ t_{xy} &= t_{yz} = t_{zx} = 0 \end{aligned} \tag{9.15.4}$$

When we speak of propagation of waves in a bar we need to modify (9.15.3) and (9.15.4) by setting $t_{yy} = t_{zz} = 0$ and using the theorem of correspondence. In this case, the Laplace transform of t_{xx}, $\bar{t}_{xx}$, takes the form

$$\bar{t}_{xx} = \bar{E}\,\bar{e}_{xx}$$

where $\bar{E}$ is given by $(9.10.12)_1$. In a special class of materials (commonly used in analytical work) in which $\lambda_e/\mu_e = \lambda_v/\mu_v$, $\bar{E}$ takes a simple form so that the expression of t_{xx} is identical to (9.15.4), with $\lambda_e + 2\mu_e$ and $\lambda_v + 2\mu_v$, respectively replaced by

$$E \equiv \frac{(3\lambda_e + 2\mu_e)\mu_e}{\lambda_e + \mu_e}, \qquad E_v \equiv \frac{(3\lambda_v + 2\mu_v)\mu_v}{\lambda_v + \mu_v} \tag{9.15.5}$$

A plane harmonic wave solution of (9.15.2) will have the form

$$u = U \exp\,[i(\omega t - \lambda x)] \tag{9.15.6}$$

where U, ω, and λ are constants. Substitution of (9.15.6) into (9.15.2) gives a quadratic equation in ω and λ.

$$\omega^2 - 2ik\lambda^2\omega - \lambda^2 c_1^2 = 0 \tag{9.15.7}$$

We study two distinct problems: (1) the wavelength $L = 2\pi/\lambda$, hence λ is real; and (2) the frequency ω is real.

1. The Wavelength Is Real. In this case, the solution of the frequency equation is

$$\omega = ik\lambda^2 \pm \Omega_v \qquad \Omega_v \equiv (\lambda^2 c_1^2 - k^2\lambda^4)^{1/2} \tag{9.15.8}$$

The general solution has, therefore, the form

$$u = \exp(-k\lambda^2 t - i\lambda x)[U_1 \exp(i\Omega_v t) + U_2 \exp(-i\Omega_v t)] \tag{9.15.9}$$

where U_1 and U_2 are constants. We observe several features of this solution:

(a) For $k = 0$ we recover the plane harmonic wave solution of undamped elastic waves, that is, equation (9.15.6) with $\omega/\lambda = c_1$. Purely elastic waves propagate with the constant phase velocity c_1 (phase velocity $\equiv c \equiv Re\,\omega/Re\,\lambda$) of irrotational waves and the wave amplitude remains unchanged in time and space for all wavelengths $L = 2\pi/\lambda$ and for all frequencies.

(b) Viscoelastic waves are damped out in time as is apparent from the presence of $\exp(-k\lambda^2 t)$ in (9.15.9) with $k > 0$.

(c) The frequency of the viscoelastic waves is dependent on the wavelength in a nonlinear manner (see equation (9.15.8)). This means that different wavelengths will propagate with different phase velocities. This phenomenon is called *dispersion*. Viscoelastic waves are, therefore, dispersive.

(d) The oscillatory character of the waves is destroyed whenever the internal viscosity is large enough to satisfy

$$k \geq c_1/\lambda \tag{9.15.10}$$

Thus, when k takes the value $k_{cr} \equiv c_1/\lambda$, or exceeds it, the waves will not be harmonic in time.

(e) For large wavelengths, that is, $k^2\lambda^2/c_1^2 \ll 1$, we can approximate Ω_v by

$$\Omega_v = \lambda c_1 + O(k^2\lambda^2/c_1^2)$$

so that (9.15.9) may be approximated by

$$u = \exp(-k\lambda^2 t)\{U_1 \exp[i\lambda(c_1 t - x)] + U_2 \exp[-i\lambda(c_1 t + x)]\} \tag{9.15.11}$$

Therefore, waves having large wavelengths propagate forward and backward with the phase velocity of irrotational waves. The amplitude of these

waves decreases exponentially in time with an exponent proportional to the square of the wavelength.

2. *The Frequency ω Is Real.* In this case the solution of (9.15.8) for λ is

$$\lambda^2 = \frac{\omega^2}{c_1^2(1 + ik_1\omega)} \tag{9.15.12}$$

$$k_1 \equiv \frac{\lambda_v + 2\mu_v}{\lambda_e + 2\mu_e} \quad \text{or} \quad k_1 = \frac{E_v}{E} \text{ (beam)}$$

Now write

$$\lambda = \pm (\Omega_1 - i\alpha_1) \tag{9.15.13}$$

where

$$\Omega_1 \equiv \frac{\omega}{\sqrt{2}\, c_1} \left[\frac{(1 + k_1^2\omega^2)^{1/2} + 1}{1 + k_1^2\omega^2}\right]^{1/2} \tag{9.15.14}$$

$$\alpha_1 \equiv \frac{\omega}{\sqrt{2}\, c_1} \left[\frac{(1 + k_1^2\omega^2)^{1/2} - 1}{1 + k_1^2\omega^2}\right]^{1/2}$$

$$c/c_1 \equiv \omega/c_1\Omega_1 = (\sqrt{2}/k_1\omega)(1 + k_1^2\omega^2)^{1/2}[(1 + k_1^2\omega^2)^{1/2} - 1]^{1/2}$$

where c is the phase velocity. From these expressions, it is clear that for small values of $k_1\omega$ we have

$$\Omega_1 = \frac{\omega}{c_1}[1 - \tfrac{3}{8}k_1^2\omega^2 + O(k_1^4\omega^4)]$$

$$\alpha_1 = \frac{k_1\omega^2}{2c_1}[1 + O(k_1^2\omega^2)] \tag{9.15.15}$$

$$c/c_1 = 1 + \tfrac{3}{8}k_1^2\omega^2 + O(k_1^4\omega^4)$$

Thus, for small $k_1\omega$ the phase velocity $c = \omega/\Omega_1$ is almost the same as that of a purely elastic solid, c_1. However, the waves attenuate exponentially with an attenuation factor α_1 proportional to $k_1\omega^2/2c_1$. For high frequencies, on the other hand, $k_1\omega \gg 1$ and (9.15.14) give

$$\Omega_1 = \alpha_1 = (\omega/2k_1c_1^2)^{1/2} \tag{9.15.16}$$

$$c/c_1 = (2k_1\omega)^{1/2}$$

Since the phase velocity is a nonlinear function of ω, the waves are dispersive. For dispersive waves, the *group velocity*, c_g, defined by

$$c_g \equiv \frac{\partial\omega}{\partial\lambda}$$

is a characteristic velocity that represents, roughly, the speed with which energy is propagated [13] Since $c \equiv \omega/\lambda$, we also have

$$c_g = c\left(1 - \frac{\omega}{c}\frac{dc}{d\omega}\right)^{-1} \tag{9.15.17}$$

Using $(9.15.14)_3$ for the present case we get

$$c_g/c_1 = (2c/c_1)(1 + k_1^2\omega^2)[3 + k_1^2\omega^2 - (1 + k_1^2\omega^2)^{1/2}]^{-1} \tag{9.15.18}$$

The phase velocity c, group velocity c_g, and the attenuation factor α_1 are plotted in Figure 9 15.1 as a function of $k_1\omega$. For very small $k_1\omega$ we get $c_g = c_1$, and for very large $k_1\omega$, $c_g = 2c = 2c_1\sqrt{2k_1\omega}$. This, of course, is physically absurd, since it implies that a wave packet consisting of infinitesimally short wavelengths will propagate with an infinite velocity. We note that as $k_1\omega$ becomes large, α_1 also becomes very large, indicating that the high-frequency waves are quickly attenuated. Thus c_g approaching infinity with $k_1\omega$ may be insignificant, physically.

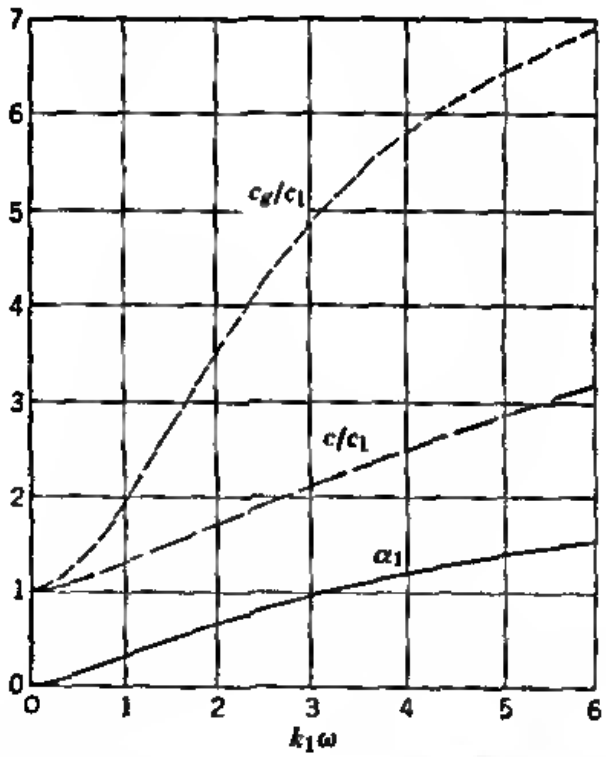


Fig. 9.15.1 Phase velocity c, group velocity c_g, and attenuation coefficient α_1 for a Kelvin-Voigt solid (after Davies [1956]).

(ii) Maxwell Solid. For this solid the constitutive equations are given by (9.11.7). Considerable simplicity results for the special material in which

[13] For a review and development of ideas on group velocity, see Whitham [1961]

$\alpha_m = 0$.[14] For the one-dimensional motion, we have all $t_{kl} = 0$ except $t_{11} \equiv t_{xx}$ for which we have the constitutive equation

$$\dot{t}_{xx} + 2\beta_m t_{xx} = (\lambda_m + 2\mu_m)\frac{\partial^2 u}{\partial x \partial t} \tag{9.15.19}$$

Upon substituting

$$t_{xx} = T \exp[i(\omega t - \lambda x)]$$
$$u = U \exp[i(\omega t - \lambda x)] \tag{9.15.20}$$

with T and U constant, (9.15.19) gives

$$T = \frac{(\lambda_m + 2\mu_m)\omega\lambda U}{i\omega + 2\beta_m} \tag{9.15.21}$$

Carrying (9.15.20) into the equation of one dimensional motion

$$\frac{\partial t_{xx}}{\partial x} - \rho\frac{\partial^2 u}{\partial t^2} = 0$$

we get

$$-iT\lambda + \rho\omega^2 U = 0 \tag{9.15.22}$$

Substituting from (9.15.21) we obtain the frequency equation

$$\omega(\omega^2 - 2i\beta_m\omega - c_m{}^2\lambda^2) = 0 \tag{9.15.23}$$

where

$$c_m \equiv \left(\frac{\lambda_m + 2\mu_m}{\rho}\right)^{1/2} \tag{9.15.24}$$

is a velocity corresponding to the velocity of irrotational elastic waves.

We proceed to study the roots of (9.15.23) for real λ and real ω separately:

1. The Wavelength Is Real. In this case, the roots of (9.15.23) are

$$\omega = 0, \qquad \omega = i\beta_m \pm \omega_m \tag{9.15.25}$$

where

$$\omega_m \equiv (c_m{}^2\lambda^2 - \beta_m{}^2)^{1/2} \tag{9.15.26}$$

The displacement field is thus given by

$$u = \exp(-\beta_m t - i\lambda x)[U_1 \exp(i\omega_m t) + U_2 \exp(-i\omega_m t)] + U_3 \exp(-i\lambda x) \tag{9.15.27}$$

[14] Alternatively, the state of strain is given by (9.10.13). This corresponds to the usual approximation for beam theories.

Excluding the standing wave solution (coefficient U_3), special features of this solution are:

(a) Plane harmonic waves are damped in time. The coefficient of damping is dependent on the wavelength for small wavelengths, $(c_m^2\lambda^2 > \beta_m^2)$

(b) The oscillatory character of the waves disappears for certain wavelengths, $\lambda \leq \lambda_{cr}$, where

$$\lambda_{cr} \equiv \beta_m/c_m \tag{9 15 28}$$

(c) For large wavelengths $\lambda \gg \lambda_{cr}$, ω_m and hence u may be approximated by

$$\omega_m = c_m\lambda + 0(\lambda_{cr}^2/\lambda^2) \tag{9.15 29}$$

$$u = \exp(-\beta_m t - i\lambda x)[U_1 \exp(ic_m\lambda t) + U_2 \exp(-ic_m\lambda t)] + U_3 \exp(-i\lambda x) \tag{9.15.30}$$

(d) The elastic wave solution is obtained by taking $\beta_m = 0$, $c_m = c_1$, and $U_3 \equiv 0$. This latter condition is due to the fact that the constitutive equation (9.15 19), in this case, is the rate of Hooke's law.

2. The Frequency ω Is Real Through equation (9.15.23) we solve for

$$\lambda = \pm(\Omega_2 - i\alpha_2) \tag{9 15 31}$$

where

$$\Omega_2 = \frac{\omega}{\sqrt{2}\,c_m}[(1 + 4\beta_m^2\omega^{-2})^{1/2} + 1]^{1/2}$$
$$\alpha_2 = \frac{\omega}{\sqrt{2}\,c_m}[(1 + 4\beta_m^2\omega^{-2})^{1/2} - 1]^{1/2} \tag{9.15 32}$$

The phase velocity c is given by

$$\frac{c}{c_m} = \frac{\omega}{c_m\Omega_2} = \sqrt{2}\,[(1 + 4\beta_m^2\omega^{-2})^{1/2} + 1]^{-1/2} \tag{9.15.33}$$

From these results, we observe that when ω is large, as compared with $2\beta_m$, the attenuation factor, $\alpha_2 c \simeq \beta_m/c_m$, is independent of the frequency. The phase velocity for high frequencies is c_m which, for $\lambda_m + 2\mu_m = \lambda_e + 2\mu_e$, is the same as that of the elastic solid. At low frequencies $(2\beta_m/\omega \gg 1)$ from (9.15 32) and (9.15.33), we find that

$$\alpha_2 \simeq \sqrt{\beta_m\omega}/c_m, \qquad c/c_m \simeq \sqrt{\omega/\beta_m} \tag{9.15.34}$$

which is similar to the high frequency behavior of the Kelvin-Voigt model. As $\omega \to 0$ we have $\alpha_2 \to 0$, $c \to 0$ expressing the fact that although these waves do not attenuate, they do not move

The group velocity c_g is calculated from (9.15.17)

$$\frac{c_g}{c_m} = 2\frac{c}{c_m}\left[1 + (1 + 4\beta_m{}^2\omega^{-2})^{-1/2}\right]^{-1} \tag{9.15.35}$$

For low frequencies, $c_g/c_m = 2\sqrt{\omega/\beta_m}$, and for high frequencies $c_g/c_m = 1$.

The phase velocity c, group velocity c_g, and attenuation factor α_2 are plotted against $\omega/2\beta_m$ in Figure 9.15.2. The group velocity shows a maximum at $\omega/2\beta_m \simeq 0.57$ with $(c_g/c_m)_{\max} = 1.08$ indicating that the wave front can travel with a velocity exceeding c_m by 8 percent.

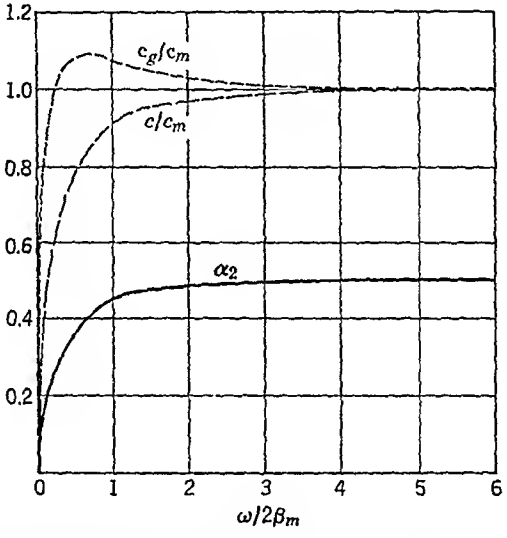


Fig. 9.15.2 Phase velocity c, group velocity c_g, and attenuation coefficient α_2 for a Maxwell solid (after Davies [1956]).

It must be pointed out here that neither the Kelvin-Voigt nor the Maxwell model approximates linear viscoelastic materials in all ranges of frequency. In an attempt to obtain better correlations with experiments, other models have been studied. Hillier [1949] and Kolsky [1953] discussed the propagation of waves in a *standard linear solid*. This model is described by a constitutive equation of the form

$$\left(\alpha_m + \gamma_1\frac{\partial}{\partial t}\right)t_{rr}\,\delta_{kl} + 2\left(\beta_m + \gamma_2\frac{\partial}{\partial t}\right)t_{kl} = \left(\lambda_e + \lambda_v\frac{\partial}{\partial t}\right)\tilde{e}_{rr}\,\delta_{kl} + 2\left(\mu_e + \mu_v\frac{\partial}{\partial t}\right)\tilde{e}_{kl} \tag{9.15.36}$$

which combines features of both the Kelvin-Voigt and Maxwell models.

These attempts have been fruitful in throwing light on the behavior of viscoelastic waves regardless of their poor quantitative predictions in a reasonable frequency range. Experimental results of Kolsky (cf. Section 9.10) are plotted in Figure 9.10.3 to indicate the situation for polythene. The attempts to correlate the theoretical results, based on such simple models as discussed earlier, are unsuccessful. Recent work in this area is, therefore, moving in the direction of using the general viscoelastic model expressed by (9.11.9), and carrying out computations on electronic computers, or also approximating the attenuation factor and the phase velocity on the basis of experimental work. One such discussion is to be found in Section 9.16.

9.16. HALF SPACE SUBJECT TO IMPULSIVE PRESSURE

In this section we discuss a boundary-value problem concerning the half space $x > 0$ subject to impulsive pressure loading at its boundary $x = 0$ The solution of this problem, with minor modification, is also the solution of the problem of a semi-infinite bar subject to an impact loading at its end $x = 0$

Propagation of viscoelastic waves in one-space direction is governed by an equation obtained from (9.11.9) by setting $u_2 = u_3 = 0$ and $u_1(x_1, t) = u(x, t)$ where u is independent of x_2 and x_3, and depends on $x_1 \equiv x$ only. Thus

$$(9.16.1) \qquad (\lambda_e + 2\mu_e)\frac{\partial^2 u}{\partial x^2} + \int_0^t [\lambda_v(t-s) + 2\mu_v(t-s)]\frac{\partial^3 u(s)}{\partial x^2 \partial s}\,ds = \rho\frac{\partial^2 u}{\partial t^2}$$

where we also assumed that the body forces vanish In this case, for the stress components, we have

$$t_{xx} = (\lambda_e + 2\mu_e)\frac{\partial u}{\partial x} + \int_0^t [\lambda_v(t-s) + 2\mu_v(t-s)]\frac{\partial^2 u(s)}{\partial x \partial s}\,ds$$

$$(9.16.2) \qquad t_{yy} = t_{zz} = \lambda_e\frac{\partial u}{\partial x} + \int_0^t \lambda_v(t-s)\frac{\partial^2 u(s)}{\partial x \partial s}\,ds$$

$$t_{xy} = t_{yz} = t_{zx} = 0$$

When we speak of propagation of waves in bars we need to modify (9 16.1) and (9.16.2), as explained in Section 9.15, by replacing $\lambda_e + 2\mu_e$ and $\lambda_v + 2\mu_v$, respectively, by

$$(9.16.3) \qquad E \equiv \frac{(3\lambda_e + 2\mu_e)\mu_e}{\lambda_e + \mu_e}, \qquad E_v \equiv \frac{(3\lambda_v + 2\mu_v)\mu_v}{\lambda_v + \mu_v}$$

and setting $t_{yy} = t_{zz} = 0$. The half space $x > 0$ is taken to be at rest initially.

The waves may be generated by the application of any one of the following boundary conditions

(a) a uniform displacement $u(0, t) = u_0(t)$

(b) a uniform velocity $\dot{u}(0, t) = v_0(t)$

(c) a uniform pressure $t_{xx}(0, t) = -p_0(t)$

at the boundary, $x = 0$.

The solution of these problems may be obtained by the use of Laplace or Fourier transforms. We give examples employing both. Let $\bar{u}(x, \zeta)$ be the Laplace transform of $u(x, t)$. Then the Laplace transforms of (9.16.1) and $(9.16.2)_1$ with $u(x, 0) = \dot{u}(x, 0) = 0$ give

$$c^2(\zeta)\frac{d^2\bar{u}}{dx^2} - \zeta^2\bar{u} = 0 \tag{9.16.4}$$

$$\bar{t}_{xx} = \rho c^2(\zeta)\frac{d\bar{u}}{dx} \tag{9.16.5}$$

where

$$c^2(\zeta) = {c_1}^2[1 + \zeta\bar{a}(\zeta)] \tag{9.16.6}$$

$${c_1}^2 \equiv \frac{\lambda_e + 2\mu_e}{\rho}, \qquad \bar{a}(\zeta) \equiv \frac{\bar{\lambda}_v + 2\bar{\mu}_v}{\lambda_e + 2\mu_e} \tag{9.16.7}$$

The solutions of (9.16.4) subject to the boundary conditions of problems (a) to (c) are therefore given by

$$\bar{u}(x, \zeta) = \bar{u}_0(\zeta)\exp\left[-\frac{x\zeta}{c(\zeta)}\right] \qquad \text{problem (a)} \tag{9.16.8}$$

$$\bar{u}(x, \zeta) = \bar{v}_0(\zeta)\zeta^{-1}\exp\left[-\frac{x\zeta}{c(\zeta)}\right] \qquad \text{problem (b)} \tag{9.16.9}$$

$$\bar{u}(x, \zeta) = \frac{\bar{p}_0(\zeta)}{\rho\zeta c(\zeta)}\exp\left[-\frac{x\zeta}{c(\zeta)}\right] \qquad \text{problem (c)} \tag{9.16.10}$$

The inverse Laplace transforms of these give the solutions in the time domain. These are however, difficult to obtain except for some special situations. We now give two of these cases.

(i) ***Kelvin–Voigt Material.*** In this case $\bar{a}(\zeta)$ is a constant

$$\bar{a}(\zeta) = a = \frac{\lambda_v + 2\mu_v}{\lambda_e + 2\mu_e} \tag{9.16.11}$$

so that

$$c^2(\zeta) = c_1{}^2[1 + a\zeta] \tag{9.16.12}$$

We proceed to invert problem **(b)** for this case, as follows. The inverse of $\bar{v}_0(\zeta)$ is $v_0(t)$, so that the inverse of (9.16.9) according to the convolution theorem has the form

$$u(x, t) = a\int_0^t v_0(\sigma)G(x, t-\sigma)\,d\sigma \tag{9.16.13}$$

where $G(x, t)$ is a function whose Laplace transform is given by

$$\bar{G}(x, \zeta) = \frac{1}{a\zeta}\exp\left[-\frac{x\zeta}{c_1\sqrt{1+a\zeta}}\right] \tag{9.16.14}$$

To invert this, we introduce

$$\bar{B}(y, s) \equiv \bar{G}(x, \zeta) = s^{-1}\exp\left(-ys/\sqrt{1+s}\right) \tag{9.16.15}$$
$$s \equiv a\zeta, \qquad y \equiv x/ac_1$$

The inversion of $\bar{B}$ is obtained as follows (Morrison [1956]) According to a general result in the theory of Laplace transforms (Carslaw and Jaeger [1948, p. 259]), if $\bar{f}(s)$ and $\bar{\varphi}(s, \eta)$ are the Laplace transforms of $f(t)$ and $\varphi(t, \eta)$, respectively, and

$$\exp\left[-\eta h(s)\right]/g(s) = \bar{\varphi}(s, \eta) \tag{9.16.16}$$

then

$$L^{-1}\{\bar{f}[h(s)]/g(s)\} = \int_0^\infty \varphi(t, \eta)f(\eta)\,d\eta \tag{9.16.17}$$

Now let

$$\bar{f}(s) \equiv \frac{1}{\sqrt{s}}\exp\left(-y\sqrt{s}\right), \qquad g(s) \equiv \sqrt{s}$$

$$h(s) \equiv (s-1)^2/s$$

$$\bar{\varphi}(s, \eta) = \frac{1}{\sqrt{s}}\exp\left[-\frac{\eta(s-1)^2}{s}\right] = e^{2\eta}e^{-\eta s}\frac{\exp\left(-\eta/s\right)}{\sqrt{s}}$$

Then clearly

$$\bar{f}[h(s)]/g(s) = \frac{1}{s-1}\exp\left[-\frac{y(s-1)}{\sqrt{s}}\right] \tag{9.16.18}$$
$$\equiv \bar{B}(y, s-1)$$

Hence if we find this inverse, by the shifting theorem

$$L^{-1}\{\bar{B}(y, s-1)\} = e^{\tau}B(y, \tau) = aG(x, t)e^{t/a} \tag{9.16.19}$$

$$\tau \equiv t/a$$

we get the desired inversion. From a transform table (Churchill [1958, p. 328]), we have

$$f(\tau) = (\pi\tau)^{-1/2} \exp\left(-\frac{y^2}{4\tau}\right)$$

$$L^{-1}\left\{\frac{1}{\sqrt{s}} \exp\left(-\frac{\eta}{s}\right)\right\} = (\pi\tau)^{-1/2} \cos\left(2\sqrt{\eta\tau}\right)$$

Hence by the shifting theorem

$$\varphi(\tau, \eta) = \frac{\cos\sqrt{4\eta(\tau-\eta)}}{\sqrt{\pi(\tau-\eta)}} e^{2\eta} H(\tau \sim \eta)$$

where $H(\eta)$ is the Heaviside unit function defined as

$$H(\eta) = \begin{cases} 1 & \text{when } \eta > 0 \\ 0 & \text{when } \eta < 0 \end{cases}$$

Now, carrying $f(\eta)$ and $\varphi(\tau, \eta)$ into (9.16.17) for $s > 0$, we get

$$L^{-1}\{\bar{B}(y, s-1)\} = \frac{1}{\pi}\int_0^{\tau} \frac{\cos\sqrt{4\eta(\tau-\eta)}}{\sqrt{\eta(\tau-\eta)}} \exp\left(2\eta - \frac{y^2}{4\eta}\right) d\eta$$

According to (9.16.19) we therefore have

$$aG(x, t) \equiv V(y, \tau) \tag{9.16.20}$$

$$= \frac{1}{\pi} e^{-\tau} \int_0^{\tau} \frac{\cos\sqrt{4\eta(\tau-\eta)}}{\sqrt{\eta(\tau-\eta)}} \exp\left(2\eta - \frac{y^2}{4\eta}\right) d\eta$$

$$\tau \equiv t/a, \qquad y \equiv x/ac_1$$

We note that $V(y, \tau)$ is none other than the particle velocity produced at (x, t) for a suddenly applied and maintained unit velocity (Heaviside unit function) at the plane $x = 0$ at time $t = 0+$. To obtain the solution of the same problem for an arbitrary time-dependent applied velocity, we carry (9.16.20) into the convolution theorem (9.16.13). The particle velocity $v(x, t) = \dot{u}(x, t)$ follows from this by differentiation, that is,

$$v(x, t) = av_0(0)G(x, t) + a\int_0^t G(x, t-\sigma)\dot{v}_0(\sigma)\, d\sigma \tag{9.16.21}$$

The stress t_{xx} can be calculated from using $u(x, t)$ so obtained and the constitutive equations for the Kelvin-Voigt solids Alternatively, we may invert

$$\bar{t}_{xx}(x, \zeta) = -\rho c_1(1 + a\zeta)^{1/2}\bar{v}_0(\zeta) \exp\left[-x\zeta/c_1\sqrt{1 + a\zeta}\right] \tag{9 16 22}$$

We pursue the method used above to determine the stress produced by a sudden application of a constant velocity v_0 to $x = 0$ at time $t = 0+$. In this case $\bar{v}_0(\zeta) = v_0/\zeta$, and (9.16.22) reduces to

$$-\frac{\bar{t}_{xx}(x, \zeta)}{\rho c_1 a v_0} = \frac{\sqrt{1+s}}{s} \exp\left(-\frac{ys}{\sqrt{1+s}}\right) \tag{9.16 23}$$

$$s \equiv a\zeta, \qquad y \equiv x/c_1 a$$

To obtain the inverse of this we make use of

$$\bar{\varphi}(s, \eta) \equiv \exp\left[-\eta\left(s + \frac{1}{s} - 2\right)\right] = e^{2\eta}e^{-\eta s}e^{-\eta/s}$$

$$\bar{f}(s) \equiv \frac{1}{\sqrt{s}} \exp(-y\sqrt{s})$$

Now, formally

$$L^{-1}\{\exp(-\eta/s)\} = \delta(\tau) - \sqrt{\eta/\tau}\, J_1(\sqrt{4\eta\tau}), \qquad \tau \equiv t/a$$

where $\delta(\tau)$ is the Dirac delta function and J_1 is the Bessel function of the first kind of order 1. The shifting theorem gives

$$\varphi(\tau, \eta) = e^{2\eta}\{\delta(\tau - \eta) - \sqrt{\eta/(\tau - \eta)}\, J_1[\sqrt{4\eta(\tau - \eta)}]\}H(\tau - \eta)$$

Using (9.16.17) for $s > 0$, we get

$$L^{-1}\left\{\frac{\sqrt{s}}{s-1} \exp\left[-\frac{y(s-1)}{\sqrt{s}}\right]\right\} = \frac{1}{\sqrt{\pi\tau}} \exp\left(2\tau - \frac{y^2}{4\tau}\right) - \int_0^\tau \frac{J_1[\sqrt{4\eta(\tau - \eta)}]}{\sqrt{\pi(\tau - \eta)}} \exp\left(2\eta - \frac{y^2}{4\eta}\right) d\eta$$

The shifting theorem, replacing s by $s - 1$, now gives

$$-\frac{t_{xx}(x, t)}{\rho c_1 v_0} = \Sigma(y, \tau) = e^{\tau}\left\{\frac{1}{\sqrt{\pi\tau}} \exp\left(-\frac{y^2}{4\tau}\right) - \int_0^\tau \frac{J_1\sqrt{4\eta(\tau - \eta)}}{\sqrt{\pi(\tau - \eta)}} \exp\left[-2(\tau - \eta) - \frac{y^2}{4\eta}\right] d\eta\right\} \tag{9.16.24}$$

At the origin $x = 0$, this reduces to

$$-\frac{t_{xx}(0, t)}{c_1 v_0} = \frac{1}{\sqrt{\pi\tau}} e^{-\tau} \operatorname{erf}(\sqrt{\tau}) \tag{9.16.25}$$

where erf τ is the error function.

The particle velocity $V(y, \tau)$ given by (9.16.20) and the non-dimensional stress $\Sigma(y, \tau)$ given by (9.16.24) due to a sudden application of unit velocity at $x = 0$ are plotted in Figures 9.16.1 and 9.16.2.

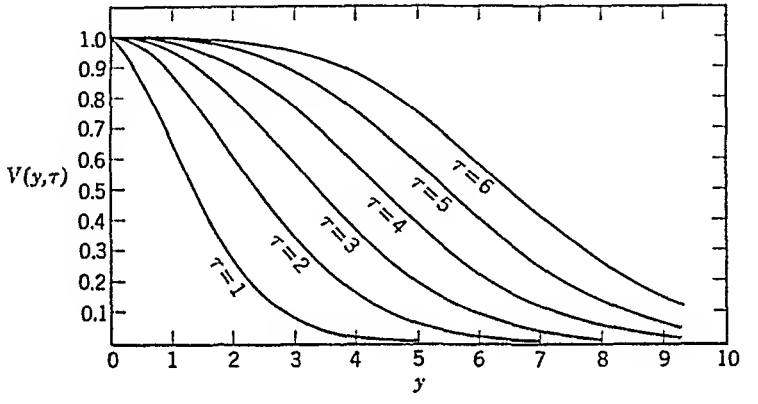


Fig. 9.16.1 Particle velocity in a semi-infinite Voigt bar due to a sudden application of unit velocity at $y = 0$ (after Morrison [1956]).

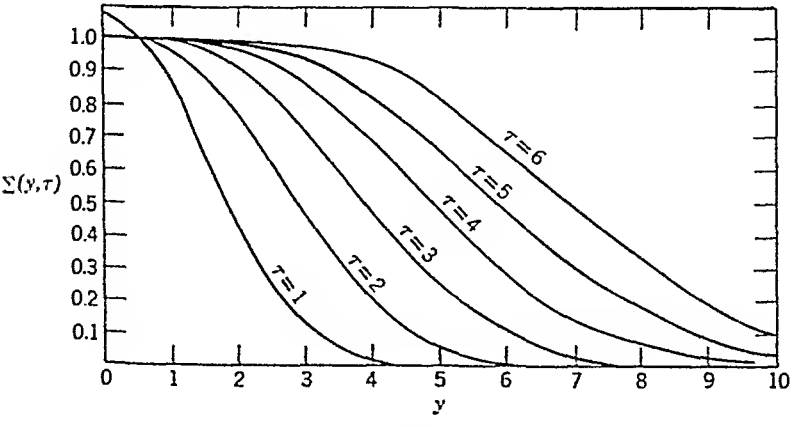


Fig. 9.16.2 Nondimensional stress in a semi-infinite Voigt bar due to a sudden application of constant velocity at $y = 0$ (after Morrison [1956]).

(ii) Maxwell Solid. For this model, the Laplace transform of the stress constitutive equation for t_{xx} in the case of a one-dimensional strain is

$$\bar{t}_{xx} = [\lambda(\zeta) + 2\mu(\zeta)]\frac{\partial \bar{u}}{\partial x} \tag{9.16 26}$$

where $\lambda(\zeta)$ and $\mu(\zeta)$ are given by (9.12 12) The propagation of waves in a bar, arising from impact, has been studied. For beam theories, when appropriate combinations of elastic and viscous moduli are made,[15] we can write

$$\bar{t}_{xx} = \rho c^2(\zeta)\frac{\partial \bar{u}}{\partial x} \tag{9 16.27}$$

where

$$c(\zeta) \equiv c_m[\zeta/(\zeta + 2\beta_m)]^{1/2}, \qquad c_m^2 \equiv E_m/\rho \tag{9.16 28}$$

The stress for problem (b) follows from (9.16.10) and (9.16.27)

$$-\bar{t}_{xx}(x, \zeta)/\rho c_m = \zeta \bar{v}_0(\zeta)[\zeta(\zeta + 2\beta_m)]^{-1/2} \times \exp\left[-y\sqrt{\zeta(\zeta + 2\beta_m)}\right] \qquad \text{problem (b)} \tag{9.16 29}$$

where

$$y \equiv x/c_m \tag{9.16.30}$$

The inverse of the coefficient of $\zeta\bar{v}_0(\zeta)$ on the right-hand side of (9 16 29) can be read from tables (cf. Churchill [1958, p 329]). Hence, by using the convolution theorem, we get[16]

$$-t_{xx}(x, t)/\rho c_m = \{v_0(0)e^{-\beta_m t}\mathrm{I}_0(\beta_m\sqrt{t^2 - y^2}) + \int_y^t \dot{v}_0(t - \sigma)e^{-\beta_m \sigma}\mathrm{I}_0(\beta_m\sqrt{\sigma^2 - y^2})\, d\sigma\}H(t - y) \tag{9.16.31}$$

In the absence of dissipation, the purely elastic stress t_e is given by

$$t_e(x, t) = -\rho c_m v_0(t - y)H(t - y) \tag{9.16.32}$$

In terms of the non-dimensional parameters

$$\tau \equiv \beta_m t \qquad \xi \equiv \beta_m y = \beta_m x/c_m \tag{9.16.33}$$

[15] For this, one takes $\alpha = 0$ and replaces $\lambda + 2\mu$ by E_m.

[16] Lee and Kanter [1953] Glauz and Lee [1954] treated the same problem for a four-parameter model viscoelastic material Morrison [1956] obtained the solution of this problem for various types of models See also Lee and Morrison [1956] Berry and Hunter [1956] studied other types of edge conditions, different models, and bars having finite length. For a finite bar suddenly set into motion, see Eringen and Dunkin [1959]

the ratio of dissipative and elastic stresses t_{xx}/t_e for the case of $v_0(t) = v_0 =$ constant is therefore given by

$$t_{xx}/t_e = e^{-\tau} I_0(\sqrt{\tau^2 - \xi^2}) \tag{9.16.34}$$

Following Davies [1956] this is plotted on Figure (9.16.3) as a function of the non-dimensional time τ and non-dimensional position ξ. At the leading edge of the pulse, we have $\tau = \xi$ so that the wave front travels with a velocity c_m as if no dissipation were present, a result irreconcilable with the group velocity analysis.

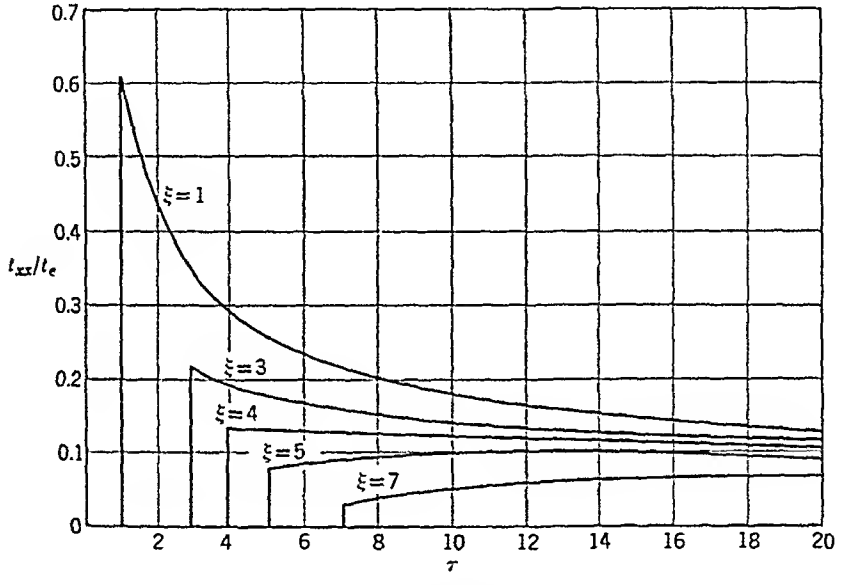


Fig. 9.16.3 Variation of stress in a Maxwell solid (rod) with non-dimensional time at various locations ξ. The rod is set in motion with constant velocity applied to one end $\xi = 0$ at time $\tau = 0$.

As we see, there are no fundamental difficulties in dealing with these problems other than calculating some tedious inversion integrals. Below we give a third example which employs some experimental findings for inversion.

(*iii*) ***Fourier Transform Technique.*** The use of the Fourier transform in place of the Laplace transform gives identical results for (9.16.8) to (9.16.10) with ζ replaced by $-i\omega$. The inversion is then written in the

usual manner For example, the solution of problem (c), (9.16.10), reads

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\bar{p}_0(\omega)}{\rho\omega c(\omega)} \exp\left[-i\omega t + i\frac{x\omega}{c(\omega)}\right] d\omega, \tag{9.16.35}$$

$$\operatorname{Im}\left[\frac{\omega}{c(\omega)}\right] > 0, \qquad x > 0$$

where a bar superposed on letters indicates the Fourier transform, for example,

$$\bar{p}_0(\omega) \equiv \int_{-\infty}^{\infty} p_0(t) e^{i\omega t}\, dt \tag{9.16.36}$$

and

$$c^2(\omega) \equiv c_1{}^2[1 - i\omega\tilde{a}(\omega)]$$

$$\bar{a}(\omega) \equiv \frac{\bar{\lambda}_e(\omega) + 2\bar{\mu}_e(\omega)}{\lambda_e + 2\mu_e} \tag{9.16.37}$$

We decompose $\omega/c(\omega)$ into real and imaginary parts as

$$\frac{\omega}{c(\omega)} = \frac{\omega}{\gamma(\omega)} + i\alpha(\omega), \qquad \alpha(\omega) > 0 \tag{9 16 38}$$

where $\alpha(\omega)$ and $\gamma(\omega)$ are real and

$$\alpha(-\omega) = \alpha(\omega) > 0, \qquad \gamma(\omega) = \gamma(-\omega) > 0, \qquad 0 < \omega < \infty \tag{9.16 39}$$

From (9.16.39) and (9 16.38) we find that

$$\gamma(\omega) = |c(\omega)| \sec\frac{\Delta}{2}$$

$$\alpha(\omega) = \frac{\omega}{\gamma}\tan\frac{\Delta}{2} \tag{9 16.40}$$

where

$$\tan\Delta = \omega\tilde{a}(\omega) \tag{9.16 41}$$

Using (9.16.38) in the expression for the velocity $v(x,t) = \dot{u}(x,t)$ calculated from (9.15 35), we have

$$v(x,t) = \frac{1}{2\pi\rho}\int_{-\infty}^{\infty} \bar{p}_0(\omega)(\gamma^{-1} + i\alpha\omega^{-1}) \exp[-\alpha(\omega)x - i\omega(t - x/\gamma(\omega))]\, d\omega \tag{9.16 42}$$

From this it is clear that $\alpha(\omega)$ plays the role of the *attenuation factor*, and $\gamma(\omega)$ plays that of the *phase velocity* for harmonic waves with frequency ω progressing in the direction of the positive x-axis.

According to the experiments of Hillier and Kolsky [1949] and Hillier [1949] on polyethylene at 10°C, the attenuation was linear in ω over a frequency range 1 kc/s $< \omega/2\pi <$ 16 kc/s. The corresponding phase velocity increased rapidly from 3.5×10^4 cm/sec at low frequencies to 10^5 cm/sec at 2 kc/s, then remained constant from 2 kc/s to 16 kc/s. Thus over a large frequency range $\gamma(\omega) = c_0 = \text{const.}$, $\alpha(\omega) = \alpha_0|\omega|$ when α_0 is a constant. By choosing

$$\gamma(\omega) \cong c_0 > 0\,, \qquad \alpha(\omega) = \alpha_0|\omega| > 0 \tag{9.16.43}$$

where according to (9.16.40)

$$\alpha_0 = c_0^{-1} \tan\frac{\Delta}{2} \tag{9.16.44}$$

the inversion (9.16.42) can be carried out for the case of impact $p_0(t) = P\delta(t)$ or $\bar{p}_0(\omega) = P$. Here P is the constant pressure applied to $x = 0$ at $t = 0$. Hence[17]

$$v(x, t) = \frac{P}{\pi\rho\alpha_0 c_0 x} \cdot \frac{1 + \alpha_0 c_0 \tau}{1 + \tau^2} \tag{9.16.45}$$

where τ is a dimensionless time variable

$$\tau \equiv \frac{t - (x/c_0)}{\alpha_0 x} \tag{9.16.46}$$

This result compares reasonably well with Kolsky's experimental pulse obtained after 50 cm of travel in polyethylene (Figure 9.16.4). Here $v/v_{\max}$ is plotted versus τ after matching the maxima of the theory and experiment. The values $c_0 = 9.5 \times 10^4$ cm/sec. and $\tan(\Delta/2) = 0.081$ are taken from Kolsky's experiment, in conjunction with (9.16.43) and (9.16.44).

The results so obtained are in reasonable correlation with those of experiments on polyethylene. However, for perspex we have a much smaller value, $\tan(\Delta/2) = 0.02$, and the theory fails to predict the asymmetry of the experimental pulses found by Kolsky. Calculations based on the assumption that $\tan\Delta = \text{const.}$ and employing some 30 Fourier harmonics were also carried out by Kolsky [1956] and Hunter [1960] and good agreement was found with the experiment.

From the preceding discussion, it is clear that the general Boltzmann-Volterra theory requires tedious calculations. Thus, while the theory is sound, its practical application requires much numerical work.

[17] Hunter [1960], [1967].

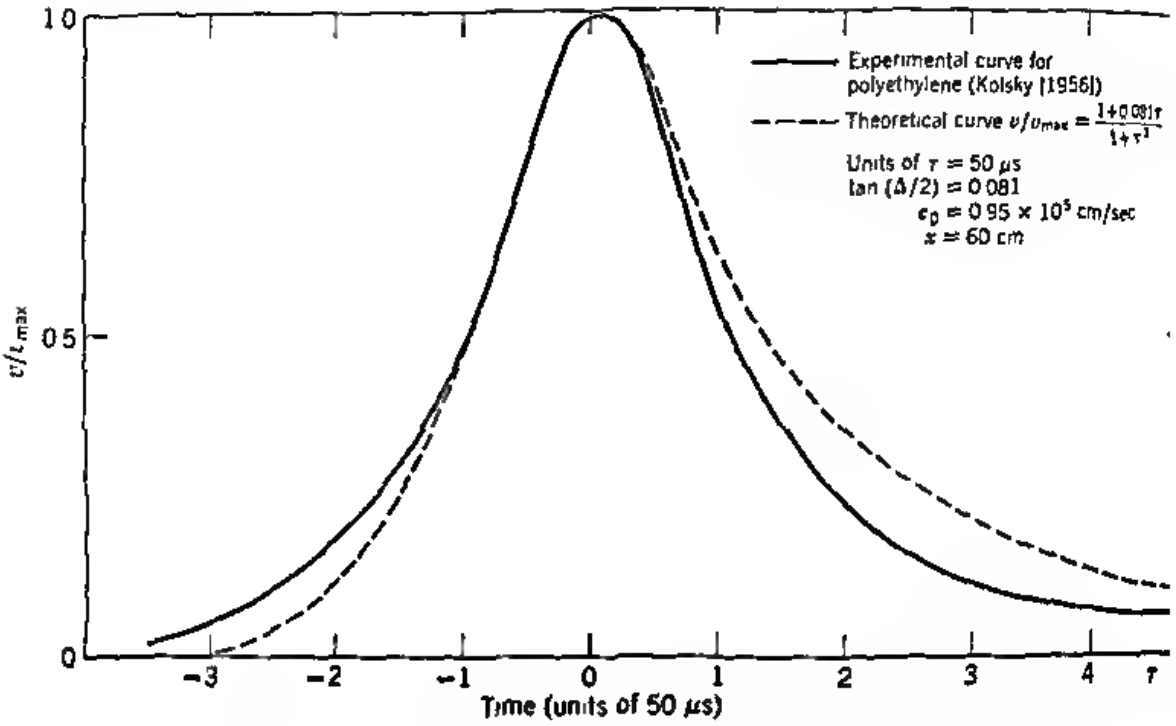


Fig. 9.16.4 Velocity profiles after 60 cm. travel in polyethylene (after Hunter [1967])

9.17. BLAST IN A SPHERICAL CAVITY IN A VISCOELASTIC SOLID

In Section (6 12) we gave the solution of this problem for a purely elastic medium Using the theorem of correspondence, we can obtain the solution of this problem for a viscoelastic solid. The Laplace transform of the displacement field given by (6 12.4) and (6 12.9) may be written

$$\bar{u}_r = \frac{\partial}{\partial r}\left[\frac{1}{r}\exp\left(\frac{a-r}{c_1}\zeta\right)\bar{F}(\zeta)\right] \tag{9.17.1}$$

where

$$\bar{F}(\zeta) \equiv -\frac{a}{\rho}\frac{\bar{p}(\zeta)}{(\zeta+\alpha)^2+\beta^2}$$

$$\alpha \equiv \frac{2}{a\sqrt{\rho}}\frac{\mu_e}{(\lambda_e+2\mu_e)^{1/2}} \tag{9 17.2}$$

$$\beta \equiv \frac{2}{a\sqrt{\rho}}\left[\frac{(\lambda_e+\mu_e)\mu_e}{\lambda_e+2\mu_e}\right]^{1/2}$$

$$c_1 \equiv \frac{(\lambda_e+2\mu_e)^{1/2}}{\sqrt{\rho}}$$

According to the theorem of correspondence, we now replace λ_e and μ_e by $\lambda(\zeta)$ and $\mu(\zeta)$, respectively, to obtain the Laplace transform of the corresponding solution in viscoelasticity. Thus we write

$$\bar{u}_r = \frac{\partial}{\partial r}\left\{\frac{1}{r}\exp\left[\frac{(a-r)\zeta}{c(\zeta)}\right]\bar{F}(\zeta)\right\} \tag{9.17.3}$$

where

$$\begin{aligned}
\bar{F}(\zeta) &\equiv -\frac{a}{\rho}\frac{\bar{p}(\zeta)}{[\zeta+\alpha(\zeta)]^2+\beta^2(\zeta)}\\
\alpha(\zeta) &\equiv \frac{2}{a\sqrt{\rho}}\frac{\mu(\zeta)}{[\lambda(\zeta)+2\mu(\zeta)]^{1/2}}\\
\beta(\zeta) &\equiv \frac{2}{a\sqrt{\rho}}\frac{\{[\lambda(\zeta)+\mu(\zeta)]\mu(\zeta)\}^{1/2}}{[\lambda(\zeta)+2\mu(\zeta)]^{1/2}}\\
c(\zeta) &\equiv \frac{[\lambda(\zeta)+2\mu(\zeta)]^{1/2}}{\sqrt{\rho}}\\
\lambda(\zeta) &\equiv \lambda_e+\zeta\bar{\lambda}_v(\zeta), \qquad \mu(\zeta)\equiv\mu_e+\zeta\bar{\mu}_v(\zeta)
\end{aligned} \tag{9.17.4}$$

If the waves are to proceed towards $r>0$ and vanish at $r=\infty$, we must have $|\arg[\zeta/c(\zeta)]| \le \pi/2$ when ζ is in the half plane of convergence of $\bar{u}_r$.

It is clear that the inversion of (9.17.3) cannot be obtained for arbitrary viscosities $\lambda_v(t)$ and $\mu_v(t)$. We, therefore, proceed with the Maxwell solid that has the same relaxation time $1/2\beta_m$ for both dilatation and shear, that is,

$$\begin{aligned}
\lambda(\zeta) &= \lambda_m\zeta/(\zeta+2\beta_m)\\
\mu(\zeta) &= \mu_m\zeta/(\zeta+2\beta_m)
\end{aligned}$$

Substitution of these into (9.17.3) and (9.17.4) gives

$$\bar{u}_r(r,\zeta) = \bar{p}(\zeta)\bar{G}(r,\zeta) \tag{9.17.5}$$

where

$$\begin{aligned}
\bar{G}(r,\zeta) &\equiv \frac{a}{\rho r^2}\left(1+\frac{2\beta_m}{\zeta}\right)\bar{\phi}(r,\sqrt{\zeta(\zeta+2\beta_m)})\\
\bar{\phi}(r,s) &\equiv \left(1+\frac{r}{c_m}s\right)[(s+\alpha)^2+\beta^2]^{-1}\exp\left(-t_0 s\right)
\end{aligned}$$

$$\begin{aligned}
\alpha &\equiv \frac{2}{a\sqrt{\rho}}\mu_m(\lambda_m+2\mu_m)^{-1/2}\\
\beta &\equiv \frac{2}{a\sqrt{\rho}}\left[\frac{\mu_m(\lambda_m+\mu_m)}{(\lambda_m+2\mu_m)}\right]^{1/2}\\
c_m &\equiv \left(\frac{\lambda_m+2\mu_m}{\rho}\right)^{1/2}, \quad t_0 \equiv \frac{r-a}{c_m}
\end{aligned} \tag{9.17.6}$$

The inverse transform $\phi(r, t)$ of $\bar{\phi}(r, s)$ may be found by using tables of Laplace transforms (Churchill [1958, p. 324]).

$$\phi(r, t) = \Phi(r, t)H\left(t - t_0\right) \tag{9 17.7}$$

where $H(t)$ is the Heaviside unit function and

$$\begin{aligned}\Phi(r, t) = \Big\{&\frac{r}{c_m}\cos\left[\beta\left(t - t_0\right)\right] \\ &+ \frac{1}{\beta}\left(1 - \frac{\alpha r}{c_m}\right)\sin\left[\beta\left(t - t_0\right)\right]\Big\} \\ &\times \exp\left[-\alpha\left(t - t_0\right)\right]\end{aligned} \tag{9 17 8}$$

To obtain the inverse Laplace transform of $\bar{G}(r, \zeta)$ we write it as

$$\bar{G}(r, \zeta) = \frac{a}{\rho r^2}\left(1 + \frac{2\beta_m}{\zeta}\right)\bar{\phi}(r, \sqrt{(\zeta + \beta_m)^2 - \beta_m^2})$$

and apply the shifting theorem and the result (12) (p 228, Erdélyi, et al. [1954]) This gives

$$\begin{aligned}G(r, t) = \frac{a}{\rho r^2}H(t - t_0)\Big\{&e^{-\beta_m t}\Phi(r, t) \\ &+ \beta_m e^{-\beta_m t}\int_0^{\sqrt{t^2 - t_0^2}}\Phi(r, \sqrt{t^2 - \tau^2})I_1(\beta_m\tau)\,d\tau \\ &+ 2\beta_m\int_{t_0}^{t} e^{-\beta_m\tau}\Phi(r, \tau)\,d\tau \\ &+ 2\beta_m^2\int_{t_0}^{t}\int_0^{\sqrt{\tau^2 - t_0^2}} e^{-\beta_m\tau}\Phi(r, \sqrt{\tau^2 - \sigma^2})I_1(\beta_m\sigma)\,d\sigma\,d\tau\Big\}\end{aligned} \tag{9 17 9}$$

This represents the displacement field due to unit impulsive loading, that is, the Green function. The inverse transform of (9.17.5) is now given by

$$u_r(r, t) = \int_0^t G(r, t - \tau)p(\tau)\,d\tau \tag{9.17.10}$$

For an impulse P, that is, $p(t) = P\,\delta(t)$ where $\delta(t)$ is the Dirac delta function, we get

$$u_r(r, t) = PG(r, t) \tag{9.17.11}$$

By setting $\beta_m = 0$ we recover the displacement field exterior to a spherical cavity in an elastic medium

$$u_r = \frac{Pa}{\rho r^2}\,\Phi(r, t)H\left(t - \frac{r-a}{c_m}\right) \tag{9.17.12}$$

which may be brought into agreement with the results of Section 6.12 for an impulsive loading after a few manipulations and putting the velocity of the wave front $c_m = c_1$. The above problem for a Kelvin-Voigt solid has been treated by Mattice and Lieber [1954] who calculated the displacements produced by blasts of various forms. Other works on the subject are due to Bland [1957], Berry [1958], Lockett [1961], and Chao and Achenbach [1964].

PROBLEMS

9.1. In a nonlinear, Kelvin-Voigt solid, dissipative stress ${}_D\mathbf{T}$ is approximated by a second-order polynomial in $\dot{\mathbf{C}}$. If the solid is thermodynamically admissible, what final form must ${}_D\mathbf{T}$ take?

9.2. Determine the ultimate form of the stress constitutive equations of an isotropic, Kelvin-Voigt solid in which the stress, $\mathbf{t}$, is of second degree in $\mathbf{C}$ and $\dot{\mathbf{C}}$.

9.3. If in Problem 9.2, the viscoelastic solid is incompressible, what forms do the constitutive equations take?

9.4. An isotropic, Kelvin-Voigt solid is subjected to a steady simple shear. Determine the stress tensor.

9.5. A linear, isotropic Kelvin-Voigt material is subjected to a stress field of the following types:

(*a*) Uniform stress of duration, t_o;

(*b*) Uniform oscillatory stress field.

Determine the strain field for each case.

9.6. An isotropic, incompressible, Kelvin-Voigt bar having a circular cross section is subjected to a constant twist. Calculate the axial load and the twisting moment in terms of the rate of shearing.

9.7. A linear, isotropic Kelvin-Voigt bar having circular cross section is undergoing free, torsional oscillations. Determine the displacement field.

9.8. A linear, isotropic Kelvin-Voigt cylinder having circular cross section is subject to a uniform external pressure. Determine the displacement and stress fields.

9.9. A linear, isotropic Kelvin-Voigt body of infinite extent is subjected to a constant concentrated force. Determine the displacement field.

9.10. A linear, isotropic, Kelvin-Voigt half space is subjected to a uniform twist over a circular region on its boundary. Determine the stress field.

9.11. A plane wave advancing normal to a flat, free boundary of a linear, isotropic, Kelvin-Voigt half space is reflected. Find the ratio of the amplitude of the reflected wave to that of the incident wave.

9.12. Find the amplitude decay of a spherical wave advancing in an infinite, isotropic, linear Kelvin-Voigt solid.

9.13. A linear Maxwell solid is subjected to a uniform stress at time, $t=0$ Determine the stress field at subsequent times

9.14. Find the solution of any one of the Problems 9 4 to 9 11 for a Maxwell solid.

9.15. Determine the stress field whose flux is nil.

9.16. A linear Maxwell solid is subject to a uniform strain field of the following types.

(*a*) Constant strain of duration, t_*;

(*b*) Uniform oscillating strains

Determine the stress fields for each case.

9.17. A linear, rate-dependent, isotropic, viscoelastic material is subject to a steady, uniform, oscillatory strain field. Determine the stress field

9.18. A linear, rate-dependent, isotropic, viscoelastic material is called a standard solid if the stress constitutive equations contain only the rates up to and including the first-order stress and strain rates. Such a solid is being subjected to either one of the following strain fields:

(*a*) Uniform strain field of duration, t_*;

(*b*) Oscillating, uniform strain field

Find the stress fields.

9.19. A semi-finite, circular bar made of a standard solid (Problem 9.18) is subjected to an impact load at one end Determine the axial displacement in the bar.

9.20. A linear, isotropic, Boltzmann-Volterra solid is subject to an oscillating *uniform stress field. Determine the strain* field

9.21. A linear, isotropic, Boltzmann-Volterra solid possesses kernels of the exponential type

$$\lambda_*(t)=\lambda_1 e^{-\alpha t}, \quad \mu_*(t)=\mu_1 e^{-\alpha t}$$

where λ_1, μ_1 and $\alpha>0$ are constant. An infinite circular cylindrical bar *made of this material is subjected to an impact load at one of its ends* Determine the axial stress field

9.22. One end of the Boltzmann-Volterra bar described in (9 21) is oscillating with a constant frequency ω_*. Determine the displacement field

9.23. A thick cylindrical rocket made of a linear isotropic Boltzmann-Volterra solid is rotating with an angular velocity ω_0. Determine the displacement of the external surface of the cylinder.

9.24. A linear isotropic Boltzmann-Volterra cylinder is subject to a constant pressure applied at time $t = 0$. Initially the cylinder is at rest. Determine the stress field at subsequent times.

9.25. The solid described in (9.24) is subjected to an axi-symmetric external loading. Determine the quasi-static solution of the field equations for the displacement field.

9.26. (*Short Term Paper.*) Construct the constitutive equations of a thermo-viscoelastic material in which the history of deformations and temperature is important.

9.27. (*Short Term Paper.*) Study literature and give a discussion of the experimental work on viscoelastic materials. Limit your discussion to either the linear or the nonlinear case.

APPENDIX A

Vectors

A1. INTRODUCTION

An account of vector analysis is presented here to provide the necessary foundation for reading this book. While most students by this time have had some exposure to vectors it is felt that a uniformity in notation and a recapitulation of the basic notions are necessary for the proper use of index notation and for a smooth transition to tensors.

Definitions of various types of vectors and addition of vectors as well as the multiplication of a vector by a scalar are presented in Section A2. In Section A3 we discuss linear dependence.

Basc vectors and cartesian coordinates constitute the topics of Section A4, and in Section A5 we introduce the index notation, summation convention, Kronecker delta, and permutation symbols. These are necessary for the manipulation of Cartesian tensors. The scalar product (Section A6), the vector product (Section A7), the scalar triple product (Section A8), and the vector triple product (Section A9) are defined in vector notation and represented in indcx notation. Several of their properties are discussed. In Section A10, we introduce the reciprocal base vectors in rectilinear coordinates. The covariant and contravariant components of vectors can now be distinguished (Section A11), and with the use of the Euclidean metric tensor (Section A12) lengths and angles can be calculated. In Section A13 we present a brief account of vector functions and differentiation. The linear operators, gradient, divergence, and curl, which are of frequent occurrence in our study, are presented in the same section. The last section of this Appendix gives the integral theorems of Green-Gauss and Stokes.

For a medium intersected by a discontinuity surface, these theorems are modified. Results so obtained are fundamental in the derivations of local-balance laws and jump conditions of continuum mechanics.

A2. VECTORS, ADDITION OF VECTORS

DEFINITION 1. *A vector is a quantity which is characterized by* (a) *Magnitude: its length or measure,* (b) *Direction or line of action: the line on which it is located; and* (c) *Sense: forward or backward, positive or negative.*

EXAMPLE 1. Displacement, velocity, and force are vectors

A straight-line displacement of a point from A to B is characterized by specifying the length $|\overline{AB}|$ (=magnitude), the direction of line L on which the motion takes place, and the sense, which is positive from A to B (Figure A2 1).

We shall indicate vectors by arrows on drawings, and by boldface letters or sometimes by an arrow on top of letters in writing, that is, $\mathbf{a}$, $\overrightarrow{AB}$, $\vec{a}$. The magnitude of a vector will be expressed as a or $|\mathbf{a}|$ or $|\overrightarrow{AB}|$.

DEFINITION 2. *A vector whose one end is restricted to a set of points is called a bound vector. If both ends are free it is called a free vector.*

DEFINITION 3. *A vector that is restricted to lie in a definite fixed line is called a line vector.*

EXAMPLE. Forces acting on a rigid body are line vectors.

DEFINITION 4 *A vector having zero magnitude and arbitrary direction and sense is called a zero or null vector* It is indicated by **0**.

DEFINITION 5 (*Equality of Vectors*). *Two vectors are equal if and only if they have identical magnitudes, parallel lines of actions, and identical senses.*

This is expressed as $\mathbf{a} = \mathbf{b}$ Note that the lines of action of the two vectors need not coincide. See Figure A2.2

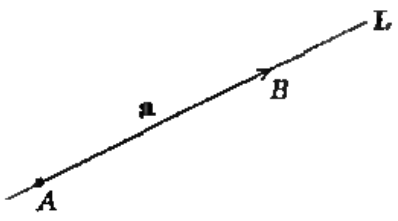


Fig. A2.1

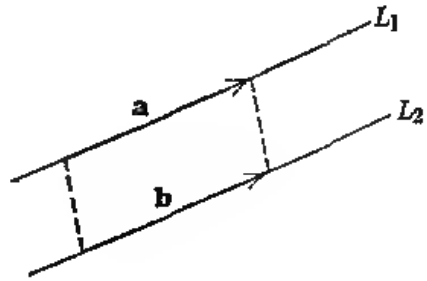


Fig. A2.2

DEFINITION 6. *To add* **b** *to* **a** *we use the parallelogram rule: Draw* **a** *through a point* O *and transport* **b** *parallel to itself so that its beginning point will coincide with the end of* **a**. *The vector* **c** *joining* O *to the end of* **b** *is the sum* (Figure A2.3). Addition is indicated as

$$\mathbf{c} = \mathbf{a} + \mathbf{b} \tag{A2.1}$$

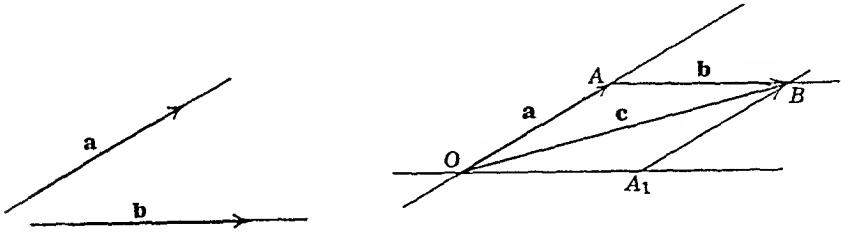


Fig. A2.3

THEOREM. *Vector addition obeys commutative and associative laws.*

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} \qquad \text{(commutative law)} \tag{A2.2}$$

$$(\mathbf{a} + \mathbf{b}) + \mathbf{d} = \mathbf{a} + (\mathbf{b} + \mathbf{d}) \qquad \text{(associative law)} \tag{A2.3}$$

Proof of (A2.2) is seen from Figure A2.3 by following the route OAB or OA_1B to obtain **c**.

Proof of (A2.3) can be seen from Figure A2.4.

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = (\mathbf{a} + \mathbf{b}) + \mathbf{d}$$

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = \mathbf{a} + (\mathbf{b} + \mathbf{d})$$

Hence the proof of the theorem.

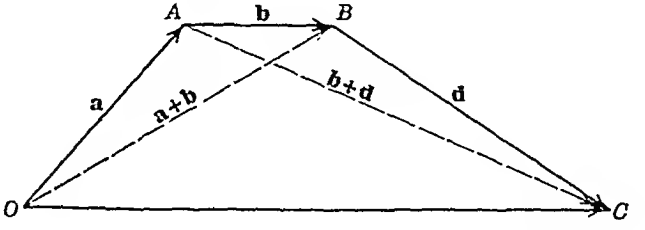


Fig. A2.4

Addition of n vectors ($n = 5$) is indicated on Figure A2.5.

DEFINITION 7 (*Multiplication of a Vector by a Real Number*). *Let* m *be a real number. Then* $m\mathbf{a}$ *is a vector whose magnitude is* m *times as long as* **a**, *whose line of action is the same as* **a**, *whose sense is positive if* $m > 0$, *negative if* $m < 0$ *and zero if* $m = 0$.

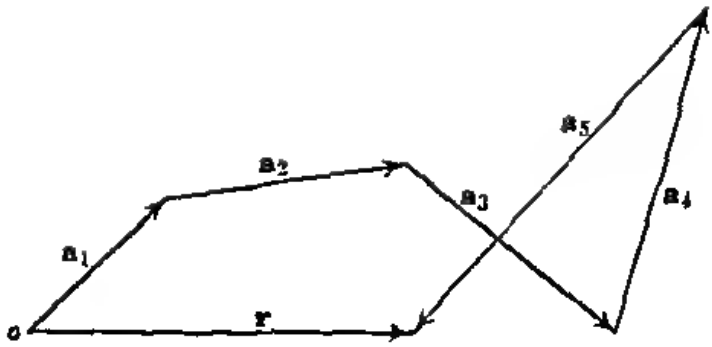


Fig. A2.5

From the properties of real numbers m and n it follows that

$$\begin{aligned} m\mathbf{a} &= \mathbf{a}m && \text{(commutative law)} \\ (mn)\mathbf{a} &= m(n\mathbf{a}) && \text{(associative law)} \\ (m+n)\mathbf{a} &= m\mathbf{a} + n\mathbf{a} && \text{(distributive law No. 1)} \\ m(\mathbf{a}+\mathbf{b}) &= m\mathbf{a} + m\mathbf{b} && \text{(distributive law No. 2)} \end{aligned}$$

Subtraction now follows from Figure A2 6

$$\mathbf{c} = \mathbf{a} + (-1)\mathbf{b} = \mathbf{a} - \mathbf{b}$$

Also

$$\mathbf{a} - \mathbf{a} = 0, \qquad 0\mathbf{a} = 0$$

Hence, so far as the operations of addition, subtraction and multiplication by real numbers are concerned, vectors may be treated formally in accordance with the rules of ordinary algebra. We note, however, that signs like $=$, $+$, $-$, have different meanings than the signs of ordinary algebra.

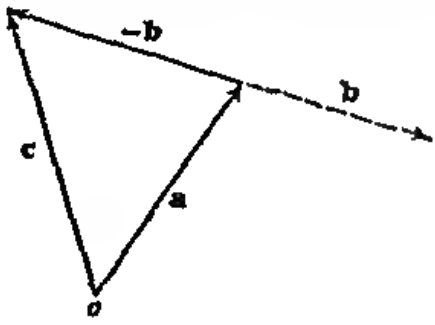


Fig. A2.6

A3. LINEAR DEPENDENCE

DEFINITION 1. *The n vectors* $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ *are linearly dependent if and only if there exist real numbers* $m_1, m_2, \ldots, m_n$ *not all zero such that*

$$m_1\mathbf{a}_1 + m_2\mathbf{a}_2 + \cdots + m_n\mathbf{a}_n = 0 \tag{A3.1}$$

otherwise the vectors are linearly independent.

EXAMPLE 1. One vector is linearly dependent if and only if it is a zero vector. For $m\mathbf{a} = \mathbf{0}$, $m \neq 0$ then $\mathbf{a} = \mathbf{0}$.

EXAMPLE 2. Two parallel (or collinear) vectors are linearly dependent. For two parallel vectors $\mathbf{a}$ and $\mathbf{b}$ we can write $\mathbf{b} = k\mathbf{a}$, $k \neq 0$. By writing $k = -m_1/m_2$ we find $m_1\mathbf{a} + m_2\mathbf{b} = 0$, $m_1 \neq 0$, and $m_2 \neq 0$.

DEFINITION 2. *The totality of vectors* $m\mathbf{a}$ *for an arbitrary real* m *and* $\mathbf{a} \neq \mathbf{0}$ *forms a one-dimensional real linear vector space.* The reason for this is clear, for every point on the line can be represented by some position vector $m\mathbf{a}$.

EXAMPLE 3. Three vectors on a plane are linearly dependent. For: let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be coplanar, then obviously according to Figure A3.1 $\mathbf{c} = m\mathbf{a} + n\mathbf{b}$. Writing $m = -m_1/m_3$, $n = -m_2/m_3$ not all m_1, m_2, m_3 zero, we obtain $m_1\mathbf{a} + m_2\mathbf{b} + m_3\mathbf{c} = \mathbf{0}$.

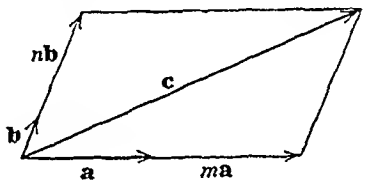


Fig. A3.1

DEFINITION 3. *The formula* $m\mathbf{a} + n\mathbf{b}$ *where* $\mathbf{a}$ *and* $\mathbf{b}$ *are two linearly independent vectors and* m *and* n *are arbitrary real numbers, defines a two-dimensional vector space.*
Similarly

DEFINITION 4. *An* n*-dimensional linear vector space is defined by the relation* $m_1\mathbf{a}_1 + m_2\mathbf{a}_2 + \cdots + m_n\mathbf{a}_n$ *if and only if* $\mathbf{a}_1, \ldots, \mathbf{a}_n$ *are linearly independent and* $m_1, \ldots, m_n$ *are arbitrary real numbers.*
In an n-dimensional space $n + 1$ vectors are always linearly dependent.

A4. BASE VECTORS, CARTESIAN COORDINATES

From the previous section we learned that in an n-dimensional linear vector space there are n linearly independent vectors. In particular, for $n = 3$ we have a three-dimensional space. In this space we can express any arbitrary vector as a linear combination of three linearly independent vectors. It is convenient to choose the latter vectors of unit magnitudes.

DEFINITION 1. *Three linearly independent vectors,* $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$, *whose directions are fixed in space when used to represent any arbitrary vector, are called rectilinear base vectors. If they have unit magnitudes, they are*

named rectilinear unit base vectors or simply rectilinear unit vectors. If the directions of the base vectors change in space, then they are not rectilinear.

In three dimensions, any vector **a** can be expressed as

$$\mathbf{a} = a^1\mathbf{e}_1 + a^2\mathbf{e}_2 + a^3\mathbf{e}_3 \tag{A4.1}$$

where a^1, a^2, a^3 are called the components of the vector **a** with respect to the rectilinear base vectors $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$. Superscripts of a^1, a^2, and a^3 are not exponents but mere identification tags. Equation (A4.1) also follows from Figure A4.1. For

$$\overrightarrow{OA} = \overrightarrow{OB} + \overrightarrow{OA}_3, \quad \overrightarrow{OB} = \overrightarrow{OA}_1 + \overrightarrow{OA}_2$$

Hence

$$\overrightarrow{OA} = \overrightarrow{OA}_1 + \overrightarrow{OA}_2 + \overrightarrow{OA}_3$$

Selecting base vectors of unit length, we may write:

$$\overrightarrow{OA}_1 = |\overrightarrow{OA}_1|\mathbf{e}_1, \quad \overrightarrow{OA}_2 = |\overrightarrow{OA}_2|\mathbf{e}_2, \quad \overrightarrow{OA}_3 = |\overrightarrow{OA}_3|\mathbf{e}_3$$

Hence substitution gives (A4.1) if we write

$$|\overrightarrow{OA}_1| = a^1, \quad |\overrightarrow{OA}_2| = a^2, \quad |\overrightarrow{OA}_3| = a^3$$

Thus a^i are the lengths of the parallel projections of **a** onto the direction lines of $\mathbf{e}_i$

DEFINITION 2 *Base vectors* $\mathbf{e}_1$, $\mathbf{e}_2$, *and* $\mathbf{e}_3$ *are called a right-handed system when a right-handed screw advancing in the* $\mathbf{e}_3$*-direction rotates* $\mathbf{e}_1$ *onto* $\mathbf{e}_2$ *by rotating through an angle less than* 180°. *Otherwise they are a left-handed system.*

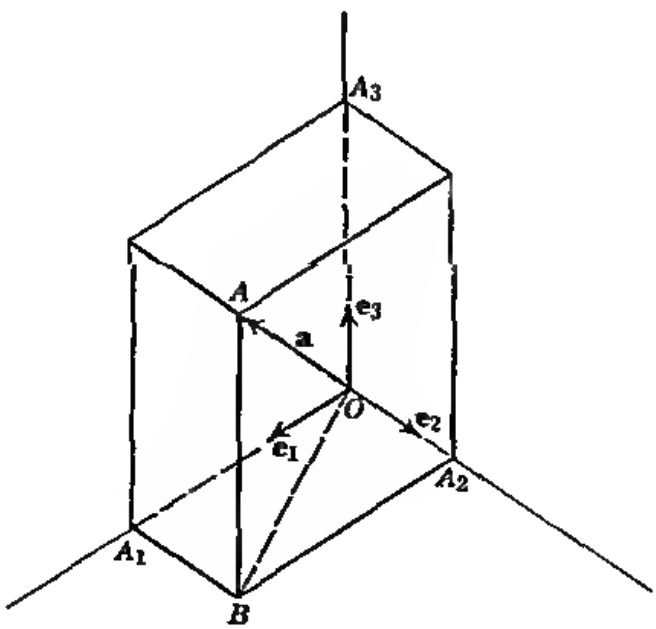


Fig. A4.1

EXAMPLE. System $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ of Figure A4.1 is right-handed.

DEFINITION 3. *When the base vectors are mutually orthogonal they are called orthogonal base vectors. If, in addition, they are of unit magnitude they are named orthonormal base vectors.*

We use $\mathbf{i}_1$, $\mathbf{i}_2$, and $\mathbf{i}_3$ to represent rectilinear orthonormal systems, which we call *rectangular* base vectors. We note that when base vectors are orthonormal, parallel projection becomes a normal projection (Figures A.4.2a,b).

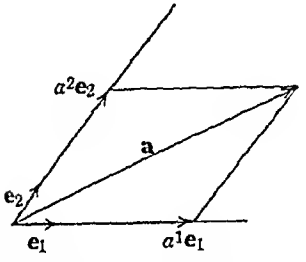


Fig. A4.2a Parallel projection.

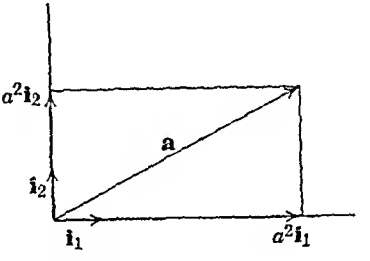


Fig. A4.2b Normal projection.

DEFINITION 4. *If* $\mathbf{a}$ *is the position vector* $\overrightarrow{OA}$ *of a point A then* a^i *with respect to* $\mathbf{e}_i$ *are called the rectilinear coordinates, and* a^i *with respect to* $\mathbf{i}_i$ *are called rectangular coordinates.*

We usually mark direction lines of $\mathbf{e}_i$ by x^i and express components of $\overrightarrow{OA}$ by x^1, x^2, x^3 (Figure A4.3).

$$\overrightarrow{OA} = x^1\mathbf{e}_1 + x^2\mathbf{e}_2 + x^3\mathbf{e}_3$$

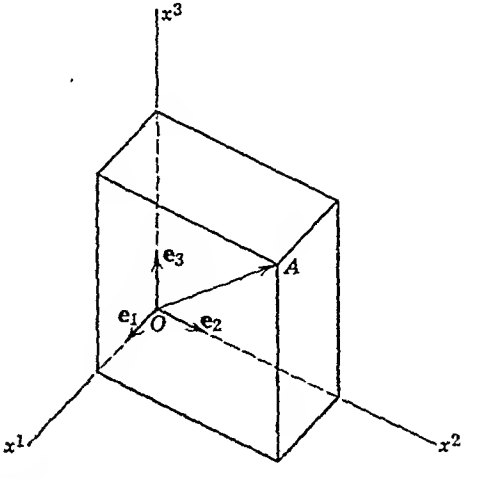


Fig. A4.3 Cartesian coordinates.

A5. RANGE, SUMMATION CONVENTION, KRONECKER DELTA, PERMUTATION SYMBOLS

The following conventions are useful as a shorthand:

DEFINITION 1 (*Range*). *When an index occurs unrepeated in a term, it is understood to take the values* 1, 2, . . , *N, where N is a specified integer called the range.*

EXAMPLE 1. $a_i = b_i (i = 1, 2, 3)$ means: $a_1 = b_1$, $a_2 = b_2$, $a_3 = b_3$; when the range is 3.

DEFINITION 2 (*Summation Convention*) *When an index appears twice in a term, summation with respect to that index over the range of the index is understood.*

EXAMPLE 2. $\mathbf{a} = a^1\mathbf{e}_1 + a^2\mathbf{e}_2 + a^3\mathbf{e}_3 = a^i\mathbf{e}_i, \qquad (i = 1, 2, 3)$

EXAMPLE 3. $a_{ij}b^j = a_{i1}b^1 + a_{i2}b^2 + a_{i3}b^3, \qquad (i = 1, 2, 3)$
The repeated index is called a *dummy index*. The use of other letters in place of dummy indices does not alter the expression.

DEFINITION 3. The *Kronecker Deltas* $\delta_{ij}, \delta^i_{\ j}, \delta_i^{\ j}$, and δ^{ij} *are double-indexed symbols defined as* 1 *when* $i = j$, *and zero when* $i \neq j$.

EXAMPLE 4. $a_i = \delta_{ij}b_j = b_i$, for in the summation $\delta_{ij}b_j$ the only term which is not zero is the term for which $i = j$.

EXAMPLE 5. $a_i = \delta_{ij}a_j$

DEFINITION 4. *The permutation symbols* e_{ijk} *or* e^{ijk} *are triple-indexed symbols defined as:* (a) *zero when at least two indices are equal;* (b) 1, *if the sequence of numbers* i, j, k *is the sequence* 1, 2, 3 *or an even permutation of the sequence; and* (c) -1, *if the sequence of* i, j, k *is an odd permutation of the sequence* 1, 2, 3.

Hence

$$e_{123} = e_{231} = e_{312} = 1, \qquad e_{132} = e_{213} = e_{321} = -1, \qquad \text{all other } e_{ijk} = 0$$

Similarly for e^{ijk}.

EXAMPLE 6. $a_i = e_{ijk}b_jc_k$ is equivalent to $a_1 = b_2c_3 - b_3c_2$, $a_2 = b_3c_1 - b_1c_3$, $a_3 = b_1c_2 - b_2c_1$.

A6. SCALAR PRODUCT

DEFINITION 1. *The scalar product* $\mathbf{a} \cdot \mathbf{b}$ *of two vectors* $\mathbf{a}$ *and* $\mathbf{b}$, *is defined as the product of their magnitudes times the cosine of the angle between them* (Figure A6.1).

$$\mathbf{a} \cdot \mathbf{b} = ab \cos(\mathbf{a}, \mathbf{b}) = |\mathbf{a}|\,|\overrightarrow{OA}| = |\mathbf{b}|\,|\overrightarrow{OB}| \tag{A6.1}$$

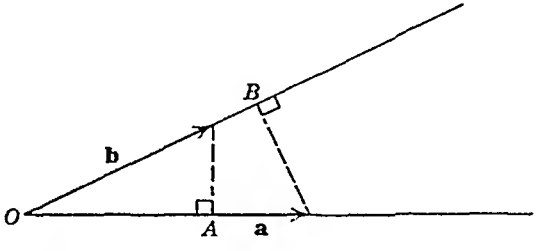


Fig. A6.1 Scalar product.

THEOREM 1. $\mathbf{a} \cdot \mathbf{b} = 0$ *if and only if either* $\mathbf{a} = 0$ *or* $\mathbf{b} = 0$ *or* $\mathbf{a}$ *is perpendicular to* $\mathbf{b}$. The proof immediately follows from (A6.1).

THEOREM 2. *The scalar product is commutative, distributive, and associative.*

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} \qquad \text{(commutative law)} \tag{A6.2}$$

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} \qquad \text{(distributive law)} \tag{A6.3}$$

$$m(\mathbf{a} \cdot \mathbf{b}) = (m\mathbf{a}) \cdot \mathbf{b} \qquad \text{(associative law)} \tag{A6.4}$$

THEOREM 3. *The rectangular base vectors* $\mathbf{i}_j$ *satisfy*

$$\mathbf{i}_j \cdot \mathbf{i}_k = \delta_{jk} \tag{A6.5}$$

The proof of this follows from (A6.1). For

$$\mathbf{i}_1 \cdot \mathbf{i}_1 = \mathbf{i}_2 \cdot \mathbf{i}_2 = \mathbf{i}_3 \cdot \mathbf{i}_3 = 1, \qquad \mathbf{i}_1 \cdot \mathbf{i}_2 = \mathbf{i}_2 \cdot \mathbf{i}_3 = \mathbf{i}_3 \cdot \mathbf{i}_1 = 0$$

This is precisely the same as (A6.5).

THEOREM 4. *If* a^i *and* b^i *are the rectangular components of* $\mathbf{a}$ *and* $\mathbf{b}$, *respectively, then*

$$\mathbf{a} \cdot \mathbf{b} = a^i b^i = a^1 b^1 + a^2 b^2 + a^3 b^3 \tag{A6.6}$$

For

$$\mathbf{a} \cdot \mathbf{b} = a^j \mathbf{i}_j \cdot b^k \mathbf{i}_k = a^j b^k \mathbf{i}_j \cdot \mathbf{i}_k = a^j b^k \delta_{jk}$$

This is a double summation over j and k in which all terms with $j \neq k$ are zero in view of the definition of the Kronecker delta, and the terms for which $j = k$ we have $\delta_{jk} = 1$. Thus

$$\mathbf{a} \cdot \mathbf{b} = a^j b^j$$

THEOREM 5. *The length a of vector* $\mathbf{a}$ *is given by*

$$a = (\mathbf{a} \cdot \mathbf{a})^{1/2} = (a^1 a^1 + a^2 a^2 + a^3 a^3)^{1/2} \tag{A6.7}$$

hence the *Theorem of Pythagoras.*

THEOREM 6

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{a^j b^j}{(a^j a^j)^{1/2} (b^k b^k)^{1/2}} \tag{A6.8}$$

The proof follows by equating (A6.1) to (A6.6) and using (A6.7) for the magnitudes of $\mathbf{a}$ and $\mathbf{b}$. We note that (A6.6) and (A6.8) are *not* valid for nonorthonormal base vectors.

A7. VECTOR PRODUCT

DEFINITION 1 *The vector product* $\mathbf{a} \times \mathbf{b}$ *of two vectors* $\mathbf{a}$ *and* $\mathbf{b}$ *is defined as the vector:*

$$\mathbf{a} \times \mathbf{b} = ab \sin(\mathbf{a}, \mathbf{b})\mathbf{e} \tag{A7.1}$$

where $\mathbf{e}$ *is the unit vector perpendicular to both* $\mathbf{a}$ *and* $\mathbf{b}$, *and forming a right-hand system in the order* $\mathbf{a}$, $\mathbf{b}$, $\mathbf{e}$ (Figure A7.1).

THEOREM 1. *If* $\mathbf{a} \times \mathbf{b} = 0$, *then* $\mathbf{a} = 0$ *or* $\mathbf{b} = 0$, *or* $\mathbf{a}$ *is parallel to* $\mathbf{b}$. The proof follows from (A7.1)

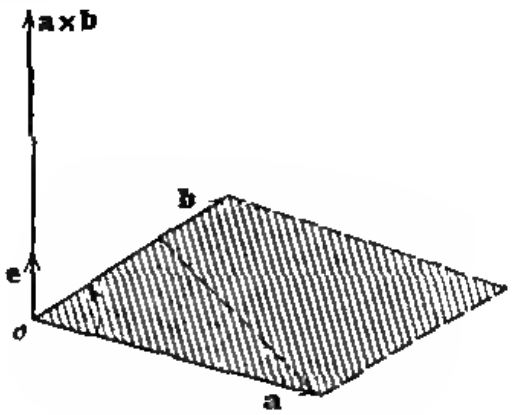


Fig. A7.1 Vector product.

THEOREM 2. *Vector multiplication is associative and distributive.*

$$(\text{A7.2}) \qquad (m\mathbf{a}) \times (n\mathbf{b}) = mn(\mathbf{a} \times \mathbf{b}) \qquad \text{(associative law)}$$

$$(\text{A7.3}) \qquad \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c} \qquad \text{(distributive law)}$$

The proof of (A7.2) is seen immediately from (A7.1). The proof of (A7.3) is obtained by applying (A7.1) to both sides of (A7.3).

THEOREM 3. *Vector multiplication is not commutative.*
The proof follows from the fact that if we change the order of multiplication, the scalar factor in (A7.1) is not changed but e is reversed.
Hence

$$(\text{A7.4}) \qquad \mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$

EXAMPLE 1

$$(\text{A7.5}) \qquad (\mathbf{a} + \mathbf{b}) \times (\mathbf{c} + \mathbf{d}) = \mathbf{a} \times \mathbf{c} + \mathbf{a} \times \mathbf{d} + \mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{d}$$

EXAMPLE 2. In general

$$(\text{A7.6}) \qquad (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} \neq \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$$

We thus conclude that the rules of ordinary algebra are valid for the vector product provided that the order of the factors is not altered.

THEOREM 4. *If base vectors* $\mathbf{i}_j$ *are orthonormal and right-handed, then*

$$(\text{A7.7}) \qquad \mathbf{i}_i \times \mathbf{i}_j = e_{ijk}\,\mathbf{i}_k$$

Proof: Applying (A7.1) to the right-handed orthonormal base vectors $\mathbf{i}_1$, $\mathbf{i}_2$, $\mathbf{i}_3$, we see that

$$\mathbf{i}_1 \times \mathbf{i}_2 = \mathbf{i}_3 \qquad \mathbf{i}_2 \times \mathbf{i}_3 = \mathbf{i}_1 \qquad \mathbf{i}_3 \times \mathbf{i}_1 = \mathbf{i}_2$$

$$\mathbf{i}_1 \times \mathbf{i}_1 = 0 \qquad \mathbf{i}_2 \times \mathbf{i}_2 = 0 \qquad \mathbf{i}_3 \times \mathbf{i}_3 = 0$$

These expressions are condensed in (A7.7).

THEOREM 5. *If* a^i *and* b^i *are rectangular components of* $\mathbf{a}$ *and* $\mathbf{b}$, *then*

$$(\text{A7.8}) \qquad \mathbf{a} \times \mathbf{b} = e_{ijk}a^j b^k \mathbf{i}_i = (a^2b^3 - a^3b^2)\mathbf{i}_1 + (a^3b^1 - a^1b^3)\mathbf{i}_2 + (a^1b^2 - a^2b^1)\mathbf{i}_3$$

Proof:

$$\mathbf{a} \times \mathbf{b} = a^j\mathbf{i}_j \times b^k\mathbf{i}_k = a^j b^k \mathbf{i}_j \times \mathbf{i}_k = e_{ijk}a^j b^k \mathbf{i}_i$$

in view of (A7.7). Hence the proof. We notice that (A7.8) can be written as a determinant:

$$\text{(A7.9)} \qquad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i}_1 & \mathbf{i}_2 & \mathbf{i}_3 \\ a^1 & a^2 & a^3 \\ b^1 & b^2 & b^3 \end{vmatrix} = e_{ijk} a^j b^k \mathbf{i}_i$$

Hence we have found a way of expanding determinants.

EXAMPLE 3

$$\text{(A7.10)} \qquad \begin{vmatrix} a^1 & a^2 & a^3 \\ b^1 & b^2 & b^3 \\ c^1 & c^2 & c^3 \end{vmatrix} = e_{ijk} a^i b^j c^k$$

This is seen immediately from (A7.9) and also from expanding the left and the right-hand side of (A7.10).

Representation of Areas by Vectors If we observe that the magnitude of $\mathbf{a} \times \mathbf{b}$ given by the coefficient of $\mathbf{e}$ in (A7.1) is the area of the parallelogram built on $\mathbf{a}$ and $\mathbf{b}$, then we find a vector representation of areas:

We associate a vector with a given plane area A, whose boundary curve is traced in a right-handed screw direction when the screw is advancing in the direction of this vector. We choose the direction of this vector perpendicular to the plane of the area and its magnitude equal to the surface area (Figure A7.2).

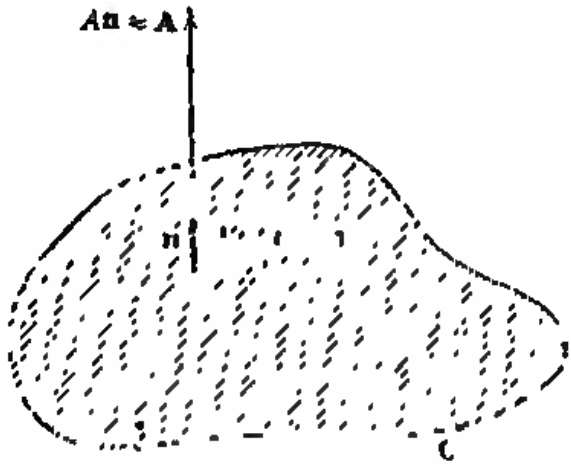


Fig. A7.2 Area vector

Thus, the area of the parallelogram, whose sides are the vectors $\mathbf{a}$ and $\mathbf{b}$ (see Figure A7.1) may be represented by $\mathbf{a} \times \mathbf{b}$.

Generalizing, we can represent any plane area in three dimensions by a vector. It is obvious that any nonflat surface area cannot be represented by a vector unless we are dealing with an infinitesimal area.

A8. SCALAR TRIPLE PRODUCT

The scalar triple product of three vectors **a**, **b**, **c** is written as $\mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$ or [abc]. No ambiguity can arise from omitting the parentheses since $\mathbf{a} \times (\mathbf{b} \cdot \mathbf{c})$ is meaningless.

THEOREM 1. *The product* $\mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$ *is numerically equal to the volume V of a parallelepiped having* **a**, **b**, **c**, *as concurrent edges. Its sign is positive or negative according as* **a**, **b**, **c** *form a right-handed or left-handed system* (Figure A8.1).

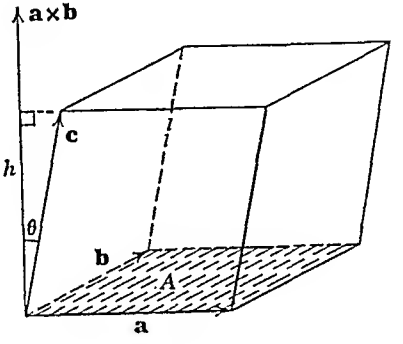


Fig. A8.1

The proof follows from multiplying the area $A = ab \sin(\mathbf{a}, \mathbf{b}) = |\mathbf{a} \times \mathbf{b}|$ by the corresponding altitude $h = c \cos \theta$. Thus

$$V = |\mathbf{a} \times \mathbf{b}|\,|\mathbf{c}| \cos \theta$$

The angle between $\mathbf{a} \times \mathbf{b}$ and **c** is θ or $\pi - \theta$ accordingly as **a**, **b**, **c** is a right-handed or left-handed system. Thus

$$\text{(A8.1)} \qquad \mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = \begin{cases} V \\ -V \end{cases} \quad \text{for } \mathbf{a}, \mathbf{b}, \mathbf{c} \quad \begin{cases} \text{right-handed} \\ \text{left-handed} \end{cases}$$

THEOREM 1. *The volume* $V = 0$ *if either one of* **a**, **b**, **c** *is zero,* **a** *is parallel to* **b**, *or* $\mathbf{a} \times \mathbf{b}$ *is perpendicular to* **c**.

All of these cases are included in **a**, **b**, **c** being coplanar (parallel to the same plane).

THEOREM 2. *If* a^i, b^i, c^i *are rectangular components of* **a**, **b**, **c**, *then*

$$\text{(A8.2)} \qquad \mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = \begin{vmatrix} a^1 & a^2 & a^3 \\ b^1 & b^2 & b^3 \\ c^1 & c^2 & c^3 \end{vmatrix} = e_{ijk} a^i b^j c^k$$

The proof is obtained by taking the scalar product of $\mathbf{c} = c^i\mathbf{i}_i$ by expression (A7.9) of $\mathbf{a} \times \mathbf{b}$

$$\mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = e_{ijk}a^j b^k \mathbf{i}_i \cdot c^l \mathbf{i}_l = e_{ijk}a^j b^k c^l \delta_{il} = e_{ijk}a^j b^k c^l$$

This is the same as (A8.2).

THEOREM 3. *The interchange of the dot and cross symbols does not alter the scalar triple product.*

$$\mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \tag{A8.3}$$

The proof is immediate from the rule of interchange of rows in a determinant.

A9. VECTOR TRIPLE PRODUCT

The vector triple product $\mathbf{r} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ is a vector perpendicular to $\mathbf{a}$ and $(\mathbf{b} \times \mathbf{c})$; hence it is in the plane of $\mathbf{b}$, $\mathbf{c}$.

Writing $\mathbf{q} = \mathbf{b} \times \mathbf{c}$ we find that

$$\begin{aligned} r^1 &= a^2q^3 - a^3q^2 = a^2(b^1c^2 - c^1b^2) - a^3(b^3c^1 - c^3b^1) \\ &= b^1(a^1c^1 + a^2c^2 + a^3c^3) - c^1(a^1b^1 + a^2b^2 + a^3b^3) \\ r^1 &= b^1(\mathbf{a} \cdot \mathbf{c}) - c^1(\mathbf{a} \cdot \mathbf{b}) \end{aligned}$$

Similarly r^2 and r^3 are obtained

$$\begin{aligned} r^2 &= b^2(\mathbf{a} \cdot \mathbf{c}) - c^2(\mathbf{a} \cdot \mathbf{b}) \\ r^3 &= b^3(\mathbf{a} \cdot \mathbf{c}) - c^3(\mathbf{a} \cdot \mathbf{b}) \end{aligned}$$

Therefore we find that

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\mathbf{c} \tag{A9.1}$$

a useful relation that can be memorized by reading *cab—bac*.

THEOREM. *Cross product is not associative.*

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} \neq \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$$

This can be seen easily by observing that the left side is coplanar with $\mathbf{a}$, $\mathbf{b}$ while the right is coplanar with $\mathbf{b}$, $\mathbf{c}$. We can also prove that

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} + (\mathbf{b} \times \mathbf{c}) \times \mathbf{a} + (\mathbf{c} \times \mathbf{a}) \times \mathbf{b} = 0 \tag{A9.2}$$

A10. RECIPROCAL BASE VECTORS

DEFINITION 1. *Two sets of base vectors* $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ *and* $\mathbf{e}^1, \mathbf{e}^2, \mathbf{e}^3$ *are called reciprocal if and only if*

$$\begin{aligned} &\mathbf{e}_1 \cdot \mathbf{e}^1 = 1 \quad \mathbf{e}_1 \cdot \mathbf{e}^2 = 0 \quad \mathbf{e}_1 \cdot \mathbf{e}^3 = 0 \\ &\mathbf{e}_2 \cdot \mathbf{e}^1 = 0 \quad \mathbf{e}_2 \cdot \mathbf{e}^2 = 1 \quad \mathbf{e}_2 \cdot \mathbf{e}^3 = 0 \\ &\mathbf{e}_3 \cdot \mathbf{e}^1 = 0 \quad \mathbf{e}_3 \cdot \mathbf{e}^2 = 0 \quad \mathbf{e}_3 \cdot \mathbf{e}^3 = 1 \end{aligned} \tag{A10.1}$$

These equations may be written shortly:

$$\mathbf{e}_i \cdot \mathbf{e}^j = \delta^j_i \tag{A10.2}$$

THEOREM 1. *Reciprocal base vectors* $\mathbf{e}_i$ *and* $\mathbf{e}^j$ *are related to each other by*

$$\mathbf{e}^1 = E^{-1}\mathbf{e}_2 \times \mathbf{e}_3, \qquad \mathbf{e}^2 = E^{-1}\mathbf{e}_3 \times \mathbf{e}_1, \qquad \mathbf{e}^3 = E^{-1}\mathbf{e}_1 \times \mathbf{e}_2 \tag{A10.3}$$

$$\mathbf{e}_1 = E\mathbf{e}^2 \times \mathbf{e}^3, \qquad \mathbf{e}_2 = E\mathbf{e}^3 \times \mathbf{e}^1, \qquad \mathbf{e}_3 = E\mathbf{e}^1 \times \mathbf{e}^2 \tag{A10.4}$$

where

$$E \equiv [\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3] = 1/[\mathbf{e}^1\mathbf{e}^2\mathbf{e}^3] \tag{A10.5}$$

Proof: Equations $\mathbf{e}_2 \cdot \mathbf{e}^1 = 0$, $\mathbf{e}_3 \cdot \mathbf{e}^1 = 0$ of the first column of (A10.1) express the fact that $\mathbf{e}^1$ is perpendicular to $\mathbf{e}^2$ and $\mathbf{e}^3$. Hence we write

$$\mathbf{e}^1 = m\mathbf{e}_2 \times \mathbf{e}_3$$

Taking the scalar product of this with $\mathbf{e}_1$ and using $\mathbf{e}_1 \cdot \mathbf{e}^1 = 1$ of the first column of (A10.1), we obtain $m = 1/[\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3] = E^{-1}$. Similarly, using equations of second and third columns of (A10.1), we obtain the other two equations of (A10.3).

Proof of (A10.4) follows, using equations on each row of (A10.1) with the same argument. If we multiply scalarly the first equations of (A10.3) and (A10.4) we obtain the identity $1 = 1$ proving the relation (A10.5).

Equation (A10.5) expresses the fact that the volumes of parallelepipeds built on $\mathbf{e}_i$ and $\mathbf{e}^i$ are reciprocal to each other in value. In view of (A10.5) both box products have the same sign. Hence, *two reciprocal sets are either both right-handed or both left-handed.*

Finally, we note that we can express (A10.3) and (A10.4) in the compact forms:

$$E\varepsilon_{ijk}\mathbf{e}^k = \mathbf{e}_i \times \mathbf{e}_j \tag{A10.6}$$

$$E^{-1}\varepsilon^{ijk}\mathbf{e}_k = \mathbf{e}^i \times \mathbf{e}^j \tag{A10.7}$$

DEFINITION 2. *When a set of base vectors and their reciprocals are identical, we say that the bases are self-reciprocal.* In this case, equation (A10.2) becomes

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij} \tag{A10 8}$$

This equation characterizes an orthonormal triple of unit vectors. Hence

THEOREM 2. *A set of base vectors are self-reciprocal if and only if they are mutually orthogonal unit vectors.*

These triple vectors are right-handed if $[\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3] = 1$, and left-handed if $[\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3] = -1$. We conclude then that when we use triple orthonormal vectors there is no need to distinguish their reciprocals. Hence no need for superscripts. This is the case with the $\mathbf{i}_i$-systems.

Geometrically, a clear understanding is reached by using a two-dimensional vector space. Let the two given unit vectors $\mathbf{e}_1$, $\mathbf{e}_2$ lie on the x^1- and the x^2-axes. We take the x_1- and the x_2-axes perpendicular to x^2 and x^1, respectively, at the origin O (Figure A10.1). We draw perpendiculars $\overline{AA_1}$ and $\overline{BB_1}$ from the end points of A and B of $\mathbf{e}_1$ and $\mathbf{e}_2$. Let A_1 and B_1 be the intersections of $\overline{AA_1}$ and $\overline{BB_1}$ with the x_1- and the x_2-axes, respectively.

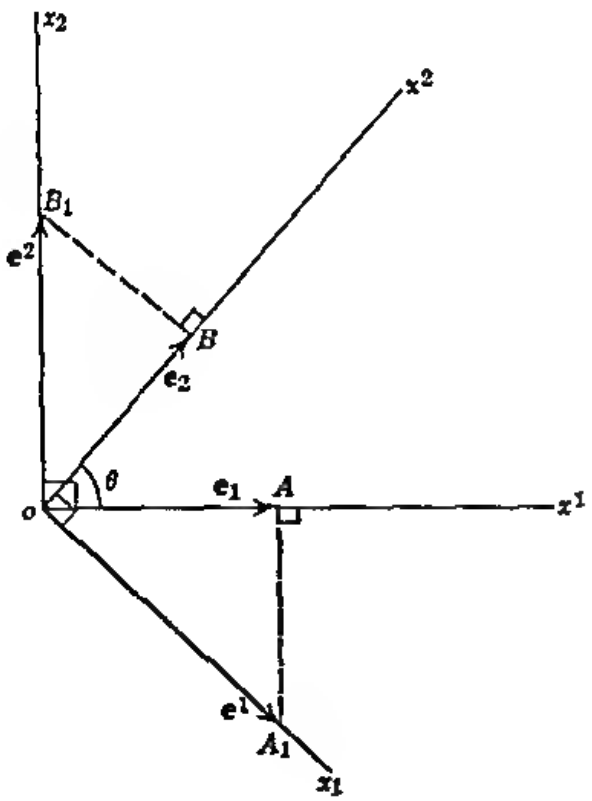


Fig. A10.1 Reciprocal basis.

We want to show that

$$\mathbf{e}^1 = \overrightarrow{OA}_1, \qquad \mathbf{e}^2 = \overrightarrow{OB}_1$$

are reciprocal to $\mathbf{e}_1$ and $\mathbf{e}_2$. Since Ox_2 is perpendicular to Ox^1 and Ox_1 is perpendicular to Ox^2, we have

$$\mathbf{e}_1 \cdot \mathbf{e}^2 = 0, \qquad \mathbf{e}_2 \cdot \mathbf{e}^1 = 0$$

Further

$$\mathbf{e}_1 \cdot \mathbf{e}^1 = |\overrightarrow{OA}|\,|\overrightarrow{OA}_1| \cos\left(\frac{\pi}{2} - \theta\right) = |\overrightarrow{OA}|\,|\overrightarrow{OA}_1| \sin\theta$$

$$= |\overrightarrow{OA}|^2 = 1$$

Similarly

$$\mathbf{e}_2 \cdot \mathbf{e}^2 = 1$$

Hence the proof.

Notice that the scale for the magnitudes of $\mathbf{e}^i$ is different from those of $\mathbf{e}_i$. That is, if $\mathbf{e}_i$ are of unit magnitudes, then $\mathbf{e}^i$ will have magnitudes greater than unity. In fact

$$|\mathbf{e}^1| = |\mathbf{e}_1|/\sin(\mathbf{e}_1, \mathbf{e}_2), \qquad |\mathbf{e}^2| = |\mathbf{e}_2|/\sin(\mathbf{e}_1, \mathbf{e}_2)$$

Here $\sin(\mathbf{e}_1, \mathbf{e}_2) \neq 0$, since otherwise $\mathbf{e}_1$ and $\mathbf{e}_2$ would be linearly dependent, and therefore contradictory to the original hypothesis that the vector space under consideration is two-dimensional.

In the special case of $\sin(\mathbf{e}_1, \mathbf{e}_2) = 1$, $\mathbf{e}_i$ and $\mathbf{e}^i$ coincide, leading to the $\mathbf{i}_i$-system.

A11. COVARIANT AND CONTRAVARIANT COMPONENTS OF A VECTOR

DEFINITION 1. *Components of a vector with respect to base vectors* $\mathbf{e}_i$ *are called contravariant, and with respect to reciprocal basis* $\mathbf{e}^i$ *are called covariant.*

We write

(A11.1) $$\mathbf{a} = a^i \mathbf{e}_i$$

(A11.2) $$\mathbf{a} = a_i \mathbf{e}^i$$

where a^i are contravariant and a_i are covariant components of the vector a. Scalar multiplication of both sides of (A11.1) and (A11.2) with $\mathbf{e}^j$ and $\mathbf{e}_j$, respectively, and the use of (A10.2) lead to

(A11.3) $$a^i = \mathbf{a} \cdot \mathbf{e}^i$$

(A11.4) $$a_i = \mathbf{a} \cdot \mathbf{e}_i$$

For two dimensions with $|\mathbf{e}_i| = 1$, these components are shown in Figure (A11.1).

We note that in transferring vector expressions into expressions in terms of components, the same base vectors must be used, for example, $\mathbf{a} = 0$ if and only if $a^i = 0$, or $a_i = 0$, $(m\mathbf{a})_i = ma_i$, and $(m\mathbf{a})^i = ma^i$. Similarly

$$(\mathbf{a} + \mathbf{b})_i = a_i + b_i, \qquad (\mathbf{a} + \mathbf{b})^i = a^i + b^i \tag{A11.5}$$

$$\mathbf{a} = \mathbf{b} \quad \text{if and only if} \quad a^i = b^i \quad \text{or} \quad a_i = b_i$$

The sum $a^i + b_i$ is *not* a component of the vector $\mathbf{a} + \mathbf{b}$ unless a^i and b_i are rectangular components. In the latter case, of course, there is no distinction between covariant and contravariant components of a vector—hence no need for superscripts.

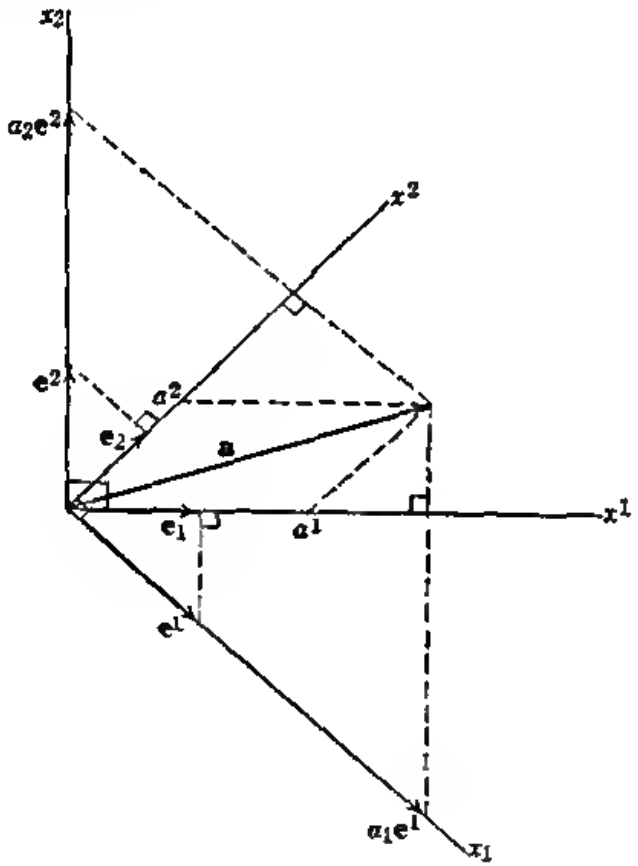


Fig. A11.1

A12. EUCLIDEAN METRIC TENSOR

DEFINITION 1. *Covariant g_{ij}, contravariant g^{ij}, and mixed $g_i^{\ j}$ components of the Euclidean metric tensor are defined by*

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j, \qquad g^{ij} = \mathbf{e}^i \cdot \mathbf{e}^j, \qquad g_i^{\ j} = \delta_i^{\ j} = \mathbf{e}_i \cdot \mathbf{e}^j \tag{A12.1}$$

THEOREM 1. *Components of the Euclidean metric tensor are symmetric.*

$$g_{ij} = g_{ji}, \qquad g^{ij} = g^{ji} \tag{A12.2}$$

The proof follows from (A12.1) and the fact that the scalar product is commutative.

In three dimensions, we have six independent components for each metric tensor. When these quantities are known, we can calculate the length of a vector, the angle between two vectors, and the scalar and vector products of two vectors. They make measurement possible, and, consequently, are named *Euclidean metric tensors.*

THEOREM 2. *The scalar product of* $\mathbf{a}$ *and* $\mathbf{b}$ *in rectilinear coordinates may be calculated by any one of the following formulas:*

$$\mathbf{a} \cdot \mathbf{b} = a^i b_i = a_i b^i \tag{A12.3}$$

$$\mathbf{a} \cdot \mathbf{b} = g_{ij} a^i b^j = g^{ij} a_i b_j \tag{A12.4}$$

Proof.

$$\mathbf{a} \cdot \mathbf{b} = a^i \mathbf{e}_i \cdot b_j \mathbf{e}^j = a^i b_j \delta_i{}^j = a^i b_i$$

$$\mathbf{a} \cdot \mathbf{b} = a^i \mathbf{e}_i \cdot b^j \mathbf{e}_j = a^i b^j \mathbf{e}_i \cdot \mathbf{e}_j = a^i b^j g_{ij}$$

Similarly, the second equations of (A12.3) and (A12.4) are proved.

THEOREM 3. *The square of the length of* $\mathbf{a}$ *and cos* $(\mathbf{a}, \mathbf{b})$ *in rectilinear coordinates may be calculated by any one of the following formulas:*

$$a^2 = \mathbf{a} \cdot \mathbf{a} = a_i a^i = g_{ij} a^i a^j = g^{ij} a_i a_j \tag{A12.5}$$

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{a_i b^i}{(a_j a^j)^{1/2} (b_k b^k)^{1/2}} = \frac{g_{ij} a^i b^j}{(g_{kl} a^k a^l)^{1/2} (g_{mn} b^m b^n)^{1/2}} = \frac{g^{ij} a_i b_j}{(g^{kl} a_k a_l)^{1/2} (g^{mn} b_m b_n)^{1/2}} \tag{A12.6}$$

The proof of (A12.5) and (A12.6) follows from (A12.3), (A12.4) and the definition (A6.1) of the scalar product.

THEOREM 4. *The covariant and contravariant components of* $\mathbf{a}$ *are related to each other by*

$$a_i = g_{ij} a^j \tag{A12.7}$$

$$a^i = g^{ij} a_j \tag{A12.8}$$

Proof:

$$a_i = \mathbf{a} \cdot \mathbf{e}_i = a^j \mathbf{e}_j \cdot \mathbf{e}_i = g_{ij} a^j$$

$$a^i = \mathbf{a} \cdot \mathbf{e}^i = a_j \mathbf{e}^j \cdot \mathbf{e}^i = g^{ij} a_j$$

Equations (A12.7) and (A12.8) enable us to calculate one set of components (for example, covariant) when the other set (for example, contravariant) and the components of the Euclidean metric tensor are known. Actually, we need to know only one set of g_{ij} or g^{ij}, for we can calculate the other set in terms of the first. This is shown to be the case by Theorem 6

THEOREM 5 *Reciprocal base vectors are related to each other by*

$$\mathbf{e}_i = g_{ij}\mathbf{e}^j \tag{A12.9}$$

$$\mathbf{e}^i = g^{ij}\mathbf{e}_j \tag{A12.10}$$

Proof:

$$\mathbf{a} = a_j\mathbf{e}^j = (\mathbf{a}\cdot\mathbf{e}_j)\mathbf{e}^j$$

Now take $\mathbf{e}_i$ in place of a

$$\mathbf{e}_i = (\mathbf{e}_i\cdot\mathbf{e}_j)\mathbf{e}^j = g_{ij}\mathbf{e}^j$$

Similarly, (A12.10) is proved.

THEOREM 6. *Contravariant components g^{ij} and covariant components g_{ij} of the Euclidean metric tensor are related to each other by*

$$g^{ij} = \frac{\text{cofactor of } g_{ij} \text{ in } |g_{ij}|}{|g_{ij}|} = \text{reduced cofactor}^1 \tag{A12.11}$$

$$|g_{ij}| = \begin{vmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{vmatrix} \tag{A12.12}$$

Proof:

$$\delta_i^j = \mathbf{e}_i\cdot\mathbf{e}^j = g_{ik}\mathbf{e}^k\cdot\mathbf{e}^j$$

$$\delta_i^j = g^{jk}\mathbf{e}_k\cdot\mathbf{e}_i$$

In the first expression we used (A12.9) and in the second (A12.10). Hence

$$\delta_i^j = g_{ik}g^{kj}, \qquad \delta_i^j = g^{jk}g_{ki} \tag{A12.13}$$

These equations are identical since g_{ij} and g^{ij} are symmetric Equation (A12.13) proves (A12.11), since the former is nothing but the expansion of the determinant (A12.12) with respect to the ith row divided by the value of $|g_{ij}|$. When $i = j$, this is equal to unity, since the sum of g_{ik} times their reduced cofactors is equal to unity. When $i \neq j$, then we multiply the

[1] The cofactor of an element g_{ij} of the determinant $|g_{ij}|$ is the subdeterminant obtained by striking the row and column in which g_{ij} stands, times $(-1)^{i+j}$

elements of the ith row with the reduced cofactors of the jth row and sum. In accordance with the theory of determinants, this gives zero. These and other properties of determinants are all included in (A12.13). From (A12.13) it also follows that

$$|g_{ij}| \cdot |g^{ji}| = 1 \quad \text{or} \quad |g_{ij}| = 1/|g^{ij}| \tag{A12.14}$$

Further we note that

$$g \equiv |g_{ij}| \equiv \det(\mathbf{e}_i \cdot \mathbf{e}_j) = [\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3]^2 = E^2 \tag{A12.15}$$

$$|g^{ij}| \equiv \det(\mathbf{e}^i \cdot \mathbf{e}^j) = [\mathbf{e}^1\mathbf{e}^2\mathbf{e}^3]^2 = E^{-2} = 1/g \tag{A12.16}$$

The first of these may be proved as follows

$$\det(\mathbf{e}_l \cdot \mathbf{e}_j) = e^{ijk}(\mathbf{e}_i \cdot \mathbf{e}_1)(\mathbf{e}_j \cdot \mathbf{e}_2)(\mathbf{e}_k \cdot \mathbf{e}_3)$$

according to (A7.10). But the cross product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is given by equation (A10.7)

$$\mathbf{a} \times \mathbf{b} = E^{-1}e^{ijk}a_ib_j\mathbf{e}_k$$

Selecting

$$a_l = \mathbf{e}_i \cdot \mathbf{e}_1, \qquad b_j = \mathbf{e}_j \cdot \mathbf{e}_2 \quad \text{or} \quad \mathbf{a} = \mathbf{e}_1, \qquad \mathbf{b} = \mathbf{e}_2$$

we see that

$$\det(\mathbf{e}_l \cdot \mathbf{e}_j) = E\mathbf{e}_1 \times \mathbf{e}_2 \cdot \mathbf{e}_3 = E^2$$

which proves the assertion. A similar proof is valid for (A12.16).

DEFINITION 2. *Permutation symbols* ε_{ijk} *and* ε^{ijk} *are defined as*

$$\varepsilon_{ijk} = Ee_{ijk} = \sqrt{g}e_{ijk}, \qquad \varepsilon^{ijk} = E^{-1}e^{ijk} = e^{ijk}/\sqrt{g} \tag{A12.17}$$

THEOREM 7. *Expressions for the cross product of* $\mathbf{a}$ *and* $\mathbf{b}$ *in rectilinear coordinates in terms of contravariant and covariant components of* $\mathbf{a}$ *and* $\mathbf{b}$ *are given by*

$$\mathbf{a} \times \mathbf{b} = Ee_{ijk}a^ib^j\mathbf{e}^k = \varepsilon_{ijk}a^ib^j\mathbf{e}^k \tag{A12.18}$$

$$\mathbf{a} \times \mathbf{b} = E^{-1}e^{ijk}a_ib_j\mathbf{e}_k = \varepsilon^{ijk}a_lb_j\mathbf{e}_k \tag{A12.19}$$

To prove (A12.18), we take the cross product of $\mathbf{a} = a^i\mathbf{e}_i$, $\mathbf{b} = b^j\mathbf{e}_j$, that is,

$$\mathbf{a} \times \mathbf{b} = a^ib^j\mathbf{e}_i \times \mathbf{e}_j$$

Upon using (A10.6), we obtain (A12.18). Proof of (A12.19) follows from the cross product of $\mathbf{a} = a_i\mathbf{e}^i$ by $\mathbf{b} = b_j\mathbf{e}^j$ and from using (A10.7). For rectangular coordinates, $E = 1$, which agrees with (A7.9). Note that the summation symbol appears once as a subscript and once as a superscript.

This is vitally important when the coordinates are *not* rectangular. Generalization of the g-, δ-, e-, and ϵ-systems and various theorems presented in this chapter to n-dimensional linear vector spaces is straightforward.

We now give an application to illustrate the power of the above notation.

EXAMPLE 1. Solve the system of linear equations

$$b^i = a_j^i x^j \qquad (i, j = 1, 2, \ldots, n) \tag{A12 20}$$

Solution:

Let α_j^i be reduced cofactor of a_j^i and the determinant $|a_j^i| \neq 0$. Thus we have

$$a_j^i \alpha_k^j = \delta_k^i, \qquad \alpha_j^i a_k^j = \delta_k^i \tag{A12.21}$$

Multiply both sides of (A12.20) by α_i^k and sum over i:

$$b^i \alpha_i^k = a_j^i \alpha_i^k x^j = \delta_j^k x^j = x^k$$

which is the solution of (A12.20).

A13. VECTOR FUNCTIONS, DIFFERENTIATION

DEFINITION 1 *A vector function* $\mathbf{u}(t)$ *of a scalar variable* t, *with* t *in an interval* $a \leq t \leq b$, *is defined uniquely if for a given* t *in this interval there is a single vector* $\mathbf{u}$. The vector function $\mathbf{u}(t)$ is well defined when the component functions $u_k(t)$ are well defined as functions in the usual sense, that is,

$$\mathbf{u}(t) = u_k(t)\mathbf{i}_k \tag{A13.1}$$

DEFINITION 2. *The derivative of* $\mathbf{u}(t)$ *with respect to* t *is defined in the same fashion as the derivative of a function, that is,*

$$\frac{d\mathbf{u}}{dt} = \lim_{\Delta t \to 0} \frac{\mathbf{u}(t + \Delta t) - \mathbf{u}(t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{u}}{\Delta t}$$

If we fix the origin of $\mathbf{u}(t)$ for all t, then as t varies the end points of $\mathbf{u}(t)$ will trace a curve $\mathscr{C}$ (Figure A13.1) Observe that the vector $\Delta \mathbf{u} = \mathbf{u}(t + \Delta t) - \mathbf{u}(t)$ as $\Delta t \to 0$ approaches the tangent at the end point of $\mathbf{u}(t)$.

The derivative of a vector can be expressed in terms of the derivatives of its components as

$$\frac{d\mathbf{u}}{dt} = \frac{du_k}{dt}\mathbf{i}_k \tag{A13.2}$$

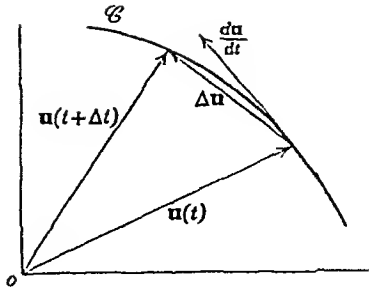


Fig. A13.1

It is easy to see that we can immediately apply our knowledge of differential calculus to a vector function $\mathbf{u}(t)$. Here, instead of one scalar function, we are dealing with three, $u_k(t)$. From component representation, it is simple to pass back to the vector forms. For instance, we can show that

$$\frac{d}{dt}(\lambda\mathbf{u}) = \lambda\frac{d\mathbf{u}}{dt} + \frac{d\lambda}{dt}\mathbf{u}$$

$$\frac{d}{dt}(\mathbf{u}+\mathbf{v}) = \frac{d\mathbf{u}}{dt} + \frac{d\mathbf{v}}{dt}$$

$$\frac{d}{dt}(\mathbf{u}\cdot\mathbf{v}) = \frac{d\mathbf{u}}{dt}\cdot\mathbf{v} + \mathbf{u}\cdot\frac{d\mathbf{v}}{dt}$$

$$\frac{d}{dt}(\mathbf{u}\times\mathbf{v}) = \frac{d\mathbf{u}}{dt}\times\mathbf{v} + \mathbf{u}\times\frac{d\mathbf{v}}{dt}$$

etc., where λ is a scalar function.

DEFINITION 3. *A scalar function $f(\mathbf{x})$ of a vector variable may be defined uniquely if for a given* x *in a region of space there is a single scalar f.*

In three dimensions, we have

$$\mathbf{x} = x_k\mathbf{i}_k \qquad (k = 1, 2, 3)$$

so that a scalar function $f(\mathbf{x})$ is a function of three variables $f(x_1, x_2, x_3)$.

A vector function $\mathbf{u}(\mathbf{x})$ of a vector variable is similarly defined, and the knowledge of it is equivalent to the knowledge of its component functions $u_k(\mathbf{x})$, that is,

(A13.3) $$\mathbf{u}(\mathbf{x}) = u_k(\mathbf{x})\mathbf{i}_k$$

Vector functions of two or more vectors are similarly introduced, for example,

$$\mathbf{u}(x, y) = u_k(x, y)\mathbf{i}_k \tag{A13 4}$$

Certain linear operators are of frequent occurrence in the study of vector functions. Here we define these operators in rectangular coordinates. Their expressions in curvilinear coordinates are given in Appendix C5.

DEFINITION 4. *The gradient of a scalar function* $f(\mathbf{x})$ *is defined by*

$$\text{grad } f \equiv \frac{\partial f}{\partial x_k}\mathbf{i}_k \tag{A13.5}$$

DEFINITION 5. *The divergence of a vector function* $\mathbf{u}(\mathbf{x})$ *is defined by*

$$\text{div } \mathbf{u} \equiv \frac{\partial u_k}{\partial x_k} \tag{A13.6}$$

DEFINITION 6. *The curl of a vector function* $\mathbf{u}(\mathbf{x})$ *is defined by*

$$\text{curl } \mathbf{u} \equiv e_{ijk}\frac{\partial u_k}{\partial x_j}\mathbf{i}_i \tag{A13.7}$$

If we formally introduce the operator

$$\nabla \equiv \mathbf{i}_k \frac{\partial}{\partial x_k} \tag{A13.8}$$

we see that operationally

$$\text{grad } f \equiv \nabla f = \mathbf{i}_k \frac{\partial f}{\partial x_k}$$

$$\text{div } \mathbf{u} \equiv \nabla \cdot \mathbf{u} = \mathbf{i}_k \frac{\partial}{\partial x_k} \cdot \mathbf{u} = \frac{\partial u_k}{\partial x_k} \tag{A13.9}$$

$$\text{curl } \mathbf{u} \equiv \nabla \times \mathbf{u} = \mathbf{i}_k \frac{\partial}{\partial x_k} \times \mathbf{u} = e_{ijk}\frac{\partial u_k}{\partial x_j}\mathbf{i}_i$$

Also frequently encountered is the *Laplacian*, defined by

$$\nabla^2 f \equiv \text{div grad } f = \frac{\partial^2 f}{\partial x_k \partial x_k} = \frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2} + \frac{\partial^2 f}{\partial x_3^2} \tag{A13.10}$$

A14. INTEGRAL THEOREMS

We present two important integral theorems of vector analysis without proof here.[2] These theorems are then extended to a region containing discontinuity surfaces.

THEOREM 1 (*Green-Gauss*)

$$\int_{\mathscr{V}} \operatorname{div} \mathbf{u}\, dv = \oint_{\mathscr{S}} \mathbf{u} \cdot \mathbf{n}\, da \tag{A14.1}$$

where $\mathscr{S}$ is the closed surface bounding $\mathscr{V}$ and $\mathbf{n}$ is the unit outward normal to $\mathscr{S}$ subject to certain regularity conditions (cf. Brand [1947, p. 218] or Kellogg [1929]).

THEOREM 2 (*Stokes*)

$$\int_{\mathscr{S}} \operatorname{curl} \mathbf{A} \cdot \mathbf{n}\, da = \oint_{\mathscr{C}} \mathbf{A} \cdot d\mathbf{p} \tag{A14.2}$$

where $\mathbf{n}$ is the unit normal vector to $\mathscr{S}$ closed by the directed boundary curve $\mathscr{C}$. The positive direction on $\mathscr{C}$ is taken so that a right-hand screw advances in the direction of $\mathbf{n}$ when the screw rotates in the positive (counter-clockwise) direction of $\mathscr{C}$.

When $\mathscr{V}$ contains a *discontinuity surface* σ across which $\mathbf{u}$ undergoes a jump (Fig. A14.1), (A14.1) is modified to

$$\int_{\mathscr{V}-\sigma} \operatorname{div} \mathbf{u}\, dv + \int_{\sigma} [\mathbf{u}] \cdot \mathbf{n}\, da = \oint_{\mathscr{S}-\sigma} \mathbf{u} \cdot \mathbf{n}\, da \qquad \text{(Green-Gauss)} \tag{A14.3}$$

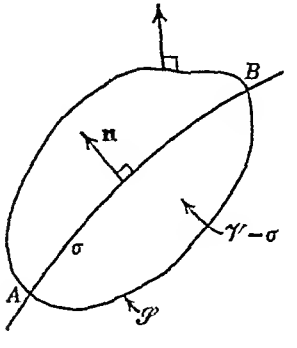


Fig. A14.1

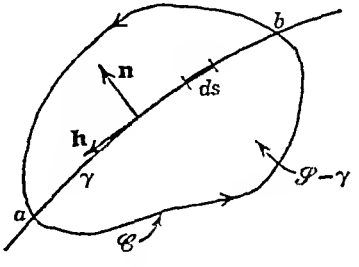


Fig. A14.2

[2] For proof, see books on vector analysis (cf. Brand [1947, Ch. VI]).

and when the surface $\mathscr{S}$ in (A14.2) contains a discontinuity line γ across which A is changed suddenly (Figure A14.2), (A14.2) is modified to

$$\text{(A14.4)} \qquad \int_{\mathscr{S}-\gamma} \operatorname{curl} \mathbf{A} \cdot \mathbf{n}\, da + \int_{\gamma} [\mathbf{A}] \cdot \mathbf{h}\, ds = \oint_{\mathscr{C}-\gamma} \mathbf{A} \cdot d\mathbf{p}$$

where the positive direction of the unit tangent vector h is shown in Figure A14.2. Also, quantities enclosed by a boldface bracket indicate the *jump*, that is, the difference between their values from the positive and negative sides of the discontinuity surface or line, for example,

$$[\mathbf{u}] = \mathbf{u}^+ - \mathbf{u}^-$$

where $\mathbf{u}^+$ and $\mathbf{u}^-$ are the values of u immediately near $\mathscr{S}$ at the positive side of the normal n to $\mathscr{S}$ and the opposite side, respectively. The same convention applies to [A] for the discontinuity line γ. To prove (A14.3) we apply the Green-Gauss theorem (A14.1) to $\mathscr{V}^+$ and $\mathscr{V}^-$ on each side of σ. Hence

$$\int_{\mathscr{V}^+} \operatorname{div} \mathbf{u}\, dv = \int_{\mathscr{S}^+ + \sigma^+} \mathbf{u} \cdot \mathbf{n}\, da$$

$$\int_{\mathscr{V}^-} \operatorname{div} \mathbf{u}\, dv = \int_{\mathscr{S}^- + \sigma^-} \mathbf{u} \cdot \mathbf{n}\, da$$

Upon adding these two expressions, we get

$$\int_{\mathscr{V}^+ + \mathscr{V}^-} \operatorname{div} \mathbf{u}\, dv = \int_{\mathscr{S}^+ + \mathscr{S}^-} \mathbf{u} \cdot \mathbf{n}\, da + \int_{\sigma^+} \mathbf{u}^+ \cdot \mathbf{n}^+\, da + \int_{\sigma^-} \mathbf{u}^- \cdot \mathbf{n}^-\, da$$

where $\mathbf{n}^+$ and $\mathbf{n}^-$ are the exterior normals to σ^+ and σ^-, respectively (Figure A14.3). But we have

$$\mathbf{n}^+ = -\mathbf{n}^- = -\mathbf{n}$$

so that

$$\int_{\sigma^+} \mathbf{u}^+ \cdot \mathbf{n}^+\, da + \int_{\sigma^-} \mathbf{u}^- \cdot \mathbf{n}^-\, da = \int_{\sigma} (\mathbf{u}^- - \mathbf{u}^+) \cdot \mathbf{n}\, da = -\int_{\sigma} [\mathbf{u}] \cdot \mathbf{n}\, da$$

Carrying this into the foregoing expression and writing $\mathscr{V} - \sigma$ for $\mathscr{V}^+ + \mathscr{V}^-$ and $\mathscr{S} - \sigma = \mathscr{S}^+ + \mathscr{S}^-$, we obtain[3] (A14.3).

[3] Note that $\mathscr{S} - \sigma$ in A14.3 excludes a curve on the surface $\mathscr{S}$ whose intersections with the plane of Figure A14.1 are marked as *A* and *B*

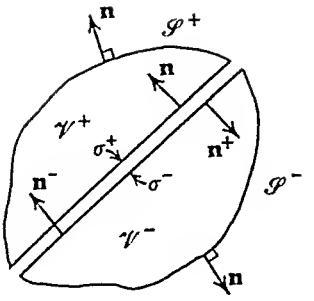


Fig. A14.3

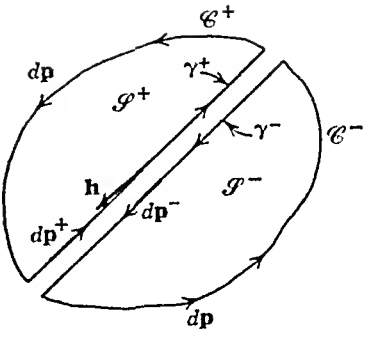


Fig. A14.4

To prove (A14.4) we apply (A14.2) to the surfaces on both sides of the discontinuity line γ.

$$\int_{\mathscr{S}^+} \text{curl A} \cdot \mathbf{n}\, da = \int_{\mathscr{C}^+} \mathbf{A} \cdot d\mathbf{p} + \int_{\gamma^+} \mathbf{A} \cdot d\mathbf{p}$$

$$\int_{\mathscr{S}^-} \text{curl A} \cdot \mathbf{n}\, da = \int_{\mathscr{C}^-} \mathbf{A} \cdot d\mathbf{p} + \int_{\gamma^-} \mathbf{A} \cdot d\mathbf{p}$$

Adding we have

$$\int_{\mathscr{S}^+ + \mathscr{S}^-} \text{curl A} \cdot \mathbf{n}\, da = \int_{\mathscr{C}^+ - \mathscr{C}^-} \mathbf{A} \cdot d\mathbf{p} + \int_{\gamma^+} \mathbf{A}^+ \cdot d\mathbf{p}^+ + \int_{\gamma^-} \mathbf{A}^- \cdot d\mathbf{p}^-$$

But on γ^+ and γ^- we have (Figure A14.4)

$$d\mathbf{p}^+ = -\mathbf{h}\, ds \qquad d\mathbf{p}^- = \mathbf{h}\, ds$$

so that

$$\int_{\gamma^+} \mathbf{A}^+ \cdot d\mathbf{p} + \int_{\gamma^-} \mathbf{A}^- \cdot d\mathbf{p}^- = \int_{\gamma} (\mathbf{A}^- - \mathbf{A}^+) \cdot \mathbf{h}\, ds = -\int_{\gamma} [\mathbf{A}] \cdot \mathbf{h}\, ds$$

Carrying this into the foregoing expression and writing $\mathscr{S}^+ + \mathscr{S}^- = \mathscr{S} - \gamma$ and $\mathscr{C}^+ + \mathscr{C}^- = \mathscr{C} - \gamma$ we obtain[4] (A14.4).

PROBLEMS

A1. Geometrically and analytically, prove that

$$|\mathbf{a}| + |\mathbf{b}| \geq |\mathbf{a} + \mathbf{b}|, \quad |\mathbf{a}| - |\mathbf{b}| \leq |\mathbf{a} - \mathbf{b}|$$

[4] Again $\mathscr{C}$ does not include two points a, b which are the intersections of γ with $\mathscr{C}$.

A2. What are the necessary and sufficient conditions for the end points of four vectors having the same origin to lie in the same plane?

A3. Calculate the sum of vectors drawn from the center of a rectangular polygon to its vertices

A4. Prove that

$$\sum_{i=1}^{n} a_i b_i \leq \left(\sum_{i=1}^{n} a_i^2\right)^{1/2} \left(\sum_{j=1}^{n} b_i^2\right)^{1/2}$$

A5. Prove the identities among the three vectors, **a**, **b**, and **c**.

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{c}) = \mathbf{b} \cdot \mathbf{c} - (\mathbf{a} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{c}),$$

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) = [\mathbf{abc}]^2$$

A6. Express the following vector expressions in terms of their rectangular components. Use the index notation.

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}, \quad (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}),$$

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d})$$

A7. If a_k, b_k, and c_k are covariant components of three vectors in rectilinear coordinates, determine the angle between the vector $d^h \equiv \varepsilon^{klm} a_l b_m$ and c_k. What is the angle between d^k and the reciprocal vector c^k.

A8. If $\dot{\mathbf{a}} = \boldsymbol{\omega} \times \mathbf{a}$, $\dot{\mathbf{b}} = \boldsymbol{\omega} \times \mathbf{b}$, show that $\dfrac{d}{dt}(\mathbf{a} \times \mathbf{b}) = \boldsymbol{\omega} \times (\mathbf{a} \times \mathbf{b})$

Express this result in index notation in terms of the rectangular components of the vectors involved.

A9. Show that, in three dimensions

$$e^{ijk} e_{lmn} = \delta^i{}_m \delta^j{}_n - \delta^i{}_n \delta^j{}_m$$

$$e^{ijk} e_{jkn} = 2\delta^i{}_n$$

$$e^{ijk} e_{ijk} = 6$$

A10. Verify the identity

$$e^{ijk} e_{mnp} a^{mnp} = a^{ijk} - a^{ikj} + a^{jki} - a^{jik} + a^{kij} - a^{kji}$$

A11. Write in full the expressions

$$e^{ijk} e_{kmn} a^{mn}, \quad e^{ijk} e_{jkm} a^m$$

A12. Prove that for a 3×3 determinant

$$\det(a^r{}_s) = \tfrac{1}{6} e_{ijk} e^{mnp} a^i{}_m a^j{}_n a^k{}_p$$

If $\det(a^r{}_s)$ is a $n \times n$ determinant, what would be the expression replacing this?

A13. If the element of a determinant, $|a^i{}_j|$, is a function of x^k prove that

$$\frac{\partial |a^i{}_j|}{\partial x^k} = A^i{}_j \frac{\partial a^j{}_i}{\partial x^k}$$

where $A^i{}_j$ is the cofactor of $a^j{}_i$.

A14. Express the following identities in terms of the rectangular components of the vectors. Use the index notation.

$$\nabla(ab), \quad \nabla \cdot (a\mathbf{u}), \quad \nabla \times (a\mathbf{u}),$$

$$\nabla \cdot (\nabla a), \quad \nabla \cdot (\nabla \times \mathbf{u}), \quad \nabla \cdot (\mathbf{u} \times \mathbf{v}),$$

$$\nabla \times (\mathbf{u} \times \mathbf{v}), \quad \nabla \times (\nabla \times \mathbf{u}), \quad \nabla(\mathbf{u} \cdot \mathbf{v})$$

A15. By use of index notation, prove the identities

(*a*) $\nabla(ab) = a\nabla b + b\nabla a$
(*b*) $\nabla \cdot (a\mathbf{u}) = a\nabla \cdot \mathbf{u} + \mathbf{u} \cdot \nabla a$
(*c*) $\nabla \times (a\mathbf{u}) = a\nabla \times \mathbf{u} + \nabla a \times \mathbf{u}$
(*d*) $\nabla(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \times (\nabla \times \mathbf{v}) + \mathbf{v} \times (\nabla \times \mathbf{u}) + (\mathbf{u} \cdot \nabla)\mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{u}$
(*e*) $\nabla \times (\nabla \times \mathbf{u}) = \nabla(\nabla \cdot \mathbf{u}) - \nabla^2 \mathbf{u}$
(*f*) $\nabla \times (\mathbf{u} \times \mathbf{v}) = (\mathbf{v} \cdot \nabla)\mathbf{u} - \mathbf{v}(\nabla \cdot \mathbf{u}) + \mathbf{u}(\nabla \cdot \mathbf{v}) - (\mathbf{u} \cdot \nabla)\mathbf{v}$

A16. If $\mathbf{r}$ is the position vector of a point, prove

$$\nabla \cdot (\mathbf{r}/r^3) = 0, \quad \nabla \times [f(r)\mathbf{r}] = 0$$

A17. Prove that u, v, and w satisfy $f(u, v, w) = 0$ if and only if $\nabla u \cdot \nabla v \times \nabla w = 0$. Express this latter expression in terms of the rectangular components of the vectors involved.

A18. If $\mathbf{a}$ is a constant vector, show that

$$\oint_{\mathscr{C}} \mathbf{a} \cdot d\mathbf{p} = 0, \quad \oint_{\mathscr{C}} \mathbf{a} \times d\mathbf{p} = 0$$

where $d\mathbf{p} = dx_k \mathbf{i}_k$, and $\mathscr{C}$ is a closed, regular curve.

A19. Calculate the following integrals over the surface of a sphere $\mathscr{S}$:

$$\oint_{\mathscr{S}} x\mathbf{i} \cdot \mathbf{n}\, da, \quad \oint_{\mathscr{S}} \frac{d\mathbf{p}}{p}$$

where $\mathbf{p}$ is the position vector of a point on $\mathscr{S}$.

A20. Prove Green's theorem

$$\int_{\mathscr{V}} (u\nabla^2 v - v\nabla^2 u)dv = \int_{\mathscr{S}} (u\nabla v - v\nabla u) \cdot d\mathbf{a}$$

Discuss the conditions for the validity of this theorem.

A21. Calculate the following integrals over an open sphere $\mathscr{S}$, $x_k x_k = 1$, $x_3 > 0$,

$$I_1 = \int_{\mathscr{S}} (\mathbf{n} \times \nabla) \cdot \mathbf{u}\, da, \quad I_2 = \int_{\mathscr{S}} (\mathbf{n} \times \nabla) \times \mathbf{u}\, da$$

for the cases

(*a*) $\mathbf{u} = 2x_2\mathbf{i}_1 - x_1\mathbf{i}_2 + x_3\mathbf{i}_3$
(*b*) $\mathbf{u} = [(x_3)^2 - 1]\,\mathbf{i}_3$

A22. If f is a solution of $\nabla^2 f = \rho$, show that

$$f(\mathbf{x}) = \frac{1}{4\pi} \oint_{\mathscr{S}} \left[\frac{1}{r} \nabla f - f \nabla \frac{1}{r}\right] \cdot d\mathbf{a} - \frac{1}{4\pi} \int_{\mathscr{V}} \frac{\rho}{r}\, dv$$

where r is the distance between $\mathbf{x}$, in $\mathscr{V}$, and the integration point.

A23. If ϕ is a solution of the wave equation, $\nabla^2\phi - \frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2} = 0$ show that

$$\phi(\mathbf{x}, t) = -\frac{1}{4\pi} \oint_{\mathscr{S}} \left\{[\phi] \frac{\partial}{\partial n}\left(\frac{1}{r}\right) - \frac{1}{cr}\frac{\partial r}{\partial n}\left[\frac{\partial \phi}{\partial t}\right] - \frac{1}{r}\left[\frac{\partial \phi}{\partial n}\right]\right\} da$$

where r is the distance from $\mathbf{x}$ inside of $\mathscr{S}$ to a typical point of $\mathscr{S}$. Here, $\frac{\partial}{\partial n}$ denotes differentiation along the outward normal to $\mathscr{S}$ and square brackets indicate the *retarded values*, that is, for $f(\mathbf{x}, t)$

$$[f] \equiv f\left(\mathbf{x}, t - \frac{r}{c}\right)$$

If $\mathbf{x}$ is outside the surface $\mathscr{S}$, then $\phi(\mathbf{x},t) = 0$.

APPENDIX B

Matrix Algebra

B1. INTRODUCTION

A short account of matrix algebra is included here to provide the background needed in studying this book. After the presentation of elementary algebraic operations in Sections B2 and B3, we give an account of matrix polynomials in Section B4. The Cayley-Hamilton theorem is of central importance in the reduction of polynomial constitutive equations. This is discussed in the same section. With Section B5 we begin a discussion of the theory of invariants. In Section B6 we present the invariants of any number of vectors, and in Section B7 that of a second-order tensor. The integrity basis of two matrices is discussed in Section B8. Section B9 is devoted to the reduction of matrix polynomials in one and two matrices.

The literature on matrices is extensive. A few books on the subject are: MacDuffee [1943], Michal [1947], Frazer, Duncan, and Collar [1947]. On classical group theory, we have the important book of Weyl [1946] and on algebraic invariants, Grace and Young [1903], Turnbull [1960], and Gurevich [1964]. For a useful account on invariant theory, see Spencer [1966].

B2. DEFINITIONS, ALGEBRAIC OPERATIONS

A collection of $m \times n$ numbers arranged in a rectangular tabular form having *m-rows* and *n-columns* is called a *matrix* with m-rows and n-columns. If a_{ij} is the element in the ith row and the jth column, then this

table is in the form

$$\|a_{ij}\| = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots\cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \tag{B2.1}$$

When $m = n$, we say that the matrix is *square*. When $m = 1$, the matrix is called a *row matrix*, and when $n = 1$, it is called a *column matrix*. Thus

$$\|a_{ij}\| \equiv \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \quad \text{(square matrix)} \tag{B2.2}$$

$$\|a_{ij}\| = [a_{11} \quad \cdots \quad a_{1n}] \quad \text{(row matrix)} \tag{B2.3}$$

$$\|a_{ij}\| = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} \quad \text{(column matrix)} \tag{B2.4}$$

In these matrices, we make the convention that the first index i indicates the number of the row and the second the number of the column. Thus, the element a_{35} in (B2.1) stands in row 3 and column 5.

In place of the elements a_{ij} in (B2.1) we may use elements with superscripts, for example, a^{ij}; or superscripts and subscripts, such as $a^i{}_j$. In the latter case, the ambiguity is removed if we displace the subscript and superscript as $a^i{}_j$ or $a_j{}^i$. To simplify the writing in place of $\|a_{ij}\|$ we also employ boldface letters a when no ambiguity arises as to the character of a.

DEFINITION 1. *Let* a *and* b *be two matrices of the same type, that is, they have the same number of rows and columns. The sum* a + b *is defined by*

$$\mathbf{a} + \mathbf{b} = \|a_{ij} + b_{ij}\| \tag{B2.5}$$

The tabular form of (B2.5) is

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12}, & \ldots, & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22}, & \ldots, & a_{2n} + b_{2n} \\ \cdots\cdots\cdots\cdots & \cdots\cdots\cdots\cdots & & \\ a_{m1} + b_{m1} & a_{m2} + b_{m2}, & \ldots, & a_{mn} + b_{mn} \end{bmatrix}$$

THEOREM 1. *If* a, b *and* c *are any matrices of the same type, then*

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} \quad \text{(commutative law)} \tag{B2.6}$$

$$(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c}) \quad \text{(associative law)} \tag{B2.7}$$

The proof is clear from Definition 1 of this section.

DEFINITION 2. *A matrix having all elements zero is called the zero matrix.*

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \tag{B2.8}$$

A square matrix has two diagonals, one ranges from the first element a_{11}, placed on the upper-left corner, to the last a_{nn}, placed on the lower-right corner. This diagonal is called the *main diagonal.* The other diagonal is traced from a_{1n} to a_{n1} and is called the second diagonal.

DEFINITION 3. *A square matrix with zero elements everywhere except on the main diagonal where the elements are all unity is called the unit matrix. Thus*

$$\mathbf{I} = \|\delta_{ij}\| = \begin{bmatrix} 1 & 0 & & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & & \cdots & 0 & 1 \end{bmatrix} \tag{B2.9}$$

DEFINITION 4. *Multiplication of a matrix* $\mathbf{b}$ *by a scalar* λ *is defined by*

$$\lambda\mathbf{b} = \|\lambda b_{ij}\| \tag{B2.10}$$

By adding $\mathbf{a}$ to $\lambda\mathbf{b}$ we shall have

$$\mathbf{a} + \lambda\mathbf{b} = \|a_{ij} + \lambda b_{ij}\| \tag{B2.11}$$

when $\lambda = -1$ we obtain *subtraction,*

$$\mathbf{a} - \mathbf{b} = \|a_{ij} - b_{ij}\|$$

Multiplication of two matrices is defined only when the number of rows of one matrix is equal to the number of columns of the other. Thus an $m \times n$ matrix $\mathbf{a}$ (m-rows, n-columns) may be multiplied by an $n \times p$ matrix $\mathbf{b}$.

DEFINITION 5. *The product* $\mathbf{c} = \mathbf{ab}$ *of two matrices* $\mathbf{a}$ *and* $\mathbf{b}$ *is defined by*

$$\mathbf{c} \equiv \mathbf{ab} = \|a_{ir}b_{rj}\| \tag{B2.12}$$

From this it is clear that the element $c_{ij} \equiv (\mathbf{ab})_{ij}$ of $\mathbf{ab}$ placed on ith row and jth column is given by

$$c_{ij} = (\mathbf{ab})_{ij} = a_{ir}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \tag{B2.13}$$

The following results are obtained from these

THEOREM 2

$$\mathbf{Ia} = \mathbf{aI} = \mathbf{a} \tag{B2.14}$$

$$\mathbf{0a} = \mathbf{a0} = \mathbf{0}$$

We also observe that the product of two matrices may vanish when neither of the matrices is the zero matrix. Thus, for example,

$$\mathbf{a} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{then} \quad \mathbf{ab} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{0}$$

The proof of the following theorem follows from Definition 5.

THEOREM 3 *If* **a**, **b**, *and* **c** *are three matrices, then*

$$\begin{aligned} (\mathbf{a} + \mathbf{b})\mathbf{c} &= \mathbf{ac} + \mathbf{bc} \\ \mathbf{c}(\mathbf{a} + \mathbf{b}) &= \mathbf{ca} + \mathbf{cb} \end{aligned} \qquad \text{(distributive law)} \tag{B2 15}$$

We note that in general the commutative law does not apply, that is,

$$\mathbf{ab} \neq \mathbf{ba} \tag{B2.16}$$

DEFINITION 6. *The transpose of a matrix* **a** *is obtained by interchanging the rows and columns in* **a**. Thus if **a** is as in (B2 1), its transpose $\mathbf{a}^T$ is

$$\mathbf{a}^T = \|a^T_{ij}\| = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \cdots & \cdots & \cdots & \cdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix} \tag{B2 17}$$

Using the definitions of addition, multiplication, and transposition of matrices, we can prove that

THEOREM 4. *If* **a** *and* **b** *are two matrices, and* λ *is a scalar, then,*

$$(\mathbf{a} + \mathbf{b})^T = \mathbf{a}^T + \mathbf{b}^T \tag{B2.18}$$

$$(\lambda\mathbf{a})^T = \lambda\mathbf{a}^T \tag{B2.19}$$

$$(\mathbf{ab})^T = \mathbf{b}^T\mathbf{a}^T \tag{B2.20}$$

We draw the attention of the reader particularly to the order of multiplication in (B2.20).

DEFINITION 7. *The trace of a square matrix* **a** *is the sum of its elements on the main diagonal.*

$$\text{tr } \mathbf{a} \equiv a_{ii} \tag{B2.21}$$

The trace of the product of two or three matrices is given by

$$\text{tr } (\mathbf{ab}) = a_{ij}b_{ji}$$

$$\text{tr } (\mathbf{abc}) = a_{ij}b_{jk}c_{ki}$$

It is clear that

$$\text{tr}\,(\mathbf{ab}) = \text{tr}\,(\mathbf{ba})$$
$$\text{tr}\,(\mathbf{abc}) = \text{tr}\,(\mathbf{bca}) = \text{tr}\,(\mathbf{cab}) \tag{B2.22}$$

Hence

THEOREM 5. *The trace of a matrix product is unaltered by a cyclic interchange of the factors of the product.*

Since the operation of transposition does not affect the elements of a matrix lying on its main diagonal, we see

THEOREM 6. *The trace of a matrix product is equal to the trace of the transpose of the product.*

$$\text{tr}\,(\mathbf{abc}) = \text{tr}\,(\mathbf{a}^T\mathbf{b}^T\mathbf{c}^T) \tag{B2.23}$$

Extension of (B2.22) and (B2.23) to any number of matrices is obvious.

B3. DETERMINANTS, INVERSES OF MATRICES

A set of n linear algebraic equations in n unknowns $x_1, x_2, \ldots, x_n$ may be expressed as

$$a_{ij}x_j = y_i \qquad (i = 1, 2, \ldots, n) \tag{B3.1}$$

Given y_i, the solution of (B3.1) may be obtained by the familiar method of determinants.

Given a square matrix $\mathbf{a}$, its determinant is expressed as

$$a \equiv |a_{ij}| \equiv \det a_{ij} \equiv \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \tag{B3.2}$$

For the multiplication of two determinants $|a_{ij}|$ and $|b_{ij}|$ we have the familiar rule

$$|a_{ij}|\,|b_{ij}| = |c_{ij}| \tag{B3.3}$$

where

$$c_{ij} = a_{ir}b_{rj} \tag{B3.4}$$

Thus the element c_{ij} in the ith row and jth column of the product of $|a_{ij}|$ and $|b_{ij}|$ is $c_{ij} = a_{ir}b_{rj}$. The expansion of a determinant $|a_{ij}|$ in terms of its *cofactors*, A_{ij}, is made by

$$\begin{aligned} a_{ik}A_{kj} &= a\delta_{ij} \\ a_{ki}A_{jk} &= a\delta_{ij} \end{aligned} \tag{B3.5}$$

where δ_{ij} is the Kronecker delta.

The reduced cofactor α_{ij} is defined by

$$\alpha_{ij} \equiv \frac{A_{ij}}{a} \tag{B3.6}$$

provided $a \neq 0$.

We now see that (B3.5) can also be expressed as

$$\begin{aligned} a_{ik}\alpha_{kj} &= \delta_{ij} \\ a_{ki}\alpha_{jk} &= \delta_{ij} \end{aligned} \tag{B3.7}$$

The linear equations (B3.1) can now formally be solved by multiplying both sides by α_{ki} and using (B3.7), that is,

$$\alpha_{ki}a_{ij}x_j = \alpha_{ki}y_i$$

In view of $(B3.7)_1$, this is equivalent to

$$\delta_{kj}x_j = \alpha_{ki}y_i$$

or

$$x_k = \alpha_{ki}y_i \tag{B3.8}$$

which is the solution of (B3.1).

In matrix notation (B3.1) may be expressed as

$$\mathbf{ax} = \mathbf{y} \tag{B3.9}$$

Now if we can find an inverse matrix $\mathbf{a}^{-1}$ such that

$$\mathbf{a}^{-1}\mathbf{a} = \mathbf{I} \tag{B3.10}$$

from (B3.9) through multiplication by $\mathbf{a}^{-1}$, then we get

$$\mathbf{a}^{-1}\mathbf{ax} = \mathbf{a}^{-1}\mathbf{y}$$

or, in view of (B3.10) and (B2.14),

$$\mathbf{x} = \mathbf{a}^{-1}\mathbf{y} \tag{B3.11}$$

which is the desired solution. Comparing this with (B3.6) we are led to

DEFINITION 1. *The inverse matrix* $\mathbf{a}^{-1}$ *of a matrix* $\mathbf{a}$ *is defined by*

$$a_{ij}^{-1} = \alpha_{ij} = \text{reduced cofactor of } a_{ij} = \frac{A_{ij}}{a} \tag{B3.12}$$

It is clear from (B3.7) that

$$\mathbf{a}\mathbf{a}^{-1} = \mathbf{a}^{-1}\mathbf{a} = \mathbf{I} \tag{B3.13}$$

The following theorems can be proven by using (B3.13) and (B2.20)

THEOREMS

$$(\mathbf{a}^{-1})^T = (\mathbf{a}^T)^{-1} \tag{B3.14}$$

$$(\mathbf{a}\mathbf{b})^{-1} = \mathbf{b}^{-1}\mathbf{a}^{-1} \tag{B3.15}$$

B4. MATRIX POLYNOMIALS, CAYLEY-HAMILTON THEOREM

In invariant theory, certain results of matrix polynomials are of great importance. Just as in the case of ordinary polynomials of scalars, we can form polynomials with matrices. For example,

DEFINITION 1. *An* n*th-degree polynomial in a matrix* $\mathbf{a}$ *may be defined by*

$$\mathbf{P}(\mathbf{a}) \equiv \alpha_0\mathbf{a}^n + \alpha_1\mathbf{a}^{n-1} + \cdots + \alpha_{n-1}\mathbf{a} + \alpha_n\mathbf{I}, \qquad \alpha_0 \neq 0 \tag{B4.1}$$

where $\alpha_0, \alpha_1, \ldots, \alpha_n$ *are constants.*

Similarly, polynomials in two or more matrices may be formed, for example

$$\mathbf{P}(\mathbf{a}, \mathbf{b}) \equiv \sum_{r,s} \alpha_{rs}\mathbf{a}^r\mathbf{b}^s \tag{B4.2}$$

where the highest number $r + s$ in the sum is the degree of the polynomial, and the *degree* of $\mathbf{a}$ is the highest number r in the sum.

DEFINITION 2. *The characteristic matrix of* $\mathbf{a}$ *is defined by* $\lambda\mathbf{I} - \mathbf{a}$ *where* λ *is a scalar.*

If $\mathbf{a}$ is an $n \times n$ matrix, then we see that

$$f(\lambda) \equiv \det(\lambda\mathbf{I} - \mathbf{a}) = \lambda^n + \alpha_1\lambda^{n-1} + \cdots + \alpha_{n-1}\lambda + \alpha_n \tag{B4.3}$$

is a polynomial in λ. Since $f(0) = \det(-\mathbf{a})$, we have

$$\alpha_n = (-)^n \det \mathbf{a} \tag{B4.4}$$

The remaining coefficients $\alpha_1, \alpha_2, \ldots, \alpha_{n-1}$ can be calculated in terms of the traces of powers of $\mathbf{a}$. For example, it can be verified that for a 3×3 matrix $\mathbf{a}$ we have

$$f(\lambda) = \det(-\lambda\mathbf{I}\mathbf{a}) = \lambda^3 - (\operatorname{tr}\mathbf{a})\lambda^2 + \tfrac{1}{2}[(\operatorname{tr}\mathbf{a})^2 - \operatorname{tr}\mathbf{a}^2]\lambda - \det\mathbf{a} \tag{B4.5}$$

For the general case (B4.1), by a long argument, it can be shown that $\alpha_1, \alpha_2, \ldots, \alpha_n$ can be computed from the recurrence formula

$$\begin{aligned}
\alpha_1 &= -\operatorname{tr}\mathbf{a} \\
\alpha_2 &= -\tfrac{1}{2}(\alpha_1 \operatorname{tr}\mathbf{a} + \operatorname{tr}\mathbf{a}^2) \\
\alpha_3 &= -\tfrac{1}{3}(\alpha_2 \operatorname{tr}\mathbf{a} + \alpha_1 \operatorname{tr}\mathbf{a}^2 + \operatorname{tr}\mathbf{a}^3) \qquad \text{(B4.6)} \\
&\cdots\cdots \quad \cdots\cdots \quad \cdots\cdots \\
\alpha_n &= -\frac{1}{n}(\alpha_{n-1} \operatorname{tr}\mathbf{a} + \alpha_{n-2} \operatorname{tr}\mathbf{a}^2 + \cdots + \alpha_1 \operatorname{tr}\mathbf{a}^{n-1} + \operatorname{tr}\mathbf{a}^n)
\end{aligned}$$

DEFINITION 3. *The algebraic equation*

$$f(\lambda) \equiv \det(\lambda\mathbf{I} - \mathbf{a}) = 0 \tag{B4.7}$$

is called the characteristic equation of the matrix $\mathbf{a}$. *The roots of this equation are called the characteristic roots (or proper numbers) of* $\mathbf{a}$.

An important theorem in the theory of matrix polynomials and in invariant theory is

THE CAYLEY-HAMILTON THEOREM. *Let*

$$f(\lambda) = \lambda^n + \alpha_1\lambda^{n-1} + \cdots + \alpha_{n-1}\lambda + \alpha_n$$

be the characteristic function of a matrix $\mathbf{a}$, *and* $\mathbf{I}$ *and* $\mathbf{0}$ *be, respectively, the unit and the zero matrices of the same order as* $\mathbf{a}$. *Then the matrix* $\mathbf{a}$ *satisfies the matrix polynomial equation*

$$\mathbf{a}^n + \alpha_1\mathbf{a}^{n-1} + \cdots + \alpha_{n-1}\mathbf{a} + \alpha_n\mathbf{I} = \mathbf{0} \tag{B4.8}$$

For a proof, see MacDuffee [1943, p. 77].

For a 3×3 matrix, (B4.8) has the explicit form

$$\mathbf{a}^3 - (\operatorname{tr}\mathbf{a})\mathbf{a}^2 + \tfrac{1}{2}[(\operatorname{tr}\mathbf{a})^2 - \operatorname{tr}\mathbf{a}^2]\mathbf{a} - (\det\mathbf{a})\mathbf{I} = \mathbf{0} \tag{B4.9}$$

Taking the trace of this equation, we see that

$$\operatorname{tr}\mathbf{a}^3 - \tfrac{3}{2}\operatorname{tr}\mathbf{a}^2 \operatorname{tr}\mathbf{a} + \tfrac{1}{2}(\operatorname{tr}\mathbf{a})^3 - 3\det\mathbf{a} = 0 \tag{B4.10}$$

so that $\det\mathbf{a}$ can be expressed in terms of $\operatorname{tr}\mathbf{a}$, $\operatorname{tr}\mathbf{a}^2$, and $\operatorname{tr}\mathbf{a}^3$, a fact which is also clear from (B4.6).

Here, we present an application of the Cayley-Hamilton theorem in the calculation of the inverse matrix $\mathbf{a}^{-1}$. Upon multiplication of (B4.8) by $\mathbf{a}^{-1}$, we find

$$\mathbf{a}^{-1} = -\frac{1}{\alpha_n}(\mathbf{a}^{n-1} + \alpha_1\mathbf{a}^{n-2} + \cdots + \alpha_{n-2}\mathbf{a} + \alpha_{n-1}\mathbf{I}) \tag{B4.11}$$

Consequently, by calculating $\alpha_1, \alpha_2, \ldots, \alpha_{n-1}$ from (B4.6) and the powers of a up to degree $n-1$, we obtain the inverse matrix $\mathbf{a}^{-1}$.

Other important applications of the Cayley-Hamilton theorem are given in the following section.

B5. THEORY OF INVARIANTS

In this section we give a brief account on the invariants of matrix polynomials. We shall deal only with polynomials in (a) one or two square matrices, (b) one square matrix and one vector, and (c) many vectors.

Let a and b denote two 3×3 square matrices and U and V two vectors with rectangular components U_i and V_i, respectively; that is,

$$\begin{aligned} \mathbf{a} &= \|a_{ij}\|, \qquad \mathbf{b} = \|b_{ij}\| \\ \mathbf{U} &= U_j\mathbf{i}_j, \qquad \mathbf{V} = V_j\mathbf{i}_j \end{aligned} \tag{B5.1}$$

From U_j we can form a skew-symmetric matrix (or bivector) $\mathbf{u}$ (or $u_{ij} = -u_{ji}$) by

$$u_{ij} = e_{ijk}U_k, \qquad U_k = \tfrac{1}{2}e_{ijk}u_{ij} \tag{B5.2}$$

where e_{ijk} is the permutation symbol. It is, therefore, clear that by replacing U and V by expressions of the type $(B5.2)_2$, we reduce the problem of invariants of matrix polynomials in vectors or vectors and matrices to those of invariants of matrices.

We shall be concerned only with invariance under orthogonal transformations. If x_i is a set of rectangular coordinates, then the transformation

$$x'_i = Q_{ij}x_j \tag{B5.3}$$

determines a new set of rectangular coordinates x'_i if Q_{ij} has the property

$$Q_{ki}Q_{kj} = \delta_{ij} \tag{B5.4}$$

It is immediate from this that

$$Q_{ik}Q_{jk} = \delta_{ij} \quad \text{and} \quad \det Q_{ij} = \pm 1 \tag{B5.5}$$

The matrix Q satisfying (B5.4) and (B5.5) is said to be an *orthogonal transformation*. The set of all orthogonal transformations forms a group.

This is the *full group* of orthogonal transformations. The set of orthogonal transformations with positive determinants also forms a group. This group is called the *proper group* of orthogonal transformations. For this group we have

$$Q_{ik}Q_{jk} = Q_{ki}Q_{kj} = \delta_{ij}, \qquad \det Q_{ij} = +1 \tag{B5.6}$$

Under transformation of the coordinates, a vector U and a tensor a transform in accordance with

$$U'_i = Q_{ij}U_j \tag{B5.7}$$

$$a'_{ij} = Q_{ik}Q_{jl}a_{kl} \tag{B5.8}$$

where U'_i and a'_{ij} are the components of U and a in rectangular coordinates x'_i. We note that if the bivector u_{ij} associated with U_i transforms as a second-order tensor, that is, if

$$u'_{ij} = Q_{ik}Q_{jl}u_{kl} \tag{B5.9}$$

then the vector U'_k associated with u'_{ij} by

$$U'_k = \tfrac{1}{2}e_{ijk}u'_{ij}$$

transforms according to

$$U'_i = |Q_{rs}|Q_{ij}U_j \tag{B5.10}$$

A vector of this kind is called an *axial vector.*

DEFINITION 1. *A function*

$$f(U_1, V_1, \ldots; a_{ij}, b_{ij}, \ldots)$$

of the components of a set of vectors U, V, . . . *and matrices* a, b, . . . *is said to be an invariant of these vectors and matrices under a given group of transformations* Q_{ij} *if*

$$f(U'_i, V'_i, \ldots; a'_{ij}, b'_{ij}, \ldots) = |Q_{ij}|^{\alpha} f(U_i, V_i, \ldots; a_{ij}, b_{ij}, \ldots) \tag{B5.11}$$

for every transformation Q_{ij} *of the group. In the case* $\alpha = 0$, *f is an absolute invariant, and in the case* $\alpha \neq 0$, *it is a relative invariant with weight* α.

We shall deal with absolute invariants unless otherwise stated.

EXAMPLE 1. The scalar product of two vectors U_iV_i is an absolute invariant. For we have

$$U'_iV'_i = Q_{ij}U_jQ_{ik}V_k = \delta_{jk}U_jV_k = U_iV_i \tag{B5.12}$$

EXAMPLE 2. The trace of a matrix a is an absolute invariant

$$a'_{ii} = Q_{ik}Q_{il}a_{kl} = \delta_{kl}a_{kl} = a_{kk} \tag{B5.13}$$

EXAMPLE 3. The determinant of a matrix a is an absolute invariant. For

$$|a'_{ij}| = |Q_{ik}Q_{jl}a_{kl}| = |Q_{ik}|^2|a_{kl}| = |a_{kl}| \tag{B5.14}$$

DEFINITION 2. *If* (B5.4) *and* (B5.5) *are valid for all members of the full orthogonal group* Q_{ij}, *then the invariant function* f *is called isotropic. If it includes only the members of the full orthogonal group for which* $\det Q_{ij} = +1$ *we say that* f *is hemitropic.* In this case, reflection of the x_i axis with respect to the plane $x_i = 0$ is not allowed.

DEFINITION 3. *An invariant is said to be even if it is unchanged under all orthogonal transformations; it is called odd if it changes sign under an orthogonal transformation with a negative determinant.*

Even invariants are absolute invariants for the full and proper orthogonal groups of transformations. Odd invariants are absolute invariants for the proper group, and relative invariants for the full group. For example, the scalar products U_iV_i, V_iW_i, W_iU_i are even invariants of three vectors **U**, **V**, and **W**, while the triple scalar product $e_{ijk}U_iV_jW_k$ is an odd invariant.

DEFINITION 4. *If the function* f *in* (B5.11) *is a polynomial in the components of vectors and tensors, then it is said to be a polynomial invariant.*

In this book we deal only with polynomial invariants. The knowledge of the smallest number of polynomial invariants enables one to reduce a matrix polynomial of any degree to its ultimate form. A few cases used in this book are presented in Sections 6 to 8.

DEFINITION 5. *A set of invariants that can be used to express any polynomial invariant as a polynomial in members of the given set is called an integrity basis. An integrity basis is minimal if it contains the smallest possible number of members.*

The main problem of invariant theory is to determine the minimal integrity basis.

B.6. INTEGRITY BASIS FOR VECTORS

In this section, we present a discussion of the integrity bases for absolute vectors. We confine our attention to vectors in three dimensional space. For simplicity, we begin with the discussion of three independent vectors $\mathbf{U}^{(r)}$, $(r = 1, 2, 3)$ and extend it subsequently to any M vectors $M > 3$.

1, Three Vectors. A scalar-valued function $f(\mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \mathbf{U}^{(3)})$ *of three vectors* $\mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \mathbf{U}^{(3)}$ *is isotropic if and only if it can be represented as a function of the scalar products* $\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)}$ $(r, s = 1, 2, 3)$.

From the previous article we know that the inner products

$$\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)} \qquad (r, s = 1, 2, 3) \tag{B6.1}$$

are invariant under the full orthogonal group of transformations. We would like to show that every invariant function $f(\mathbf{U}^{(r)})$ can be expressed in terms of (B6.1).

According to the definition (B5.11), if $f(\mathbf{U}^{(r)})$ is a scalar invariant function, $\mathbf{U}^{(r)}$, under the group of orthogonal transformations, Q_{ij}, we must have

$$f(U_i^{(1)}, U_i^{(2)}, U_i^{(3)}) = f(V_i^{(1)}, V_i^{(2)}, V_i^{(3)}) \tag{B6.2}$$

where

$$V_i^{(r)} = Q_{ij} U_j^{(r)} \tag{B6.3}$$

and $\mathbf{Q}$ is subject to

$$Q_{ik} Q_{jk} = Q_{ki} Q_{kj} = \delta_{ij}, \det Q_{ij} = \pm 1 \tag{B6.4}$$

Because of the restrictions (B6.4) on the nine components of Q_{ij}, only three are independent. Thus (B6.2) can be considered to be a function of these three independent variables. Since the left-hand side of (B6.2) is independent of Q_{ij}, the partial derivatives of (B6.2), with respect to these three independent variables, must vanish. We accomplish this differentiation by use of the lagrange's multipliers, λ_{kl}, that is,

$$\frac{\partial}{\partial Q_{kl}} [f(V_i^{(r)}) - \tfrac{1}{2}\lambda_{mn}(Q_{mi}Q_{ni} - \delta_{mn})] = 0$$

where $\lambda_{mn} = \lambda_{nm}$ are independent of $\mathbf{Q}$. Upon carrying out the above differentiation and using (B6.3), we get

$$\frac{\partial f}{\partial V_i^{(r)}} U_l^{(r)} - \lambda_{kj} Q_{jl} = 0$$

If we multiply this by Q_{ml} and use (B6.3) and (B6.4), we find

$$V_m^{(r)} \frac{\partial f}{\partial V_k^{(r)}} = \lambda_{km} \tag{B6.5}$$

Since $\lambda_{km} = \lambda_{mk}$, it follows that

$$V_m^{(r)} \frac{\partial f}{\partial V_k^{(r)}} - V_k^{(r)} \frac{\partial f}{\partial V_m^{(r)}} = 0 \tag{B6.6}$$

The characteristic equations of this system are

$$\frac{dV_k^{(r)}}{V_m^{(r)}} = -\frac{dV_m^{(s)}}{V_k^{(s)}} = ds; \qquad r, s \text{ not summed}$$

From this we form

$$V_k^{(r)} dV_k^{(s)} + V_k^{(s)} dV_k^{(r)} = 0$$

whose integral is

$$V_i^{(r)} V_i^{(s)} = U_i^{(r)} U_i^{(s)} \equiv J^{rs}, \qquad (r, s = 1, 2, 3) \tag{B6.7}$$

Thus, the general solution of the system (B6.6) is

$$f = f(J^{rs}) = f(\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)}), \qquad (r, s = 1, 2, 3) \tag{B6.8}$$

By direct substitution we can verify that

$$e_{ijk} V_i^{(r)} V_j^{(s)} V_k^{(t)} \tag{B6.9}$$

is also a solution of (B6.6). This is also clear from the fact that

$$(e_{ijk} V_i^{(r)} V_j^{(s)} V_j^{(t)})^2 = \det(U^{(r)} \cdot U^{(s)}), \quad (r, s, t = 1, 2, 3) \tag{B6.10}$$

Thus, in general, f may also depend on (B6.9). This, of course, is the case when f is hemitropic.

When f is an isotropic function, we have $\det Q_{ij} = \pm 1$, and the dependence on odd powers of (B6.9) disappears. None of the six inner products can be expressed in terms of the others, hence the integrity basis (B6.1) is minimal.

If we substitute (B6.8) into (B6.5), we obtain

$$2V_k^{(r)} V_m^{(s)} \frac{\partial f}{\partial J^{rs}} = \lambda_{km} \tag{B6.11}$$

If f is a polynomial in $\mathbf{U}^{(r)}$, it is also a polynomial in $\mathbf{V}^{(r)}$. From (B6.5), it follows that λ_{km} is a polynomial in $\mathbf{V}^{(r)}$. Now (B6.11) shows that when f is a polynomial in J^{rs}, this condition is satisfied. Conversely, we can show that when f is an isotropic polynomial in $\mathbf{U}^{(r)}$, then it is also a polynomial in J^{rs} (cf. Weyl [1939, Ch. II, Art. 9]). When f is a hemitropic polynomial, then it must have a form

$$f = P_1(\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)}) + [e_{ijk} U_i^{(1)} U_j^{(2)} U_k^{(3)}] P_2(\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)}), \qquad (r, s = 1, 2, 3, 4) \tag{B6.12}$$

where P_1 and P_2 are polynomials in the inner products, $\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)}$, since all the other ones in (B6.9) and their powers can be expressed in terms of $e_{ijk} U_j^{(1)} U_j^{(2)} U_k^{(3)}$ and (B6.1) through the use of (B6.10).

The relations of the type, (B6.10), among the invariants are called *syzygies*. By use of such relations, we can eliminate some of the odd invariants (B6.9) from our list, thus obtaining a minimal integrity basis for vectors.

2. More Than Three Vectors. The theorem given above can be shown to be valid for four or more vectors. To see this, suppose that we have four vectors $\mathbf{U}^{(r)}$ $(r = 1, 2, 3, 4)$ of which $\mathbf{U}^{(1)}$, $\mathbf{U}^{(2)}$, $\mathbf{U}^{(3)}$ are linearly independent, that is,

$$\det(\mathbf{U}^{(\alpha)} \cdot \mathbf{U}^{(\beta)}) \neq 0 \qquad (\alpha, \beta = 1, 2, 3) \tag{B6.13}$$

In three dimensions, the fourth vector, $\mathbf{U}^{(4)}$, can always be expressed in terms of $\mathbf{U}^{(r)}$, that is

$$\mathbf{U}^{(4)} = A_r \mathbf{U}^{(r)}$$

By taking the scalar product of this with $U^{(s)}$, we get

$$\mathbf{U}^{(4)} \cdot \mathbf{U}^{(s)} = A_r \mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)})$$

This set of three linear equations in the three unknowns, A_r, possesses a unique solution since (B6.13) is satisfied. The solution is of the form

$$A_r = A_r(\mathbf{U}^{(4)} \cdot \mathbf{U}^{(s)}, \mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)}, \det \mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)})$$

Clearly, then, the scalar function, f, of the four vectors, $\mathbf{U}^{(r)}$, $\mathbf{U}^{(4)}$, is expressible as

$$f(\mathbf{U}^{(r)}, \mathbf{U}^{(4)}) = f(\mathbf{U}^{(r)}, \mathbf{A}_r \mathbf{U}^{(r)}) = g(\mathbf{U}^{(r)}) \tag{B6.14}$$

since A_r is an invariant and expressible in terms of the inner products of the four vectors and the $\det(\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)})$. Thus, the problem is reduced to the previous case of a scalar function of three vectors.

It is clear that this procedure is valid for any number of vectors. The integrity basis (B6 1) and (B6 9) for $n > 3$ is not, however, minimal and we must create such syzygies as (B6.10) to reduce the set to a minimal basis.

Summarizing the foregoing results, we have

THEOREM (CAUCHY). *A scalar-valued function* $f(\mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \ldots, \mathbf{U}^{(m)})$ *of m vectors* $\mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \ldots, \mathbf{U}^{(m)}$ *is isotropic if and only if it can be expressed as a function of all possible scalar products.*

$$\mathbf{U}^{(r)} \cdot \mathbf{U}^{(s)} \qquad (r, s = 1, 2, \ldots m) \tag{B6.15}$$

It is hemitropic if it can be expressed as a function of (B6.15) *and all possible triple products.*[1]

$$e_{ijk} U_i^{(r)} U_j^{(s)} U_k^{(t)} \qquad (r, s, t = 1, 2, \ldots, m) \tag{B6.16}$$

The odd invariants (B6 15) are not all independent and are related to each other by (B6.10).

[1] This theorem was first proved by Cauchy [1850, p. 615]. For n-dimensions, see also Weyl [1946, Ch. II, Art. 9] and Spencer [1966] The present proof is different from various previous approaches and is believed to be new.

B7. INTEGRITY BASIS FOR A SECOND-ORDER TENSOR

In three dimensions, a second-order symmetric tensor **a** has three invariants

$$\text{(B7.1)} \qquad I_1(\mathbf{a}) \equiv \operatorname{tr} \mathbf{a}, \qquad I_2(\mathbf{a}) \equiv \operatorname{tr} \mathbf{a}^2, \qquad I_3(\mathbf{a}) \equiv \operatorname{tr} \mathbf{a}^3$$

Suppose that the invariants of another second-order symmetric tensor **b** coincide with (B7.1), that is,

$$\text{(B7.2)} \qquad I_j(\mathbf{a}) = I_j(\mathbf{b})$$

we would like to show that

THEOREM. *A scalar-valued function* $f(\mathbf{a})$ *of a symmetric tensor* **a** *is form invariant if and only if it can be expressed as a function of the invariants* $I_j(\mathbf{a})$, *that is,*

$$\text{(B7.3)} \qquad f(\mathbf{a}) = f(\mathbf{Q}\mathbf{a}\mathbf{Q}^T)$$

is valid if and only if

$$\text{(B7.4)} \qquad f = f(I_1, I_2, I_3)$$

To prove this, we recall that in view of (B7.2) the characteristic equations for **a** and **b** are the same (cf. (B4.9), (B4.10)), that is,

$$\text{(B7.5)} \qquad \lambda^3 - I_a\lambda^2 + II_a\lambda - III_a = 0$$

where

$$\text{(B7.6)} \qquad \begin{aligned} &I_a \equiv \operatorname{tr} \mathbf{a}, \qquad II_a \equiv \tfrac{1}{2}[(\operatorname{tr} \mathbf{a})^2 - \operatorname{tr} \mathbf{a}] \\ &III_a \equiv \det \mathbf{a} \equiv \tfrac{1}{3}[\operatorname{tr} \mathbf{a}^3 - \tfrac{3}{2} \operatorname{tr} \mathbf{a}^2 \operatorname{tr} \mathbf{a} + \tfrac{1}{2}(\operatorname{tr} \mathbf{a})^3] \end{aligned}$$

are called the *principal invariants* of **a**. The three roots a_j $(j = 1, 2, 3)$ of (B7.5) are called the proper numbers of the matrix **a**. In view of (B7.2) the proper numbers of **a** and **b** are therefore identical.

$$\text{(B7.7)} \qquad a_j = b_j \qquad (j = 1, 2, 3)$$

The characteristic equation (B7.5) is the same as (B4.7). This latter equation in turn is equivalent to a set of linear equations

$$\text{(B7.8)} \qquad (\mathbf{a} - \lambda\mathbf{I})\mathbf{e} = \mathbf{0}$$

where **e** is any vector. For each root $\lambda = a_i$ we have a set of three base vectors $\mathbf{e}_i$ such that

$$\text{(B7.9)} \qquad \mathbf{a}\mathbf{e}_i = a_i\mathbf{e}_i$$

where underlined indices are **not summed**. Similarly, for **b** we will have another basis $\mathbf{f}_i$ such that

$$\mathbf{b}\mathbf{f}_i = b_{\underline{i}}\mathbf{f}_{\underline{i}} = a_{\underline{i}}\mathbf{f}_{\underline{i}} \tag{B7.10}$$

The set $\mathbf{e}_i$ ($i = 1, 2, 3$) can be shown to be an orthogonal set as well as the set $\mathbf{f}_i$ (cf. Section 1.10) Consequently, an orthogonal transformation **Q** exists such that

$$\mathbf{Q}\mathbf{e}_i = \mathbf{f}_i \tag{B7.11}$$

Through (B7.9) to (B7.11) we see that

$$\mathbf{Q}\mathbf{a}\mathbf{e}_i = \mathbf{a}\mathbf{Q}\mathbf{e}_i = a_{\underline{i}}\mathbf{f}_{\underline{i}} = b_{\underline{i}}\mathbf{f}_{\underline{i}} = \mathbf{b}\mathbf{f}_i = \mathbf{b}\mathbf{Q}\mathbf{e}_i$$

Hence

$$\mathbf{Q}\mathbf{a} = \mathbf{b}\mathbf{Q} \quad \text{or} \quad \mathbf{b} = \mathbf{Q}\mathbf{a}\mathbf{Q}^T \tag{B7.12}$$

Conversely $(B7.12)_2$ implies (B7.2).

We have therefore shown that the invariants $\mathbf{I}_i$ of two tensors coincide if and only if an orthogonal tensor **Q** exists such that $(B7.12)_2$ holds. It follows from the definition (B7.3) that $f(\mathbf{a}) = f(\mathbf{b})$ whenever **a** and **b** have the same invariants. This, of course, is the proof of our theorem.

Using a theorem (*Weyl [1946, Ch. II, Art. 3]*), it can be shown that if f is a polynomial in the proper numbers a_i, then f is a polynomial in $\mathbf{I}_j$. Since a_i are the elements of the diagonal matrix corresponding to **a** in the basis $\mathbf{e}_i$, we see that

COROLLARY. *If* $f(\mathbf{a})$ *is a polynomial invariant in* **a**, *then it can be expressed as a polynomial in the invariants* $\mathbf{I}_j$.

From the above theorem, it is clear that $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ or $\mathbf{I}_a$, $\mathbf{II}_a$, and $\mathbf{III}_a$ constitute a minimal basis for a second-order symmetric tensor **a**.

B8. INTEGRITY BASIS OF TWO MATRICES

The scalar invariants of two matrices **a** and **b** can be shown to be[1]

$$\begin{gathered}
\operatorname{tr}\mathbf{a}, \quad \operatorname{tr}\mathbf{a}^2, \quad \operatorname{tr}\mathbf{a}^3 \\
\operatorname{tr}\mathbf{b}, \quad \operatorname{tr}\mathbf{b}^2, \quad \operatorname{tr}\mathbf{b}^3 \\
\operatorname{tr}\mathbf{ab}, \quad \operatorname{tr}\mathbf{a}^2\mathbf{b}, \quad \operatorname{tr}\mathbf{ab}^2, \quad \operatorname{tr}\mathbf{a}^2\mathbf{b}^2 \\
\operatorname{tr}\mathbf{aba}^2\mathbf{b}^2, \quad \operatorname{tr}\mathbf{a}^2\mathbf{bab}^2
\end{gathered} \tag{B8.1}$$

[1] The scalar invariants of any finite number of matrices were studied by Spencer and Rivlin [1959, 1960]. For a recent assessment of the work in this area, see Spencer [1966].

When a and b are *symmetric*, the list of invariants reduces to

$$\begin{matrix} \operatorname{tr} \mathbf{a}, & \operatorname{tr} \mathbf{a}^2, & \operatorname{tr} \mathbf{a}^3 \\ \operatorname{tr} \mathbf{b}, & \operatorname{tr} \mathbf{b}^2, & \operatorname{tr} \mathbf{b}^3 \\ \operatorname{tr} \mathbf{ab}, & \operatorname{tr} \mathbf{a}^2\mathbf{b}, & \operatorname{tr} \mathbf{ab}^2, \quad \operatorname{tr} \mathbf{a}^2\mathbf{b}^2 \end{matrix} \tag{B8.2}$$

When a is *antisymmetric* and b is *symmetric*, the list of invariants is

$$\begin{matrix} \operatorname{tr} \mathbf{a}^2, & \operatorname{tr} \mathbf{b}, & \operatorname{tr} \mathbf{b}^2, & \operatorname{tr} \mathbf{b}^3 \\ \operatorname{tr} \mathbf{a}^2\mathbf{b}, & \operatorname{tr} \mathbf{a}^2\mathbf{b}^2, & \operatorname{tr} \mathbf{ab}\, \mathbf{a}^2\mathbf{b}^2 \end{matrix} \tag{B8.3}$$

Since a vector A has the bivector equivalent

$$a_{ij} = e_{ijk} A_k$$

we see that the list (B8.3) is also the list of the scalar invariants of a vector A and a second-order symmetric tensor b.

B9. REDUCTION OF MATRIX POLYNOMIALS

DEFINITION 1. *A tensor function* $\mathbf{f}(\mathbf{a}, \mathbf{b}, \ldots, \mathbf{U}, \mathbf{V}, \ldots)$ *of the second-order tensor variables* a, b, . . . *and vectors* U, V, . . . *is said to be isotropic if*

$$\mathbf{Q}\mathbf{f}(\mathbf{a}, \mathbf{b}, \ldots; \mathbf{U}, \mathbf{V}, \ldots)\, \mathbf{Q}^T = \mathbf{f}(\mathbf{Q}\mathbf{a}\mathbf{Q}^T, \mathbf{Q}\mathbf{b}\mathbf{Q}^T, \ldots; \mathbf{Q}\mathbf{U}, \mathbf{Q}\mathbf{V}, \ldots) \tag{B9.1}$$

holds for every orthogonal transformation Q *satisfying* (B5.4) *and* (B5.5). *It is said to be hemitropic if* Q *satisfies* (B5.6).

When f is an isotropic matrix polynomial in the variables a, b, . . . ; U, V, . . . , reduction is possible to simpler forms. To illustrate this, let us consider an isotropic matrix polynomial P(a) of a single symmetric matrix a. By definition P(a) is subject to

$$\mathbf{Q}\mathbf{P}(\mathbf{a})\mathbf{Q}^T = \mathbf{P}(\mathbf{Q}\mathbf{a}\mathbf{Q}^T) \tag{B9.2}$$

THEOREM 1. *A matrix polynomial* P(a) *of a single* 3 × 3 *symmetric matrix* a *is expressible as*

$$\mathbf{P}(\mathbf{a}) = \kappa_0 \mathbf{I} + \kappa_1 \mathbf{a} + \kappa_2 \mathbf{a}^2 \tag{B9.3}$$

where κ_0, κ_1, and κ_2 are polynomials in the invariants of a

$$\operatorname{tr} \mathbf{a}, \quad \operatorname{tr} \mathbf{a}^2, \quad \operatorname{tr} \mathbf{a}^3 \tag{B9.4}$$

To prove this theorem, we consider the matrix polynomial of degree n in the 3×3 symmetric matrix $\mathbf{a}$; that is,

$$\text{(B9.5)} \qquad \mathbf{P}(\mathbf{a}) \equiv \alpha_0\mathbf{a}^n + \alpha_1\mathbf{a}^{n-1} + \cdots + \alpha_{n-1}\mathbf{a} + \alpha_n\mathbf{I}, \qquad \alpha_0 \neq 0$$

with α_r being some scalars. From the Cayley-Hamilton theorem (B4.9), it is clear that $\mathbf{a}^3$ is expressible, in terms of $\mathbf{a}^2$, $\mathbf{a}$, and $\mathbf{I}$, with coefficients that are polynomials in the invariants (B9.4). Similarly, by multiplying (B4.9) with $\mathbf{a}$ we see that we can express $\mathbf{a}^4$ in terms of $\mathbf{a}^3$, $\mathbf{a}^2$, and $\mathbf{a}$, or, in view of (B4.9) (by replacing $\mathbf{a}^3$ by its equivalent) in terms of $\mathbf{a}^2$, $\mathbf{a}$, and $\mathbf{I}$, where the coefficients of these terms are polynomials in the invariants of (B9.4). Continuing in this fashion, we see that every term $\mathbf{a}^k$, $(k > 2)$ in (B9.5) can be expressed as a linear combination of $\mathbf{a}^2$, $\mathbf{a}$, and $\mathbf{I}$, with coefficients that are polynomials in (B9.4). Hence the proof of the theorem

THEOREM 2. *A hemitropic matrix polynomial* $\mathbf{P}(\mathbf{a}, \mathbf{b})$ *of two matrices can be expressed in the form*[2]

$$\begin{aligned}\text{(B9.6)} \qquad \mathbf{P} = {} & \kappa_0\mathbf{I} + \kappa_1\mathbf{a} + \kappa_2\mathbf{a}^2 + \kappa_3\mathbf{b} + \kappa_4\mathbf{b}^2 \\ & + \kappa_5\mathbf{ab} + \kappa_6\mathbf{ba} + \kappa_7\mathbf{a}^2\mathbf{b} + \kappa_8\mathbf{ba}^2 \\ & + \kappa_9\mathbf{ab}^2 + \kappa_{10}\mathbf{b}^2\mathbf{a} + \kappa_{11}\mathbf{a}^2\mathbf{b}^2 + \kappa_{12}\mathbf{b}^2\mathbf{a}^2 \\ & + \kappa_{13}\mathbf{aba}^2 + \kappa_{14}\mathbf{bab}^2 + \kappa_{15}\mathbf{ab}^2\mathbf{a}^2 + \kappa_{16}\mathbf{ba}^2\mathbf{b}^2\end{aligned}$$

where κ_0 to κ_{16} are polynomials in the invariants (B8.1).

COROLLARY 1. *When* $\mathbf{P}$, $\mathbf{a}$, *and* $\mathbf{b}$ *are symmetric* (B9.6) *reduces to*

$$\begin{aligned}\text{(B9.7)} \qquad \mathbf{P} = {} & \kappa_0\mathbf{I} + \kappa_1\mathbf{a} + \kappa_2\mathbf{a}^2 + \kappa_3\mathbf{b} + \kappa_4\mathbf{b}^2 \\ & + \kappa_5(\mathbf{ab} + \mathbf{ba}) + \kappa_6(\mathbf{a}^2\mathbf{b} + \mathbf{ba}^2) \\ & + \kappa_7(\mathbf{ab}^2 + \mathbf{b}^2\mathbf{a}) + \kappa_8(\mathbf{a}^2\mathbf{b}^2 + \mathbf{b}^2\mathbf{a}^2)\end{aligned}$$

where κ_0 *to* κ_8 *are polynomials in the invariants* (B8.2).

COROLLARY 2. *When* $\mathbf{P}$, $\mathbf{a}$, *are antisymmetric and* $\mathbf{b}$ *is symmetric then* (B9.6) *reduces to*

$$\begin{aligned}\text{(B9.8)} \qquad \mathbf{P} = {} & \kappa_1\mathbf{a} + \kappa_2(\mathbf{ab} + \mathbf{ba}) + \kappa_3(\mathbf{ab}^2 - \mathbf{b}^2\mathbf{a}) + \kappa_4(\mathbf{a}^2\mathbf{b} - \mathbf{ba}^2) \\ & + \kappa_5(\mathbf{a}^2\mathbf{b}^2 - \mathbf{b}^2\mathbf{a}^2) + \kappa_6(\mathbf{ba}^2\mathbf{b}^2 - \mathbf{b}^2\mathbf{a}^2\mathbf{b})\end{aligned}$$

where κ_1 *to* κ_6 *are polynomials in the invariants* (B8.3).

[2] For proof of this, see Spencer and Rivlin [1959, 1960] See also Green and Adkins [1960, Appendix).

PROBLEMS

1. Perform the following matrix operations:

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \pm \begin{bmatrix} c_1 & d_1 \\ c_2 & d_2 \end{bmatrix}, \qquad \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} c_1 & d_1 \\ c_2 & d_2 \end{bmatrix}, \qquad \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} d_1 & e_1 & f_1 \\ d_2 & e_2 & f_2 \\ d_3 & e_3 & f_3 \end{bmatrix}$$

2. Give the solution of the homogeneous equations

$$x_1 + ax_2 + b^2 x_3 = 0$$

$$x_1 + \qquad\qquad x_3 = 0$$

$$bx_1 - ax_2 - (a+b)x_3 = 0$$

3. Find the reciprocal of the matrices

$$\|a_{ij}\| = \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & -2 \\ 0 & 1 & 2 & 1 \end{bmatrix}, \qquad \|a_{ij}\| = \begin{bmatrix} 15 & 11 & 6 & -9 & -15 \\ 1 & 3 & 9 & -3 & -8 \\ 7 & 6 & 6 & -3 & -11 \\ 7 & 7 & 5 & -3 & -11 \\ 17 & 12 & 5 & -10 & -16 \end{bmatrix}$$

4. Find the eigenvalues of the matrices:

$$\begin{bmatrix} u & x & y & z \\ -x & u & -z & y \\ -y & z & u & -x \\ -z & -y & x & u \end{bmatrix}, \qquad \begin{bmatrix} 7 & 4 & 3 & 2 \\ -3 & -2 & -5 & 2 \\ -6 & -4 & 0 & -4 \\ -6 & -4 & 1 & -5 \end{bmatrix}$$

5. If x_i in the quadratic form $q \equiv a_{ij}x_i x_j$, with a constant matrix, a, is subjected to a linear transformation, $x_i = b_{ij}y_j$, with a constant matrix, b, prove that the resulting form has the matrix $\mathrm{b}^T\mathrm{a}\,\mathrm{b}$.

6. Reduce the following quadratic forms to sums of squares with constant coefficients:

$$A \equiv 2x_1^2 + 2x_2^2 - 15x_3^2 + 8x_1x_2 - 12x_2x_3 - 12x_1x_3$$

$$B \equiv (6+a^2)(x_1^2 + x_3^2) + (9+a^2)x_2^2 + 2(a-4)x_1x_2 + 2(3a-2)x_1x_3 - 2(5a+2)x_2x_3$$

7. Prove the Cayley-Hamilton theorem.

8. Prove the following identities among the 3×3 matrices, a, b, and c:

(*a*) $\mathbf{aba^2 + a^2ba = aba\ tra + a^2trab + a(tra^2b - tra\ trab)}$

$\mathbf{- b\ det\ a + I\ det\ a\ trb}$

(*b*) $\mathbf{abc + bca + cab + bac + acb + cba = (bc + cb)\ tra}$

$\mathbf{+ (ca + ac)\ trb + (ab + ba)\ trc + a(trbc - trb\ trc)}$

$\mathbf{+ b(trca - trc\ tra) + c(trab - tra\ trb)}$

$\mathbf{+ I(tra\ trb\ trc - tra\ trbc - trb\ trca - trc\ trab}$

$\mathbf{+ trabc + trcba)}$

9. Show that for 3×3 matrices, a, b, and c, the following identities hold (Jordan and Eringen, [1964a])

(*a*) $\mathbf{aba + a^2b + ba^2 = a(trab - tra\ trb) + \tfrac{1}{2}b\,[tra^2 - (tra)^2]}$

$\mathbf{+ (ab + ba)\ tra + a^2trb + I\ (tra^2b}$

$\mathbf{- tra\ trab + \tfrac{1}{2}\ trb[(tra)^2 - tra^2]\}}$

(*b*) $\mathbf{aba^2 + a^2ba = aba\ tra + a^2\ trab + a(tra^2b - tra\ trab)}$

$\mathbf{-b\ det\ a + I\ det\ a\ trb}$

(*c*) $\mathbf{a^2bca + acba^2 + abca^2 + a^2cba + acaba + abaca}$

$\mathbf{= (abca + acba)tra + (aca^2 + a^2ca)trb + (a^2ba + aba^2)trc}$

$\mathbf{+ a^2(trbc - trb\ trc) + aba(trca - trc\ tra) + aca(trab - tra\ trb)}$

$\mathbf{+ a^2(tra\ trb\ trc - tra\ trbc - trb\ trca - trc\ trab + trabc + trcba)}$

10. Prove the identities among 3×3 matrices, a, b, and c (Jordan & Eringen, [1964a]):

(*a*) $\mathbf{abaca + acaba = aba\ trac + aca\ trab + a(trabac - trab\ trac)}$

$\mathbf{+ [bc + cb - b\ trc - c\ trb + I(trb\ trc - trbc)]\ det\ a}$

(*b*) $\mathbf{a^2bca + acba^2 = -(abca^2 + a^2cba) - aba\ trac - aca\ trab}$

$\mathbf{- a(trabac - trab\ trac) - det\ a[bc + cb - b\ trc - c\ trb}$

$\mathbf{+ I(trb\ trc - trbc)] + (abca + acba)tra}$

$\mathbf{+ (aca^2 + a^2ca)trb + (a^2ba + aba^2)trc}$

$\mathbf{+ a^3(trbc - trb\ trc) + aba(trca - trc\ tra)}$

$\mathbf{+ aca(trab - tra\ trb) + a^2(tra\ trb\ trc - tra\ trbc}$

$\mathbf{- trb\ trca - trc\ trab + trabc + trcba)}$

11. If a, b, and c are antisymmetric 3×3 matrices and G is a 3×3 symmetric matrix, prove that (Jordan & Eringen, [1964a]):

(*a*) $\mathrm{a^2bc + cba^2 = a^2\ trbc - \tfrac{1}{2}(ab + ba)trac + \tfrac{1}{2}(cb + bc)tra^2 + \tfrac{1}{2}\,I(trac\ trab - tra^2\ trcb)}$

(*b*) $\mathrm{a^2b^2c - cb^2a^2 = \tfrac{1}{4}[(cab - bac)trab + (abc - cba)trab - (ab^2 - b^2a)trac - (ba^2 - a^2b)trcb - (ca^2 - a^2c)trb^2 - (cb^2 - b^2c)tra^2]}$

(*c*) $\mathrm{acba^2 - a^2bca = \tfrac{1}{2}(acb - bca)tra^2 + \tfrac{1}{2}(ba^2 - a^2b)trac}$

(*d*) $\mathrm{cbGa^2 + a^2Gbc = (ab + ba)\,(tracG - \tfrac{1}{2}\,trac\,trG) + \tfrac{1}{2}(bcG + Gcb)\,tra^2 + \tfrac{1}{2}(a^2G + Ga^2)trcb + \tfrac{1}{2}\,G\,tra^2\,trcb + I\,trab(tracG - \tfrac{1}{2}\,trac\,trG)}$

12. If a, b, c and G are as prescribed in Problem 11, show that the following identities hold (Jordan & Eringen, [1964a]):

(*a*) $\mathrm{(ab^2cG + Gcb^2a) + (cb^2aG + Gab^2c) = \tfrac{1}{2}[(Gab + baG) + (Gba + abG)]trbc + \tfrac{1}{2}[(Gbc + cbG) + (Gcb + bcG)]trab - (Gb^2 + b^2G)trac - G(trab\ trbc - trb^2\ trac)}$

(*b*) $\mathrm{(Ga^2bc + cba^2G) + (Gcba^2 + a^2bcG) = \tfrac{1}{2}[(Gcb + bcG) + (Gbc + cbG)]tra^2 - \tfrac{1}{2}[(Gab + baG) + (Gba + abG)]trac + (Ga^2 + a^2G)trbc - G(trac\ trab - tra^2\ trcb)}$

(*c*) $\mathrm{(cGab^2 + b^2aGc) + (cGb^2a + ab^2Gc) = \tfrac{1}{2}(aGc + cGa)trb^2 + \tfrac{1}{2}(bGc + cGb)trab}$

13. Find the independent scalar invariants of one vector and one 3×3 matrix.

14. Give the complete list of independent scalar invariants of one vector and two 3×3 matrices.

15. Make all reductions in a symmetric 3×3 hemitropic matrix polynomial of

(*a*) Three vectors;
(*b*) A vector and two symmetric 3×3 matrices;
(*c*) Two vectors and one symmetric 3×3 matrix;
(*d*) Three symmetric 3×3 matrices.

16. (Short Term Paper) Study the literature and write an account of the determination of the scalar invariants of N vectors and M 3×3 symmetric matrices.

17. (Short Term Paper) Study the literature and give an account of the reduction of vector and matrix polynomials that are hemitropic in N vectors and M 3×3 matrices.

APPENDIX C

Tensor Analysis

C1. INTRODUCTION

In this appendix, we give a short account of tensor analysis necessary in the systematic derivation of the basic equations of continuum mechanics in curvilinear coordinates. Although the theory developed in this book is based on rectangular coordinates, the conversion of the field equations to curvilinear coordinates without use of tensors is a tedious task. The account given here provides a minimum knowledge for performing this task. However, for those who are not concerned with the curvilinear expressions of these equations, the study of this appendix is not essential, except, possibly, for the compatibility conditions of strain.[1]

The background laid on vector algebra and analysis in Appendix A should provide a smooth transition for this appendix.

Tensors are a generalization of vector concepts. They provide powerful tools for the formulation of physical laws in a systematic fashion. In addition, with regard to making the transition to curvilinear coordinates a simple matter, the theory itself suggests various extensions of these laws.

Curvilinear coordinates are presented in Section C2, along with an introduction and illustration of base vectors and metric tensors. In Section C3 we define tensors and briefly study their algebra. Section C4 is devoted to physical components of tensors. In Section C5 we give an account of

[1] An alternative method of derivation of the compatibility conditions is mentioned in Section 1.14 without the use of tensor calculus. For infinitesimal strain, another method exists, cf. Sokolnikoff [1956, Art. 10].

tensor analysis. The concept of covariant differentiation, differential operators, gradient, divergence, and curl are defined and illustrated by examples. Theorems of Green-Gauss and Stokes, essential in the derivation local balance laws, are presented. The last section, C6, presents an account of the Riemann-Christoffel tensor, which is important in the derivation of the compatibility conditions of strain.

There are a few monographs (of various complexity) on the subject. For beginners, we recommend Michal [1947], Sokolnikoff [1951], and Thomas [1961]. For an intermediate level, we recommend Eisenhart [1926], and Synge and Schild [1949] Advanced readers should consult Schouten [1951] and Yano [1957]

C2. CURVILINEAR COORDINATES

Let z^1, z^2, z^3, or z^k ($k = 1, 2, 3$) be rectangular coordinates of a geometrical point, and x^1, x^2, x^3, or x^i ($i = 1, 2, 3$) be three variables. If between z^k and x^i we can establish a correspondence, then we say that there exists a coordinate transformation between z^k and x^i. This can be expressed in the form of three functions

$$z^k = z^k(x^1, x^2, x^3) \qquad (k = 1, 2, 3) \tag{C2.1}$$

If this correspondence is one-to-one, then there exists a unique inverse of (C2.1) in the form

$$x^k = x^k(z^1, z^2, z^3) \qquad (k = 1, 2, 3) \tag{C2.2}$$

It can be shown that such a unique inverse exists in some neighborhood of z^k if the Jacobian

$$J = \det\left(\frac{\partial z^k}{\partial x^i}\right) = \begin{vmatrix} \partial z^1/\partial x^1 & \partial z^1/\partial x^2 & \partial z^1/\partial x^3 \\ \partial z^2/\partial x^1 & \partial z^2/\partial x^2 & \partial z^2/\partial x^3 \\ \partial z^3/\partial x^1 & \partial z^3/\partial x^2 & \partial z^3/\partial x^3 \end{vmatrix} \neq 0 \tag{C2.3}$$

For a fixed set of values of z^1, z^2, z^3, the transformation (C2.1) gives three non-coincident surfaces, called *curvilinear surfaces*, which intersect each other at a single point P with a fixed x^1, x^2, x^3. This point can therefore be marked with the values of x^k called *curvilinear coordinates* of P (Figure C2.1). The intersection of any two of the surfaces (C2.1) gives a line, through P, called a *curvilinear coordinate line*.

EXAMPLE: CYLINDRICAL COORDINATES. The cylindrical coordinates x^k are defined by their relations to rectangular coordinates z^i.

$$z^1 = x^1 \cos x^2, \qquad z^2 = x^1 \sin x^2, \qquad z^3 = x^3 \tag{C2.4}$$

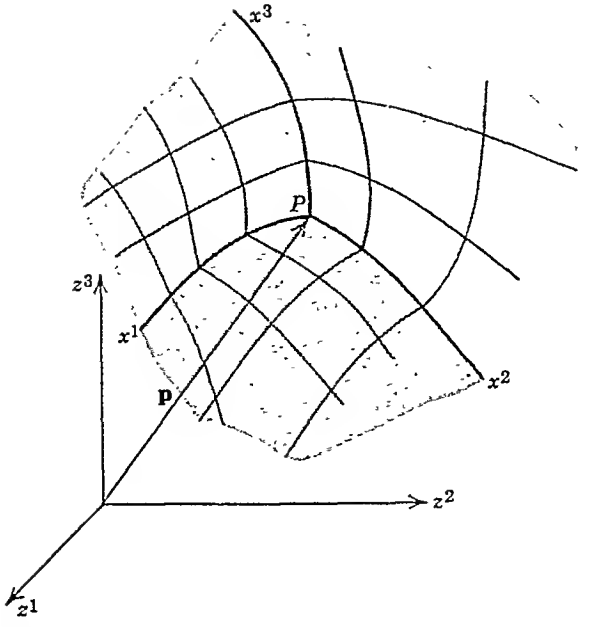


Fig. C2.1 Curvilinear coordinates.

The Jacobian J in this case is

$$J = \begin{vmatrix} \cos x^2 & -x^1 \sin x^2 & 0 \\ \sin x^2 & x^1 \cos x^2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = x^1$$

Hence a unique inverse to (C2.4) exists everywhere except at $x^1 = 0$, and this is

$$\text{(C2.5)} \qquad x^1 = \sqrt{(z^1)^2 + (z^2)^2}, \qquad x^2 = \arctan\,(z^2/z^1), \qquad x^3 = z^3$$

The coordinate surfaces are circular cylinders having the x^3-axis as their axis, vertical planes through the x^3-axis, and planes perpendicular to the x^3-axis (Figure C2.2).

The position vector $\mathbf{p}$ of a point P has the rectangular coordinates z^k, that is,

$$\text{(C2.6)} \qquad \mathbf{p} = z^k \mathbf{i}_k$$

where $\mathbf{i}_k$ ($k = 1, 2, 3$) are the unit rectangular base vectors. The repeated indices in diagonal positions (one as superscript and one as subscript) represent summation over the range ($k = 1, 2, 3$) of the indices. In dealing with curvilinear tensors, this is a much simpler convention than the one

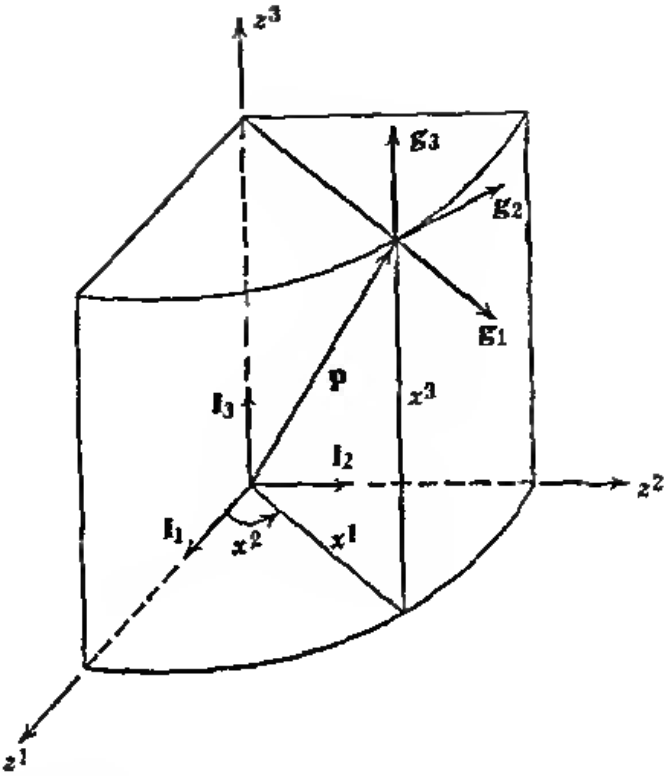


Fig. C2.2 Cylindrical coordinates.

used in Appendix A. When coordinates are rectangular, however, no need arises for the use of the summation convention in this form, and we may revert superscripts back to subscript positions if we wish.

Base vectors $\mathbf{g}_k(x^1, x^2, x^3)$ *are defined by*

$$\mathbf{g}_k = \frac{\partial \mathbf{p}}{\partial x^k} = \frac{\partial z^m}{\partial x^k}\mathbf{i}_m \tag{C2 7}$$

By multiplying both sides of (C2.7) by $\dfrac{\partial x^k}{\partial z^n}$ we also obtain

$$\mathbf{i}_n = \frac{\partial x^k}{\partial z^n}\mathbf{g}_k \tag{C2 8}$$

Just as with the rectangular base vectors $\mathbf{i}_k$, the curvilinear base vectors $\mathbf{g}_k$ are tangent to the curvilinear coordinate lines This is clear from (C2.7) and Figure C2 2. For cylindrical coordinates through (C2.4) and (C2.7) we find that

$$\begin{aligned} \mathbf{g}_1 &= (\cos x^2)\mathbf{i}_1 + (\sin x^2)\mathbf{i}_2 \\ \mathbf{g}_2 &= -(x^1 \sin x^2)\mathbf{i}_1 + (x^1 \cos x^2)\mathbf{i}_2 \\ \mathbf{g}_3 &= \mathbf{i}_3 \end{aligned} \tag{C2.9}$$

An *infinitesimal* vector $d\mathbf{p}$ can be expressed as

$$d\mathbf{p} = \frac{\partial \mathbf{p}}{\partial x^k}\, dx^k = \mathbf{g}_k\, dx^k \tag{C2.10}$$

We may use (C2.10) to calculate the length of such vectors and the angle between any two of them. For example, the square of the arc length, ds^2, of $d\mathbf{p}$ is obtained through

$$\begin{aligned} ds^2 = d\mathbf{p} \cdot d\mathbf{p} &= (\mathbf{g}_k\, dx^k) \cdot (\mathbf{g}_l\, dx^l) \\ &= g_{kl}(\mathbf{x})\, dx^k\, dx^l \end{aligned} \tag{C2.11}$$

where

$$g_{kl} \equiv \mathbf{g}_k \cdot \mathbf{g}_l = \frac{\partial z^m}{\partial x^k} \frac{\partial z^n}{\partial x^l}\, \delta_{mn} \tag{C2.12}$$

is called the *metric tensor*. This name is justified through the fact that when g_{kl} is known we can calculate the length of any vector and the angle between two vectors.

In general, the curvilinear coordinates may not be orthogonal, that is,

$$\mathbf{g}_k \cdot \mathbf{g}_l = g_{kl} \neq 0 \quad \text{for} \quad k \neq l \tag{C2.13}$$

Note that vanishing g_{kl} $(k \neq l)$ is necessary and sufficient for orthogonality of the curvilinear coordinates.

The reciprocal base vectors $\mathbf{g}^k(\mathbf{x})$ may be constructed by finding the solution of the nine equations

$$\mathbf{g}^k \cdot \mathbf{g}_l = \delta^k{}_l \tag{C2.14}$$

where $\delta^k{}_l$ is the Kronecker delta. It can be verified that the unique solution of (C2.14) is

$$\mathbf{g}^k = g^{kl}(\mathbf{x})\mathbf{g}_l \tag{C2.15}$$

where g^{kl} is the reduced cofactor in the determinant of g_{kl}, that is,

$$g^{kl}(\mathbf{x}) = \frac{\text{cofactor } g_{kl}}{g}, \qquad g \equiv \det g_{kl} \tag{C2.16}$$

From (C2.15), by taking the scalar products with $\mathbf{g}^m$ and $\mathbf{g}_m$, we obtain

$$g^{km} = \mathbf{g}^k \cdot \mathbf{g}^m, \qquad \delta^k{}_m = g^{kl} g_{lm} \tag{C2.17}$$

For cylindrical coordinates using (C2.9) in (C2.12) and (C2.16), we find that

$$\|g_{kl}\| = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(C2.18)

$$\|g^{kl}\| = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/(x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The magnitudes of the base vectors $\mathbf{g}_k$ and $\mathbf{g}^k$, respectively, are

$$|\mathbf{g}_k| = \sqrt{g_{\underline{k}\underline{k}}}, \qquad |\mathbf{g}^k| = \sqrt{g^{\underline{k}\underline{k}}} \tag{C2.19}$$

where underscores are placed under the indices to suspend the summation. This convention is used throughout this book.

We notice that in cylindrical coordinates, $g_{12} = g_{23} = g_{31} = 0$. Hence, these coordinates are orthogonal. Moreover, we have $g_{11} = 1/g^{11} = 1$, $g_{22} = 1/g^{22} = (x^1)^2$, $g_{33} = 1/g^{33} = 1$, so that the magnitudes of the vectors $\mathbf{g}_1$ and $\mathbf{g}^1$, $\mathbf{g}_3$ and $\mathbf{g}^3$ are unity. However, $\mathbf{g}_2$ has the magnitude x^1 and $\mathbf{g}^2$, $1/x^1$.

The base vectors $\mathbf{g}_k$ and $\mathbf{g}^k$ may be used to express any vector $\mathbf{v}$ in terms of its parallel projections. Hence, we may write

$$\mathbf{v} = v^k \mathbf{g}_k = v_k \mathbf{g}^k \tag{C2.20}$$

The components v^k and v_k of the same vector $\mathbf{v}$ are in general not identical since $\mathbf{g}_k$ and $\mathbf{g}^k$ are neither of unit magnitude nor orthogonal. When $\mathbf{g}_k$ is a rectangular base vector so is $\mathbf{g}^k$, and the distinction between v^k and v_k disappears. However, in curvilinear coordinates, they are different, and, in fact, by scalar multiplication by $\mathbf{g}^l$ and $\mathbf{g}_l$ from (C2.20) we obtain

$$v^k = g^{kl} v_l, \qquad v_k = g_{kl} v^l \tag{C2.21}$$

Hence, we see that when either v_k or v^k is known we can determine the other by (C2.21) as long as the metric tensor g_{kl} is known. This process is known as *raising* and *lowering* indices.

C3. VECTORS AND TENSORS

DEFINITION 1 *A function $\phi(x^1, x^2, x^3)$ is called an absolute scalar if upon transformation of coordinates it does not change its original value, that is,*

$$\phi(x^1(x'), x^2(x'), x^3(x')) \equiv \phi'(x'^1, x'^2, x'^3) = \phi(x^1, x^2, x^3) \tag{C3.1}$$

The temperature at a point in a body is an absolute scalar.

DEFINITION 2. *The quantities* $A^k(\mathbf{x})$ *and* $A_k(\mathbf{x})$ *are called, respectively, contravariant and covariant components of a vector (or simply contravariant and covariant vectors) if, upon transformation of coordinates, they transform respectively according to the rules*

$$A'^k(\mathbf{x}') = A^m(\mathbf{x}) \frac{\partial x'^k}{\partial x^m} \qquad \text{(contravariant)} \tag{C3.2}$$

$$A'_k(\mathbf{x}') = A_m(\mathbf{x}) \frac{\partial x^m}{\partial x'^k} \qquad \text{(covariant)} \tag{C3.3}$$

For example, the differential vector dx^k is a contravariant vector since according to the chain rule

$$dx'^k = \frac{\partial x'^k}{\partial x^m} dx^m$$

which agrees with (C3.2) when $A^m \equiv dx^m$. Similarly, the partial derivatives of a scalar $\partial\phi/\partial x^k$ are covariant vectors since

$$\frac{\partial \phi}{\partial x'^k} = \frac{\partial \phi}{\partial x^m} \frac{\partial x^m}{\partial x'^k}$$

and this agrees with (C3.3).

DEFINITION 3. *The quantities* $A^{kl}(\mathbf{x})$, $A_{kl}(\mathbf{x})$ *and* $A^k{}_l(\mathbf{x})$ *are called, respectively, contravariant, covariant, and mixed tensors of order two if they transform according to the rules*

$$A'^{kl}(\mathbf{x}') = A^{mn}(\mathbf{x}) \frac{\partial x'^k}{\partial x^m} \frac{\partial x'^l}{\partial x^n} \qquad \text{(contravariant)} \tag{C3.4}$$

$$A'_{kl}(\mathbf{x}') = A_{mn}(\mathbf{x}) \frac{\partial x^m}{\partial x'^k} \frac{\partial x^n}{\partial x'^l} \qquad \text{(covariant)} \tag{C3.5}$$

$$A'^k{}_l(\mathbf{x}') = A^m{}_n(\mathbf{x}) \frac{\partial x'^k}{\partial x^m} \frac{\partial x^n}{\partial x'^l} \qquad \text{(mixed)} \tag{C3.6}$$

Examples of these tensors are g^{kl}, g_{kl}, and $\delta^k{}_l$.

Higher-order tensors are similarly defined. The scalars, the vectors, and the tensors so defined are called absolute. We now define more generally relative tensors. Relative scalars, vectors, and tensors of weight N are defined in a fashion similar to the foregoing except that on the right side of these expressions we have the Nth power of $\det(\partial x'^r/\partial x^s)$, for example,

$$\phi'(\mathbf{x}') = \left|\frac{\partial x^r}{\partial x'^s}\right|^N \phi(\mathbf{x}) \qquad \text{(relative scalar)}$$

(C3 7) $$A'_k(\mathbf{x}') = \left|\frac{\partial x^r}{\partial x'^s}\right|^N A_m(\mathbf{x})\frac{\partial x^m}{\partial x'^k} \qquad \text{(relative covariant vector)}$$

$$A'^k{}_l(\mathbf{x}') = \left|\frac{\partial x^r}{\partial x'^s}\right|^N A^m{}_n(\mathbf{x})\frac{\partial x'^k}{\partial x^m}\frac{\partial x^n}{\partial x'^l} \qquad \text{(relative mixed tensor)}$$

Addition. Two tensors of the same order and type may be added or subtracted to obtain another tensor of the same order and type, for example,

$$C^k{}_{lm} = A^k{}_{lm} + B^k{}_{lm}$$

Multiplication. The outer product of two tensors is obtained by simply multiplying their components, for example,

$$C_{lm}{}^k = A_{lm}B^k$$

Note that the product has an order that is the sum of the orders of the original tensors. In the above example, A_{lm} and B^k are second and first order, respectively. Hence $C_{lm}{}^k$ is third order.

Contraction. In a tensor, by equating a superscript to a subscript, we obtain a summation. The operation is called *contraction*; for example,

$$C_{ml}{}^m$$

A contraction produces a tensor of order two lower than the orginal tensor

The following two theorems are useful in the recognition of tensors

THEOREM 1. *If A_k is an arbitrary covariant vector and A_kX^k is invariant,* that is,

$$A_kX^k = A'_kX'^k$$

then X^k is a contravariant vector.

THEOREM 2. *If A^k is an arbitrary contravariant vector and $B_k = X_{kl}A^l$ is a covariant vector, then X_{kl} is a covariant tensor of the second order.*

A corollary to this theorem is that if A^k and B^l are two arbitrary contravariant vectors and $X_{kl}A^kB^l$ is invariant, then X_{kl} is a covariant tensor of the second order. If $X_{kl}A^kA^l$ is invariant then $X_{kl} + X_{lk}$ is a covariant tensor of the second order. For example, according to (C2.11) we have

$$ds^2 = g_{kl}\,dx^k\,dx^l$$

where the left side is invariant and dx^k is a contravariant vector. Hence

$g_{kl} + g_{lk}$ (or since $g_{kl} = g_{lk}$) g_{kl} is a covariant tensor of the second order. We see that this tensor transforms according to

$$g'_{kl}(\mathbf{x}') = g_{mn}(\mathbf{x}) \frac{\partial x^m}{\partial x'^k} \frac{\partial x^n}{\partial x'^l} \tag{C3.8}$$

Raising and Lowering Indices. By use of g^{kl} and g_{kl} we raise and lower the indices of a tensor to obtain *associated tensors*, for example,

$$\begin{aligned} A^k_{\cdot l} &= g^{km} A_{ml} \\ A^{k}_{l\cdot} &= g^{km} A_{lm} \end{aligned} \tag{C3.9}$$

where a dot is inserted in place of a displaced index. Often this is ignored in writing, so long as we keep the horizontal positions of the indices distinct. Thus $A^k_{\cdot l} \equiv A^k_{\ l}$ is not necessarily the same as $A^{k}_{l\cdot} \equiv A^{k}_{l}$. Both indices of A_{mn} may be raised by repeating the process

$$A^{kl} = g^{km} g^{ln} A_{mn} = g^{km} A_m^{\ l} = g^{ln} A_n^{\ k} \tag{C3.10}$$

When A_{kl} is symmetrical, that is, $A_{kl} = A_{lk}$, so is A^{kl}. In this case, $A^k_{\ l} = A^{k}_{l}$ and the relative horizontal positions of indices become unimportant.

The lowering of indices follows a similar procedure through the use of g_{kl}, for example,

$$\begin{aligned} A^k_{\ l} &= g_{lm} A^{km}, \qquad A^{k}_{l} = g_{lm} A^{mk} \\ A_{kl} &= g_{lm} A_k^{\ m} = g_{km} A^m_{\ l} = g_{km} g_{ln} A^{mn} \end{aligned} \tag{C3.11}$$

We see, therefore, that by first raising the indices of a tensor we obtain an associated tensor. The lowering process when applied to this tensor brings back the original tensor.

C4. PHYSICAL COMPONENTS

A vector $\mathbf{u}$ referred to bases $\mathbf{g}_k$ (and $\mathbf{g}^k$) is expressible in terms of its contravariant (covariant) components by

$$\mathbf{u} = u^k \mathbf{g}_k, \qquad \mathbf{u} = u_k \mathbf{g}^k \tag{C4.1}$$

Since all members of $\mathbf{g}_k$ and $\mathbf{g}^k$ are not, in general, of unit magnitude we see that all members u^k and u_k will not have the same physical dimensions. For example, if $\mathbf{u}$ is a displacement vector then it has the dimension L. Referred to cylindrical coordinates

$$|\mathbf{g}_1| = \sqrt{g_{11}} = 1, \qquad |\mathbf{g}_2| = \sqrt{g_{22}} = x^1, \qquad |\mathbf{g}_3| = \sqrt{g_{33}} = 1$$

so that the dimensions of the components u^1 and u^3 are L, but the dimension of u^2 is $L/L = 1$. Thus, there is a need for finding the physical components of vectors and tensors. This is accomplished by taking the parallel projections of vectors on unit vectors lying along the coordinate curves. We define the physical component $u^{(k)}$ of the vector u^k by

$$\mathbf{u} \equiv u^{(k)}\mathbf{e}_k \tag{C4.2}$$

where $\mathbf{e}_k$ are the unit vectors defined by

$$\mathbf{e}_k \equiv \frac{\mathbf{g}_k}{\sqrt{g_{kk}}} \tag{C4.3}$$

Through $(C4.1)_1$ and (C4.3) we see that

$$\mathbf{u} = u^k\mathbf{g}_k = u^{(k)}\mathbf{e}_k \tag{C4.4}$$

or comparing the components

$$u^{(k)} = u^k\sqrt{g_{kk}}, \qquad u^k = \frac{u^{(k)}}{\sqrt{g_{kk}}} \tag{C4.5}$$

If we want to replace u_k by its physical component, then all we need is to lower the index of u^k, that is,

$$u_k = g_{kl}u^l = \sum_l g_{kl}\frac{u^{(l)}}{\sqrt{g_{ll}}}$$

where we inserted the summation sign again since the index l is repeated more than twice, thus bringing an ambiguity in the order of summations.

An equally consistent definition of physical components may be made by parallel projections on unit vectors lying along $\mathbf{g}^k$. To be consistent with a convention we will always select $\mathbf{e}_k$ lying on $\mathbf{g}_k$.

The physical components of second- and higher-order tensors may be found by their relations to vectors and scalars. In nonorthogonal coordinates, several different types of physical components arise. We consider *here only the case of symmetric second-order tensors with components in* orthogonal coordinates. Suppose that a second-order mixed tensor $t^k{}_l$ is related to a contravariant vector t^k through

$$t^k = t^k{}_l n^l$$

where n^l is also a contravariant vector. Now replace t^k and n^l by their physical components as given by expressions of the form $(C4.5_2)$

$$\frac{t^{(k)}}{\sqrt{g_{kk}}} = \sum_l t^k{}_l \frac{n^{(l)}}{\sqrt{g_{ll}}}$$

or

$$t^{(k)} = \sum_l t^k{}_l \frac{\sqrt{g_{\underline{kk}}}}{\sqrt{g_{\underline{ll}}}} n^{(l)}$$

The physical components $t^{(k)}{}_{(l)}$ of $t^k{}_l$ may then be defined by

$$t^{(k)}{}_{(l)} = t^k{}_l \frac{\sqrt{g_{\underline{kk}}}}{\sqrt{g_{\underline{ll}}}} \tag{C4.6}$$

When the coordinates are orthogonal

$$g^{\underline{kk}} = 1/g_{\underline{kk}}; \qquad g_{kl} = 0, \qquad k \neq l$$

and t_{kl} symmetric we can easily show that

$$t^{(k)}{}_{(l)} = \frac{t_{kl}}{\sqrt{g_{\underline{kk}} g_{\underline{ll}}}} = t^{kl} \sqrt{g_{\underline{kk}} g_{\underline{ll}}} = t^k{}_l \frac{\sqrt{g_{\underline{kk}}}}{\sqrt{g_{\underline{ll}}}} \tag{C4.7}$$

For nonorthogonal coordinates and/or nonsymmetric tensors see the discussion given in Eringen [1962, p. 438].

C5. TENSOR CALCULUS

The partial derivative of a vector in rectangular coordinates is obtained by

$$\frac{\partial \mathbf{u}}{\partial z^l} = \frac{\partial (u^k \mathbf{i}_k)}{\partial z^l} = \frac{\partial u^k}{\partial z^l} \mathbf{i}_k$$

since the rectangular base vectors are constant vectors. However, the same for curvilinear coordinates requires the calculation of the partial derivatives of $\mathbf{g}_k$ for

$$\frac{\partial \mathbf{u}}{\partial x^l} = \frac{\partial (u^k \mathbf{g}_k)}{\partial x^l} = \frac{\partial u^k}{\partial x^l} \mathbf{g}_k + u^k \frac{\partial \mathbf{g}_k}{\partial x^l}$$

Through (C2.7) we can calculate

$$\frac{\partial \mathbf{g}_k}{\partial x^l} = \frac{\partial}{\partial x_l} \left(\frac{\partial z^n}{\partial x_k} \mathbf{i}_n \right) = \frac{\partial^2 z^n}{\partial x^k \partial x^l} \mathbf{i}_n$$

Upon replacing $\mathbf{i}_n$ by (C2.8) we find

$$\frac{\partial \mathbf{g}_k}{\partial x^l} = \begin{Bmatrix} m \\ kl \end{Bmatrix} \mathbf{g}_m \tag{C5.1}$$

where

$$\left\{ \begin{matrix} m \\ kl \end{matrix} \right\} \equiv \frac{\partial^2 z^n}{\partial x^k \partial x^l} \frac{\partial x^m}{\partial z^n} \tag{C5 2}$$

are known as the Christoffel symbols of the second kind. The *Christoffel symbols of the first kind* are also of frequent occurrence; they are defined by

$$[kl, m] \equiv g_{mn} \left\{ \begin{matrix} n \\ kl \end{matrix} \right\} \quad \text{or} \quad \left\{ \begin{matrix} m \\ kl \end{matrix} \right\} \equiv g^{mn}[kl, n] \tag{C5.3}$$

Using (C2 12), we can show that

$$[kl, m] = \frac{1}{2} \left(\frac{\partial g_{km}}{\partial x^l} + \frac{\partial g_{lm}}{\partial x^k} - \frac{\partial g_{kl}}{\partial x^m} \right) \tag{C5.4}$$

Both symbols are symmetric in two indices

$$\left\{ \begin{matrix} m \\ kl \end{matrix} \right\} = \left\{ \begin{matrix} m \\ lk \end{matrix} \right\}, \qquad [kl, m] = [lk, m] \tag{C5.5}$$

The Christoffel symbols are *not* tensors.

By use of (C2.15) we also find

$$\frac{\partial \mathbf{g}^m}{\partial x^l} = - \left\{ \begin{matrix} m \\ lk \end{matrix} \right\} \mathbf{g}^k \tag{C5.6}$$

By use of (C5.1) it is now possible to find an expression for the partial derivative of a vector. Hence,

$$\frac{\partial \mathbf{u}}{\partial x^k} = \frac{\partial}{\partial x^k} (u^m \mathbf{g}_m) = \frac{\partial u^m}{\partial x^k} \mathbf{g}_m + u^m \frac{\partial \mathbf{g}_m}{\partial x^k} = \left(\frac{\partial u^m}{\partial x_k} + \left\{ \begin{matrix} m \\ kl \end{matrix} \right\} u^l \right) \mathbf{g}_m$$

This may be written, in short,

$$\frac{\partial \mathbf{u}}{\partial x^k} = u^m{}_{,k} \mathbf{g}_m \tag{C5.7}$$

thus, defining the *covariant partial derivative* of a contravariant vector

$$u^m{}_{,k} \equiv \frac{\partial u^m}{\partial x^k} + \left\{ \begin{matrix} m \\ kl \end{matrix} \right\} u^l \tag{C5.8}$$

Similarly, by differentiating $\mathbf{u} = u_m \mathbf{g}^m$ and using (C5.6), we obtain the covariant partial derivative of a covariant vector

$$u_{m,k} \equiv \frac{\partial u_m}{\partial x^k} - \left\{ \begin{matrix} l \\ mk \end{matrix} \right\} u_l \tag{C5.9}$$

so that

$$\frac{\partial \mathbf{u}}{\partial x^k} = u_{m;k}\mathbf{g}^m \tag{C5.10}$$

In rectangular coordinates, the Christoffel symbols vanish, thus reducing covariant partial differentiation to the usual partial differentiation.

The covariant derivative of a contravariant vector is a mixed second-order tensor, and that of a covariant vector is a covariant second-order tensor.

Christoffel symbols and the covariant derivative of a contravariant vector u^k in cylindrical coordinates are calculated through (C2.18), (C5.3), and (C5.8)

$$[12, 2] = [21, 2] = x^1, \qquad [22, 1] = -x^1, \qquad \text{all other } [kl, m] = 0$$

$$\left\{\begin{matrix}2\\12\end{matrix}\right\} = \left\{\begin{matrix}2\\21\end{matrix}\right\} = \frac{1}{x^1}, \qquad \left\{\begin{matrix}1\\22\end{matrix}\right\} = -x^1, \qquad \text{all other } \left\{\begin{matrix}m\\kl\end{matrix}\right\} = 0 \tag{C5.11}$$

$$u^1{}_{;1} = \frac{\partial u^1}{\partial x^1}, \qquad u^1{}_{;2} = \frac{\partial u^1}{\partial x^1} - x^1 u^2, \qquad u^1{}_{;3} = \frac{\partial u^1}{\partial x^3}$$

$$u^2{}_{;1} = \frac{\partial u^2}{\partial x^1} + \frac{1}{x^1}u^2, \qquad u^2{}_{;2} = \frac{\partial u^2}{\partial x^2} + \frac{1}{x^1}u^1, \qquad u^2{}_{;3} = \frac{\partial u^2}{\partial x^3} \tag{C5.12}$$

$$u^3{}_{;k} = \frac{\partial u^3}{\partial x^k} \qquad (k = 1, 2, 3)$$

The covariant partial derivatives of higher-order tensors are defined in a similar fashion, for example,

$$A^{kl}{}_{;m} \equiv \frac{\partial A^{kl}}{\partial x^m} + \left\{\begin{matrix}k\\mn\end{matrix}\right\}A^{nl} + \left\{\begin{matrix}l\\mn\end{matrix}\right\}A^{kn}$$

$$A^k{}_{l;m} \equiv \frac{\partial A^k{}_l}{\partial x^m} - \left\{\begin{matrix}n\\lm\end{matrix}\right\}A^k{}_n + \left\{\begin{matrix}k\\mn\end{matrix}\right\}A^{nl} \tag{C5.13}$$

$$A_{kl;m} \equiv \frac{\partial A_{kl}}{\partial x^m} - \left\{\begin{matrix}n\\km\end{matrix}\right\}A_{nl} - \left\{\begin{matrix}n\\lm\end{matrix}\right\}A_{kn}$$

These covariant partial derivatives are third-order tensors.

For a relative scalar ϕ, a vector u^k, and a tensor $A^k{}_l$ of weight N, the covariant derivatives include an extra term, for example,

$$\phi_{,k} \equiv \frac{\partial \phi}{\partial x^k} - N\begin{Bmatrix} r \\ kr \end{Bmatrix}\phi \quad \text{(relative scalar)}$$

(C5.14)
$$u^m{}_{,k} \equiv \frac{\partial u^m}{\partial x^k} + \begin{Bmatrix} m \\ kl \end{Bmatrix}u^l - N\begin{Bmatrix} r \\ kr \end{Bmatrix}u^m \quad \text{(relative vector)}$$

$$A^k{}_{l,m} \equiv \frac{\partial A^k{}_l}{\partial x^m} - \begin{Bmatrix} n \\ lm \end{Bmatrix}A^k{}_n + \begin{Bmatrix} k \\ mn \end{Bmatrix}A^n{}_l - N\begin{Bmatrix} r \\ mr \end{Bmatrix}A^k{}_l$$

(relative mixed tensor)

Applying covariant partial differentiation to g^{kl} and $\delta^k{}_l$ we find that

(C5.15)
$$g_{kl,m} = g^{kl}{}_{,m} = \delta^k{}_{l,m} = 0$$

This theorem is known as *Ricci's theorem. Therefore, in the process of covariant differentiation,* g_{kl}, g^{kl}, and $\delta^k{}_l$ are not affected, that is,

(C5.16)
$$A^k{}_{,l} = (g^{km}A_m)_{,l} = g^{km}A_{m,l}$$
$$A_{k,l} = (g_{km}A^m)_{,l} = g_{km}A^m{}_{,l}$$

We can easily see that

$$(A^kB_{lm})_{,r} = A^k{}_{,r}B_{lm} + A^kB_{lm,r}$$

A useful result that can be proved by differentiation is

(C5.17)
$$\frac{\partial}{\partial x^k}(\log \sqrt{g}) = \begin{Bmatrix} m \\ mk \end{Bmatrix}; \qquad g \equiv \det g_{kl}$$

This result can also be obtained through Ricci's theorem in the form $(\sqrt{g})_{,k} = 0$ and by using $(C5.14)_1$ and noting that $\sqrt{g}$ is a relative scalar of weight one.

The differential operators *gradient*, of an absolute scalar ϕ and the *divergence* and *curl* of an absolute vector A are defined by

(C5.18)
$$\text{grad } \phi \equiv \frac{\partial \phi}{\partial x^k}\mathbf{g}_k$$
$$\text{div } \mathbf{A} = A^k{}_{,k}$$
$$\text{curl } \mathbf{A} = \varepsilon^{klm}A_{m,l}\mathbf{g}_k$$

where

(C5.19)
$$\varepsilon^{klm} \equiv \frac{e^{klm}}{\sqrt{g}}$$

is a third-order absolute tensor known as the ε*-symbol,* and e^{klm} *is the usual* permutation symbol. Also used is the covariant ε-symbol

(C5.20)
$$\varepsilon_{klm} \equiv e_{klm}\sqrt{g}$$

Sometimes it is convenient to use the operator ∇ defined by

$$\nabla \equiv \mathrm{g}^k \frac{\partial}{\partial x^k} \tag{C5.21}$$

By use of this, we can show that

$$\begin{aligned}
\text{grad } \phi &\equiv \nabla\phi = \mathrm{g}^k \frac{\partial \phi}{\partial x^k} \\
\text{div } \mathrm{A} &\equiv \nabla \cdot \mathrm{A} = \mathrm{g}^k \cdot \frac{\partial}{\partial x^k}(A^l \mathrm{g}_l) = \mathrm{g}^k \cdot \mathrm{g}_l A^l{}_{;k} \\
&= A^k{}_{;k} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^k}(\sqrt{g}\, A^k) \qquad \text{(C5.22)} \\
\text{curl } \mathrm{A} &\equiv \nabla \times \mathrm{A} = \mathrm{g}^k \frac{\partial}{\partial x^k} \times (A_l \mathrm{g}^l) = \mathrm{g}^k \times \mathrm{g}^l A_{l;k} \\
&\equiv \epsilon^{klm} A_{m;l} \mathrm{g}_k
\end{aligned}$$

The last expression of div A is obtained as follows:

$$\begin{aligned}
A^k{}_{;k} &= \frac{\partial A^k}{\partial x^k} + \left\{ \begin{matrix} k \\ km \end{matrix} \right\} A^m = \frac{\partial A^k}{\partial x^k} + A^k \frac{\partial}{\partial x^k}(\log \sqrt{g}) \\
&= \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^k}(\sqrt{g}\, A^k)
\end{aligned}$$

where we used (C5.17). For the *Laplacian* ∇^2 in curvilinear coordinates, we have

$$\begin{aligned}
\nabla^2\phi \equiv \text{div grad } \phi &= \left(g^{kl} \frac{\partial \phi}{\partial x^l} \right)_{;k} = g^{kl} \left(\frac{\partial \phi}{\partial x^l} \right)_{;k} \\
&= \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^k} \left(\sqrt{g}\, g^{kl} \frac{\partial \phi}{\partial x^l} \right) \qquad \text{(C5.23)}
\end{aligned}$$

In *orthogonal curvilinear coordinates* the expressions of these operators are:

$$ds^2 = g_{11}(dx^1)^2 + g_{22}(dx^2)^2 + g_{33}(dx^3)^2$$

$$g^{\underline{kk}} = \frac{1}{g_{\underline{kk}}} \qquad \mathrm{g}^k = g^{\underline{kk}} \mathrm{g}_k, \qquad g = g_{11} g_{22} g_{33}$$

$$\begin{aligned}
&\left\{ \begin{matrix} l \\ \underline{k}\underline{k} \end{matrix} \right\} = -\frac{1}{2g_{\underline{ll}}} \frac{\partial g_{\underline{kk}}}{\partial x^l}, \qquad \left\{ \begin{matrix} \underline{k} \\ kl \end{matrix} \right\} = \frac{\partial}{\partial x^l}(\log \sqrt{g_{\underline{kk}}}) \qquad \text{(C5.24)} \\
&\left\{ \begin{matrix} \underline{k} \\ \underline{k}\underline{k} \end{matrix} \right\} = \frac{\partial}{\partial x^k}(\log \sqrt{g_{\underline{kk}}}), \qquad \left\{ \begin{matrix} k \\ lm \end{matrix} \right\} = 0 \qquad (k \neq l \neq m)
\end{aligned}$$

$$\text{grad } \phi = \frac{1}{\sqrt{g_{11}}}\frac{\partial \phi}{\partial x^1}\mathbf{e}_1 + \frac{1}{\sqrt{g_{22}}}\frac{\partial \phi}{\partial x^2}\mathbf{e}_2 + \frac{1}{\sqrt{g_{33}}}\frac{\partial \phi}{\partial x^3}\mathbf{e}_3$$

$$\text{div } \mathbf{A} = (g_{11}g_{22}g_{33})^{-1/2}\left[\frac{\partial}{\partial x^1}\left(\sqrt{g_{22}g_{33}}\,A^{(1)}\right) + \frac{\partial}{\partial x^2}\left(\sqrt{g_{33}g_{11}}\,A^{(2)}\right) + \frac{\partial}{\partial x^3}\left(\sqrt{g_{11}g_{22}}\,A^{(3)}\right)\right]$$

$$\text{curl } \mathbf{A} = (g_{22}g_{33})^{-1/2}\left[\frac{\partial}{\partial x^2}\left(\sqrt{g_{33}}\,A^{(3)}\right) - \frac{\partial}{\partial x^3}\left(\sqrt{g_{22}}\,A^{(2)}\right)\right]\mathbf{e}_1$$

(C5 25)

$$+(g_{33}g_{11})^{-1/2}\left[\frac{\partial}{\partial x^3}\left(\sqrt{g_{11}}\,A^{(1)}\right) - \frac{\partial}{\partial x^1}\left(\sqrt{g_{33}}\,A^{(3)}\right)\right]\mathbf{e}_2$$

$$+(g_{11}g_{22})^{-1/2}\left[\frac{\partial}{\partial x^1}\left(\sqrt{g_{22}}\,A^{(2)}\right) - \frac{\partial}{\partial x^2}\left(\sqrt{g_{11}}\,A^{(1)}\right)\right]\mathbf{e}_3$$

$$\nabla^2\phi = (g_{11}g_{22}g_{33})^{-1/2}\left[\frac{\partial}{\partial x^1}\left(\frac{\sqrt{g_{22}g_{33}}}{\sqrt{g_{11}}}\frac{\partial \phi}{\partial x^1}\right) + \frac{\partial}{\partial x^2}\left(\frac{\sqrt{g_{33}g_{11}}}{\sqrt{g_{22}}}\frac{\partial \phi}{\partial x^2}\right) + \frac{\partial}{\partial x^3}\left(\frac{\sqrt{g_{11}g_{22}}}{\sqrt{g_{33}}}\frac{\partial \phi}{\partial x^3}\right)\right]$$

Cylindrical coordinates (r, θ, z) are defined in terms of rectangular coordinates z^k [we use $x^1 \equiv r$, $x^2 \equiv \theta$, $x^3 \equiv z$, $A^{(1)} \equiv A_r$, $A^{(2)} \equiv A_\theta$, $A^{(3)} \equiv A_z$

$$z^1 = r\cos\theta, \qquad z^2 = r\sin\theta, \qquad z^3 = z$$

$$ds^2 = dr^2 + r^2\,d\theta^2 + dz^2$$

$$g_{11} = g^{11} = g_{33} = g^{33} = 1, \qquad g_{22} = \frac{1}{g^{22}} = r^2$$

$$\begin{Bmatrix} 2 \\ 12 \end{Bmatrix} = \begin{Bmatrix} 2 \\ 21 \end{Bmatrix} = \frac{1}{r}, \qquad \begin{Bmatrix} 1 \\ 22 \end{Bmatrix} = -r \qquad \text{all other } \begin{Bmatrix} k \\ lm \end{Bmatrix} = 0$$

$$\text{grad } \phi = \frac{\partial \phi}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial \phi}{\partial \theta}\mathbf{e}_\theta + \frac{\partial \phi}{\partial z}\mathbf{e}_z$$

$$\text{div } \mathbf{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\theta}{\partial \theta} + \frac{\partial A_z}{\partial z}$$

(C5.26)

$$\text{curl } \mathbf{A} = \left(\frac{1}{r}\frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z}\right)\mathbf{e}_r + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right)\mathbf{e}_\theta + \left[\frac{1}{r}\frac{\partial}{\partial r}(rA_\theta) - \frac{1}{r}\frac{\partial A_r}{\partial \theta}\right]\mathbf{e}_z$$

$$\nabla^2\phi = \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2}$$

Spherical coordinates (r, θ, φ) are defined by (Figure C5.1)

$$z^1 = r\sin\theta\cos\varphi, \qquad z^2 = r\sin\theta\sin\varphi, \qquad z^3 = r\cos\theta$$

$$g_{11} = g^{11} = 1, \qquad g_{22} = \frac{1}{g^{22}} = r^2, \qquad g_{33} = \frac{1}{g^{33}} = r^2\sin^2\theta$$

$$g_{kl} = 0 \qquad (k \neq l)$$

$$\begin{Bmatrix} 1 \\ 22 \end{Bmatrix} = -r, \qquad \begin{Bmatrix} 1 \\ 33 \end{Bmatrix} = -r\sin^2\theta, \qquad \begin{Bmatrix} 2 \\ 12 \end{Bmatrix} = \frac{1}{r}$$

$$\begin{Bmatrix} 2 \\ 33 \end{Bmatrix} = -\sin\theta\cos\theta, \qquad \begin{Bmatrix} 3 \\ 13 \end{Bmatrix} = \frac{1}{r}$$

$$\begin{Bmatrix} 3 \\ 23 \end{Bmatrix} = \cot\theta \quad \text{all other } \begin{Bmatrix} k \\ lm \end{Bmatrix} = 0$$

(C5.27)
$$\operatorname{grad}\phi = \frac{\partial\phi}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\mathbf{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial\phi}{\partial\varphi}\mathbf{e}_\varphi$$

$$\operatorname{div}\mathbf{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(A_\theta\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\varphi}{\partial\varphi}$$

$$\operatorname{curl}\mathbf{A} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(A_\varphi\sin\theta) - \frac{\partial A_\theta}{\partial\varphi}\right]\mathbf{e}_r + \left[\frac{1}{r\sin\theta}\frac{\partial A_r}{\partial\varphi} - \frac{1}{r}\frac{\partial}{\partial r}(rA_\varphi)\right]\mathbf{e}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]\mathbf{e}_\varphi$$

$$\nabla^2\phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\varphi^2}$$

Green-Gauss and Stokes theorems of vector analysis are expressed as

(C5.28)
$$\int_{\mathscr{V}-\sigma}\operatorname{div}\mathbf{u}\,dv + \int_\sigma[\mathbf{u}]\cdot\mathbf{n}\,da = \oint_{\mathscr{S}-\sigma}\mathbf{u}\cdot\mathbf{n}\,da \qquad \text{(Green-Gauss)}$$

(C5.29)
$$\int_{\mathscr{S}-\gamma}\operatorname{curl}\mathbf{A}\cdot\mathbf{n}\,da + \int_\gamma[\mathbf{A}]\cdot\mathbf{h}\,ds = \oint_{\mathscr{C}-\gamma}\mathbf{A}\cdot d\mathbf{p} \qquad \text{(Stokes)}$$

where $\mathbf{n}$ is the exterior normal to the closed surface $\mathscr{S}$ of the volume $\mathscr{V}$ in (C5.28), and in (C5.29) it is the normal to an open surface $\mathscr{S}$ with an oriented boundary curve $\mathscr{C}$. The positive direction of $\mathscr{C}$ is counterclockwise, that is, a right-hand screw advancing in the direction of $\mathbf{n}$ rotates in the positive direction of $\mathscr{C}$. The above theorems are given for a medium $\mathscr{V} + \mathscr{S}$ containing a discontinuity surface σ. In the case of Stokes's theorem, γ is the intersection curve of such a surface with $\mathscr{S}$. The orientations are shown in Figures C5.2 and C5.3.

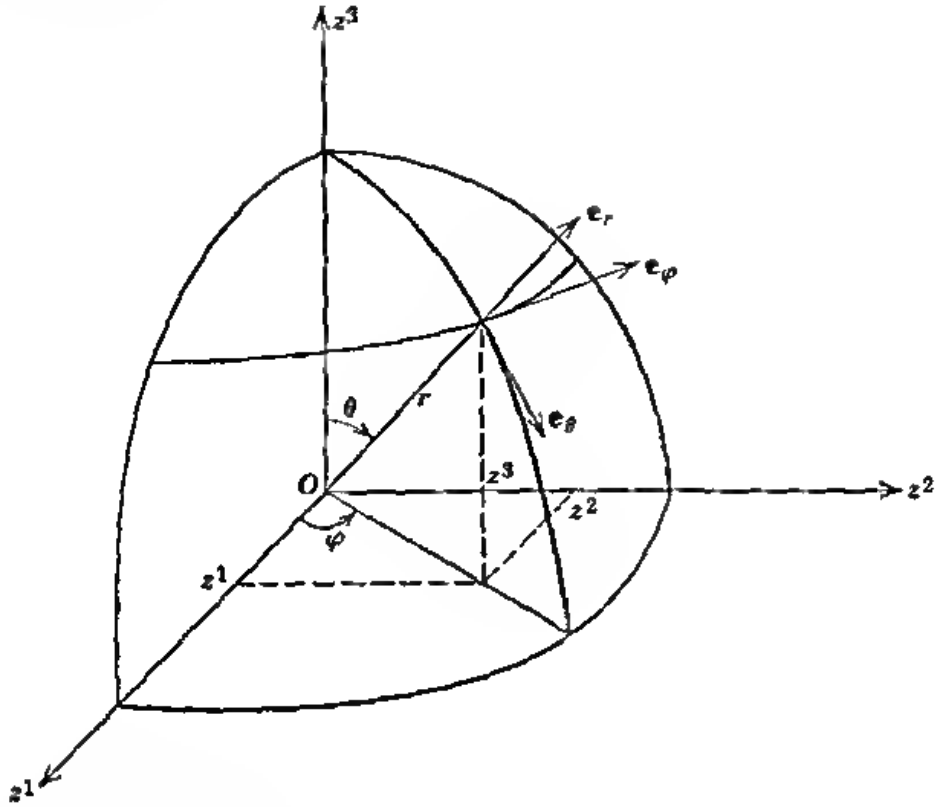


Fig. C5.1 Spherical coordinates.

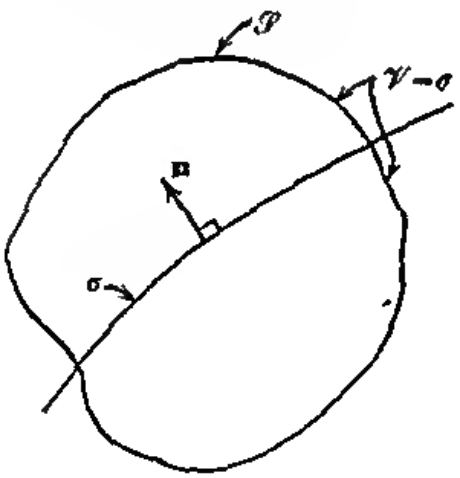


Fig. C5.2

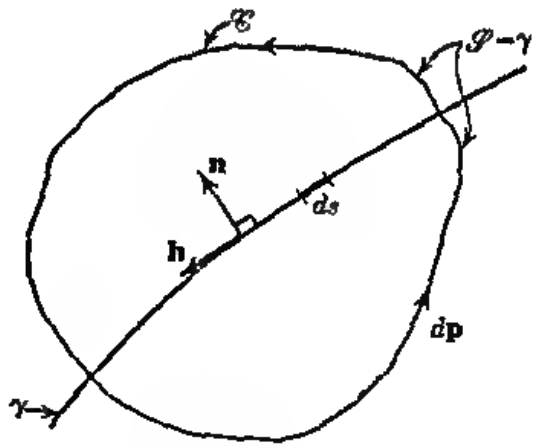


Fig. C5.3.

In component notation, the foregoing equations read

$$\text{(C5 30)} \qquad \int_{\mathscr{V}-\sigma} \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^k} (\sqrt{g}\, u^k)\, dv + \int_{\sigma} [u^k] n_k \, da = \oint_{\mathscr{S}-\sigma} u^k n_k \, da$$

$$\text{(C5.31)} \qquad \int_{\mathscr{S}-\gamma} \varepsilon^{klm} A_{m,l} n_k \, da + \int_{\gamma} [A^k] h_k \, ds = \oint_{\mathscr{C}-\gamma} A_k \, dx^k$$

C6. THE RIEMANN-CHRISTOFFEL CURVATURE TENSOR

According to a theorem of calculus, the order of mixed partial differentiation is not important, that is,

$$\frac{\partial^2\phi}{\partial x_1 \partial x_2} = \frac{\partial^2\phi}{\partial x_2 \partial x_1}$$

It is natural to ask the question: Under what conditions do the second-order covariant partial derivatives of a tensor commute? For a covariant vector A_k, for example, when can we write

$$A_{k;lm} = A_{k;ml}$$

The answer is found by forming both sides of this equation and subtracting. We have

$$A_{k;l} = A_{k,l} - \left\{ \begin{matrix} r \\ kl \end{matrix} \right\} A_r \tag{C6.1}$$

Consequently

$$A_{k;lm} = (A_{k;l})_{,m} - \left\{ \begin{matrix} r \\ km \end{matrix} \right\} A_{r;l} - \left\{ \begin{matrix} r \\ lm \end{matrix} \right\} A_{k;r}$$

or

$$\begin{aligned} A_{k;lm} = A_{k,lm} - \left\{ \begin{matrix} r \\ kl \end{matrix} \right\}_{,m} A_r - \left\{ \begin{matrix} r \\ kl \end{matrix} \right\} A_{r,m} - \left\{ \begin{matrix} r \\ km \end{matrix} \right\} A_{r,l} \\ + \left\{ \begin{matrix} r \\ km \end{matrix} \right\} \left\{ \begin{matrix} s \\ rl \end{matrix} \right\} A_s - \left\{ \begin{matrix} r \\ lm \end{matrix} \right\} A_{k,r} + \left\{ \begin{matrix} r \\ lm \end{matrix} \right\} \left\{ \begin{matrix} s \\ kr \end{matrix} \right\} A_s \end{aligned} \tag{C6.2}$$

Interchanging the indices l and m and subtracting the result from (C6.2) we get

$$A_{k;lm} - A_{k;ml} = R^r{}_{klm} A_r \tag{C6.3}$$

where

$$R^r{}_{klm} \equiv \left\{ \begin{matrix} r \\ km \end{matrix} \right\}_{,l} - \left\{ \begin{matrix} r \\ kl \end{matrix} \right\}_{,m} + \left\{ \begin{matrix} s \\ km \end{matrix} \right\} \left\{ \begin{matrix} r \\ sl \end{matrix} \right\} - \left\{ \begin{matrix} s \\ kl \end{matrix} \right\} \left\{ \begin{matrix} r \\ sm \end{matrix} \right\} \tag{C6.4}$$

is obviously a tensor of fourth order, once contravariant and three times covariant. This tensor is known as the *Riemann-Christoffel* tensor. Clearly, $R^r{}_{klm}$ is independent of the choice of A_r, and it is formed only from the metric tensor g_{kl} and its partial derivatives.

To distinguish Riemann-Christoffel tensors formed from other second-order tensors that can be used as metrics, sometimes we write $R^{(g)r}{}_{klm}$.

It is now clear that

THEOREM. *Cross covariant partial derivatives of any vector are equal if and only if the Riemann-Christoffel tensor vanishes identically.*

By lowering the contravariant index, we obtain a covariant fourth-order tensor called the *curvature tensor.* Thus

$$R_{klmn} = g_{kr}R^r_{lmn} = \tfrac{1}{2}(g_{kn,lm} + g_{lm,kn} - g_{km,ln} - g_{ln,km}) + g^{rs}([lm, s][kn, r] - [ln, s][km, r]) \tag{C6.5}$$

Extension of (C6.3) to higher-order tensors is made in the same manner as above. For example,

$$A_{kl,mn} - A_{kl,nm} = A_{kr}R^r_{lmn} + A_{rl}R^r_{kmn} \tag{C6.6}$$

Both tensors R^k_{lmn} and R_{klmn} possess certain symmetry properties. It can be shown that in an N-dimensional space there are $N^2(N^2-1)/12$ nonvanishing components of R_{klmn}. In three dimensions, the nonvanishing components are R_{1212}, R_{1313}, R_{2323}, R_{1213}, R_{2123}, and R_{3132}. In two dimensions, the only nonvanishing component is R_{1212}.

By taking the covariant derivatives of R^k_{lmn} and using (C5.13) and (C6.4) we can show that

$$\begin{aligned} R^k_{lmn,r} + R^k_{lnr,m} + R^k_{lrm,n} &= 0 \\ R_{klmn,r} + R_{klnr,m} + R_{klrm,n} &= 0 \end{aligned} \tag{C6.7}$$

These are known as *Bianchi's identities.*

The Riemann-Christoffel tensor is a measure of the curvature of the space. Because of this, when this tensor vanishes in a space that space is called *flat. In euclidean space the Riemann-Christoffel tensor vanishes identically.* Thus, the euclidean space is a flat space.

PROBLEMS

1. The parabolic cylinder coordinates, x^k, are related to the rectangular coordinates, z^k, by

$$z^1 = a(x^2 - x^1), \qquad z^2 = 2a\sqrt{x^1x^2}, \qquad z^3 = x^3$$

where a is a constant.

(*a*) Sketch the coordinate surfaces and lines.

(*b*) Determine the base vectors and metric tensors.

(*c*) Express the physical components of a vector in terms of its tensor components.

2. The prolate spheroidal coordinates, x^k, are defined through their relations to the rectangular coordinates, z^k:

$$z^1 = a\{[(x^1)^2 - 1][1 - (x^2)^2]\}^{1/2} \cos x^3$$

$$z^2 = a\{[(x^1)^2 - 1][1 - (x^2)^2]\}^{1/2} \sin x^3, \qquad z^3 = ax^1x^2$$

where a is a constant.

(*a*) Sketch the coordinate surfaces and lines.

(*b*) Determine the base vectors and metric tensors.

(*c*) Find the expressions of the tensor components of a vector in terms of its physical components.

3. The rectangular coordinates are transformed into parabolic cylinder coordinates (see Problem 1). Find the explicit form of the transformation law among:

(*a*) Absolute vectors;

(*b*) Absolute second-order tensors;

(*c*) Relative second-order tensors of degree one.

4. Find the explicit forms of the laws of transformation for absolute vectors and second-order tensors when the cylindrical coordinates are transformed into prolate spheroidal coordinates.

5. Find the relations between metric tensors in cylindrical and parabolic cylinder coordinates.

6. Find the relations of the contravariant components of a vector in spherical and prolate spheroidal coordinates.

7. In cylindrical coordinates (r, θ, z), a second-order absolute tensor, $\mathbf{c}$, has the physical components expressed by the matrix:

$$\|c^{(k)}{}_{(l)}\| = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -K \\ 0 & -Kr^2 & K^2r^2 + 1 \end{bmatrix}, \qquad K \equiv \text{const.}$$

Find the components of this tensor in spheroidal coordinates.

8. If t_{kl} is an absolute covariant tensor, show that $\det(t_{kl})$ is an invariant of weight 2. In this regard, what can be said about $\det(t^k{}_l)$?

9. If t_{kl} is an absolute tensor, prove that the cofactors of the $\det(t_{kl})$ are relative tensors of weight 2.

10. If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are three arbitrary vectors, show that

$$[\mathbf{a}\,\mathbf{b}\,\mathbf{c}]^2 = \begin{vmatrix} a_k a^k & a_k b^k & a_k c^k \\ b_k a^k & b_k b^k & b_k c^k \\ c_k a^k & c_k b^k & c_k c^k \end{vmatrix}$$

11. If A^i, B^i, C^i are arbitrary contravariant vectors and $X^i{}_{lm}$ are some quantities satisfying

$$X^i{}_{lm}A^lB^m \equiv C^i$$

Prove that $X^i{}_{lm}$ is an absolute tensor once contravariant and twice covariant.

12. Prove that the alternating tensors e^{ilm} and e_{ilm} are relative tensors of weight $+1$ and -1, respectively.

13. Prove that the Kronecker deltas defined by

$$\delta^{rst}_{ilm} \equiv e^{rst}e_{ilm}, \qquad \delta^{rs}_{il} \equiv \delta^{rsm}_{ilm}, \qquad \delta^r{}_i \equiv \tfrac{1}{2}\delta^{rs}_{is},$$

are absolute tensors. Prove the following identities:

$$\delta^{rst}_{ilm} = \begin{vmatrix} \delta^r{}_i & \delta^r{}_l & \delta^r{}_m \\ \delta^s{}_i & \delta^s{}_l & \delta^s{}_m \\ \delta^t{}_i & \delta^t{}_l & \delta^t{}_m \end{vmatrix}, \qquad \delta^{rs}_{il} = \begin{vmatrix} \delta^r{}_i & \delta^r{}_l \\ \delta^s{}_i & \delta^s{}_l \end{vmatrix}$$

14. If w^i is an absolute contravariant vector, prove that

$$w_{kl} = e_{klm}w^m$$

is a skew-symmetric second-order tensor of weight -1. Solve w^m from these equations.

15. The toroidal coordinates, x^i, are defined by their relations to the rectangular coordinates, z^i.

$$z^1 = \frac{a \sinh x^1 \cos x^2}{\cosh x^1 - \cos x^2}, \qquad z^2 = \frac{a \sinh x^1 \sin x^3}{\cosh x^1 - \cos x^2}$$

$$z^3 = \frac{a \sin x^2}{\cosh x^1 - \cos x^2}, \qquad a \equiv \text{const.}$$

(*a*) Determine the base vectors and metric tensor.
(*b*) Find the components of the Christoffel symbols of the first and the second kind.
(*c*) Determine the explicit form of the covariant derivative of a contravariant vector.

16. Find the Christoffel symbols in parabolic cylinder coordinates.

17. Express the covariant derivative of a contravariant vector of weight 1 in prolate spheroidal coordinates.

18. Find the expressions of the linear operators gradient, divergence, curl, and laplacian in one of the following coordinates:
(*a*) Parabolic cylinder;
(*b*) Prolate spheroidal;
(*c*) Toroidal.

19. Find the components of the acceleration vector, $\mathbf{a} \equiv \dot{\mathbf{v}}$, in one of the three coordinates mentioned in Problem 18. Here $\mathbf{v} = \mathbf{v}(\mathbf{x}, t)$ is the velocity vector and $\mathbf{x} = \mathbf{x}(\mathbf{X}, t)$ is the equation of motion of a particle, $\mathbf{X}$.

20. Find the components of the vorticity, $\mathbf{w} = \nabla \times \mathbf{v}$, in one of the coordinates mentioned in Problem 18, in terms of the physical components of $\mathbf{v}$.

21. The equation of a surface S in Gaussian form is given by

$$x^k = x^k(u^1, u^2), \qquad (k = 1, 2, 3)$$

where x^k are any curvilinear coordinates and u^α, $(\alpha = 1, 2)$, are Gaussian coordinates covering the surface S.

(*a*) Find the arc length on S.
(*b*) Find the angle between two directions on S.
(*c*) Calculate the area element of S.

22. The arc length on a surface S may be calculated by

$$ds^2 = a_{\alpha\beta}(u^1, u^2)\, du^\alpha\, du^\beta$$

where $a_{\alpha\beta}$ is the metric tensor on S.

(*a*) Find the Riemann tensor $R_{\alpha\beta\gamma\delta}$ of S.
(*b*) For a surface of revolution defined by

$$z^1 = u^1 \cos u^2, \qquad z^2 = u^1 \sin u^2, \qquad z^3 = f(u^3)$$

show that

$$K \equiv \frac{R_{1212}}{\det(a_{\alpha\beta})} = \frac{f'f''}{u^1[1 + (f')^2]^2}$$

23. Find the expression $K \equiv R_{1212}/\det(a_{\alpha\beta})$, in terms of $a_{\alpha\beta}$.

24. If $\mathbf{n}$ is the normal to a surface S, show that

$$d\mathbf{n} \cdot d\mathbf{r} = -b_{\alpha\beta}\, du^\alpha\, du^\beta \qquad \text{on } S$$

where $b_{\alpha\beta} u^\alpha u^\beta \equiv b$ is called the *second fundamental form* with

$$b_{\alpha\beta} \equiv -\frac{1}{2}\left(\frac{\partial \mathbf{n}}{\partial u^\alpha} \cdot \frac{\partial \mathbf{r}}{\partial u^\beta} + \frac{\partial \mathbf{n}}{\partial u^\beta} \cdot \frac{\partial \mathbf{r}}{\partial u^\alpha}\right)$$

25. If $b_{\alpha\beta}$ are given as in Problem 24, show that

(*a*) $b_{\alpha\beta} = x^k{}_{;\alpha\beta}\, n_k$
(*b*) $n^k{}_{;\alpha} = -a^{\beta\gamma}\, b_{\beta\alpha}\, x^k{}_{,\gamma}$ (Weingarten)
(*c*) $b_{\alpha\beta;\gamma} - b_{\alpha\gamma;\beta} = 0$ (Codazzi)
(*d*) $b_{\rho\beta} b_{\alpha\gamma} - b_{\rho\gamma} b_{\alpha\beta} = R_{\rho\alpha\beta\gamma}$ (Gauss)

26. (Short Term Paper) Study literature and give the expressions of gradient, divergence, curl, laplacian, and the acceleration, v, in one of the following coordinates:

(*a*) Oblate spheroidal;
(*b*) Ellipsoidal;
(*c*) Bipolar,
(*d*) Conical;
(*e*) Paraboloidal;
(*f*) Helical.

27. (Short Term Paper) Study literature and present an account of the lie derivative of vectors and tensors

28. (Short Term Paper) Study literature and present a short account of the Einstein theory of gravitation

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(Parenthetical numbers following the entries indicate the sections in which text references are made.)

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